1 **(a)** $2^3 \times 3^2 \times 7$ **[B2]**

M1 for a correct partial factorisation (e.g., 8×63 or 2×252) or a correct prime factor tree/ladder.

(b) 14 **[B2]**

M1 for identifying the needed powers 2×7 or writing $2^4 \times 3^2 \times 7^2$.

[Total Marks: 4]

2 **(a)** $2^2 \times 3 \times 7$ **[B2]**

M1 for $2, 2, 3, 7$ correctly identified in a factor tree or division ladder

(b) 252 **[B2]**

M1 for writing $126 = 2 \times 3^2 \times 7$ or writing multiples of 84 and 126

[Total Marks: 4]

3 **(a)** **(i)** 12 or 36 **[B1]**

(ii) 64 **[B1]**

(iii) 23 **[B1]**

(b) Prime factors: $36 = 2^2 \times 3^2$ and $64 = 2^6$ **[M1]**

LCM $= 2^6 \times 3^2 = 576$ **[A1]**

[Total Marks: 5]

4 **(a)** 13 **[B1]**

(b) $\sqrt{7}$ **[B1]**

(c) -5 **[B1]**

(d) 0.25 **[B1]**

(e) 64 **[B1]**

[Total Marks: 5]

5 **(a)** **(i)** 23 **[B1]**

(ii) 8 or 27 or 64 **[B1]**

(iii) 64 **[B1]**

(b) $2^2 \times 3^2$ (or $2 \times 2 \times 3 \times 3$) **[B2]**

Partial working: Correct factor tree or repeated division showing 2 and 3 as prime factors **[M1]**

[Total Marks: 5]

6 **(a)** **(i)** 2 or 23 **[B1]**

(ii) 27 or 64 **[B1]**

(iii) 64 **[B1]**

(iv) 15 **[B1]**

(b) $2^3 \times 3^2$ **[B2]**

If 0 scored, M1 for $2 \times 2 \times 2 \times 3 \times 3$ or a correct factor tree/division **[M1]**

[Total Marks: 6]

7 **(a)** Factor tree or repeated division showing prime factors 2, 2, 2, 3, 7 **[M1]**

$2^3 \times 3 \times 7$ **[A1]**

(b) $2^2 \times 3 \times 7$ or matching list of factors **[M1]**

84 **[A1]**

(c) $2^3 \times 3^2 \times 7$ or matching list of multiples **[M1]**

504 **[A1]**

[Total Marks: 6]

8 **(a)** **(i)** 13 **[B1]**

(ii) 8 or -1 **[B1]**

(iii) $\sqrt{30}$ or π **[B1]**

(iv) -5 or -1 **[B1]**

(b) $\frac{5}{4}$ or $1\frac{1}{4}$ **[B2]**

M1 for $\frac{1}{0.8}$ or $\frac{10}{8}$ or $\frac{8}{10}$ seen.

[Total Marks: 6]

9 **(a)** 36 (or $2^2 \times 3^2$) **[B2]**

Partial working: Identifies 2^2 and 3^2 as common prime factors **[M1]**

(b) $2^3 \times 3^4 \times 5 \times 7$ **[B2]**

Partial working: One incorrect power or missing prime factor **[B1]**

(c) 66 **[B3]**

Partial working: Identifies that C must contain the factor 11 from the LCM **[M1]**

Partial working: Identifies that C must contain exactly $2^1 \times 3^1$ to satisfy the HCF condition **[M1]**

[Total Marks: 7]

10 **(a)** Irrational **[B1]**

(b) Any valid example, e.g. $\sqrt{2}$, $\sqrt{3}$, π (if $y^2 = \pi^2$ which is irrational, so this is wrong)

Only square roots of non-square rational numbers, e.g. $\sqrt{2}$ **[B1]**

(c) **(i)** 53 and Yes **[B2]**

M1 for substituting $n = 3$: $3^2 + 3 + 41$

(ii) Substituting $n = 41$ yields $41^2 + 41 + 41$ **[M1]**

This factorises to $41(41 + 1 + 1) = 41 \times 43$, which has factors other than 1 and itself, therefore it is not prime. **[A1]**

[Total Marks: 6]

11 **(a)** 2 **[B3]**

M2 for $540 = 2^2 \times 3^3 \times 5$ and $180 = 2^2 \times 3^2 \times 5$ written as prime products

or M1 for either 540 or 180 correctly written as a product of prime factors.

(b) $y = 4$ **[B1]**

$z = 3$ **[B1]**

(c) 15 **[B2]**

M1 for identifying the needed powers to reach multiples of 3, e.g. $3^1 \times 5^1$.

[Total Marks: 7]

12 **(a)** 40 **[B2]**

If 0 scored, allow [M1] for setting up the equation $12x = 4 \times 120$ or finding prime factors of 120 indicating $2^3 \times 5$.

(b) **(i)** $2^4 \times 3^2 \times 7$ **[B2]**

If 0 scored, allow [M1] for a correct factor tree or continuous division with at least 2 correct prime division steps.

(ii) 216 **[B3]**

If 0 scored, allow [M1] for writing $72 = 2^3 \times 3^2$.

Allow a further [M1] for deducing that N must contain $2^3 \times 3^3$ to be the smallest perfect cube with the correct HCF (power of 2 is exactly 3, power of 3 is at least 3).

[Total Marks: 7]

13 **(a)** **(i)** pq **[B1]**

(ii) p^3q^3 **[B1]**

(b) $pq = 15$, so the prime numbers are 3 and 5 **[M1]**

$m > n \implies p^3q > pq^3 \implies p^2 > q^2 \implies p > q$ **[M1]**

$p = 5$ and $q = 3$ **[A1]**

(c) $m \times n = (p^3q)(pq^3) = p^4q^4$ **[M1]**

$= (p^2q^2)^2$ which is the square of an integer, hence a square number **[A1]**

(d) Substitutes values: $m = 5^3 \times 3^1$ **[M1]**

Recognises the power of 3 needs to be a multiple of 3, so $k = 3^2 = 9$ **[A1]**

[Total Marks: 9]

14 (a) $p \times q = \text{LCM} \times \text{HCF}$ or $70 \times q = 1260 \times 14$ **[M1]**

$q = \dfrac{17640}{70}$ **[M1]**

252 **[A1]**

(b) Identifying $72 = 2^3 \times 3^2$ or $10080 = 2^5 \times 3^2 \times 5 \times 7$ **[M1]**

Comparing powers of 2 for LCM to establish $x = 5$ **[M1]**

$x = 5$ **[A1]**

$y = 2$ **[A1]**

(c) Factorising expression to $n(n + 1)$ **[M1]**

Valid statement explaining that n and $n + 1$ are consecutive integers, therefore one must be a multiple of 2 (even), making their product always even **[A1]**

[Total Marks: 9]

15 $x = 2$ **[B2]**

$y = 3$ **[B2]**

If 0 scored, award M1 for expressing $180 = 2^2 \times 3^2 \times 5$ or $12600 = 2^3 \times 3^2 \times 5^2 \times 7$

[Total Marks: 4]

16 (a) 36 (or $2^2 \times 3^2$) **[B1]**

(b) $2^3 \times 3^3 \times 5 \times 7$ **[B2]**

If 0 scored, allow [M1] for a product with at least three correct prime bases and powers, or for the correct evaluation 7560.

(c) 10 **[B2]**

If 0 scored, allow [M1] for identifying required missing powers 2×5 or $2^1 \times 5^1$.

[Total Marks: 5]

17 (a) Identifies HCF as $2^1 \times 3^1$ **[M1]**

$x = 1$ **[A1]**

(b) Prime factorises $8100 = 81 \times 100 = 3^4 \times 2^2 \times 5^2$ **[M1]**

Compares highest powers of 3 between A (3^3) and C (3^y) to the LCM (3^4) **[M1]**

$y = 4$ **[A1]**

(c) $\sqrt{11}$ **[B1]**

4π **[B1]**

[Total Marks: 7]

18 (a) (i) Any correct rational number (e.g., 0.45 or $\frac{9}{20}$) **[B1]**

(ii) Any correct irrational number (e.g., $\sqrt{17}$ or $\pi + 1$) **[B1]**

(b) Two different prime numbers identified **[M1]**

Correct sum equating to a square number (e.g., $2 + 7 = 9$ or $13 + 3 = 16$) **[A1]**

(c) 21 **[B1]**

(d) 12 or a^2b^2 seen in the final answer **[M1]**

$12a^2b^2$ **[A1]**

[Total Marks: 7]

19 (a) 12 **[B2]**

If 0 scored, M1 for $84 = 2^2 \times 3 \times 7$ and $120 = 2^3 \times 3 \times 5$ OR for a common factor of 4 or 6 identified **[M1]**

(b) 840 **[B2]**

If 0 scored, M1 for $2^3 \times 3 \times 5 \times 7$ **[M1]**

(c) $\frac{5}{4}$ or $1\frac{1}{4}$ **[B2]**

If 0 scored, M1 for $\frac{1}{0.8}$ or $\frac{10}{8}$ seen **[M1]**

(d) Irrational, because 120 is not a perfect square (or $\sqrt{120}$ cannot be written as a simple fraction) **[B2]**

If 0 scored, B1 for Irrational with no/incorrect reason **[B1]**

[Total Marks: 8]

20 (a) (i) $x = 2$, $y = 8$, $z = 10$ **[B2]**

Partial working: Lists correctly prime numbers $x < 10$ and cube numbers $y > 1$ **[M1]**

(ii) $x = 7$, $y = 8$, $z = 15$ OR $x = 2$, $y = 64$, $z = 66$ **[B2]**

Partial working: Identifies another valid combination of a prime and cube summing to a triangle number **[M1]**

(b) (i) e.g. $p = \sqrt{2}$, $q = -\sqrt{2}$ OR $p = 1 + \sqrt{3}$, $q = 1 - \sqrt{3}$ **[B1]**

(ii) e.g. $p = \sqrt{2}$, $q = \sqrt{8}$ OR $p = \sqrt{3}$, $q = 2\sqrt{3}$ **[B1]**

[Total Marks: 6]

21 **(a)** $x = 2$ **[B1]**

$y = 2$ **[B1]**

M1 for $84 = 2^2 \times 3 \times 7$ OR $3528 = 2^3 \times 3^2 \times 7^2$ seen **[M1]**

(b) 1764 **[B1]**

(c) 10:21:00 **[B3]**

M2 for LCM of 45 and 72 = 360 seconds (or 6 minutes) **[M2]**

If M2 not scored, M1 for listing multiples of 45 and 72 with at least one common multiple, OR $45 = 3^2 \times 5$ and $72 = 2^3 \times 3^2$ **[M1]**

[Total Marks: 7]

22 **(a)** Correct elements in all four regions: 2 in the intersection, $4, 6, 8, 10$ in A only, $3, 5, 7$ in B only, and $1, 9$ outside both circles **[B2]**

If 0 scored, allow [B1] for 2 or 3 correct regions

(b) $\{2\}$ or 2 **[B1]**

(c) 2 **[B1]**

[Total Marks: 4]

23 **(a)** 2, 3, 5, 7 **[B1]**

(b) 1, 2, 3, 4, 6, 12 **[B1]**

(c) 8 **[B1]**

(d) 2, 3 **[B1]**

(e) 8 or 9 or 10 **[B1]**

[Total Marks: 5]

24 **(a)** $\{3, 5, 7\}$ **[B1]**

(b) 4 **[B2]**

M1 for correctly identifying $(A \cup B)' = \{4, 6, 8, 10\}$ or $A \cup B = \{1, 2, 3, 5, 7, 9\}$.

(c) $\in$ **[B1]**

(d) $\subset$ or $\subseteq$ **[B1]**

[Total Marks: 5]

25 **(a)** Intersection: $3, 5, 7$ **[B1]**

Remainder of diagram correct: A only region has 2; B only region has $1, 9$; Outer region has $4, 6, 8, 10$ **[B1]**

(b) 2 **[B1]**

(c) 6 **[B1]**

(d) $A \cap B'$ **[B1]**

[Total Marks: 5]

26 **(a)** $1, 2, 3, 6, 9$ **[B1]**

(b) $2, 3$ **[B1]**

(c) 3 **[B2]**

Partial working: Correctly identifies elements in the union $\{1, 2, 3, 5, 6, 7, 9\}$ OR elements outside $\{4, 8, 10\}$ **[M1]**

(d) $\notin$ **[B1]**

[Total Marks: 5]

27 **(a)** $2, 3, 5, 7$ **[B1]**

(b) Listing elements of $A \cup B$ as $1, 2, 3, 5, 7, 10$ **[M1]**

6 **[A1]**

(c) **(i)** $\notin$ **[B1]**

(ii) $\subseteq$ **[B1]**

[Total Marks: 5]

28 **(a)** 4 **[B2]**

M1 for $6 + 3 + 7 + 1 + 2 + 4 + 8 + x = 35$ or $31 + x = 35$.

(b) 9 **[B1]**

(c) $A' \cap B \cap C$ or $B \cap C \cap A'$ **[B1]**

(d) 16 **[B2]**

M1 for correctly identifying $n(B \cap C) = 6$ or summing $12 + 4$.

[Total Marks: 6]

29 **(a)** 7 **[B3]**

Partial working: Correct simplified equation $7x + 3 = 52$ **[M2]**

Partial working: Sets up unsimplified sum $(2x + 1) + x + (3x - 2) + (x + 4) = 52$ **[M1]**

(b) 33 **[B2]**

Partial working: Evaluates correct sub-regions for $x = 7$ (e.g. $15 + 7 + 11$) OR identifies correct algebraic expression $4x + 5$ **[M1]**

(c) Region strictly inside A but outside B is shaded (the left crescent) **[B1]**

[Total Marks: 6]

30 **(a)** Identifies correct boundary values -2 and 3 **[M1]**

$-2 < x < 3$ **[A1]**

(b) Identifying $R' = \{x : x \leq 1\}$ leading to intersection $-2 < x \leq 1$ **[M1]**

$-1, 0, 1$ **[A1]**

(c) Recognising that $P \cap S = P$ implies $P \subseteq S$ **[M1]**

$k \geq 5$ **[A1]**

(d) Identifies one correct distinct interval, e.g., $-4 \leq x \leq -1$ or $2 \leq x < 3$ **[M1]**

$-4 \leq x \leq -1$ and $2 \leq x < 3$ (accept 'or' between them) **[A1]**

[Total Marks: 8]

31 **(a)** Intersection: x and Outer region: 4 **[B1]**

P only region: $28 - x$ AND C only region: $22 - x$ **[B1]**

(b) $(28 - x) + x + (22 - x) + 4 = 45$ oe **[M1]**

$x = 9$ **[A1]**

(c) 19 **[B1]**

[Total Marks: 5]

32 **(a)** Summing all regions: $(x + 5) + 2x + x + 3 + 4 + 2 + 5 + (x + 1) = 45$ (or equivalent) **[M1]**

$5x + 20 = 45 \implies 5x = 25$ **[M1]**

$x = 5$ **[A1]**

(b) 34 **[B1]**

(c) Shading exactly the region representing P only (the top part of circle P, excluding the intersections with Q and R) **[B1]**

[Total Marks: 5]

33 **(a)** 5 in outer region **[B1]**

8 in intersection **[B1]**

17 in F only AND 10 in S only **[B1]**

(b) 17 **[B1]**

(c) $\dfrac{27}{40}$ **[B2]**

M1 for $17 + 10$ or 27 seen **[M1]**

(d) $\dfrac{8}{25} \times \dfrac{7}{24}$ **[M1]**

$\dfrac{7}{75}$ **[A1]**

[Total Marks: 7]

34 **(a)** $(x+2)+2+3+(x-1)+4+2x+x+1=36$ **[M1]**

$5x+11=36$ or $5x=25$ **[M1]**

5 **[A1]**

(b) Substituting $x=5$ into $(x+2)+2+x+3$ **[M1]**

17 **[A1]**

(c) Identifying the correct regions: $(2x)+3$ **[M1]**

13 **[A1]**

(d) The region strictly inside F, but outside both T and S shaded. (The 'Football only' region) **[B1]**

[Total Marks: 8]

35 **(a)** Intersection of all three: 8 AND Outer region: 3 **[B1]**

Two-set only intersections: 12, 7, and 10 in correct regions **[B1]**

Single set regions: x (in C), $2x$ (in D), and x (in B) **[B1]**

(b) $x+2x+x+8+12+7+10+3=100$ oe **[M1]**

$x=15$ **[A1]**

(c) Identifies the regions required: Only C + Only D + ($C \cap D$ only) OR uses $x+2x+12$ **[M1]**

57 **[A1]**

[Total Marks: 7]

36 **(a)** 4 in center intersection **[B1]**

8, 5, 7 in the two-sport intersections **[B1]**

13, 9, 9 in the one-sport regions AND 5 in outer region **[B1]**

(b) 5 **[B1]**

(c) 17 **[B1]**

(d) $B \cap T' \cap S'$ or $B \cap (T \cup S)'$ **[B2]**

M1 for $B \cap \ldots$ **[M1]**

[Total Marks: 7]

37 **(a)** $2x$ in the T only region **[B1]**

x in the intersection and $3x$ in the C only region **[B1]**

2 in the region outside both circles **[B1]**

(b) 8 **[B2]**

M1 for $2x+x+3x+2=50$ or $6x=48$.

(c) 16 **[B1]**

(d) Region outside T shaded, as well as the entirety of C (the only unshaded region is 'T only') **[B2]**

B1 for shading T' correctly OR shading C correctly.

[Total Marks: 8]

38 (a) Places 15 in the central intersection **[B1]**

Places 15 in $B \cap T$ only, 13 in $T \cap C$ only, and 7 in $B \cap C$ only **[B1]**

Places x, $2x$, $x + 5$, and 8 in the respective remaining regions (B only, T only, C only, and outside the sets) **[B1]**

(b) $x + 2x + (x + 5) + 15 + 15 + 13 + 7 + 8 = 103 \implies 4x + 63 = 103$ (or equivalent valid method) **[M1]**

$x = 10$ **[A1]**

(c) 85 **[B1]**

(d) $\dfrac{13}{63}$ **[B2]**

If 0 scored, allow [M1] for correctly identifying the denominator $n(T) = 63$ or the numerator $n(T \cap C \cap B') = 13$

[Total Marks: 8]

39 (a) All 8 regions correctly populated **[B3]**

Partial working: 5 to 7 regions correctly populated **[B2]**

Partial working: 3 to 4 regions correctly populated (e.g. places $12, 15, x$ correctly) **[B1]**

(Correct values are: Center = 12. $M \cap D$ only = 13. $D \cap A$ only = 20. $M \cap A$ only = 16. M only = 9. D only = 20. A only = x. Outside = 15)

(b) 15 **[B2]**

Partial working: Forms correct sum $9 + 20 + x + 13 + 16 + 20 + 12 + 15 = 120$ OR $105 + x = 120$ **[M1]**

(c) 70 **[B2]**

Partial working: Identifies the 20 students in only Drama and Art OR correctly calculates $50 + 20$ **[M1]**

(d) Region strictly inside M but outside both D and A is shaded (the top crescent / "only M" region) **[B1]**

[Total Marks: 8]

40 (a) 11 **[B1]**

(b) 61 **[B2]**

M1 for 64 or 3 seen.

[Total Marks: 3]

41 **(a)** 8×9 **[M1]**

72 **[A1]**

(b) 5 **[B1]**

[Total Marks: 3]

42 **(a)** 21 **[B2]**

M1 for 13 or 8

(b) 80 **[B2]**

M1 for 16 or 5

[Total Marks: 4]

43 **(a)** Calculates $3^3 = 27$ and $\sqrt{144} = 12$ **[M1]**

15 **[A1]**

(b) Calculates $2^4 = 16$ and $\sqrt[3]{125} = 5$ **[M1]**

80 **[A1]**

[Total Marks: 4]

44 **(a)** 14 **[B1]**

(b) 59 **[B2]**

If 0 scored, M1 for 64 or 5 seen **[M1]**

(c) 5 **[B2]**

If 0 scored, M1 for $169 - 144$ or 25 seen **[M1]**

[Total Marks: 5]

45 **(a)** 14 **[B1]**

(b) 48 **[B2]**

If 0 scored, allow [M1] for correctly evaluating either $\sqrt[3]{27} = 3$ or $2^4 = 16$

(c) 8.8 or $\frac{44}{5}$ **[B2]**

If 0 scored, allow [M1] for a correct numerator of 44 or a correct denominator of 5

[Total Marks: 5]

46 **(a)** $\frac{37}{7}$ or $5\frac{2}{7}$ **[B2]**

M1 for $25 + 12$ or $8 - 1$ seen.

(b) 4 **[B2]**

M1 for 9×9 or $3^2 \times 3^2$ or 81 seen.

[Total Marks: 4]

47 **(a)** Numerator correctly evaluated to $64 + 15 = 79$ **[B1]**

Denominator correctly evaluated to 4 **[B1]**

19.75 **[B1]**

(b) Substitution leading to $25 - (-8)$ **[M1]**

33 **[A1]**

[Total Marks: 5]

48 **(a)** 5 **[B1]**

(b) 150 **[B2]**

If 0 scored, allow [M1] for 6×5^2 or 6×25

(c) 121 **[B2]**

If 0 scored, allow [M1] for evaluating $3^3 - 4^2 = 11$

[Total Marks: 5]

49 **(a)** $\frac{15 + 10}{25}$ **[M1]**

$\left(\frac{25}{25}\right)^{-2}$ or 1^{-2} **[M1]**

1 **[A1]**

(b) $x^{\frac{2}{3}} = 4$ **[M1]**

$x = 4^{\frac{3}{2}}$ or $(\sqrt{4})^3$ or $x^2 = 4^3 = 64$ **[M1]**

8 **[A1]**

[Total Marks: 6]

50 **(a)** $3x + 4 = 64$ **[M1]**

$3x = 60$ **[M1]**

20 **[A1]**

(b) $x^2 = 64 + 36 = 100$ **[M1]**

$10, -10$ **[A1]**

(c) $3\sqrt{x} = 21$ or $\sqrt{x} = 7$ **[M1]**

49 **[A1]**

[Total Marks: 7]

51 (a) 125 **[B2]**

If 0 scored, M1 for evaluating $16 - 9 - 2$ to get 5 **[M1]**

(b) -2 **[B3]**

M1 for $\sqrt{64}$ or 8 **[M1]**

M1 for $5 - 9$ or -4 **[M1]**

(c) 729 **[B3]**

M1 for $2\sqrt[3]{y} = 6$ or $\sqrt[3]{y} = 3$ **[M1]**

M1 for $y = 27$ **[M1]**

[Total Marks: 8]

52 (a) 5.3990299... **[B2]**

M1 for numerator evaluated to 163.91... or denominator evaluated to 30.359...

(b) 5.399 **[B1]**

(c) 6 **[B2]**

M1 for evaluating $6^4 = 1296$ and $7^4 = 2401$

[Total Marks: 5]

53 (a) (i) $1.49315\ldots$ **[B2]**

If 0 scored, M1 for 3.329 seen or 72.25 and 68.921 **[M1]**

(ii) 1.49 **[B1]**

(b) 96 **[B2]**

If 0 scored, M1 for 23.04 or 0.24 seen **[M1]**

[Total Marks: 5]

54 (a) $\sqrt{151.83}$ or $12.32\ldots$ or 29.791 **[M1]**

$0.4136\ldots$ **[A1]**

(b) $p \times q = 3402$ **[M1]**

$\sqrt[3]{3402}$ or $15.0401\ldots$ **[M1]**

15.0 **[A1]**

[Total Marks: 5]

55 **(a)** 1.59 **[B2]**

M1 for $23.02\ldots$ or 14.44 seen.

(b) 1734 **[B3]**

M1 for $\sqrt[3]{4913}$ or 17 seen.

M1 for 6×17^2.

[Total Marks: 5]

56 **(a)** 1.5 **[B1]**

(b) Rearranges to $2x^4 = 1250$ leading to $x^4 = 625$ **[M1]**

$x = 5$ **[B1]**

$x = -5$ **[B1]**

(c) Correct substitution: $\sqrt{\dfrac{3(4.2)^2 + (-5.7)}{1.8}}$ **[M1]**

Evaluates numerator to 47.22 OR fraction to $26.2333\ldots$ **[M1]**

5.12 **[A1]**

[Total Marks: 7]

57 **(a)** 0.239 **[B2]**

If 0 scored, allow [M1] for evaluating the numerator as 19.679... or the denominator as 82.464...

(b) 64 **[B3]**

If 0 scored, allow [M1] for isolating x^2 correctly, e.g., $x^2 = 16^3$ or $x^2 = 4096$

Allow a further [M1] for $x = \sqrt{4096}$

(c) 7.44 or 7.443... **[B3]**

If 0 scored, allow [M1] for finding the volume of the cube as 12^3 or 1728

Allow a further [M1] for correctly rearranging the sphere volume formula to solve for r, e.g., $r^3 = \dfrac{1728 \times 3}{4\pi}$ or $r = \sqrt[3]{\dfrac{1296}{\pi}}$

[Total Marks: 8]

58 **(a)** $\dfrac{5}{8}$ 64% 0.65 $\dfrac{2}{3}$ **[B2]**

M1 for correct conversions to decimals seen: $0.666\ldots$, 0.64, 0.625

(b) $\dfrac{17}{20}$ **[B1]**

[Total Marks: 3]

59 **(a)** $\frac{7}{9}$ **[B1]**

(b) $\frac{36}{99}$ **[M1]**

$\frac{4}{11}$ **[A1]**

[Total Marks: 3]

60 $\frac{10}{3} \div \frac{14}{5}$ **[M1]**

$\frac{10}{3} \times \frac{5}{14}$ or $\frac{25}{21}$ **[M1]**

$\frac{5}{6}$ **[A1]**

[Total Marks: 3]

61 **(a)** $\frac{9}{20}$ **[B1]**

(b) 0.36 **[B1]**

(c) $\frac{204}{100}$ or $2\frac{4}{100}$ **[M1]**

$2\frac{1}{25}$ **[A1]**

[Total Marks: 4]

62 **(a)** $\frac{9}{20}$ **[B1]**

(b) 0.375 **[B1]**

(c) 0.4 $\frac{5}{12}$ 42% $\frac{3}{7}$ **[B2]**

If 0 scored, allow [M1] for converting at least two of the fractions or percentages to decimals correctly, e.g., $0.416\ldots$, 0.42, $0.428\ldots$

[Total Marks: 4]

63 **(a)** $\frac{75}{1000}$ or equivalent unsimplified fraction **[M1]**

$\frac{3}{40}$ **[A1]**

(b) 3.4 **[B1]**

(c) M1 for converting at least two of the other forms to decimals or percentages correctly (e.g. $\frac{5}{8} = 0.625$, $\frac{7}{11} = 0.636\ldots$, $62\% = 0.62$) **[M1]**

0.6 62% $\frac{5}{8}$ $\frac{7}{11}$ **[A1]**

[Total Marks: 5]

64 Length multiplier $= \frac{4}{3}$ **[B1]**

Area multiplier $= \frac{7}{6}$ **[B1]**

$\frac{4}{3} \times$ Width multiplier $= \frac{7}{6}$ oe **[M1]**

Width multiplier $= \frac{7}{6} \times \frac{3}{4} = \frac{7}{8}$ **[M1]**

$x = 12.5$ **[A1]**

[Total Marks: 5]

65 (a) 60 **[B3]**

M1 for $60\% = \frac{3}{5}$ or $\frac{1}{5} = 20\%$ seen

M1 for $\frac{3}{5} - \frac{1}{5} = \frac{2}{5}$ or $60\% - 20\% = 40\%$

M1 for $\frac{24}{2} \times 5$ or $\frac{24}{40} \times 100$

(b) 400 **[B3]**

M1 for $100\% - 30\% = 70\%$ or $0.7x$ seen

M1 for $\frac{3}{4} \times 0.7x = 210$ or $0.525x = 210$ or equivalent equation

[Total Marks: 6]

66 (a) Total number of counters $= x + (2x + 5) + (3x - 1) = 6x + 4$ **[M1]**

$\frac{2x+5}{6x+4} = \frac{35}{100}$ or $\frac{7}{20}$ **[M1]**

$20(2x + 5) = 7(6x + 4)$ leading to $40x + 100 = 42x + 28$ **[M1]**

36 **[A1]**

(b) Substituting $x = 36$ to find green counters $= 3(36) - 1 = 107$ and total counters $= 6(36) + 4 = 220$ **[M1]**

$\frac{107}{220} \times 100$ **[M1]**

$48\frac{7}{11}$ **[A1]**

[Total Marks: 7]

67 (a) $1000x = 245.4545\ldots$ and $10x = 2.4545\ldots$ or $100x = 24.5454\ldots$ and $x = 0.24545\ldots$ **[M1]**

$990x = 243$ or $99x = 24.3$ **[M1]**

$\frac{27}{110}$ **[A1]**

(b) Identifies that the denominator $42 = 2 \times 3 \times 7$ (or states that 42 has prime factors other than 2 and 5) **[B1]**

Terminating decimals can only have prime factors of 2 and/or 5 in their denominator. Since the fraction is in its simplest form, the factors of 3 and 7 do not cancel out, so it must recur. **[B1]**

(c) $\dfrac{10x+y}{99} = \dfrac{k}{11}$ or $\dfrac{9k}{99}$ (or states that the two-digit number xy is a multiple of 9) **[M1]**

9 **[A1]**

[Total Marks: 7]

68 (a) States appropriate pair of equations to eliminate the recurring part, e.g., $100x = 27.77\ldots$ and $10x = 2.77\ldots$ **[M1]**

Subtracts to get $90x = 25$ leading to $x = \dfrac{25}{90}$, and correctly simplifies this to $\dfrac{5}{18}$ **[A1]**

(b) (i) 36 **[B3]**

If 0 scored, allow [M1] for calculating the fraction or percentage of the whole school for either boys or girls, e.g., $\dfrac{2}{5} \times 45\%$ or $\dfrac{3}{5} \times 30\%$

Allow [M1] for evaluating either as 18%

(ii) 400 **[B2]**

If 0 scored, allow [M1] for a correct algebraic setup or division, e.g., $\dfrac{144}{0.36}$ or $36\% = 144$

[Total Marks: 7]

69 (a) $x - \dfrac{1}{3}x + 12$ or $\dfrac{2}{3}x$ **[M1]**

$\dfrac{2}{3}x + 12$ **[A1]**

(b) Expression after 2nd stop: $\dfrac{3}{4}\left(\dfrac{2}{3}x + 12\right)$ or $0.75\left(\dfrac{2}{3}x + 12\right)$ **[M1]**

Expands correctly to $\dfrac{1}{2}x + 9$ **[M1]**

Equates to final condition: $\dfrac{1}{2}x + 9 = x - 15$ **[M1]**

48 **[A1]**

(c) Final number of passengers $= 48 - 15 = 33$ **[B1]**

$\dfrac{33}{48} \times 100$ or $\dfrac{11}{16} \times 100$ or $\dfrac{275}{4}$ **[M1]**

$68\dfrac{3}{4}$ **[A1]**

[Total Marks: 9]

70 (a) 315 **[B1]**

(b) 27.5 **[B2]**

M1 for $\frac{3}{8} \times 100 = 37.5\%$ or for $100 - (35 + 37.5)$

(c) $\frac{9}{160}$ **[B2]**

M1 for $\frac{15}{100} \times \frac{3}{8}$ or $\frac{47.25}{840}$

[Total Marks: 5]

71 (a) Multiplier for 1st hour: $1 - 0.24 = 0.76$ (or 76%) **[B1]**

Multiplier for 2nd hour: $1 - \frac{2}{5} = \frac{3}{5} = 0.6$ (or 60%) **[B1]**

$0.76 \times 0.6 \times V = 45.6$ or $0.456V = 45.6$ **[M1]**

100 **[A1]**

(b) Total lost $= 100 - 45.6 = 54.4$ or $1 - 0.456 = 0.544$ **[M1]**

$\frac{68}{125}$ **[A1]**

[Total Marks: 6]

72 (a) $100\% - 45\% - 37.5\% = 17.5\%$ oe **[M1]**

$\frac{140}{0.175}$ oe **[M1]**

800 **[A1]**

(b) Junior students $= \frac{3}{8} \times 800 = 300$ **[M1]**

$1.5G + G = 300$ oe **[M1]**

120 **[A1]**

[Total Marks: 6]

73 (a) Converts at least two of the non-decimals to decimals correctly (e.g. 0.71 and 0.75) **[M1]**

$71\%, 0.714, \frac{5}{7}, \frac{3}{4}$ **[A1]**

(b) $\frac{1}{2}\left(\frac{2}{5} + \frac{5}{8}\right)$ or equivalent **[M1]**

$\frac{41}{80}$ **[A1]**

(c) $9x - 4x = 35$ or states the multiplier is $35 \div 5 = 7$ **[M1]**

$\frac{28}{63}$ **[A1]**

[Total Marks: 6]

74 **(a)** $\frac{2}{5} = 40\%$ or 0.4, and total percentage of Sales and Production $= 40\% + 28\% = 68\%$ **[M1]**

$100\% - 68\% = 32\%$ and equating $0.32 \times$ Total $= 24$ or equivalent **[M1]**

75 **[A1]**

(b) Number of women in Sales $= 0.60 \times 30 = 18$ **[M1]**

Number of women in Production $= \frac{3}{7} \times 21 = 9$ (or men $= 12$, so women $= 9$) **[M1]**

Total women $= 18 + 9 + 12 = 39$ **[M1]**

52 **[A1]**

[Total Marks: 7]

75 **(a)** 369 **[B2]**

If 0 scored, allow [M1] for 450×0.82 or 450×0.18

(b) 740 **[B3]**

If 0 scored, allow [M2] for $\frac{629}{1 - 0.15}$ or $\frac{629}{0.85}$

If 0 scored, allow [M1] for identifying that $85\% = \$629$

(c) 4 **[B2]**

If 0 scored, allow [M1] for a correct calculation showing the compound effect, e.g., $1.2 \times 0.8 = 0.96$ or choosing a starting value of 100 and reaching 96

[Total Marks: 7]

76 (i) $>$ **[B1]**

(ii) $>$ **[B1]**

[Total Marks: 2]

77 Converts to a common format (e.g. decimals: $0.625, 0.62, 0.615, 0.6$) **[M1]**

$\frac{3}{5}, 61.5\%, 0.62, \frac{5}{8}$ **[A1]**

[Total Marks: 2]

78 **(a)** $>$ **[B1]**

(b) $=$ **[B1]**

(c) $<$ **[B1]**

[Total Marks: 3]

79 **(a)** **(i)** $<$ **[B1]**

(ii) $=$ **[B1]**

(b) M1 for converting at least three values into a comparable form (e.g. percentages: 55%, $54.54\%\ldots$, 54%, $55.55\%\ldots$) **[M1]**

$54\%, \frac{6}{11}, 0.55, \frac{5}{9}$ **[A1]**

[Total Marks: 4]

80 (a) Converting to comparable decimals: 0.75, 0.72, 0.74, $0.714\ldots$, $0.744\ldots$ (at least three correct) **[M1]**

$\frac{5}{7},\ 0.72,\ 74\%,\ 0.7\dot{4},\ \frac{3}{4}$ **[A1]**

(b) (i) $>$ **[B1]**

(ii) $=$ **[B1]**

[Total Marks: 4]

81 (a) (i) $=$ **[B1]**

(ii) $>$ **[B1]**

(iii) $>$ **[B1]**

(b) $0.3 \quad 33\% \quad \frac{1}{3} \quad \frac{3}{8}$ **[B2]**

If 0 scored, allow [M1] for converting at least two of the values correctly to decimals, e.g. 0.33, $0.333\ldots$, 0.375

[Total Marks: 5]

82 (a) $p^2 \quad p \quad \sqrt{p} \quad \frac{1}{p}$ **[B2]**

M1 for evaluating the expressions with a chosen test value for p (e.g., $p = 0.25$) OR for any 3 expressions in the correct relative order.

(b) (i) $>$ **[B1]**

(ii) $<$ **[B1]**

[Total Marks: 4]

83 (a) $x^2, x, \sqrt{x}, \frac{1}{x}$ **[B2]**

(B1 for 3 in correct relative order or if reversed)

(b) $x^{-2}, \frac{1}{x}, x, x^3$ **[B2]**

(B1 for 3 in correct relative order or if reversed)

[Total Marks: 4]

84 **(a)** M1 for correctly calculating at least two values: $x^2 = 0.16$, $x^3 = -0.064$, $\frac{1}{x} = -2.5$ **[M1]**

$\frac{1}{x}, x, x^3, x^2$ **[A1]**

(b) M1 for squaring both terms to get 75 and 80, OR expressing as $\sqrt{75}$ and $\sqrt{80}$ **[M1]**

$<$ **[A1]**

(c) M1 for identifying a common denominator, e.g. cross multiplying to get $9m < 35$ **[M1]**

M1 for $35 < 9(m+1)$ leading to $26 < 9m$ **[M1]**

3 **[A1]**

[Total Marks: 7]

85 **(a)** $p^2 \quad p \quad \sqrt{p} \quad \frac{1}{p}$ **[B2]**

If 0 scored, allow [M1] for a correct order of any three of the terms or for successfully substituting a test value such as $p = \frac{1}{4}$ to evaluate all four expressions.

(b) **(i)** $<$ **[B1]**

(ii) $<$ **[B1]**

(c) c, b, a or $5^{20}, 3^{30}, 2^{50}$ **[A1]**

Shows index manipulation to obtain a common power, e.g. $(2^5)^{10}$, $(3^3)^{10}$ and $(5^2)^{10}$ **[M1]**

Evaluates the bases correctly as 32, 27 and 25 **[M1]**

[Total Marks: 7]

86 **(a)** Testing a valid value such as $x = \frac{1}{4}$ to obtain $\frac{1}{16}$, $\frac{1}{2}$, 4, $\frac{1}{4}$ or comparing powers of x **[M1]**

x^2, x, $\sqrt{x}$, $\frac{1}{x}$ **[A1]**

(b) Substituting a valid test value such as $k = -2$ to get -8, $-\frac{1}{2}$, 4, $-\frac{1}{4}$ **[M1]**

Correct ordering of three negative terms: $k^3 < \frac{1}{k} < -\frac{1}{k^2}$ **[M1]**

k^3, $\frac{1}{k}$, $-\frac{1}{k^2}$, k^2 **[A1]**

(c) Rearranging to $n^2 - n \leq 0$ or $n(n-1) \leq 0$ **[M1]**

Identifying critical values $n = 0$ and $n = 1$ **[M1]**

$0 < n \leq 1$ **[A1]**

[Total Marks: 8]

87 $0.12 \quad 12.1\% \quad \frac{4}{33} \quad \sqrt{0.015}$ **[B2]**

M1 for correct decimal equivalents seen for at least two of the non-decimal values (e.g.,

0.121, 0.1212..., 0.1224...).

[Total Marks: 2]

88 Converts all to a common unit, e.g. grams: 0.52 g, 0.04 g, 4.5 g, 0.6 g **[M2]**

(M1 for 2 correct conversions to a common unit)

4.5×10^3 mg, 6×10^{-7} tonnes, 5.2×10^{-4} kg, 0.04 g **[A1]**

[Total Marks: 3]

89 (a) B1 for at least two correct evaluated values ($a = 3100$, $b = 3130.49\ldots$, $c = 3125$, $d = 3136$) **[B1]**

B1 for all four evaluated correctly **[B1]**

d, b, c, a **[A1]**

(b) 15 **[B1]**

[Total Marks: 4]

90 (a) 5^{-3} 0.00805 $\frac{1}{120}$ 8.5×10^{-3} **[B2]**

If 0 scored, allow [M1] for evaluating at least two correctly as standard decimals: 0.008, 0.0085, 0.00833...

(b) $11.5 \leq m < 12.5$ **[B2]**

If 0 scored, allow [M1] for identifying the bounds 11.5 and 12.5

[Total Marks: 4]

91 (a) Converts at least two values to decimals or percentages correctly (e.g. 0.875, 0.86, 0.833...) **[M1]**

$\frac{5}{6}$, 86%, 0.865, $\frac{7}{8}$ **[A1]**

(b) Finds a common denominator (e.g. 6, 12, etc.) **[M1]**

Any correct fraction (e.g. $\frac{5}{12}$, $\frac{2}{5}$) **[A1]**

[Total Marks: 4]

92 (a) Evaluating decimals: $p = 0.0048$, $q \approx 0.004878$, $r \approx 0.004796$, $s = 0.00485$, $t = 0.0049$ (at least three correct) **[M1]**

Correct ordering of decimal values **[M1]**

t, q, s, p, r (accept original numerical expressions) **[A1]**

(b) Any correct value in standard form strictly greater than $4.7958\cdots \times 10^{-3}$ and less than or equal to 4.8×10^{-3} (e.g. 4.798×10^{-3} or 4.8×10^{-3}) **[B1]**

(c) Always true **[B1]**

Since $q \approx 0.004878$ and $s = 0.00485$, $q > s$; therefore any $y \geq q$ must also be strictly greater than s **[R1]**

[Total Marks: 6]

93 (a) $<$ **[B1]**

(b) Evaluates at least two expressions correctly (0.25, 0.707..., 2) **[M1]**

a^2, a, $\sqrt{a}$, $\dfrac{1}{a}$ **[A1]**

(c) False **[B1]**

Provides a valid counterexample, e.g. let $p = 2$ and $q = -3$. Then $2 > -3$, but $\dfrac{1}{2} > -\dfrac{1}{3}$. **[B1]**

[Total Marks: 5]

94 $\left(-\dfrac{1}{4}\right)^2$ or $\dfrac{1}{16}$ **[M1]**

$4 \times \dfrac{1}{16}$ or $\dfrac{1}{4}$ **[M1]**

$-3\dfrac{1}{4}$ or $-\dfrac{13}{4}$ or -3.25 **[A1]**

[Total Marks: 3]

95 (a) 9 **[B1]**

(b) 25 **[B1]**

(c) 27 **[B2]**

If 0 scored, M1 for -4 or -3 seen **[M1]**

[Total Marks: 4]

96 (a) 34 **[B2]**

M1 for $-2 \times -3 = 6$ or $12 - (-6) + 16$ or $18 + 16$ seen

(b) 1 **[B2]**

M1 for $\dfrac{1}{6} \times \dfrac{3}{2} = \dfrac{3}{12}$ or $\dfrac{1}{4}$ seen

[Total Marks: 4]

97 (a) Evaluates multiplication and division correctly to get $+12$ and -5 **[M1]**

19 **[A1]**

(b) Correct division seen (e.g. $\frac{2}{5} \times \frac{7}{4}$ or $\frac{14}{20}$ or $\frac{7}{10}$) **[M1]**

$\frac{3}{5}$ **[A1]**

[Total Marks: 4]

98 (a) $15 - 3 \times 6$ or $15 - 18$ **[M1]**

-3 **[A1]**

(b) $\frac{9}{4} \times \frac{8}{3}$ **[M1]**

6 **[A1]**

$4\frac{1}{2}$ or $\frac{9}{2}$ **[A1]**

[Total Marks: 5]

99 (a) 28 **[B2]**

If 0 scored, allow [M1] for correctly evaluating the bracket to -4 or for showing $12 - (-16)$

(b) 30 **[B2]**

If 0 scored, allow [M1] for an equivalent fraction such as $\frac{120}{4}$ or $\frac{12}{0.4}$

(c) 3.5 **[B1]**

[Total Marks: 5]

100 (a) $1\frac{13}{30}$ **[B3]**

If 0 scored, allow [M1] for correctly evaluating the multiplication as $\frac{18}{30}$ or $\frac{3}{5}$

Allow a further [M1] for finding a correct common denominator to add the fractions, e.g., $\frac{25}{30} + \frac{18}{30}$ or giving the improper fraction $\frac{43}{30}$

(b) 6 **[B3]**

If 0 scored, allow [M1] for correctly evaluating the bracket as $\frac{3}{8}$

Allow a further [M1] for converting to an improper fraction and correctly multiplying, e.g., $\frac{9}{4} \times \frac{8}{3}$

[Total Marks: 6]

101 (a) $\frac{8}{6} - \frac{5}{6}$ or $\frac{1}{2}$ **[M1]**

$\frac{12}{5} \div$ their $\frac{1}{2}$ **[M1]**

$\frac{12}{5} \times \frac{2}{1}$ or equivalent multiplication **[M1]**

$4\frac{4}{5}$ **[A1]**

(b) 0.54 or $\frac{45 \times 12}{600}$ or equivalent **[M1]**

$54 \div 6$ or $0.54 \div 0.06$ **[M1]**

9 **[A1]**

[Total Marks: 7]

102 Denominator: $2 - 0.4 = 1.6$ or $\frac{8}{5}$ **[B1]**

Right hand side: $-1\frac{1}{4} = -\frac{5}{4}$ **[B1]**

$x + \frac{1}{3} = -\frac{5}{4} \times \frac{8}{5}$ **[M1]**

$x + \frac{1}{3} = -2$ **[A1]**

$x = -\frac{7}{3}$ or $-2\frac{1}{3}$ **[A1]**

[Total Marks: 5]

103 Numerator multiplication: $\frac{7}{3} \times \frac{9}{7} = 3$ **[M1]**

Evaluates root as $\frac{5}{3}$ and cube as $\frac{1}{8}$ **[M1]**

Denominator division: $\frac{1}{8} \times \frac{16}{3} = \frac{2}{3}$ **[M1]**

Numerator subtraction: $3 - \frac{5}{3} = \frac{4}{3}$ **[M1]**

2 **[A1]**

[Total Marks: 5]

104 Converts mixed numbers to improper fractions and multiplies correctly: $\frac{9}{4} \times \frac{4}{3} = 3$ (or $\frac{12}{4}$ or $\frac{36}{12}$) **[M1]**

Evaluates the numerator correctly: $3\frac{2}{3} + 3 = 6\frac{2}{3}$ (or $\frac{20}{3}$) **[M1]**

Evaluates the denominator correctly using a common denominator: $\frac{27}{6} - \frac{11}{6} = \frac{16}{6} = \frac{8}{3}$ **[M1]**

Sets up the correct division for the final step: $\frac{20}{3} \div \frac{8}{3} = \frac{20}{3} \times \frac{3}{8}$ (or $\frac{20}{8}$) **[M1]**

$\frac{5}{2}$ or $2\frac{1}{2}$ **[A1]**

[Total Marks: 5]

105 **(a)** $6\frac{2}{5}$ **[B4]**

M1 for changing mixed numbers to improper fractions, e.g., $\frac{5}{4}$ or $\frac{8}{3}$

M1 for writing fractions with a common denominator to subtract, e.g., $\frac{15}{12} - \frac{10}{12}$

M1 for $\frac{8}{3} \times \frac{12}{5}$ or $\frac{32}{5}$

(b) $38\frac{1}{3}$ **[B4]**

M1 for $83\frac{1}{3} \div 12\frac{1}{2}$

M1 for $\frac{250}{3} \times \frac{2}{25}$ or width $= \frac{20}{3}$ $(6\frac{2}{3})$

M1 for $2 \times \left(12\frac{1}{2} + \frac{20}{3}\right)$ or $2 \times \left(\frac{75}{6} + \frac{40}{6}\right)$

[Total Marks: 8]

106 **(a)** Finds the temperature at 14:00 $(-4.5 + 11.2 = 6.7°\text{C})$ or sets up the full expression $(-4.5 + 11.2 - 8.1)$ **[M1]**

-1.4 **[A1]**

(b) Evaluates numerator to 19.63... or denominator to 5.115 **[M1]**

3.84 **[A1]**

[Total Marks: 4]

107 Net rate of fill $= 0.75 - 0.45 = 0.3$ litres/min oe **[M1]**

Total time $= \frac{85.5}{0.3} = 285$ minutes **[M1]**

$285 \div 60 = 4$ with remainder 45 **[M1]**

4 hours 45 minutes **[A1]**

[Total Marks: 4]

108 **(a)** 37.5 **[B2]**

M1 for 15×2.8 or 42 seen

(b) 23 **[B3]**

M1 for $37.5 - 1.2m = 9.9$ or $37.5 - 9.9$ or 27.6 seen

M1 for $27.6 \div 1.2$

[Total Marks: 5]

109 **(a)** $12.5 - (3.2 \times 8)$ or $12.5 - 25.6$ **[M1]**

-13.1 **[A1]**

(b) $(-4.5 \times -2) - 18.3$ **[M1]**

Final temperature of Sample B $= -9.3$ **[A1]**

Difference $= |-9.3 - (-13.1)|$ or $13.1 - 9.3$ **[M1]**

3.8 **[A1]**

[Total Marks: 6]

110 **(a)** $\frac{2}{5} \times 1000$ or 400 **[M1]**

18.5×20 or 370 **[M1]**

770 **[A1]**

(b) Target volume $= 500$ or volume to lose $= 770 - 500 = 270$ **[M1]**

$270 \div 4.5$ **[M1]**

60 **[A1]**

(c) Volume to fill $= 1000 - 500 = 500$ **[M1]**

1 hour 20 minutes $= 80$ minutes **[M1]**

$500 \div 80$ leading to 6.25 **[A1]**

[Total Marks: 9]

111 **(a)** Number of bottles $= 360$ **[B1]**

Volume left over $= 0.5$ **[B2]**

If 0 scored for the volume left over, allow [M1] for $450.5 - (360 \times 1.25)$

(b) 4 hours 54 minutes **[B4]**

If 0 scored, allow [M1] for calculating the volume of the pool: $24.5 \times 12 \times 1.8 = 529.2$ m^3

Allow a further [M1] for converting the volume to litres: $529.2 \times 1000 = 529200$ litres

Allow a further [M1] for calculating total minutes: $\frac{529200}{1800} = 294$ minutes

(c) -8.5 **[B3]**

If 0 scored, allow [M1] for correctly interpreting 6 hours 20 minutes as a fraction or decimal of hours, e.g., $6\frac{1}{3}$ hours or $\frac{19}{3}$ hours

Allow a further [M1] for calculating the total temperature rise: $1.5 \times \frac{19}{3} = 9.5$

[Total Marks: 10]

112 **(a)** 1 **[B1]**

(b) 9 **[B1]**

(c) x^3 **[B1]**

[Total Marks: 3]

113 **(a)** 5 **[B1]**

(b) $\frac{1}{36}$ **[B1]**

(c) t^8 **[B1]**

[Total Marks: 3]

114 **(a)** m^7 **[B1]**

(b) $\frac{1}{25}$ or 0.04 **[B1]**

(c) 7 **[B1]**

[Total Marks: 3]

115 **(a)** $\frac{1}{9}$ **[B1]**

(b) $\frac{2}{3}$ **[B1]**

(c) x^2 **[B1]**

(d) y^{12} **[B1]**

[Total Marks: 4]

116 **(a)** Adds powers: $-3 + 5 = 2$ or 2^2 **[M1]**

4 **[A1]**

(b) 9 **[B1]**

(c) 1 **[B1]**

[Total Marks: 4]

117 **(a)** $\frac{1}{8}$ **[B2]**

M1 for $(\sqrt[4]{16})^{-3}$ or $(2)^{-3}$ or $\frac{1}{16^{\frac{3}{4}}}$

(b) $4w^8y^{-2}$ or $\frac{4w^8}{y^2}$ **[B3]**

B1 for 4

B1 for w^8

B1 for y^{-2} or $\frac{1}{y^2}$

[Total Marks: 5]

118 **(a)** 27 or x^3 or y^6 seen **[M1]**

$27x^3y^6$ **[A1]**

(b) Writes 16 as 2^4 **[M1]**

$(2^4)^x = 2^{4x}$ **[M1]**

$4x = 10 \implies x = 2.5$ or $\frac{5}{2}$ **[A1]**

[Total Marks: 5]

119 **(a)** $16x^6y^2$ **[B2]**

If 0 scored, allow [M1] for a product containing two of 16, x^6, y^2

(b) $\frac{3a^4}{b^3}$ or $3a^4b^{-3}$ **[B2]**

If 0 scored, allow [M1] for a term containing two of 3, a^4, b^{-3}

(c) $\frac{1}{9}$ **[B2]**

If 0 scored, allow [M1] for evaluating the cube root to 3 or for showing $\frac{1}{27^{\frac{2}{3}}}$ or 9

[Total Marks: 6]

120 **(a)** **(i)** 16 **[B2]**

If 0 scored, M1 for $(\sqrt[3]{64})^2$ or 4^2 or $4096^{\frac{1}{3}}$ seen **[M1]**

(ii) $\frac{1}{16}$ or 0.0625 **[B1]**

(b) **(i)** $\frac{5a^4}{b^4}$ or $5a^4b^{-4}$ **[B2]**

If 0 scored, M1 for $\frac{15}{3} = 5$ or a^{6-2} or b^{3-7} **[M1]**

(ii) $2m^3n^{-1}$ or $\frac{2m^3}{n}$ **[B2]**

If 0 scored, M1 for $4^{\frac{1}{2}} = 2$ or $m^{6\times\frac{1}{2}}$ or $n^{-2\times\frac{1}{2}}$ **[M1]**

[Total Marks: 7]

121 **(a)** $16^{\frac{1}{4}} = 2$ or equivalent correct first step **[M1]**

$\frac{1}{8}$ **[A1]**

(b) 25 **[B1]**

x^6y^{-4} or $\dfrac{x^6}{y^4}$ **[B1]**

(c) Expresses terms with a common base, e.g. $4^x = 2^{2x}$ or $32 = 2^5$ **[M1]**

Sets up a correct linear equation from indices, e.g. $3x - 1 + 2x = 5$ **[M1]**

$x = 1.2$ or $x = \dfrac{6}{5}$ **[A1]**

[Total Marks: 7]

122 (a) Deals with negative index: $\frac{1}{125^{\frac{2}{3}}}$ **[M1]**

Takes cube root: $\sqrt[3]{125} = 5$ or 5^2 **[M1]**

$\frac{1}{25}$ or 0.04 **[A1]**

(b) Takes 4th root of numerator and denominator: $\frac{2a^2}{3b^3}$ **[M1]**

Cubes the expression: $\left(\frac{2a^2}{3b^3}\right)^3$ **[M1]**

$\frac{8a^6}{27b^9}$ **[A1]**

[Total Marks: 6]

123 (a) $\dfrac{4y^4}{5x^2}$ **[B3]**

B1 for reciprocal $\left(\dfrac{16y^8}{25x^4}\right)^{\frac{1}{2}}$ seen

B1 for $\sqrt{16} = 4$ and $\sqrt{25} = 5$

B1 for $(y^8)^{\frac{1}{2}} = y^4$ and $(x^4)^{\frac{1}{2}} = x^2$

(b) -6 **[B4]**

M1 for changing to base 2: $4 = 2^2$ and $8 = 2^3$

M1 for $2^{3x-1} \times 2^{2(x+2)} = 2^{3(x-3)}$

M1 for equating powers: $3x - 1 + 2x + 4 = 3x - 9$

[Total Marks: 7]

124 (a) Inverts fraction to remove negative power or correctly applies negative power to all terms, e.g. $\left(\dfrac{y^4}{16x^8}\right)^{\frac{1}{2}}$ or $\dfrac{16^{-\frac{1}{2}}x^{-4}}{y^{-2}}$ **[M1]**

Correctly evaluates the coefficient $16^{-\frac{1}{2}} = \dfrac{1}{4}$ or correctly halves powers **[M1]**

$\dfrac{y^2}{4x^4}$ or $0.25x^{-4}y^2$ **[A1]**

(b) Recognises that $4^x = 2^{2x}$ or substitutes $u = 4^x$ **[M1]**

Expresses 2^{2x+1} as 2×2^{2x} or $2(4^x)$ **[M1]**

Simplifies to a single exponential term, e.g. $-2^{2x} = -8$ or $u - 2u = -8$ **[M1]**

$x = 1.5$ or $x = \dfrac{3}{2}$ **[A1]**

[Total Marks: 7]

125 (a) $\dfrac{27y^6}{8x^9}$ or $3.375y^6x^{-9}$ **[B3]**

If 0 scored, allow [M1] for taking the reciprocal to remove the negative power: $\left(\dfrac{81y^8}{16x^{12}}\right)^{\frac{3}{4}}$

Allow a further [M1] for finding the 4th root of the expression: $\left(\dfrac{3y^2}{2x^3}\right)^3$

(b) 6 **[B3]**

If 0 scored, allow [M1] for expressing both sides with a common base 2, e.g. $(2^3)^x = (2^2)^{x+3}$

Allow a further [M1] for equating the powers to set up the linear equation $3x = 2(x + 3)$

(c) 1 **[B3]**

If 0 scored, allow [M1] for expressing all terms as powers of 5, e.g. $5^1 \times (5^2)^k = 5^3$

Allow a further [M1] for correctly forming the equation for the indices $1 + 2k = 3$

[Total Marks: 9]

126 (a) -1.5 or $-\dfrac{3}{2}$ **[B3]**

If 0 scored, M2 for $5^k = (5^3)^{-\frac{1}{2}}$ or $5^k = 5^{-\frac{3}{2}}$ **[M2]**

If 0 or M2 not scored, M1 for $\sqrt{125} = 125^{\frac{1}{2}}$ or $\dfrac{1}{125^{0.5}}$ or $125 = 5^3$ **[M1]**

(b) $\dfrac{27y^6}{8x^9}$ or $3.375y^6x^{-9}$ **[B3]**

If 0 scored, M2 for $\left(\dfrac{81y^8}{16x^{12}}\right)^{\frac{3}{4}}$ or $\dfrac{16^{-\frac{3}{4}}x^{-9}}{81^{-\frac{3}{4}}y^{-6}}$ **[M2]**

If 0 or M2 not scored, M1 for finding the 4th root: $\dfrac{2}{3}$, x^3, or y^2 correctly applied to at least one term **[M1]**

(c) 1.5 or $\dfrac{3}{2}$ **[B3]**

If 0 scored, M2 for $2x - 1 + 2x = 5$ or $4x - 1 = 5$ **[M2]**

If 0 or M2 not scored, M1 for writing with common base 2: $4^x = 2^{2x}$ or $32 = 2^5$ **[M1]**

[Total Marks: 9]

127 **(a)** 4.07×10^{-3} **[B1]**

(b) 15×10^{10} **[M1]**

1.5×10^{11} **[A1]**

[Total Marks: 3]

128 **(a)** 4.5×10^{5} **[B1]**

(b) 0.000308 **[B1]**

(c) 8×10^{8} **[B1]**

[Total Marks: 3]

129 **(a)** 4.52×10^{5} **[B1]**

(b) 0.00607 **[B1]**

(c) 1.2×10^{10} **[B2]**

If 0 scored, M1 for 12×10^{9} **[M1]**

[Total Marks: 4]

130 **(a)** 7.14×10^{-3} **[B1]**

(b) 50200 **[B1]**

(c) 1.8×10^{10} **[B2]**

M1 for 18×10^{9} or 18000000000 seen

[Total Marks: 4]

131 **(a)** **(i)** 3.4×10^{-4} **[B1]**

(ii) 52000 **[B1]**

(b) 8×10^{8} **[B2]**

If B0, M1 for $2 \times 4 = 8$ or $10^{3} \times 10^{5} = 10^{8}$ seen.

[Total Marks: 4]

132 **(a)** 4.52×10^{-4} **[B1]**

(b) 371000 **[B1]**

(c) 8×10^{7} **[B2]**

If 0 scored, allow [M1] for 80000000 or 8×10^{k} where $k \neq 7$

[Total Marks: 4]

133 **(a)** $597 \times 10^{22} - 7.35 \times 10^{22}$ or $5.97 \times 10^{24} - 0.0735 \times 10^{24}$ **[M1]**

5.8965×10^{24} **[A1]**

(b) Sets up the correct division: $\dfrac{1.99 \times 10^{30}}{5.97 \times 10^{24}}$ **[M1]**

Evaluates correctly, e.g., $0.3333... \times 10^{6}$ **[M1]**

3.33×10^{5} **[A1]**

[Total Marks: 5]

134 **(a)** $85 \times 10^{5} - 4 \times 10^{5}$ or $8.5 \times 10^{6} - 0.4 \times 10^{6}$ **[M1]**

8.1×10^{6} **[A1]**

(b) $\dfrac{2 \times 10^{9}}{5 \times 10^{4}} = 0.4 \times 10^{5}$ or 4×10^{4} **[M1]**

$\sqrt{4 \times 10^{4}} = \sqrt{4} \times \sqrt{10^{4}}$ **[M1]**

2×10^{2} **[A1]**

[Total Marks: 5]

135 **(a)** 4.5×10^{2} **[B3]**

If 0 scored, allow [M1] for $\dfrac{3.6}{8} \times 10^{3}$ or 0.45×10^{3}

Allow a further [M1] for correctly converting to standard form, or for seeing 450

(b) 5.7×10^{8} **[B2]**

If 0 scored, allow [M1] for $(50 \times 10^{7}) + (7 \times 10^{7})$ or 570000000 or digits 57

(c) $3.2 \times 10^{2m+n+1}$ **[B2]**

If 0 scored, allow [M1] for $(4 \times 10^{2m}) \times (8 \times 10^{n})$ or $32 \times 10^{2m+n}$

[Total Marks: 7]

136 **(a)** 3×10^{8} **[B3]**

If 0 scored, M1 for $5.4 \times 10^{5} + 0.6 \times 10^{5}$ or $54 \times 10^{4} + 6 \times 10^{4}$ or 6.0×10^{5} seen **[M1]**

M1 for $\dfrac{\text{their } 6.0 \times 10^{5}}{2 \times 10^{-3}}$ **[M1]**

(b) 5×10^{4} **[B2]**

If 0 scored, M1 for $\sqrt{25 \times 10^{8}}$ or 50000 seen **[M1]**

(c) $5 \times 10^{2x-y-1}$ **[B3]**

If 0 scored, M1 for $p^{2} = 4 \times 10^{2x}$ **[M1]**

M1 for $\dfrac{4 \times 10^{2x}}{8 \times 10^{y}}$ or $0.5 \times 10^{2x-y}$ **[M1]**

[Total Marks: 8]

137 **(a)** $4 \times 10^{m+n+1}$ or $4.0 \times 10^{m+n+1}$ **[B2]**

M1 for $40 \times 10^{m+n}$ seen

(b) $3.125 \times 10^{2m-n}$ **[B3]**

M1 for $(5 \times 10^m)^2$ or 25×10^{2m} seen

M1 for $\frac{25}{8} \times 10^{2m-n}$

(c) 2 **[B3]**

M1 for $8 \times 10^n = 5.08 \times 10^m - 5 \times 10^m$

M1 for $8 \times 10^n = 0.08 \times 10^m$ or $8 \times 10^n = 8 \times 10^{m-2}$

[Total Marks: 8]

138 **(a)** 500 **[B2]**

M1 for $1.5 \times 10^8 \div 3.0 \times 10^5$

(b) 6.0×10^3 or 6×10^3 **[B2]**

M1 for $1.5 \times 10^8 \div 2.5 \times 10^4$ or 6000 seen

[Total Marks: 4]

139 **(a)** $1.23 \times 4.87 \times 10^{24}$ **[M1]**

5.99×10^{24} **[A1]**

(b) $\frac{1.99 \times 10^{30}}{4.87 \times 10^{24}}$ **[M1]**

4.09×10^5 **[A1]**

[Total Marks: 4]

140 **(a)** $\frac{1.99 \times 10^{30}}{1.90 \times 10^{27}}$ **[M1]**

$1047.36\ldots$ or $1.047\cdots \times 10^3$ **[A1]**

1.05×10^3 **[A1]**

(b) $(4.5 \times 10^{13}) \times 1000$ or $(4.5 \times 10^{13}) \times 10^3$ **[M1]**

4.5×10^{16} **[A1]**

[Total Marks: 5]

141 **(a)** 3.61×10^7 **[B2]**

If B0, M1 for $3.4 \times 10^7 + 0.21 \times 10^7$ or $34 \times 10^6 + 2.1 \times 10^6$ or 36100000 **[M1]**

(b) Converts minutes to seconds: $5 \times 60 = 300$ **[M1]**

Sets up division: $\frac{300}{4 \times 10^{-9}}$ **[M1]**

7.5×10^{10} **[A1]**

[Total Marks: 5]

142 **(a)** 6.04×10^{24} **[B2]**

If 0 scored, allow [M1] for 6.0435×10^{24} or 604.35×10^{22} or digits 604

(b) 81 **[B2]**

If 0 scored, allow [M1] for $\dfrac{5.97 \times 10^{24}}{7.35 \times 10^{22}}$ or 81.2...

(c) 1.79 **[B2]**

If 0 scored, allow [M1] for $300 \times 5.97 \times 10^{24}$ or 1.791×10^{27} or digits 179

[Total Marks: 6]

143 **(a)** 5.8965×10^{24} **[B2]**

If 0 scored, M1 for $5.97 \times 10^{24} - 0.0735 \times 10^{24}$ or $597 \times 10^{22} - 7.35 \times 10^{22}$ seen **[M1]**

(b) 5×10^2 or 5.00×10^2 **[B2]**

If 0 scored, M1 for $\dfrac{1.50 \times 10^8}{3.00 \times 10^5}$ or 500 **[M1]**

(c) 5.10×10^{14} **[B3]**

If 0 scored, M2 for $4 \times \pi \times (6.37 \times 10^6)^2$ **[M2]**

If 0 or M2 not scored, M1 for $(6.37 \times 10^6)^2$ or $4.057... \times 10^{13}$ **[M1]**

[Total Marks: 7]

144 **(a)** $\pi \times (4.5 \times 10^2)^2 \times (1.2 \times 10^3)$ **[M1]**

$7.63407 \cdots \times 10^8$ **[A1]**

7.634×10^8 **[A1]**

(b) Volume in $m^3 = \dfrac{7.634 \times 10^8}{10^6} = 763.4\ m^3$ (or equivalent rate conversion $1.5 \times 10^2\ cm^3/\text{s}$) **[M1]**

Time in seconds $= \dfrac{763.4}{1.5 \times 10^{-4}} = 5.089 \cdots \times 10^6$ **[M1]**

Time in hours $= \dfrac{5.089 \cdots \times 10^6}{3600} = 1413.7 \ldots$ **[M1]**

1.41×10^3 **[A1]**

[Total Marks: 7]

145 **(a)** 23.50 **[B1]**

(b) 0.0407 **[B1]**

[Total Marks: 2]

146 **(a)** **(i)** 47.54 **[B1]**

(ii) 48 **[B1]**

(iii) 48 **[B1]**

[Total Marks: 3]

147 **(a)** 780 **[B1]**

(b) 0.036 **[B1]**

(c) 4.10 **[B1]**

[Total Marks: 3]

148 **(a)** 48.98 **[B1]**

(b) 0.0031 **[B1]**

(c) 600 **[A1]**

M1 for $\frac{8 \times 30}{0.4}$

[Total Marks: 4]

149 **(a)** **(i)** 34.57 **[B1]**

(ii) 0.070 **[B1]**

(b) $\frac{60 \times 4}{0.5}$ **[M1]**

480 **[B1]**

[Total Marks: 4]

150 **(a)** 0.048 **[B1]**

(b) 380 **[B1]**

(c) $\frac{60 \times 4}{20 - 3}$ or $\frac{240}{17}$ **[M1]**

14 or 14.1 (from $\frac{240}{17}$) or 15 (if 17 is estimated as 20) **[A1]**

[Total Marks: 4]

151 **(a)** **(i)** 203.50 **[B1]**

(ii) 203 **[B1]**

(iii) 200 **[B1]**

(b) 560 **[A1]**

If 0 scored, M1 for $\frac{7 \times 40}{0.5}$ **[M1]**

[Total Marks: 5]

152 $\frac{60 \times 3}{0.2}$ **[M1]**

$\frac{180}{0.2}$ **[M1]**

900 **[A1]**

[Total Marks: 3]

153 (a) $\frac{8 \times 30}{0.4}$ **[M1]**

600 **[A1]**

(b) $\sqrt{100}$ or 10, and 20, and 0.5 **[M1]**

$\frac{10 + 20}{0.5^2}$ or $\frac{30}{0.25}$ **[M1]**

120 **[A1]**

[Total Marks: 5]

154 (a) $\frac{\sqrt{100} \times 4^2}{0.2}$ **[M1]**

$\frac{10 \times 16}{0.2}$ or $\frac{160}{0.2}$ **[M1]**

800 **[B1]**

(b) 500×200 **[M1]**

100000 **[B1]**

[Total Marks: 5]

155 400, 0.05 and 0.2 seen [B2]
(or B1 for any two correct rounding to 1 sf)

$\frac{400 \times 0.05}{0.2} = \frac{20}{0.2} = 100$ **[M1]**

10 **[A1]**

[Total Marks: 4]

156 (a) 2000 **[A3]**

If 0 scored, M1 for rounding the numbers to 4×10^6, 0.05 and 20 **[M1]**

M1 for evaluating the numerator to 2000 and the denominator to 1 **[M1]**

(b) Greater (than) **[B1]**

Valid reason, e.g. the numerator was rounded up and the denominator was rounded down. **[B1]**

[Total Marks: 5]

157 (a) 1600 **[A1]**

M1 for $\frac{\sqrt[3]{8} \times 20^2}{0.5}$

M1 for 2×400 or 800 seen in numerator

(b) 3.2×10^6 **[A1]**

M1 for $4 \times 10^5 \times 2^3$

M1 for 32×10^5 or 3200000 seen

[Total Marks: 6]

158 (a) $\sqrt{100} \times 4^3$ or 10×64 $(= 640)$ **[M1]**

0.2×50 $(= 10)$ **[M1]**

$\frac{10 \times 64}{10} = 64$ **[A1]**

(b) Uses $a = 4$, $x = 0.05$, $b = 1$, $c = 0.8$ **[M1]**

$\frac{4 \times 0.05}{1 - 0.8}$ or $\frac{0.2}{0.2}$ **[M1]**

1 **[A1]**

[Total Marks: 6]

159 (a) 40, 5, and 1000 seen **[M1]**

$\frac{40 \times 5^2}{\sqrt[3]{1000}}$ or $\frac{40 \times 25}{10}$ **[M1]**

100 **[A1]**

(b) Radius ≈ 4 and Height ≈ 10 **[B1]**

Volume $= 3 \times 4^2 \times 10$ **[M1]**

480 **[A1]**

(c) Distance ≈ 4000 and Speed ≈ 600 **[B1]**

$\frac{4000}{600}$ or $\frac{20}{3}$ or 6.66... hours **[M1]**

6 hours and 40 minutes **[A1]**

[Total Marks: 9]

160 **(a)** $\frac{1}{3} \times \pi \times 5.42^2 \times 12.8$ **[M1]**

393.8 **[A1]**

(b) Volume of one sphere $= \frac{4}{3} \times \pi \times 0.85^3$ or $2.57\ldots$ seen **[M1]**

$393.75\cdots \div 2.57\ldots$ or $153.06\ldots$ **[M1]**

153 **[A1]**

[Total Marks: 5]

161 **(a)** $125 \times 74 = 9250$ **[M1]**

9300 **[A1]**

(b) $x^2 = 9250$ **[M1]**

$x = \sqrt{9250}$ (96.176...) **[M1]**

96.2 **[A1]**

[Total Marks: 5]

162 **(a)** 404 **[A2]**

If 0 scored, M1 for $\frac{1}{3} \times \pi \times 5.82^2 \times 11.4$ **[M1]**

(b) 360 **[A2]**

If 0 scored, M1 for $\frac{1}{3} \times 3 \times 6^2 \times 10$ **[M1]**

(c) 70 **[A2]**

If 0 scored, M1 for $850 \div 12$ or 70.83... seen **[M1]**

[Total Marks: 6]

163 **(a)** 87.1 **[B2]**

M1 for 87.098... seen or $\frac{78.38...}{0.9}$

(b) **(i)** 27 **[A1]**

M1 for $\frac{4}{3} \times \pi \times 1.85^3$ or 26.5... seen

(ii) 35 hours 22 minutes **[A1]**

M1 for their exact volume (or 27) $\times 1000$ (e.g. 26522 litres)

M1 for $\frac{\text{their } 26522}{12.5}$ $(= 2121.76$ mins$)$

[Total Marks: 7]

164 **(a)** **(i)** 70.35465... **[B1]**

Accept an answer containing at least 5 decimal places.

(ii) 70.4 **[B1]**

(b) $\frac{30^2 - 10^2}{\sqrt{100}}$ or $\frac{900 - 100}{10}$ **[M1]**

80 **[B1]**

[Total Marks: 4]

165 (a) $r \approx 4$, $h \approx 20$, density ≈ 8 **[M1]**

Estimated Mass $= 3 \times 4^2 \times 20 \times 8$ **[M1]**

7680 **[A1]**

(b) Volume $= \pi \times (4.15)^2 \times 19.8 = 1071.3\ldots$ **[M1]**

Mass $= 1071.3 \cdots \times 8.13 = 8710.19\ldots$ **[M1]**

8710 **[A1]**

[Total Marks: 6]

166 (a) 3 : 5 : 7 **[B1]**

(b) 80 **[B2]**

If 0 scored, M1 for $\frac{240}{8 + 10 + 12} \times 10$ **[M1]**

[Total Marks: 3]

167 (a) 2 : 1 **[B1]**

(b) $\frac{240}{2 + 3 + 5} \times (5 - 2)$ or $120 - 48$ **[M1]**

72 **[A1]**

[Total Marks: 3]

168 (a) M1 for 1.2 kg = 1200 g or 450 g = 0.45 kg

3 : 8 **[A1]**

(b) M1 for $3 + 4 + 5$ or 12 seen

140 **[A1]**

[Total Marks: 4]

169 (a) Total parts $= 2 + 3 + 5 = 10$ **[M1]**

Ben's share $= \frac{3}{10} \times 840 = 252$ **[A1]**

(b) $\frac{5}{10}$ **[M1]**

$\frac{1}{2}$ **[A1]**

[Total Marks: 4]

170 (a) $3 + 4 + 5 = 12$ **[M1]**

$\frac{5}{12} \times 120 = 50$ **[A1]**

(b) $\frac{3}{12} \times 120$ or 3×10 **[M1]**

30 **[A1]**

(c) 4 : 5 **[B1]**

[Total Marks: 5]

171 (a) 45 : 120 **[M1]**

3 : 8 **[A1]**

(b) $3.6 \times 100 = 360$ cm **[M1]**

$\frac{360}{2+3+7} \times 7$ **[M1]**

210 **[A1]**

[Total Marks: 5]

172 (a) $\frac{m+30}{f} = 3$ or $m + 30 = 3f$ **[B1]**

$\frac{m}{f+20} = 1$ or $m = f + 20$ **[B1]**

(b) M1 for substituting to eliminate one variable, e.g., $f + 20 + 30 = 3f$ or $m + 30 = 3(m - 20)$

M1 for correctly simplifying their equation to $2f = 50$ or $2m = 90$

$m = 45$, $f = 25$ **[A1]**

(c) 9 : 5 **[B1]**

[Total Marks: 6]

173 (a) 15 : 20 : 8 **[B2]**

If 0 scored, M1 for $3 \times 5 : 4 \times 5$ or $5 \times 4 : 2 \times 4$ **[M1]**

(b) 24 **[B2]**

If 0 scored, M1 for $\frac{129}{15+20+8} \times 8$ **[M1]**

(c) 6 **[B3]**

If 0 scored, M1 for $\dfrac{60-y}{24-y}=3$ or $\dfrac{60-y}{24-y}=\dfrac{3}{1}$ **[M1]**

M1 for $60-y=72-3y$ (or equivalent linear equation) **[M1]**

[Total Marks: 7]

174 **(a)** 4.5×50000 or 225000 seen **[M1]**

2.25 **[A1]**

(b) Area scale is 1 cm^2 : 0.25 km^2 or map area ratio $1^2 : 50000^2$ **[M1]**

$14.2\div 0.25$ or $14.2\times 10^{10}\div 50000^2$ **[M1]**

56.8 **[A1]**

[Total Marks: 5]

175 **(a)** Actual distance $= 8.4\times 250\,000 = 2\,100\,000$ cm **[M1]**

Distance in m $= 21\,000$ or distance in km $= \dfrac{2\,100\,000}{100\,000}$ **[M1]**

21 **[A1]**

(b) Area scale factor $= 1^2 : (250\,000)^2 = 1 : 6.25\times 10^{10}$ **[M1]**

35 $\text{km}^2 = 35\times 10^{10}$ cm^2 **[M1]**

Map area $= \dfrac{35\times 10^{10}}{6.25\times 10^{10}} = 5.6$ **[A1]**

[Total Marks: 6]

176 **(a)** 16.4×250000 or 4100000 **[M1]**

41 **[A1]**

(b) Scale factor for area is $(250000)^2$ or 1 km $= 10^5$ cm so 1 $\text{km}^2 = 10^{10}$ cm^2 **[M1]**

$\dfrac{35\times 10^{10}}{250000^2}$ or $\dfrac{35}{2.5^2}$ **[M1]**

5.6 **[A1]**

(c) 400000 **[B1]**

[Total Marks: 6]

177 **(a)** 29 **[B2]**

If 0 scored, M1 for $14.5\times 1000\times 100\div 50000$ or equivalent **[M1]**

(b) 3.1 **[B2]**

If 0 scored, M1 for $12.4\times\left(\dfrac{50000}{100000}\right)^2$ or 12.4×0.25 **[M1]**

(c) Calculates requires 360 g of flour and 225 g of butter **[B2]**

If B0 scored, M1 for $\frac{240}{8} \times 12$ or $\frac{150}{8} \times 12$ **[M1]**

No, because he needs 225 g of butter but only has 200 g **[B1]**

[Total Marks: 7]

178 **(a)** $16.8 \times 250\,000 = 4\,200\,000$ cm **[M1]**

42 **[A1]**

(b) Scale of area $= 1^2 : (2.5)^2$ OR 1 cm^2 represents 6.25 km^2 **[M1]**

8.96 **[A1]**

(c) 14 cm^2 represents 56 $km^2 \implies$ 1 cm^2 represents 4 km^2 **[M1]**

Linear ratio $= \sqrt{1}$ cm represents $\sqrt{4}$ km $\implies$ 1 cm to 2 km **[M1]**

200 000 **[A1]**

[Total Marks: 7]

179 **(a)** M1 for $5 - 3 = 2$ parts

M1 for $84 \div 2$ or 42 seen

336 **[A1]**

(b) **(i)** M1 for 8.5×25000 or 212500 seen

2.125 **[A1]**

(ii) M1 for 1 cm represents 0.25 km or 1 km represents 4 cm

M1 for area scale factor 1 cm^2 represents 0.0625 km^2 or $3.5 \times 10^{10} \div 25000^2$

56 **[A1]**

[Total Marks: 8]

180 **(a)** Let total workers be $8x$. Males $= 5x$, Females $= 3x$. **[M1]**

Full-time males $= \frac{3}{5} \times 5x = 3x$. Part-time males $= \frac{2}{5} \times 5x = 2x$. **[M1]**

Full-time females $= \frac{4}{5} \times 3x = 2.4x$. Part-time females $= \frac{1}{5} \times 3x = 0.6x$. **[M1]**

Total full-time $= 3x + 2.4x = 5.4x$. Total part-time $= 2x + 0.6x = 2.6x$. Ratio is $5.4 : 2.6 = 27 : 13$. **[A1]**

(b) Difference $= 3x - 0.6x = 2.4x$ **[M1]**

$2.4x = 42 \implies x = 17.5$ **[M1]**

Total workers $= 8x = 8 \times 17.5 = 140$ **[A1]**

[Total Marks: 7]

181 **(a)** $5 - 3 = 2$ parts represent 42 workers **[M1]**

1 part is $\dfrac{42}{2} = 21$ **[M1]**

$8 \times 21 = 168$ **[A1]**

(b) Day shift workers: $5 \times 21 = 105$. Night shift workers: $3 \times 21 = 63$ **[M1]**

Day shift males: $\dfrac{3}{7} \times 105 = 45$. Day shift females: $105 - 45 = 60$ **[M1]**

Night shift males: $\dfrac{2}{9} \times 63 = 14$. Night shift females: $63 - 14 = 49$ **[M1]**

Total males $= 45 + 14 = 59$. Total females $= 60 + 49 = 109$. Ratio is $59 : 109$ **[A1]**

[Total Marks: 7]

182 (a) Let the number of boys be $3x$ and girls be $5x$.
Forming equation: $\dfrac{3x-4}{5x+8} = \dfrac{1}{3}$ [M2]
(or M1 for either $3x - 4$ or $5x + 8$ seen in a ratio)

$9x - 12 = 5x + 8$ **[M1]**

15 **[A1]**

(b) Common ratio $A : B : C = 15 : 20 : 24$ **[M1]**

Difference in parts $24 - 15 = 9$, so 9 parts = 45 kg
$\implies$ 1 part = 5 kg **[M1]**

Total mass $= (15 + 20 + 24) \times 5 = 295$ **[A1]**

[Total Marks: 7]

183 (a) Alex : Ben $= 15 : 20$ and Ben : Chris $= 20 : 24$ **[M1]**

$15 : 20 : 24$ **[A1]**

(b) Difference in parts $= 24 - 15 = 9$ parts $\implies$ 9 parts $= 16\,200$ **[M1]**

Total parts $= 15 + 20 + 24 = 59$ **[M1]**

$P = \dfrac{16\,200}{9} \times 59 = 106\,200$ **[A1]**

(c) Alex's share $= \dfrac{15}{59} \times 106\,200 = 27\,000$ **[B1]**

$27\,000 \times \left(1 + \dfrac{4.2}{100}\right)^5$ **[M1]**

33 167 **[A1]**

[Total Marks: 8]

184 Time taken = 1.5 hours (or 1 hour 30 minutes) **[B1]**

$120 \div 1.5 = 80$ **[B1]**

[Total Marks: 2]

185 1 hour 12 minutes = 1.2 hours or 72 minutes **[B1]**

$\frac{102}{1.2}$ or $\frac{102}{72} \times 60$ **[M1]**

85 **[A1]**

[Total Marks: 3]

186 Time = 1.5 hours **[B1]**

Average Speed = $\frac{120}{1.5}$ **[M1]**

80 **[A1]**

[Total Marks: 3]

187 (a) M1 for $108 \div 1.5$ or $\frac{108}{90} \times 60$

72 **[A1]**

(b) M1 for $72 \times \frac{1000}{3600}$ or $72 \div 3.6$

20 **[A1]**

[Total Marks: 4]

188 (a) $\frac{120}{1.5}$ or $\frac{120}{90} \times 60$ **[M1]**

80 **[A1]**

(b) $\frac{8}{100} \times 350$ or 8×3.5 **[M1]**

28 **[A1]**

[Total Marks: 4]

189 (a) Time = 2.25 or $\frac{9}{4}$ hours **[M1]**

$180 \div 2.25 = 80$ **[A1]**

(b) Time in seconds = $450 \div 1.5 = 300$ **[M1]**

$300 \div 60 = 5$ **[A1]**

[Total Marks: 4]

190 (a) M1 for $\frac{30}{v} + \frac{40}{v+5} = 4$

M1 for $30(v+5) + 40v = 4v(v+5)$ (removing fractions)

A1 for obtaining $2v^2 - 25v - 75 = 0$ with no errors or omitted steps in working

(b) M2 for $(2v + 5)(v - 15) = 0$ or correct substitution into the quadratic formula
(M1 for $(2v + a)(v + b) = 0$ where $ab = -75$ or $2b + a = -25$)

$v = 15$ and $v = -2.5$ **[A1]**

(c) 2 **[B1]**

[Total Marks: 7]

191 Time $= 4\frac{25}{60}$ hours or 4.416... hours **[M1]**

Average Speed $= \frac{385}{4.416...}$ **[M1]**

87.2 (correct to 1 decimal place) **[A1]**

[Total Marks: 3]

192 (a) 45×1000 or $\frac{45}{60}$ **[M1]**

$\frac{45000}{60} = 750$ **[A1]**

(b) Base area $= 150 \times 80 = 12000$ cm^2 **[M1]**

Rate of height increase $= \frac{\text{Volume per second}}{\text{Base area}} = \frac{750}{12000}$ **[M1]**

0.0625 **[A1]**

[Total Marks: 5]

193 (a) $\frac{350}{100} \times 6.4$ **[M1]**

22.4 **[A1]**

(b) Cost in euros $= 22.4 \times 1.35 = 30.24$ **[M1]**

Cost in dollars $= 30.24 \div 0.85$ **[M1]**

35.58 **[A1]**

[Total Marks: 5]

194 (a) $150 \times 60 \times 50$ or 450000 seen or $1.5 \times 0.6 \times 0.5 = 0.45$ m^3 **[M1]**

450 **[A1]**

(b) $\frac{450}{24}$ or 18.75 **[M1]**

0.75×60 or 45 **[M1]**

18 minutes 45 seconds **[A1]**

[Total Marks: 5]

195 **(a)** M1 for 52×1000 or 52000 seen

M1 for their $52000 \div 1.6$ (implies 32500 seconds)

9 hours 2 minutes **[A1]**

(b) M1 for 52×2.15 or 111.8 seen

M1 for their 111.8×0.84

93.91 **[A1]**

[Total Marks: 6]

196 **(a)** Fuel used $= \frac{325}{100} \times 8.4$ or 27.3 **[M1]**

27.3×1.35 **[M1]**

36.855 or 36.86 **[A1]**

(b) 24 minutes = 0.4 hours or $\frac{2}{5}$ hours **[B1]**

Distance $= 75 \times 4.4$ **[M1]**

330 **[A1]**

[Total Marks: 6]

197 **(a)** Volume in $m^3 = 1.8 \times 1.5 \times 1.2 = 3.24$ **[M1]**

$3.24 \times 1000 = 3240$ litres **[A1]**

(b) Time in seconds $= \frac{3240}{2.5} = 1296$ **[M1]**

Time in minutes $= \frac{1296}{60} = 21.6$ **[A1]**

(c) Rate $= \frac{\text{Volume}}{\text{Time}} = \frac{3240}{45}$ **[M1]**

$x = 72$ **[A1]**

[Total Marks: 6]

198 **(a)** Volume $= 2.5 \times 1.4 \times 1.2$ or 4.2 (m^3) **[M1]**

Volume in litres $= 4.2 \times 1000$ or 4200 **[M1]**

Time in seconds $= \frac{4200}{0.35}$ or 12000 **[M1]**

3 h 20 min **[A1]**

(b) $\frac{780}{0.84}$ or 928.57... **[M1]**

$928.57 - 850$ **[M1]**

78.57 **[A1]**

[Total Marks: 7]

199 **(a)** Time A to B $= \dfrac{d}{v}$ **[M1]**

Time B to C $= \dfrac{2d}{v+15}$ **[M1]**

Total Time $T = \dfrac{d}{v} + \dfrac{2d}{v+15} = \dfrac{d(v+15)+2dv}{v(v+15)} = \dfrac{dv+15d+2dv}{v(v+15)} = \dfrac{3dv+15d}{v(v+15)}$ **[A1]**

(b) Substitute $d = 120$ and $T = 4.5$ into the equation from part (a): $4.5 = \dfrac{3(120)v+15(120)}{v(v+15)}$ **[M1]**

$4.5(v^2+15v) = 360v + 1800$ **[M1]**

$4.5v^2 + 67.5v - 360v - 1800 = 0$ **[M1]**

$4.5v^2 - 292.5v - 1800 = 0 \implies divideby1.5toget3\mathrm{v}^2 - 195v - 1200 = 0$ **[A1]**

[Total Marks: 7]

200 **(a)** $\dfrac{x}{12} + \dfrac{x+5}{15} = \dfrac{31}{12}$ or $2\dfrac{35}{60}$ [M2]
(M1 for either $\dfrac{x}{12}$ or $\dfrac{x+5}{15}$ seen as a time)

$\dfrac{5x+4(x+5)}{60} = \dfrac{155}{60}$ or equivalent correct clearing of fractions **[M1]**

$x = 15$ **[A1]**

(b) Total distance $= 15 + (15+5) = 35$ **[B1]**

Average speed $= \dfrac{35}{\frac{31}{12}}$ **[M1]**

13.5 **[A1]**

[Total Marks: 7]

201 **(a)** $\dfrac{2400}{v}$ **[B1]**

(b) $\dfrac{2400}{v} + \dfrac{2400}{v+40} = 10.5$ **[M1]**

$2400(v+40) + 2400v = 10.5v(v+40)$ **[M1]**

$2400v + 96000 + 2400v = 10.5v^2 + 420v$ **[M1]**

$10.5v^2 - 4380v - 96000 = 0$ **[A1]**

Multiply by $\dfrac{2}{3}$ to get $7v^2 - 2920v - 192000 = 0$ **[A1]**

(c) $v = \dfrac{-(-2920) \pm \sqrt{(-2920)^2 - 4(7)(-192000)}}{2(7)}$ **[M1]**

$v = 475.25...$ or $v = -57.68...$ (discard negative) **[A1]**

475 (correct to nearest integer) **[A1]**

[Total Marks: 7]

202 **(a)** $\dfrac{30}{x} + \dfrac{18}{x-4} = 2.5$ **[M1]**

$\dfrac{30(x-4)+18x}{x(x-4)} = 2.5$ **[M1]**

$48x - 120 = 2.5(x^2 - 4x)$ **[M1]**

$48x - 120 = 2.5x^2 - 10x \implies 2.5x^2 - 58x + 120 = 0 \implies 5x^2 - 116x + 240 = 0$ **[A1]**

(b) $x = \dfrac{-(-116) \pm \sqrt{(-116)^2 - 4(5)(240)}}{2(5)}$ **[M1]**

$x = \dfrac{116 \pm \sqrt{13456 - 4800}}{10} = \dfrac{116 \pm \sqrt{8656}}{10}$ **[M1]**

$x = 20.90$ or $x = 2.30$ **[A1]**

[Total Marks: 7]

203 **(a)** $\dfrac{15}{x} + \dfrac{12}{x-4} = 3.5$ **[M1]**

Multiply by $2x(x-4)$ to clear denominators (or equivalent): $30(x-4) + 24x = 7x(x-4)$ **[M1]**

$54x - 120 = 7x^2 - 28x \implies 7x^2 - 82x + 120 = 0$ **[A1]**

(b) **(i)** $(7x - 12)(x - 10) = 0$ OR $\dfrac{82 \pm \sqrt{(-82)^2 - 4(7)(120)}}{14}$ **[M2]**

If M0, allow M1 for $(7x \pm a)(x \pm b) = 0$ where $ab = 120$ or $7b \pm a = -82$

$x = 10$, $x = \dfrac{12}{7}$ (or 1.71) **[A1]**

(ii) Identifies valid speed $x = 10$ (since $x - 4 > 0 \implies x > 4$) **[B1]**

Time $= \dfrac{15}{10} = 1.5$ **[B1]**

[Total Marks: 8]

204 **(a)** $\dfrac{150}{x}$ **[B1]**

(b) $\dfrac{150}{x} + \dfrac{90}{x+15} = 3$ **[M1]**

$\dfrac{150(x+15)+90x}{x(x+15)} = 3$ or $150(x+15) + 90x = 3x(x+15)$ **[M1]**

$150x + 2250 + 90x = 3x^2 + 45x$ **[M1]**

$3x^2 - 195x - 2250 = 0$ leading to $x^2 - 65x - 750 = 0$ **[A1]**

(c) $(x - 75)(x + 10) = 0$ or $\dfrac{65 \pm \sqrt{(-65)^2 - 4(1)(-750)}}{2}$ **[M2]**

If M0 scored, allow [M1] for $(x \pm 75)(x \pm 10) = 0$ or for correct substitution into the quadratic formula.

75 **[A1]**

[Total Marks: 8]

205 Profit = \$400 − \$320 = \$80 or $\frac{400}{320} \times 100$

[M1]

25 **[A1]**

[Total Marks: 2]

206 **(a)** $\frac{15}{100} \times 80$ or 12 **[M1]**

68 **[A1]**

(b) $\frac{4000 - 3200}{4000} \times 100$ or $\frac{800}{4000} \times 100$ **[M1]**

20 **[A1]**

[Total Marks: 4]

207 **(a)** $\frac{24}{60} \times 100$ **[M1]**

40 **[A1]**

(b) $\frac{15}{100} \times 80 = 12$ OR 80×0.85 **[M1]**

68 **[A1]**

[Total Marks: 4]

208 **(a)** $\frac{15}{100} \times 320$ or $10\% = 32, 5\% = 16$ **[M1]**

48 **[A1]**

(b) $\frac{20}{100} \times 45 = 9$ or $\frac{80}{100} \times 45$ **[M1]**

$45 - 9$ (if first method used) **[M1]**

36 **[A1]**

[Total Marks: 5]

209 **(a)** Discount $= 0.15 \times 120 = 18$ OR 0.85×120 **[M1]**

102 **[A1]**

(b) Let cost price be C. $1.2C = 120$ **[M1]**

$C = \frac{120}{1.2}$ **[M1]**

100 **[A1]**

[Total Marks: 5]

210 **(a)** $\frac{15}{100} \times 240$ or 0.15×240 **[M1]**

36 **[A1]**

(b) $\frac{40-32}{40} \times 100$ or $\frac{8}{40} \times 100$ or $100 - \frac{32}{40} \times 100$ **[M1]**

20 **[A1]**

(c) $120\% = 60$ or $1.2x = 60$ **[M1]**

50 **[A1]**

[Total Marks: 6]

211 **(a)** M1 for $15 \times (1 - 0.20)$ or $15 - 3$

12 **[A1]**

(b) M2 for $56 \div 0.8$ or $\frac{56}{80} \times 100$
(M1 for $80\% = 56$ or equivalent)

70 **[A1]**

(c) M1 for $\frac{42-30}{30} \times 100$ or $\frac{12}{30} \times 100$

40 **[A1]**

[Total Marks: 7]

212 **(a)** $4500 \times \left(1 + \frac{3.2}{100}\right)^5$ **[M2]**

OR 4500×1.032^5 **[M2]**

(M1 for 4500×1.032^k where $k > 1$)

5267.57... **[A1]**

(b) $\frac{408-320}{320} \times 100$ or $\frac{88}{320} \times 100$ **[M1]**

27.5 **[A1]**

[Total Marks: 5]

213 **(a)** $45\,000 \times 1.042$ OR $45\,000 + 45\,000 \times 0.042$ **[M1]**

46 890 **[A1]**

(b) $100\% - 24\% = 76\%$ **[B1]**

$\frac{342}{0.76}$ OR $\frac{342}{76} \times 100$ **[M1]**

450 **[A1]**

[Total Marks: 5]

214 **(a)** $4500 \times (1 + \frac{2.4}{100})^5$ or 4500×1.024^5 **[M2]**

If M0, M1 for 4500×1.024 or 4608 seen.

5066.57 or 5066.56... **[A1]**

(b) Interest earned $= 5066.57 - 4500 = 566.57$ (accept unrounded) **[M1]**

$\frac{4500 \times r \times 5}{100} = 566.57$ OR $225r = 566.57$ **[M1]**

2.52 or 2.518... **[A1]**

[Total Marks: 6]

215 **(a)** $8000 \times \left(1 + \frac{3.5}{100}\right)^4$ or 8000×1.035^4 **[M2]**

If M0 scored, allow [M1] for 8000×1.035^k where $k > 1$

9180.18 **[A1]**

(b) $x \times \left(1 - \frac{4}{100}\right)^3 = 45000$ or $x \times 0.96^3 = 45000$ **[M1]**

$\frac{45000}{0.96^3}$ **[M1]**

50863 **[A1]**

[Total Marks: 6]

216 **(a)** $4500 \times \left(1 + \frac{2.4}{100}\right)^5$ or 4500×1.024^5 **[M2]**

[M1] for 4500×1.024^n where $n > 1$

5066.55 **[A1]**

(b) $100\% - 18\% = 82\%$ or 0.82 **[B1]**

$\frac{557.60}{0.82}$ or $x \times 0.82 = 557.60$ **[M1]**

680 **[A1]**

[Total Marks: 6]

217 **(a)** M2 for $8000 \times \left(1 + \frac{3.5}{100}\right)^4$

(M1 for 8000×1.035^k where $k > 1$)

9180.19 **[A1]**

(b) M2 for $24000 \times 0.88^n < 10000$ or evaluating at least two consecutive trials (e.g. $n = 6$ and $n = 7$)
(M1 for 24000×0.88^n seen)

A1 for $n = 7$ (Value at year 6 is \$11145.69, Value at year 7 is \$9808.21)

(c) M2 for $150000 \div 1.25$

(M1 for 125% = 150000 or equivalent)

120000 **[A1]**

[Total Marks: 9]

218 **(a)** $4000 \times \left(1 + \frac{2.5}{100}\right)^5 - 4000$ [M2]

(M1 for 4000×1.025^5)

525.63 or 525.64 **[A1]**

(b) Interest $= 4896 - 4000 = 896$ **[B1]**

$896 = 4000 \times \frac{3.2}{100} \times t$ or $t = \frac{896}{128}$ **[M1]**

7 **[A1]**

(c) $P \times 1.018^6 = 6676.70$ **[M1]**

$P = \frac{6676.70}{1.018^6}$ **[M1]**

6000 **[A1]**

[Total Marks: 9]

219 **(a)** $(1.25 \times 10^6) \times 1.045^{10}$ **[M1]**

1941214.7... **[A1]**

1.94×10^6 **[A1]**

(b) $299.25 = 86\%$ or $\frac{299.25}{1 - 0.14}$ **[M1]**

348 **[A1]**

(c) Let V be the original value. $V \times 0.85 \times 0.90^2 = 17212.50$ **[M1]**

$V = \frac{17212.50}{0.85 \times 0.81} = 25000$ **[A1]**

[Total Marks: 7]

220 **(a)** $1\,250\,000 \times (1 + \frac{x}{100})^3 = 1\,367\,631$ **[M1]**

$(1 + \frac{x}{100})^3 = 1.0941048$ **[M1]**

$1 + \frac{x}{100} = \sqrt[3]{1.0941048} = 1.0304\cdots \implies x = 3.04\cdots \approx 3$ **[A1]**

(b) Target population $= 2 \times 1\,250\,000 = 2\,500\,000$ **[B1]**

$1\,250\,000 \times (1.03)^n > 2\,500\,000 \implies 1.03^n > 2$ **[M1]**

Using logs: $n \log(1.03) > \log(2)$ OR trial and improvement yielding n between 23 and 24 **[M1]**

$n > 23.449\ldots$, so it takes 24 years. Year $= 2015 + 24 = 2039$ **[A1]**

[Total Marks: 7]

221 **(a)** $p \times 1.2 = 960$ or $\dfrac{960}{1.2}$ **[M1]**

800 **[A1]**

(b) Selling price $= 1.15q$ **[B1]**

Marked Price $\times 0.9 = 1.15q$ or $\dfrac{1.15q}{0.9}$ **[M1]**

$\dfrac{23q}{18}$ or $\dfrac{23}{18}q$ **[A1]**

(c) $500 \times \left(1 + \dfrac{r}{100}\right)^{10} = 740.12$ or $\left(1 + \dfrac{r}{100}\right)^{10} = \dfrac{740.12}{500}$ **[M1]**

$1 + \dfrac{r}{100} = \sqrt[10]{1.48024}$ or 1.04 **[M1]**

4 **[A1]**

[Total Marks: 8]

222 **(a)** $5000 \times \left(1 + \dfrac{2.5}{100}\right)^3$ **[M1]**

5384.45 **[A1]**

(b) $\left(1 + \dfrac{r}{100}\right)^5 = 1.30$ **[M1]**

$1 + \dfrac{r}{100} = \sqrt[5]{1.30}$ **[M1]**

5.39 (or $5.387\ldots$) **[A1]**

(c) $V \times (1 - 0.12)^3 = 17\,036.80$ **[M1]**

$V = \dfrac{17\,036.80}{0.88^3}$ **[M1]**

25 000 **[A1]**

[Total Marks: 8]

223 **(a)** Expression for 3 years depreciation: e.g. $V \times (1 - 0.15) \times (1 - 0.08)^2$ or $V \times 0.85 \times 0.92^2$ **[M2]**

[M1] for $V \times 0.85$ or $V \times 0.92^2$ seen

$V \times 0.71944 = 17986$ or $V = \dfrac{17986}{0.85 \times 0.92^2}$ **[M1]**

25 000 **[A1]**

(b) $\dfrac{25000 - 17986}{25000} \times 100$ or $100\% - (0.85 \times 0.92^2 \times 100)\%$ **[M1]**

28.1 **[A1]**

(c) $P \times \left(1 + \dfrac{r}{100}\right)^{12} = 2P$ or $\left(1 + \dfrac{r}{100}\right)^{12} = 2$ **[M1]**

$1 + \dfrac{r}{100} = \sqrt[12]{2}$ or 1.059... **[M1]**

5.95 (correct to 3 significant figures) **[A1]**

[Total Marks: 9]

224 **(a)** $V \times (1 - 0.15)^3 = 12282.50$ or $V \times 0.85^3 = 12282.50$ [M2]
(M1 for $V \times 0.85^k = 12282.50$ where $k > 1$ or $\dfrac{12282.50}{0.85}$ seen)

20 000 **[A1]**

(b) $15000 \times \left(1 + \dfrac{x}{100}\right)^5 = 21500$ or $\left(1 + \dfrac{x}{100}\right)^5 = \dfrac{215}{150}$ **[M1]**

$\sqrt[5]{\dfrac{215}{150}} - 1$ or $1.0746... - 1$ **[M1]**

7.5 **[A1]**

(c) Multiplier $= 1.04 \times 0.98 \times 1.05$ or 1.07016 [M2]
(M1 for two correct multipliers used, e.g. 1.04×0.98)

Overall percentage increase $= (1.07016 - 1) \times 100$ **[M1]**

7.02 **[A1]**

[Total Marks: 10]

225 **(a)** M2 for $\left(1 + \dfrac{r}{100}\right)^6 = 2$ or equivalent
(M1 for $\left(1 + \dfrac{r}{100}\right)^6$ seen)

12.2 **[A1]**

(b) M2 for $X \times 1.40 \times 0.85 = 142.80$ or $1.19X = 142.80$
(M1 for $X \times 1.40$ or Selling Price $\times 0.85 = 142.80$)

120 **[A1]**

(c) M1 for $P \times \dfrac{x}{100} \times 5 = 250$ (implies $Px = 5000$)

M2 for $P\left(1 + \dfrac{x}{100}\right)^2 - P = 102.50$
(M1 for $P\left(1 + \dfrac{x}{100}\right)^2 = P + 102.50$)

M1 for correctly substituting to form an equation in one variable, e.g.
$\dfrac{5000}{x}\left(\dfrac{2x}{100} + \dfrac{x^2}{10000}\right) = 102.50$

$x = 5$ **[A1]**

$P = 1000$ **[A1]**

[Total Marks: 11]

226 Numerator $\approx 1.907878...$, Denominator $= 1.025$ **[M1]**

1.861344784 **[A1]**

[Total Marks: 2]

227 **(a)** 3.068822... **[B2]**

If 0 scored, allow [B1] for seeing 15.36 or 2.485... or 0.81

(b) 3.069 [B1] (strict FT from their (a) if given to at least 4 d.p.)

[Total Marks: 3]

228 **(a)** Numerator $\sqrt{22.01} = 4.69148\ldots$ or Denominator $12.167 - 1.4 = 10.767$ seen **[M1]**

0.435727839 **[A1]**

(b) 0.436 **[B1]**

[Total Marks: 3]

229 **(a)** M1 for $\sqrt{158.79}$ or $12.60119\ldots$ or 4.48 seen

2.812765825 (accept to at least 4 decimal places) **[A1]**

(b) 2.81 **[B1ft]**

(c) 3.75×10^8 **[B1]**

[Total Marks: 4]

230 **(a)** 64.79 or 14.861 or ≈ 3.855 seen **[M1]**

16.80675747 (allow up to 2 missing trailing digits depending on calculator model) **[A1]**

(b) 16.8 **[B1]**

(c) 6 hours 27 minutes **[B1]**

[Total Marks: 4]

231 **(a)** 4.423... or $\dfrac{15.98...}{3.612}$ **[M1]**

4.42 or 4.424 **[A1]**

(b) 1.4862... − 0.8191... or 0.6671... **[M1]**

0.667 **[A1]**

[Total Marks: 4]

232 **(a)** 4.8 **[M1]**

4.80 **[A1]**

(b) $2 + \dfrac{18}{60}$ or $\dfrac{138}{60}$ **[M1]**

2.3 **[A1]**

[Total Marks: 4]

233 **(a)** $150.0625 + 4.4412...$ or $154.5037...$ **[M1]**

154.50 **[A1]**

(b) $\dfrac{1.3639...}{0.9510...}$ or $1.434...$ **[M1]**

1.43 **[A1]**

[Total Marks: 4]

234 **(a)** 8.40 **[B1]**

(b) 5 hours 9 minutes **[B1]**

(c) $280 \div 85$ **[M1]**

$3.294...$ **[A1]**

3 hours 18 minutes **[A1]**

[Total Marks: 5]

235 **(a)** **(i)** Numerator 4.94 or denominator $1.14...$ seen **[M1]**

$4.332666...$ **[A1]**

(ii) 4.33 **[B1]**

(b) 0.35×60 **[M1]**

5 h 21 min **[A1]**

[Total Marks: 5]

236 **(a)** $\pi \times 4.2^2$ or $55.417...$ **[M1]**

55 **[A1]**

(b) $23 \div 60$ or $0.3833...$ seen **[M1]**

Time $= 4 + \dfrac{23}{60}$ or $4.3833...$ hours **[M1]**

Distance $= 78 \times 4.3833...$ **[M1]**

341.9 **[A1]**

[Total Marks: 6]

237 **(a)** Time $= 3 + \dfrac{40}{60}$ or $\dfrac{11}{3}$ hours **[M1]**

Cost $= 45 + 32.50 \times \dfrac{11}{3}$ **[M1]**

164.17 **[A1]**

(b) $230.25 - 45 = 185.25$ **[M1]**

$\dfrac{185.25}{32.50} = 5.7$ hours **[M1]**

5 hours 42 minutes **[A1]**

[Total Marks: 6]

238 **(a)** M1 for $\frac{4}{3} \times \pi \times 4.85^3$

477.8 **[A1]**

(b) M1 for $\frac{1}{3} \times \pi \times 1.2^2 \times 3.5$ (implies $5.2778\ldots$)

M1 for $\frac{\text{their } 477.8}{\text{their } 5.2778\ldots}$

A1 for $90.5\ldots$

90 **[B1]**

(c) 0.584 (accept 0.582 to 0.585) **[B1ft]**

[Total Marks: 7]

239 **(a)** Time $= 24\frac{40}{60}$ or 24.666... or $\frac{74}{3}$ **[M1]**

$\frac{74}{3} \times 18.60 = 458.8$ **[M1]**

458.80 (must have trailing zero) **[A1]**

(b) 479.45 **[B1]**

(c) 3.4×150 or 510 minutes **[M1]**

$510 \div 60$ or 8.5 **[M1]**

8 hours 30 minutes **[A1]**

[Total Marks: 7]

240 **(a)** $\frac{4}{3}\pi(1.4)^3$ **[M1]**

11.49404... **[A1]**

(b) their 11.494... $\div 0.035$ **[M1]**

328.4... or 328 (minutes) **[A1]**

their 328.4... $\div 60$ (converting to hours) **[M1]**

5 hours 28 minutes **[A1]**

[Total Marks: 6]

241 **(a)** 14.4×10^3 or 14400 **[M1]**

1.44×10^4 **[A1]**

(b) $\frac{1}{2.5066...} \times 0.7788...$ or $0.398942... \times 0.778800...$ or 0.310696... **[M1]**

0.3107 **[A1]**

(c) $1.5 = 2\pi\sqrt{\dfrac{l}{9.81}} \implies \dfrac{1.5}{2\pi} = \sqrt{\dfrac{l}{9.81}} \implies 0.2387... = \sqrt{\dfrac{l}{9.81}}$ **[M1]**

$l = 9.81 \times (0.2387...)^2 \implies \mathrm{l} = 0.55909...$ **[M1]**

0.56 **[A1]**

[Total Marks: 6]

242 **(a)** Numerator $= 340000 + 27000 = 367000$ or 3.67×10^5 **[M1]**

$3.67 \times 10^5 \div (8.1 \times 10^{-3}) = 45308641.98...$ **[M1]**

4.53×10^7 or 4.531×10^7 **[A1]**

(b) Upper bound for $r = 5.25$ and Upper bound for $h = 12.45$ **[M1]**

$V = \dfrac{1}{3} \times \pi \times 5.25^2 \times 12.45$ **[M1]**

359.34... or 359 **[A1]**

[Total Marks: 6]

243 **(a)** M1 for $2\pi\sqrt{\dfrac{0.85}{9.81}}$

1.850 **[A1]**

(b) M1 for $\left(\dfrac{T}{2\pi}\right)^2 = \dfrac{l}{g}$ or $\dfrac{T^2}{4\pi^2} = \dfrac{l}{g}$

$\dfrac{gT^2}{4\pi^2}$ **[A1]**

(c) M1 for their $l = \dfrac{9.81 \times 2.5^2}{4\pi^2}$

A1 for $l = 1.5532\ldots$ metres

155.3 **[A1]**

[Total Marks: 7]

244 **(a)** $\dfrac{1}{3} \times \pi \times 3.2^2 \times (3 \times 8.45 - 3.2)$ **[M1]**

237.5 **[A1]**

(b) $\dfrac{3V}{\pi h^2} = 3R - h$ **[M1]**

$\dfrac{3V}{\pi h^2} + h = 3R$ **[M1]**

$R = \dfrac{V}{\pi h^2} + \dfrac{h}{3}$ or $R = \dfrac{3V + \pi h^3}{3\pi h^2}$ **[A1]**

(c) $\dfrac{500}{\pi \times 5.6^2} + \dfrac{5.6}{3}$ **[M1]**

6.94 **[A1]**

[Total Marks: 7]

245 **(a)** $\dfrac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{6.80 \times 10^{6}}$ or $5.8558... \times 10^{7}$ **[M1]**

$\sqrt{5.8558... \times 10^{7}}$ or 7652.36... **[M1]**

7650 **[A1]**

(b) Total distance $= 2 \times \pi \times 6.80 \times 10^{6}$ or 42725660... **[M1]**

Time in seconds $= \dfrac{42725660...}{7652.36...} = 5583.32...$ **[M1]**

$5583 \div 3600 = 1.55...$ (implies 1 hour) and remainder 1983 seconds **[M1]**

1 hour 33 minutes 3 seconds **[A1]**

[Total Marks: 7]

246 **(a)** $\dfrac{1}{3} \times \pi \times 2.31^2 \times (3 \times 8.74 - 2.31)$ **[M1]**

133.607... **[A1]**

133.6 **[A1]**

(b) Rounding initial values: $R = 8.7$ and $h = 2.3$ **[B1]**

$\dfrac{1}{3} \times \pi \times 2.3^2 \times (3 \times 8.7 - 2.3)$ or 131.84... **[M1]**

Difference $= 133.607... - 131.844... = 1.76...$ **[M1]**

1.8 **[A1]**

[Total Marks: 7]

247 **(a)** 2 hours 35 minutes **[B1]**

(b) M1 for 15 35 or 17 70 seen

18 10 **[A1]**

[Total Marks: 3]

248 **(a)** 2 hours 15 minutes **[B1]**

(b) Adds 2 hours 35 minutes to 16 45 (e.g., 18 80) **[M1]**

19 20 **[A1]**

[Total Marks: 3]

249 Arrival time in London local time $= 21 : 45 + 6$ hours 50 minutes **[M1]**

$04 : 35$ (next day) **[M1]**

Dubai local time = 04 : 35 + 4 hours = 08 : 35 **[A1]**

[Total Marks: 3]

250 **(a)** London departure time = 10 30 or Dubai arrival time = 23 15 **[M1]**

8 hours 45 minutes **[A1]**

(b) London departure time = 18 45 or Dubai arrival time = 06 05 (Saturday) **[M1]**

02 05 Saturday **[A1]**

[Total Marks: 4]

251 **(a)** 22 50 + 14 h 35 min **[M1]**

13 25 **[A1]**

(b) 0.4×60 **[M1]**

3 h 24 min **[A1]**

[Total Marks: 4]

252 **(a)** Time from 09 45 to 10 00 = 15 minutes, and 10 00 to 11 20 = 1 hour 20 minutes **[M1]**

1 hour 35 minutes **[A1]**

(b) 14 55 + 4 hours = 18 55 **[M1]**

19 05 **[A1]**

[Total Marks: 4]

253 **(a)** 115 minutes = 1 hour 55 minutes **[M1]**

21 30 (or 21 : 30) **[A1]**

(b) Arrival time in New York = 21 40 + 7 h 15 m = 04 55 (next day) **[M1]**

Arrival time in Paris = 04 55 + 6 hours **[M1]**

10 55 (or 10 : 55) **[A1]**

[Total Marks: 5]

254 Flight arrival time in Dubai local time: 14 30 + 14 h 15 min = 28 45 (or 04 45 next day) **[M1]**

Adds time zone difference: 28 45 + 7 hours (or 04 45 + 7 hours) **[M1]**

11 45 **[A1]**

[Total Marks: 3]

255 **(a)** $3 \times 10^5 \times 60 \times 60 \times 24 \times 365$ **[M1]**

9.46×10^{12} or 9.4608×10^{12} **[A1]**

(b) $\dfrac{4.5 \times 10^{13}}{9.4608 \times 10^{12}}$ **[M1]**

4.76 **[A1]**

[Total Marks: 4]

256 (a) $11:03 - 08:15$ **[M1]**

2 hours 48 minutes **[A1]**

(b) 48 minutes $= \dfrac{48}{60}$ hours $= 0.8$ hours **[M1]**

Average speed $= \dfrac{420}{2.8}$ **[M1]**

150 **[A1]**

[Total Marks: 5]

257 (a) M1 for arrival time in Dubai (10 00 Tuesday) OR departure time in London (22 15 Monday) seen

M1 for a valid time zone adjustment and flight duration addition

06 00, Tuesday **[A1]**

(b) M1 for $1.5 \times 365 \times 24 \times 60$

A1 for 788400

7.884×10^5 **[B1]**

[Total Marks: 6]

258 (a) London departure time $= 16\,30 - 14$ h 15 min or $02\,15$ **[M1]**

Tokyo departure time $= 02\,15 + 9$ h **[M1]**

$11\,15$ **[A1]**

(b) 1 min 45 s = 105 seconds or 8 hours = 480 minutes **[B1]**

$(8 \times 3600) \div 105$ or $480 \div 1.75$ **[M1]**

274 **[A1]**

[Total Marks: 6]

259 (a) London departure time in Dubai local time $= 13\,45 + 4$ hours $= 17\,45$ **[M1]**

Time from $17\,45$ to $00\,15 = (24\,00 - 17\,45) + 15$ minutes = 6 hours 15 minutes +15 minutes **[M1]**

6 hours 30 minutes **[A1]**

(b) Flight time converted to decimal hours = 6.5 **[B1]**

Speed $= \dfrac{5486}{6.5}$ **[M1]**

844 **[A1]**

[Total Marks: 6]

260 **(a)** Time $= \dfrac{42.2}{12.5}$ or 3.376 hours **[M1]**

$0.376 \times 60 = 22.56$ minutes **[M1]**

$0.56 \times 60 = 33.6$ seconds **[M1]**

3 hours, 22 minutes, 34 seconds **[A1]**

(b) Correct difference calculation, e.g. 12 hours 64 minutes −08 hours 35 minutes **[M1]**

4 hours 29 minutes **[A1]**

[Total Marks: 6]

261 **(a)** M1 for $30 \times 24 \times 60$ or 43200 minutes seen

M1 for $\dfrac{43200}{94.5}$ or $457.142\ldots$

457 **[A1]**

(b) M1 for 100×94.5 or 9450 minutes

M1 for converting minutes into hours and days (e.g., 157.5 hours OR 6 days 13 hours 30 minutes)

M1 for adding their 6 days and 13 hours 30 minutes to 05 50 on 1st March

19 20, 7th March **[A1]**

[Total Marks: 7]

262 **(a)** Total minutes $= \dfrac{1\,000\,000}{80} = 12\,500$ **[M1]**

Total hours $= 12\,500 \div 60 = 208$ with remainder 20 (so 20 minutes) **[M1]**

Total days $= 208 \div 24 = 8$ with remainder 16 (so 16 hours) **[M1]**

8 days, 16 hours, 20 minutes **[A1]**

(b) Production start in Tokyo time = 08 00 + 14 hours = 22 00 on Monday **[M1]**

Add 8 days

$22\,00 on Tuesday (next week) [M1]$

Add 16 hours

$22\,00 + 16 hours = 38\,00$

$14\,00 on Wednesday [M1]$

Add 20 minutes

Wednesday,14 20[*A*1]

[Total Marks: 7]

263 **(a)** $920 \div 24$ **[M1]**

38.333... hours **[A1]**

38 hours 20 minutes **[A1]**

(b) Adds their time from part (a) to Wednesday 17 50 **[M1]**

Wednesday 17 50 + 20 minutes = 18 10 **[M1]**

Wednesday 18 10 + 38 hours = Thursday 18 10 + 14 hours **[M1]**

Day: Friday **[A1]**

Time: 08 10 **[A1]**

[Total Marks: 7]

264 **(a)** $\frac{6600}{800}$ or 8.25 **[M1]**

8 hours 15 minutes **[A1]**

(b) Arrival time in City A time = 05 55 **[M1]**

00 55 Wednesday **[A1]**

(c) Departure from City B = 15 15 (local time) or 20 15 (City A time) **[M1]**

Arrival in City B time = 23 00 **[M1]**

04 00 Thursday **[A1]**

[Total Marks: 7]

265 **(a)** $\frac{35}{x} + \frac{42}{x-5}$ **[B1]**

(b) 4 hours 33 minutes $= 4 + \frac{33}{60}$ or $\frac{91}{20}$ hours **[B1]**

$\frac{35}{x} + \frac{42}{x-5} = \frac{91}{20}$ **[M1]**

Multiplies by common denominator: $20 \times 35(x-5) + 20 \times 42x = 91x(x-5)$ **[M1]**

$700x - 3500 + 840x = 91x^2 - 455x$ **[M1]**

$1540x - 3500 = 91x^2 - 455x \implies 91x^2 - 1995x + 3500 = 0$ **[M1]**

Divides by 7 to reach $13x^2 - 285x + 500 = 0$ without errors **[A1]**

(c) Uses quadratic formula or factors: $(13x - 25)(x - 20) = 0$ **[M1]**

$x = 20$ or $x = \frac{25}{13}$ (or 1.92...) **[A1]**

$x = 20$ only (rejects $\frac{25}{13}$ as speed $x - 5$ must be positive) **[A1]**

[Total Marks: 8]

266 **(a)** Minutes in a year $= 365 \times 24 \times 60$ **[M1]**

525600 minutes **[M1]**

Number of orbits $= \frac{525600}{92} = 5713.043\ldots$ **[M1]**

5713 **[A1]**

(b) **(i)** $\frac{11440}{880} = 13$ **[B1]**

(ii) Time difference $= 9 - (-4) = 13$ hours (Q is 13 hours ahead) **[M1]**

Departure time in City Q's local time = 22 : 45 (Monday) +13 hours = 11 : 45 (Tuesday) **[M1]**

Arrival time in City Q's local time = 11 : 45 (Tuesday) +13 hours (flight time) = 00 : 45 (Wednesday) **[A1]**

[Total Marks: 8]

267 **(a)** 15×92.75 or 1380 min 675 s **[M1]**

23 h 11 min 15 s **[A1]**

(b) Arrival time in City A local time = 09 45 + 8 h 20 min or 18 05 **[M1]**

26 05 − 18 05 **[M1]**

8 **[A1]**

(c) Distance travelled by Train P at 08:30 $= 90 \times 0.5 = 45$ km **[M1]**

Relative speed of approaching trains $= 90 + 120 = 210$ km/h **[M1]**

Time taken to meet from 08:30 $= (360 - 45) \div 210 = 1.5$ hours **[M1]**

10 00 **[A1]**

[Total Marks: 9]

268 **(a)** 1680 **[B1]**

(b) M1 for $1680 - 1050$ or 630 seen

140 **[A1]**

[Total Marks: 3]

269 $255 \div 0.85$ **[M1]**

\$300 **[A1]**

\$100 **[A1]**

[Total Marks: 3]

270 $(3 \times 1.20) + (4 \times 2.50)$ or $3.60 + 10.00$ or 13.60 **[M1]**

$20 - 13.60$ **[M1]**

6.40 **[A1]**

[Total Marks: 3]

271 $300 \times 1.50 = 450$ **[M1]**

$450 - 210 = 240$ **[M1]**

$240 \div 1.50 = 160$ **[A1]**

[Total Marks: 3]

272 **(a)** 400×0.85 **[M1]**

340 **[A1]**

(b) $\frac{102}{0.85}$ or $\frac{10200}{85}$ **[M1]**

120 **[A1]**

[Total Marks: 4]

273 **(a)** Cost of pens $= 4 \times 2.35 = 9.40$ **[M1]**

Cost of notebooks $= 3 \times 4.50 = 13.50$ **[M1]**

Change $= 50 - (9.40 + 13.50) = 27.10$ **[A1]**

(b) 54 **[B1]**

[Total Marks: 4]

274 **(a)** 850×148 **[M1]**

125800 **[A1]**

(b) $125800 - (45000 + 31500)$ or 49300 **[M1]**

$\frac{49300}{152}$ or $324.342...$ **[M1]**

324.34 **[A1]**

[Total Marks: 5]

275 **(a)** $1\,250\,000 \div 145$ **[M1]**

8620.69 **[A1]**

(b) Total cost $= 8620.69 + 600 = 9220.69$ **[M1]**

Profit $= 12\,500 - 9220.69 = 3279.31$ (or $\dfrac{12500 - 9220.69}{9220.69}$) **[M1]**

35.6 (accept $35.56\ldots$) **[A1]**

[Total Marks: 5]

276 **(a)** 450×1.14 **[M1]**

513 **[A1]**

(b) $1.32 \div 1.14$ **[M1]**

1.1578... **[A1]**

1.16 **[A1]**

[Total Marks: 5]

277 **(a)** M1 for 680×1.32 (implies 897.60)

M1 for their $897.60 - 850$
OR
M1 for $850 \div 1.32$ (implies $643.939\ldots$)
M1 for (680 − their $643.939\ldots$) $\times 1.32$

47.60 (accept 47.6) **[A1]**

(b) M1 for 850×1.04 or $850 \times 0.04 + 850$ (implies 884)

M1 for their 884×0.85

751.40 (accept 751.4) **[A1]**

[Total Marks: 6]

278 **(a)** Amount in euros $= 450 \times 1.18 = 531$ **[M1]**

Remaining euros $= 531 - 285 = 246$ **[M1]**

Amount back in pounds $= 246 \div 1.22$ **[M1]**

201.639... rounding to 201.64 **[A1]**

(b) Sale price $= 850 \times 0.85$ $(= 722.5)$ **[M1]**

Final price $= 722.5 \times 0.95 = 686.375$ which rounds to 686.38 **[A1]**

[Total Marks: 6]

279 **(a)** $\dfrac{704}{1.28} = 550$ $(£)$ **[M1]**

550×1.14 **[M1]**

627 **[A1]**

(b) Zurich price in pounds $= \dfrac{995}{1.14} = £872.807...$ (or London in CHF $= 850 \times 1.14 = 969$) **[M1]**

Difference in dollars $= (872.807... - 850) \times 1.28$ (or $\dfrac{995 - 969}{1.14} \times 1.28$) **[M1]**

29.19 **[A1]**

[Total Marks: 6]

280 **(a)** 850×0.98 or $850 - (850 \times 0.02)$ **[M1]**

833 **[A1]**

149940 **[A1]**

(b) $149940 - 115000$ $(= 34940)$ **[M1]**

their $34940 - 500$ $(= 34440)$ **[M1]**

their $34440 \div 184$ **[M1]**

187.17 (accept 187.17...) **[A1]**

[Total Marks: 7]

281 **(a)** $\frac{17400 + 1200}{1.24}$ or $\frac{18600}{1.24}$ **[M1]**

15000 **[A1]**

(b) $17100 - 15000$ or 2100 **[M1]**

$\frac{2100}{15000} \times 100$ **[M1]**

14 **[A1]**

(c) $2100 \times (1 - 0.02)$ or 2100×0.98 or 2058 **[M1]**

2058×1.16 **[M1]**

2387.28 **[A1]**

[Total Marks: 8]

282 Total revenue in USD $= 1000 \times 4.20 = 4200$ **[B1]**

Convert revenue to EUR: $4200 \div 1.20$ $(= 3500)$ **[M1]**

Equate to cost with profit multiplier: Total Cost $\times 1.40 = 3500$ or $3500 \div 1.40$ $(= 2500)$ **[M1]**

Form equation for cost: $1000y + 250 = 2500$ **[M1]**

$y = 2.25$ **[A1]**

[Total Marks: 5]

283 **(a)** 800×0.85 or 680 **[M1]**

$680 - 420$ or 260 **[M1]**

$260 \div 0.82$ **[M1]**

317.07 **[A1]**

(b) $61.20 \div (1 - 0.15)$ or $61.20 \div 0.85$ **[M1]**

72 **[A1]**

[Total Marks: 6]

284 **(a)** Cost of Brand A per 100g $= 12.50 \div 5 = 2.50$ (or per gram $= 0.025$) **[M1]**

Cost of Brand B per 100g $= 8.40 \div 3 = 2.80$ (or per gram $= 0.028$) **[M1]**

Brand A with correct supporting working **[A1]**

(b) $x + y = 2000$ **[M1]**

Cost equation: $0.025x + 0.028y = 54.10$ (or equivalent, e.g. $\frac{12.5}{500}x + \frac{8.4}{300}y = 54.1$) **[M1]**

Substitute $y = 2000 - x$: $0.025x + 0.028(2000 - x) = 54.10$ **[M1]**

$0.025x + 56 - 0.028x = 54.10 \implies -0.003x = -1.9 \implies x = 633.3...$ and $y = 1366.6...$
Accept $x = 633$ and $y = 1367$ (or exact fractions $\frac{1900}{3}$ and $\frac{4100}{3}$) **[A1]**

[Total Marks: 7]

285 **(a)** Amount after commission $= 2500 \times (1 - 0.015) = 2462.50$ **[M1]**

2462.50×112 **[M1]**

275 800 **[A1]**

(b) Remaining yen $= 275\,800 - 198\,400 = 77\,400$ **[M1]**

$\frac{77\,400}{115} = 673.043...$ **[M1]**

673.04 **[A1]**

(c) $\frac{77\,400}{112} - 673.043...$ or $691.071... - 673.043...$ **[M1]**

18.03 **[A1]**

[Total Marks: 8]

286 **(a)** $\frac{400}{x}$ **[B1]**

(b) $\frac{450}{x - 0.05} - \frac{400}{x} = 100$ **[M1]**

$450x - 400(x - 0.05) = 100x(x - 0.05)$ **[M1]**

$450x - 400x + 20 = 100x^2 - 5x$ **[M1]**

$50x + 20 = 100x^2 - 5x \implies 100x^2 - 55x - 20 = 0$ **[A1]**

Dividing by 5 yields $20x^2 - 11x - 4 = 0$ (must show working to reach final equation) **[A1]**

(c) $(5x - 4)(4x + 1) = 0$ or use of quadratic formula $\frac{11 \pm \sqrt{(-11)^2 - 4(20)(-4)}}{2(20)}$ **[M2]**

If M2 not scored, allow [M1] for $(5x \pm a)(4x \pm b) = 0$ where $ab = 4$ or $a(4) \pm b(5) = -11$, or correct substitution into formula but with one error.

0.8 (with -0.25 rejected or ignored) **[A1]**

[Total Marks: 8]

287 **(a)** M1 for $21500 - 24000x = 740$

M1 for $24000x = 20760$

0.865 **[A1]**

(b) M1 for 780×1.28 (implies 998.4)

M1 for their $998.4 \div (1 - 0.025)$ or their $998.4 \div 0.975$

1024 **[A1]**

(c) M1 for setting up a correct equation, e.g. $R \times (1 - 0.08) \times k = R \times 0.828$ or $0.92k = 0.828$

M1 for $k = \dfrac{0.828}{0.92}$

0.9 **[A1]**

[Total Marks: 9]

288 **(a)** 840×1.15 or 966 **[M1]**

$\dfrac{821.10}{966}$ or 0.85 **[M1]**

15 **[A1]**

(b) $\dfrac{2430000}{450}$ or 5400 **[M1]**

$\dfrac{5400}{7.2}$ or 750 **[M1]**

90 **[A1]**

(c) 840×1.025 or 861 **[M1]**

$\dfrac{861}{1.32}$ or 652.272... **[M1]**

652.27 **[A1]**

[Total Marks: 9]

289 **(a)** **(a)** $8000(1 + \dfrac{p}{100})^3 = 27000$ **[M1]**

$1 + \dfrac{p}{100} = \sqrt[3]{\dfrac{27000}{8000}} = \dfrac{3}{2}$ **[M1]**

50 **[A1]**

(b) $27000 \times (1.5)^2$ or $8000 \times (1.5)^5$ **[M1]**

60 750 **[A1]**

[Total Marks: 5]

290 **(a)** $\left(1 + \dfrac{r}{100}\right)^2 = 1.44$ or $x^2 = 1.44$ **[M1]**

$1 + \frac{r}{100} = 1.2$ or $x = 1.2$ **[M1]**

20 **[A1]**

(b) Number of doubling periods $= 45 \div 15 = 3$ or 80×2^3 **[M1]**

640 **[A1]**

[Total Marks: 5]

291 M1 for $24000 \times (1 - 0.08)^5$ or 24000×0.92^5

A1 for $15817.96\ldots$

15818 **[B1]**

[Total Marks: 3]

292 $125\,000 \times (1 + 0.018)^8$ or $125\,000 \times 1.018^8$ [M2]
(If not M2, award M1 for 1.018^8 or $125\,000 \times 1.018^k$ where $k > 1$)

144 183.6... **[A1]**

144 184 **[B1]**

[Total Marks: 3]

293 $26000 \times (1 - 0.15)^4$ or 26000×0.85^4 **[M2]**

If M0, allow [M1] for 26000×0.85^k where $k > 1$

13572 **[A1]**

[Total Marks: 3]

294 $450 \times \left(1 + \frac{6.5}{100}\right)^{12}$ **[M1]**

450×1.065^{12} **[M1]**

958.01 (accept 958) **[A1]**

[Total Marks: 3]

295 $32000 \times (1 - 0.12)^5$ oe **[M2]**

For 32000×0.88^n where $n > 1$, award **[M1]**

16887 **[A1]**

[Total Marks: 3]

296 (a) $65000 \times \left(1 + \frac{2.8}{100}\right)^5$ or 65000×1.028^5 **[M1]**

74624 **[A1]**

(b) $\frac{74624 - 65000}{65000} \times 100$ or $(1.028^5 - 1) \times 100$ **[M1]**

14.8 **[A1]**

[Total Marks: 4]

297 **(a)** $42\,000 \times 1.025^6$ $(= 48\,707.12...)$ **[M1]**

48 700 **[A1]**

(b) $(1.025^6 - 1) \times 100$ or $\frac{48\,707.12... - 42\,000}{42\,000} \times 100$ **[M1]**

16.0 (15.969...) **[A1]**

[Total Marks: 4]

298 **(a)** 1500×1.08^{12} **[M1]**

3777 **[A1]**

(b) $1500 \times 1.08^n > 3000$ or $1.08^n = 2$ **[M1]**

Trial with values (e.g., $1.08^9 = 1.999...$ and $1.08^{10} = 2.158...$) or uses logarithms (e.g., $n = \frac{\log(2)}{\log(1.08)} = 9.006...$) **[M1]**

10 **[A1]**

[Total Marks: 5]

299 **(a)** **(a)** $85000(1 + \frac{r}{100})^{12} = 112000$ **[M1]**

$1 + \frac{r}{100} = \left(\frac{112}{85}\right)^{\frac{1}{12}}$ **[M1]**

2.33 (from 2.327...) **[A1]**

(b) $85000 \times \left(1 + \frac{2.327...}{100}\right)^{20}$ or $112000 \times \left(1 + \frac{2.327...}{100}\right)^{8}$ **[M1]**

135 000 **[A1]**

[Total Marks: 5]

300 **(a)** $28\,000 \times \left(1 - \frac{r}{100}\right)^4 = 16\,150$ **[M1]**

$\left(1 - \frac{r}{100}\right) = \sqrt[4]{\frac{16\,150}{28\,000}}$ or $\sqrt[4]{0.57678...}$ **[M1]**

12.9 (or 12.88 to 12.89) **[A1]**

(b) $16\,150 \times (1 - 0.1289...)^3$ or $28\,000 \times (1 - 0.1289...)^7$ **[M1]**

10 677 **[A1]**

[Total Marks: 5]

301 **(a)** $45000 \times \left(1 - \frac{x}{100}\right)^4 = 26000$ **[M1]**

$1 - \frac{x}{100} = \sqrt[4]{\frac{26000}{45000}}$ or $1 - \frac{x}{100} = 0.8711...$ **[M1]**

12.9 **[A1]**

(b) $45000 \times (0.8711...)^n < 10000$ or $(0.8711...)^n < \frac{2}{9}$ **[M1]**

Testing values or using logarithms: $n = 10 \implies \$11342$ and $n = 11 \implies \$9881$ **[M1]**

11 **[A1]**

[Total Marks: 6]

302 **(a)** M1 for $125000 \times \left(1 + \frac{2.4}{100}\right)^{10}$ or 125000×1.024^{10}

A1 for $158461.5\ldots$

158500 **[B1]**

(b) M1 for $85000 \times \left(1 - \frac{r}{100}\right)^8 = 68000$

M1 for $1 - \frac{r}{100} = \sqrt[8]{\frac{68000}{85000}}$ or $0.8^{\frac{1}{8}}$

2.75 (accept 2.750 to 2.751) **[A1]**

[Total Marks: 6]

303 **(a)** $36\,000 \times 0.86^4$ $(= 19\,692.29...)$ **[M1]**

19 692 **[A1]**

(b) $36\,000 \times 0.86^n < 10\,000$ or $0.86^n = \frac{10\,000}{36\,000}$ **[M1]**

Applies logarithms or systematic trials: $n = \frac{\log(10/36)}{\log(0.86)}$ or $0.86^8 = 0.2992...$,

$0.86^9 = 0.2573...$ **[M1]**

$n = 8.49...$ **[A1]**

9 **[A1]**

[Total Marks: 6]

304 **(a)** $28\,500 \times \left(1 - \frac{12}{100}\right)^5$ **[M1]**

$28\,500 \times 0.88^5$ **[M1]**

15 040 **[A1]**

(b) $35\,000 \times \left(1 - \frac{r}{100}\right)^4 = 15\,200$ **[M1]**

$\left(1 - \frac{r}{100}\right)^4 = \frac{152}{350}$ **[M1]**

$1 - \frac{r}{100} = \sqrt[4]{\frac{152}{350}} = 0.8123\ldots$ **[M1]**

18.8 (accept 18.76 to 18.77) **[A1]**

[Total Marks: 7]

305 **(a)** Let $k = 1 - \frac{r}{100}$. Then $M_0k^2 = 80$ and $M_0k^5 = 40.96$ **[M1]**

$k^3 = \frac{40.96}{80} = 0.512 \implies k = \sqrt[3]{0.512} = 0.8$ **[M1]**

$1 - \frac{r}{100} = 0.8 \implies r = 20$ **[A1]**

(b) $M_0 \times 0.8^2 = 80$ or $M_0 \times 0.64 = 80$ **[M1]**

125 **[A1]**

(c) $125 \times 0.8^n < 10$ or $0.8^n < 0.08$ **[M1]**

Testing values: $n = 11 \implies 0.8^{11} = 0.0858...$ and $n = 12 \implies 0.8^{12} = 0.0687...$ **[M1]**

12 **[A1]**

[Total Marks: 8]

306 **(a)** M1 for $500 \times (1 - 0.12)^7$ or 500×0.88^7

204.3 (accept 204 or $204.33\ldots$) **[A1]**

(b) M1 for $500 \times 0.88^d < 50$ or $0.88^d = 0.1$

M1 for evaluating trials (e.g. $0.88^{18} = 0.1001\ldots$ and $0.88^{19} = 0.0881\ldots$) OR using logarithms $d > \frac{\log(0.1)}{\log(0.88)}$

19 **[A1]**

(c) M1 for $\left(1 - \frac{r}{100}\right)^{15} = 0.5$

M1 for $\sqrt[15]{0.5}$ or $0.5^{\frac{1}{15}}$

4.52 (accept 4.515 to 4.516) **[A1]**

[Total Marks: 8]

307 **(a)** $M_0k^{-2} = 256$ and $M_0k^{-5} = 108$ **[M1]**

Divides equations: $\frac{M_0k^{-2}}{M_0k^{-5}} = \frac{256}{108} \implies k^3 = \frac{64}{27}$ **[M1]**

$k = \sqrt[3]{\frac{64}{27}} = \frac{4}{3}$ **[A1]**

(b) $M_0 \times \left(\frac{4}{3}\right)^{-2} = 256$ or $M_0 = 256 \times \frac{16}{9}$ **[M1]**

455.1 (455.111... or $\frac{4096}{9}$) **[A1]**

(c) $0.1M_0 = M_0\left(\frac{4}{3}\right)^{-t}$ or $\left(\frac{3}{4}\right)^{t} = 0.1$ **[M1]**

$t = \frac{\log(0.1)}{\log(0.75)}$ or $t = \frac{\log(10)}{\log(4/3)}$ **[M1]**

8.00 (8.0039...) **[A1]**

[Total Marks: 8]

308 **(a)** $4000 \times \left(1 + \frac{4.5}{100}\right)^{10}$ or 4000×1.045^{10} **[M1]**

6211.88 (or 6211.87...) **[A1]**

(b) $6211.88 + 500$ or 6711.88 **[B1]**

$4000 \times \left(1 + \frac{r}{100}\right)^{10} = 6711.88...$ **[M1]**

$1 + \frac{r}{100} = \sqrt[10]{\frac{6711.88...}{4000}}$ **[M1]**

5.31 (or 5.31...) **[A1]**

(c) $2500 \times 2^{\frac{20}{8}}$ or $2500 \times 2^{2.5}$ **[M1]**

14 142 (or 14 142.13...) **[A1]**

[Total Marks: 8]

309 **(a)** $2500 \times \left(1 + \frac{r}{100}\right)^{6} = 9100$ **[M1]**

$\left(1 + \frac{r}{100}\right)^{6} = 3.64$ **[M1]**

$1 + \frac{r}{100} = 3.64^{\frac{1}{6}} = 1.2403\ldots$ **[M1]**

24.0 (accept 24.03...) **[A1]**

(b) 2500×1.2403^{10} or 9100×1.2403^{4} **[M1]**

21 500 **[A1]**

(c) $2500 \times 1.2403^{n} > 50\,000 \implies 1.2403^{n} > 20$ **[M1]**

$n \times \log(1.2403) > \log(20) \implies n > 13.91\ldots$ (or equivalent trial and error) **[M1]**

14 **[A1]**

[Total Marks: 9]

310 $5\sqrt{3} - 3\sqrt{3} + 2\sqrt{3}$ **[M1]**

$4\sqrt{3}$ **[A1]**

[Total Marks: 2]

311 $5\sqrt{3}$ or $4\sqrt{3}$ or $3\sqrt{3}$ seen **[M1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 2]

312 $6\sqrt{2}$ or $5\sqrt{2}$ or $3\sqrt{2}$ seen **[M1]**

$8\sqrt{2}$ **[A1]**

[Total Marks: 2]

313 $5\sqrt{3}$ or $3\sqrt{3}$ or $2\sqrt{3}$ seen **[M1]**

$4\sqrt{3}$ **[A1]**

[Total Marks: 2]

314 $\sqrt{25 \times 3} + \sqrt{16 \times 3} - \sqrt{9 \times 3} = 5\sqrt{3} + 4\sqrt{3} - 3\sqrt{3}$ **[M1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 2]

315 (a) $3\sqrt{5}$ **[B1]**

(b) $2\sqrt{5}$ or $4\sqrt{5}$ seen **[M1]**

$6\sqrt{5}$ **[A1]**

[Total Marks: 3]

316 $\sqrt{75} = 5\sqrt{3}$ **[B1]**

$\sqrt{27} = 3\sqrt{3}$ or $\sqrt{12} = 2\sqrt{3}$ **[B1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 3]

317 $\sqrt{48} = 4\sqrt{3}$ **[B1]**

$\sqrt{12} = 2\sqrt{3}$ **[B1]**

$4\sqrt{3} - 2\sqrt{3} = 2\sqrt{3}$ **[B1]**

[Total Marks: 3]

318 **(a)** M1 for $5\sqrt{3}$ or $2\sqrt{3}$

$7\sqrt{3}$ **[A1]**

(b) M1 for $\dfrac{14 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$ or $\dfrac{14\sqrt{7}}{7}$

$2\sqrt{7}$ **[A1]**

[Total Marks: 4]

319 **(a)** $5\sqrt{3}$ or $2\sqrt{3}$ seen **[M1]**

$3\sqrt{3}$ **[A1]**

(b) $\dfrac{18\sqrt{6}}{6}$ **[M1]**

$3\sqrt{6}$ **[A1]**

[Total Marks: 4]

320 **(a)** $\sqrt{25 \times 3}$ or $\sqrt{25}\sqrt{3}$ **[M1]**

$5\sqrt{3}$ **[A1]**

(b) $6\sqrt{2} - 2\sqrt{2}$ **[M1]**

$4\sqrt{2}$ **[A1]**

[Total Marks: 4]

321 **(a)** **(i)** $3\sqrt{5} + 2\sqrt{5}$ **[M1]**

$5\sqrt{5}$ **[A1]**

(ii) Multiplies numerator and denominator by $\sqrt{3}$: $\dfrac{18\sqrt{3}}{3}$ **[M1]**

$6\sqrt{3}$ **[A1]**

[Total Marks: 4]

322 **(a)** $3\sqrt{5}$ **[B1]**

(b) $6\sqrt{2} + 5\sqrt{2} - 3\sqrt{2}$ **[M1]**

$8\sqrt{2}$ **[A1]**

(c) $\dfrac{21\sqrt{7}}{\sqrt{7}\sqrt{7}}$ or $\dfrac{21\sqrt{7}}{7}$ **[M1]**

$3\sqrt{7}$ **[A1]**

[Total Marks: 5]

323 **(a)** $8 + 4\sqrt{3} - 2\sqrt{3} - 3$ **[M1]**

$5 + 2\sqrt{3}$ **[A1]**

(b) $\frac{21\sqrt{7}}{7}$ **[M1]**

$3\sqrt{7}$ **[A1]**

[Total Marks: 4]

324 (a) $9 - 3\sqrt{5} - 3\sqrt{5} + 5$ (or $9 - 6\sqrt{5} + 5$) **[M1]**

$14 - 6\sqrt{5}$ **[A1]**

(b) $\frac{14 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Total Marks: 4]

325 (a) $12 + 3\sqrt{5} - 8\sqrt{5} - 10$ **[M1]**

$2 - 5\sqrt{5}$ **[A1]**

(b) Multiplies numerator and denominator by $\sqrt{7}$ to get $\frac{14\sqrt{7}}{7}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Total Marks: 4]

326 (a) (a) $12 + 3\sqrt{5} - 8\sqrt{5} - 2(5)$ or $12 - 5\sqrt{5} - 10$ **[M1]**

$2 - 5\sqrt{5}$ **[A1]**

(b) $\frac{15 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}}$ or $\frac{15\sqrt{3}}{3}$ **[M1]**

$5\sqrt{3}$ **[A1]**

[Total Marks: 4]

327 (a) $12 - 3\sqrt{2} + 4\sqrt{2} - 2$ **[M1]**

$10 + \sqrt{2}$ **[A1]**

(b) Multiply numerator and denominator by $(5 + \sqrt{3})$ **[M1]**

Denominator evaluates to $25 - 3 = 22$ **[M1]**

$5 + \sqrt{3}$ **[A1]**

[Total Marks: 5]

328 (a) M1 for $9 + 3\sqrt{2} + 3\sqrt{2} + 2$ (allow one numerical error in expansion)

$11 + 6\sqrt{2}$ **[A1]**

(b) M1 for multiplying numerator and denominator by $(3+\sqrt{5})$

M1 for denominator evaluated to 4 (from $3^2-(\sqrt{5})^2$)

$3+\sqrt{5}$ **[A1]**

[Total Marks: 5]

329 (a) $20-9$ or $4(5)-9$ **[M1]**

11 **[A1]**

(b) Multiplies numerator and denominator by $(3+\sqrt{2})$ **[M1]**

Denominator of $9-2$ or 7 seen **[M1]**

$6+2\sqrt{2}$ **[A1]**

[Total Marks: 5]

330 (a) $12+15\sqrt{2}-2\sqrt{2}-5$ **[M1]**

$7+13\sqrt{2}$ **[A1]**

(b) Multiplies numerator and denominator by $(4+\sqrt{3})$ **[M1]**

Denominator of $16-3$ or 13 **[M1]**

$8+2\sqrt{3}$ or $2(4+\sqrt{3})$ **[A1]**

[Total Marks: 5]

331 (a) $9-3\sqrt{5}-3\sqrt{5}+5$ **[M1]**

$14-6\sqrt{5}$ **[A1]**

(b) Multiplies numerator and denominator by $(4+\sqrt{5})$ **[M1]**

$\dfrac{11(4+\sqrt{5})}{16-5}$ or denominator 11 seen **[M1]**

$4+\sqrt{5}$ **[A1]**

[Total Marks: 5]

332 (a) $\dfrac{15\sqrt{5}}{5}$ **[M1]**

$3\sqrt{5}$ **[A1]**

(b) $10-2\sqrt{3}+5\sqrt{3}-3$ **[M1]**

$7+3\sqrt{3}$ **[A2]**

[Total Marks: 5]

333 (a) $6-3\sqrt{5}+2\sqrt{5}-5$ **[M1]**

$1 - \sqrt{5}$ **[A1]**

(b) $\dfrac{14 \times \sqrt{7}}{\sqrt{7} \times \sqrt{7}}$ **[M1]**

$\dfrac{14\sqrt{7}}{7}$ **[M1]**

$2\sqrt{7}$ **[A1]**

[Total Marks: 5]

334 **(a)** $15 + 5\sqrt{7} - 6\sqrt{7} - 2(7)$ **[M1]**

$1 - \sqrt{7}$ **[A1]**

(b) Multiplies numerator and denominator by $(5 + \sqrt{3})$: $\dfrac{22(5 + \sqrt{3})}{(5 - \sqrt{3})(5 + \sqrt{3})}$ **[M1]**

Denominator becomes $25 - 3 = 22$ **[M1]**

$5 + \sqrt{3}$ **[A1]**

[Total Marks: 5]

335 **(a)** $20 + 8\sqrt{2} - 15\sqrt{2} - 6(2)$ **[M1]**

$8 - 7\sqrt{2}$ **[A1]**

(b) $\dfrac{11(4 + \sqrt{5})}{(4 - \sqrt{5})(4 + \sqrt{5})}$ **[M1]**

$\dfrac{11(4 + \sqrt{5})}{16 - 5}$ or $\dfrac{11(4 + \sqrt{5})}{11}$ **[M1]**

$4 + \sqrt{5}$ **[A1]**

(c) $3x\sqrt{3} - x\sqrt{3} = 18$ **[M1]**

$2x\sqrt{3} = 18$ or $x\sqrt{3} = 9$ **[M1]**

$x = \dfrac{9}{\sqrt{3}} = 3\sqrt{3}$ **[A1]**

[Total Marks: 8]

336 $\times \dfrac{3 + \sqrt{7}}{3 + \sqrt{7}}$ **[M1]**

Numerator $15 + 5\sqrt{7} + 3\sqrt{7} + 7$ or $22 + 8\sqrt{7}$ **[B1]**

Denominator $9 - 7$ or 2 **[B1]**

$11 + 4\sqrt{7}$ **[A1]**

[Total Marks: 4]

337 M1 for rearranging to group x terms, e.g., $x\sqrt{3} - x = 2 + \sqrt{12}$

M1 for factorising x, e.g., $x(\sqrt{3}-1) = 2+2\sqrt{3}$

M1 for multiplying by the conjugate to rationalise: $\dfrac{2+2\sqrt{3}}{\sqrt{3}-1} \times \dfrac{\sqrt{3}+1}{\sqrt{3}+1}$

M1 for correctly expanding the numerator to $8+4\sqrt{3}$ or evaluating the denominator to 2

$4+2\sqrt{3}$ **[A1]**

[Total Marks: 5]

338 Forms an equation for the area: $(a+b\sqrt{3})(2-\sqrt{3}) = 1+4\sqrt{3}$ **[M1]**

Expands the brackets: $2a - a\sqrt{3} + 2b\sqrt{3} - 3b = 1+4\sqrt{3}$ **[M1]**

Equates rational and irrational parts to form simultaneous equations: $2a-3b=1$ and $-a+2b=4$ **[M1]**

Attempts to solve the simultaneous equations (e.g. multiplies second equation by 2: $-2a+4b=8$ and adds to first to get $b=9$) **[M1]**

$a=14$, $b=9$ **[A1]**

[Total Marks: 5]

339 (a) Multiplies numerator and denominator by $(2+\sqrt{3})$ **[M1]**

Numerator expands to $10+5\sqrt{3}+2\sqrt{3}+3$ **[M1]**

Denominator is $4-3=1$ leading to final answer $13+7\sqrt{3}$ **[A1]**

(b) $x = \dfrac{14}{3-\sqrt{2}}$ **[M1]**

Multiplies numerator and denominator by $(3+\sqrt{2})$ to get $\dfrac{14(3+\sqrt{2})}{9-2}$ **[M1]**

$x = 6+2\sqrt{2}$ **[A1]**

[Total Marks: 6]

340 (a) Multiply numerator and denominator by $2+\sqrt{3}$ **[M1]**

Numerator becomes $(5+\sqrt{3})(2+\sqrt{3}) = 10+5\sqrt{3}+2\sqrt{3}+3$ (giving $13+7\sqrt{3}$) **[M1]**

$13+7\sqrt{3}$ **[A1]**

(b) Length $= \dfrac{12+4\sqrt{5}}{3+\sqrt{5}}$ **[M1]**

Multiply numerator and denominator by $3-\sqrt{5}$ **[M1]**

Numerator becomes $(12+4\sqrt{5})(3-\sqrt{5}) = 36-12\sqrt{5}+12\sqrt{5}-20 = 16$ **[M1]**

4 **[A1]**

[Total Marks: 7]

341 (a) Multiplies numerator and denominator by $(5+\sqrt{3})$ **[M1]**

Denominator evaluates to $25 - 3$ or 22 **[M1]**

$5 + \sqrt{3}$ **[A1]**

(b) $x(\sqrt{2} - 1) = \sqrt{8}$ or $x(\sqrt{2} - 1) = 2\sqrt{2}$ **[M1]**

$x = \dfrac{2\sqrt{2}}{\sqrt{2} - 1}$ **[M1]**

Multiplies numerator and denominator by $(\sqrt{2} + 1)$ **[M1]**

$4 + 2\sqrt{2}$ **[A1]**

[Total Marks: 7]

342 **(a)** $\dfrac{1}{2} \times (2 + \sqrt{3}) \times h = 5 + 3\sqrt{3}$ **[M1]**

$h = \dfrac{10 + 6\sqrt{3}}{2 + \sqrt{3}}$ **[M1]**

Multiply numerator and denominator by $(2 - \sqrt{3})$ **[M1]**

$h = \dfrac{20 - 10\sqrt{3} + 12\sqrt{3} - 18}{4 - 3} = 2 + 2\sqrt{3}$ **[A1]**

(b) $y^2 = (2 + \sqrt{3})^2 + (2 + 2\sqrt{3})^2$ **[M1]**

$y^2 = (4 + 4\sqrt{3} + 3) + (4 + 8\sqrt{3} + 12)$ **[M1]**

$23 + 12\sqrt{3}$ **[A1]**

[Total Marks: 7]

343 **(a)** $\dfrac{5 + 2\sqrt{3}}{2 + \sqrt{3}}$ **[M1]**

Multiplies numerator and denominator by $(2 - \sqrt{3})$ **[M1]**

$4 - \sqrt{3}$ **[A1]**

(b) $x\sqrt{5} - 2x = 2 + \sqrt{20}$ or $x(\sqrt{5} - 2) = \ldots$ **[M1]**

$x = \dfrac{2 + 2\sqrt{5}}{\sqrt{5} - 2}$ **[M1]**

Multiplies numerator and denominator by $(\sqrt{5} + 2)$ **[M1]**

$14 + 6\sqrt{5}$ **[A1]**

[Total Marks: 7]

344 **(a)** $\dfrac{14 - 2\sqrt{5}}{3 + \sqrt{5}}$ **[M1]**

Multiplies numerator and denominator by $(3 - \sqrt{5})$ **[M1]**

Numerator expands to $42 - 14\sqrt{5} - 6\sqrt{5} + 10$ (or $52 - 20\sqrt{5}$) **[M1]**

$13 - 5\sqrt{5}$ **[A1]**

(b) Uses Pythagoras' theorem: $(3 + \sqrt{5})^2 + (13 - 5\sqrt{5})^2$ **[M1]**

$14 + 6\sqrt{5}$ or $294 - 130\sqrt{5}$ seen **[M1]**

$308 - 124\sqrt{5}$ with no errors seen **[A1]**

[Total Marks: 7]

345 **(a)** Multiplies numerator and denominator by $(3 + \sqrt{3})$: $\dfrac{6(3+\sqrt{3})}{(3-\sqrt{3})(3+\sqrt{3})}$ **[M1]**

Expands denominator correctly: $9 - 3$ or 6 **[M1]**

$\dfrac{6(3+\sqrt{3})}{6}$ **[M1]**

$3 + \sqrt{3}$ **[A1]**

(b) $\dfrac{2\sqrt{6}+3\sqrt{6}}{\sqrt{3}}$ or $\dfrac{5\sqrt{6}}{\sqrt{3}}$ or $5\sqrt{2}$ **[M1]**

$\dfrac{10\sqrt{2}}{2}$ or $5\sqrt{2}$ **[M1]**

0 (or $0\sqrt{2} + 0\sqrt{3}$) **[A1]**

[Total Marks: 7]

346 **(a)** $\dfrac{1}{2}(\sqrt{7}-1)(\sqrt{7}+2)$ **[M1]**

Expands brackets correctly: $7 + 2\sqrt{7} - \sqrt{7} - 2 = 5 + \sqrt{7}$ **[M1]**

$\dfrac{5+\sqrt{7}}{2}$ **[A1]**

(b) $AC^2 = (\sqrt{7}-1)^2 + (\sqrt{7}+2)^2$ **[M1]**

$(\sqrt{7}-1)^2 = 8 - 2\sqrt{7}$ and $(\sqrt{7}+2)^2 = 11 + 4\sqrt{7}$ **[M1]**

$AC^2 = 19 + 2\sqrt{7}$ **[A1]**

$\sqrt{19 + 2\sqrt{7}}$ **[A1]**

[Total Marks: 7]

347 **(a)** Area $= \dfrac{1}{2} \times$ base $\times$ height $\implies 5 + 2\sqrt{2} = \dfrac{1}{2} \times (3 + \sqrt{2}) \times h$ **[M1]**

$h = \dfrac{2(5+2\sqrt{2})}{3+\sqrt{2}}$ **[A1]**

(b) Multiply numerator and denominator by $(3 - \sqrt{2})$: $\dfrac{2(5+2\sqrt{2})(3-\sqrt{2})}{(3+\sqrt{2})(3-\sqrt{2})}$ **[M1]**

Denominator: $3^2 - (\sqrt{2})^2 = 9 - 2 = 7$ **[B1]**

Numerator: $2(15 - 5\sqrt{2} + 6\sqrt{2} - 4) = 2(11 + \sqrt{2}) = 22 + 2\sqrt{2}$ **[M1]**

Height $= \dfrac{22+2\sqrt{2}}{7}$ **[A1]**

(c) $\dfrac{1(\sqrt{7}+2)}{(\sqrt{7}-2)(\sqrt{7}+2)}$ **[M1]**

$\dfrac{\sqrt{7}+2}{7-4} = \dfrac{\sqrt{7}+2}{3}$ **[A1]**

[Total Marks: 8]

348 **(a)** $\dfrac{1}{2} \times (\sqrt{6}-\sqrt{2})(\sqrt{6}+\sqrt{2})$ **[M1]**

$\dfrac{1}{2}(6-2)$ **[M1]**

2 **[A1]**

(b) $(\text{Hypotenuse})^2 = (\sqrt{6}-\sqrt{2})^2 + (\sqrt{6}+\sqrt{2})^2$ **[M1]**

$(6-2\sqrt{12}+2) + (6+2\sqrt{12}+2)$ or $8-2\sqrt{12}+8+2\sqrt{12}$ **[M1]**

$(\text{Hypotenuse})^2 = 16$ **[A1]**

$\text{Hypotenuse} = \sqrt{16} = 4$ **[A1]**

(c) $\text{Area scale factor} = \dfrac{18}{2} = 9$ **[M1]**

$\text{Length scale factor} = \sqrt{9} = 3$ **[M1]**

$4 \times 3 = 12$ **[A1]**

[Total Marks: 10]

349 M1 for $3(-4)^2$ or $-5(-4)(2)$ or 48 or +40

88 **[A1]**

[Total Marks: 2]

350 Correct substitution: $5(-2)^2 - 3(4)$ or $5(4) - 12$ **[M1]**

8 **[A1]**

[Total Marks: 2]

351 $3(4) - 2(-5)$ or $12 - -10$ or $12 + 10$ **[M1]**

22 **[A1]**

[Total Marks: 2]

352 Substitutes values correctly, e.g. $3(-3)^2 - 4(-5)$ **[M1]**

47 **[A1]**

[Total Marks: 2]

353 Substitution of all three values correctly e.g. $\frac{3(5)-(-4)^2}{2(0.5)}$ **[M1]**

-1 **[A1]**

[Total Marks: 2]

354 Substitute the given values into the formula: $v = 5 + (-2)(3)$ **[M1]**

$v = 5 - 6$ **[M1]**

-1 **[A1]**

[Total Marks: 3]

355 **(a)** $4(-2) - 3(5)$ or $-8 - 15$ **[M1]**

-23 **[A1]**

(b) $\frac{5^2-7}{2(-3)}$ or $\frac{25-7}{-6}$ **[M1]**

-3 **[A1]**

[Total Marks: 4]

356 **(a)** -6 or -20 seen **[M1]**

-26 **[A1]**

(b) $(-3)^2 = 9$ or $4(-3) = -12$ seen **[M1]**

-3 **[A1]**

[Total Marks: 4]

357 **(a)** $P = 3(4)^2 - 2(4)(-3) + 5(-3)$ or $48 + 24 - 15$ seen **[M1]**

57 **[A1]**

(b) $19 = 3(-2)^2 - 2(-2)b + 5b$ or $19 = 12 + 9b$ **[M1]**

$\frac{7}{9}$ **[A1]**

[Total Marks: 4]

358 **(a)** Substitutes values correctly: $5(3) - 2(-4)$ **[M1]**

$15 + 8 = 23$ **[A1]**

(b) Substitutes values: $\frac{(-4)^2-(2)^3}{2(3)}$ **[M1]**

Evaluates numerator: $16 - 8 = 8$

$\frac{8}{6} = \frac{4}{3}$ or $1\frac{1}{3}$ **[A1]**

[Total Marks: 4]

359 **(a)** **(a)** $40 + 25(3.5)$ or $40 + 87.5$ **[M1]**

127.50 **[A1]**

(b) $-12 + 9.8 \times 3$ **[M1]**

$-12 + 29.4$ **[M1]**

17.4 **[A1]**

[Total Marks: 5]

360 **(a)** $40 + 25h$ **[B1]**

(b) $(-2)^3 = -8$ and $(-3)^2 = 9$ **[M1]**

$\sqrt{\frac{1}{16}} = \frac{1}{4}$ **[B1]**

-68 **[A1]**

[Total Marks: 4]

361 Evaluates $p^{-2} = \left(\frac{1}{2}\right)^{-2} = 4$ **[B1]**

Evaluates $q^3 = (-2)^3 = -8$ **[B1]**

Evaluates $\sqrt[3]{r} = \sqrt[3]{-\frac{1}{8}} = -\frac{1}{2}$ **[B1]**

8 **[A1]**

[Total Marks: 4]

362 **(a)** $d = \sqrt{(-8-4)^2 + (2--3)^2}$ or $\sqrt{144+25}$ **[M1]**

13 **[A1]**

(b) $17 = \sqrt{(3-k)^2 + (-7-8)^2}$ **[M1]**

$289 = (3-k)^2 + 225$ **[M1]**

$(3-k)^2 = 64$ or $k^2 - 6k - 55 = 0$ **[M1]**

$k = -5$ and $k = 11$ **[A1]**

[Total Marks: 6]

363 **(a)** Forms three equations: $a + b + c = 4$, $4a + 2b + c = 13$, $9a + 3b + c = 28$ **[M1]**

Eliminates c correctly to obtain two linear equations: $3a + b = 9$ and $5a + b = 15$ **[M1]**

$a = 3$ **[A1]**

$b = 0$ **[A1]**

$c = 1$ **[A1]**

(b) $[3(k+1)^2 + 1] - [3k^2 + 1] = 57$ or $6k + 3 = 57$ **[M1]**

9 **[A1]**

[Total Marks: 7]

364 (a) States $A = \text{Curved Surface Area} + \text{Base Area} = 2\pi rh + \pi r^2$ **[M1]**

Factorises πr out to obtain $\pi r(2h + r)$ **[A1]**

(b) (i) Substitutes $r = 4$ and $h = 6$: $A = \pi(4)(2(6) + 4)$ **[M1]**

$A = 4\pi(12 + 4) = 64\pi$ **[A1]**

(ii) Rearranges volume formula: $h = \dfrac{V}{\pi r^2}$ **[M1]**

Substitutes h into area formula: $A = 2\pi r\left(\dfrac{V}{\pi r^2}\right) + \pi r^2$ **[M1]**

Simplifies first term: $\dfrac{2\pi rV}{\pi r^2} = \dfrac{2V}{r}$, resulting in $A = \dfrac{2V}{r} + \pi r^2$ **[A1]**

[Total Marks: 7]

365 (a) Substitute $x = -2$ and $y = 4$: $P = \dfrac{2(-2)^2 - 3(4)}{-2 - 4}$ **[M1]**

$P = \dfrac{2(4) - 12}{-6} = \dfrac{8 - 12}{-6}$ **[M1]**

$\dfrac{2}{3}$ (or equivalent) **[A1]**

(b) Multiply both sides by $(x - y)$: $P(x - y) = 2x^2 - 3y$ **[M1]**

Expand: $Px - Py = 2x^2 - 3y$ **[M1]**

Collect terms with y on one side: $3y - Py = 2x^2 - Px$ **[M1]**

Factorise and divide: $y(3 - P) = 2x^2 - Px \implies y = \dfrac{2x^2 - Px}{3 - P}$ (or equivalent) **[A1]**

[Total Marks: 7]

366 (a) (i) $\dfrac{6^2 - (-2)^2}{6(-2)}$ **[M1]**

$-\dfrac{8}{3}$ **[A1]**

(ii) $\dfrac{0.5^2 - 4^2}{0.5(4)}$ **[M1]**

$\dfrac{-15.75}{2}$ **[M1]**

$-\dfrac{63}{8}$ **[A1]**

(b) $\dfrac{x^2-2^2}{2x}=\dfrac{3}{2}$ **[M1]**

$2(x^2-4)=6x$ (or $x^2-3x-4=0$) **[M1]**

4 **[A1]**

[Total Marks: 8]

367 **(a)** $C+Rh$ or $C+hR$ **[B1]**

(b) M1 for $45+32.50\times 3.5$ or $45+113.75$

158.75 **[A1]**

[Total Marks: 3]

368 Substitutes values correctly: $\dfrac{\sqrt{12.5-3.84}}{(-1.2)^3}$ **[M1]**

Evaluates numerator to 2.94... or denominator to -1.728 **[M1]**

-1.70 (accept $-1.703...$ rounded strictly to 3 sig figs) **[A1]**

[Total Marks: 3]

369 $\pi\times(4.5)^2\times 8.2+\dfrac{2}{3}\times\pi\times(4.5)^3$ **[M1]**

521.6... + 190.8... or 712.5... **[M1]**

713 **[A1]**

[Total Marks: 3]

370 **(a)** $45+32h$ (or equivalent) **[B1]**

(b) 157 **[B1]**

(c) Correct substitution: $(-3)^3-4(-3)(2.5)$ or $-27-(-30)$ **[M1]**

3 **[A1]**

[Total Marks: 4]

371 **(a)** $V=\dfrac{1}{3}\pi(6.4)^2(12.6)$ **[M1]**

540.5 **[A1]**

(b) $800=\dfrac{1}{3}\pi r^2(14)$ **[M1]**

$r^2=\dfrac{2400}{14\pi}$ or $54.567\ldots$ **[M1]**

7.39 **[A1]**

[Total Marks: 5]

372 **(a)** $85(7) + 0.24(450)$ **[M1]**

703 **[A1]**

(b) $1475.60 = 85(14) + 0.24m$ **[M1]**

$1475.60 - 1190 = 0.24m$ (or $285.60 = 0.24m$) **[M1]**

1190 **[A1]**

[Total Marks: 5]

373 **(a)** Substitute $h = 3.5$: $C = 45 + 35(3.5)$ **[M1]**

167.5 **[A1]**

(b) Substitute $C = 325$: $325 = 45 + 35h$ **[M1]**

$280 = 35h$ **[M1]**

8 **[A1]**

[Total Marks: 5]

374 **(a)** $A = \pi \times 5 \times \left(5 + \sqrt{5^2 + 12^2}\right)$ or 90π seen **[M1]**

283 (282.74...) **[A1]**

(b) $96\pi = \pi(6)\left(6 + \sqrt{6^2 + h^2}\right)$ **[M1]**

$16 = 6 + \sqrt{36 + h^2}$ or $\sqrt{36 + h^2} = 10$ **[M1]**

$36 + h^2 = 100$ or $h^2 = 64$ **[M1]**

8 **[A1]**

[Total Marks: 6]

375 **(a)** Substitutes $h = 4.2$ and $a = 6.5$ into formula: $V = \frac{1}{6}\pi(4.2)(3(6.5)^2 + 4.2^2)$ **[M1]**

$V = 317.53... = 317.5$ **[A1]**

(b) Multiplies by 6 and divides by πh: $\frac{6V}{\pi h} = 3a^2 + h^2$ **[M1]**

Subtracts h^2: $\frac{6V}{\pi h} - h^2 = 3a^2$ **[M1]**

$a^2 = \frac{2V}{\pi h} - \frac{h^2}{3}$ or $a^2 = \frac{1}{3}\left(\frac{6V}{\pi h} - h^2\right)$ **[A1]**

(c) $a = \sqrt{\frac{2(150)}{\pi(3)} - \frac{3^2}{3}} = \sqrt{31.83... - 3} = \sqrt{28.83...} = 5.37$ **[B1]**

[Total Marks: 6]

376 Numerator correctly evaluated: $-4.5 - (-3.2)^2 = -4.5 - 10.24 = -14.74$ **[B1]**

Denominator correctly evaluated: $1.8 - \sqrt{0.64} = 1.8 - 0.8 = 1$ **[B1]**

-14.74 **[B1]**

[Total Marks: 3]

377 **(a)** **(a)** $5.2(4.5) + 0.5(-9.8)(4.5)^2$ **[M1]**

-75.825 **[A1]**

(b) $-9.8 = 4.9t + \frac{1}{2}(-9.8)t^2$ **[M1]**

$4.9t^2 - 4.9t - 9.8 = 0$ or $t^2 - t - 2 = 0$ **[M1]**

$t = 2$ (rejects -1 as time must be positive) **[A1]**

[Total Marks: 5]

378 **(a)** M1 for correctly substituting the values into the formula: $\frac{1}{3}\pi(12.5)(4.2^2 + 4.2 \times 7.8 + 7.8^2)$

M1 for 111.24 or 463.5π or 1456.1...

1460 **[A1]**

(b) M1 for correctly substituting the values into the square root part: $\sqrt{(7.8 - 4.2)^2 + 12.5^2}$

M1 for $\sqrt{169.21}$ or 13.008...

736.9 **[A1]**

[Total Marks: 6]

379 At least 3 terms correct from $6x^2 + 15x - 4x - 10$ **[M1]**

$6x^2 + 11x - 10$ **[A1]**

[Total Marks: 2]

380 **(a)** Expansion to $6x - 3 - 2x - 8$ **[M1]**

$4x - 11$ **[A1]**

(b) Extracting a partial common factor, e.g. $5xy(3x - 4y)$ or $x(15xy - 20y^2)$ or $5(3x^2y - 4xy^2)$ **[M1]**

$5xy(3x - 4y)$ **[A1]**

[Total Marks: 4]

381 **(a)** M1 for $8a - 12b$ or $-3a + 15b$ seen

$5a + 3b$ **[A1]**

(b) M1 for a correct partial factorisation, e.g., $6x(2xy - 3y^2)$ or $2xy(6x - 9y)$

$6xy(2x - 3y)$ **[A1]**

[Total Marks: 4]

382 **(a)** $8x - 12y - 3x + 15y$ **[M1]**

$5x + 3y$ **[A1]**

(b) $7ab$ extracted as a common factor **[M1]**

$7ab(2a - 3b)$ **[A1]**

[Total Marks: 4]

383 **(a)** $12x - 8 - 2x + 10$ **[M1]**

$10x + 2$ **[A1]**

(b) $5ab(3a - 2b)$ **[B2]**

If not B2, then B1 for $5a(3ab - 2b^2)$ or $5b(3a^2 - 2ab)$ or $ab(15a - 10b)$ **[B1]**

[Total Marks: 4]

384 **(a)** $6x - 15 - 2x - 8$ **[M1]**

$4x - 23$ **[A1]**

(b) $4xy(2x - 3y)$ or $4x(2xy - 3y^2)$ or $4y(2x^2 - 3xy)$ or $xy(8x - 12y)$ **[M1]**

$4xy(2x - 3y)$ **[A1]**

[Total Marks: 4]

385 **(a)** Collects at least two like terms correctly: $-5x^2$, $-5xy$, or -5 **[M1]**

$-5x^2 - 5xy - 5$ **[A1]**

(b) Extracts partial common factor correctly, e.g. $7a(2a^2b - 3ab^2)$ or $a^2b(14a - 21b)$ **[M1]**

$7a^2b(2a - 3b)$ **[A1]**

[Total Marks: 4]

386 **(a)** Correct expansion: $6x - 15 - 4x + 8$ **[M1]**

$2x - 7$ **[A1]**

(b) $6xy(2x - 3y)$ **[B2]**

Correct partial factorisation e.g., $6x(2xy - 3y^2)$ or $2xy(6x - 9y)$ **[M1]**

[Total Marks: 4]

387 **(a)** $5a$ or $+3b$ **[B1]**

$5a + 3b$ **[A1]**

(b) $6xy(2x - 3y)$ **[B2]**

If B0, allow [M1] for a correct partial factorisation, e.g. $6x(2xy - 3y^2)$ or $3xy(4x - 6y)$

[Total Marks: 4]

388 **(a)** Expansion to give $6x - 15$ and $-4x + 12$ **[M1]**

$2x - 3$ **[A1]**

(b) $7xy(2x + 3y)$ **[B2]**

Partial factorisation, e.g. $7x(2xy + 3y^2)$ or $xy(14x + 21y)$ **[B1]**

[Total Marks: 4]

389 **(a)** $3(4x^2 - 9y^2)$ or $(6x - 9y)(2x + 3y)$ or $(2x - 3y)(6x + 9y)$ **[M1]**

$3(2x - 3y)(2x + 3y)$ **[A1]**

(b) $2a(x - 3y) + b(x - 3y)$ or $x(2a + b) - 3y(2a + b)$ **[M1]**

$(2a + b)(x - 3y)$ **[A1]**

[Total Marks: 4]

390 **(a)** $2x^2 + 10x - 3x - 15$ **[M1]**

$2x^2 + 7x - 15$ **[A1]**

(b) $(y - 4x)(y + 4x)$ **[B1]**

(c) $(p + q)(p - 5q)$ **[B2]**

If B0, allow [M1] for $(p + aq)(p + bq)$ where $ab = -5$ or $a + b = -4$

[Total Marks: 5]

391 **(a)** $2x^2 + 10x - 3x - 15$ **[M1]**

$2x^2 + 7x - 15$ **[A1]**

(b) $c(a - 3b) + 2d(a - 3b)$ or $a(c + 2d) - 3b(c + 2d)$ **[M1]**

$(a - 3b)(c + 2d)$ **[A1]**

(c) $(4x - 5y)(4x + 5y)$ **[B1]**

[Total Marks: 5]

392 **(a)** M1 for 3 correct terms out of $2x^2 + 8x - 3x - 12$

$2x^2 + 5x - 12$ **[A1]**

(b) M1 for $x(a - b)$ or $-3y(a - b)$ seen

$(x - 3y)(a - b)$ **[A1]**

(c) $(7x - 4y)(7x + 4y)$ **[B2]**

If 0 scored, allow M1 for $(7x - 4y)(7x - 4y)$ or $(7x + 4y)(7x + 4y)$

[Total Marks: 6]

393 **(a)** $(x + a)(x + b)$ where $ab = 15$ or $a + b = -8$ **[M1]**

$(x - 3)(x - 5)$ **[A1]**

(b) $(2x + a)(x + b)$ where $ab = -12$ or $2b + a = -5$ **[M1]**

$(2x + 3)(x - 4)$ **[A1]**

(c) $a(x - 3) + 2b(x - 3)$ or $x(a + 2b) - 3(a + 2b)$ **[M1]**

$(a + 2b)(x - 3)$ **[A1]**

[Total Marks: 6]

394 **(a)** $(3x - 2)(x + 5) = 3x^2 + 13x - 10$ **[B1]**

$(2x - 3)^2 = 4x^2 - 12x + 9$ **[B1]**

$-x^2 + 25x - 19$ **[B1]**

(b) **(i)** $3x(x^2 - 4y^2)$ **[M1]**

$3x(x - 2y)(x + 2y)$ **[A1]**

(ii) $(3x + a)(2x + b)$ where $ab = -10$ or $3b + 2a = -11$ **[M1]**

$(3x + 2)(2x - 5)$ **[A1]**

[Total Marks: 6]

395 **(a)** $6x^2 + 2x - 15x - 5$ (at least 3 correct terms) **[M1]**

$6x^2 - 13x - 5$ **[A1]**

(b) $(x + a)(x + b)$ where $ab = -18$ or $a + b = -7$ **[M1]**

$(x - 9)(x + 2)$ **[A1]**

(c) $2(9x^2 - 25y^2)$ **[M1]**

$(3x - 5y)(3x + 5y)$ seen **[M1]**

$2(3x - 5y)(3x + 5y)$ **[A1]**

[Total Marks: 7]

396 **(a)** Correct expansion of any two brackets, e.g. $2x^2 - x - 6$ or $2x^2 - 7x - 15$ or $x^2 - 7x + 10$ **[M1]**

Correctly multiplies their 3-term quadratic by the remaining bracket to get at least 4 correct terms **[M1]**

$2x^3 - 11x^2 - x + 30$ **[A1]**

(b) Extracts common factor x to give $x(12x^2 + 17x - 5)$ **[M1]**

Correctly factorises the quadratic expression to $(4x - 1)(3x + 5)$ **[M1]**

$x(4x - 1)(3x + 5)$ **[A1]**

[Total Marks: 6]

397 **(a)** Taking out common factor to reach $2(9x^2 - 25y^2)$ **[M1]**

Factorising difference of squares to reach $(3x - 5y)(3x + 5y)$ **[M1]**

$2(3x - 5y)(3x + 5y)$ **[A1]**

(b) Taking out common factor x to reach $x(3x^2 + 10x - 8)$ **[M1]**

Factorising quadratic to reach $(3x - 2)(x + 4)$ **[M1]**

$x(3x - 2)(x + 4)$ **[A1]**

[Total Marks: 6]

398 **(a)** Correct expansion of two brackets, e.g. $(x^2 + x - 6)(2x - 1)$ or $(x - 2)(2x^2 + 5x - 3)$ **[M1]**

Multiplication of quadratic by third bracket yielding 6 terms, e.g.
$2x^3 - x^2 + 2x^2 - x - 12x + 6$ **[M1]**

$2x^3 + x^2 - 13x + 6$ **[A1]**

(b) **(i)** $(3x + a)(2x + b)$ where $ab = -10$ or $2a + 3b = -11$ **[M1]**

$(3x + 2)(2x - 5)$ **[A1]**

(ii) $a(x - y) + 3c(x - y)$ or $x(a + 3c) - y(a + 3c)$ **[M1]**

$(a + 3c)(x - y)$ **[A1]**

[Total Marks: 7]

399 **(a)** $x(c - 5d) + 3y(c - 5d)$ or $c(x + 3y) - 5d(x + 3y)$ **[M1]**

$(x + 3y)(c - 5d)$ **[A1]**

(b) $2x(x^2 - 9)$ **[M1]**

$2x(x - 3)(x + 3)$ **[A1]**

(c) $2x(6x^2 - 7x - 5)$ **[M1]**

$2x(3x - 5)(2x + 1)$ **[A2]**

If not A2, then A1 for $(3x - 5)(2x + 1)$ or $2x(3x + 5)(2x - 1)$ **[A1]**

[Total Marks: 7]

400 **(a)** Expands any two brackets, e.g. $(2x^2 - 7x - 4)(x + 3)$ **[M1]**

Multiplies their quadratic by the third bracket, e.g. $2x^3 - 7x^2 - 4x + 6x^2 - 21x - 12$ **[M1]**

$2x^3 - x^2 - 25x - 12$ **[A1]**

(b) $2a(x+y)-3b(y+x)$ or $x(2a-3b)+y(2a-3b)$ **[M1]**

$(x+y)(2a-3b)$ **[A1]**

(c) $(2x+5)(3x-4)$ **[B2]**

If B0, allow [M1] for $(2x+a)(3x+b)$ where $ab=-20$ or $3a+2b=7$

[Total Marks: 7]

401 (a) Correctly expands any two brackets, e.g. $(2x^2+5x-3)(3x-2)$ **[M1]**

Multiplies by the third bracket correctly: $6x^3-4x^2+15x^2-10x-9x+6$ **[M1]**

$6x^3+11x^2-19x+6$ **[A1]**

(b) $2a(2x-3y)-3b(2x-3y)$ or $2x(2a-3b)-3y(2a-3b)$ **[M1]**

$(2a-3b)(2x-3y)$ **[A1]**

(c) Uses difference of two squares: $(x-y)(x+y)=45$ **[M1]**

$5(x+y)=45 \implies x+y=9$ **[M1]**

$(x+y)^2=9^2=81$ **[A1]**

[Total Marks: 8]

402 (a) $(2a-b)(3x-y)$ **[B2]**

Correct partial factorisation e.g., $2a(3x-y)-b(3x-y)$ or $3x(2a-b)-y(2a-b)$ **[M1]**

(b) (i) Extracts common factor of 2: $2(25x^2-4y^2)$ **[M1]**

Factorises difference of two squares: $(5x-2y)(5x+2y)$ **[M1]**

$2(5x-2y)(5x+2y)$ **[A1]**

(ii) Extracts common factor of x: $x(2x^2+5x-12)$ **[M1]**

Factorises quadratic: $(2x-3)(x+4)$ **[M1]**

$x(2x-3)(x+4)$ **[A1]**

[Total Marks: 8]

403 (a) Correct expansion of any two brackets, e.g., $(2x^2-5x-3)$ or $(2x^2+9x+4)$ or (x^2+x-12) **[M1]**

Correct expansion of their quadratic with the third bracket, producing at least 5 terms, e.g., $2x^3+8x^2-5x^2-20x-3x-12$ **[M1]**

$2x^3+3x^2-23x-12$ **[A1]**

(b) $(2x+3)(x-5)$ **[B2]**

Accept $(2x+a)(x+b)$ where $ab=-15$ or $2b+a=-7$ **[M1]**

[Total Marks: 5]

404 (a) Four terms with at least three correct from $2x^2+10x-3x-15$ **[M1]**

$2x^2 + 7x - 15$ **[A1]**

(b) $(x + a)(x + b)$ where $ab = -18$ or $a + b = -7$ **[M1]**

$(x - 9)(x + 2)$ **[A1]**

(c) $(2x + a)(2x - a)$ or $(cx + 5y)(cx - 5y)$ **[M1]**

$(2x - 5y)(2x + 5y)$ **[A1]**

[Total Marks: 6]

405 (a) $(x - 4)(2x^2 + 7x + 6)$ or $(2x^2 - 5x - 12)(x + 2)$ or $(x^2 - 2x - 8)(2x + 3)$ **[M1]**

$2x^3 + 7x^2 + 6x - 8x^2 - 28x - 24$ (allow one error in expansion) **[M1]**

$2x^3 - x^2 - 22x - 24$ **[A1]**

(b) $(4x - 7y)(4x + 7y)$ **[B2]**

If not B2, then B1 for $(4x - ky)(4x + ky)$ or $(kx - 7y)(kx + 7y)$ where $k \neq 0$ **[B1]**

(c) $(3x + 2)(x - 4)$ **[B2]**

If not B2, then B1 for $(3x + a)(x + b)$ where $ab = -8$ or $3b + a = -10$ **[B1]**

[Total Marks: 7]

406 (a) Expands any two brackets correctly, e.g., $(2x^2 - 5x - 12)(x + 1)$ or $(x - 4)(2x^2 + 5x + 3)$ or $(x^2 - 3x - 4)(2x + 3)$ **[M1]**

Multiplies their resulting quadratic by the third bracket, allowing one error **[M1]**

$2x^3 - 3x^2 - 17x - 12$ **[A1]**

(b) $(3x + a)(2x + b)$ or $(6x + c)(x + d)$ where $ab = -10$ or $cd = -10$ **[M1]**

$(3x + 2)(2x - 5)$ **[A1]**

(c) $x^2(x + 3) - 4(x + 3)$ or $x(x^2 - 4) + 3(x^2 - 4)$ **[M1]**

$(x^2 - 4)(x + 3)$ **[M1]**

$(x - 2)(x + 2)(x + 3)$ **[A1]**

[Total Marks: 8]

407 (a) M1 for correctly expanding any pair of brackets, e.g., $2x^2 - 3x - 2$ or $2x^2 + 7x + 3$ or $x^2 + x - 6$

M1 for multiplying their quadratic expression by the third bracket, allowing one error in terms

$2x^3 + 3x^2 - 11x - 6$ **[A1]**

(b) M1 for $(3x + a)(2x + b)$ where $ab = -10$ or $3b + 2a = -11$

$(3x + 2)(2x - 5)$ **[A1]**

(c) M1 for partial factorisation extracting x: $x(3x^2 + 5x - 2)$

M1 for factorising the quadratic factor: $(3x - 1)(x + 2)$

$x(3x-1)(x+2)$ **[A1]**

[Total Marks: 8]

408 **(a)** $(x^2+x-6)(2x-1)$ or $(x-2)(2x^2+5x-3)$ or $(x+3)(2x^2-5x+2)$ **[M1]**

$2x^3-x^2+2x^2-x-12x+6$ (allow one error in expansion) **[M1]**

$2x^3+x^2-13x+6$ **[A1]**

(b) $(3x \pm a)(2x \pm b)$ where $ab=10$ or $6x^2-15x+4x-10$ **[M1]**

$(3x+2)(2x-5)$ **[A1]**

(c) $2x(x^2-9)$ **[M1]**

$(x-3)(x+3)$ seen **[M1]**

$2x(x-3)(x+3)$ **[A1]**

[Total Marks: 8]

409 Correct common denominator of 15 with unsimplified numerator $3(2x)-5(x-1)$ **[M1]**

$\dfrac{x+5}{15}$ **[A1]**

[Total Marks: 2]

410 M1 for a correct common denominator with at least one correct numerator, e.g.,
$\dfrac{5(2x-1)+3(x+4)}{15}$ or $\dfrac{10x-5}{15}+\dfrac{3x+12}{15}$

$\dfrac{13x+7}{15}$ **[A1]**

[Total Marks: 2]

411 $\dfrac{5(3x)-4(x-2)}{20}$ or $\dfrac{15x}{20}-\dfrac{4x-8}{20}$ **[M1]**

$\dfrac{11x+8}{20}$ **[A1]**

[Total Marks: 2]

412 Uses common denominator $(x+2)(3x-1)$ **[B1]**

Correct unsimplified numerator $5(3x-1)+2(x+2)$ **[M1]**

$\dfrac{17x-1}{(x+2)(3x-1)}$ or $\dfrac{17x-1}{3x^2+5x-2}$ **[A1]**

[Total Marks: 3]

413 Writes fractions over a common denominator of 20 **[M1]**

Correct unsimplified numerator: $5(x+3)-4(2x-1)$ or $5x+15-8x+4$ **[M1]**

$\frac{19-3x}{20}$ or $\frac{-3x+19}{20}$ **[A1]**

[Total Marks: 3]

414 Common denominator of 12 and correct numerator formulation: $\frac{3(3x)-4(x-2)}{12}$ **[M1]**

Numerator expanded correctly to $9x-4x+8$ **[M1]**

$\frac{5x+8}{12}$ **[A1]**

[Total Marks: 3]

415 $\frac{5(2x)-3(x-4)}{15}$ or $\frac{10x}{15}-\frac{3(x-4)}{15}$ **[M1]**

$10x-3x+12$ or better as numerator (allow one sign error for M1, e.g., $10x-3x-12$) **[M1]**

$\frac{7x+12}{15}$ **[A1]**

[Total Marks: 3]

416 (a) Uses a common denominator of 12, e.g. $\frac{4(2x)+3(x-1)}{12}$ **[M1]**

$\frac{11x-3}{12}$ **[A1]**

(b) $3x^2$ or $\frac{1}{y^2}$ seen **[M1]**

$\frac{3x^2}{y^2}$ or $3x^2y^{-2}$ **[A1]**

[Total Marks: 4]

417 (a) Common denominator 20 with numerator $5(3x)-4\times 2(x-3)$ or $15x-8x+24$ **[M1]**

$\frac{7x+24}{20}$ **[A1]**

(b) $\frac{6a^3b}{5c^2}\times\frac{10bc}{9a^2}$ **[M1]**

$\frac{4ab^2}{3c}$ **[A1]**

[Total Marks: 4]

418 (a) $\frac{15x+8x}{20}$ **[M1]**

$\frac{23x}{20}$ **[A1]**

(b) $\frac{5a^2b}{12} \times \frac{9}{10ab^2}$ **[M1]**

$\frac{3a}{8b}$ **[A1]**

[Total Marks: 4]

419 $(x-4)(x+4)$ seen **[B1]**

$(x-4)(x-1)$ seen **[B1]**

$\frac{x+4}{x-1}$ **[A1]**

[Total Marks: 3]

420 $3(x^2-4)$ or $3(x-2)(x+2)$ **[B1]**

$(x+5)(x-2)$ [B2] (or [M1] for $(x+a)(x+b)$ where $ab=-10$ or $a+b=3$)

$\frac{3(x+2)}{x+5}$ or $\frac{3x+6}{x+5}$ **[A1]**

[Total Marks: 4]

421 (a) $\frac{3(x+5)-2(x-2)}{(x-2)(x+5)}$ **[M1]**

$3x+15-2x+4$ correctly expanded on numerator **[M1]**

$\frac{x+19}{(x-2)(x+5)}$ or $\frac{x+19}{x^2+3x-10}$ **[A1]**

(b) $(x-4)(x+4)$ seen for numerator **[M1]**

$(x-4)(x+2)$ seen for denominator **[M1]**

$\frac{x+4}{x+2}$ **[A1]**

[Total Marks: 6]

422 (a) M1 for factorising the numerator correctly: $(x-3)(x+3)$

M1 for factorising the denominator correctly: $(2x-1)(x+3)$

$\frac{x-3}{2x-1}$ **[A1]**

(b) M1 for the correct common denominator $(x-2)(x+1)$

M1 for the correct unsimplified numerator $4(x+1)-3(x-2)$

$\frac{x+10}{(x-2)(x+1)}$ or $\frac{x+10}{x^2-x-2}$ **[A1]**

[Total Marks: 6]

423 (a) Common denominator $(x-3)(x+2)$ seen **[M1]**

Numerator $4(x+2)-3(x-3)$ or $4x+8-3x+9$ **[M1]**

$\dfrac{x+17}{(x-3)(x+2)}$ or $\dfrac{x+17}{x^2-x-6}$ **[A1]**

(b) Numerator factorised as $(x-3)(x+3)$ **[B1]**

Denominator factorised as $(x+2)(x+3)$ **[B1]**

$\dfrac{x-3}{x+2}$ **[B1]**

[Total Marks: 6]

424 Factorises denominator to $(2x-3)(2x+3)$ **[B1]**

Attempts to factorise numerator to $(2x+a)(x+b)$ where $ab=-12$ or $2b+a=5$ **[M1]**

Correctly factorises numerator to $(2x-3)(x+4)$ **[A1]**

$\dfrac{x+4}{2x+3}$ **[A1]**

[Total Marks: 4]

425 $(3x+1)(x-2)$ **[B2]**

If not B2, then B1 for $(3x+a)(x+b)$ where $ab=-2$ or $3b+a=-5$ **[B1]**

$(x-2)(x+2)$ **[B1]**

$(x-2)(x+1)$ **[B1]**

$\dfrac{3x^2-5x-2}{x^2-4}\times\dfrac{x^2-x-2}{3x+1}$ or equivalent correct inverted product **[M1]**

$\dfrac{(x-2)(x+1)}{x+2}$ or $\dfrac{x^2-x-2}{x+2}$ **[A1]**

[Total Marks: 6]

426 (a) Factorises $2x^2-5x-3=(2x+1)(x-3)$ and $4x^2-1=(2x-1)(2x+1)$ **[B1]**

Factorises $x^2-9=(x-3)(x+3)$ and $2x^2+5x-3=(2x-1)(x+3)$ **[B1]**

Inverts second fraction and multiplies: $\dfrac{(2x+1)(x-3)}{(2x-1)(2x+1)}\times\dfrac{(2x-1)(x+3)}{(x-3)(x+3)}$ **[M1]**

1 **[A1]**

(b) Factorises denominators: $(x-2)(x+2)$ and $(x+1)(x+2)$ **[M1]**

Common denominator $(x-2)(x+2)(x+1)$ with numerator $3(x+1)-2(x-2)$ **[M1]**

$\dfrac{x+7}{(x-2)(x+2)(x+1)}$ **[A1]**

[Total Marks: 7]

427 (a) Factorising numerator: $(2x+1)(x-3)$ **[B1]**

Factorising denominator: $(2x-3)(2x+3)$ **[B1]**

Note: no common factor cancels; the expression is already in its simplest form

Final answer: $\frac{2x^2-5x-3}{4x^2-9}$ or $\frac{(2x+1)(x-3)}{(2x-3)(2x+3)}$ **[A1]**

(b) Recognising $x^2-x-2=(x-2)(x+1)$ **[M1]**

Writing all over a common denominator of $(x-2)(x+1)$: $\frac{3(x+1)+4(x-2)-2x}{(x-2)(x+1)}$ **[M2]**

Simplifying numerator: $3x+3+4x-8-2x=5x-5$ **[M1]**

$\frac{5(x-1)}{(x-2)(x+1)}$ **[A1]**

[Total Marks: 7]

428 Factorising numerator to $(x-3)(x+3)$ **[B1]**

Factorising denominator to $(2x-1)(x+3)$ **[B1]**

$\frac{x-3}{2x-1}$ **[A1]**

[Total Marks: 3]

429 Denominator of $(x-2)(2x+1)$ **[B1]**

Correct unsimplified numerator: $5(2x+1)-3(x-2)$ **[M1]**

$\frac{7x+11}{(x-2)(2x+1)}$ **[A1]**

[Total Marks: 3]

430 Numerator factorised as $(2x-3)(x-3)$ **[B2]**

Allow [M1] for $(2x+a)(x+b)$ where $ab=9$ or $2b+a=-9$

Denominator factorised as $2(x-3)(x+3)$ or $2(x^2-9)$ or $(2x-6)(x+3)$ **[B1]**

$\frac{2x-3}{2(x+3)}$ or $\frac{2x-3}{2x+6}$ **[A1]**

[Total Marks: 4]

431 $2(x-2)(x+2)$ **[B2]**

If not B2, then B1 for $2(x^2-4)$ or $(2x-4)(x+2)$ **[B1]**

$(x-2)(x+3)$ **[B2]**

If not B2, then B1 for $(x+a)(x+b)$ where $ab=-6$ or $a+b=1$ **[B1]**

$\frac{2(x+2)}{x+3}$ or $\frac{2x+4}{x+3}$ **[A1]**

[Total Marks: 5]

432 **(a)** Common denominator $(2x-1)(x+4)$ seen **[M1]**

Numerator: $(x+3)(x+4)-(x-2)(2x-1)$ **[M1]**

$(x^2+7x+12)-(2x^2-5x+2)$ correctly simplified to $-x^2+12x+10$ **[A1]**

(b) Factorising numerator to $(x-4)(x+4)$ or denominator to $x(x+4)$ **[M1]**

$\dfrac{x-4}{x}$ **[A1]**

[Total Marks: 5]

433 Factorising x^2-25 to $(x-5)(x+5)$ and x^2-3x to $x(x-3)$ **[B1]**

Factorising x^2+6x+5 to $(x+1)(x+5)$ and $2x-6$ to $2(x-3)$ **[B1]**

Showing multiplication by the reciprocal, e.g. $\dfrac{(x-5)(x+5)}{x(x-3)} \times \dfrac{2(x-3)}{(x+1)(x+5)}$ **[M1]**

$\dfrac{2(x-5)}{x(x+1)}$ (or equivalent expanded form $\dfrac{2x-10}{x^2+x}$) **[A1]**

[Total Marks: 4]

434 **(a)** M1 for factorising $2x^2-8x$ to $2x(x-4)$

M1 for factorising x^2-16 to $(x-4)(x+4)$

M1 for inverting the second fraction and factorising, i.e., multiplying by $\dfrac{3(x+4)}{6x}$

1 **[A1]**

(b) M1 for writing the denominator x^2-9 as $(x-3)(x+3)$

M1 for common denominator $(x-3)(x+3)$ and correct numerator $5(x+3)+2x+1$

$\dfrac{7x+16}{(x-3)(x+3)}$ or $\dfrac{7x+16}{x^2-9}$ **[A1]**

[Total Marks: 7]

435 **(a)** $\dfrac{4(2x-1)-3(x-3)}{(x-3)(2x-1)}$ **[M1]**

Numerator expanded correctly to $8x-4-3x+9$ **[M1]**

$\dfrac{5x+5}{(x-3)(2x-1)}$ or $\dfrac{5(x+1)}{(x-3)(2x-1)}$ **[A1]**

(b) Multiplies by reciprocal: $\times\dfrac{4x+2}{x+5}$ **[M1]**

Factorises x^2-25 to $(x-5)(x+5)$ or $4x+2$ to $2(2x+1)$ **[M1]**

Factorises denominator $2x^2-9x-5$ to $(2x+1)(x-5)$ **[M1]**

2 **[A1]**

[Total Marks: 7]

436 **(a)** Uses common denominator $(x-2)(x+2)$ or x^2-4 **[B1]**

Correct unsimplified numerator $3(x+2)-(x+5)$ **[M1]**

$\dfrac{2x+1}{(x-2)(x+2)}$ or $\dfrac{2x+1}{x^2-4}$ **[A1]**

(b) Factorises $4x^2-25$ as $(2x-5)(2x+5)$ **[B1]**

Factorises denominators as $x(2x-5)$ and $2x(2x+5)$ **[B2]**

Allow [B1] if only one denominator is factorised correctly

Correct cancellation leading to final answer $3x$ **[A1]**

[Total Marks: 7]

437 **(a)** $\dfrac{5(2x+3)+4(x-2)}{(x-2)(2x+3)}$ **[M1]**

$10x+15+4x-8$ **[M1]**

$\dfrac{14x+7}{(x-2)(2x+3)}$ or $\dfrac{7(2x+1)}{(x-2)(2x+3)}$ **[A1]**

(b) $\dfrac{2x^2-5x-12}{4x^2-9} \times \dfrac{2x-3}{x-4}$ **[M1]**

$(2x+3)(x-4)$ **[B1]**

$(2x-3)(2x+3)$ **[B1]**

1 **[A1]**

[Total Marks: 7]

438 **(a)** $(2x-1)(x+3)$ seen for numerator **[M1]**

$(2x-1)(2x+1)$ seen for denominator **[M1]**

Identifies and cancels the common factor $(2x-1)$ **[M1]**

$\dfrac{x+3}{2x+1}$ **[A1]**

(b) Factorises denominator: $x^2-4x+3=(x-1)(x-3)$ **[M1]**

Writes with a common denominator: $\dfrac{5(x-1)+(x+2)}{(x-1)(x-3)}$ **[M1]**

Simplifies numerator: $5x-5+x+2=6x-3$ **[M1]**

$\dfrac{3(2x-1)}{(x-1)(x-3)}$ or $\dfrac{6x-3}{x^2-4x+3}$ **[A1]**

[Total Marks: 8]

439 If incorrect, award [B1] for any two of 12, x^8, or y^4 correct in a product

$12x^8y^4$ **[B2]**

[Total Marks: 2]

440 **(a)** $12x^{-3}$ or $\dfrac{12}{x^3}$ **[B1]**

(b) $\dfrac{2x^3}{3}$ or $\dfrac{2}{3}x^3$ **[B2]**

Correct cube root of coefficient $\dfrac{2}{3}$ OR correct index x^3 seen **[M1]**

[Total Marks: 3]

441 **(a)** $20x$ **[B1]**

(b) Numerator $4x^8$ or denominator y^6 seen **[M1]**

$\dfrac{4x^8}{y^6}$ **[A1]**

(c) -3 **[B1]**

[Total Marks: 4]

442 **(a)** M1 for any two of 12, x^8 or y^4 seen in a product

$12x^8y^4$ **[A1]**

(b) M1 for $\dfrac{1}{16^{\frac{1}{4}}}$ or 2^{-1} or $\dfrac{1}{\sqrt[4]{16}}$

$\dfrac{1}{2}$ **[A1]**

[Total Marks: 4]

443 **(a)** 15 or x^2 in a product **[M1]**

$15x^2$ **[A1]**

(b) $\sqrt[4]{16}$ or 2^3 or $4096^{\frac{1}{4}}$ **[M1]**

8 **[A1]**

[Total Marks: 4]

444 **(a)** y^3 **[B1]**

(b) $\dfrac{1}{25}$ or 0.04 **[B1]**

(c) $8x^{12}$ **[B2]**

If B0, allow [M1] for $8x^k$ or kx^{12} where $k \neq 0$

[Total Marks: 4]

445 **(a)** Coefficient 6 or power $-3 + \dfrac{1}{2}$ seen **[M1]**

$6x^{-\frac{5}{2}}$ or $\dfrac{6}{x^{\frac{5}{2}}}$ **[A1]**

(b) Two of 27, y^{12}, or denominator 8 correct **[M1]**

$\dfrac{27y^{12}}{8}$ **[A1]**

[Total Marks: 4]

446 **(a)** -3 **[B1]**

(b) $2y^3$ **[B2]**

If not B2, then B1 for 2 or y^3 seen **[B1]**

(c) $\dfrac{1}{8}$ **[B1]**

[Total Marks: 4]

447 **(a)** w^{11} **[B1]**

(b) 3^2 or 9 seen, or x^6 seen **[M1]**

$9x^6$ **[A1]**

(c) $\dfrac{15}{5} = 3$ or $y^{8-(-2)}$ **[M1]**

$3y^{10}$ **[A1]**

[Total Marks: 5]

448 **(a)** p^{11} **[B1]**

(b) $9q^6$ **[B2]**

Partial correct answer $9q^k$ or cq^6 **[B1]**

(c) $(\sqrt[3]{27})^2$ or 3^2 or $\sqrt[3]{729}$ **[M1]**

9 **[A1]**

[Total Marks: 5]

449 **(a)** $4x^3y^4$ **[B2]**

If not B2, then B1 for two of 4, x^3, y^4 correct **[B1]**

(b) $(3^3)^{x-1} = (3^2)^{2x+1}$ or $3^{3(x-1)} = 3^{2(2x+1)}$ **[M1]**

$3x - 3 = 4x + 2$ **[M1]**

-5 **[A1]**

[Total Marks: 5]

450 **(a)** $\frac{1}{4}$ or 0.25 **[B2]**

Correct intermediate step e.g., 2^{-2} or $\left(\frac{1}{32}\right)^{\frac{2}{5}}$ or $32^{\frac{1}{5}} = 2$ **[M1]**

(b) Writes 125 as 5^3 **[M1]**

Equates indices: $x + 2 = 3x$ **[M1]**

1 **[A1]**

[Total Marks: 5]

451 **(a)** $(\sqrt[3]{125})^2$ or 5^2 or $125^2 = 15625$ and $\sqrt[3]{15625}$ **[M1]**

25 **[A1]**

(b) 3 **[B1]**

x^6 **[B1]**

y^{-8} or $\frac{1}{y^8}$ **[B1]**

Note: Award [B3] for a fully correct final answer of $\frac{3x^6}{y^8}$ or $3x^6y^{-8}$

[Total Marks: 5]

452 **(a)** $\frac{1}{4}$ or 0.25 **[B1]**

(b) 25 or $\frac{1}{25}$ seen **[M1]**

1 **[A1]**

(c) $125 = 5^3$ seen or implied **[B1]**

$2x - 1 = 3x$ **[M1]**

$x = -1$ **[A1]**

[Total Marks: 6]

453 **(a)** $4 \times \frac{3}{2} = 6$ or $x^{-3+\frac{1}{2}}$ **[M1]**

$6x^{-\frac{5}{2}}$ or $\frac{6}{x^{\frac{5}{2}}}$ **[A1]**

(b) $3^{2x+1} = 3^3$ or $2x + 1 = 3$ **[M1]**

1 **[A1]**

(c) $8^{\frac{1}{3}} = 2$ or $(a^6)^{\frac{1}{3}} = a^2$ or $(b^3)^{\frac{1}{3}} = b$ **[M1]**

$\frac{2a^2}{b}$ **[A1]**

[Total Marks: 6]

454 **(a)** Numerator $8x^{15}$ **[B1]**

Denominator $27y^6$ **[B1]**

$\dfrac{8x^{15}}{27y^6}$ **[A1]**

(b) Writing both sides with a common base of 3, e.g. $3^{2x} = 3^{3(x+1)}$ **[M1]**

$2x = 3x + 3$ **[M1]**

-3 **[A1]**

[Total Marks: 6]

455 **(a)** $\left(\dfrac{64}{27}\right)^{\frac{2}{3}}$ or $\left(\dfrac{3}{4}\right)^{-2}$ or $\dfrac{4}{3}$ seen **[M1]**

$\dfrac{16}{9}$ **[A1]**

(b) Writes both sides with base 2: $(2^3)^{2x-1} = (2^5)^{x+2}$ **[M1]**

Equates exponents: $3(2x - 1) = 5(x + 2)$ **[M1]**

$6x - 3 = 5x + 10$ **[M1]**

13 **[A1]**

[Total Marks: 6]

456 **(a)** Takes the reciprocal to deal with the negative index: $\left(\dfrac{64y^{12}}{27x^9}\right)^{\frac{1}{3}}$ or evaluates cube root of constants correctly **[M1]**

Applies the power of $\dfrac{1}{3}$ to variables: $y^{12\times\frac{1}{3}}$ and $x^{9\times\frac{1}{3}}$ **[M1]**

$\dfrac{4y^4}{3x^3}$ **[A1]**

(b) Writes 32 as 2^5 **[B1]**

Equates indices: $3x - 1 = 5x$ **[M1]**

-0.5 or $-\dfrac{1}{2}$ **[A1]**

[Total Marks: 6]

457 **(a)** 8 **[B1]**

x^9 **[B1]**

y^6 as denominator or y^{-6} **[B1]**

(b) $8 = 2^3$ seen or implied by $8^x = 2^{3x}$ **[M1]**

$x + 3 = 3x$ **[M1]**

1.5 or $\frac{3}{2}$ **[A1]**

[Total Marks: 6]

458 **(a)** $(3^2)^{2x} = (3^3)^{x+1}$ or $3^{4x} = 3^{3x+3}$ **[M1]**

$4x = 3x + 3$ **[M1]**

3 **[A1]**

(b) Inverts fraction to remove negative power: $\left(\frac{81b^{12}}{16a^8}\right)^{\frac{3}{4}}$ **[M1]**

Applies power $\frac{1}{4}$ (fourth root) to all terms: $\frac{3b^3}{2a^2}$ **[M1]**

Cubes the result to get final answer: $\frac{27b^9}{8a^6}$ or $3.375b^9a^{-6}$ **[A1]**

[Total Marks: 6]

459 **(a)** Expressing both sides with base 2: $(2^4)^{2x-1} = (2^3)^{x+2}$ **[M1]**

Equating powers: $4(2x - 1) = 3(x + 2)$ or $8x - 4 = 3x + 6$ **[M1]**

2 **[A1]**

(b) Taking the reciprocal to remove negative index: $\left(\frac{16y^8}{81x^{12}}\right)^{\frac{3}{4}}$ **[M1]**

Evaluating the fractional power of the numbers: $16^{\frac{3}{4}} = 8$ and $81^{\frac{3}{4}} = 27$ **[M1]**

$\frac{8y^6}{27x^9}$ **[A1]**

[Total Marks: 6]

460 **(a)** Simplifies the first fraction to $8x^2y^{-5}$ or $\frac{8x^2}{y^5}$ **[M1]**

Multiplies by the reciprocal of the second fraction: $\times\frac{y^4}{4x^3}$ **[M1]**

$\frac{2}{xy}$ or $2x^{-1}y^{-1}$ **[A1]**

(b) Writes 25 as 5^2 and 125 as 5^3 **[M1]**

Applies exponent rules correctly to get a single base on the left: $5^{2x-1+2(x+2)}$ **[M1]**

Equates indices to form a linear equation: $4x + 3 = 3x$ **[M1]**

-3 **[A1]**

[Total Marks: 7]

461 **(a)** $12x^{\frac{3}{2}}$ seen as numerator **[M1]**

$2x^k$ for any k or $cx^{\frac{7}{6}}$ for any c **[M1]**

$2x^{\frac{7}{6}}$ **[A1]**

(b) Converting all terms to base 3: $3^{2(2x-1)} \times 3^{x+1} = 3^{3x}$ **[M1]**

Applying exponent multiplication law: $3^{4x-2+x+1}$ or 3^{5x-1} **[M1]**

Equating indices: $5x - 1 = 3x$ **[M1]**

0.5 or $\frac{1}{2}$ **[A1]**

[Total Marks: 7]

462 **(a)** $(9x^{-4})^{\frac{1}{2}} = 3x^{-2}$ **[B1]**

Numerator simplified to $12x^{-\frac{1}{2}}$ **[M1]**

Subtracts indices correctly: $-\frac{1}{2} - \left(-\frac{3}{2}\right) = 1$ **[M1]**

$2x$ **[A1]**

(b) Expresses all terms as powers of 3: $\frac{3^{2k-1}}{3^{3k}} = 3^{\frac{1}{2}}$ **[M1]**

Equates indices: $(2k-1) - 3k = \frac{1}{2}$ or $-k - 1 = \frac{1}{2}$ **[M1]**

$-\frac{3}{2}$ or -1.5 **[A1]**

[Total Marks: 7]

463 **(a)** M1 for simplifying the numerator: $30x^{\frac{1}{2}}$

M1 for correctly handling the division: $6x^{\frac{1}{2}-\frac{3}{2}}$

$6x^{-1}$ or $\frac{6}{x}$ **[A1]**

(b) M1 for reciprocating to eliminate the negative index: $x^{\frac{2}{3}} = \frac{9}{4}$

M1 for isolating x: $x = \left(\frac{9}{4}\right)^{\frac{3}{2}}$

$\frac{27}{8}$ or $3\frac{3}{8}$ **[A1]**

(c) M1 for converting to base 3: $3^{2x} \times 3^{2x-1} = 3^{3(x+1)}$

M1 for applying index laws on the LHS to get 3^{4x-1}

M1 for equating powers correctly: $4x - 1 = 3x + 3$

4 **[A1]**

[Total Marks: 10]

464 **(a)** Any two of 10, x^6, y^2 correct **[M1]**

$10x^6y^2$ **[A1]**

(b) $\frac{3a^2}{b^3}$ or $3a^2b^{-3}$ **[B2]**

If 0 scored, award 1 mark for any two of 3, a^2, b^3 (or b^{-3}) correct in a product or quotient **[M1]**

(c) $\frac{1}{4}$ or 0.25 **[B1]**

[Total Marks: 5]

465 **(a)** M1 for $16^{\frac{3}{4}} = 8$ seen

M1 for $x^{12\times\frac{3}{4}} = x^9$ or $y^{8\times\frac{3}{4}} = y^6$ seen

$\frac{8x^9}{y^6}$ or $8x^9y^{-6}$ **[A1]**

(b) M1 for expressing both sides with a common base of 2: $2^{2(2x-1)}$ and $2^{3(x+2)}$

M1 for equating the exponents: $4x - 2 = 3x + 6$

8 **[A1]**

[Total Marks: 6]

466 **(a)** Expressing both sides with a base of 2: $2^{2(2x-1)}$ and 2^{5x} **[M1]**

Equating indices: $4x - 2 = 5x$ **[M1]**

$x = -2$ **[A1]**

(b) Any two of 4, x^3, y^{-4} (or $\frac{1}{y^4}$) correct **[M1]**

$4x^3y^{-4}$ or $\frac{4x^3}{y^4}$ **[A1]**

(c) $x = 64^{\frac{2}{3}}$ or $(\sqrt[3]{64})^2$ or 4^2 **[M1]**

16 **[A1]**

[Total Marks: 7]

467 **(a)** $4x^2$ or y^{-3} or $\frac{1}{y^3}$ seen **[M1]**

$\frac{4x^2}{y^3}$ or $4x^2y^{-3}$ **[A1]**

(b) 3^{2x} or $3^{3(x-1)}$ seen **[B1]**

$3\sqrt{3} = 3^{1.5}$ or $3^{\frac{3}{2}}$ seen **[B1]**

$2x + 3x - 3 = 1.5$ or equivalent equation in x **[M1]**

$x = 0.9$ or $\frac{9}{10}$ **[A1]**

[Total Marks: 6]

468 **(a)** Expressing all terms with a base of 2, e.g. $2^{2x} \times 2^{3(2x-1)} = 2^4$ **[M1]**

Correct linear equation for the exponents: $2x + 6x - 3 = 4$ or $8x = 7$ **[M1]**

$\frac{7}{8}$ or 0.875 **[A1]**

(b) Simplifying the numerator to $24x^1$ (or $24x$) **[M1]**

Correct manipulation of the coefficients $\frac{24}{8} = 3$ or indices $x^{1-(-3)}$ **[M1]**

$3x^4$ **[A1]**

[Total Marks: 6]

469 **(a)** $\left(\frac{27y^6}{8x^9}\right)^{\frac{2}{3}}$ or $\frac{4x^6}{9y^4}$ (inverted or squared/cubed root correctly applied to some terms) **[M1]**

$\left(\frac{3y^2}{2x^3}\right)^2$ **[M1]**

$\frac{9y^4}{4x^6}$ or $2.25y^4x^{-6}$ **[A1]**

(b) $\frac{5^{a-2} \times 5^{2b}}{5^{3(a+1)}} = 5^4$ (expressing all terms as powers of 5) **[M1]**

$a - 2 + 2b - 3(a + 1) = 4$ **[M1]**

$-2a + 2b - 5 = 4$ or $-2a + 2b = 9$ **[A1]**

Accept any pair of values satisfying the equation, e.g. $a = 0, b = 4.5$ or $a = -4.5, b = 0$ or $b = a + 4.5$ **[A1]**

[Total Marks: 7]

470 **(a)** $3x^2y^{-6}$ or $\frac{3x^2}{y^6}$ **[B3]**

Coefficient 3 seen **[M1]**

Correct subtraction of indices $x^{\frac{3}{2}-\left(-\frac{1}{2}\right)}$ leading to x^2 OR y^{-2-4} leading to y^{-6} **[M1]**

(b) Expresses all terms as powers of 2: $(2^2)^{2x-1} \times (2^3)^{x+1} = (2^5)^x$ **[M1]**

Correctly adds powers on the LHS: $2(2x - 1) + 3(x + 1)$ or $4x - 2 + 3x + 3$ **[M1]**

Forms linear equation equating LHS and RHS powers: $7x + 1 = 5x$ **[M1]**

$-\frac{1}{2}$ or -0.5 **[A1]**

[Total Marks: 7]

471 **(a)** $\frac{2}{5}a^4b^{-6}$ or $\frac{2a^4}{5b^6}$ (correctly simplifying the first fraction) **[B1]**

$\left(\frac{2}{a^2b}\right)^2$ or $\frac{4}{a^4b^2}$ or $4a^{-4}b^{-2}$ (correctly simplifying the second fraction) **[B2]**

If 0 scored for the second fraction, award 1 mark for applying the negative index: $\frac{a^{-4}b^{-2}}{2^{-2}}$ **[M1]**

$\frac{8}{5}b^{-8}$ or $\frac{8}{5b^8}$ **[A1]**

(b) $3^{2x} \times 3^y = 3^7$ yielding $2x + y = 7$ **[B1]**

$2^{3x} \div 2^{2y} = 2^7$ yielding $3x - 2y = 7$ **[B1]**

Correct method to solve their simultaneous linear equations, e.g., substituting $y = 7 - 2x$ to get $3x - 2(7 - 2x) = 7$ or matching coefficients **[M1]**

$7x = 21 \implies x = 3$ **[A1]**

$y = 1$ **[A1]**

[Total Marks: 9]

472 $3(3x - 1) = 4(x + 5)$ or $9x - 3 = 4x + 20$ **[M1]**

$5x = 23$ **[M1]**

4.6 or $\frac{23}{5}$ **[A1]**

[Total Marks: 3]

473 Multiplies all terms by 12 (or forms a single fraction): $3(2x - 3) + 4(x + 1) = 24$ **[M1]**

Expands brackets correctly: $6x - 9 + 4x + 4 = 24$ **[M1]**

2.9 or $\frac{29}{10}$ **[A1]**

[Total Marks: 3]

474 Valid method to eliminate one variable, e.g. multiplying the first equation by 2 to get $4x - 2y = 14$ and adding the equations **[M1]**

$x = 4$ **[A1]**

$y = 1$ **[A1]**

[Total Marks: 3]

475 Multiplies one or both equations to equate coefficients of x or y (e.g. $4x - 10y = 44$) **[M1]**

Adds or subtracts equations to eliminate one variable (e.g. $13y = -39$) **[M1]**

$x = 3.5$, $y = -3$ **[A1]**

[Total Marks: 3]

476 **(a)** $4x - 12 = 2x + 10$ **[M1]**

$x = 11$ **[A1]**

(b) $5y = 2x - 1$ **[M1]**

$\dfrac{5y + 1}{2}$ **[A1]**

[Total Marks: 4]

477 **(a)** $6x - 3 = 5x + 8$ **[M1]**

11 **[A1]**

(b) $y + 7 = 4x$ or $\dfrac{y}{4} = x - \dfrac{7}{4}$ **[M1]**

$\dfrac{y + 7}{4}$ or $\dfrac{y}{4} + \dfrac{7}{4}$ **[A1]**

[Total Marks: 4]

478 **(a)** M1 for expanding brackets correctly: $6x - 15 = 4x + 4$

M1 for correctly isolating x terms, e.g., $6x - 4x = 4 + 15$ or $2x = 19$

9.5 or $\dfrac{19}{2}$ **[A1]**

(b) M1 for multiplying by 3 to get $3V = \pi r^2 h$

$\dfrac{3V}{\pi r^2}$ **[A1]**

[Total Marks: 5]

479 **(a)** $6x - 3 = 5x + 8$ **[M1]**

11 **[A1]**

(b) $5y = 2x + 3$ **[M1]**

$2x = 5y - 3$ **[M1]**

$\dfrac{5y - 3}{2}$ **[A1]**

[Total Marks: 5]

480 **(a)** $8x - 12 = 5x + 9$ **[M1]**

$3x = 21 \implies x = 7$ **[A1]**

(b) Correctly eliminates one variable: e.g. $9x + 6y = 48$ and $10x - 6y = 28$ **[M1]**

$19x = 76 \implies x = 4$ **[A1]**

$y = 2$ **[A1]**

[Total Marks: 5]

481 **(a)** Expands brackets correctly to $6x - 3 = 4x + 11$ **[M1]**

7 **[A1]**

(b) Multiplies second equation by 2 to get $2x - 2y = 4$ OR substitutes $x = y + 2$ into first equation **[M1]**

Correctly finds $x = 4$ **[A1]**

Correctly finds $y = 2$ **[A1]**

[Total Marks: 5]

482 **(a)** $5x - 3 = 2x + 12$ **[M1]**

5 **[A1]**

(b) Multiply by 12 to get $4(x + 2) + 3(2x - 1) = 36$ (or equivalent common denominator method) **[M1]**

$4x + 8 + 6x - 3 = 36$ **[M1]**

$\frac{31}{10}$ or 3.1 **[A1]**

[Total Marks: 5]

483 **(a)** Expands bracket to obtain $6x - 15$ **[M1]**

Rearranges terms to isolate x, e.g., $6x - 4x = 7 + 15$ or $2x = 22$ **[M1]**

11 **[A1]**

(b) Multiplies by 5 to obtain $5y = 2x + 3$ **[M1]**

$\frac{5y - 3}{2}$ **[A1]**

[Total Marks: 5]

484 **(a)** $8x - 4 = 3x + 11$ **[M1]**

$5x = 15$ **[M1]**

3 **[A1]**

(b) $2y = 3x - 5$ **[M1]**

$2y + 5 = 3x$ **[M1]**

$\frac{2y + 5}{3}$ or $\frac{2y}{3} + \frac{5}{3}$ **[A1]**

[Total Marks: 6]

485 **(a)** $5x - 10 = 3x + 14$ **[M1]**

$2x = 24$ **[M1]**

12 **[A1]**

(b) $2A = h(a+b)$ or $\dfrac{A}{h} = \dfrac{a+b}{2}$ **[M1]**

$\dfrac{2A}{h} = a+b$ or $2A = ha + hb$ **[M1]**

$\dfrac{2A}{h} - a$ or $\dfrac{2A - ah}{h}$ **[A1]**

[Total Marks: 6]

486 (a) $6x - 15 = 4x + 7$ **[M1]**

$2x = 22$ **[M1]**

11 **[A1]**

(b) Equates coefficients, e.g. multiplies first equation by 2 and second equation by 3 to get $4x + 6y = 26$ and $15x - 6y = 12$ **[M1]**

Adds equations to eliminate y, e.g. $19x = 38$, so $x = 2$ **[A1]**

$y = 3$ **[A1]**

[Total Marks: 6]

487 (a) Expands brackets: $4x - 12 = 2x + 7$ **[M1]**

Rearranges to collect terms: $2x = 19$ **[M1]**

9.5 or $\dfrac{19}{2}$ **[A1]**

(b) Multiplies first equation by 2 to equate coefficients of y: $10x - 4y = 32$ (or equivalent valid method) **[M1]**

Adds equations to eliminate y: $13x = 39 \implies x = 3$ **[A1]**

Substitutes $x = 3$ into either equation to find y: $15 - 2y = 16 \implies y = -0.5$ **[A1]**

[Total Marks: 6]

488 $y(x+4) = 3x - 2$ **[M1]**

$xy - 3x = -4y - 2$ or $3x - xy = 4y + 2$ **[M1]**

$x(y-3) = -4y - 2$ or $x(3-y) = 4y + 2$ **[M1]**

$\dfrac{4y+2}{3-y}$ or $\dfrac{-4y-2}{y-3}$ **[A1]**

[Total Marks: 4]

489 $y(x-5) = 2x + 3$ or $xy - 5y = 2x + 3$ **[M1]**

$xy - 2x = 5y + 3$ or $2x - xy = -5y - 3$ **[M1]**

$x(y-2) = 5y + 3$ or $x(2-y) = -5y - 3$ **[M1]**

$\dfrac{5y+3}{y-2}$ or $\dfrac{-5y-3}{2-y}$ **[A1]**

[Total Marks: 4]

490 Multiplies by the denominator: $y(2x+1) = 3x - 5$ **[M1]**

Expands the bracket: $2xy + y = 3x - 5$ **[M1]**

Rearranges to collect all terms involving x on one side: $y + 5 = 3x - 2xy$ (or $2xy - 3x = -y - 5$) **[M1]**

$\frac{y+5}{3-2y}$ or $\frac{-y-5}{2y-3}$ **[A1]**

[Total Marks: 4]

491 (a) Multiplies equations to equate coefficients of x or y, e.g., $9x + 6y = 36$ and $10x - 6y = 2$ **[M1]**

Adds or subtracts equations to eliminate a variable **[M1]**

$x = 2$ and $y = 3$ **[A1]**

(b) $(2x \pm a)(x \pm b) = 0$ where $ab = 12$ **[M1]**

$(2x+3)(x-4) = 0$ **[M1]**

$x = 4$ and $x = -1.5$ (or $-\frac{3}{2}$) **[A1]**

[Total Marks: 6]

492 (a) Correct method to equate coefficients (e.g., $10x + 15y = 65$ and $10x - 4y = 8$, leading to $19y = 57$) or correctly substitute **[M1]**

$y = 3$ **[A1]**

$x = 2$ **[A1]**

(b) $y(x+2) = 3x - 1$ **[M1]**

$xy + 2y = 3x - 1$ **[M1]**

$xy - 3x = -1 - 2y$ or $2y + 1 = 3x - xy$ **[M1]**

$\frac{-1-2y}{y-3}$ or $\frac{2y+1}{3-y}$ **[A1]**

[Total Marks: 7]

493 (a) $y(x+4) = 3x - 2$ **[M1]**

$xy + 4y = 3x - 2$ **[M1]**

$xy - 3x = -4y - 2$ or $x(y-3) = -(4y+2)$ **[M1]**

$\frac{-4y-2}{y-3}$ or $\frac{4y+2}{3-y}$ **[A1]**

(b) $5(x+1) = 2(x-2)$ **[M1]**

$5x + 5 = 2x - 4$ **[M1]**

-3 **[A1]**

[Total Marks: 7]

494 **(a)** Multiplies equations to equate coefficients, e.g., $9x + 6y = 36$ and $10x - 6y = 2$ **[M1]**

Adds or subtracts equations to eliminate one variable, e.g., $19x = 38$ **[M1]**

$x = 2$ **[A1]**

$y = 3$ **[A1]**

(b) Finds correct product and sum components, e.g., $-10x$ and $3x$ **[M1]**

Correctly factorises to $(2x + 3)(x - 5) = 0$ **[M1]**

$x = -1.5, x = 5$ **[A1]**

[Total Marks: 7]

495 **(a)** Multiply the second equation by 2 to get $8x + 2y = 36$ (or correct substitution method) **[M1]**

$x = 5$ **[A1]**

$y = -2$ **[A1]**

(b) $T(p - 2) = 3p + 4$ **[M1]**

$Tp - 2T = 3p + 4$ **[M1]**

$Tp - 3p = 2T + 4$ or $p(T - 3) = 2T + 4$ **[M1]**

$\frac{2T + 4}{T - 3}$ or $\frac{-2T - 4}{3 - T}$ **[A1]**

[Total Marks: 7]

496 **(a)** Multiplies the first equation by 2 to give $4x - 2y = 18$ (or rearranges to $y = 2x - 9$) **[M1]**

Adds the equations to eliminate y, yielding $7x = 28$ (or correctly substitutes to form an equation in one variable) **[M1]**

$x = 4$ **[A1]**

$y = -1$ **[A1]**

(b) Multiplies throughout by a common denominator (e.g., 12) to obtain $4(2x) - 3(x - 4) = 30$ **[M1]**

Expands brackets correctly: $8x - 3x + 12 = 30$ **[M1]**

$5x = 18$ **[M1]**

3.6 or $\frac{18}{5}$ **[A1]**

[Total Marks: 8]

497 **(a)** Multiply second equation by 2: $4x - 10y = 18$ **[M1]**

Subtract equations to eliminate x: $13y = -13$ **[M1]**

$y = -1$ **[A1]**

$x = 2$ **[A1]**

(b) $(3x + 2)(x - 4) = 0$ **[M2]**

If M0, M1 for $(3x \pm a)(x \pm b)$ where $ab = 8$ or $a \pm 3b = -10$

$x = -\frac{2}{3}$ **[A1]**

$x = 4$ **[A1]**

[Total Marks: 8]

498 Forms common denominator or multiplies through: $2(x - 2) + 1(x - 1) = (x - 1)(x - 2)$ **[M1]**

Expands brackets correctly: $2x - 4 + x - 1 = x^2 - 3x + 2$ **[M1]**

Rearranges to a 3-term quadratic: $x^2 - 6x + 7 = 0$ **[A1]**

Applies quadratic formula: $\frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(7)}}{2(1)}$ or completes the square: $(x - 3)^2 - 9 + 7 = 0$ **[M1]**

Correct unsimplified surd form: $\frac{6 \pm \sqrt{8}}{2}$ or $(x - 3)^2 = 2$ **[M1]**

$3 \pm \sqrt{2}$ **[A1]**

[Total Marks: 6]

499 (a) Multiplies through by $(x - 1)(2x + 1)$: $4(2x + 1) - 3(x - 1) = 2(x - 1)(2x + 1)$ **[M1]**

Expands left-hand side: $8x + 4 - 3x + 3 = 5x + 7$ **[B1]**

Expands right-hand side: $2(2x^2 - x - 1) = 4x^2 - 2x - 2$ **[B1]**

Rearranges cleanly to $4x^2 - 7x - 9 = 0$ with no errors **[A1]**

(b) Substitutes into quadratic formula: $x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(4)(-9)}}{2(4)}$ **[M1]**

Evaluates discriminant: $49 + 144 = 193$ **[A1]**

$\frac{7 + \sqrt{193}}{8}$ and $\frac{7 - \sqrt{193}}{8}$ (or $\frac{7 \pm \sqrt{193}}{8}$) **[A1]**

[Total Marks: 7]

500 (a) Squares both sides and clears denominator: $y^2(t + 2) = 3t - 5$ **[M1]**

Collects t terms on one side: $2y^2 + 5 = 3t - ty^2$ or $t(y^2 - 3) = -5 - 2y^2$ **[M1]**

$t = \frac{2y^2 + 5}{3 - y^2}$ or $t = \frac{-5 - 2y^2}{y^2 - 3}$ **[A1]**

(b) One correct root seen: -1.69, -1.00, or 2.69 **[B1]**

All three roots correct to 3 significant figures: -1.69, -1.00 (or -1), 2.69 **[B2]**

[Total Marks: 6]

501 **(a)** B1 for $x + y = 150$

B1 for $12x + 8y = 1480$

M1 for a valid method to solve the equations (e.g., multiplying the first equation by 8 to get $8x + 8y = 1200$ and subtracting)

$x = 70$ and $y = 80$ **[A1]**

(b) M1 for isolating the square root: $\dfrac{T}{2\pi} = \sqrt{\dfrac{m}{k}}$

M1 for squaring both sides: $\dfrac{T^2}{4\pi^2} = \dfrac{m}{k}$

$\dfrac{kT^2}{4\pi^2}$ **[A1]**

[Total Marks: 7]

502 **(a)** $p(x + 4) = 3x - 1$ **[M1]**

$px + 4p = 3x - 1$ **[M1]**

$3x - px = 4p + 1$ (or equivalent step isolating x terms on one side) **[M1]**

$\dfrac{4p + 1}{3 - p}$ (or $\dfrac{-4p - 1}{p - 3}$) **[A1]**

(b) $(2x \pm a)(x \pm b) = 0$ where $ab = -12$ or $2b \pm a = -5$ **[M1]**

$(2x + 3)(x - 4) = 0$ **[M1]**

$x = -1.5$, $x = 4$ **[A1]**

[Total Marks: 7]

503 **(a)** $2(3x + 2 + x - 1) = 42$ or $3x + 2 + x - 1 = 21$ **[M1]**

$4x + 1 = 21$ **[M1]**

5 **[A1]**

(b) $p^2 = \dfrac{x - 3}{x}$ **[M1]**

$p^2 x = x - 3$ **[M1]**

$p^2 x - x = -3$ or $x - p^2 x = 3$ or $x(p^2 - 1) = -3$ **[M1]**

$\dfrac{3}{1 - p^2}$ or $\dfrac{-3}{p^2 - 1}$ **[A1]**

[Total Marks: 7]

504 **(a)** Multiplies by denominator: $y(x - 3) = 2x + 5$ **[M1]**

Expands and collects x terms on one side: $xy - 2x = 3y + 5$ **[M1]**

$\frac{3y+5}{y-2}$ or $\frac{-3y-5}{2-y}$ **[A1]**

(b) Multiplies throughout by common denominator $3(x-2)$ to reach $(x+1)(x-2)-6=3(x-2)$ **[M1]**

Expands brackets correctly: $x^2-2x+x-2-6=3x-6$ **[M1]**

Rearranges to form a correct 3-term quadratic equation: $x^2-4x-2=0$ **[A1]**

Uses the quadratic formula $\frac{-(-4)\pm\sqrt{(-4)^2-4(1)(-2)}}{2(1)}$ **[M1]**

4.45 **[A1]**

-0.45 **[A1]**

[Total Marks: 9]

505 **(a)** Sets up equation for area: $(2x+3)(x-1)=45$ **[M1]**

Expands and simplifies to given result: $2x^2-2x+3x-3=45 \implies 2x^2+x-48=0$ **[A1]**

(b) Uses quadratic formula with correct values: $\frac{-1\pm\sqrt{1^2-4(2)(-48)}}{2(2)}$ **[M1]**

Correct discriminant: $\frac{-1\pm\sqrt{385}}{4}$ **[M1]**

4.655... and $-5.155...$ **[A1]**

4.66 and -5.16 **[A1]**

(c) Multiplies by denominator: $y(x-3)=ax+5$ **[M1]**

Expands brackets: $xy-3y=ax+5$ **[M1]**

Collects terms in x on one side: $xy-ax=3y+5$ **[M1]**

Factorises x and divides: $\frac{3y+5}{y-a}$ (or equivalent) **[A1]**

[Total Marks: 10]

506 $\frac{3(x-1)+1(x+2)}{(x+2)(x-1)}=2$ or $3(x-1)+1(x+2)=2(x+2)(x-1)$ **[M1]**

$3x-3+x+2$ or $4x-1$ **[M1]**

$2x^2-2x-3=0$ or $-2x^2+2x+3=0$ **[A2]**

$\frac{-(-2)\pm\sqrt{(-2)^2-4(2)(-3)}}{2(2)}$ or GDC equivalent to solve the 3-term quadratic **[M1]**

1.82 and -0.82 **[A1]**

[Total Marks: 6]

507 Multiplies by the common denominator to get $3(2x+1)+4(x-2)=2(x-2)(2x+1)$ [M2]
(allow M1 for $\frac{3(2x+1)+4(x-2)}{(x-2)(2x+1)}=2$)

Expands brackets to get $10x - 5 = 2(2x^2 - 3x - 2)$ or $10x - 5 = 4x^2 - 6x - 4$ **[M1]**

Rearranges to form a correct 3-term quadratic equation equal to zero: $4x^2 - 16x + 1 = 0$ **[A1]**

Substitutes correctly into the quadratic formula: $x = \dfrac{-(-16) \pm \sqrt{(-16)^2 - 4(4)(1)}}{2(4)}$ **[M1]**

$x = 3.94$ and $x = 0.06$ **[A1]**

[Total Marks: 6]

508 Multiplies by the common denominator to get $5(2x+1) - 3(x-2) = 2(x-2)(2x+1)$ **[M2]**

If not M2, M1 for a correct single fraction $\dfrac{5(2x+1) - 3(x-2)}{(x-2)(2x+1)}$ seen

$4x^2 - 13x - 15 = 0$ **[A2]**

If not A2, A1 for correctly expanding sides to $7x + 11$ or $4x^2 - 6x - 4$

$x = \dfrac{-(-13) \pm \sqrt{(-13)^2 - 4(4)(-15)}}{2(4)}$ **[M1]**

4.15 **[A1]**

-0.90 **[A1]**

[Total Marks: 7]

509 (a) $\dfrac{3(x+3) - 2(x-1)}{(x-1)(x+3)} = 1$ **[M1]**

$3x + 9 - 2x + 2 = (x-1)(x+3)$ **[M1]**

$x + 11 = x^2 + 2x - 3$ **[M1]**

$x^2 + x - 14 = 0$ established with no errors seen **[A1]**

(b) $\dfrac{-1 \pm \sqrt{1^2 - 4(1)(-14)}}{2(1)}$ **[M1]**

$\dfrac{-1 \pm \sqrt{57}}{2}$ **[M1]**

$3.27, -4.27$ **[A1]**

[Total Marks: 7]

510 (a) (i) Forms a correct initial equation involving time: $\dfrac{120}{x} - \dfrac{120}{x+10} = \dfrac{36}{60}$ **[M1]**

Simplifies the time difference to $\dfrac{3}{5}$ and multiplies by a common denominator: $600(x+10) - 600x = 3x(x+10)$ **[M1]**

$6000 = 3x^2 + 30x$ leading correctly to $x^2 + 10x - 2000 = 0$ with no errors **[A1]**

(ii) $(x+50)(x-40) = 0$ **[M1]**

$x = 40, x = -50$ **[A1]**

(b) Uses the quadratic formula with correct substitution: $x = \dfrac{-(-7) \pm \sqrt{(-7)^2 - 4(3)(-5)}}{2(3)}$ **[M1]**

Simplifies to $\dfrac{7 \pm \sqrt{109}}{6}$ **[M1]**

$x = 2.91$ or $x = -0.57$ **[A1]**

[Total Marks: 8]

511 **(a)** Constructs initial time equation: $\dfrac{120}{x} + \dfrac{150}{x+10} = 4$ **[M1]**

Multiplies by common denominator $x(x+10)$ to reach $120(x+10) + 150x = 4x(x+10)$ **[M1]**

Expands brackets correctly: $120x + 1200 + 150x = 4x^2 + 40x$ **[M1]**

Rearranges to $4x^2 - 230x - 1200 = 0$ and divides by 2 to yield $2x^2 - 115x - 600 = 0$ with no errors seen **[A1]**

(b) Substitutes values into the quadratic formula $\dfrac{115 \pm \sqrt{(-115)^2 - 4(2)(-600)}}{4}$ **[M1]**

62.3 **[A1]**

(c) -1.84 **[B1]**

2.36 **[B1]**

[Total Marks: 8]

512 **(a)** $\dfrac{40}{x}$ **[B1]**

(b) Converts 45 minutes to hours: $\dfrac{45}{60}$ or $\dfrac{3}{4}$ (or 0.75) **[B1]**

Forms a correct fractional equation: $\dfrac{40}{x} - \dfrac{40}{x+20} = \dfrac{3}{4}$ **[M1]**

Finds a common denominator or multiplies through to clear fractions:
$40(x+20) - 40x = \dfrac{3}{4}x(x+20)$ **[M1]**

Expands correctly and simplifies to the given quadratic with no errors: $800 = \dfrac{3}{4}(x^2 + 20x)$

$3200 = 3\text{x}^2 + 60x$

$3\text{x}^2 + 60x - 3200 = 0$ [A1]

(c) Correct substitution into the quadratic formula: $x = \dfrac{-60 \pm \sqrt{60^2 - 4(3)(-3200)}}{2(3)}$ **[M1]**

Correct simplification of the discriminant: $\dfrac{-60 \pm \sqrt{42000}}{6}$ **[M1]**

24.16 and -44.16 (both correct) **[A1]**

[Total Marks: 8]

513 **(a)** M1 for multiplying by the common denominator to obtain $5(x-3)+2(x+2)=(x+2)(x-3)$

M1 for expanding the brackets correctly: $5x-15+2x+4=x^2-x-6$

M1 for rearranging into a correct three-term quadratic equation, e.g., $x^2-8x+5=0$

M1 for correct use of the quadratic formula, e.g., $\dfrac{-(-8)\pm\sqrt{(-8)^2-4(1)(5)}}{2(1)}$

7.32 and 0.68 **[A1]**

(b) M1 for multiplying by the denominator: $y(5x+1)=2x-3$

M1 for expanding the bracket and collecting x terms on one side: $5xy-2x=-y-3$ or $2x-5xy=y+3$

M1 for factorising x: $x(5y-2)=-y-3$ or $x(2-5y)=y+3$

$\dfrac{-y-3}{5y-2}$ or $\dfrac{y+3}{2-5y}$ **[A1]**

[Total Marks: 9]

514 **(a)** $x(x-6)=3(x+2)$ **[M1]**

$x^2-6x=3x+6$ **[M1]**

$x^2-9x-6=0$ **[A1]**

Uses the quadratic formula, e.g., $x=\dfrac{-(-9)\pm\sqrt{(-9)^2-4(1)(-6)}}{2(1)}$ **[M1]**

$\dfrac{9\pm\sqrt{105}}{2}$ **[A1]**

(b) $y(3x+1)=2x-5$ **[M1]**

$3xy+y=2x-5$ **[M1]**

Isolates terms containing x on one side, e.g., $3xy-2x=-y-5$ or $2x-3xy=y+5$ **[M1]**

$\dfrac{-y-5}{3y-2}$ or $\dfrac{y+5}{2-3y}$ **[A1]**

[Total Marks: 9]

515 **(a)** $3(2x+1)+2(x-1)=2(x-1)(2x+1)$ **[M1]**

$6x+3+2x-2=4x^2-2x-2$ **[M1]**

$4x^2-10x-3=0$ **[A1]**

$\dfrac{10\pm\sqrt{(-10)^2-4(4)(-3)}}{2(4)}$ **[M1]**

2.77, -0.27 **[A1]**

(b) $x^2+(x+2)^2=340$ (or equivalent setup for consecutive even numbers) **[M1]**

$2x^2+4x-336=0$ or $x^2+2x-168=0$ **[A1]**

$(x+14)(x-12)=0$ or correct application of quadratic formula **[M1]**

12 and 14 (both correct) **[A1]**

[Total Marks: 9]

516 **(a)** $\frac{100}{x} - \frac{100}{x+15} = \frac{1}{2}$ (or equivalent e.g., $\frac{100}{x} = \frac{100}{x+15} + 0.5$) **[M1]**

$200(x+15) - 200x = x(x+15)$ **[M1]**

$200x + 3000 - 200x = x^2 + 15x$ leading to $x^2 + 15x - 3000 = 0$ **[A1]**

(b) $\frac{-15 \pm \sqrt{15^2 - 4(1)(-3000)}}{2(1)}$ **[M2]**

Allow [M1] for correctly substituting values into the quadratic formula or evaluating the discriminant $\sqrt{12225}$

47.78 **[A1]**

-62.78 **[A1]**

(c) Speed of car $= 47.78 + 15 = 62.78$ km/h **[M1]**

Time $= \frac{100}{62.78} = 1.592...$ hours **[M1]**

1 hour 36 minutes **[A1]**

[Total Marks: 10]

517 **(a)** Multiplies by the denominator to obtain $y(x+4) = 3x - 2$ **[M1]**

Expands the bracket to obtain $xy + 4y = 3x - 2$ **[M1]**

Collects terms in x on one side, e.g., $xy - 3x = -4y - 2$ or $3x - xy = 4y + 2$ **[M1]**

Factorises x out and divides: $\frac{-4y-2}{y-3}$ or $\frac{4y+2}{3-y}$ **[A1]**

(b) Writes with common denominator or multiplies through to eliminate fractions:
$5(2x+1) - 3(x-2) = 2(x-2)(2x+1)$ **[M1]**

Correctly expands LHS: $10x + 5 - 3x + 6$ **[M1]**

Correctly expands RHS: $2(2x^2 - 3x - 2)$ or $4x^2 - 6x - 4$ **[M1]**

Rearranges to form a correct 3-term quadratic equation, e.g., $4x^2 - 13x - 15 = 0$ **[A1]**

Correctly substitutes into the quadratic formula: $\frac{-(-13) \pm \sqrt{(-13)^2 - 4(4)(-15)}}{2(4)}$ **[M1]**

$x = 4.15$ and $x = -0.90$ **[A1]**

[Total Marks: 10]

518 **(a)** $y(x+2) = 3x - 1$ **[M1]**

$yx + 2y = 3x - 1$ **[M1]**

$2y + 1 = 3x - yx$ or $yx - 3x = -2y - 1$ **[M1]**

$x = \frac{2y+1}{3-y}$ or $x = \frac{-2y-1}{y-3}$ **[A1]**

(b) Multiply throughout by $(x-1)(x+2)$: $2(x+2) + 3(x-1) = (x-1)(x+2)$ **[M1]**

Expand terms: $2x + 4 + 3x - 3 = x^2 + x - 2$ **[M1]**

Collect to form a quadratic equation: $x^2 - 4x - 3 = 0$ **[A1]**

Use the quadratic formula: $x = \dfrac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(-3)}}{2(1)}$ **[M1]**

$x = \dfrac{4 \pm \sqrt{28}}{2}$ **[M1]**

$x = 4.65$ and $x = -0.65$ **[A1]**

[Total Marks: 10]

519 **(a)** $\dfrac{150}{x}$ **[B1]**

(b) $\dfrac{150}{x - 15} - \dfrac{150}{x} = 0.5$ (or $\dfrac{30}{60}$ or $\dfrac{1}{2}$) **[M1]**

$150x - 150(x - 15) = 0.5x(x - 15)$ **[M1]**

$150x - 150x + 2250 = 0.5x^2 - 7.5x$ **[M1]**

$4500 = x^2 - 15x \implies x^2 - 15x - 4500 = 0$ with no errors seen **[A1]**

(c) $(x - 75)(x + 60) = 0$ or $\dfrac{-(-15) \pm \sqrt{(-15)^2 - 4(1)(-4500)}}{2(1)}$ **[M2]**

(or $M1$ for correct factors $(x + a)(x + b)$ where $ab = -4500$ or $a + b = -15$, or for correct substitution into quadratic formula)

75 (rejects negative solution -60) **[A1]**

(d) $P(q - 1) = 2q + 3$ **[M1]**

$Pq - P = 2q + 3$ **[M1]**

$Pq - 2q = P + 3$ **[M1]**

$\dfrac{P + 3}{P - 2}$ or $\dfrac{-P - 3}{2 - P}$ **[A1]**

[Total Marks: 12]

520 Rearranges to collect x terms on one side and constants on the other, e.g., $-8 < 4x$ or $-4x < 8$ **[M1]**

$x > -2$ or $-2 < x$ **[A1]**

[Total Marks: 2]

521 **(a)** $3x - 6 > 5x + 4$ **[M1]**

$x < -5$ **[A1]**

(b) -6 **[B1]**

[Total Marks: 3]

522 Expands brackets correctly: $12x - 4 - 5x \geq 2x + 16$ **[M1]**

Rearranges to collect x terms: $7x - 2x \geq 16 + 4$ or $5x \geq 20$ **[M1]**

$x \geq 4$ **[A1]**

[Total Marks: 3]

523 (a) $3x - 12 < 5x - 2$

[M1]

$-10 < 2x \implies x > -5$ (or $-5 < x$)

[A1]

(b) -4

[B1]

[Total Marks: 3]

524 (a) $6x - 3 > 4x + 7$ **[M1]**

$2x > 10$ leading to $x > 5$ **[A1]**

(b) Open circle placed at 5 with an arrow pointing to the right **[B1]**

(c) 6 **[B1]**

[Total Marks: 4]

525 (a) $-6 < 3x \leq 12$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) $-1, 0, 1, 2, 3, 4$ **[B2]**

(or [B1] for 5 correct with no incorrect, or all 6 correct with one incorrect)

[Total Marks: 4]

526 (a) $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) Open circle at -2 and closed circle at 4 **[B1]**

Thick line joining the two circles **[B1]**

[Total Marks: 4]

527 (a) $4x \leq 20$ **[M1]**

$x \leq 5$ **[A1]**

(b) $3x + 6 < 20$ or $x + 2 < 6.66\ldots$ **[M1]**

4 **[A1]**

[Total Marks: 4]

528 **(a)** $2x < 12$ **[M1]**

$x < 6$ **[A1]**

(b) Recognises open circle means strict inequality ($>$ or $<$) and arrow to right means greater than **[M1]**

$x > -2$ **[A1]**

[Total Marks: 4]

529 **(a)** M1 for a correct first step, e.g., adding 1 to all parts to get $-4 < 2x \leq 8$

M1 for a correct second step, e.g., dividing all parts by 2

$-2 < x \leq 4$ **[A1]**

(b) M1 for an open circle at -2 and a closed solid circle at 4

A1 for a continuous thick line drawn between the two circles

[Total Marks: 5]

530 **(a)** Expands brackets correctly: $3x - 6 \leq 5x + 8$ **[M1]**

Rearranges to isolate x, e.g. $-14 \leq 2x$ or $-2x \leq 14$ **[M1]**

$x \geq -7$ **[A1]**

(b) Places a closed, solid circle at -7 **[B1]**

Draws a line with an arrow pointing to the right from the circle **[B1]**

[Total Marks: 5]

531 **(a)** Adds 1 to all three parts: $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) Open circle at -2 and closed (filled) circle at 4 **[B1]**

Horizontal line segment connecting the circles at -2 and 4 **[B1]**

(c) -1 **[B1]**

[Total Marks: 5]

532 **(a)** $-9 > 2x$ or $-2x > 9$ **[M1]**

$x < -4.5$ **[A1]**

(b) $-4 < 2n \leq 4$ **[M1]**

$-2 < n \leq 2$ **[M1]**

$-1, 0, 1, 2$ **[A1]**

(c) $x > -1$ (accept any valid variable) **[B1]**

[Total Marks: 6]

533 **(a)** $-15 \leq 3x$ or $-3x \leq 15$ **[M1]**

$x \geq -5$ or $-5 \leq x$ **[A1]**

(b) $-7 < 2x \leq 8$ **[M1]**

$-3.5 < x \leq 4$ **[A1]**

(c) Largest $= 4$ **[B1]**

Smallest $= -3$ **[B1]**

[Total Marks: 6]

534 **(a)** $-3x < 9$ or $2 - 11 < 3x$ **[M1]**

$x > -3$ **[A1]**

(b) $8x - 4 \leq 5x + 17$ **[M1]**

$3x \leq 21$ **[M1]**

$x \leq 7$ **[A1]**

(c) $-3 < n \leq 7$ **[M1]**

$-2, -1, 0, 1, 2, 3, 4, 5, 6, 7$ **[A1]**

[Total Marks: 7]

535 **(a)** Subtracts $3x$ and 3 to reach $\pm 4x \leq \pm 8$ or similar **[M1]**

$x \geq -2$ **[A1]**

(b) Solid circle (or equivalent clear inclusive mark) placed at -2 **[B1]**

Thick line or arrow pointing to the right starting from -2 **[B1]**

(c) Adds 5 to all parts to get $3 < 3x \leq 9$ **[M1]**

Divides by 3 to get $1 < x \leq 3$ **[A1]**

$2, 3$ **[A1]**

[Total Marks: 7]

536 $y \geq 1$ **[B1]**

$y \leq x$ **[B1]**

$x + y \leq 6$ **[B1]**

[Total Marks: 3]

537 $y \leq 5x - 4$ or equivalent **[B1]**

$2y \geq x + 1$ or equivalent **[B1]**

$x + y \leq 8$ or equivalent **[B1]**

[Total Marks: 3]

538 $x \geq 2$ **[B1]**

$y > x$ **[B1]**

$x + y \leq 8$ (or $y \leq -x + 8$) **[B1]**

[Total Marks: 3]

539 $x \geq 1$

[B1]

$y \geq 2$

[B1]

$x + y < 6$ (or $y < -x + 6$)

[B1]

[Total Marks: 3]

540 (a) $-6 < 3x \leq 12$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) Open circle at -2 **[B1]**

Closed circle at 4 and a line joining the two circles **[B1]**

[Total Marks: 4]

541 (a) $-6 < 2x \leq 6$ (Allow M1 for one correct side e.g. $-6 < 2x$ or $2x \leq 6$) **[M2]**

$-3 < x \leq 3$ **[A1]**

(b) Open circle at -3 and closed circle at 3 **[B1]**

A single continuous line drawn connecting the two circles **[B1]**

[Total Marks: 5]

542 (a) (a) Adds 1 to all three parts: $-4 \leq 2x < 7$ **[M1]**

Divides all three parts by 2 **[M1]**

$-2 \leq x < 3.5$ or $-2 \leq x < \frac{7}{2}$ **[A1]**

(b) Award [B1] for at least three correct integers and no incorrect integers, or all correct integers with only one incorrect integer

$-2, -1, 0, 1, 2, 3$ **[B2]**

[Total Marks: 5]

543 **(a)** $x < 4$ **[B1]**

$y \geq -x + 3$ **[B1]**

$y \leq \frac{1}{2}x + 3$ (or equivalent, e.g. $2y \leq x + 6$) **[B1]**

(b) Substitutes $x = 3$ into the inequalities: $b \geq -3 + 3 = 0$ and $b \leq 0.5(3) + 3 = 4.5$ **[M1]**

Deduces the range $0 \leq b \leq 4.5$ **[A1]**

$0, 1, 2, 3, 4$ **[A1]**

[Total Marks: 6]

544 **(a)** $y \geq 0$ **[B1]**

Equation of sloping boundary identified as $x + y = 6$ or $y = -x + 6$ **[B1]**

$x + y < 6$ or $y < -x + 6$ (correct strict inequality symbol) **[B2]**

(b) Identifies valid integer coordinate $(1, 4)$ (or tests boundary lattice points such as $(1, 4), (2, 3), (3, 2)$) **[M1]**

14 **[A1]**

[Total Marks: 6]

545 $x \geq 1$ **[B1]**

$y \geq 2$ **[B1]**

$x + y < 6$ or $y < -x + 6$ **[B1]**

[Total Marks: 3]

546 **(a)** $y \geq 1$ **[B1]**

$y \leq x$ **[B1]**

$x + y \leq 6$ or $y \leq -x + 6$ **[B1]**

(b) Tests a valid integer coordinate close to the boundary, e.g., $(5, 1)$, $(4, 2)$, or $(3, 3)$ **[M1]**

17 (from $x = 5, y = 1$) **[A1]**

[Total Marks: 5]

547 Identifies horizontal line $y = 2$ **[M1]**

States $y \geq 2$ (accept $y > 2$) **[A1]**

Identifies dashed line passing through $(0, 6)$ and $(4, 2)$ **[M1]**

Finds equation $y = -x + 6$ and states $y < -x + 6$ **[A1]**

Identifies solid line passing through $(0, 2)$ and $(1, 5)$ **[M1]**

Finds equation $y = 3x + 2$ and states $y \leq 3x + 2$ **[A1]**

[Total Marks: 6]

548 **(a)** Multiplies by 15 to clear denominators: $3(3x - 2) - 5(2x - 5) \leq 30$ **[M1]**

$9x - 6 - 10x + 25 \leq 30$ **[M1]**

$-x + 19 \leq 30$ **[M1]**

$x \geq -11$ **[A1]**

(b) Identifies that y can only take integer values of 2 or 3 (since $2y > x + y > 2 \implies y > 1$) **[M1]**

Finds correct points for one of the valid y values: e.g. $(1, 2)$ for $y = 2$ OR $(0, 3), (1, 3), (2, 3)$ for $y = 3$ **[B1]**

States all four correct points: $(1, 2), (0, 3), (1, 3), (2, 3)$ with no extras **[A1]**

[Total Marks: 7]

549 **(a)** Rearranging terms to get $3x \leq 15$ **[M1]**

$x \leq 5$ **[A1]**

(b) $1, 2, 3, 4, 5$ **[B2]**

If 0 scored, award 1 mark for $0, 1, 2, 3, 4, 5$ OR for omitting exactly one correct value **[B1]**

[Total Marks: 4]

550 $y \geq 2$ **[B1]**

$y \leq 2x$ **[B1]**

$x + y \leq 6$ **[B1]**

[Total Marks: 3]

551 $x \geq 1$ **[B1]**

$y \leq 2x$ **[B1]**

$x + y \leq 6$ **[B1]**

(Note: Accept strict inequalities $<, >$)

[Total Marks: 3]

552 **(a)** Subtracting 2 from all parts: $-9 < 3x \leq 9$ **[M1]**

Dividing by 3: $-3 < x$ and $x \leq 3$ **[M1]**

$-3 < x \leq 3$ **[A1]**

(b) 3 **[B1]**

[Total Marks: 4]

553 **(a)** Adding 1 to all parts: $-6 \leq 3x < 12$ **[M2, M1 for one side correct]**

Dividing by 3: $-2 \leq x < 4$ **[A1]**

(b) Identifies at least 3 correct integers or includes 4 but excludes -2 **[M1]**

$-2, -1, 0, 1, 2, 3$ **[A1]**

[Total Marks: 5]

554 **(a)** $y \geq 1$ **[B1]**

$y \leq x + 1$ **[B1]**

$y < 5 - x$ (or $x + y < 5$) **[B1]**

All three inequality signs correct (with strict inequality for the dashed line) **[B1]**

(b) Identifies $(1, 2)$ as a valid integer coordinate point inside R **[M1]**

$(1, 2)$ **[A1]**

[Total Marks: 6]

555 **(a)** Subtracts 1 from all parts: $-6 < 2x \leq 10$ **[M1]**

Divides by 2 across all parts **[M1]**

$-3 < x \leq 5$ **[A1]**

(b) Identifies at least two correct integers from the set **[M1]**

$-2, -1, 0$ **[A1]**

(c) Finds the correct roots of the quadratic $2x^2 - 7x + 3 = 0$ as 0.5 and 3 **[B1]**

$x < 0.5$ or $x > 3$ **[B1]**

[Total Marks: 7]

556 **(a)** $y \geq 1$ **[B1]**

$y \leq 2x - 2$ **[B2]**

(or $M1$ for finding the boundary equation $y = 2x - 2$)

$x + y \leq 7$ (or $y \leq -x + 7$) **[B2]**

(or $M1$ for finding the boundary equation $x + y = 7$)

(b) Identifies a valid integer coordinate inside or on the boundary of R that maximizes the expression, e.g., $(6, 1)$, $(5, 2)$, or $(4, 3)$ **[M1]**

13 (from $2(6) + 1 = 13$) **[A1]**

[Total Marks: 7]

557 **(a)** $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) $-1, 0, 1, 2, 3, 4$ **[B2]**

1 omission or 1 extra integer **[B1]**

(c) **(i)** $2(3x + 2) + 2x < 40$ or $(3x + 2) + x < 20$ **[M1]**

$8x + 4 < 40$

8x ¡ 36

x ¡ $4.5 with outerrors [A1]$

(ii) Identifies maximum integer $x = 4$ **[M1]**

Length = 14, Width = 4, Area = 56 **[A1]**

[Total Marks: 8]

558 **(a)** $y \geq 0.5x + 0.5$ or $2y \geq x + 1$ (Accept equivalent solid boundary) **[B1]**

$y \leq 6x - 5$ (Accept equivalent solid boundary) **[B1]**

$y < -\frac{4}{3}x + \frac{29}{3}$ or $4x + 3y < 29$ (Must be strict inequality for dashed line) **[B1]**

(b) Identifies a valid integer point near the boundary that maximizes the sum, e.g., $(2, 6), (3, 5)$, or $(4, 4)$. Note: $(2, 7)$ is invalid as it lies on the dashed boundary **[M1]**

8 **[A1]**

[Total Marks: 5]

559 **(a)** $y \geq 2$ **[B1]**

$y \leq x$ **[B1]**

$x + y < 8$ **[B1]**

(b) Identifies $(4, 3)$ or $(5, 2)$ as integer coordinates within the region R. Note that $(4, 4)$ is excluded due to the strict inequality $x + y < 8$. **[M1]**

Evaluates $2x + 3y$ and correctly identifies $(4, 3)$ as yielding the maximum value (17). **[A1]**

[Total Marks: 5]

560 **(a)** $x \geq 15$ **[B1]**

$x + y \leq 40$ **[B1]**

$y \geq \frac{1}{2}x$ or $2y \geq x$ **[B1]**

(b) Identifies the objective function for profit: $P = 400x + 150y$ **[M1]**

Tests the valid integer vertex $(26, 14)$ or evaluates the profit equation showing profit is maximised at the highest possible integer value for x **[M1]**

12500 **[A1]**

[Total Marks: 6]

561 **(a)** $x \geq 2$ **[B1]**

$y \geq 1$ **[B1]**

$x + y < 7$ or $y < -x + 7$ **[B1]**

(b) Finding critical values -0.45 and 4.45 (allow ≈ -0.449 and ≈ 4.449) **[B2]**

(Allow B1 for setting $x^2 - 4x - 2 = 0$ and finding one correct root or correct quadratic formula setup)

$-0.45 \leq x \leq 4.45$ **[B1]**

[Total Marks: 6]

562 **(a)** Draw correct solid line $y = 2$

[B1]

Draw correct solid line $y = x + 2$

[B1]

Draw correct solid line $2x + y = 8$

[B1]

Correct shading of unwanted regions, leaving the triangle with vertices $(0, 2)$, $(3, 2)$, and $(2, 4)$ unshaded and clearly representing R.

[B1]

(b) Identify integer coordinates $(2, 4)$ or $(3, 2)$ as yielding the maximum value

[M1]

14

[A1]

[Total Marks: 6]

563 **(a)** $2[(2x + 5) + (x - 1)] < 50$ or $6x + 8 < 50$ **[M1]**

$6x < 42 \implies x < 7$ **[A1]**

(b) $(2x + 5)(x - 1) \geq 42$ **[M1]**

$2x^2 + 3x - 5 \geq 42 \implies 2x^2 + 3x - 47 \geq 0$ **[A1]**

(c) Finds positive root of $2x^2 + 3x - 47 = 0$ as 4.155... (4.16) **[M1]**

Solves quadratic inequality for $x > 1$: $x \geq 4.16$ **[A1]**

$4.16 \leq x < 7$ **[A1]**

[Total Marks: 7]

564 **(a)** $y \geq 2$ **[B1]**

$y \leq x$ **[B1]**

$x + y \leq 8$ **[B1]**

(b) Identifies at least two valid integer coordinate pairs within the region R (e.g., $(6, 2)$, $(5, 3)$, $(4, 4)$) **[M1]**

Evaluates $2x + 3y$ for at least one critical integer point, e.g., $2(4) + 3(4) = 20$ or $2(5) + 3(3) = 19$ **[M1]**

20 **[A1]**

(c) Identifies the correct total number of integer points in R is 9, or identifies the 3 correct points where $x = y$: $(2, 2), (3, 3), (4, 4)$ **[M1]**

$\frac{3}{9}$ or $\frac{1}{3}$ **[A1]**

[Total Marks: 8]

565 (a) $-4 < 2x \leq 8$ **[M1]**

$-2 < x \leq 4$ **[A1]**

(b) Multiplies throughout by 12 (or uses common denominator): $4(2x - 5) - 3(x - 3) > 12$ **[M1]**

Expands and simplifies: $8x - 20 - 3x + 9 > 12 \implies 5x - 11 > 12 \implies 5x > 23$ **[M1]**

5 **[A1]**

(c) Constructs correct inequality: $3.20 + 1.50k > 5.00 + 1.10k$ **[M1]**

Solves for k: $0.40k > 1.80 \implies k > 4.5$ **[M1]**

5 **[A1]**

[Total Marks: 8]

566 (a) $x + y \leq 500$ **[B1]**

$y \geq 150$ **[B1]**

$5x + 10y \geq 3000$ (or simplified to $x + 2y \geq 600$) **[B1]**

(b) Line drawn correctly for $x + y = 500$ (passing through $(0, 500)$ and $(500, 0)$) **[B1]**

Line drawn correctly for $y = 150$ (horizontal line) **[B1]**

Line drawn correctly for $x + 2y = 600$ (passing through $(0, 300)$ and $(600, 0)$) **[B1]**

Correct region identified and all unwanted regions properly shaded (leaving quadrilateral with vertices $(0, 300), (0, 500), (350, 150), (300, 150)$ unshaded) **[B1]**

(c) Identifies the vertex $(350, 150)$ as giving the maximum x value **[M1]**

Children's tickets = 350, Adult tickets = 150 **[A1]**

[Total Marks: 9]

567 (a) $x + y \leq 60$ **[B1]**

$x \geq 10$ **[B1]**

$y \geq 0.5x$ (or $2y \geq x$ or $y \geq \frac{1}{2}x$) **[B1]**

(b) Draws the line $x + y = 60$ correctly (from $(0, 60)$ to $(60, 0)$) **[B1]**

Draws the line $x = 10$ correctly (vertical line at 10) **[B1]**

Draws the line $y = 0.5x$ correctly (from $(0, 0)$ passing through $(20, 10)$, $(40, 20)$, etc.) **[B1]**

Shades the unwanted regions correctly, leaving the correct triangular region R unshaded **[B1]**

(c) Identifies $(40, 20)$ as the vertex for maximum profit (or tests multiple vertices) **[M1]**

2100 **[A1]**

[Total Marks: 9]

568 (a) $x + y \leq 60$ **[B1]**

$x + 2y \leq 80$ **[B1]**

$y \geq 10$ **[B1]**

(b) B1 for correctly drawing the line $x + y = 60$

B1 for correctly drawing the line $x + 2y = 80$

B1 for correctly drawing the line $y = 10$

B1 for correctly shading the unwanted regions (above the two diagonal lines, and below the horizontal line) to leave the correct valid region unshaded

(c) M1 for a valid method to find the maximum profit, e.g., testing the vertex $(40, 20)$ in the objective function $5x + 8y$

360 **[A1]**

[Total Marks: 9]

569 (a) Forms left inequality: $3(2x - 3) < 4(x + 5)$ or equivalent **[M1]**

Simplifies to $6x - 9 < 4x + 20$

x ¡ $\frac{29}{2}$ (or 14.5) [A1]

Forms right inequality: $2(x + 5) \leq 3(3x - 1)$ or equivalent **[M1]**

Simplifies to $2x + 10 \leq 9x - 3$

x $\geq \frac{13}{7}$ [A1]

$\frac{13}{7} \leq x < \frac{29}{2}$ (accept $1.86 \leq x < 14.5$ or equivalent double inequality) **[A1]**

(b) (i) $25 < 12 + 0.07x$ (or $12 + 0.07x > 25$) **[B1]**

(ii) $13 < 0.07x$ **[M1]**

$x > 185.71\ldots$ **[A1]**

186 **[A1]**

[Total Marks: 9]

570 **(a)** $x + y \leq 60$ **[B1]**

$y \geq 10$ **[B1]**

$x \geq 15$ **[B1]**

$y \leq 2x$ **[B1]**

(b) Line $x + y = 60$ drawn correctly **[B1]**

Line $y = 10$ drawn correctly **[B1]**

Line $x = 15$ drawn correctly **[B1]**

Line $y = 2x$ drawn correctly **[B1]**

Correct region R indicated with all four unwanted regions shaded **[B1]**

(c) Identifies $(20, 40)$ as the vertex for maximum profit **[M1]**

Mountain bicycles = 20 and Road bicycles = 40 **[A1]**

Maximum profit = 2600 **[A1]**

[Total Marks: 12]

571 **(a)** 27 **[B1]**

(b) Identifies the common difference of 4, e.g. writes $4n$ **[M1]**

$4n + 3$ **[A1]**

[Total Marks: 3]

572 **(a)** 29 **[B1]**

(b) M1 for $6n + k$ where k is an integer

$6n - 1$ **[A1]**

(c) States that 150 is even but all terms in the sequence are odd OR solves $6n - 1 = 150$ to get $n = 25.1...$ and states n must be an integer **[B1]**

[Total Marks: 4]

573 **(a)** 33 **[B1]**

(b) $7n + k$ for any integer k **[M1]**

$7n - 2$ **[A1]**

(c) $7n - 2 = 150$

$7\text{n} = 152 which does not give an integer value for \text{n}. [B1]$

[Total Marks: 4]

574 **(a)** 29 **[B1]**

(b) $6n + c$ or finding common difference of 6 **[M1]**

$6n - 1$ **[A1]**

(c) Sets $6n - 1 = 150 \implies 6n = 151$ and states that 151 is not divisible by 6 (or n is not an integer) **[B1]**

[Total Marks: 4]

575 **(a)** Common difference is 4 **[M1]**

$4n + 1$ **[A1]**

(b) Identifies common ratio is 3 **[M1]**

$2 \times 3^{n-1}$ **[A1]**

[Total Marks: 4]

576 **(a)** $23, 27$ **[B1]**

(b) $4n + k$ (for any k) **[M1]**

$4n + 3$ **[A1]**

(c) Equates $4n + 3 = 102$ and attempts to solve for n **[M1]**

$n = 24.75$, which is not a whole number/integer **[A1]**

[Total Marks: 5]

577 **(a)** **(i)** 21 and 25 **[B1]**

(ii) Identifies common difference of 4 leading to $4n$ **[M1]**

$4n + 1$ **[A1]**

(b) Identifies common ratio of 3 **[M1]**

$2 \times 3^{n-1}$ **[A1]**

[Total Marks: 5]

578 **(a)** 17 **[B1]**

(b) $4n + c$ where c is a constant **[M1]**

$4n + 1$ **[A1]**

(c) $4n + 1 = 141$ or $141 - 1 = 140$ **[M1]**

35 **[A1]**

[Total Marks: 5]

579 **(a)** Column 4 correctly completed: $4, 4, 8$ **[B1]**

Column 5 correctly completed: $5, 5, 10$ **[B1]**

(b) $2n$ **[B1]**

(c) $2k = 94$ **[M1]**

47 **[A1]**

[Total Marks: 5]

580 **(a)** Identifying the second differences are constant 4 **[M1]**

Deducing the n^2 coefficient is 2, or setting up $2n^2 + bn + c$ **[M1]**

$2n^2 - n - 2$ **[A1]**

(b) $2n^2 - n - 2 = 433$ **[M1]**

$2n^2 - n - 435 = 0$ leading to correct factorisation $(2n + 29)(n - 15) = 0$ or correct substitution into the quadratic formula **[M1]**

15 (must reject negative solution if given) **[A1]**

[Total Marks: 6]

581 **(a)** 80 **[B1]**

(b) Second difference = 4 seen, or coefficient of n^2 is 2 **[M1]**

Subtracts $2n^2$ from sequence to obtain linear sequence $3, 4, 5, 6, 7$ **[M1]**

$2n^2 + n + 2$ **[A1]**

(c) $U_6 = 3 \times 2^5 = 96$ **[B1]**

$80 - 96 = -16$ **[B1]**

[Total Marks: 6]

582 **(a)** First differences: $10, 20, 34, 52$ **[M1]**

Second differences: $10, 14, 18$
Third differences: $4, 4$
Since the third difference is a non-zero constant, the sequence is cubic. **[A1]**

(b) Uses $6a = 4 \implies a = \dfrac{2}{3}$ **[M1]**

Uses $12a + 2b = 10 \implies 12(\dfrac{2}{3}) + 2b = 10 \implies 8 + 2b = 10 \implies b = 1$ **[M1]**

Uses $7a + 3b + c = 10 \implies 7(\dfrac{2}{3}) + 3(1) + c = 10 \implies c = 10 - 3 - \dfrac{14}{3} = \dfrac{7}{3}$ **[M1]**

Uses $a + b + c + d = 4 \implies \dfrac{2}{3} + 1 + \dfrac{7}{3} + d = 4 \implies 4 + d = 4 \implies d = 0$ **[M1]**

$\dfrac{2}{3}n^3 + n^2 + \dfrac{7}{3}n$ **[A1]**

[Total Marks: 7]

583 Finds the first differences $6, 8, 10, 12$ and a constant second difference of 2 **[M1]**

Deduces that n^2 is part of the nth term expression **[M1]**

$n^2 + 3n - 1$ **[A1]**

[Total Marks: 3]

584 (a) 79 **[B1]**

(b) Calculates second differences to be 4, or equates $2a = 4$ **[M1]**

Deduces that the sequence contains $2n^2$ **[M1]**

$2n^2 + n + 1$ **[A1]**

[Total Marks: 4]

585 (a) $a + b = 5$ **[B1]**

$4a + 2b = 14$ **[B1]**

(b) M1 for a correct method to eliminate one variable, e.g. multiplying the first equation by 2 to give $2a + 2b = 10$ and subtracting from the second equation

$2a = 4$ leading to $a = 2$ **[A1]**

$b = 3$ **[A1]**

[Total Marks: 5]

586 (a) One correct term evaluated **[M1]**

$3, 3, 5$ **[A1]**

(b) Second differences are 4 **[M1]**

Identifies $2n^2$ as part of the expression **[M1]**

$2n^2 + n - 1$ **[A1]**

[Total Marks: 5]

587 (a) Finds first differences of $7, 13, 19$ and constant second differences of 6 **[M1]**

Deduces the quadratic term is $3n^2$ **[M1]**

$3n^2 - 2n + 1$ **[A1]**

(b) 35 **[B1]**

(c) $2^n + 3 = 131$ leading to $2^n = 128$ **[M1]**

7 **[A1]**

[Total Marks: 6]

588 (a) Finds first differences: $3, 7, 11$ and second differences: $4, 4$ **[M1]**

Deduces the term involves $2n^2$ **[M1]**

$2n^2 - 3$ **[A1]**

(b) $3^n - 2 \approx 1000 \implies 3^n \approx 1002$ **[M1]**

Uses logarithms or trial and improvement: $n = 6 \implies 3^6 - 2 = 729 - 2 = 727$; $n = 7 \implies 3^7 - 2 = 2187 - 2 = 2185$ **[M1]**

727 **[A1]**

[Total Marks: 6]

589 **(a)** 77 **[B1]**

(b) Finds first differences $(7, 11, 15, 19)$ and constant second difference of 4 **[M1]**

Deduces the coefficient of n^2 is 2 **[A1]**

Subtracts $2n^2$ from the sequence terms to identify the linear sequence $n - 1$ **[M1]**

$2n^2 + n - 1$ **[A1]**

(c) Sets up the equation $2k^2 + k - 1 = 464$ **[M1]**

Rearranges to form a quadratic equal to zero: $2k^2 + k - 465 = 0$ **[M1]**

15 **[A1]**

[Total Marks: 7]

590 **(a)** Recognises the relationship to cube numbers $(1, 8, 27, 64, 125)$ **[M1]**

$n^3 + 1$ **[A1]**

(b) Identifies second difference is 2, or relates to square numbers **[M1]**

$n^2 + 2n$ or $(n + 1)^2 - 1$ **[A1]**

(c) $\dfrac{n^3 + 1}{n^2 + 2n}$ **[B1]**

(d) Substitutes $n = 12$ into their expression for sequence R, yielding $\dfrac{1728 + 1}{144 + 24}$ or $\dfrac{1729}{168}$ **[M1]**

Simplifies the fraction correctly to $\dfrac{247}{24}$ **[A1]**

[Total Marks: 7]

591 **(a)** 217 **[B1]**

(b) M1 for finding the first differences $(7, 19, 37, 61)$ and second differences $(12, 18, 24)$

M1 for finding third differences are constant at 6, implying an n^3 term (or recognizing the sequence is $1^3 + 1, 2^3 + 1, 3^3 + 1, ...$)

$n^3 + 1$ **[A1]**

(c) M1 for setting up the equation $k^3 + 1 = 8001$

M1 for $k^3 = 8000$

$k = 20$ **[A1]**

[Total Marks: 7]

592 **(a)** Identifies second difference of 2 or relates to square numbers **[M1]**

$n^2 + 1$ **[A1]**

(b) n^3 **[B1]**

(c) Recognises the relationship that Sequence C is Sequence B minus Sequence A **[M1]**

$n^3 - n^2 - 1$ **[A1]**

(d) Sets up the correct equation: $n^2 + 1 = 10001$ or $n^2 = 10000$ **[M1]**

100 **[A1]**

[Total Marks: 7]

593 **(a)** Finds third differences are 6 (first diffs: $9, 25, 47, 75$; second diffs: $16, 22, 28$; third diffs: 6) **[M1]**

Identifies n^3 as the cubic term **[B1]**

Subtracts n^3 from the sequence to get a quadratic sequence $(-2, 0, 6, 16, 30)$ and finds its second differences are 4 to give $2n^2$ **[M1]**

$n^3 + 2n^2 - 4n$ **[A1]**

(b) $5 \times 3^{k-1} = 3645$

$3^{k-1} = 729$ [M1]

7 **[A1]**

(c) $\frac{2}{3}$ or 0.667 **[B1]**

[Total Marks: 7]

594 **(a)** Finds first differences $(8, 20, 38, 62)$, second differences $(12, 18, 24)$, and constant third difference $= 6$ **[M1]**

Deduces coefficient of n^3 is $\frac{6}{6} = 1$ **[M1]**

Subtracts n^3 from A_n to obtain $1, 2, 3, 4, 5$ **[M1]**

$n^3 + n$ **[A1]**

(b) $B_n = [(n+1)^3 + (n+1)] - (n^3 + n)$ **[M1]**

Expands $(n^3 + 3n^2 + 3n + 1 + n + 1) - n^3 - n = 3n^2 + 3n + 2$ correctly **[A1]**

(c) $3n^2 + 3n - 270 = 0$ or $n^2 + n - 90 = 0$ **[M1]**

9 (rejecting $n = -10$) **[A1]**

[Total Marks: 8]

595 **(a)** Finds first differences $14, 38, 74$ and second differences $24, 36$ **[M1]**

Finds third difference of 12 leading to a cubic term of $2n^3$ **[M1]**

Evaluates the sequence of residuals $T_n - 2n^3$ to find a constant difference of -1 **[M1]**

$2n^3 - 1$ **[A1]**

(b) **(i)** $V_1 = 1$, $V_2 = \frac{5}{4}$, $V_3 = \frac{5}{4}$ **[B2]**

Award B1 if any two are correct **[B1]**

(ii) Substitutes $n = 5$ to find $V_5 = \frac{25+1}{32} = \frac{13}{16}$ **[B1]**

Substitutes $n = 4$ to find $V_4 = \frac{16+1}{16} = \frac{17}{16}$ **[B1]**

Evaluates $\frac{13}{16} - \frac{17}{16} = -\frac{1}{4}$ (accept -0.25) **[A1]**

[Total Marks: 9]

596 **(a)** Correct first differences $1, 9, 23, 43$ and second differences $8, 14, 20$ and third differences 6 **[M1]**

Deducing the cubic term coefficient is $a = 6 \div 6 = 1$ **[B1]**

Setting up equations to solve for other coefficients, e.g. $12a + 2b = 8$ **[M1]**

$n^3 - 2n^2 + 5$ **[A1]**

(b) $2a + b = 10$ **[B1]**

$8a + b = 34$ **[B1]**

Correct method to solve simultaneous equations, e.g. subtracting to get $6a = 24$ **[M1]**

$a = 4, b = 2$ **[A1]**

(c) Substituting $n = 12$ into their $4 \times 2^n + 2$ **[M1]**

16386 **[A1]**

[Total Marks: 10]

597 **(a)** $4n + 1$ **[B2]**

If $4n + k$ seen, award **[M1]**

(b) **(i)** 182 **[B1]**

(ii) Identifies third differences are constant (6), so term contains n^3 **[M1]**

Subtracts n^3 terms $(1, 8, 27, 64, 125)$ from the sequence **[M1]**

Remainder sequence is $1, -2, -7, -14, -23$ **[M1]**

Deduces the remainder is $-n^2 + 2$ **[M1]**

$n^3 - n^2 + 2$ **[A1]**

(c) **(i)** $\frac{25}{182}$ **[B1]**

(ii) Recognises the pattern is Sequence A divided by Sequence B **[M1]**

$\dfrac{4n+1}{n^3-n^2+2}$ **[A1]**

[Total Marks: 10]

598 **(a)** $y = kx^2$ or $36 = k(3^2)$ **[M1]**

$4x^2$ **[A1]**

(b) 100 **[B1]**

[Total Marks: 3]

599 $y = k(x+2)$ **[M1]**

$20 = k(3+2)$ leading to $k = 4$ **[M1]**

36 **[A1]**

[Total Marks: 3]

600 **(a)** M1 for $y = kx^2$ or $36 = k(3)^2$

$4x^2$ **[A1]**

(b) M1 for $64 = 4x^2$ leading to $x^2 = 16$

4 and -4 (or ± 4) **[A1]**

[Total Marks: 4]

601 **(a)** $y = k(x+2)^2$ and substitutes $x = 3, y = 75$ **[M1]**

$y = 3(x+2)^2$ **[A1]**

(b) $y = 3(6+2)^2$ **[M1]**

192 **[A1]**

[Total Marks: 4]

602 **(a)** Sets up equation $y = kx$ and substitutes $x = 5, y = 15$ to get $15 = 5k$ **[M1]**

$y = 3x$ **[A1]**

(b) Substitutes $x = 12$ into their equation from part (a), e.g. $y = 3 \times 12$ **[M1]**

36 **[A1]**

[Total Marks: 4]

603 **(a)** Substitutes into the proportion equation to find the constant, e.g. $135 = k(1+2)^3$ **[M1]**

$5(x+2)^3$ **[A1]**

(b) 40 **[B1]**

(c) Sets up the equation $320 = 5(x+2)^3$ and simplifies to $(x+2)^3 = 64$ **[M1]**

2 **[A1]**

[Total Marks: 5]

604 **(a)** $y = kx^2$ leading to $36 = k(3)^2$ **[M1]**

$4x^2$ **[A1]**

(b) 100 **[B1]**

(c) $196 = 4x^2$ leading to $x^2 = 49$ **[M1]**

7 and -7 **[A1]**

[Total Marks: 5]

605 **(a)** $y = kx^3$ **[M1]**

$24 = k(2)^3 \implies k = 3$ **[M1]**

$3x^3$ **[A1]**

(b) $y = 3(5)^3 = 3(125)$ **[M1]**

375 **[A1]**

(c) $81 = 3x^3$ **[M1]**

$x^3 = 27$ **[M1]**

3 **[A1]**

[Total Marks: 5]

606 **(a)** Recognises inverse square proportion, e.g. tests p^2q **[M1]**

$2^2 \times 100 = 400$, $4^2 \times 25 = 400 \implies k = 400$ **[M1]**

$\dfrac{400}{p^2}$ **[A1]**

(b) $q = \dfrac{400}{5^2} = \dfrac{400}{25}$ **[M1]**

16 **[A1]**

(c) $1 = \dfrac{400}{p^2}$ **[M1]**

$p^2 = 400$ **[M1]**

20 (Accept ± 20) **[A1]**

[Total Marks: 5]

607 **(a)** $R = \dfrac{kL}{d^2}$ **[M1]**

$9 = \dfrac{12k}{2^2}$ giving $k = 3$ **[M1]**

$R = \dfrac{3L}{d^2}$ **[A1]**

(b) $7.5 = \dfrac{3 \times 40}{d^2}$ or $d^2 = 16$ **[M1]**

4 **[A1]**

(c) New resistance $R_{\text{new}} = \dfrac{k(1.25L)}{(0.5d)^2} = \dfrac{1.25}{0.25}R = 5R$ **[M1]**

400 **[A1]**

[Total Marks: 7]

608 **(a)** Sets up $P = k_1\sqrt{Q}$ and substitutes
$Q = 9, P = 12 \implies 12 = k_1\sqrt{9} \implies 12 = 3k_1 \implies k_1 = 4$, so $P = 4\sqrt{Q}$ **[M1]**

Sets up $Q = \dfrac{k_2}{R^3}$ and substitutes $R = 2, Q = 2 \implies 2 = \dfrac{k_2}{2^3} \implies 2 = \dfrac{k_2}{8} \implies k_2 = 16$, so $Q = \dfrac{16}{R^3}$ **[M1]**

Substitutes Q into the P equation: $P = 4\sqrt{\dfrac{16}{R^3}}$ **[M1]**

Simplifies the square root: $P = 4 \times \dfrac{4}{\sqrt{R^3}}$ **[M1]**

$P = \dfrac{16}{R^{1.5}}$ or $P = 16R^{-1.5}$ or $P = \dfrac{16}{R\sqrt{R}}$ **[A1]**

(b) Substitutes $R = 4$ into their formula: $P = \dfrac{16}{4\sqrt{4}}$ or $P = \dfrac{16}{8}$ **[M1]**

2 **[A1]**

[Total Marks: 7]

609 **(a)** M1 for setting up $A = kB^2$ and finding $k = 5$ (i.e., $A = 5B^2$)

M1 for setting up $B = \dfrac{m}{C+2}$ and finding $m = 12$ (i.e., $B = \dfrac{12}{C+2}$)

M1 for substituting $C = 4$ into their equation for B to get $B = 2$

20 **[A1]**

(b) M1 for substituting their expression for B into their equation for A: $A = 5\left(\dfrac{12}{C+2}\right)^2$

M1 for expanding or rearranging to isolate $(C+2)^2$: $(C+2)^2 = \dfrac{720}{A}$

M1 for taking the positive square root: $C + 2 = \sqrt{\dfrac{720}{A}}$

$\sqrt{\dfrac{720}{A}} - 2$ (or equivalent valid form e.g. $\dfrac{12\sqrt{5}}{\sqrt{A}} - 2$) **[A1]**

[Total Marks: 8]

610 **(a)** Checks constant product $q\sqrt{p} = 18 \times 2 = 36$, $12 \times 3 = 36$, $7.2 \times 5 = 36$ **[M1]**

$q \propto \dfrac{1}{\sqrt{p}}$ or $p \propto \dfrac{1}{q^2}$ **[A1]**

(b) $q = \dfrac{k}{\sqrt{p}}$ with $k = 18\sqrt{4}$ **[M1]**

$q = \dfrac{36}{\sqrt{p}}$ **[A1]**

(c) $2.4 = \dfrac{36}{\sqrt{p}}$ or $\sqrt{p} = 15$ **[M1]**

225 **[A1]**

[Total Marks: 6]

611 **(a)** Sets up equation $y = \dfrac{k}{x^2}$ **[M1]**

Substitutes $x = 4, y = 3$ to get $3 = \dfrac{k}{4^2} \implies k = 3 \times 16 = 48$ **[M1]**

$y = \dfrac{48}{x^2}$ **[A1]**

(b) Sets $0.75 = \dfrac{48}{x^2}$ **[M1]**

Rearranges to $x^2 = \dfrac{48}{0.75} = 64$ **[M1]**

$x = 8, x = -8$ (both required for final mark) **[A1]**

[Total Marks: 6]

612 **(a)** $P = \dfrac{k}{Q^2}$ **[M1]**

$1.5 = \dfrac{k}{4^2}$ **[M1]**

$\dfrac{24}{Q^2}$ **[A1]**

(b) 0.24 or $\dfrac{6}{25}$ **[B1]**

(c) $24 = \dfrac{24}{Q^2}$ or $Q^2 = 1$ **[M1]**

1 **[A1]**

[Total Marks: 6]

613 **(a)** **(i)** $P = \dfrac{k}{\sqrt{Q}}$ leading to $12 = \dfrac{k}{\sqrt{9}}$ **[M1]**

$k = 36$ **[B1]**

3 **[A1]**

(ii) $1.5 = \dfrac{36}{\sqrt{Q}}$ leading to $\sqrt{Q} = 24$ **[M1]**

576 **[A1]**

(b) Identifying that the new value of P involves dividing by $\sqrt{4}$ or 2 **[M1]**

50 **[A1]**

[Total Marks: 7]

614 **(a)** $M = \dfrac{k}{\sqrt{N}}$ or $15 = \dfrac{k}{\sqrt{16}}$ **[M1]**

$\dfrac{60}{\sqrt{N}}$ **[A1]**

(b) $\dfrac{60}{\sqrt{100}}$ or $\dfrac{60}{10}$ **[M1]**

6 **[A1]**

(c) $1.2 = \dfrac{60}{\sqrt{N}}$ **[M1]**

$\sqrt{N} = \dfrac{60}{1.2}$ or $\sqrt{N} = 50$ **[M1]**

2500 **[A1]**

[Total Marks: 7]

615 **(a)** M1 for recognising that y is inversely proportional to x^2 (e.g. testing $x^2y = c$)

M1 for a correct substitution to find k, e.g., $k = 4^2 \times 9 = 144$

$\dfrac{144}{x^2}$ **[A1]**

(b) M1 for $2.25 = \dfrac{144}{x^2}$ or $x^2 = 64$

8 **[A1]**

(c) M1 for substituting $3x$ into their equation: $y_{new} = \dfrac{144}{(3x)^2} = \dfrac{144}{9x^2}$

$\dfrac{1}{9}$ (or 0.111...) **[A1]**

[Total Marks: 7]

616 **(a)** Sets up the correct proportional relationship, e.g. $p = \dfrac{k}{\sqrt{q}}$ or $5 = \dfrac{k}{\sqrt{144}}$ **[M1]**

$p = \dfrac{60}{\sqrt{q}}$ **[A1]**

(b) Substitutes $p = 15$ into their equation, e.g. $15 = \dfrac{60}{\sqrt{q}}$ or $\sqrt{q} = 4$ **[M1]**

16 **[A1]**

(c) Recognises that increasing by 300% means the new value of q is $4q$ **[B1]**

Substitutes $4q$ into the equation to find the new p: $p_{new} = \dfrac{60}{\sqrt{4q}} = \dfrac{60}{2\sqrt{q}} = \dfrac{1}{2}p$ **[M1]**

50 **[A1]**

[Total Marks: 7]

617 **(a)** Substitutes into the proportion equation to find the constant, e.g. $4 = \dfrac{k}{\sqrt{36}}$ **[M1]**

$\dfrac{24}{\sqrt{Q}}$ **[A1]**

(b) 2 **[B1]**

(c) Sets up the equation $1.5 = \dfrac{24}{\sqrt{Q}}$ leading to $\sqrt{Q} = 16$ **[M1]**

256 **[A1]**

(d) Identifies the new value of Q is $4Q$, or uses numerical values (e.g. Q increases from 36 to 144) **[M1]**

Evaluates the new value of P as $\dfrac{1}{2}P$, or calculates the new P is 2 when original was 4 **[M1]**

50 **[A1]**

[Total Marks: 8]

618 **(a)** Tests products or ratios, e.g., shows $x^2y = 300$ for all pairs or identifies $k = 300$ **[M1]**

Identifies $y \propto \dfrac{1}{x^2}$ as the correct model **[A1]**

$\dfrac{300}{x^2}$ **[A1]**

(b) Uses 1.5 multiplier and finds $(1.5)^3$ or 3.375 **[M1]**

Identifies increase factor as $3.375 - 1$ or 2.375 **[M1]**

237.5 **[A1]**

[Total Marks: 6]

619 **(a)** Uses new $x = 1.2x$ to write $y_{new} = k(1.2x)^3$ **[M1]**

Calculates $1.2^3 = 1.728$ **[M1]**

72.8 **[A1]**

(b) Tests one incorrect model and shows the constant is not equal, e.g. for $A \propto v$, calculates $\dfrac{12}{2} = 6$ but $\dfrac{48}{4} = 12$ **[M1]**

Tests $A \propto v^2$ and shows $\dfrac{A}{v^2}$ is constant by calculating $\dfrac{12}{4} = 3$, $\dfrac{48}{16} = 3$, and $\dfrac{300}{100} = 3$ **[M1]**

$A \propto v^2$ **[A1]**

[Total Marks: 6]

620 **(a)** **(i)** Recognises that x^2y is constant (128), or identifies the model as $y \propto \frac{1}{x^2}$ **[M1]**

Sets up the equation $y = \frac{k}{x^2}$ and correctly calculates $k = 128$ **[M1]**

$\frac{128}{x^2}$ **[A1]**

(ii) 1.28 **[B1]**

(b) Sets up the correct scale factor equation for B, e.g. $m^3 = 1.728$ or $\left(1 + \frac{x}{100}\right)^3 = 1.728$ **[M1]**

Calculates the multiplier $m = \sqrt[3]{1.728} = 1.2$ **[M1]**

20 **[A1]**

[Total Marks: 7]

621 **(a)** Forms equation: $y = \frac{k}{\sqrt{x}}$ **[M1]**

Uses multiplier 1.44 for the increase in x: New $y = \frac{k}{\sqrt{1.44x}}$ **[M1]**

Simplifies denominator: $\sqrt{1.44} = 1.2$, so New $y = \frac{1}{1.2} \times \frac{k}{\sqrt{x}}$ (or $\frac{5}{6}y$) **[M1]**

Calculates percentage change: $\left(\frac{5}{6} - 1\right) \times 100$ **[M1]**

-16.7 (or 16.7 decrease) **[A1]**

(b) Deduces relationship between P and R: $P = k_1\left(\frac{k_2}{R}\right)^3 = \frac{K}{R^3}$ (or $P \propto \frac{1}{R^3}$) **[M1]**

Substitutes given values to find constant: $10 = \frac{K}{2^3} \implies K = 80$ **[M1]**

Substitutes $R = 4$ into their equation: $P = \frac{80}{4^3} = \frac{80}{64}$ **[M1]**

1.25 (or $\frac{5}{4}$) **[A1]**

[Total Marks: 7]

622 **(a)** Tests a non-linear variation model and sets up an equation, e.g. $y = \frac{k}{x^2}$ **[M1]**

Correctly identifies the constant k by substituting a coordinate pair, e.g. $100 = \frac{k}{2^2} \implies k = 400$ **[M1]**

$y = \frac{400}{x^2}$ **[A1]**

(b) **(i)** Sets up $z = kx$ and finds k: $10 = 4k \implies k = 2.5$ **[M1]**

$\frac{z}{2.5}$ or $\frac{2z}{5}$ or $0.4z$ **[A1]**

(ii) Substitutes their expression for x into the equation from part (a): $y = \frac{400}{(\frac{2z}{5})^2}$ or $y = \frac{400}{(0.4z)^2}$ **[M1]**

Squares the fraction or decimal correctly: $y = \frac{400}{\frac{4z^2}{25}}$ or $y = \frac{400}{0.16z^2}$ **[M1]**

$\frac{2500}{z^2}$ **[A1]**

[Total Marks: 8]

623 (a) Demonstrating that $y \times x^3$ is constant for at least two pairs (e.g., $40 \times 2^3 = 320$ and $5 \times 4^3 = 320$) **[M1]**

$y \propto \frac{1}{x^3}$ **[B1]**

$\frac{320}{x^3}$ **[A1]**

(b) $2560 = \frac{320}{x^3}$ leading to $x^3 = \frac{1}{8}$ **[M1]**

$\frac{1}{2}$ or 0.5 **[A1]**

(c) $x = \frac{k_2}{w}$ leading to $2 = \frac{k_2}{5}$ **[M1]**

$x = \frac{10}{w}$ **[A1]**

Substituting into their equation for y: $y = \frac{320}{(\frac{10}{w})^3}$ **[M1]**

$0.32w^3$ or $\frac{8}{25}w^3$ **[A1]**

[Total Marks: 9]

624 (a) 4 **[B1]**

(b) 2 **[B1]**

[Total Marks: 2]

625 (a) 4 **[B1]**

(b) Identifies period is $180°$ OR uses $360 \div 180$ **[M1]**

2 **[A1]**

[Total Marks: 3]

626 **(a)** -5 **[B1]**

(b) M1 for substituting $(2, 7)$ and their c into the equation, e.g. $7 = a(2)^2 - 5$

3 **[A1]**

[Total Marks: 3]

627 **(a)** 4 **[B1]**

(b) $a = 4$ **[B1]**

$b = 3$ **[B1]**

[Total Marks: 3]

628 **(a)** $a = 3$ **[B1]**

$b = 2$ **[B1]**

(b) $(90, -3)$ **[B1]**

[Total Marks: 3]

629 **(a)** $a = 4$ **[B1]**

$b = 2$ **[B1]**

(b) 180 **[B1]**

[Total Marks: 3]

630 **(a)** **(i)** B **[B1]**

(ii) A **[B1]**

(iii) D **[B1]**

(iv) C **[B1]**

[Total Marks: 4]

631 **(a)** 5 **[B1]**

(b) 180 **[B1]**

(c) $a = 5$ **[B1]**

$b = 2$ **[B1]**

[Total Marks: 4]

632 **(a)** 4 **[B1]**

(b) 180 **[B1]**

(c) $a = 4$ **[B1]**

$b = 2$ **[B1]**

[Total Marks: 4]

633 **(a)** $a = 5$ **[B1]**

$b = 3$ **[B1]**

(b) $-24 = k(2^3)$ or $8k = -24$ **[M1]**

-3 **[A1]**

[Total Marks: 4]

634 **(a)** $a = 4$ **[B1]**

$b = 3$ **[B1]**

(b) 120 **[B1]**

(c) 6 **[B1]**

[Total Marks: 4]

635 **(a)** Amplitude = 3 **[B1]**

Period = $180°$ **[B1]**

(b) $a = 3$ **[B1]**

$b = 2$ **[B1]**

(c) 4 **[B1]**

[Total Marks: 5]

636 **(a)** -8 **[B1]**

(b) Substitutes an x-intercept from the graph, e.g. $4^2 + 4b - 8 = 0$ **[M1]**

-2 **[A1]**

(c) $(1, -9)$ [B2]
(If incorrect, B1 for $x = 1$ or $y = -9$)

[Total Marks: 5]

637 A: $y = 2^x$ **[B1]**

B: $y = \dfrac{1}{x}$ **[B1]**

C: $y = -x^2$ **[B1]**

D: $y = x^3$ **[B1]**

[Total Marks: 4]

638 **(a)** x-coordinate of minimum is $3 \implies -\dfrac{b}{2} = 3$ **[M1]**

$b = -6$ **[A1]**

Substitute $x = 3, y = -4, b = -6$ to find c: $-4 = 3^2 - 6(3) + c \implies c = 5$ **[A1]**

(b) $x^2 - 6x + 5 = 0$ **[M1]**

$(1, 0)$ and $(5, 0)$ **[A1]**

[Total Marks: 5]

639 **(a)** 3 **[B1]**

(b) 2 **[B1]**

(c) M1 for setting up $3\sin(2x) = 1.5$ leading to $\sin(2x) = 0.5$

B1 for one correct value, e.g. $15°$ or $75°$

$15°$ and $75°$ **[A1]**

[Total Marks: 5]

640 $c = -5$ **[B1]**

Uses line of symmetry $x = 2$ to form equation $-\dfrac{b}{2a} = 2 \implies b = -4a$ **[M1]**

Substitutes $(2, 3)$ and $c = -5$ into equation: $4a + 2b - 5 = 3$ **[M1]**

Solves simultaneous equations for a: $4a + 2(-4a) = 8 \implies -4a = 8 \implies a = -2$ **[M1]**

$b = -4(-2) = 8$ **[A1]**

[Total Marks: 5]

641 **(a)** Uses horizontal asymptote or $(0, 3)$ to find $c = 2$ **[B1]**

Substitutes $(2, 11)$ into $f(x) = a^x + 2$: $a^2 + 2 = 11$ **[M1]**

$a = 3$ **[A1]**

(b) $\dfrac{7}{3}$ (or $2\dfrac{1}{3}$) **[B1]**

(c) $f(x) > 2$ or $y > 2$ **[B2]**

(Award [B1] for $f(x) \geq 2$ or > 2 without variable)

[Total Marks: 6]

642 **(a)** Substitute $(-1, 3)$: $(-1)^3 + a(-1)^2 + b(-1) = 3$

a - b = 4[$B1$]

Substitute $(2,0)$: $(2)^3 + a(2)^2 + b(2) = 0$

2a + b = -4[$B1$]

(b) Valid method to solve simultaneous equations (e.g. addition: $3a = 0$) **[M1]**

$a = 0$ and $b = -4$ **[A1]**

(c) $x^3 - 4x = 0$ **[M1]**

$x(x^2 - 4) = 0$

x(x-2)(x+2) = 0[$M1$]

$(0,0)$ and $(-2,0)$ **[A1]**

[Total Marks: 7]

643 **(a)** Substitutes $(3,0)$ into $f(x)$ to get $a(3)^3 + b(3)^2 = 0 \implies 27a + 9b = 0$, leading to $3a + b = 0$. **[B1]**

(b) Substitutes $(2,-4)$ to form a second equation: $8a + 4b = -4$ (or $2a + b = -1$) **[M1]**

Solves simultaneous equations (e.g. substitutes $b = -3a$ into $2a + b = -1$) **[M1]**

$a = 1, b = -3$ **[A1]**

(c) Identifies the turning points at $y = 0$ and $y = -4$ as critical values **[M1]**

$k < -4$ or $k > 0$ **[A1]**

[Total Marks: 6]

644 **(a)** 0 **[B1]**

(b) $(6,0)$ **[B1]**

(c) Uses the vertex form $y = a(x-3)^2 + 18$ OR substitutes $(3,18)$ into $18 = 9a + 3b$ (knowing $c = 0$) **[M1]**

Uses a second point e.g., $(6,0)$ to set up $0 = 36a + 6b$ OR substitutes $(0,0)$ into vertex form: $0 = a(-3)^2 + 18$ **[M1]**

$a = -2, b = 12$ **[A1]**

(d) Identifies that $y = k$ must intersect the curve strictly between $y = 0$ and $y = 18$ on the positive x-axis **[M1]**

$0 < k < 18$ **[A1]**

[Total Marks: 7]

645 **(a)** 0 **[B1]**

(b) Sets up equation for $x = 3$: $27a + 9b + 3c = 0$ (or $9a + 3b + c = 0$) **[M1]**

Sets up equation for $x = -3$: $-27a + 9b - 3c = 0$ (or $-9a + 3b - c = 0$) **[M1]**

Adds or subtracts equations correctly to deduce $18b = 0 \implies b = 0$, and substitutes to conclude $9a + c = 0$ **[A1]**

(c) Substitutes $x = 1, y = -16$ into the equation to get $a + c = -16$ **[M1]**

Solves $9a + c = 0$ and $a + c = -16$ simultaneously **[M1]**

$a = 2$, $c = -18$ **[A1]**

(d) $r = -5$ **[B1]**

$q = 0$ **[B1]**

Sets up $4p - 5 = 7$ to find $p = 3$ **[B1]**

[Total Marks: 10]

646 **(a)** 4 **[B1]**

(b) 2 **[B1]**

(c) $4\sin(2 \times 30°)$ or $4\sin(60°)$ **[M1]**

3.46 **[A1]**

[Total Marks: 4]

647 **(a)** $a = 4$ **[B1]**

$b = 2$ **[B1]**

(b) 5 **[B1]**

(c) 4 **[B1]**

[Total Marks: 4]

648 **(a)** 2 **[B1]**

(b) Substitutes $(1, 6)$ into equation: $6 = \frac{a}{1} + 2$ **[M1]**

4 **[A1]**

(c) Substitutes $(2, 4)$ into $y = k^x$: $4 = k^2$ **[M1]**

2 **[A1]**

[Total Marks: 5]

649 **(a)** 3 **[B1]**

(b) $7 = \frac{a}{2} + 3$ **[M1]**

$a = 8$ **[A1]**

(c) $\frac{8}{x} + 3 = 0$ **[M1]**

$\left(-\frac{8}{3}, 0\right)$ or $(-2.67, 0)$ **[A1]**

[Total Marks: 5]

650 **(a)** 4 **[B1]**

(b) Recognises the period is $120°$ or sets $b \times 120 = 360$ **[M1]**

3 **[A1]**

(c) Recognises there are 2 solutions per period (or sketches horizontal line) **[M1]**

6 **[A1]**

[Total Marks: 5]

651 **(a)** -4 **[B1]**

(b) Uses x-intercepts to set up equation: $a(-1)^2 + b(-1) - 4 = 0$ or $a(x+1)(x-4)$ **[M1]**

Substitute y-intercept $(0, -4)$ into factored form: $a(1)(-4) = -4 \implies a = 1$ **[M1]**

$b = -3$ **[A1]**

(c) Identifies x-coordinate of minimum as $x = \dfrac{-1+4}{2} = 1.5$ (or uses $x = \dfrac{-b}{2a}$) **[M1]**

$(1.5, -6.25)$ **[A1]**

[Total Marks: 6]

652 **(a)** Uses $P(3,8)$ in $y = a^x$: $a^3 = 8$ **[M1]**

$a = 2$ **[A1]**

Uses $P(3,8)$ in $y = \dfrac{b}{x}$: $8 = \dfrac{b}{3} \implies b = 24$ **[B1]**

(b) $(0, 1)$ **[B1]**

(c) Gradient $= \dfrac{8-1}{3-0}$ **[M1]**

$\dfrac{7}{3}$ (or 2.33 or $2\dfrac{1}{3}$) **[A1]**

[Total Marks: 6]

653 **(a)** **(i)** Graph C (or C) **[B1]**

(ii) Graph B (or B) **[B1]**

(iii) Graph D (or D) **[B1]**

(b) $p^3 = 125$ **[M1]**

$p = 5$ **[A1]**

(c) $\dfrac{q}{0.5} = 8$ **[M1]**

$q = 4$ **[A1]**

[Total Marks: 7]

654 **(a)** $c = -5$ **[B1]**

Uses the roots to write $y = k(x+1)(x-5)$ **[M1]**

Expands to $x^2 - 4x - 5$ to find $a = 1$ and $b = -4$ **[A1]**

(b) $x = 2$ **[B1]**

$(2, -9)$ **[B1]**

(c) $2 = p \times q^0 \implies p = 2$ **[B1]**

$18 = 2 \times q^2 \implies q^2 = 9$ **[M1]**

$q = 3$ **[A1]**

[Total Marks: 8]

655 **(a)** 4 **[B1]**

(b) Sets up equation using $(-1, 0)$: $(-1)^3 + a(-1)^2 + b(-1) + 4 = 0 \implies a - b = -3$ **[M1]**

Sets up equation using $(2, 0)$: $(2)^3 + a(2)^2 + b(2) + 4 = 0 \implies 4a + 2b = -12$ OR $2a + b = -6$ **[M1]**

Correct method to solve simultaneous equations (e.g. adding the equations) **[M1]**

$a = -3$ and $b = 0$ **[A1]**

(c) 3 **[B1]**

[Total Marks: 6]

656 **(a)** 4 **[B1]**

(b) M1 for substituting $(-1, 0)$ to form an equation: $-1 + b - c + 4 = 0$ (or $b - c = -3$)

M1 for substituting $(2, 0)$ to form a second equation: $8 + 4b + 2c + 4 = 0$ (or $2b + c = -6$)

M1 for a valid method to solve their simultaneous equations

OR M2 for expanding $(x+1)(x-2)^2$

$b = -3$ **[A1]**

$c = 0$ **[A1]**

(c) 1 **[B1]**

[Total Marks: 6]

657 **(a)** 12 **[B1]**

(b) Substitution of $(-2, 0)$ or $(2, 0)$ into the equation: $0 = -8 + 4b - 2c + 12$ or $0 = 8 + 4b + 2c + 12$ **[M1]**

Forming two correct simultaneous equations, e.g., $2b - c = -2$ and $2b + c = -10$ **[M1]**

$b = -3$ **[A1]**

$c = -4$ **[A1]**

(c) Setting up an equation for the roots, e.g., $(x-2)(x+2)(x-p) = x^3 - 3x^2 - 4x + 12$ OR

$p^3 - 3p^2 - 4p + 12 = 0$ **[M1]**

$p = 3$ **[A1]**

[Total Marks: 7]

658 **(a)** $8a + 2b = -16$ or $4a + b = -8$ **[M1]**

$-27a - 3b = 9$ or $-9a - b = 3$ **[M1]**

$a = 1, b = -12$ **[A1]**

(b) -2 **[B1]**

16 **[B1]**

(c) $k > 16$ **[B1]**

$k < -16$ **[B1]**

[Total Marks: 7]

659 **(a)** $10 = a(2)^3 + b(2) \implies 8a + 2b = 10$ or $4a + b = 5$ **[M1]**

$4 = a(-1)^3 + b(-1) \implies -a - b = 4$ **[M1]**

Attempt to solve simultaneous equations (e.g., adding to eliminate b) **[M1]**

$a = 3$, $b = -7$ **[A1]**

(b) $3x^3 - 7x = 0$ **[M1]**

$x(3x^2 - 7) = 0 \implies x = \pm\sqrt{\dfrac{7}{3}}$ **[M1]**

Length $AB = 2 \times \sqrt{\dfrac{7}{3}} = \dfrac{2\sqrt{21}}{3}$ (or equivalent exact form) **[A1]**

[Total Marks: 7]

660 **(a)** Writes $g(x) = (x + 2)(x - 1)(x - 3)$ **[M1]**

Expands correctly: $(x^2 + x - 2)(x - 3) = x^3 - 2x^2 - 5x + 6$ **[M1]**

$b = -2$, $c = -5$, $d = 6$ **[A1]**

(b) $(0, 6)$ **[B1]**

(c) Local maximum at $(-0.79, 8.21)$ or $x = -0.786...$ seen **[M1]**

Local maximum $x = -0.79$ **[A1]**

Local minimum $x = 2.12$ $(2.1196...)$ **[A1]**

[Total Marks: 7]

661 **(a)** -3 **[B1]**

(b) Substitutes $(0, -1)$ into the equation: $a \cdot b^0 - 3 = -1$ **[M1]**

$a = 2$ **[A1]**

Substitutes $(1, 3)$ into the updated equation: $2 \cdot b^1 - 3 = 3$ **[M1]**

$b = 3$ **[A1]**

(c) Identifies that $g(1) = f(1)$, so $g(1) = 3$ **[M1]**

$\frac{k}{1} = 3 \implies k = 3$ **[A1]**

[Total Marks: 7]

662 **(a)** Forms equation using roots: $f(x) = k(x + 3)(x - 1)(x - 4)$ **[M1]**

Substitutes $(0, -24)$ to find k: $k(3)(-1)(-4) = -24$ **[M1]**

$12k = -24 \implies k = -2$ **[A1]**

Expands brackets: $-2(x^2 + 2x - 3)(x - 4) = -2(x^3 - 4x^2 + 2x^2 - 8x - 3x + 12)$ **[M1]**

Simplifies to: $-2(x^3 - 2x^2 - 11x + 12) = -2x^3 + 4x^2 + 22x - 24$ **[M1]**

$a = -2, b = 4, c = 22, d = -24$ **[A1]**

(b) Recognises translation moves the y-intercept up by 10 (calculates $-24 + 10$) **[M1]**

$(0, -14)$ **[A1]**

[Total Marks: 8]

663 **(a)** Sets up simultaneous equations: $8 = a + b$ and $3.5 = \frac{a}{4} + b$ **[M1]**

Subtracts to eliminate b: $4.5 = \frac{3a}{4} \implies a = 6$ **[A1]**

$b = 2$ **[A1]**

(b) $y = 2$ **[B1]**

(c) **(i)** $x - 3 = \frac{6}{x} + 2$ **[M1]**

$x - 5 = \frac{6}{x} \implies x^2 - 5x = 6 \implies x^2 - 5x - 6 = 0$ **[A1]**

(ii) Solves $(x - 6)(x + 1) = 0$ to get $x = 6$ (rejects $x = -1$ as $x > 0$) **[M1]**

$(6, 3)$ **[A1]**

[Total Marks: 8]

664 **(a)** **(i)** $f(-1) = (-1)^3 + b(-1)^2 + c(-1) + 4 = 0$ **[M1]**

$-1 + b - c + 4 = 0$ leading to $b - c = -3$ **[A1]**

(ii) Use point $(2, 0)$ to form equation: $2^3 + b(2)^2 + c(2) + 4 = 0$ **[M1]**

$8 + 4b + 2c + 4 = 0$ leading to $2b + c = -6$ **[A1]**

Attempt to solve simultaneous equations $b - c = -3$ and $2b + c = -6$ **[M1]**

$b = -3$, $c = 0$ **[A1]**

(b) $p \cdot q^0 = 2 \implies p = 2$ **[B1]**

$2 \cdot q^3 = 54 \implies q^3 = 27$ **[M1]**

$q = 3$ **[A1]**

[Total Marks: 9]

665 **(a)** $r = 5$ **[B1]**

Uses vertex $(2, 9)$ to form an equation, e.g. $9 = p(2)^2 + q(2) + 5$ or uses vertex form $y = p(x - 2)^2 + 9$ **[M1]**

Forms a second equation using symmetry, e.g.

$$\frac{-q}{2p} = 2$$

q = -4p, *orsubstitutes*(0,5)*intovertexform*5 = p(0-2)^2 + 9 [M1]

$p = -1$ and $q = 4$ **[A1]**

(b) Equating $f(x)$ to zero: $-x^2 + 4x + 5 = 0$

$x^2 - 4x - 5 = 0$ [M1]

$(5, 0)$ **[A1]**

(c) $a = 5$ **[B1]**

First positive intercept occurs when $5b = 180$ **[M1]**

$b = 36$ **[A1]**

[Total Marks: 9]

666 **(a)** B1 for a smooth U-shaped parabola passing through $(0, 1)$, $(2, -3)$, and $(4, 1)$

B1 for correct endpoints plotted or stopped cleanly at $(-1, 6)$ and $(5, 6)$

(b) $(2, -3)$ **[B1]**

(c) 0.27 **[B1]**

3.73 **[B1]**

[Total Marks: 5]

667 **(a)** All three values 4, −6, 0 correctly entered **[B2]**

(Award [B1] for two values correct)

(b) All 7 points plotted accurately **[B1]**

Smooth parabolic curve drawn through all plotted points with minimum below $y = -6$ **[B1]**

(c) $(1.25, -6.125)$ or $(1.25, -6.13)$ **[B1]**

[Total Marks: 5]

668 **(a)** Correct inverted U-shape parabola **[B1]**

Maximum approximately at $(3, 4)$ and crosses the x-axis at 1 and 5 **[B1]**

(b) $x = 1$ **[B1]**

$x = 5$ **[B1]**

(c) $(3, 4)$ **[B2]**

If B0, M1 for $x = 3$ or $y = 4$ identified

[Total Marks: 6]

669 (a) Correct shape (inverted parabola) **[B1]**

Graph passes approximately through $(-0.5, 0)$, $(3, 0)$, $(0, 3)$ and vertex roughly at $(1.25, 6.1)$ **[B1]**

(b) $(1.25, 6.125)$ (Accept 6.13 or 6.1) **[B2]**

If B0, M1 for identifying correct x-coordinate 1.25 or correct derivative $-4x + 5 = 0$

(c) $x = -0.5$ **[B1]**

$x = 3$ **[B1]**

[Total Marks: 6]

670 (a) Correct basic cubic shape with a local maximum and local minimum **[B1]**

Graph crosses y-axis at 2, has a maximum at $(0, 2)$ and a minimum at $(2, -2)$ **[B1]**

(b) Uses GDC to find roots **[M1]**

$-0.732, 1, 2.73$ (accept to 3 s.f.) **[A1]**

(c) Identifies correct turning point from GDC **[M1]**

$(2, -2)$ **[A1]**

[Total Marks: 6]

671 (a) Correct U-shaped parabola passing through the y-axis at -3 **[B1]**

Vertices and endpoints approximately in the correct positions **[B1]**

(b) -0.5 **[B1]**

3 **[B1]**

(c) $x = 1.25$ **[B1]**

$y = -6.125$ **[B1]**

[Total Marks: 6]

672 (a) B1 for a smooth cubic curve with y-intercept at $(0, 1)$ and correct general shape

B1 for correct turning points in the second and fourth quadrants and correct x-intercepts near -3.05, 0.52, and 2.53

(b) Local maximum $= (-1.63, 3.18)$ **[B1]**

Local minimum $= (1.63, -1.18)$ **[B1]**

(c) $-1.18 < k < 3.18$ **[B2]**

If 0 scored, allow B1 for identifying the boundary values -1.18 and 3.18

[Total Marks: 6]

673 **(a)** Bell-shaped curve for $y = f(x)$ with local maximum at $(0, 4)$, symmetric about y-axis, passing through $(\pm 1, 2)$ and $(\pm 2, 0.8)$ **[B2]**

Straight line for $y = g(x)$ passing through $(0, -1)$ and $(2, 3)$ **[B1]**

(b) $x = 1.29$ (1.2896...) **[B1]**

$y = 1.58$ (1.5792...) **[B1]**

(c) $y = 0$ **[B1]**

[Total Marks: 6]

674 **(a)** Correct cubic shape with a local maximum and a local minimum **[B1]**

Local maximum approximately at $(-1, 3)$ and local minimum approximately at $(1, -1)$ **[B1]**

(b) $(-1, 3)$ **[B2]**

If B0, M1 for $x = -1$ or $y = 3$

(c) Correct straight line passing through $(0, -1)$ and $(2, 3)$ **[B1]**

(d) $x = 2$ **[B1]**

$x = 0.414$ and $x = -2.41$ **[B1]**

[Total Marks: 7]

675 **(a)** Straight line for $g(x)$ passing through $(0, -2)$ and $(2, 0)$ **[B1]**

Two correct hyperbolic branches for $f(x)$ **[B1]**

Correct asymptotic behaviour near $x = -1$ **[B1]**

(b) $x = -1$ **[B1]**

(c) Sets up $\dfrac{2}{x+1} = x - 2$ or uses intersection feature on GDC **[M1]**

One correct coordinate pair: $(-1.56, -3.56)$ or $(2.56, 0.562)$ (to 3 s.f.) **[A1]**

Second correct coordinate pair **[A1]**

[Total Marks: 7]

676 **(a)** Correct basic cubic shape **[B1]**

Local maximum near $x = 0$, local minimum near $x = 2.5$, and passing through $(0, 2)$ **[B1]**

(b) 1 **[B1]**

-0.562 (or $-0.561...$) **[B1]**

3.56 (or 3.561...) **[B1]**

(c) $x = 2.54$ (or 2.535...) **[B1]**

$y = -4.88$ (or $-4.879...$) **[B1]**

(d) Straight line passing through $(0, 1)$ with a negative gradient **[B1]**

(e) -0.468 (allow -0.47) **[B1]**

0.743 (allow 0.74) **[B1]**

3.73 **[B1]**

[Total Marks: 11]

677 **(a)** Correct general cubic shape starting bottom left and ending top right **[B1]**

Local maximum correctly positioned at approximately $(-1, 4)$ **[B1]**

Local minimum correctly positioned at $(1, 0)$ **[B1]**

(b) $(-1, 4)$ **[B2]**

If B0, M1 for attempting to find stationary points e.g. evaluating max on calculator or $3x^2 - 3 = 0$

(c) $x = -2$ **[B1]**

$x = 1$ **[B1]**

(d) Straight line with positive gradient **[B1]**

Line passing through $(0, 1)$ and $(-0.5, 0)$ **[B1]**

(e) Uses calculator to find intersections or equates $x^3 - 3x + 2 = 2x + 1$ **[M1]**

-2.33 (or $-2.330\ldots$) **[B1]**

0.202 (or $0.2016\ldots$) and 2.13 (or $2.128\ldots$) **[B2, or B1 for 1 correct]**

[Total Marks: 12]

678 **(a)** B1 for $y = f(x)$ starting at $(0, 0)$, rising to a single peak just above 1.5, and decaying towards the x-axis

B1 for $y = g(x)$ drawn as a straight line from $(0, 0.2)$ to $(5, 2.7)$

B1 for clearly showing two points of intersection in the first quadrant

(b) $(1.44, 1.59)$ **[B2]**

If 0 scored, allow B1 for $x = 1.44$ or $y = 1.59$

(c) M1 for finding the x-coordinates of the points of intersection: 0.08 (or $0.082\ldots$) and 2.46 (or $2.456\ldots$)

$0.08 < x < 2.46$ **[A1]**

[Total Marks: 7]

679 **(a)** One correct zero seen: -1.33, 1.20, or 3.13 **[B1]**

All three zeros correct to 2 decimal places: -1.33, 1.20, 3.13 **[B2]**

(b) Local maximum $= (-0.29, 5.30)$ (or $(-0.29, 5.3)$) **[B1]**

Local minimum $= (2.29, -3.30)$ (or $(2.29, -3.3)$) **[B1]**

(c) Identifies boundary values -3.30 and 5.30 **[M1]**

$-3.30 < k < 5.30$ **[A1]**

[Total Marks: 7]

680 **(a)** Graph touches the origin $(0, 0)$ **[B1]**

Graph reaches a maximum in the first quadrant and then approaches the x-axis asymptotically **[B1]**

Curve sketched with reasonable accuracy from $x = -1$ to $x = 5$, with $f(-1) = 2$ **[B1]**

(b) Uses maximum feature on GDC **[M1]**

$(2.89, 1.13)$ (accept $x = 2.88$ to 2.89 and $y = 1.12$ to 1.13) **[A1]**

(c) Draws line $y = 0.5$ on GDC or sets up equation **[M1]**

$-0.589, 0.835, 5.38$ (all three required, accept to 3 s.f.) **[A1]**

(d) Recognises the horizontal line $y = k$ must lie between the local minimum (0) and local maximum (1.13) **[M1]**

$0 < k < 1.13$ **[A1]**

[Total Marks: 9]

681 **(a)** Correct W-shaped curve passing through the origin **[B1]**

Two minima (one in the third quadrant, one in the first quadrant) and one maximum roughly in correct places **[B1]**

Graph drawn between $x = -2$ and $x = 3$ with correct relative heights at the endpoints **[B1]**

(b) $(-1, -19)$ **[B1]**

$(2, 8)$ **[B1]**

(c) $(1, 13)$ **[B1]**

(d) Uses the y-values of the local extremum in the first quadrant to set bounds **[M1]**

$8 < k < 13$ **[A1]**

(e) $(1, 18)$ **[B1]**

[Total Marks: 9]

682 **(a)** Correct shape with two distinct branches **[B1]**

Left branch passing through $(0, -2)$ and tending downwards as x

2^- [B1]

Right branch passing through $(4, 6)$ and tending upwards as x

2^+ [B1]

(b) $x = 2$ **[B1]**

(c) $(0, -2)$ **[B2]**

If B0, M1 for either coordinate correct

(d) $(4, 6)$ **[B2]**

If B0, M1 for either coordinate correct

(e) Identifies the critical values -2 and 6 from the local extrema **[M1]**

$-2 < k < 6$ **[A1]**

[Total Marks: 10]

683 (a) Left branch in the 2nd and 3rd quadrants, passing through x-axis and asymptotic to y-axis downwards **[B1]**

Right branch in 1st quadrant with a clear minimum, asymptotic to y-axis upwards **[B1]**

Both branches curving upwards as x

$\pm\infty$ [B1]

(b) $(1, 3)$ **[B2]**

If B0, M1 for evaluating turning point on calculator or setting derivative to 0

(c) -1.26 (or $-1.259\ldots$) **[B1]**

(d) $x = 0$ **[B1]**

(e) Recognises need to find intersections of graph with $y = 4$ or rearranging to $x^3 - 4x + 2 = 0$ **[M1]**

-2.21 (or $-2.214\ldots$) **[B1]**

0.539 (or $0.5391\ldots$) and 1.68 (or $1.675\ldots$) **[B1]**

[Total Marks: 10]

684 (a) $g(3) = 3^2 + 1 = 10$ **[M1]**

$f(10) = 2(10) - 5 = 15$ **[A1]**

(b) $y = 2x - 5$ OR $x = 2y - 5$ **[M1]**

$\dfrac{x + 5}{2}$ **[A1]**

[Total Marks: 4]

685 (a) 11 **[B1]**

(b) $\{-7, -4, 2, 11\}$ **[B1]**

(c) $x = 3y - 4$ or $y + 4 = 3x$ **[M1]**

$\dfrac{x+4}{3}$ **[A1]**

[Total Marks: 4]

686 **(a)** 10 **[B1]**

(b) 5 **[B1]**

(c) $y = 3x - 2 \implies y + 2 = 3x$ **[M1]**

$\dfrac{x+2}{3}$ **[A1]**

[Total Marks: 4]

687 **(a)** **(i)** -14 **[B1]**

(ii) $f(2) = 3(2) - 2 = 4$ **[M1]**

20 **[A1]**

(b) $y = 3x - 2 \implies y + 2 = 3x$ **[M1]**

$\dfrac{x+2}{3}$ **[A1]**

[Total Marks: 5]

688 **(a)** -13 **[B1]**

(b) $y = 4x - 5$ or $x = 4y - 5$ **[M1]**

$\dfrac{x+5}{4}$ **[A1]**

(c) $4\left(\dfrac{x}{3} + 2\right) - 5$ **[M1]**

$\dfrac{4x}{3} + 3$ or $\dfrac{4x+9}{3}$ **[A1]**

[Total Marks: 5]

689 **(a)** 18 **[B1]**

(b) $y = 5x - 2$ or $x = \dfrac{y+2}{5}$ **[M1]**

$\dfrac{x+2}{5}$ **[A1]**

(c) $g(2) = 2^2 + 3 = 7$ **[M1]**

$f(7) = 5(7) - 2 = 33$ **[A1]**

[Total Marks: 5]

690 **(a)** 7 **[B1]**

(b) Let $y = 4x - 5$ leading to $y + 5 = 4x$ or equivalent **[M1]**

$\dfrac{x+5}{4}$ **[A1]**

(c) $g(2) = 6$ **[M1]**

19 **[A1]**

[Total Marks: 5]

691 **(a)** 10 **[B1]**

(b) $y = 3x - 2 \implies 3x = y + 2$ **[M1]**

$\dfrac{x+2}{3}$ **[A1]**

(c) $f(2) = 3(2) - 2 = 4$ **[M1]**

$g(4) = 4^2 + 4 = 20$ **[A1]**

[Total Marks: 5]

692 **(a)** -19 **[B1]**

(b) $f(2) = 4(2) - 7 = 1$ **[M1]**

$g(1) = 2(1)^2 + 1 = 3$ **[A1]**

(c) $y = 4x - 7$ rearranged to $4x = y + 7$ or $x = 4y - 7$ **[M1]**

$\dfrac{x+7}{4}$ **[A1]**

[Total Marks: 5]

693 **(a)** 13 **[B1]**

(b) Finds $f(2) = 1$ **[M1]**

-2 **[A1]**

(c) Sets $y = 5 - 2x$ and rearranges to $2x = 5 - y$ **[M1]**

$\dfrac{5-x}{2}$ **[A1]**

[Total Marks: 5]

694 **(a)** 5 **[B1]**

(b) $y = 4x - 3 \implies x = \dfrac{y+3}{4}$ or $x = 4y - 3$ **[M1]**

$\dfrac{x+3}{4}$ **[A1]**

(c) $f(1) = 4(1) - 3 = 1$ **[M1]**

$g(1) = 2 - 5(1) = -3$ **[A1]**

[Total Marks: 5]

695 **(a)** Calculates $g(3) = 2(3)^2 + 1 = 19$ **[B1]**

Substitutes their 19 into $f(x)$: $4(19) - 3 = 73$ **[B1]**

(b) Let $y = 4x - 3$ and attempts to rearrange for x (e.g., $y + 3 = 4x$) **[M1]**

$\dfrac{x+3}{4}$ **[A1]**

(c) Sets $4x - 3 = 17$ **[M1]**

$x = 5$ **[A1]**

[Total Marks: 6]

696 **(a)** 10 **[B1]**

(b) Sets $y = 3x - 2$ or $x = 3y - 2$ OR clearly shows inverse operations $(x + 2) \div 3$ **[M1]**

$\dfrac{x+2}{3}$ **[A1]**

(c) Substitutes $f(x)$ into $g(x)$: $(3x - 2)^2 + 1$ **[M1]**

Correct expansion of $(3x - 2)^2$: $9x^2 - 6x - 6x + 4$ OR $9x^2 - 12x + 4$ **[M1]**

$9x^2 - 12x + 5$ **[A1]**

[Total Marks: 6]

697 **(a)** $\dfrac{5}{(2x+1)-3}$ **[M1]**

$\dfrac{5}{2x-2}$ or $\dfrac{5}{2(x-1)}$ **[A1]**

(b) $y(x - 3) = 5$ or $xy - 3y = 5$ **[M1]**

$\dfrac{5}{x} + 3$ or $\dfrac{5+3x}{x}$ **[A1]**

(c) $2\left(\dfrac{5}{x-3}\right) + 1 = 6$ or $\dfrac{10}{x-3} = 5$ **[M1]**

5 **[A1]**

[Total Marks: 6]

698 **(a)** $y(x - 1) = 3x + 2$ **[M1]**

$xy - 3x = y + 2$ or $x(y - 3) = y + 2$ **[M1]**

$\dfrac{x+2}{x-3}$ **[A1]**

(b) $\dfrac{3x+2}{x-1} = x - 2$ **[M1]**

$3x + 2 = (x - 1)(x - 2)$ leading to $3x + 2 = x^2 - 3x + 2$ **[M1]**

$x^2 - 6x = 0$ or $x(x - 6) = 0$ **[M1]**

0 and 6 **[A1]**

[Total Marks: 7]

699 **(a)** 1 **[B1]**

(b) $y = 2x + 3 \implies 2x = y - 3$ **[M1]**

$\dfrac{x - 3}{2}$ **[A1]**

(c) $hk(x) = \dfrac{12}{(2x + 3) - 1}$ **[M1]**

$\dfrac{12}{2x + 2} = \dfrac{6}{x + 1}$ **[A1]**

(d) $k(2) = 2(2) + 3 = 7$ **[B1]**

$\dfrac{12}{x - 1} = 7 \implies 12 = 7(x - 1)$ **[M1]**

$12 = 7x - 7 \implies 19 = 7x \implies x = \dfrac{19}{7}$ (or $2\dfrac{5}{7}$) **[A1]**

[Total Marks: 8]

700 **(a)** Sets $y = \dfrac{4}{2x - 3}$ and multiplies by denominator to obtain $y(2x - 3) = 4$ **[M1]**

Expands and rearranges to collect terms in x: $2xy = 4 + 3y$ **[M1]**

$\dfrac{4 + 3x}{2x}$ **[A1]**

(b) Finds $g(-3) = 2$ **[B1]**

Equates and attempts to solve: $\dfrac{4}{2x - 3} = 2 \implies 4x - 6 = 4$ **[M1]**

2.5 (or $\dfrac{5}{2}$) **[A1]**

(c) Substitutes $f(x)$ into $g(x)$: $\dfrac{4}{2x - 3} + 5$ **[M1]**

Expresses over a common denominator: $\dfrac{4 + 5(2x - 3)}{2x - 3}$ **[M1]**

$\dfrac{10x - 11}{2x - 3}$ **[A1]**

[Total Marks: 9]

701 **(a)** Let $y = 5 - 3x$ leading to $3x = 5 - y$ or equivalent **[M1]**

$\dfrac{5 - x}{3}$ **[A1]**

(b) $x^2 - 2 = 34$

$x^2 = 36$ [M1]

$x = 6, x = -6$ **[A1]**

(c) $5 - 3\left(\dfrac{x-3}{2}\right)$ **[M1]**

$\dfrac{10 - 3(x-3)}{2}$ or equivalent common denominator step **[M1]**

$\dfrac{19 - 3x}{2}$ **[A1]**

(d) $h(2) = 5 - 3(2) = -1$ **[M1]**

-1 **[A1]**

[Total Marks: 9]

702 **(a)** $y = \dfrac{x+3}{x-1} \implies xy - y = x + 3$ **[M1]**

$xy - x = y + 3 \implies x(y-1) = y + 3$ **[M1]**

$\dfrac{x+3}{x-1}$ **[A1]**

(b) $\dfrac{\dfrac{x+3}{x-1} + 3}{\dfrac{x+3}{x-1} - 1}$ **[M1]**

$\dfrac{\dfrac{x+3+3(x-1)}{x-1}}{\dfrac{x+3-(x-1)}{x-1}}$ (or equivalent correct manipulation using a common denominator) **[M1]**

$\dfrac{x+3+3x-3}{x+3-x+1} = \dfrac{4x}{4} = x$ with no mathematical errors seen **[A1]**

(c) $f(3) = \dfrac{3+3}{3-1} = 3$ **[B1]**

$2x^2 - 5 = 3$ **[M1]**

$2x^2 = 8 \implies x^2 = 4$ **[M1]**

$2, -2$ **[A1]**

[Total Marks: 10]

703 **(a)** $y(x-1) = 3x + 2$ or $x(y-1) = 3y + 2$ **[M1]**

$xy - 3x = y + 2$ or $x(y-3) = y + 2$ **[M1]**

$\dfrac{x+2}{x-3}$ **[A1]**

(b) $fg(x) = \dfrac{3\left(\dfrac{2}{x+1}\right) + 2}{\dfrac{2}{x+1} - 1}$ **[M1]**

Numerator becomes $\frac{6+2(x+1)}{x+1}$ or Denominator becomes $\frac{2-(x+1)}{x+1}$ **[M1]**

$\frac{2x+8}{1-x}$ **[A1]**

(c) $\frac{3x+2}{x-1} = \frac{x+2}{x-3} \implies (3x+2)(x-3) = (x+2)(x-1)$ **[M1]**

$3x^2 - 7x - 6 = x^2 + x - 2$ **[M1]**

$2x^2 - 8x - 4 = 0$ or $x^2 - 4x - 2 = 0$ **[M1]**

$2 \pm \sqrt{6}$ **[A1]**

[Total Marks: 10]

704 **(a)** $y(x-1) = 3x + 5$ or $xy - y = 3x + 5$ **[M1]**

$xy - 3x = y + 5$ **[M1]**

$x(y-3) = y + 5$ **[M1]**

$\frac{x+5}{x-3}$ **[A1]**

(b) $\frac{3x+5}{x-1} = x$ **[M1]**

$3x + 5 = x^2 - x$ **[M1]**

$x^2 - 4x - 5 = 0$ **[M1]**

5 and -1 **[A1]**

(c) $f(2) = \frac{3(2)+5}{2-1} = 11$ **[M1]**

$11^2 + 1 = 122$ **[A1]**

[Total Marks: 10]

705 **(a)** **(i)** $\frac{4}{(2x+3)-1}$ **[M1]**

$\frac{2}{x+1}$ **[A1]**

(ii) $\frac{4}{(x^2-4)-1} - 1$ **[M1]**

$\frac{4-(x^2-5)}{x^2-5}$ **[M1]**

$\frac{9-x^2}{x^2-5}$ **[A1]**

(b) $y(x-1) = 4$ or $x - 1 = \frac{4}{y}$ **[M1]**

$xy = 4 + y$ or $x = \frac{4}{y} + 1$ **[M1]**

$\frac{x+4}{x}$ **[A1]**

(c) $2\left(\dfrac{4}{x-1}\right)+3=5$ **[M1]**

$\dfrac{8}{x-1}=2$ **[M1]**

5 **[A1]**

[Total Marks: 11]

706 **(a)** $y=\dfrac{2x+1}{3} \implies 3y=2x+1$ **[M1]**

$\dfrac{3x-1}{2}$ **[A1]**

(b) $hk(x)=\dfrac{2(5-x)+1}{3}$ **[M1]**

$\dfrac{11-2x}{3}$ **[A1]**

(c) $\dfrac{2x+1}{3}=5-x$ **[M1]**

$2x+1=15-3x \implies 5x=14$ **[M1]**

2.8 (or $\dfrac{14}{5}$) **[A1]**

[Total Marks: 7]

707 **(a)** $h(5)=\dfrac{4}{5-3}=2$ **[M1]**

$k(2)=2(2)+1=5$ **[A1]**

(b) $y(x-3)=4$ OR $x(y-3)=4$ **[M1]**

$\dfrac{4}{x}+3$ OR $\dfrac{4+3x}{x}$ **[A1]**

(c) $h(2x+1)=\dfrac{4}{(2x+1)-3}$ **[M1]**

$\dfrac{4}{2x-2}=2 \implies 4=4x-4$ **[M1]**

$x=2$ **[A1]**

[Total Marks: 7]

708 **(a)** $\dfrac{1}{5}$ **[B1]**

(b) $(2x+3)^2-4$ **[M1]**

$4x^2+6x+6x+9-4$ **[M1]**

$4x^2+12x+5$ **[A1]**

(c) $y=2x+3 \implies y-3=2x$ **[M1]**

$\dfrac{x-3}{2}$ **[A1]**

(d) 3 **[B1]**

[Total Marks: 7]

709 **(a)** 2 **[B1]**

(b) Sets $y = \dfrac{5}{x-2}$ and multiplies by $(x-2)$: $y(x-2) = 5$ **[M1]**

Rearranges to isolate x: $x - 2 = \dfrac{5}{y} \implies x = \dfrac{5}{y} + 2$ OR $xy - 2y = 5 \implies x = \dfrac{5+2y}{y}$ **[M1]**

$\dfrac{5}{x} + 2$ OR $\dfrac{5+2x}{x}$ **[A1]**

(c) Equates functions: $\dfrac{5}{x-2} = x + 2$ **[M1]**

Multiplies by $(x-2)$ to form a quadratic: $5 = (x+2)(x-2)$ **[M1]**

Expands and simplifies to $x^2 = 9$ or $x^2 - 9 = 0$ **[A1]**

3 and -3 **[A1]**

[Total Marks: 8]

710 **(a)** $(-2)^2 - (-2) - 7$ **[M1]**

-1 **[A1]**

(b) $(2x-3)^2 - (2x-3) - 7$ **[M1]**

$4x^2 - 12x + 9 - 2x + 3 - 7$ **[M1]**

$4x^2 - 14x + 5$ **[A1]**

(c) $2x - 3 = x^2 - x - 7$ **[M1]**

$x^2 - 3x - 4 = 0$ or $(x-4)(x+1) = 0$ **[M1]**

$x = 4$ and $x = -1$ **[A1]**

[Total Marks: 8]

711 **(a)** -4 and 5 seen **[B1]**

$-4 \leq f(x) \leq 5$ **[B1]**

(b) $g(3) = 7$ **[B1]**

$f(7) = 32$ **[B1]**

(c) $2(x^2 - 2x - 3) + 1$ **[M1]**

$2x^2 - 4x - 5$ **[A1]**

(d) $x = g(6)$ or $\dfrac{x-1}{2} = 6$ **[M1]**

13 **[A1]**

[Total Marks: 8]

712 **(a)** Substitution of $f(x)$ into $g(x)$: $\dfrac{5}{\dfrac{2x+1}{3}-2}$ **[M1]**

Writing the denominator as a single fraction: $\dfrac{2x+1-6}{3}$ or $\dfrac{2x-5}{3}$ **[M1]**

$\dfrac{15}{2x-5}$ **[A1]**

(b) Let $y = \dfrac{5}{x-2}$ and attempting to multiply by $(x-2)$ or swapping x and y **[M1]**

$x - 2 = \dfrac{5}{y}$ or $xy - 2y = 5$ **[M1]**

$\dfrac{5}{x} + 2$ or $\dfrac{5+2x}{x}$ **[A1]**

(c) Evaluates $g(3) = 5$ **[B1]**

Equating $f(x)$ to their $g(3)$: $\dfrac{2x+1}{3} = 5$ **[M1]**

$x = 7$ **[A1]**

[Total Marks: 9]

713 **(a)** 5 **[B1]**

(b) $y(x-3) = 2x+1$ **[M1]**

$xy - 3y = 2x + 1$
$xy - 2x = 3y + 1$
$x(y-2) = 3y + 1$ **[M1]**

$\dfrac{3x+1}{x-2}$ **[A1]**

(c) $(2x-1)^2 - 4$ **[M1]**

$4x^2 - 2x - 2x + 1 - 4$ **[M1]**

$4x^2 - 4x - 3$ **[A1]**

(d) $2x - 1 = x^2 - 4$
$x^2 - 2x - 3 = 0$ **[M1]**

$(x-3)(x+1) = 0$ **[M1]**

$x = 3, x = -1$ **[A1]**

[Total Marks: 10]

714 Substitutes $f(x)$ into f: $ff(x) = \dfrac{2\left(\dfrac{2x+3}{x-1}\right)+3}{\left(\dfrac{2x+3}{x-1}\right)-1}$ **[M1]**

Multiplies numerator and denominator by $(x-1)$: $\dfrac{2(2x+3)+3(x-1)}{(2x+3)-(x-1)}$ **[M1]**

Simplifies to $\dfrac{7x+3}{x+4}$ — note: checks algebra carefully: $4x + 6 + 3x - 3 = 7x + 3$,

$2x + 3 - x + 1 = x + 4$; wait: let us verify $f(x) = \dfrac{2x+3}{x-2}$ is self-inverse! With $f(x) = \dfrac{2x+3}{x-1}$, $ff(x) = \dfrac{7x+3}{x+4}$.

[Total Marks: 7]

715 **(a)** **(i)** 2.5 **[B1]**

(ii) $g(4-3x) = \dfrac{5}{(4-3x)-2}$ **[M1]**

$\dfrac{5}{2-3x}$ **[A1]**

(b) Minimum x-coordinate is $-\dfrac{-1}{2(1)} = 0.5$ or via completing square $(x-0.5)^2 - 7.25$ **[M1]**

$h(x) \geq -7.25$ **[A1]**

(c) $x^2 - x - 7 = 2x - 1$ leading to $x^2 - 3x - 6 = 0$ **[M1]**

$\dfrac{-(-3) \pm \sqrt{(-3)^2 - 4(1)(-6)}}{2(1)}$ **[M2]**

If M2 not scored, allow M1 for $\dfrac{3 \pm \sqrt{33}}{2}$ or correct use of formula with one error **[M1]**

-1.37 and 4.37 **[A1]**

[Total Marks: 9]

716 **(a)** $y(x-1) = 2x + 1 \implies xy - y = 2x + 1$ **[M1]**

$xy - 2x = y + 1 \implies x(y-2) = y + 1$ **[M1]**

$\dfrac{x+1}{x-2}$ **[A1]**

(b) $g(3x-2) = (3x-2)^2 - 4(3x-2)$ **[M1]**

$9x^2 - 12x + 4 - 12x + 8$ **[M1]**

$9x^2 - 24x + 12$ **[A1]**

(c) $3(x^2 - 4x) - 2 = x$ **[M1]**

$3x^2 - 13x - 2 = 0$ **[A1]**

$x = \dfrac{13 \pm \sqrt{(-13)^2 - 4(3)(-2)}}{2(3)}$ **[M1]**

$x = 4.48$ or $x = -0.15$ **[A1]**

[Total Marks: 10]

717 **(a)** Substitution of $g(x)$ into $f(x)$: $3(x^2 - x) + 2$ **[M1]**

$3x^2 - 3x + 2$ **[A1]**

(b) Substitution of $f(x)$ into $g(x)$: $(3x+2)^2 - (3x+2)$ **[M1]**

Correct expansion of $(3x+2)^2$ to $9x^2 + 12x + 4$ **[M1]**

$9x^2 + 9x + 2$ **[A1]**

(c) Sets their three-term quadratic $gf(x)$ equal to 0: $9x^2 + 9x + 2 = 0$ **[M1]**

Valid method to solve quadratic (e.g., $(3x+1)(3x+2) = 0$ or correct use of formula) **[M1]**

$x = -\frac{1}{3}$ and $x = -\frac{2}{3}$ **[A1]**

(d) $5^x = 5^{-3}$ **[M1]**

$x = -3$ **[A1]**

[Total Marks: 10]

718 **(a)** $x = 5y - 4$ or $y = 5x - 4$ **[M1]**

$\frac{x+4}{5}$ **[A1]**

(b) $\frac{3}{(5x-4)+2}$ **[M1]**

$\frac{3}{5x-2}$ **[A1]**

(c) $\frac{3}{\frac{3}{x+2}+2}$ **[M1]**

$\frac{3}{\frac{3+2(x+2)}{x+2}}$ **[M1]**

$\frac{3(x+2)}{2x+7}$ **[A1]**

(d) $\frac{3}{x+2} = 5x - 4$

$3 = (5x-4)(x+2)$ **[M1]**

$3 = 5x^2 + 10x - 4x - 8$

$5x^2 + 6x - 11 = 0$ **[M1]**

$(5x+11)(x-1) = 0$ or correct use of quadratic formula **[M1]**

$x = 1, x = -2.2$ (or $-\frac{11}{5}$) **[A1]**

[Total Marks: 11]

719 **(a)** Sets $y = \frac{2x+5}{x-3}$ and rearranges to get $y(x-3) = 2x+5$ **[M1]**

Expands and factorises to collect terms in x: $xy - 2x = 3y + 5 \implies x(y-2) = 3y + 5$ **[M1]**

$\frac{3x+5}{x-2}$ **[A1]**

(b) Substitutes $g(x)$ into $f(x)$: $\frac{2(3x-1)+5}{(3x-1)-3}$ **[M1]**

Expands and simplifies the numerator to $6x + 3$ **[M1]**

$\dfrac{6x+3}{3x-4}$ **[A1]**

(c) Sets up the equation $3\left(\dfrac{2x+5}{x-3}\right)-1=x$ **[M1]**

Eliminates the denominator to obtain $6x+15-(x-3)=x(x-3)$ (or equivalent) **[M1]**

Simplifies to form the correct 3-term quadratic equation: $x^2-8x-18=0$ **[A1]**

Correct use of the quadratic formula $x=\dfrac{-(-8)\pm\sqrt{(-8)^2-4(1)(-18)}}{2(1)}$ **[M1]**

9.83 and -1.83 **[A1]**

[Total Marks: 11]

720 **(a)** $y=\dfrac{3x+5}{x-2} \implies y(x-2)=3x+5$ **[M1]**

$xy-2y=3x+5 \implies xy-3x=2y+5 \implies x(y-3)=2y+5$ **[M1]**

$\dfrac{2x+5}{x-3}$ **[A1]**

(b) $\dfrac{3}{(3x+5)^2+2}$ **[M1]**

$\dfrac{3}{9x^2+30x+25+2}=\dfrac{3}{9x^2+30x+27}$ **[M1]**

$\dfrac{3}{3(3x^2+10x+9)}=\dfrac{1}{3x^2+10x+9}$ **[A1]**

(c) $\dfrac{3x+5}{x-2}=x \implies 3x+5=x(x-2)$ **[M1]**

$3x+5=x^2-2x \implies x^2-5x-5=0$ **[A1]**

$\dfrac{-(-5)\pm\sqrt{(-5)^2-4(1)(-5)}}{2(1)}$ **[M1]**

$\dfrac{5\pm\sqrt{25+20}}{2}=\dfrac{5\pm\sqrt{45}}{2}$ **[M1]**

5.85 and -0.85 **[A1]**

[Total Marks: 11]

721 **(a)** Sets $y=\dfrac{2x+3}{x-1}$ and multiplies: $y(x-1)=2x+3$ **[M1]**

Expands bracket and collects x terms on one side: $xy-2x=y+3$ **[M1]**

Factorises x: $x(y-2)=y+3$ **[M1]**

$\dfrac{x+3}{x-2}$ **[A1]**

(b) Substitutes $f(x)$ into itself: $\dfrac{2\left(\dfrac{2x+3}{x-1}\right)+3}{\left(\dfrac{2x+3}{x-1}\right)-1}$ **[M1]**

Forms common denominators for numerator and denominator: $\dfrac{\dfrac{2(2x+3)+3(x-1)}{x-1}}{\dfrac{(2x+3)-(x-1)}{x-1}}$ **[M1]**

Correctly simplified numerator: $4x+6+3x-3=7x+3$ OR correctly simplified denominator: $2x+3-x+1=x+4$ **[B1]**

$\dfrac{7x+3}{x+4}$ **[A1]**

(c) Calculates $f(2)=\dfrac{2(2)+3}{2-1}=7$ **[B1]**

Forms correct quadratic equation: $x^2-4x-7=0$ **[B1]**

Correct substitution into quadratic formula: $\dfrac{4\pm\sqrt{(-4)^2-4(1)(-7)}}{2}$ OR $\dfrac{4\pm\sqrt{44}}{2}$ **[M1]**

5.32 and -1.32 **[A1]**

[Total Marks: 12]

722 **(a)** 2 **[B1]**

(b) Let $y=\dfrac{x+5}{x-2}$ **[M1]**

$y(x-2)=x+5$

xy - 2y = x + 5[$M1$]

$xy-x=2y+5$

x(y - 1) = 2y + 5[$M1$]

$\dfrac{2x+5}{x-1}$ or equivalent **[A1]**

(c) $x^2+3x-10=0$ **[M1]**

$(x+5)(x-2)=0$ or correct substitution into quadratic formula **[M1]**

$x=-5, x=2$ **[A1]**

(d)

$$\frac{x+5}{x-2}=x$$

$x^2-2x=x+5$ [M1]

$x^2-3x-5=0$ **[M1]**

$x=\dfrac{-(-3)\pm\sqrt{(-3)^2-4(1)(-5)}}{2}$ **[M1]**

$x=4.19, x=-1.19$ **[A1]**

[Total Marks: 12]

723 $y=(x+3)(x-4)$ **[M1]**

x^2-x-12 **[A1]**

[Total Marks: 2]

724 $(x-4)(x+2)$ **[M1]**

x^2-2x-8 **[A1]**

[Total Marks: 2]

725 Uses $(x-p)(x-q)$ with $p=-4$ and $q=3$: $(x+4)(x-3)$ **[M1]**

x^2+x-12 **[A1]**

[Total Marks: 2]

726 Uses the intercepts to form $y=(x+4)(x-7)$ **[M1]**

$x^2-3x-28$ **[A1]**

[Total Marks: 2]

727 Uses the vertex form $y=(x-4)^2-3$ **[M1]**

Expands to $x^2-8x+16-3$ **[M1]**

$b=-8$ and $c=13$ **[A1]**

[Total Marks: 3]

728 Uses the vertex form $y=(x-3)^2-4$ OR uses $-\dfrac{b}{2(1)}=3$ **[M1]**

$b=-6$ **[A1]**

$c=5$ **[A1]**

[Total Marks: 3]

729 $y=(x-3)^2-4$ **[M1]**

$y=x^2-6x+9-4$ **[M1]**

x^2-6x+5 **[A1]**

[Total Marks: 3]

730 (a) $x^2+bx+c=(x+4)(x-2)$ **[M1]**

$b=2$, $c=-8$ **[A1]**

(b) Finds x-coordinate of vertex: $x=\dfrac{-4+2}{2}=-1$ **[M1]**

$(-1,-9)$ **[A1]**

[Total Marks: 4]

731 **(a)** Writes $f(x) = (x+3)(x-5)$ **[M1]**

Expands correctly: $x^2 - 2x - 15$ **[M1]**

$b = -2$, $c = -15$ **[A1]**

(b) $(0, -15)$ **[B1]**

[Total Marks: 4]

732 **(a)** Uses vertex form: $y = (x-3)^2 - 4$ **[M1]**

$x^2 - 6x + 5$ **[A1]**

(b) Sets $x^2 - 6x + 5 = 0 \implies (x-1)(x-5) = 0$ **[M1]**

$(1, 0)$ and $(5, 0)$ **[A1]**

[Total Marks: 4]

733 **(a)** $h = 3$ **[B1]**

$k = -2$ **[B1]**

(b) $p = (5-3)^2 - 2$ **[M1]**

$p = 2$ **[A1]**

[Total Marks: 4]

734 **(a)** $(x-3)^2$ seen **[B1]**

$(x-3)^2 - 4$ **[B1]**

(b) $y = x^2 - 6x + 9 - 4$ **[M1]**

$b = -6$ and $c = 5$ **[A1]**

(c) $(0, 5)$ **[B1]**

[Total Marks: 5]

735 **(a)** Uses vertex form: $y = (x-2)^2 - 9$ or x-coordinate of vertex $\dfrac{-p}{2} = 2$ **[M1]**

Expands to find p and q, or solves $2^2 + 2p + q = -9$ **[M1]**

$p = -4$ and $q = -5$ **[A1]**

(b) $x^2 - 4x - 5 = 0$ **[M1]**

$(x-5)(x+1) = 0$ **[M1]**

$A = (-1, 0)$ and $B = (5, 0)$ **[A1]**

[Total Marks: 6]

736 $f(x) = a(x+2)(x-5)$ **[M1]**

$a(0+2)(0-5) = -20$ or $-10a = -20$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 6x - 20$ **[A1]**

[Total Marks: 4]

737 $f(x) = a(x+2)(x-6)$ **[M1]**

$-24 = a(0+2)(0-6)$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 8x - 24$ **[A1]**

[Total Marks: 4]

738 **(a)** Writes $g(x) = a(x-2)^2 - 9$ **[M1]**

Substitutes $(5, 9)$: $9 = a(5-2)^2 - 9$ or $9a = 18$ **[M1]**

$2(x-2)^2 - 9$ **[A1]**

(b) $2(x^2 - 4x + 4) - 9$ **[M1]**

$2x^2 - 8x - 1$ **[A1]**

(c) $x = 2$ **[B1]**

[Total Marks: 6]

739 **(a)** Uses intercept form: $y = a(x+1)(x-4)$ **[M1]**

Substitutes $(0, -8)$ to find a: $-8 = a(0+1)(0-4) \implies -4a = -8 \implies a = 2$ **[M1]**

$2x^2 - 6x - 8$ **[A1]**

(b) Finds the x-coordinate of the vertex by symmetry: $x = \dfrac{-1+4}{2} = 1.5$ (or uses $x = -\dfrac{b}{2a}$) **[M1]**

Substitutes $x = 1.5$ into the equation: $y = 2(1.5)^2 - 6(1.5) - 8$ **[M1]**

$(1.5, -12.5)$ (or $(\dfrac{3}{2}, -\dfrac{25}{2})$) **[A1]**

[Total Marks: 6]

740 **(a)** $y = a(x+2)(x-4)$ **[M1]**

$16 = a(0+2)(0-4)$ **[M1]**

$a = -2$ **[A1]**

$y = -2x^2 + 4x + 16$ **[A1]**

(b) $x = \dfrac{-2+4}{2} = 1$ **[M1]**

$(1, 18)$ **[A1]**

(c) $x = 1$ **[B1]**

[Total Marks: 7]

741 **(a)** $h = -1$ **[B1]**

$k = 8$ **[B1]**

(b) $-10 = a(2 - (-1))^2 + 8$ **[M1]**

$9a = -18$ **[M1]**

-2 **[A1]**

(c) $-2(x + 1)^2 + 8$ **[M1]**

$-2(x^2 + 2x + 1) + 8$ **[M1]**

$-2x^2 - 4x + 6$ **[A1]**

[Total Marks: 8]

742 $f(x) = a(x + 1)(x - 4)$ **[M1]**

$-8 = a(0 + 1)(0 - 4)$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 6x - 8$ **[A1]**

[Total Marks: 4]

743 **(a)** Forms the equation $f(x) = a(x + 2)^2 + 8$ **[M1]**

Substitutes $(0, -4)$ into their equation:

$-4 = a(0 + 2)^2 + 8$ **[M1]**

$4a = -12 \implies a = -3$ **[A1]**

$-3x^2 - 12x - 4$ **[A1]**

(b) Sets $-3(x + 2)^2 + 8 = 0$ or $-3x^2 - 12x - 4 = 0$ **[M1]**

Rearranges to $(x + 2)^2 = \dfrac{8}{3}$ OR correctly substitutes into the quadratic formula:

$x = \dfrac{-(-12) \pm \sqrt{(-12)^2 - 4(-3)(-4)}}{2(-3)}$ **[M1]**

$\dfrac{-6 \pm 2\sqrt{6}}{3}$ **[A1]**

[Total Marks: 7]

744 $y = a(x - 3)^2 - 5$ **[M1]**

$7 = a(1 - 3)^2 - 5 \implies 12 = 4a \implies a = 3$ **[M1]**

$3(x - 3)^2 - 5$ **[A1]**

[Total Marks: 3]

745 $y = a(x+2)^2 + 5$ **[M1]**

$-13 = a(1+2)^2 + 5$ **[M1]**

$-18 = 9a$

a = -2$[M1]$

$-2(x+2)^2 + 5$ **[A1]**

[Total Marks: 3]

746 **(a)** Uses vertex form: $y = a(x-2)^2 + 7$ **[M1]**

Substitutes $(4, -5)$ into their equation: $-5 = a(4-2)^2 + 7$ **[M1]**

$-3(x-2)^2 + 7$ **[A1]**

(b) -5 **[B1]**

[Total Marks: 4]

747 Writes the function in the factored form $y = a(x+1)(x-5)$ **[M1]**

Substitutes $(2, -18)$ into their equation:

$-18 = a(2+1)(2-5)$ **[M1]**

$a = 2$ **[A1]**

$2x^2 - 8x - 10$ **[A1]**

[Total Marks: 4]

748 Uses vertex form $y = a(x-2)^2 - 5$ **[M1]**

Substitutes $(4, 7)$ into their equation: $7 = a(4-2)^2 - 5$ **[M1]**

$4a = 12$

a = 3$[A1]$

$3x^2 - 12x + 7$ **[A1]**

[Total Marks: 4]

749 **(a)** $h = 2$ **[B1]**

$k = 5$ **[B1]**

(b) Substitutes $(4, -7)$ into equation: $-7 = a(4-2)^2 + 5$ **[M1]**

$a = -3$ **[A1]**

(c) Substitutes $x = 0$ into $y = -3(x-2)^2 + 5$ **[M1]**

$(0, -7)$ **[A1]**

[Total Marks: 6]

750 **(a)** Sets up the factored form $f(x) = a(x+2)(x-5)$ **[M1]**

Substitutes the point $(3, 10)$: $10 = a(3+2)(3-5)$ **[M1]**

Solves for a: $10 = a(5)(-2) \implies a = -1$ **[A1]**

$f(x) = -x^2 + 3x + 10$ **[A1]**

(b) Finds the x-coordinate of the vertex (e.g., midpoint of intercepts $\frac{-2+5}{2} = 1.5$ or $x = \frac{-b}{2a}$) **[M1]**

$(1.5, 12.25)$ **[A1]**

[Total Marks: 6]

751 $y = a(x+1)^2 + 8$ **[M1]**

$-10 = a(2+1)^2 + 8$ **[M1]**

$9a = -18 \implies a = -2$ **[A1]**

$-2(x^2 + 2x + 1) + 8$ **[M1]**

$a = -2$, $b = -4$, $c = 6$ **[A1]**

[Total Marks: 5]

752 **(a)** $y = a(x+3)(x-2)$ **[M1]**

$12 = a(0+3)(0-2)$ **[M1]**

$12 = -6a$

a = -2[$M1$]

$-2x^2 - 2x + 12$ **[A1]**

(b) $x = \frac{-3+2}{2}$ or $x = \frac{-(-2)}{2(-2)} = -0.5$ **[M1]**

$y = -2(-0.5)^2 - 2(-0.5) + 12 = 12.5$ **[M1]**

$(-0.5, 12.5)$ **[A1]**

[Total Marks: 6]

753 **(a)** Sets up equation using x-intercepts: $f(x) = a(x+1.5)(x-4)$ OR $f(x) = a(2x+3)(x-4)$ **[M1]**

Substitutes $(2, -14)$ to find a: $-14 = a(2+1.5)(2-4)$ **[M1]**

Solves for a: $a = 2$ **[A1]**

$2x^2 - 5x - 12$ **[A1]**

(b) Identifies x-coordinate of vertex as midpoint of roots: $x = \frac{-1.5+4}{2} = 1.25$ OR uses $x = \frac{-b}{2a}$ **[M1]**

Substitutes $x = 1.25$ into their function: $y = 2(1.25)^2 - 5(1.25) - 12$ **[M1]**

$(1.25, -15.125)$ OR $\left(\frac{5}{4}, -\frac{121}{8}\right)$ **[A1]**

[Total Marks: 7]

754 **(a)** **(i)** $h = -4$ **[B1]**

$k = 20$ **[B1]**

(ii) $0 = a(-2+4)^2 + 20$ **[M1]**

$a = -5$ **[A1]**

(iii) -6 **[B1]**

(iv) $f(x) = -5(x+4)^2 + 20 = -5(x^2 + 8x + 16) + 20$ **[M1]**

$-5x^2 - 40x - 60$ **[A1]**

[Total Marks: 7]

755 **(a)** Writes $h(x) = a(x+2)(x-4)$ **[M1]**

Substitutes $(1, 13.5)$: $13.5 = a(3)(-3) \implies -9a = 13.5$ **[M1]**

$a = -1.5$ (or $-\frac{3}{2}$) **[A1]**

Expands $-1.5(x^2 - 2x - 8)$ to obtain $b = 3$ and $c = 12$ **[A1]**

(b) x-coordinate of vertex $= \frac{-2+4}{2} = 1$ **[M1]**

$(1, 13.5)$ **[A1]**

(c) $h(x) \leq 13.5$ or $y \leq 13.5$ **[B1]**

[Total Marks: 7]

756 **(a)** Sets up the equation in factored form: $f(x) = a(x+1)(x-5)$ **[M1]**

Substitutes $(0, 10)$ to find a: $10 = a(0+1)(0-5)$ **[M1]**

$a = -2$ **[A1]**

$-2x^2 + 8x + 10$ **[A1]**

(b) Finds the x-coordinate of the vertex using symmetry $x = \frac{-1+5}{2} = 2$ OR uses $x = -\frac{b}{2a}$ **[M1]**

$(2, 18)$ **[A1]**

(c) Translates the vertex $(2, 18)$

(4, 15)$ORAppliestransformation$-2((x-2)-2)^2 + 18 − 3 [M1]

$-2(x-4)^2 + 15$ **[A1]**

[Total Marks: 8]

757 **(a)** Uses the vertex coordinates to write $f(x) = a(x+2)^2 + 8$ **[M1]**

Substitutes the y-intercept $(0, 5)$: $5 = a(0+2)^2 + 8$ **[M1]**

$f(x) = -0.75(x+2)^2 + 8$ or equivalent using fractions like $-\frac{3}{4}$ **[A1]**

(b) Expands $(x+2)^2$ as $x^2 + 4x + 4$ **[M1]**

$f(x) = -0.75x^2 - 3x + 5$ **[A1]**

(c) Equates function to zero: $-0.75(x+2)^2 + 8 = 0$ or $-0.75x^2 - 3x + 5 = 0$ **[M1]**

Rearranges to $(x+2)^2 = \frac{32}{3}$ or correctly applies the quadratic formula **[M1]**

$x = 1.27$ and $x = -5.27$ **[A1]**

[Total Marks: 8]

758 **(a)** $c = 15$ **[B1]**

$y = a(x+2)(x-5)$ **[M1]**

$15 = a(0+2)(0-5)$ **[M1]**

$a = -1.5$ **[A1]**

$b = 4.5$ **[A1]**

(b) $x = \frac{-2+5}{2} = 1.5$ **[M1]**

$y = -1.5(1.5)^2 + 4.5(1.5) + 15$ **[M1]**

$(1.5, 18.375)$ or $\left(\frac{3}{2}, \frac{147}{8}\right)$ **[A1]**

(c) $k > 18.375$ **[B1]**

[Total Marks: 9]

759 **(a)** Uses vertex form: $y = a(x+2)^2 + 18$ **[M1]**

Substitutes $(0, 10)$ to find a: $10 = a(0+2)^2 + 18 \implies 4a = -8 \implies a = -2$ **[M1]**

$-2x^2 - 8x + 10$ **[A1]**

(b) Sets $f(x) = 0$: $-2x^2 - 8x + 10 = 0$ **[M1]**

$x^2 + 4x - 5 = 0 \implies (x+5)(x-1) = 0 \implies x = -5$ or $x = 1$ **[M1]**

6 **[A1]**

(c) Equates line and curve: $-2x^2 - 8x + 10 = -2x + k$ **[M1]**

Rearranges to form a quadratic equation: $2x^2 + 6x + (k-10) = 0$ (or equivalent) **[M1]**

Sets discriminant $\Delta = 0$ for a tangent: $6^2 - 4(2)(k-10) = 0$ **[M1]**

14.5 (or $\frac{29}{2}$) **[A1]**

[Total Marks: 10]

760 **(a)** $h = 4$ and $k = -5$ **[B1]**

$8.5 = a(1-4)^2 - 5$ **[M1]**

$a = 1.5$ **[A1]**

(b) $1.5(x-4)^2 - 5 = 0$ or $(x-4)^2 = \dfrac{10}{3}$ **[M1]**

$x = 4 \pm \sqrt{\dfrac{10}{3}}$ **[M1]**

$x = 2.17$ and $x = 5.83$ **[A1]**

(c) $1.5(x-4)^2 - 5 = 2x - 7$ **[M1]**

$3x^2 - 28x + 52 = 0$ **[A1]**

$x = 2.56$ and $x = 6.77$ **[A1]**

(d) New vertex at $(1, 4)$ **[M1]**

$1.5(x-1)^2 + 4$ **[A1]**

[Total Marks: 11]

761 **(a)** Forms equation using x-intercepts: $f(x) = a(x+1)(x-3)$ OR using vertex $f(x) = a(x-1)^2 - 8$ **[M1]**

Substitutes appropriate point to find a: $a(1+1)(1-3) = -8 \implies -4a = -8 \implies a = 2$ **[M1]**

Expands to show final form: $2(x^2 - 2x - 3) = 2x^2 - 4x - 6$ **[A1]**

(b) $2x^2 - 4x - 6 = 10 \implies 2x^2 - 4x - 16 = 0$ **[M1]**

Factors correctly to $(x-4)(x+2) = 0$ OR uses quadratic formula correctly **[M1]**

$x = 4$ and $x = -2$ **[A1]**

(c) Sets equations equal: $2x^2 - 4x - 6 = mx - 8$ leading to $2x^2 - (4+m)x + 2 = 0$ **[M1]**

States condition for tangent (discriminant $= 0$): $(-(4+m))^2 - 4(2)(2) = 0$ **[M1]**

Simplifies to $(4+m)^2 = 16$ or $m^2 + 8m = 0$ **[M1]**

$m = 0$, $m = -8$ **[A1]**

(d) Recognises minimum y-value must be strictly greater than 0: $-8 + k > 0$ **[M1]**

$k > 8$ **[A1]**

[Total Marks: 12]

762 **(a)** $x = 2$ **[B1]**

(b) $y = 1$ **[B1]**

[Total Marks: 2]

763 **(a)** $x = 4$ **[B1]**

(b) $y = 3$ **[B1]**

[Total Marks: 2]

764 **(a)** -3 **[B1]**

(b) -2 **[B1]**

[Total Marks: 2]

765 **(a)** $x = 2$ **[B1]**

(b) $y = 1$ **[B1]**

(c) $x = -2$ **[B1]**

[Total Marks: 3]

766 **(a)** $x = 2$ **[B1]**

(b) $y = 1$ **[B1]**

(c) $f(0) = 0$ **[B1]**

$x = 0$ **[B1]**

[Total Marks: 4]

767 **(a)** $x = 3$ **[B1]**

(b) $y = 1$ **[B1]**

(c) **(i)** 4 **[B1]**

(ii) 2 **[B1]**

[Total Marks: 4]

768 **(a)** **(i)** $x = 1$ **[B1]**

(ii) $y = 3$ **[B1]**

(b) $\dfrac{4}{x-1} + 3 = 0$ **[M1]**

$(-\dfrac{1}{3}, 0)$ **[A1]**

(c) $(0, -1)$ **[B1]**

[Total Marks: 5]

769 **(a)** $x = 45°$ **[B1]**

$x = 135°$ **[B1]**

(b) Draws dashed vertical lines at $x = 45°$ and $x = 135°$ **[B1]**

Correct tangent curve shape passing through $(0°, 0)$, $(90°, 0)$, and $(180°, 0)$ **[B1]**

Branches approach asymptotes cleanly without touching or curling back **[B1]**

[Total Marks: 5]

770 **(a)** $x = 90°$ **[B1]**

$x = 270°$ **[B1]**

(b) There is a vertical asymptote at $x = 90°$ (or the curve never touches $x = 90°$) **[B1]**

(c) $x = 125°$ **[B1]**

$x = 305°$ **[B1]**

(d) 225 **[B1]**

[Total Marks: 6]

771 **(a)** $90°$ **[B1]**

$270°$ **[B1]**

(b) Vertical asymptote: $x = -2.5$ (or $-\frac{5}{2}$) **[B1]**

Horizontal asymptote: $y = 2$ **[B1]**

(c) $4x - 1 = 0$ **[M1]**

0.25 (or $\frac{1}{4}$) **[A1]**

[Total Marks: 6]

772 **(a)** Vertical asymptote: $x = -2$ **[B1]**

Horizontal asymptote: $y = 1$ **[B1]**

(b) Substitute $y = 0$ to find x-intercept: $0 = \frac{3}{x+2} + 1 \implies x = -5$ **[M1]**

x-axis at $(-5, 0)$ and y-axis at $(0, 2.5)$ **[A1]**

(c) Two correct branches of the hyperbola sketched **[B1]**

Graph approaches $x = -2$ and $y = 1$ correctly without touching or crossing them **[B1]**

Graph passes through $(-5, 0)$ and $(0, 2.5)$ approximately **[B1]**

[Total Marks: 7]

773 **(a)** $x = -2$ **[B1]**

(b) $y = -4$ **[B1]**

(c) Sets $x = 0$ to find the y-intercept: $y = \frac{3}{0+2} - 4$ **[M1]**

y-intercept is $(0, -2.5)$ **[A1]**

Sets $y = 0$ to find the x-intercept: $0 = \dfrac{3}{x+2} - 4$ **[M1]**

$4 = \dfrac{3}{x+2} \implies 4(x+2) = 3 \implies 4x + 8 = 3$ **[M1]**

x-intercept is $(-1.25, 0)$ **[A1]**

(d) $x = 90°$ **[B1]**

$x = 270°$ **[B1]**

[Total Marks: 9]

774 (a) $b = 2$ **[B1]**

$c = -1$ **[B1]**

(b) Substitutes $(5, 0)$ into the equation: $\dfrac{a}{5-2} - 1 = 0$ **[M1]**

$a = 3$ **[A1]**

(c) Sets up the equation $\dfrac{3}{x-2} - 1 = -x + 5$ or uses GDC to find intersection of $y = f(x)$ and $y = -x + 5$ **[M1]**

$x = 3$ and $x = 5$ **[A1]**

[Total Marks: 6]

775 (a) $q = 2$ **[B1]**

$r = -1$ **[B1]**

(b) Substitutes $(3, 3)$, $q = 2$ and $r = -1$ into $f(x)$: $3 = \dfrac{p}{3-2} - 1$ **[M1]**

$p = 4$ **[A1]**

(c) Sets $f(x) = 0$

$\dfrac{4}{x-2} - 1 = 0$ [M1]

$$\frac{4}{x-2} = 1$$

x - 2 = 4[$M1$]

$(6, 0)$ **[A1]**

[Total Marks: 7]

776 (a) Two correct vertical asymptotes at $x = 90°$ and $x = 270°$ **[B1]**

Three correct branches passing through $(0, 0)$, $(180, 0)$, and $(360, 0)$ with correct general curvature **[B1]**

(b) $x = 90°$ **[B1]**

$x = 270°$ **[B1]**

(c) $\tan x = -1.5$ **[M1]**

$x = 123.7°$ **[A1]**

$x = 303.7°$ **[A1]**

[Total Marks: 7]

777 **(a)** $x = -0.5$ (or $-\frac{1}{2}$) **[B1]**

$x = 3$ **[B1]**

(b) $x = 2$ **[B1]**

(c) As x

$\pm\infty$, $h(x)$ increases or decreases without bound (does not approach a constant value) [B1]

Mentions that the degree of the numerator (2) is greater than the degree of the denominator (1), or that the curve approaches an oblique/sloping line $y = 2x - 1$ **[B1]**

(d) Finds local maximum at $(0.27, 4.07)$ and local minimum at $(3.73, 1.93)$ using GDC **[M1]**

$1.93 < k < 4.07$ **[A1]**

[Total Marks: 7]

778 **(a)** **(i)** $x = 1.5$ **[B1]**

(ii) $y = 2$ **[B1]**

(b) $y = 2$ is a horizontal asymptote so the curve never reaches $y = 2$ **[B1]**

(c) Vertical asymptote: $x = -1.5$ **[B1]**

Horizontal asymptote: $y = 2$ **[B1]**

(d) $\frac{5}{x - 1.5} + 2 = 3.4$ or $\frac{5}{x - 1.5} = 1.4$ **[M1]**

$x - 1.5 = \frac{5}{1.4}$ **[M1]**

5.07 **[A1]**

[Total Marks: 8]

779 **(a)** $q = 5$ **[B1]**

$r = -3$ **[B1]**

(b) Substitutes $x = 7, y = -2$ and their values of q and r into the equation: $-2 = \frac{p}{7 - 5} - 3$ **[M1]**

$-2 = \frac{p}{2} - 3 \implies \frac{p}{2} = 1$ **[M1]**

$p = 2$ **[A1]**

(c) Sets $y = \dfrac{2}{x-5} - 3$ and attempts to make x the subject, or swaps x and y **[M1]**

$y + 3 = \dfrac{2}{x-5} \implies x - 5 = \dfrac{2}{y+3}$ **[M1]**

$x = \dfrac{2}{y+3} + 5$ **[M1]**

$f^{-1}(x) = \dfrac{2}{x+3} + 5$ or equivalent **[A1]**

(d) $x = -3$ **[B1]**

$y = 5$ **[B1]**

(Strict follow through from their answers in part (a): $x = r, y = q$)

[Total Marks: 11]

780 **(a)** Translation **[B1]**

by vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$ **[B1]**

(b) $f(x) + 7$ **[B1]**

[Total Marks: 3]

781 **(a)** $x^2 - 5$ **[B1]**

(b) $(x-3)^2$ **[B1]**

(c) Translation **[B1]**

by vector $\begin{pmatrix} -4 \\ -1 \end{pmatrix}$ **[B1]**

[Total Marks: 4]

782 **(a)** $g(x) = f(x) + 7$ **[M1]**

$g(x) = 2x + 2$ **[A1]**

(b) Translation **[B1]**

by vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$ **[B1]**

[Total Marks: 4]

783 **(a)** Translation **[B1]**

By vector $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ **[B1]**

(b) $f(x-3) + 4$ **[B2]**

(Award [B1] for $f(x-3)$ or +4 seen)

(c) $(-3, -3)$ **[B1]**

[Total Marks: 5]

784 **(a)** Horizontal shift 3 seen **[B1]**

$\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ **[B1]**

(b) $(x-3)^2$ seen **[B1]**

$(x-3)^2 - 2$ **[B1]**

(c) $x = -5$ **[B1]**

$(-5, -1)$ **[B1]**

[Total Marks: 6]

785 **(a)** $(-2, 1)$ **[B1]**

(b) $(5, -1)$ **[B1]**

(c) **(i)** $a = 2$ **[B1]**

$b = -5$ **[B1]**

(ii) $(5, -6)$ **[B1]**

(Strict follow through from their vector in (c)(i) applied to $(3, -1)$)

[Total Marks: 5]

786 **(a)** Writes $g(x) = f(x-2) - 5$ or $(x-2)^2 - 4(x-2) + 1 - 5$ **[M1]**

Expands correctly: $(x^2 - 4x + 4) - 4x + 8 - 4$ **[M1]**

$x^2 - 8x + 8$ **[A1]**

(b) Completes the square: $f(x) = (x-2)^2 - 3$ and $h(x) = (x+1)^2 - 3$ **[M1]**

Identifies horizontal shift of -3 units **[A1]**

$\begin{pmatrix} -3 \\ 0 \end{pmatrix}$ **[A1]**

[Total Marks: 6]

787 **(a)** $g(x) = f(x+2) = 3(x+2)^2$ **[M1]**

$3(x^2 + 4x + 4)$ **[M1]**

$3x^2 + 12x + 12$ **[A1]**

(b) $(-2, -8)$ **[B2]**

(Allow B1 for each correct coordinate)

(c) Substitutes $x = 0$ into $h(x) = 3(x+2)^2 - 8$ (or similar valid method) **[M1]**

4 **[A1]**

[Total Marks: 6]

788 **(a)** Translation by 2 units left or 3 units down **[M1]**

Correct line segments joining $(-3,-2)$, $(-1,1)$, $(1,-2)$, and $(2,-1)$ **[A1]**

(b) Translation **[B1]**

Vector $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$ **[B1]**

(c) $f(x)-5$ **[B1]**

(d) $x = 4+3 = 7$ or $y = 2+4 = 6$ **[M1]**

$(7,6)$ **[A1]**

[Total Marks: 7]

789 **(a)** $(4,-1)$ **[B1]**

(b) $f(x+3)+2$ **[B2]**

(B1 for $f(x+3)$ and B1 for $+2$)

(c) $f(-k) = 0$, so $-k = 2$ or $-k = 6$ **[M1]**

$k = -2$ and $k = -6$ **[A1]**

(d) $f(x-3)$ **[B1]**

[Total Marks: 6]

790 **(a)** Correct vertical shift applied: $(x^2-4)+5$ **[M1]**

x^2+1 **[A1]**

(b) $(-3,-4)$ **[B2]**

If 0 scored, award 1 mark for x-coordinate $= -3$ or y-coordinate $= -4$ **[B1]**

[Total Marks: 4]

791 **(a)** Translation **[B1]**

by vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$ **[B1]**

(b) Recognising that $g(x) = f(x-4)$ and substituting: $(x-4)^3 - 3(x-4)$ **[M1]**

Expanding $(x-4)^3 = x^3 - 12x^2 + 48x - 64$ **[B2]**

If 0 scored for cubic expansion, award 1 mark for $(x-4)(x^2-8x+16)$ **[M1]**

$x^3 - 12x^2 + 45x - 52$ **[A1]**

[Total Marks: 6]

792 **(a)** $g(x) = f(x+4)$ **[M1]**

$(x+4)^2 - 2(x+4) - 3$ **[M1]**

$x^2 + 8x + 16 - 2x - 8 - 3$ **[M1]**

$x^2 + 6x + 5$ **[A1]**

(b) $h(x) = x^2 - 2x - 3 - 5 = x^2 - 2x - 8$ **[M1]**

$(x-4)(x+2) = 0$ **[M1]**

$x = 4$ and $x = -2$ **[A1]**

[Total Marks: 7]

793 **(a)** Local maximum plotted at $(1, 4)$ **[B1]**

Local minimum plotted at $(3, 0)$ and y-intercept or key point $(2, 2)$ correct **[B1]**

Smooth cubic curve drawn through translated points with correct shape **[B1]**

(b) x-coordinate $= 1 - k$ **[B1]**

y-coordinate $= -1 - 2 = -3$, giving $(1-k, -3)$ **[B1]**

(c) $p = -3$ **[B1]**

$p = 1$ **[B1]**

[Total Marks: 7]

794 **(a)** $2(x^2 - 4x) + 11$ or $a = 2$ **[B1]**

$2[(x-2)^2 - 4] + 11$ or $h = -2$ **[B1]**

$2(x-2)^2 + 3$ **[A1]**

(b) Completing the square for $g(x)$: $2(x+1)^2 - 5$ OR equating $f(x-p) + q = 2x^2 + 4x - 3$ **[M1]**

Identifying the minimum point of $f(x)$ as $(2, 3)$ and the minimum point of $g(x)$ as $(-1, -5)$ **[M1]**

Horizontal shift $= -1 - 2 = -3$ **[A1]**

Vertical shift $= -5 - 3 = -8$, giving vector $\begin{pmatrix} -3 \\ -8 \end{pmatrix}$ **[A1]**

[Total Marks: 7]

795 **(a)** Finds $(x-3)^2$ as part of completing the square: $f(x) = (x-3)^2 - 9 + 14$ **[M1]**

$f(x) = (x-3)^2 + 5$ **[M1]**

Translation vector $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ **[A1]**

(b) Substitutes $x - 4$ into $g(x)$: $g(x-4) = \dfrac{3}{(x-4)+2} - 5$ **[M1]**

Adds 2 to the function: $h(x) = \dfrac{3}{x-2} - 5 + 2$ **[M1]**

$h(x) = \dfrac{3}{x-2} - 3$ or equivalent single fraction $\dfrac{9-3x}{x-2}$ **[A1]**

(c) $x = 2$ **[B1]**

$y = -3$ **[B1]**

(Strict follow through from their $h(x)$ in part (b))

[Total Marks: 8]

796 **(a)** **(i)** $\dfrac{20}{x} - 4$ **[B1]**

(ii) $\dfrac{20}{x} - 4 = 0$ **[M1]**

$x = 5$ **[A1]**

(b) $h(x) = f(x-k) = \dfrac{20}{x-k}$ **[M1]**

$\dfrac{20}{8-k} = 5$ **[M1]**

$k = 4$ **[A1]**

(c) Vertical asymptote: $x = 3$ **[B1]**

Horizontal asymptote: $y = 7$ **[B1]**

[Total Marks: 8]

797 **(a)** **(i)** $f(x+2) - 6$ **[B1]**

(ii) $2(x+2)^2 - 5(x+2) + 1 - 6$ **[M1]**

$2(x^2 + 4x + 4) - 5x - 15$ **[M1]**

$2x^2 + 3x - 7$ **[A1]**

(b) $2(x-p)^2 - 5(x-p) + 1 + q = 2x^2 - 13x + 18$ **[M1]**

$-4p - 5 = -13$ giving $p = 2$ **[A1]**

$2(2)^2 + 5(2) + 1 + q = 18$ giving $q = -1$ **[M1]**

$\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ **[A1]**

(c) $2x^2 - 5x + 1 = 2x^2 + 3x - 7$ **[M1]**

$8x = 8$ giving $x = 1$ **[A1]**

$(1, -2)$ **[A1]**

[Total Marks: 11]

798 **(a)** $\log_{10} x$ OR $\log x$ **[B1]**

(b) -2 **[B1]**

(c) $\log_{10} 45$ OR $\log 45$ **[B1]**

[Total Marks: 3]

799 **(a)** $\log 1000 = 3$ **[B1]**

(b) 0.01 or $\frac{1}{100}$ **[B1]**

(c) $\frac{\log 12}{\log 5}$ **[B1]**

[Total Marks: 3]

800 **(a)** 2 **[B1]**

(b) -3 **[B1]**

(c) x **[B1]**

[Total Marks: 3]

801 **(a)** 4 **[B1]**

(b) $\log_5 125 = 3$ **[B1]**

(c) 64 **[B1]**

[Total Marks: 3]

802 **(a)** 32 **[B1]**

(b) -2 **[B1]**

(c) $t = \log_k P$ **[B1]**

[Total Marks: 3]

803 **(a)** $\log_5 125 = 3$ **[B1]**

(b) 3 **[B1]**

(c) $x = 2^5$ **[M1]**

32 **[A1]**

[Total Marks: 4]

804 **(a)** $\log 100000 = 5$ **[B1]**

(b) 10^3 **[M1]**

1000 **[A1]**

(c) $\log x$ **[B1]**

[Total Marks: 4]

805 **(a)** **(i)** 3
3 **[B1]**

(ii) 5
5 **[B1]**

(b)
=
3
4
=
81
x=3
4
=81 **[B1]**

(c) log
logx (or
log
10
log
10
x) **[B1]**

[Total Marks: 4]

806 **(a)** **(i)** 5 **[B1]**

(ii) 0.01 or $\frac{1}{100}$ **[B1]**

(iii) 6 **[B1]**

(b) $\log x$ or $\log_{10} x$ **[B1]**

[Total Marks: 4]

807 **(a)** 3 **[B1]**

(b) 10^{-2} or $\frac{1}{100}$ **[M1]**

0.01 **[A1]**

(c) $\log_5 125$ **[B1]**

[Total Marks: 4]

808 **(a)** $\log_{10} 10000 = 4$ (or $\log 10000 = 4$) **[B1]**

(b) Uses index form $x = 5^3$ **[M1]**

125 **[A1]**

(c) $a = 7$ **[B1]**

$b = 25$ **[B1]**

[Total Marks: 5]

809 **(a)** 4 **[B1]**

(b) Writes $\frac{1}{9} = 3^{-2}$ **[M1]**

-2 **[A1]**

(c) Writes $y = \log_3 x$ as $x = 3^y$ (or $y = 3^x$) **[M1]**

3^x **[A1]**

[Total Marks: 5]

810 **(a)** **(a)** $p(1) + q = 8$ or $p(2) + q = 11$ **[B1]**

A valid method to solve simultaneous equations e.g. $(2p + q) - (p + q) = 11 - 8$ **[M1]**

$p = 3$, $q = 5$ **[A1]**

(b) $3 \log x + 5 = 23 \implies 3 \log x = 18 \implies \log x = 6$ **[M1]**

$1\,000\,000$ or 10^6 **[A1]**

(c) $x = 3 \log y + 5 \implies \log y = \dfrac{x - 5}{3}$ **[M1]**

$10^{\frac{x-5}{3}}$ **[A1]**

[Total Marks: 7]

811 **(a)** **(a)** $\log_6 y$ **[B1]**

(b) $x = \dfrac{\log 20}{\log 3}$ or $x \log 3 = \log 20$ **[M1]**

2.73 **[A1]**

[Total Marks: 3]

812 **(a)** 250×1.08^{10} **[M1]**

540 **[A1]**

(b) $250 \times 1.08^t = 1000$ **[M1]**

$1.08^t = 4 \implies t = \dfrac{\log 4}{\log 1.08}$ **[M1]**

18.0 **[A1]**

[Total Marks: 5]

813 **(a)** 15000×0.8^4 **[M1]**

6144 **[A1]**

(b) $7500 = 15000 \times 0.8^t$ **[M1]**

$0.5 = 0.8^t \implies t = \dfrac{\log 0.5}{\log 0.8}$ **[M1]**

3.11 **[A1]**

[Total Marks: 5]

814 **(a)** $25000 \times (1 - 0.12)^4$ or 25000×0.88^4 **[M1]**

14992 **[A1]**

(b) $25000 \times 0.88^n = 10000$ or $0.88^n = 0.4$ **[M1]**

$n = \dfrac{\log 0.4}{\log 0.88}$ **[M1]**

$n = 7.16...$ **[A1]**

8 **[B1]**

[Total Marks: 5]

815 **(a)** $2x = \dfrac{\log 50}{\log 3}$ **[M1]**

$2x = 3.56\ldots$ **[A1]**

1.78 **[A1]**

(b) $\log y = x - 4$ **[M1]**

$\log y + 4$ **[A1]**

[Total Marks: 5]

816 **(a)** **(a)** 5000 **[B1]**

(b) $5000 \times 1.12^t = 20000$ **[M1]**

$1.12^t = 4$ **[M1]**

$t = \dfrac{\log 4}{\log 1.12}$ **[M1]**

12.2 **[A1]**

[Total Marks: 5]

817 **(a)** 8000×1.035^n **[B1]**

(b) Sets up the inequality or equation: $8000 \times 1.035^n > 12000$ OR $1.035^n = 1.5$ **[M1]**

Uses logarithms to solve for n: $n = \dfrac{\log_{10} 1.5}{\log_{10} 1.035}$ **[M1]**

12 **[A1]**

(c) $x = \dfrac{\log_{10} 55}{\log_{10} 2}$ **[M1]**

5.78 **[A1]**

[Total Marks: 6]

818 **(a)** $x \log 3 = \log 20$ **[M1]**

$x = \dfrac{\log 20}{\log 3}$ **[M1]**

2.73 **[A1]**

(b) $100 = 500 \times 0.85^t \implies 0.85^t = 0.2$ **[M1]**

$t \log 0.85 = \log 0.2 \implies t = \dfrac{\log 0.2}{\log 0.85}$ **[M1]**

9.90 **[A1]**

[Total Marks: 6]

819 **(a)** 4500×1.08^5 **[M1]**

6612 **[A1]**

(b) $4500 \times 1.08^t = 10000$ **[M1]**

$1.08^t = \dfrac{10000}{4500}$ oe **[M1]**

$t = \dfrac{\log(20/9)}{\log 1.08}$ **[M1]**

10.38 **[A1]**

[Total Marks: 6]

820 **(a)** 4500×1.08^5 **[M1]**

6611.98 (or 6610 to 3 significant figures) **[A1]**

(b) $4500 \times 1.08^t = 10000$ **[M1]**

$1.08^t = \dfrac{10000}{4500}$ or $1.08^t = \dfrac{20}{9}$ **[M1]**

$t = \dfrac{\log\left(\dfrac{20}{9}\right)}{\log 1.08}$ **[M1]**

10.4 (10.376...) **[A1]**

[Total Marks: 6]

821 **(a)** 0.78
0.78 (accept
0.75
0.75 to

0.80
0.80) **[B1]**

(b) 3.2
3.2 (accept
3.1
3.1 to
3.3
3.3) **[B1]**

(c) (
+
1
)
log
3
=
log
20
(x+1)log3=log20 or
+
1
=
log
20
log
3
x+1=
log3
log20
[M1]

+
1
=
2.7268
...
x+1=2.7268... **[M1]**

1.73
1.73 **[A1]**

(d) 2
3
=
10
1.4
2x3=10
1.4
(or
2
3
=
25.118
...
2x3=25.118...) **[M1]**

14.1
14.1 **[A1]**

[Total Marks: 7]

822 **(a)** Substitutes $t = 5$ into the formula: 400×0.88^5 **[M1]**

211.1 **[A1]**

(b) Sets up the equation $400 \times 0.88^t = 200$ and simplifies to $0.88^t = 0.5$ **[M1]**

$t = \dfrac{\log 0.5}{\log 0.88}$ **[M1]**

5.42 **[A1]**

(c) Sets up the equation $800 \times k^{10} = 300$ **[M1]**

$k^{10} = 0.375$ **[M1]**

0.907 **[A1]**

[Total Marks: 8]

823 **(a)** 8000×1.045^5 **[M1]**

9969.46 (accept 9969 or 9970) **[A1]**

(b) $1.045^t = \dfrac{15000}{8000}$ or 1.875 **[M1]**

$t = \dfrac{\log 1.875}{\log 1.045}$ or 14.28... **[M1]**

15 **[A1]**

(c) $\log(7^{x-1}) = \log 250$ or $x - 1 = \log_7 250$ **[M1]**

$x - 1 = \dfrac{\log 250}{\log 7}$ or $x - 1 = 2.837...$ **[M1]**

3.84 **[A1]**

[Total Marks: 8]

824 **(a)** 12800 **[B1]**

(b) $800 \times 2^t = 102400 \implies 2^t = 128$ **[M1]**

$t = 7$ **[A1]**

(c) $1500 \times 3^t = 20000 \implies 3^t = \dfrac{40}{3}$ (or 13.33...) **[M1]**

Uses logarithms to solve $3^t = \dfrac{40}{3}$, e.g. $t = \dfrac{\log(40/3)}{\log 3}$ **[M1]**

2.36 (from 2.358...) **[A1]**

(d) $2000 \times 1.05^n = 4000 \implies 1.05^n = 2$ **[M1]**

$n = \dfrac{\log 2}{\log 1.05}$ (or 14.2...) **[M1]**

15 **[A1]**

[Total Marks: 9]

825 **(a)** $y = 4 \times 10^{2x} - 7 \implies y + 7 = 4 \times 10^{2x}$ **[M1]**

$\dfrac{y+7}{4} = 10^{2x} \implies 2x = \log\left(\dfrac{y+7}{4}\right)$ **[M1]**

$\dfrac{1}{2}\log\left(\dfrac{x+7}{4}\right)$ **[A1]**

(b) $4 \times 10^{2x} - 7 = 53 \implies 4 \times 10^{2x} = 60$ **[M1]**

$10^{2x} = 15 \implies 2x = \log 15$ **[M1]**

0.588 **[A1]**

[Total Marks: 6]

826 **(a)** Exponential curve passing through $(0, 1)$ and $(1, 2)$ **[B1]**

Curve passing through $(3, 8)$ and approaching the negative x-axis as an asymptote **[B1]**

(b) $\dfrac{\log x}{\log 2}$ (or $\log_2 x$) **[B1]**

(c) Logarithmic curve passing through $(1, 0)$ and $(2, 1)$ **[B1]**

Curve passing through $(8, 3)$ and approaching the negative y-axis as an asymptote **[B1]**

(d) $x = \dfrac{\log 12}{\log 2}$ **[M1]**

3.58 **[A1]**

[Total Marks: 7]

827 **(a)** **(i)** 250 **[B1]**

(ii) $250 \times 0.92^t = 50$ or $0.92^t = 0.2$ **[M1]**

$t = \dfrac{\log 0.2}{\log 0.92}$ **[M1]**

19.3 **[A1]**

(b) $800 \times 1.06^n > 2000$ or $1.06^n = 2.5$ **[M1]**

$n = \dfrac{\log 2.5}{\log 1.06}$ $(= 15.7\ldots)$ **[M1]**

16 **[A1]**

[Total Marks: 7]

828 **(a)** x-intercept $= (1, 0)$ **[B1]**

Asymptote: $x = 0$ **[B1]**

(b) 5.0 (accept 4.9 to 5.1) **[B1]**

(c) Takes logarithms of both sides: $(2x-1)\log 3 = (x+2)\log 5$ **[M1]**

Expands and collects x terms: $x(2\log 3 - \log 5) = 2\log 5 + \log 3$ **[M1]**

$x = \dfrac{2\log 5 + \log 3}{2\log 3 - \log 5}$ (or $\dfrac{\log 75}{\log 1.8}$) **[M1]**

7.34 (7.3414...) **[A1]**

[Total Marks: 7]

829 **(a)** Substitutes given values into the formula: $250 = 800 \times 10^{-15k}$ **[M1]**

Rearranges to isolate the exponential: $10^{-15k} = 0.3125$ **[M1]**

Takes logarithms to solve for k: $-15k = \log_{10}(0.3125) \implies k = \dfrac{\log_{10}(0.3125)}{-15} = 0.03367...$ **[A1]**

(b) Substitutes $m = 50$ and $k = 0.03367...$ into the formula: $50 = 800 \times 10^{-0.03367t}$ **[M1]**

Isolates the exponential and takes logarithms: $-0.03367t = \log_{10}\left(\dfrac{50}{800}\right)$ **[M1]**

35.8 (accept 35.7 to 35.8) **[A1]**

(c) Rearranges to $10^{-kt} = \dfrac{m}{800}$ and converts to logarithmic form: $-kt = \log_{10}\left(\dfrac{m}{800}\right)$ **[M1]**

$t = -\dfrac{1}{k}\log_{10}\left(\dfrac{m}{800}\right)$ OR $t = \dfrac{1}{k}\log_{10}\left(\dfrac{800}{m}\right)$ **[A1]**

[Total Marks: 8]

830 **(a)** $g(100) = \log 100 = 2$ **[M1]**

$f(2) = 5 \times 10^2 = 500$ **[A1]**

(b) $y = 5 \times 10^x$ leads to $10^x = \dfrac{y}{5}$ **[M1]**

$\log\left(\dfrac{x}{5}\right)$ or $\log x - \log 5$ **[A1]**

(c) $5 \times 10^x = 200$ leads to $10^x = 40$ **[M1]**

$x = \log 40 = 1.602...$ **[A1]**

1.60 **[B1]**

(d) $g(f(x)) = \log(5 \times 10^x)$ **[M1]**

$= \log 5 + \log 10^x = \log 5 + x$ **[M1]**

$x + \log 5$ **[A1]**

[Total Marks: 8]

831 **(a)** $5 \times 3^{-2} - 2$ **[M1]**

$-\dfrac{13}{9}$ **[A1]**

(b) $y = 5 \times 3^x - 2 \implies y + 2 = 5 \times 3^x$ **[M1]**

$3^x = \dfrac{y+2}{5}$ **[M1]**

$\dfrac{\log\left(\frac{x+2}{5}\right)}{\log 3}$ **[A1]**

(c) $5 \times 3^x - 2 = 500 \implies 3^x = 100.4$ **[M1]**

$x = \dfrac{\log 100.4}{\log 3}$ **[M1]**

4.20 **[A1]**

[Total Marks: 8]

832 **(a)** $3 = \log_a 8$ **[M1]**

$a^3 = 8 \implies a = 2$ **[A1]**

(b) **(i)** Smooth exponential curve passing through $(0, 1)$ **[B1]**

Correct shape passing through $(1, 2)$, $(2, 4)$ and $(3, 8)$ **[B1]**

(ii) $y = 0$ (accept the x-axis) **[B1]**

(c) Rearranges to $\log_2 x = 8 - x$ **[M1]**

Draws the line $y = 8 - x$ correctly on the grid (passing through $(0, 8)$ and $(8, 0)$) **[M1]**

$x = 5.5$ (accept 5.4 to 5.6) **[A1]**

[Total Marks: 8]

833 **(a)** Substitutes $x = 3$: $4 \times 10^{1.5}$ **[M1]**

126 **[A1]**

(b) Sets $y = 4 \times 10^{0.5x}$ and rearranges to $\dfrac{y}{4} = 10^{0.5x}$ **[M1]**

$\log\left(\dfrac{y}{4}\right) = 0.5x$ **[M1]**

$2\log\left(\dfrac{x}{4}\right)$ **[A1]**

(c) $4 \times 10^{0.5x} = 200$

$10^{0.5x} = 50$ [M1]

$0.5x = \log 50$ **[M1]**

3.40 **[A1]**

[Total Marks: 8]

834 **(a)** $a \times b^0 = 12 \implies a = 12$ **[B1]**

$12 \times b^3 = 96 \implies b^3 = 8$ **[M1]**

$b = 2$ **[A1]**

(b) Sets up $y = 12 \times 2^x \implies 2^x = \dfrac{y}{12}$ **[M1]**

$x \log 2 = \log\left(\dfrac{y}{12}\right)$ **[M1]**

$f^{-1}(x) = \dfrac{\log(x/12)}{\log 2}$ **[A1]**

(c) $12 \times 2^x = 500 \implies 2^x = \dfrac{125}{3}$ (41.666...) **[M1]**

$x = \dfrac{\log(125/3)}{\log 2}$ **[M1]**

5.38 **[A1]**

[Total Marks: 9]

835 **(a)**
24000
×
0.85
4
24000×0.85
4
(
=
12528.15
=12528.15) **[M1]**

12528
12528 **[A1]**

(b) $24000 \times 0.85^t = 9000$ or $0.85^t = 0.375$ **[M1]**

$t = \dfrac{\log 0.375}{\log 0.85}$ **[M1]**

6.04 **[A1]**

(c) $1.068^n = 2$ (or $1.068^n > 2$) **[M1]**

$n = \dfrac{\log 2}{\log 1.068} = 10.536\ldots$ **[M1]**

11 **[A1]**

(d) $2^y = \dfrac{x-3}{5}$ **[M1]**

$y = \log_2\left(\dfrac{x-3}{5}\right)$ or $y \log 2 = \log\left(\dfrac{x-3}{5}\right)$ **[M1]**

$log_2($\dfrac{x-3}{5}$ or
log
(
3
5
)
log
2
log2

log(
5
x3
)

[A1]

[Total Marks: 11]

836 **(a)** Substitutes $(0, 7)$ to find A: $7 = A \times b^0 \implies A = 7$ **[B1]**

Substitutes $(3, 56)$ with $A = 7$: $7 \times b^3 = 56 \implies b^3 = 8$ **[M1]**

$b = 2$ **[A1]**

(b) Sets $y = 7 \times 2^x$ **[M1]**

Rearranges to make 2^x the subject: $2^x = \dfrac{y}{7}$ **[M1]**

Converts to logarithmic form: $\log_2\left(\dfrac{x}{7}\right)$ (or equivalent correct form) **[A1]**

(c) **(i)** $\dfrac{600}{7}$ **[B1]**

(ii) $x = \dfrac{\log(600/7)}{\log 2}$ **[M1]**

6.421 (from 6.4214...) **[A1]**

(d) **(i)** Sets $4 = \log_c 81 \implies c^4 = 81$ **[M1]**

$c = 3$ **[A1]**

(ii) $\dfrac{1}{9}$ (or 3^{-2}) **[B1]**

[Total Marks: 12]

837 **(a)** x-coordinate 3 **[B1]**

y-coordinate 5 **[B1]**

(b) Base $AB = 4$ and height $= 3$ **[M1]**

12 **[A1]**

[Total Marks: 4]

838 **(a)** $(1, 2)$ **[B1]**

(b) Identifies the base length $PQ = 10$ **[M1]**

Sets up the area equation:

$$\frac{1}{2} \times 10 \times h = 35$$

h = 7[$M1$]

$(0, 9)$ **[A1]**

[Total Marks: 4]

839 **(a)** $A = (-3, 2)$ **[B1]**

$B = (2, -3)$ **[B1]**

(b) $(3, 2)$ **[B1]**

(c) $D = (-2, -3)$ **[B1]**

Area $= 5 \times 5 = 25$ (accept $D = (3, -3)$ if order $ACBD$ used, or strict $ABCD$ giving $D = (-3, -3)$ and Area $= 6 \times 5 = 30$) — note: for $A(-3, 2)$, $C(3, 2)$, $B(2, -3)$, $ABCD$ cannot be a rectangle; re-indexing vertices as $ACBD$ gives $D(-3, -3)$ and Area $= 6 \times 5 = 30$ **[B1]**

[Total Marks: 5]

840 **(a)** All three points A, B, and C plotted accurately on the grid **[B1]**

(b) Identifies the horizontal and vertical translation, e.g. $\vec{BA} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ OR correctly plots D on the grid **[M1]**

$(3, 5)$ **[A1]**

(c) Calculates the midpoint of AC or BD: $\left(\frac{-1+6}{2}, \frac{3+1}{2}\right)$ **[M1]**

$(2.5, 2)$ **[A1]**

[Total Marks: 5]

841 **(a)** $A = (-2, 3)$ **[B1]**

$C = (1, -2)$ **[B1]**

(b) Horizontal shift of 5 units or vertical shift of 5 units identified **[M1]**

$(-4, -2)$ **[A1]**

(c) Base $= 5$ and perpendicular height $= 5$ **[M1]**

25 **[A1]**

[Total Marks: 6]

842 **(a)** Uses vector shift $\overrightarrow{QR} = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$ applied to $P(-2, 1)$ **[M1]**

$(0, -4)$ **[A1]**

(b) $\left(\frac{-2+6}{2}, \frac{1+(-2)}{2}\right)$ **[M1]**

$(2, -0.5)$ or $\left(2, -\frac{1}{2}\right)$ **[A1]**

(c) Encloses $PQRS$ in bounding rectangle of width 8 and height 5 (40) and subtracts corner triangles: $40 - 2\left(\frac{1}{2} \times 6 \times 2\right) - 2\left(\frac{1}{2} \times 2 \times 5\right)$ **[M1]**

34 **[A1]**

[Total Marks: 6]

843 **(a)** All four points P, Q, R, and S plotted accurately **[B1]**

Points joined with straight line segments to form kite $PQRS$ **[B1]**

(b) $(1, 2)$ **[B1]**

(c) Diagonal lengths $PR = 9$ and $QS = 6$ **[M1]**

$\frac{1}{2} \times 9 \times 6 = 27$ **[A1]**

(d) $P' = (1, -5)$ **[B1]**

$Q' = (-4, 2)$ **[B1]**

[Total Marks: 7]

844 **(a)** Identifies the area of the enclosing square as a^2 and the three outer right-angled triangles as $\frac{1}{2}ab$, $\frac{1}{2}ab$, and $\frac{1}{2}(a-b)^2$ **[M1]**

Sets up the subtraction for the area of OPQ: Area $= a^2 - \frac{1}{2}ab - \frac{1}{2}ab - \frac{1}{2}(a-b)^2$ **[M1]**

Correctly expands brackets and simplifies algebraically to $\frac{1}{2}(a^2 - b^2)$ without errors **[A1]**

(b) Sets up the distance squared equation: $(a-b)^2 + (b-a)^2 = (4\sqrt{2})^2$

$2(a-b)^2 = 32$ [M1]

Solves to find $a - b = 4$ (accepting $a > b$) **[A1]**

Uses the area equation:

$$\frac{1}{2}(a^2 - b^2) = 24$$

$a^2 - b^2 = 48$ [B1]

Factorises to $(a-b)(a+b) = 48$, substitutes $a - b = 4$ to obtain $a + b = 12$ **[M1]**

$a = 8$ and $b = 4$ **[A1]**

[Total Marks: 8]

845 **(a)** Width $= x - (-2) = x + 2$ and Height $= y - (-4) = y + 4$ **[M1]**

$2(x+2) + 2(y+4) = 30 \implies 2x + 2y + 12 = 30 \implies x + y = 9$ **[A1]**

(b) Area $= (x+2)(y+4) = 54$ **[M1]**

Substitute $y = 9 - x$ to give $(x+2)(13-x) = 54$ **[M1]**

$x^2 - 11x + 28 = 0$ or equivalent three-term quadratic **[M1]**

$(4, 5)$ and $(7, 2)$ **[A1]**

(c) $(-2, 2)$ **[B1]**

$(7, -4)$ **[B1]**

[Total Marks: 8]

846 **(a)** Sets the x-coordinate of the midpoint to zero: $\frac{3k+k+4}{2} = 0$ **[M1]**

-1 **[A1]**

(b) Substitutes $k = -1$ to find $A(-3, 1)$ and $B(3, 5)$ **[M1]**

$AB^2 = (3-(-3))^2 + (5-1)^2 = 6^2 + 4^2 = 52$ **[M1]**

$2\sqrt{13}$ **[A1]**

(c) Lets C be $(x, 0)$ and sets up equation $AC^2 = BC^2$:
$(x+3)^2 + (0-1)^2 = (x-3)^2 + (0-5)^2$ **[M1]**

Expands both sides correctly: $x^2 + 6x + 9 + 1 = x^2 - 6x + 9 + 25$ **[M1]**

Rearranges to solve for x: $12x = 24$

x = 2[$M1$]

$(2, 0)$ **[A1]**

[Total Marks: 9]

847 **(a)** $(-6, 0)$ **[B1]**

(b) Identifies radius $OB = 6$ and angle from positive x-axis as $60°$ **[M1]**

$x = 6\cos 60° = 3$ and $y = 6\sin 60°$ $(= 3\sqrt{3})$ **[M1]**

$(3, 5.20)$ **[A1]**

(c) Calculates length $AD = 12$, and finds $AP = \frac{1}{4} \times 12 = 3$ **[M1]**

$(3, 0)$ **[A1]**

[Total Marks: 6]

848 **(a)** $A(0, 6)$ **[B1]**

$B(4, 0)$ **[B1]**

(b) $(4, 6)$ **[B1]**

(c) $2y + 3(2) = 12$ **[M1]**

$(2, 3)$ **[A1]**

(d) Base $OB = 4$ and height of $\triangle OPB = 3$ (the y-coordinate of P) **[M1]**

6 **[A1]**

[Total Marks: 7]

849 **(a)** Substitutes $y = 0$ to find $A(6, 0)$ **[B1]**

Substitutes $x = 0$ to find $B(0, 9)$ **[B1]**

(b) Recognises the change in x is -6 and change in y is 9 from A to B **[M1]**

Forms a valid ratio equation for either coordinate, e.g. x-coordinate $= 6 + \frac{2}{3}(-6)$ OR y-coordinate $= 0 + \frac{2}{3}(9)$ **[M1]**

$(2, 6)$ **[A1]**

(c) Substitutes $x = 4$ and $y = 3$ into the equation: $3(4) + 2(3)$ **[M1]**

Shows $12 + 6 = 18$ and clearly concludes 'Yes' **[A1]**

[Total Marks: 7]

850 (a) Sets $2k - 1 < 0$ and $k + 3 > 0$, giving $-3 < k < 0.5$ **[M1]**

$-2, -1, 0$ **[A1]**

(b) Substitutes coordinates of A into $y = 2x - 4$: $k + 3 = 2(2k - 1) - 4$ **[M1]**

$3k = 9 \implies k = 3$ **[A1]**

$B = (2, 4)$ **[A1]**

(c) Sets x-coordinate of midpoint to 0: $\frac{(2k-1)+(5-k)}{2} = 0$ or $k + 4 = 0$ **[M1]**

-4 **[A1]**

[Total Marks: 7]

851 (a) $h = -3$ **[B1]**

Second quadrant (accept 2nd or II) **[B1]**

(b) Width $AB = 5 - (-3) = 8$ **[M1]**

$8(k + 1) = 52$ or $k + 1 = 6.5$ **[M1]**

5.5 **[A1]**

(c) $x = \frac{-3+5}{2} = 1$ or $y = \frac{-1+5.5}{2} = 2.25$ **[M1]**

$(1, 2.25)$ **[A1]**

(d) $p = 0$ **[B1]**

$\frac{1}{2} \times 8 \times (q + 1) = 18$ or $q + 1 = 4.5$ **[M1]**

$(0, 3.5)$ **[A1]**

[Total Marks: 10]

852 $\frac{5-(-3)}{6-2}$ or equivalent **[M1]**

2 **[A1]**

[Total Marks: 2]

853 **(a)** Evidence of using $\dfrac{\text{change in } y}{\text{change in } x}$ with two correct points from the graph, e.g., $\dfrac{4-(-2)}{3-0}$ **[M1]**

2 **[A1]**

(b) $(0, -2)$ **[B1]**

[Total Marks: 3]

854 **(a)** Uses gradient formula: $\dfrac{-4-8}{4-(-2)}$ or $\dfrac{8-(-4)}{-2-4}$ **[M1]**

-2 **[A1]**

(b) Identifies coordinates of two clear points on the line, e.g., $(0, 1)$ and $(4, 4)$ **[M1]**

$\dfrac{3}{4}$ or 0.75 **[A1]**

[Total Marks: 4]

855 **(a)** Uses the gradient formula: $\dfrac{-5-7}{2-(-4)}$ **[M1]**

-2 **[A1]**

(b) Sets up the equation: $\dfrac{c-7}{0-(-4)} = -2$ **[M1]**

-1 **[A1]**

[Total Marks: 4]

856 **(a)** Uses difference in y over difference in x, e.g., $\dfrac{4-(-2)}{1-(-3)}$ **[M1]**

$\dfrac{3}{2}$ or 1.5 **[A1]**

(b) $\dfrac{-7-5}{4-(-2)}$ or $\dfrac{5-(-7)}{-2-4}$ **[M1]**

-2 **[A1]**

[Total Marks: 4]

857 **(a)** Correct vertical and horizontal change seen, e.g. $\dfrac{6}{4}$ or $\dfrac{3}{2}$ **[M1]**

1.5 (or $\dfrac{3}{2}$) **[A1]**

(b) Correct vertical and horizontal change seen, e.g. $\dfrac{-3}{6}$ **[M1]**

-0.5 (or $-\dfrac{1}{2}$) **[A1]**

(c) $\dfrac{-9-7}{5-(-3)}$ **[M1]**

-2 **[A1]**

[Total Marks: 6]

858 **(a)** **(i)** $\dfrac{3k-1-k}{5-2}$ **[M1]**

$\dfrac{2k-1}{3}$ **[A1]**

(ii) $\dfrac{2k-1}{3} = 4$ or $2k-1=12$ **[M1]**

6.5 or $\dfrac{13}{2}$ **[A1]**

[Total Marks: 4]

859 **(a)** $\dfrac{k-5}{4-(-2)} = -\dfrac{1}{2}$ **[M1]**

$k-5=-3$ **[M1]**

2 **[A1]**

(b) Gradient of $BC = \dfrac{-1-2}{10-4} = -\dfrac{1}{2}$ **[M1]**

Gradient of AB = gradient of BC and point B is common to both segments **[A1]**

(c) $\dfrac{(3p-1)-5}{p-(-2)} = 2$ **[M1]**

$3p-6=2(p+2)$ **[M1]**

10 **[A1]**

[Total Marks: 8]

860 **(a)** $\dfrac{(a+4)-3a}{5a-2a}$ or equivalent **[M1]**

$\dfrac{4-2a}{3a}$ **[A1]**

(b) States or implies gradient of PQ is $-\dfrac{1}{3}$ **[B1]**

$\dfrac{4-2a}{3a} = -\dfrac{1}{3}$ or $3(4-2a)=-3a$ **[M1]**

4 **[A1]**

(c) Gradient $AB = \dfrac{8-2}{3-(-1)}$ or $\dfrac{6}{4}$ or $\dfrac{3}{2}$ **[M1]**

$\dfrac{23-8}{c-3} = \dfrac{3}{2}$ or $\dfrac{23-2}{c-(-1)} = \dfrac{3}{2}$ **[M1]**

13 **[A1]**

[Total Marks: 8]

861 **(a)** $\dfrac{(a+6)-3a}{4-(-2)}$ **[M1]**

$\dfrac{6-2a}{6}$ **[M1]**

$\dfrac{3-a}{3}$ or $1-\dfrac{a}{3}$ **[A1]**

(b) $\dfrac{3-a}{3}=3$ oe **[M1]**

$a=-6$ **[A1]**

(c) $P=(-2,-18)$ and $Q=(4,0)$ **[M1]**

$\left(\dfrac{-2+4}{2},\dfrac{-18+0}{2}\right)$ **[M1]**

$(1,-9)$ **[A1]**

[Total Marks: 8]

862 **(a)** $\dfrac{k-2}{4}$ **[B1]**

(b) Finds an expression for the gradient of BC (e.g., $\dfrac{k}{k-5}$) and equates to the gradient of AB: $\dfrac{k-2}{4}=\dfrac{k}{k-5}$ **[M1]**

Expands brackets correctly: $(k-2)(k-5)=4k$

$\mathrm{k}^2-7k+10=4k$ [M1]

Rearranges to the given equation $k^2-11k+10=0$ with no mathematical errors seen **[A1]**

(c) Factorises $(k-1)(k-10)=0$ or uses the quadratic formula **[M1]**

1 and 10 **[A1]**

(d) Deduces that $k=10$ must be used, as $k=1$ would mean points A and C are the same, contradicting that they are distinct **[B1]**

Calculates the gradient of OB using $(5,10)$: $\dfrac{10-0}{5-0}=2$ **[B1]**

[Total Marks: 8]

863 **(a)** Calculates the gradient of AB as $\dfrac{9-3}{4-1}=2$ and equates to gradient of BC or AC, e.g., $\dfrac{y-9}{8-4}=2$ **[M1]**

$y=17$ **[A1]**

(b) Sets up the gradient equation: $\dfrac{(p^2-p-3)-3}{p-1}=2$ **[M1]**

Simplifies to form a 3-term quadratic equation: $p^2-p-6=2(p-1)$

$\mathrm{p}^2-3p-4=0$ [M1]

Factorises $(p-4)(p+1)=0$ or uses the quadratic formula correctly **[M1]**

$p = 4$ and $p = -1$ **[A1]**

(c) Sets up the gradient equation: $\dfrac{12 - m}{(m + 1) - 3} = m$ **[M1]**

Multiplies by the denominator: $12 - m = m(m - 2)$ **[M1]**

Rearranges into a 3-term quadratic: $m^2 - m - 12 = 0$ **[M1]**

$m = 4$ and $m = -3$ **[A1]**

[Total Marks: 10]

864 (a) $\dfrac{5 - 1}{4 - (-2)}$ or equivalent **[M1]**

$\dfrac{2}{3}$ **[A1]**

(b) $\dfrac{-2 - k}{8 - 3} = -1.4$ or equivalent **[M1]**

$-2 - k = -7$ **[M1]**

5 **[A1]**

[Total Marks: 5]

865 (a) $\dfrac{-9 - 5}{4 - (-3)}$ or $\dfrac{5 - (-9)}{-3 - 4}$ **[M1]**

-2 **[A1]**

(b) $\dfrac{1 - 5}{k - (-3)} = -2$ oe **[M1]**

$-4 = -2(k + 3)$ or $-2k - 6 = -4$ **[M1]**

$k = -1$ **[A1]**

[Total Marks: 5]

866 (a) Forms a correct equation for the gradient: $\dfrac{5 - 2k}{2 - (k - 3)} = -3$ OR $\dfrac{2k - 5}{k - 3 - 2} = -3$ **[M1]**

Correctly simplifies the denominator and equates: $5 - 2k = -3(5 - k)$ **[M1]**

$k = 4$ **[A1]**

(b) Evaluates coordinates of P as $(1, 8)$ and substitutes into midpoint formula: $\left(\dfrac{1 + 2}{2}, \dfrac{8 + 5}{2}\right)$ **[M1]**

$(1.5, 6.5)$ **[A1]**

[Total Marks: 5]

867 (a) Finds a correct algebraic expression for the gradient: $\dfrac{8 - 4}{k - 2 - 3k}$ or $\dfrac{4}{-2k - 2}$ **[M1]**

Equates to $-\frac{1}{2}$ and eliminates fractions: $8 = 2k + 2$ or $-8 = -2k - 2$ **[M1]**

3 **[A1]**

(b) $P(9, 4)$ and $Q(1, 8)$ **[B1]**

(c) Identifies the coordinates of R must be of the form $(0, y)$ **[B1]**

Sets up the gradient equation: $\frac{y-4}{0-9} = 1$ **[M1]**

$(0, -5)$ **[A1]**

[Total Marks: 7]

868 (a) Calculates gradient of AB: $\frac{8-(-4)}{3-(-1)} = \frac{12}{4} = 3$ **[B1]**

Sets up equation for gradient involving C, e.g., $\frac{23-8}{p-3} = 3$ or $\frac{23-(-4)}{p-(-1)} = 3$ **[M1]**

8 **[A1]**

(b) Forms expression for gradient AD: $\frac{2q-1-(-4)}{q-(-1)}$ **[M1]**

Equates to $-\frac{1}{3}$ and simplifies: $3(2q + 3) = -1(q + 1)$ **[M1]**

$-\frac{10}{7}$ or -1.43 (accept $-1.428...$) **[A1]**

[Total Marks: 6]

869 (a) Gradient $= \frac{(t^2 - 3t + 2) - 0}{t - 1}$ **[M1]**

Factorising numerator: $\frac{(t-1)(t-2)}{t-1}$ **[M1]**

Cancelling $(t - 1)$ correctly to obtain $t - 2$ **[A1]**

(b) $t - 2 = 2.8$ giving $t = 4.8$ **[M1]**

$y = 4.8^2 - 3(4.8) + 2$ **[M1]**

$(4.8, 10.64)$ **[A1]**

(c) Gradient of $QR = \frac{(t^2 - 3t + 2) - (-1.5)}{t - 3.2}$ **[M1]**

$\frac{t^2 - 3t + 3.5}{t - 3.2} = t - 2$ **[M1]**

$t^2 - 3t + 3.5 = t^2 - 5.2t + 6.4$ leading to $2.2t = 2.9$ **[M1]**

1.32 **[A1]**

[Total Marks: 10]

870 (a) $\left(\frac{2+8}{2}, \frac{-3+5}{2}\right)$ **[M1]**

$(5, 1)$ **[A1]**

(b) $\sqrt{(8-2)^2 + (5-(-3))^2}$ or $\sqrt{6^2 + 8^2}$ **[M1]**

10 **[A1]**

[Total Marks: 4]

871 **(a)** $\left(\frac{-2+4}{2}, \frac{5+(-3)}{2}\right)$ **[M1]**

$(1, 1)$ **[A1]**

(b) $(4-(-2))^2 + (-3-5)^2$ or $6^2 + (-8)^2$ **[M1]**

$\sqrt{36+64}$ or $\sqrt{100}$ **[M1]**

10 **[A1]**

[Total Marks: 5]

872 **(a)** $\left(\frac{1+7}{2}, \frac{4+(-4)}{2}\right)$ **[M1]**

$(4, 0)$ **[A1]**

(b) $(7-1)^2 + (-4-4)^2$ oe **[M1]**

$\sqrt{36+64}$ or $\sqrt{100}$ **[M1]**

10 **[A1]**

[Total Marks: 5]

873 **(a)** Uses the midpoint formula: $\left(\frac{2+8}{2}, \frac{-3+5}{2}\right)$ **[M1]**

$(5, 1)$ **[A1]**

(b) Uses the distance formula: $\sqrt{(8-2)^2 + (5-(-3))^2}$ **[M1]**

$\sqrt{6^2 + 8^2}$ or $\sqrt{36+64}$ **[M1]**

10 **[A1]**

[Total Marks: 5]

874 **(a)** $A = (-2, 1)$ **[B1]**

$B = (4, 9)$ **[B1]**

(b) $\left(\frac{-2+4}{2}, \frac{1+9}{2}\right)$ **[M1]**

$(1, 5)$ **[A1]**

(c) $\sqrt{(4-(-2))^2 + (9-1)^2}$ or $\sqrt{6^2 + 8^2}$ **[M1]**

10 **[A1]**

[Total Marks: 6]

875 **(a)** Sets up an equation for the x or y coordinate, e.g. $\frac{-2+x}{2} = 4$ or $\frac{7+y}{2} = 1$ **[M1]**

$(10, -5)$ **[A1]**

(b) $\sqrt{(k-(-2))^2 + (10-7)^2} = 5$ or $(k+2)^2 + 3^2 = 5^2$ **[M1]**

$(k+2)^2 = 16$ or $k^2 + 4k - 12 = 0$ **[M1]**

2 and -6 **[A1]**

[Total Marks: 5]

876 **(a)** Calculates length OP: $\sqrt{(-3)^2 + 4^2}$ or $\sqrt{9+16}$ **[M1]**

5 **[A1]**

(b) Recognises $O(0,0)$ is the midpoint of PQ, so sets up $\frac{-3+x}{2} = 0$ and $\frac{4+y}{2} = 0$ OR uses translation from P to O **[M1]**

$(3, -4)$ **[A1]**

(c) Sets up equation for length OR: $k^2 + (\sqrt{11})^2 = 5^2$ **[M1]**

$k^2 + 11 = 25$

$k^2 = 14$ [M1]

$-\sqrt{14}$ **[A1]**

[Total Marks: 7]

877 **(a)** $\frac{-2+x}{2} = 3$ or $\frac{4+y}{2} = -1$ **[M1]**

$(8, -6)$ **[A1]**

(b) $PQ = \sqrt{(8-(-2))^2 + (-6-4)^2}$ **[M1]**

$\sqrt{100+100}$ or $\sqrt{200}$ **[M1]**

$10\sqrt{2}$ **[A1]**

(c) Let $R = (0, y)$ and form $(0-(-2))^2 + (y-4)^2 = (0-8)^2 + (y-(-6))^2$ **[M1]**

$4 + y^2 - 8y + 16 = 64 + y^2 + 12y + 36$ **[M1]**

$-20y = 80$ giving $y = -4$ **[A1]**

$(0, -4)$ **[A1]**

[Total Marks: 9]

878 **(a)** $M = (1, 1)$ **[B1]**

$N = (3, -2)$ **[B1]**

(b) Calculates length $MN = \sqrt{(3-1)^2 + (-2-1)^2}$ **[M1]**

$MN = \sqrt{4+9} = \sqrt{13}$ **[A1]**

Calculates length $BC = \sqrt{(8-4)^2 + (-3-3)^2}$ **[M1]**

$BC = \sqrt{16+36} = \sqrt{52}$ and shows $\sqrt{52} = 2\sqrt{13}$, therefore $MN = \frac{1}{2}BC$ **[A1]**

[Total Marks: 6]

879 **(a)** Finds coordinates of the midpoint in terms of p: $\left(\frac{4p}{2}, \frac{3p-1}{2}\right)$ or $\left(2p, \frac{3p-1}{2}\right)$ **[M1]**

Substitutes into the equation of the line: $\frac{3p-1}{2} = 2p+1$ **[M1]**

-3 **[A1]**

(b) Sets up the distance squared equation: $(3p-p)^2 + (p-2-(2p+1))^2 = 6^2$ **[M1]**

Simplifies to $(2p)^2 + (-p-3)^2 = 36$ **[M1]**

Expands to $4p^2 + p^2 + 6p + 9 = 36$ and rearranges to $5p^2 + 6p - 27 = 0$ with no errors seen **[A1]**

(c) Factorises to $(5p-9)(p+3) = 0$ or uses correct quadratic formula **[M2]**

Award [M1] for a partial factorisation or one error in the quadratic formula

1.8 (or $\frac{9}{5}$) and -3 **[A1]**

[Total Marks: 9]

880 **(a)** $\left(\frac{-1+5}{2}, \frac{2+8}{2}\right)$ **[M1]**

$(2, 5)$ **[A1]**

(b) $(2-(-1))^2 + (5-2)^2$ or $\frac{1}{2}\sqrt{(5-(-1))^2 + (8-2)^2}$ oe **[M1]**

$\sqrt{9+9}$ or $\sqrt{18}$ **[M1]**

$3\sqrt{2}$ **[A1]**

(c) $\frac{2+x}{2} = \frac{5}{2}$ and $\frac{5+y}{2} = \frac{3}{2}$ oe **[M1]**

$x = 3$ and $y = -2$ to give $D(3, -2)$ **[A1]**

Length $AD = \sqrt{(3-(-1))^2 + (-2-2)^2}$ oe **[M1]**

$\sqrt{16+16}$ or $\sqrt{32}$ **[M1]**

$4\sqrt{2}$ **[A1]**

[Total Marks: 10]

881 **(a)** Uses distance formula: $\sqrt{(4-(-2))^2 + (-3-5)^2}$ **[M1]**

Simplifies to $\sqrt{6^2 + (-8)^2}$ or $\sqrt{36+64}$ **[M1]**

10 **[A1]**

(b) Uses midpoint formula: $\left(\frac{-2+4}{2}, \frac{5+(-3)}{2}\right)$ **[M1]**

$(1, 1)$ **[A1]**

[Total Marks: 5]

882 (a) $(7-3)^2 + (5-k)^2 = 5^2$ or equivalent **[M1]**

$16 + (5-k)^2 = 25$ or $(5-k)^2 = 9$ **[M1]**

$k = 2$ and $k = 8$ **[A1]**

(b) Identifies $P(3, 2)$ to use in midpoint calculation: $\left(\frac{3+7}{2}, \frac{2+5}{2}\right)$ **[M1]**

$(5, 3.5)$ **[A1]**

[Total Marks: 5]

883 (a) Sets up an equation for at least one coordinate, e.g., $\frac{-3+x}{2} = 1$ or $\frac{5+y}{2} = -2$ **[M1]**

$(5, -9)$ **[A1]**

(b) Uses distance formula with $P(-3, 5)$ and their $Q(5, -9)$: $\sqrt{(5-(-3))^2 + (-9-5)^2}$ **[M1]**

$\sqrt{8^2 + (-14)^2}$ or $\sqrt{64 + 196}$ or $\sqrt{260}$ **[M1]**

16.1 **[A1]**

[Total Marks: 5]

884 (a) $(k-(-3))^2 + (-1-5)^2 = 10^2$ oe **[M1]**

$(k+3)^2 + 36 = 100$ or $k^2 + 6k + 9 + 36 = 100$ **[M1]**

$(k+3)^2 = 64$ or $k^2 + 6k - 55 = 0$ **[M1]**

$k = 5$ or $k = -11$ **[A1]**

(b) Chooses $k = 5$ to establish coordinates of Q as $(5, -1)$ **[M1]**

$(1, 2)$ **[A1]**

[Total Marks: 6]

885 (a) $\left(\frac{-3+5}{2}, \frac{2+8}{2}\right)$ **[M1]**

$(1, 5)$ **[A1]**

(b) Radius $r = \sqrt{(1-(-3))^2 + (5-2)^2}$ or $\sqrt{4^2 + 3^2} = 5$ **[M1]**

Area $= \pi \times 5^2$ **[M1]**

78.5 **[A1]**

(c) Distance $CD = 5 \implies (k-1)^2 + (8-5)^2 = 5^2$ **[M1]**

$(k-1)^2 + 9 = 25 \implies (k-1)^2 = 16$ **[M1]**

$k = 5$ and $k = -3$ **[A1]**

[Total Marks: 8]

886 **(a)** Sets up the distance formula: $\sqrt{(3p-2p)^2+(4p-p)^2} = 5\sqrt{10}$ **[M1]**

Simplifies the terms inside the root: $\sqrt{p^2+(3p)^2} = 5\sqrt{10}$ **[M1]**

Squares both sides and simplifies: $10p^2 = 250$ **[M1]**

$p = 5$ (rejects -5 as $p > 0$) **[A1]**

(b) Evaluates coordinates of $A(10, 5)$ and $B(15, 20)$ **[B1]**

Sets up a valid expression for the coordinates of C, e.g., x-coordinate $= 10 + \frac{2}{5}(15-10)$ or y-coordinate $= 5 + \frac{2}{5}(20-5)$ **[M1]**

$(12, 11)$ **[A1]**

(c) Uses the midpoint formula, e.g. $\frac{15+x}{2} = 12$ and $\frac{20+y}{2} = 11$, or $D = 2C - B$ **[M1]**

$(9, 2)$ **[A1]**

[Total Marks: 9]

887 **(a)** $AB = \sqrt{(5-(-3))^2+(6-2)^2} = \sqrt{80}$ $(= 8.94\ldots)$ **[M1]**

$BC = \sqrt{(9-5)^2+(-2-6)^2} = \sqrt{80}$ $(= 8.94\ldots)$ **[M1]**

$AC = \sqrt{(9-(-3))^2+(-2-2)^2} = \sqrt{160}$ $(= 12.6\ldots)$ **[M1]**

$AB = BC$ (and $ACeAB$), therefore triangle ABC is isosceles **[A1]**

(b) **(i)** $\left(\frac{-3+9}{2}, \frac{2+(-2)}{2}\right)$ **[M1]**

$(3, 0)$ **[A1]**

(ii) $BM = \sqrt{(5-3)^2+(6-0)^2}$ $(= \sqrt{40})$ **[M1]**

6.32 **[A1]**

(c) $\frac{1}{2} \times \sqrt{160} \times \sqrt{40}$ or $\frac{1}{2} \times \sqrt{80} \times \sqrt{80}$ **[M1]**

40 **[A1]**

(d) $\frac{5+x}{2} = 3$ and $\frac{6+y}{2} = 0$ (or vector translation $B + \overrightarrow{AD}$) **[M1]**

$(1, -6)$ **[A1]**

[Total Marks: 12]

888 Rearranging to $5y = -3x + 20$ or $y = -\frac{3}{5}x + 4$ **[M1]**

Gradient $= -\frac{3}{5}$ oe **[A1]**

y-intercept $= 4$ **[A1]**

[Total Marks: 3]

889 **(a)** Gradient $= -2$ **[B1]**

y-intercept $= 4$ **[B1]**

$-2x + 4$ **[B1]**

(b) $2x + y = 4$ oe **[B1]**

[Total Marks: 4]

890 **(a)** Rearranges to the form $y = mx + c$, e.g. $5y = -4x + 20$ **[M1]**

Gradient $= -\frac{4}{5}$ or -0.8 **[A1]**

y-intercept $= 4$ **[A1]**

(b) $x = 3$ **[B1]**

[Total Marks: 4]

891 **(a)** Rearranges equation to $y = mx + c$ form: $5y = -3x + 15$ or $y = -\frac{3}{5}x + 3$ **[M1]**

Gradient $= -\frac{3}{5}$ **[A1]**

y-intercept $= 3$ **[A1]**

(b) Identifies parallel gradient as $-\frac{3}{5}$ **[M1]**

$-\frac{3}{5}x - 4$ **[A1]**

[Total Marks: 5]

892 **(a)** Rearranges the equation to make y the subject: $4y = 3x - 12$ or $y = \frac{3}{4}x - 3$ **[M1]**

Gradient $= \frac{3}{4}$ **[A1]**

y-intercept $= -3$ **[A1]**

(b) Identifies the gradient of M is $\frac{3}{4}$ **[M1]**

$\frac{3}{4}x + 5$ **[A1]**

[Total Marks: 5]

893 **(a)** Gradient $= \frac{3 - (-3)}{3 - 0} = 2$ **[M1]**

y-intercept $= -3$ **[B1]**

$2x - 3$ **[A1]**

(b) $x = -2$ **[B1]**

(c) $y = 2(-2) - 3 = -7$ **[M1]**

$(-2, -7)$ **[A1]**

[Total Marks: 6]

894 (a) $5y = -4x + 20$ or $y = -\frac{4}{5}x + 4$ **[M1]**

Gradient $= -\frac{4}{5}$ (or -0.8) **[A1]**

y-intercept $= 4$ **[A1]**

(b) Substitute $y = 0$: $4x = 20$ **[M1]**

$(5, 0)$ **[A1]**

(c) Gradient $= \frac{7 - (-5)}{3 - (-3)} = 2$ **[M1]**

$y = 2x + 1$ **[M1]**

$2x - y = -1$ (or $-2x + y = 1$) **[A1]**

[Total Marks: 8]

895 (a) Identifies correct y-intercept $c = 4$ and x-intercept at $x = 6$ **[B1]**

Calculates gradient: $\frac{0 - 4}{6 - 0} = -\frac{2}{3}$ **[M1]**

Forms equation $y = -\frac{2}{3}x + 4$ and rearranges to eliminate fractions **[M1]**

$2x + 3y = 12$ (or equivalent with integer coefficients) **[A1]**

(b) (i) Rearranges to isolate term in y, e.g., $\frac{y}{2} = \frac{x}{3} - 1$ **[M1]**

$\frac{2}{3}x - 2$ **[A1]**

(ii) Equates the two expressions for y: $-\frac{2}{3}x + 4 = \frac{2}{3}x - 2$
OR substitutes into $2x + 3y = 12$ **[M1]**

$(4.5, 1)$ or $\left(\frac{9}{2}, 1\right)$ **[A1]**

[Total Marks: 7]

896 (a) $\frac{(2p - 1) - (-1)}{5 - p} = 3$ **[M1]**

$2p = 3(5 - p)$ **[M1]**

$p = 3$ **[A1]**

(b) Substituting $p = 3$ to find coordinates $A(3, -1)$ or $B(5, 5)$ and using gradient 3 to find

equation **[M1]**

$y = 3x - 10$ **[A1]**

(c) Identifies x-intercept as $\frac{10}{3}$ or y-intercept as -10 **[M1]**

$\frac{1}{2} \times \frac{10}{3} \times 10$ **[M1]**

$\frac{50}{3}$ or 16.7 $(16.66\ldots)$ **[A1]**

[Total Marks: 8]

897 **(a)** Sets up the gradient expression: $\frac{k - 3k}{4 - (-2)}$ **[M1]**

$-\frac{k}{3}$ **[A1]**

(b) Substitutes a point and gradient into the linear equation: $y - k = -\frac{k}{3}(x - 4)$ or $y - 3k = -\frac{k}{3}(x - (-2))$ **[M1]**

Multiplies by 3 to eliminate the fraction: $3y - 3k = -kx + 4k$ **[M1]**

$kx + 3y = 7k$ **[A1]**

(c) Substitutes $x = 1$ and $y = 10$ into their equation from part (b): $k(1) + 3(10) = 7k$ **[M1]**

5 **[A1]**

(d) Substitutes $k = 5$ to find the line equation $5x + 3y = 35$ and sets $y = 0$: $5x = 35$ **[M1]**

$(7, 0)$ **[A1]**

[Total Marks: 9]

898 **(a)** $\frac{(3a + 2) - (a + 4)}{(a - 1) - 2a} = -4$ **[M1]**

$\frac{2a - 2}{-a - 1} = -4$ **[M1]**

$2a - 2 = 4a + 4$

-6 = 2a

a = -3[$A1$]

(b) Identifies coordinates of A as $(-6, 1)$ or B as $(-4, -7)$ **[M1]**

$y - 1 = -4(x + 6)$ or $y = -4x - 23$ **[M1]**

$4x + y = -23$ **[A1]**

(c) $4x + (3x - 2) = -23$ or $-4x - 23 = 3x - 2$ oe **[M1]**

$7x = -21$

x = -3[$A1$]

$y = -11$ **[A1]**

[Total Marks: 9]

899 **(a)** Calculates the midpoint of $AB = \left(\frac{-3+5}{2}, \frac{1+5}{2}\right) = (1, 3)$ **[B1]**

Calculates the gradient of $AB = \frac{5-1}{5-(-3)} = \frac{4}{8} = \frac{1}{2}$ **[B1]**

Finds the perpendicular gradient $= -2$ **[M1]**

Uses perpendicular gradient -2 and midpoint $(1, 3)$ to find equation, e.g., $3 = -2(1) + c$ **[M1]**

$-2x + 5$ **[A1]**

(b) Sets up simultaneous equations, e.g., substituting $y = -2x + 5$ into $x - 2y = 5$ **[M1]**

$x - 2(-2x + 5) = 5 \implies 5x - 10 = 5 \implies x = 3$ **[M1]**

$(3, -1)$ **[A1]**

(c) Uses distance formula for $A(-3, 1)$ and $P(3, -1)$ **[M1]**

$\sqrt{(3-(-3))^2 + (-1-1)^2}$ **[M1]**

$\sqrt{40}$ or $2\sqrt{10}$ **[A1]**

[Total Marks: 10]

900 **(a)** Calculates gradient: $\frac{-2-10}{5-(-3)} = \frac{-12}{8}$ **[M1]**

Gradient is -1.5 (or $-\frac{3}{2}$) **[A1]**

Substitutes a point to find c: $10 = -1.5(-3) + c$

c = 5.5*OR*y - 10 = -1.5(x - (-3))[*M*1]

$-1.5x + 5.5$ or $-\frac{3}{2}x + \frac{11}{2}$ **[A1]**

(b) Substitutes $y = 0$ into the equation: $2(0) + 3x = 15$ **[M1]**

$(5, 0)$ **[A1]**

[Total Marks: 5]

901 **(a)** Evidence of using two integer coordinates on the line, e.g. $\frac{5-(-1)}{-4-4}$ **[M1]**

$-\frac{3}{4}$ or -0.75 **[A1]**

(b) $y = -\frac{3}{4}x + 2$ or $y - 5 = -\frac{3}{4}(x + 4)$ **[M1]**

$4y = -3x + 8$ or $4y - 20 = -3x - 12$ **[M1]**

$3x + 4y = 8$ or equivalent integer form **[A1]**

[Total Marks: 5]

902 **(a)** Gradient $= \dfrac{-2-3}{6-(-4)}$ or $\dfrac{-5}{10} = -\dfrac{1}{2}$ **[M1]**

Substitution of $(-4, 3)$ or $(6, -2)$ into $y = -\dfrac{1}{2}x + c$, e.g. $3 = -\dfrac{1}{2}(-4) + c$ **[M1]**

$-\dfrac{1}{2}x + 1$ **[A1]**

(b) $C = (5, 0)$ **[B1]**

$D = (0, -3)$ **[B1]**

(c) $x = -4$ **[B1]**

[Total Marks: 6]

903 **(a)** Identifies two correct points on the line, e.g., $(0, 4)$ and $(2, 7)$ **[M1]**

Calculates the gradient $m = \dfrac{7-4}{2-0} = \dfrac{3}{2}$ **[M1]**

States correct equation in required integer form, e.g., $3x - 2y = -8$ or $2y - 3x = 8$ **[A1]**

(b) Uses $m_1 \times m_2 = -1$ to find gradient of M: $-\dfrac{2}{3}$ **[M1]**

Substitutes $(9, -1)$ into $y = -\dfrac{2}{3}x + c$ to find c: $-1 = -\dfrac{2}{3}(9) + c$ **[M1]**

$-\dfrac{2}{3}x + 5$ **[A1]**

[Total Marks: 6]

904 **(a)** Identifies the y-intercept is 4 or uses coordinates $(0, 4)$ **[B1]**

Calculates the gradient: $\dfrac{0-4}{6-0}$ or $\dfrac{-4}{6}$ **[M1]**

$-\dfrac{2}{3}x + 4$ **[A1]**

(b) Finds the coordinates of the midpoint of the segment $(0, 4)$ to $(6, 0)$: $\left(\dfrac{0+6}{2}, \dfrac{4+0}{2}\right)$ or $(3, 2)$ **[B1]**

Calculates the gradient of L_2: $\dfrac{7-2}{2-3}$ or $\dfrac{5}{-1} = -5$ **[M1]**

Uses point-gradient form for L_2: $y - 7 = -5(x - 2)$ or $y - 2 = -5(x - 3)$ **[M1]**

$5x + y = 17$ **[A1]**

[Total Marks: 7]

905 **(a)** $A = (8, 0)$ **[B1]**

$B = (0, 12)$ **[B1]**

$C = (2, 0)$ **[B1]**

(b) $3x + 2(1.5x - 3) = 24$ **[M1]**

$6x = 30$ giving $x = 5$ **[A1]**

$(5, 4.5)$ **[A1]**

(c) Area of $\triangle OAB = \frac{1}{2} \times 8 \times 12 = 48$ **[M1]**

Area of $\triangle CAP = \frac{1}{2} \times (8 - 2) \times 4.5 = 13.5$ **[M1]**

34.5 **[A1]**

(d) $5k - 4(4.5) = 14$ **[M1]**

6.4 **[A1]**

[Total Marks: 11]

906 Identifies the gradient $m = -2$ and sets up $4 = -2(3) + c$ or $y - 4 = -2(x - 3)$ **[M1]**

$-2x + 10$ **[A1]**

[Total Marks: 2]

907 Rearranges equation to find gradient, e.g., $2y = -4x + 9$ implies $m = -2$ **[M1]**

Substitutes $(3, -1)$ into $y = -2x + c$ or writes $y - (-1) = -2(x - 3)$ **[M1]**

$-2x + 5$ **[A1]**

[Total Marks: 3]

908 Gradient $m = -3$ **[B1]**

$5 = -3(-2) + c$ or $y - 5 = -3(x - (-2))$ **[M1]**

$y = -3x - 1$ **[A1]**

[Total Marks: 3]

909 (a) 3 **[B1]**

(b) Uses $m = 3$ in $y = 3x + c$ **[M1]**

Substitutes $(2, -5)$: $-5 = 3(2) + c$ **[M1]**

$3x - 11$ **[A1]**

[Total Marks: 4]

910 (a) 2 **[B1]**

(b) $1 = 2(2) + c$ or $y - 1 = 2(x - 2)$ **[M1]**

$2x - 3$ **[A1]**

(c) $5 = 2(-1) + c$ or $y - 5 = 2(x + 1)$ **[M1]**

$2x + 7$ **[A1]**

[Total Marks: 5]

911 **(a)** Rearranges the equation to make y the subject: $2y = 5x + 8$ or $y = \frac{5}{2}x + 4$ **[M1]**

$\frac{5}{2}$ or 2.5 **[A1]**

(b) Identifies the gradient of L_2 is $\frac{5}{2}$ and the y-intercept is -3 **[M1]**

$\frac{5}{2}x - 3$ (or $2.5x - 3$) **[A1]**

(c) Substitutes $(4, 1)$ and $m = \frac{5}{2}$ into linear equation: $y - 1 = \frac{5}{2}(x - 4)$ or $1 = \frac{5}{2}(4) + c$ **[M1]**

Eliminates the fraction: $2y - 2 = 5(x - 4)$ or $2y - 2 = 5x - 20$ **[M1]**

$5x - 2y = 18$ (or $-5x + 2y = -18$) **[A1]**

[Total Marks: 7]

912 **(a)** $\frac{-1 - 5}{5 - (-3)}$ or $\frac{5 - (-1)}{-3 - 5}$ **[M1]**

$-\frac{3}{4}$ **[A1]**

(b) $m = -\frac{3}{4}$ **[M1]**

$-2 = -\frac{3}{4}(4) + c$ or $y - (-2) = -\frac{3}{4}(x - 4)$ **[M1]**

$y = -\frac{3}{4}x + 1$ **[A1]**

[Total Marks: 5]

913 **(a)** Rearranges equation to $2y = -3x + 14$ or $y = -\frac{3}{2}x + 7$ **[M1]**

$-\frac{3}{2}$ or -1.5 **[A1]**

(b) Uses gradient $-\frac{3}{2}$ with point $(-4, 9)$: $9 = -\frac{3}{2}(-4) + c$ **[M1]**

Finds $c = 3$, giving $y = -\frac{3}{2}x + 3$ **[A1]**

Multiplies through by 2 and rearranges: $3x + 2y = 6$ **[M1]**

$3x + 2y = 6$ **[A1]**

[Total Marks: 6]

914 **(a)** $4y = -3x + 12$ or $y = -\frac{3}{4}x + 3$ **[M1]**

$-\frac{3}{4}$ (or -0.75) **[A1]**

(b) Gradient of $L_2 = -\frac{3}{4}$ **[B1]**

$y - 3 = -\frac{3}{4}(x - 4)$ or $3(4) + 4(3) = c$ **[M1]**

$3x + 4y = 24$ **[A1]**

(c) $A = (8, 0)$ **[B1]**

$B = (0, 6)$ **[B1]**

$(4, 3)$ **[A1]**

[Total Marks: 8]

915 (a) Finds the gradient of line P in terms of k: $m_P = \frac{4}{k}$ **[B1]**

Sets gradients equal: $\frac{4}{k} = k - 3$ **[M1]**

Forms a 3-term quadratic equation: $k^2 - 3k - 4 = 0$ **[M1]**

4 and -1 **[A1]**

(b) (i) Identifies that $k = 4$ produces a positive gradient ($m = 1$) while $k = -1$ produces a negative gradient ($m = -4$) **[M1]**

$x - 2$ **[A1]**

(ii) Substitutes $(-5, 7)$ and $m = 1$ into equation: $y - 7 = 1(x - (-5))$ **[M1]**

$x - y = -12$ or $y - x = 12$ (or equivalent integer form) **[A1]**

[Total Marks: 8]

916 Rearranges $5x - 2y = 15$ to find the gradient, e.g. $2y = 5x - 15$

y $= \frac{5}{2}x - 7.5$, so $m = \frac{5}{2}$ [M1]

Substitutes $(-4, 1)$ and $m = \frac{5}{2}$ into $y = mx + c$: $1 = \frac{5}{2}(-4) + c$ **[M1]**

$\frac{5}{2}x + 11$ or $2.5x + 11$ **[A1]**

[Total Marks: 3]

917 (a) $\frac{6 - 2}{5 - (-3)}$ oe **[M1]**

$\frac{1}{2}$ or 0.5 **[A1]**

(b) $y - (-1) = \frac{1}{2}(x - 4)$ oe **[M1]**

$2(y + 1) = x - 4$ or $2y + 2 = x - 4$ **[M1]**

$x - 2y = 6$ or $-x + 2y = -6$ **[A1]**

[Total Marks: 5]

918 **(a)** Uses the gradient formula: $\dfrac{4-1}{4-(-2)}$ **[M1]**

$\dfrac{3}{6} = \dfrac{1}{2}$ or 0.5 **[A1]**

(b) States or implies gradient $m = 0.5$ **[B1]**

Substitutes $(2, -3)$ into $y = 0.5x + c$: $-3 = 0.5(2) + c$ **[M1]**

$0.5x - 4$ (or $\dfrac{1}{2}x - 4$) **[A1]**

(c) Finds coordinates of P by setting $y = 0$: $0 = 0.5x - 4 \implies x = 8$, so $P(8, 0)$ **[B1]**

Finds coordinates of Q by setting $x = 0$: $y = -4$, so $Q(0, -4)$ **[B1]**

Applies distance formula: $\sqrt{(8-0)^2 + (0-(-4))^2}$ or $\sqrt{8^2 + (-4)^2}$ **[M1]**

8.94 (from $\sqrt{80}$) **[A1]**

[Total Marks: 9]

919 **(a)** Calculates the gradient of L_1: $m = \dfrac{-3-5}{3-(-1)}$ **[M1]**

$m = \dfrac{-8}{4} = -2$ **[A1]**

Uses parallel gradient -2 and point $C(2, 7)$ to find c: $7 = -2(2) + c$ **[M1]**

$-2x + 11$ **[A1]**

(b) Finds y-intercept $E(0, 11)$ and sets $y = 0$ to find x-intercept D: $0 = -2x + 11$

x = 5.5[$M1$]

Sets up area calculation for right-angled triangle: $\dfrac{1}{2} \times 5.5 \times 11$ **[M1]**

30.25 or $\dfrac{121}{4}$ **[A1]**

[Total Marks: 7]

920 **(a)** Calculates gradient of PQ: $\dfrac{15-3}{4-(-2)} = 2$ **[B1]**

Writes equation of line L as $y = 2x - 5$ **[M1]**

Substitutes $R(k, 2k+1)$ into $y = 2x - 5$ or equates gradient OR' with 2:
$\dfrac{(2k+1)-(-5)}{k-0} = 2$ **[M1]**

Wait: $2k + 1 = 2k - 5$ has no solution — adjust $R(k, 3k-2)$: $3k - 2 = 2k - 5$

k = -3[$A1$]

(b) Finds gradient of given line: $2y = 5x + 9$

m = 2.5($or\dfrac{5}{2}$) [B1]

Substitutes $(4, 15)$: $15 = 2.5(4) + c$

c = 5[$M1$]

$2.5x + 5$ or $\frac{5}{2}x + 5$ **[A1]**

[Total Marks: 7]

921 Gradient of $L_1 = \frac{3}{4}$ oe **[B1]**

Equation of L_2 is $y = \frac{3}{4}x + c$ or $3x - 4y = k$ **[M1]**

x-intercept of L_2 is $-\frac{4c}{3}$ (or $\frac{k}{3}$) AND y-intercept is c (or $-\frac{k}{4}$) oe **[M1]**

$(-\frac{4c}{3})^2 + c^2 = 15^2$ or $(\frac{k}{3})^2 + (-\frac{k}{4})^2 = 15^2$ oe **[M1]**

$\frac{25c^2}{9} = 225$ or $\frac{25k^2}{144} = 225$ **[M1]**

$c = \pm 9$ or $k = \pm 36$ **[A1]**

$y = \frac{3}{4}x + 9$ and $y = \frac{3}{4}x - 9$ **[A1]**

[Total Marks: 7]

922 (a) Gradient of $AB = \frac{4-2}{7-1} = \frac{1}{3}$ **[M1]**

Substitute $(10, 9)$ into $y - y_1 = m(x - x_1)$ with their gradient: $y - 9 = \frac{1}{3}(x - 10)$ **[M1]**

$y = \frac{1}{3}x + \frac{17}{3}$ or $3y = x + 17$ **[A1]**

(b) Gradient of $BC = \frac{9-4}{10-7} = \frac{5}{3}$ **[M1]**

Substitute $(1, 2)$ into $y - y_1 = m(x - x_1)$ with their gradient: $y - 2 = \frac{5}{3}(x - 1)$ **[M1]**

$y = \frac{5}{3}x + \frac{1}{3}$ or $3y = 5x + 1$ **[A1]**

(c) Equate equations from (a) and (b): $\frac{1}{3}x + \frac{17}{3} = \frac{5}{3}x + \frac{1}{3}$ **[M1]**

$4x = 16 \implies x = 4$ **[A1]**

$y = 7$ **[A1]**

[Total Marks: 9]

923 (a) Gradient of $BC = 2.5$ (or $\frac{5}{2}$) **[B1]**

$2 = 2.5(-3) + c$ or $y - 2 = 2.5(x + 3)$ **[M1]**

$2.5x + 9.5$ (or $\frac{5}{2}x + \frac{19}{2}$) **[A1]**

(b) $D = (0, 9.5)$ **[B1]**

Translation $\overrightarrow{AB} = \begin{pmatrix} 8 \\ 2 \end{pmatrix}$ applied to D **[M1]**

$C = (8, 11.5)$ **[A1]**

(c) Gradient of $AB = \dfrac{4-2}{5-(-3)} = \dfrac{1}{4}$ **[B1]**

Midpoint of $BC = \left(\dfrac{5+8}{2}, \dfrac{4+11.5}{2}\right) = (6.5, 7.75)$ **[B1]**

$y - 7.75 = \dfrac{1}{4}(x - 6.5)$ **[M1]**

$2x - 8y = -49$ (or $-2x + 8y = 49$) **[A1]**

(d) $\sqrt{(8-(-3))^2 + (11.5-2)^2}$ $(= \sqrt{211.25})$ **[M1]**

14.5 **[A1]**

[Total Marks: 12]

924 Rearranges to $5y = -2x + 15$ to find gradient of L is $-\dfrac{2}{5}$ **[M1]**

$\dfrac{5}{2}$ or 2.5 **[A1]**

[Total Marks: 2]

925 Identifies the gradient of the given line as 3 **[B1]**

Uses the perpendicular gradient $-\dfrac{1}{3}$ and substitutes $(6, 2)$ into $y = mx + c$ or $y - y_1 = m(x - x_1)$ **[M1]**

$-\dfrac{1}{3}x + 4$ **[A1]**

[Total Marks: 3]

926 (a) -2 **[B1]**

(b) Gradient of $L_2 = \dfrac{1}{2}$ **[M1]**

$1 = \dfrac{1}{2}(4) + c$ or $y - 1 = \dfrac{1}{2}(x - 4)$ **[M1]**

$y = \dfrac{1}{2}x - 1$ **[A1]**

[Total Marks: 4]

927 (a) Finds gradient of L_1: $y = \dfrac{3}{4}x - 2$ giving $m_1 = \dfrac{3}{4}$ **[M1]**

$-\dfrac{4}{3}$ **[A1]**

(b) Substitutes $(6, -5)$ and $m = -\dfrac{4}{3}$ into $y = mx + c$: $-5 = -\dfrac{4}{3}(6) + c$ **[M1]**

$-\dfrac{4}{3}x + 3$ **[A1]**

[Total Marks: 4]

928 **(a)** 2 **[B1]**

(b) $-\frac{1}{2}$ (or -0.5) **[B1]**

(c) $3 = -\frac{1}{2}(4) + c$ or $y - 3 = -\frac{1}{2}(x - 4)$ **[M1]**

$c = 5$ **[A1]**

$-\frac{1}{2}x + 5$ **[A1]**

[Total Marks: 5]

929 **(a)** Rearranges the equation to make y the subject: $y = \frac{5}{2}x + 4$ **[M1]**

$\frac{5}{2}$ or 2.5 **[A1]**

(b) $-\frac{2}{5}$ or -0.4 **[B1]**

(c) Substitutes $(10, -3)$ and $m = -\frac{2}{5}$ into linear equation: $-3 = -\frac{2}{5}(10) + c$ or $y - (-3) = -\frac{2}{5}(x - 10)$ **[M1]**

Solves for c: $-3 = -4 + c \implies c = 1$ **[M1]**

$-\frac{2}{5}x + 1$ (or $-0.4x + 1$) **[A1]**

[Total Marks: 6]

930 **(a)** Midpoint $= \left(\frac{-3+9}{2}, \frac{8+(-2)}{2}\right) = (3, 3)$ **[B1]**

Gradient $= \frac{-2-8}{9-(-3)} = -\frac{10}{12} = -\frac{5}{6}$ **[B1]**

(b) Uses perpendicular gradient $m_\perp = \frac{6}{5}$ **[M1]**

Substitutes midpoint $(3, 3)$: $3 = \frac{6}{5}(3) + c$ or $y - 3 = \frac{6}{5}(x - 3)$ **[M1]**

$6x - 5y = 3$ (or $-6x + 5y = -3$) **[A1]**

[Total Marks: 5]

931 Calculates the midpoint of $AB = \left(\frac{-4+2}{2}, \frac{7+(-1)}{2}\right) = (-1, 3)$ **[B1]**

Calculates the gradient of $AB = \frac{-1-7}{2-(-4)} = -\frac{8}{6} = -\frac{4}{3}$ **[B1]**

Finds the perpendicular gradient $= \frac{3}{4}$ **[M1]**

Substitutes the midpoint $(-1, 3)$ and gradient $\frac{3}{4}$ to find the equation, e.g.,

$y - 3 = \frac{3}{4}(x - (-1))$ **[M1]**

$3x - 4y = -15$ or $4y - 3x = 15$ or equivalent integer form **[A1]**

[Total Marks: 5]

932 **(a)** Gradient of $L = -\frac{2}{3}$ **[M1]**

$\frac{3}{2}$ (or 1.5) **[A1]**

(b) **(i)** $\left(\frac{-3+9}{2}, \frac{8+(-2)}{2}\right)$ **[M1]**

$(3, 3)$ **[A1]**

(ii) Gradient of $AB = \frac{-2-8}{9-(-3)} = -\frac{5}{6}$ **[M1]**

Gradient of perpendicular bisector $= \frac{6}{5}$ **[A1]**

$y - 3 = \frac{6}{5}(x - 3)$ **[M1]**

$6x - 5y = 3$ (or $-6x + 5y = -3$) **[A1]**

[Total Marks: 8]

933 **(a)** $\left(\frac{-3+9}{2}, \frac{8+(-2)}{2}\right)$ **[M1]**

$(3, 3)$ **[A1]**

(b) Gradient of $AB = \frac{-2-8}{9-(-3)}$ or $-\frac{10}{12}$ or $-\frac{5}{6}$ **[M1]**

Perpendicular gradient is $\frac{6}{5}$ **[M1]**

Substitute $(3, 3)$ into $y - 3 = \frac{6}{5}(x - 3)$ or $3 = \frac{6}{5}(3) + c$ **[M1]**

$6x - 5y = 3$ or $-6x + 5y = -3$ **[A1]**

[Total Marks: 6]

934 **(a)** $(0, 6)$ **[B1]**

(b) Gradient of $L_1 = -\frac{3}{2}$ **[M1]**

Gradient of $L_2 = \frac{2}{3}$ **[M1]**

$y = \frac{2}{3}x + 6$ **[A1]**

(c) Coordinates of Q: $y = \frac{2}{3}(9) + 6 = 12 \implies Q(9, 12)$ **[B1]**

Coordinates of R: $3(4) + 2y = 12 \implies y = 0 \implies R(4, 0)$ **[B1]**

Identify triangle PQR is right-angled at P OR setup valid box/shoelace method **[M1]**

Length $PQ = \sqrt{9^2 + 6^2} = \sqrt{117}$ and Length $PR = \sqrt{4^2 + (-6)^2} = \sqrt{52}$ **[M1]**

Area $= \frac{1}{2} \times \sqrt{117} \times \sqrt{52} = 39$ **[A1]**

[Total Marks: 9]

935 **(a)** Rearranges to $2y = -kx + 12$ to identify gradient $-\frac{k}{2}$ **[B1]**

(b) Uses the gradient formula: $\frac{(k+2)-5}{2-0}$ **[M1]**

$\frac{k-3}{2}$ **[A1]**

(c) Applies the perpendicularity condition $m_1 \times m_2 = -1$ **[M1]**

Forms the equation: $\left(-\frac{k}{2}\right)\left(\frac{k-3}{2}\right) = -1$ **[M1]**

Expands and rearranges to form a quadratic equation:
$-k(k-3) = -4 \implies k^2 - 3k - 4 = 0$ **[M1]**

4 and -1 **[A1]**

(d) Deduces that $k = -1$ must be chosen because $m_1 = -\frac{k}{2}$ must be positive (yielding $m_1 = \frac{1}{2}$) **[M1]**

Calculates the gradient of L_2 when $k = -1$: $m_2 = \frac{-1-3}{2} = -2$, and uses the y-intercept from point $A(0, 5)$ to write $y = -2x + 5$ **[M1]**

$2x + y = 5$ **[A1]**

[Total Marks: 10]

936 $4y = 3x - 12$ implies gradient of P is $\frac{3}{4}$ **[M1]**

Gradient of Q is $-\frac{4}{3}$ **[M1]**

Substitute $(6, -2)$ into $y = -\frac{4}{3}x + c$ or $y - (-2) = -\frac{4}{3}(x - 6)$ **[M1]**

$-\frac{4}{3}x + 6$ **[A1]**

[Total Marks: 4]

937 **(a)** $\left(\frac{-2+6}{2}, \frac{4+0}{2}\right)$ **[M1]**

$(2, 2)$ **[A1]**

(b) Gradient of $AB = \frac{0-4}{6-(-2)} = -\frac{1}{2}$ **[M1]**

Gradient of perpendicular bisector $= 2$ **[M1]**

Substitute midpoint $(2, 2)$ into $y - y_1 = m(x - x_1)$: $y - 2 = 2(x - 2)$ **[M1]**

$y = 2x - 2$ **[A1]**

[Total Marks: 6]

938 **(a)** Uses the midpoint formula: $\left(\frac{-4+2}{2}, \frac{5+(-7)}{2}\right)$ **[M1]**

$(-1, -1)$ **[A1]**

(b) Uses the gradient formula: $\frac{-7-5}{2-(-4)}$ or $\frac{5-(-7)}{-4-2}$ **[M1]**

$\frac{-12}{6} = -2$ **[A1]**

(c) Identifies the perpendicular gradient is $\frac{1}{2}$ (or 0.5) **[B1]**

Substitutes their midpoint $(-1, -1)$ and perpendicular gradient into $y = mx + c$:
$-1 = \frac{1}{2}(-1) + c$ **[M1]**

$\frac{1}{2}x - \frac{1}{2}$ (or $0.5x - 0.5$) **[A1]**

(d) Sets $y = 0$ in their equation from part (c): $0 = \frac{1}{2}x - \frac{1}{2}$ **[M1]**

$(1, 0)$ **[A1]**

[Total Marks: 9]

939 **(a)** Calculates gradient of AB: $\frac{9-1}{2-(-2)} = 2$ **[B1]**

Uses perpendicular gradient $m_{BC} = -\frac{1}{2}$ (or -0.5) **[M1]**

Substitutes $B(2, 9)$: $9 = -\frac{1}{2}(2) + c$, giving $y = -\frac{1}{2}x + 10$ (or $-0.5x + 10$) **[A1]**

(b) Substitutes $y = 0$ into equation of BC: $0 = -0.5x + 10$ **[M1]**

$(20, 0)$ **[A1]**

(c) Finds $AB = \sqrt{4^2 + 8^2} = \sqrt{80}$ and $BC = \sqrt{18^2 + (-9)^2} = \sqrt{405}$ **[M1]**

$\frac{1}{2} \times \sqrt{80} \times \sqrt{405} = 90$ **[A1]**

[Total Marks: 7]

940 **(a)** Substitutes $x = 0$ into $2y + 3x = 12$ because P is on the y-axis **[M1]**

$(0, 6)$ **[A1]**

(b) Rearranges L_1 equation to find gradient: $2y = -3x + 12 \implies y = -\frac{3}{2}x + 6$, so $m = -\frac{3}{2}$ **[B1]**

States perpendicular gradient for L_2 is $\frac{2}{3}$ **[M1]**

$\frac{2}{3}x + 6$ **[A1]**

(c) Substitutes $y = 0$ into the equation of L_2 to find the coordinates of Q **[M1]**

$0 = \frac{2}{3}x + 6 \implies x = -9$ **[A1]**

Uses the distance formula for $P(0, 6)$ and $Q(-9, 0)$: $\sqrt{(-9-0)^2 + (0-6)^2}$ **[M1]**

$\sqrt{117}$ or $3\sqrt{13}$ **[A1]**

[Total Marks: 9]

941 (a) Gradient of $AB = \frac{5-2}{5-(-1)} = \frac{1}{2}$ **[M1]**

Gradient of $BC = -2$ **[A1]**

$-2x + 15$ **[A1]**

(b) $0 = -2x + 15$ **[M1]**

$(7.5, 0)$ **[A1]**

(c) $AB = \sqrt{(5-(-1))^2 + (5-2)^2} = \sqrt{45}$ and $BC = \sqrt{(7.5-5)^2 + (0-5)^2} = \sqrt{31.25}$ **[M1]**

Area $= \frac{1}{2} \times \sqrt{45} \times \sqrt{31.25}$ **[M1]**

18.75 **[A1]**

(d) Midpoint of $AC = \left(\frac{-1+7.5}{2}, \frac{2+0}{2}\right) = (3.25, 1)$ **[B1]**

Gradient of $AC = \frac{0-2}{7.5-(-1)} = -\frac{4}{17}$, so perpendicular gradient $= \frac{17}{4}$ $(= 4.25)$ **[M1]**

$y - 1 = 4.25(0 - 3.25)$ **[M1]**

$(0, -12.8)$ **[A1]**

[Total Marks: 12]

942 (a) Hexagon **[B1]**

(b) Obtuse **[B1]**

(c) Equilateral **[B1]**

[Total Marks: 3]

943 (a) Chord **[B1]**

(b) Tangent **[B1]**

(c) Sector **[B1]**

(d) Acute **[B1]**

[Total Marks: 4]

944 **(a)** Chord **[B1]**

(b) Tangent **[B1]**

(c) Minor segment (Accept segment) **[B1]**

(d) Isosceles **[B1]**

[Total Marks: 4]

945 **(a)** Radius **[B1]**

(b) Tangent **[B1]**

(c) Diameter **[B1]**

(d) 90 **[B1]**

[Total Marks: 4]

946 **(a)** **(i)** Diameter **[B1]**

(ii) Minor segment (or segment) **[B1]**

(iii) Major sector **[B1]**

(b) Reflex angle (or 260°, reflex) **[B1]**

[Total Marks: 4]

947 **(a)** **(i)** $(n-2) \times 180 = 1080$ or $1080 \div 180$ **[M1]**

8 **[A1]**

(ii) Octagon **[B1]**

(b) Rhombus **[B1]**

(c) Cylinder **[B1]**

[Total Marks: 5]

948 **(a)** **(i)** tangent **[B1]**

(ii) chord **[B1]**

(iii) minor segment (accept segment) **[B1]**

(b) Relationship: perpendicular **[B1]**

Angle $OTQ = 90°$ **[B1]**

[Total Marks: 5]

949 **(a)** Tangent **[B1]**

(b) Radius **[B1]**

(c) 90 **[B1]**

(d) Uses $2 \times \pi \times 5$ **[M1]**

10π **[A1]**

[Total Marks: 5]

950 **(a)** Kite **[B1]**

(b) Sector (or minor sector) **[B1]**

(c) Chord **[B1]**

(d) Congruent **[B1]**

[Total Marks: 4]

951 **(a)** Kite **[B1]**

(b) Perpendicular **[B1]**

(c) Congruent **[B1]**

(d) Obtuse **[B1]**

(e) Perpendicular bisector **[B1]**

[Total Marks: 5]

952 **(a)** $180 - 156 = 24$ **[B1]**

(b) $\frac{360}{24}$ **[M1]**

15 **[A1]**

(c) Decagon **[B1]**

(d) Obtuse **[B1]**

[Total Marks: 5]

953 **(a)** Tangent **[B1]**

(b) Chord **[B1]**

(c) Segment **[B1]**

(d) Sector **[B1]**

(e) Reflex **[B1]**

[Total Marks: 5]

954 **(a)** Isosceles trapezium **[B1]**

(b) Calculates number of sides: $\frac{360^\circ}{36^\circ} = 10$ **[M1]**

Regular decagon (or decagon) **[A1]**

(c) Faces = 12 **[B1]**

Edges = 30 **[B1]**

[Total Marks: 5]

955 **(a)** trapezium **[B1]**

(b) **(i)** obtuse **[B1]**

(ii) $360 - 112 = 248$ **[B1]**

(c) **(i)** 34 **[B1]**

(ii) Angle $BCA = 180^\circ - 112^\circ - 34^\circ = 34^\circ$ (or interior angle $BCD = 68^\circ$) **[M1]**

112 **[A1]**

[Total Marks: 6]

956 **(a)** **(i)** Exterior angle $= 180^\circ - 144^\circ = 36^\circ$ **[M1]**

$\frac{360}{36} = 10$ **[A1]**

(ii) Decagon **[B1]**

(b) **(i)** Sum of interior angles $= (6-2) \times 180^\circ = 720^\circ$ **[B1]**

$x + 2x + (x + 30) + (3x + 60) + 110 + 100 = 720$ OR $7x + 300 = 720$ **[M1]**

$x = 60$ **[A1]**

(ii) Right angle (since $60 + 30 = 90$) **[B1]**

(iii) Reflex angle (since $3(60) + 60 = 240$) **[B1]**

[Total Marks: 8]

957 **(a)** $\frac{(5-2) \times 180}{5}$ or $180 - \frac{360}{5}$ **[M1]**

108 **[A1]**

(b) $\frac{(6-2) \times 180}{6}$ or $180 - \frac{360}{6}$ **[M1]**

120 **[A1]**

(c) Identifies angles around point A: 108° and 120° **[M1]**

Calculates reflex angle $= 360 - 108 - 120$ **[M1]**

132 **[A1]**

[Total Marks: 7]

958 **(a)** Similar (geometrically similar) **[B1]**

(b) In a right circular cone, the axis VB is perpendicular to the circular base planes, so $\angle VAC = 90°$ **[B1]**

Both AC and BD lie in parallel planes and are perpendicular to the same line VB in the same cross-sectional plane, so AC is parallel to BD **[B1]**

(c) Uses similarity scale factor: $\dfrac{VA}{VA+6} = \dfrac{4.5}{7.5}$ (or $\dfrac{3}{5}$) **[M1]**

$5VA = 3(VA + 6)$

2VA $= 18[M1]$

9 **[A1]**

[Total Marks: 6]

959 **(a)** Frustum **[B1]**

(b) **(i)** Similar **[B1]**

(ii) Scale factor $= \dfrac{20+10}{10} = 3$ or $\dfrac{10}{30} = \dfrac{1}{3}$ **[M1]**

$r = \dfrac{12}{3} = 4$ **[A1]**

(iii) Radius difference $= 12 - 4 = 8$ **[M1]**

$\sqrt{20^2 + 8^2}$ **[M1]**

21.5 or 21.54... **[A1]**

[Total Marks: 7]

960 **(a)** Isosceles **[B1]**

(b) $138 + 180$ or $360 - (180 - 138)$ **[M1]**

318 **[A1]**

(c) **(i)** Perpendicular bisector **[B1]**

(ii) Angle $BAC = 138 - 62 = 76°$ **[M1]**

Bearing $= 062 + (76 \div 2) = 100°$ **[A1]**

(d) Trapezium **[B1]**

[Total Marks: 7]

961 **(a)** $\dfrac{h}{h+12} = \dfrac{3}{5}$ **[M1]**

$5h = 3(h + 12)$ **[M1]**

$5h = 3h + 36 \implies 2h = 36 \implies h = 18$ **[A1]**

(b) $\dfrac{r}{r+5} = \dfrac{3}{5}$ or $\dfrac{r}{r+5} = \dfrac{18}{30}$ **[M1]**

$5r = 3r + 15 \implies 2r = 15 \implies r = 7.5$ **[A1]**

(c) Volume ratio (small cone : large cone) $= \left(\frac{3}{5}\right)^3 = \frac{27}{125}$ **[M1]**

Volume of frustum $= 125 - 27 = 98$, so $k = 98$ **[A1]**

[Total Marks: 7]

962 (a) Uses similar triangles to find the height H of the full cone: $\frac{H}{12} = \frac{H-15}{4}$ **[M1]**

$4H = 12H - 180 \implies 8H = 180 \implies H = 22.5$ cm **[A1]**

Volume of full cone $= \frac{1}{3} \times \pi \times 12^2 \times 22.5 = 1080\pi$ **[M1]**

Volume of small cone removed $= \frac{1}{3} \times \pi \times 4^2 \times (22.5 - 15) = \frac{1}{3} \times \pi \times 16 \times 7.5 = 40\pi$ **[M1]**

Volume of frustum $= 1080\pi - 40\pi = 1040\pi$ or 3267 or 3270 (3 sf) **[A1]**

(b) Identifies the area scale factor is $\frac{9}{25}$ **[B1]**

Calculates the length scale factor: $\sqrt{\frac{9}{25}} = \frac{3}{5}$ **[M1]**

New height $= 15 \times \frac{3}{5} = 9$ **[A1]**

[Total Marks: 8]

963 (a) Setting up an equation connecting interior and exterior angles, e.g., $4x + x = 180$ **[M1]**

Exterior angle $= 36°$ or Interior angle $= 144°$ **[A1]**

$n = 10$ **[A1]**

Decagon **[B1]**

(b) Faces $= 12$ **[B1]**

Edges $= 30$ **[B1]**

Vertices $= 20$ **[B1]**

(c) Finding the linear scale factor $\sqrt{\frac{16}{9}}$ or $\frac{4}{3}$ **[M1]**

Finding the volume scale factor $\left(\frac{4}{3}\right)^3$ or $\frac{64}{27}$ **[M1]**

256π **[A1]**

[Total Marks: 10]

964 (a) Angle ADE = Angle ABC because they are corresponding angles **[B1]**

Angle A is common to both triangles OR Angle AED = Angle ACB (corresponding angles) **[B1]**

(b) Length scale factor $= \frac{6}{10}$ or 0.6 **[M1]**

$DE = 15 \times 0.6 = 9$ **[A1]**

(c) $AC = \frac{7.5}{0.6} = 12.5$ **[M1]**

$EC = 12.5 - 7.5 = 5$ **[A1]**

(d) Trapezium **[B1]**

(e) Area scale factor $= 0.6^2 = 0.36$ **[M1]**

Area of triangle $ABC = \frac{18}{0.36} = 50$ **[M1]**

Area of $DBCE = 50 - 18 = 32$ **[A1]**

[Total Marks: 10]

965 **(a)** **(i)** frustum **[B1]**

(ii) tetrahedron **[B1]**

(b) Reason: the cut is parallel to the base so corresponding angles are equal **[B1]**

Scale factor $= 0.4$ (or $\frac{2}{5}$) **[B1]**

(c) $10 \times \frac{6}{15}$ **[M1]**

4 **[A1]**

(d) **(i)** Number of sides $= \frac{360}{36} = 10$ **[M1]**

decagon **[A1]**

(ii) Faces $= 12$ **[B1]**

Edges $= 30$ **[B1]**

[Total Marks: 10]

966 $048 + 180$ **[M1]**

228 **[A1]**

[Total Marks: 2]

967 Identifies the relationship between alternate angles or uses $180° + 115°$ **[M1]**

295 **[A1]**

[Total Marks: 2]

968 **(a)** $180 + 58$ **[M1]**

238 **[A1]**

(b) $180 - 84 = 096$ or 096 **[B1]**

[Total Marks: 3]

969 **(a)** $068 + 180$ **[M1]**

248 **[A1]**

(b) 270 **[B1]**

[Total Marks: 3]

970 **(a)** $072^\circ + 180^\circ$ **[M1]**

252 **[A1]**

(b) $180^\circ - 72^\circ = 108^\circ$ **[B1]**

(c) Identifies the bearing of L from S is 252° **[M1]**

$252^\circ - 215^\circ = 37$ **[A1]**

[Total Marks: 5]

971 **(a)** $128 + 180$ **[M1]**

308 **[A1]**

(b) 52 **[B1]**

(c) $180 - 90 - 52$ **[M1]**

38 **[A1]**

[Total Marks: 5]

972 **(a)** $180 + 148$ OR $360 - (180 - 148)$ **[M1]**

328 **[A1]**

(b) **(i)** 270 **[B1]**

(ii) Identifies the bearing of P from B is 090° **[B1]**

$148 - 90 = 58$ **[A1]**

[Total Marks: 5]

973 **(a)** 090
090 **[B1]**

(b) 054
+
180
054
+180

[M1]

234
234 **[A1]**

(c) Angle
=
90
54
=
36
BAC=90
54
=36
and angle
=
180
36
2
=
72
ABC=
2
180
36
=72

[M1]

Bearing
=
234
72
=
162
=234
72
=162 **[A1]**

[Total Marks: 5]

974 **(a)** $124 + 180$ **[M1]**

304 **[A1]**

(b) $\angle PQR = 90^\circ$ or $\angle QPR = 180^\circ - 124^\circ = 56^\circ$ **[M1]**

$\angle PRQ = 180^\circ - 90^\circ - 56^\circ = 34^\circ$ **[A1]**

[Total Marks: 4]

975 **(a)** Bearing of P from $Q = 115 + 180 = 295^\circ$ **[M1]**

Angle $PQR = 295^\circ - 205^\circ = 90^\circ$ **[A1]**

(b) $15^2 + 8^2$ **[M1]**

$225 + 64 = 289$ **[M1]**

17 **[A1]**

[Total Marks: 5]

976 **(a)** 065
065 (accept
063
063 to
067
067) **[B1]**

(b) Measured length
=
6
PQ=6 cm (accept
5.8
5.8 cm to
6.2
6.2 cm) **[B1]**

30
30 (accept
29
29 to
31
31) **[A1]**

(c) Line drawn from
Q on bearing
140
140
(accept
138
138
to
142
142
) **[B1]**

Point
R marked at distance
5
5 cm from
Q (accept
4.8
4.8 cm to
5.2
5.2 cm) and labelled **[B1]**

(d) 065
+
180
065

$+180$

[M1]

245

245 **[A1]**

[Total Marks: 7]

977 **(a)** Calculates reverse bearing of A from $B = 065° + 180° = 245°$ **[M1]**

$245° - 120° = 125$ **[A1]**

(b) Calculates reverse bearing of B from $C = 120° + 180° = 300°$ **[M1]**

$300° - 285° = 15$ **[A1]**

(c) **(i)** $285° - 180°$ **[M1]**

105 **[A1]**

(ii) Calculates $\angle CAB = 105° - 065° = 40°$ (or $180° - 125° - 15° = 40°$) **[M1]**

$\angle ABD = 180° - 2(40°) = 100$ **[A1]**

[Total Marks: 8]

978 **(a)** Interior angle at Q on North line $= 180° - 125° = 55°$ **[M1]**

Angle $PQR = 360° - 240° - 55° = 65°$ **[A1]**

(b) $PR^2 = 45° + 60° - 2(45)(60)\cos(65°)$ **[M2]**

$PR = 57.8$ or $57.82...$ **[A1]**

[Total Marks: 5]

979 **(a)** 150 **[B1]**

(b) $180 + 65$ OR $360 - (180 - 65)$ **[M1]**

245 **[A1]**

(c) Identifies bearing of A from B is 245° (can follow through from (b)) **[M1]**

$245 - 70 = 175$ **[A1]**

(d) 25 **[B1]**

[Total Marks: 6]

980 **(a)** Calculates the interior angle at Q on the North side: $180° - 50° = 130°$ OR bearing of P from $Q = 180° + 50° = 230°$ **[M1]**

Calculates angle PQR: $360° - 130° - 140° = 90°$ OR $230° - 140° = 90°$ **[A1]**

(b) Uses Pythagoras' theorem: $PR^2 = 40^2 + 60^2$ **[M1]**

$72.11\cdots = 72.1$ **[A1]**

(c) Uses trigonometry to find angle QPR: $\tan(QPR) = \dfrac{60}{40}$ **[M1]**

Angle $QPR = 56.3\ldots^\circ$ **[M1]**

Bearing $= 050^\circ + 56.3^\circ = 106.3^\circ$ **[A1]**

[Total Marks: 7]

981 (a) Interior angle at B (South to BA) $= 35^\circ$ OR interior angle from North line $= 180 - 35 = 145^\circ$ **[M1]**

Angle $ABC = 360 - 145 - 125 = 90^\circ$ OR $125 - 35 = 90^\circ$ **[A1]**

(b) $AC^2 = 12^2 + 16^2$ **[M1]**

$AC^2 = 144 + 256 = 400$ **[M1]**

$AC = 20$ **[A1]**

(c) $\tan(\angle BAC) = \dfrac{16}{12}$ or $\sin(\angle BAC) = \dfrac{16}{20}$ **[M1]**

$\angle BAC = 53.13^\circ$ **[A1]**

Bearing $= 035^\circ + 53.13^\circ = 088.1^\circ$ **[A1]**

[Total Marks: 8]

982 (a) Bearing of A from B is $065^\circ + 180^\circ = 245^\circ$ or Interior angle between North and AB is $180^\circ - 65^\circ = 115^\circ$ **[M1]**

$245^\circ - 155^\circ = 90^\circ$ or $360^\circ - 115^\circ - 155^\circ = 90^\circ$ **[A1]**

(b) Triangle ABC is right-angled and isosceles, so $\angle BCA = (180^\circ - 90^\circ) \div 2 = 45^\circ$ **[M1]**

Bearing of B from C $= 155^\circ + 180^\circ = 335^\circ$ **[M1]**

Bearing of A from C $= 335^\circ - 45^\circ = 290^\circ$ **[A1]**

(c) $\angle DAB = 90^\circ$ **[M1]**

Bearing of D from A $= 065^\circ + 90^\circ = 155^\circ$ **[A1]**

[Total Marks: 7]

983 (a) Calculates angle $ABC = 240^\circ - 170^\circ = 70^\circ$ (or equivalent geometric reasoning) **[B1]**

Uses the cosine rule: $AC^2 = 120^\circ + 180^2 - 2(120)(180)\cos(70^\circ)$ **[M1]**

$AC^2 = 14400 + 32400 - 43200(0.3420\ldots) = 32024.7\ldots$ **[M1]**

$178.9\cdots = 179$ (to 3 sf) **[A1]**

(b) Uses the sine rule to find angle BAC: $\dfrac{\sin(BAC)}{180} = \dfrac{\sin(70^\circ)}{178.95}$ **[M1]**

$\sin(BAC) = 0.9452\cdots \implies$ Angle $BAC = 70.9^\circ$ **[M1]**

Bearing $= 060^\circ + 70.9^\circ = 130.9^\circ$ **[A1]**

[Total Marks: 7]

984 **(a)** $\angle QPR = 118° - 42° = 76°$ **[M1]**

$QR^2 = 24^2 + 35^2 - 2(24)(35)\cos(76°)$ **[M1]**

$QR = 37.3$ or $37.34\ldots$ **[A1]**

(b) $\dfrac{\sin(\angle PQR)}{35} = \dfrac{\sin(76°)}{37.34}$ OR $\cos(\angle PQR) = \dfrac{24^2 + 37.34° - 35^2}{2(24)(37.34)}$ **[M1]**

$\sin(\angle PQR) = 0.909\ldots$ OR $\cos(\angle PQR) = 0.415\ldots$ **[M1]**

$\angle PQR = 65.4°$ or $65.42\ldots°$ **[A1]**

Bearing $= 180 + 42 - 65.4 = 156.6°$ (or $157°$) **[A1]**

[Total Marks: 7]

985 **(a)** Angle $ABC = 360° - 145° - (180° - 42°) = 77°$ **[M1]**

$\dfrac{\sin(BAC)}{BC} = \dfrac{\sin(77°)}{18}$ **[M1]**

$BC^2 = 12^2 + 18^2 - 2(12)(18)\cos(BAC)$ (Implicitly requires finding BC or BAC first. Correct approach: $\dfrac{\sin(BCA)}{12} = \dfrac{\sin(77°)}{18}$) **[M1]**

$\sin(BCA) = \dfrac{12\sin(77°)}{18} = 0.6495...$ **[A1]**

Angle $BCA = 40.5°$ **[A1]**

Angle $BAC = 180° - 77° - 40.5° = 62.5°$ **[M1]**

Bearing $= 042° + 62.5° = 104.5°$ **[A1]**

(b) Boat sails on perpendicular bisector of BC. Angle $BAC = 62.5°$. The bisector of angle BAC is not necessarily the perpendicular bisector of BC.

Line is perpendicular bisector of BC.

Let midpoint of BC be M. In $\triangle ABM$, AM is the path.

Use cosine rule to find BC: $BC^2 = 12^2 + 18^2 - 2(12)(18)\cos(62.5°)$

BC = 16.39...

$\sin(ABM)/AM = \sin(BAM)/8.19...$ (This approach is overly complex).

Simpler: Path is perpendicular bisector. Angle of BC with North is 145°.

Perpendicular to BC has bearing $145° - 90° = 055°$.

Recognise path is perpendicular to BC **[M1]**

Bearing $= 145 - 90 = 055$ (or $055°$) **[A1]**

[Total Marks: 7]

986 **(a)** Bearing of P from $A = 050 + 180 = 230°$ **[M1]**

Angle $PAB = 230° - 160° = 70°$ **[M1]**

$PB^2 = 40^2 + 60^2 - 2(40)(60)\cos(70°)$ **[M1]**

59.6 or $59.65\ldots$ **[A1]**

(b) Uses sine rule or cosine rule to find angle APB **[M1]**

$\frac{\sin(APB)}{60} = \frac{\sin(70°)}{59.65}$ or $\cos(APB) = \frac{40^2 + 59.65^2 - 60^2}{2(40)(59.65)}$ **[M1]**

Angle $APB = 70.9°$ $(70.93\ldots°)$ **[A1]**

Bearing $= 050 + 70.9 = 120.9°$ **[A1]**

[Total Marks: 8]

987 **(a)** Bearing of P from $Q = 052° + 180° = 232°$ **[M1]**

Angle $PQR = 232° - 160° = 72°$ **[A1]**

(b) Uses Cosine Rule: $PR^2 = 45^2 + 30^2 - 2(45)(30)\cos(72°)$ **[M1]**

$PR^2 = 2925 - 2700(0.3090...)$ or $2090.65...$ **[M1]**

45.7 (accept 45.72 to 45.73) **[A1]**

(c) Uses Sine Rule to find $\angle QPR$: $\frac{\sin(\angle QPR)}{30} = \frac{\sin(72°)}{45.72}$ **[M1]**

$\sin(\angle QPR) = 0.624...$ **[M1]**

$\angle QPR = 38.6°$ (or $38.61°$) **[A1]**

Bearing $= 052° + 38.6° = 090.6$ **[A1]**

[Total Marks: 9]

988 **(a)** Identifies angle $PQR = 230° - 120° = 110°$ (or uses parallel lines to find $180 - 50 = 130$ and $360 - 130 - 120 = 110$) **[B1]**

Uses Cosine rule: $PR^2 = 45^2 + 60^2 - 2(45)(60)\cos(110°)$ [M2] (M1 for correct substitution into formula)

$PR = 86.4$ or $86.44\ldots$ **[A1]**

(b) Uses Sine rule: $\frac{\sin(QPR)}{60} = \frac{\sin(110°)}{86.44}$ **[M1]**

Angle $QPR = 40.7°$ $(40.73\ldots)$ **[A1]**

Bearing $= 050° + 40.7° = 090.7°$ **[A1]**

(c) $090.7 + 180$ **[M1]**

270.7 **[A1]**

[Total Marks: 9]

989 **(a)** 270
270 **[B1]**

(b) 018.6
+
180
018.6
+180

[M1]

198.6
198.6 **[A1]**

(c) (i) 198.6
48.4
198.6
48.4

[M1]

150.2
150.2 **[A1]**

(ii) Angle
=
90
18.6
=
71.4
ADC=90
18.6
=71.4

[B1]

Angle
=
322.8
270
=
52.8
BCD=322.8
270
=52.8

[B1]

Angle
=
360
(
71.4
+
150.2
+
52.8
)
=
85.6
ABC=360
(71.4
+150.2
+52.8
)=85.6 **[A1]**

(d) Bearing of
A from
=

048.4
+
180
=
228.4
B=048.4
+180
=228.4

[M1]

Bearing of
C from
=
228.4
85.6
=
142.8
B=228.4
85.6
=142.8 **[A1]**

[Total Marks: 10]

990 Scale factor $= \frac{6+3}{6} = 1.5$ (or ratio $\frac{QR}{8} = \frac{9}{6}$) **[M1]**

12 **[A1]**

[Total Marks: 2]

991 (a) Finds scale factor: $\frac{4+6}{4} = 2.5$ **[M1]**

12.5 **[A1]**

(b) Uses area scale factor: $16 \times (2.5)^2$ **[M1]**

100 **[A1]**

[Total Marks: 4]

992 (a) Linear scale factor $= \frac{4+2}{4} = 1.5$ (or equivalent ratio $\frac{6}{4}$) **[M1]**

7.5 **[A1]**

(b) Area of $ABC = 8 \times 1.5^2 = 18$ **[M1]**

10 **[A1]**

[Total Marks: 4]

993 (a) Scale factor $= \frac{18}{6} = 3$ **[M1]**

15 **[A1]**

(b) Area scale factor $= 3^2 = 9$ **[M1]**

$\frac{72}{9} = 8$ **[A1]**

[Total Marks: 4]

994 **(a)** Volume scale factor $= \left(\frac{15}{6}\right)^3 = \frac{125}{8}$ **[M1]**

$V_B = 48 \times \frac{125}{8} = 750$ **[A1]**

(b) Area scale factor $= \left(\frac{15}{6}\right)^2 = \frac{25}{4}$ **[M1]**

$A_A = 200 \div \frac{25}{4} = 32$ **[A1]**

[Total Marks: 4]

995 **(a)** Length scale factor $= \frac{9}{3} = 3$ **[M1]**

15 **[A1]**

(b) Area scale factor $= 3^2 = 9$ **[M1]**

6 **[A1]**

[Total Marks: 4]

996 **(a)** Scale factor $= \frac{4+2}{4} = 1.5$ (or $\frac{6}{4}$ or $\frac{3}{2}$) **[M1]**

7.5 **[A1]**

(b) Area scale factor $= 1.5^2 = 2.25$ **[M1]**

Area of triangle $ABC = 8 \times 2.25 = 18$ **[M1]**

10 **[A1]**

[Total Marks: 5]

997 **(a)** Scale factor $= \frac{4+2}{4}$ or 1.5 **[M1]**

9 **[A1]**

(b) Area scale factor $= 1.5^2$ or 2.25 **[M1]**

Area of triangle $ABC = 32 \times 2.25$ or 72 **[M1]**

40 **[A1]**

[Total Marks: 5]

998 **(a)** Identifies the scale factor is $\frac{AD+DB}{AD}=\frac{6}{4}$ or 1.5 **[M1]**

9 **[A1]**

(b) Uses the area scale factor: 1.5^2 or $\frac{9}{4}$ or 2.25 **[M1]**

Calculates the area of triangle ABC: $12\times 2.25=27$ **[M1]**

15 **[A1]**

[Total Marks: 5]

999 **(a)** Finds the linear scale factor: $\frac{15}{10}=1.5$ (or $\frac{3}{2}$) **[M1]**

Finds the area scale factor: $1.5^2=2.25$ (or $\frac{9}{4}$) **[M1]**

$120\times 2.25=270$ **[A1]**

(b) Finds the volume scale factor: $1.5^3=3.375$ (or $\frac{27}{8}$) **[M1]**

Sets up the correct division: $540\div 3.375$ (or $540\times\frac{8}{27}$) **[M1]**

160 **[A1]**

[Total Marks: 6]

1000 **(a)** $\frac{EF}{6}=\frac{10}{4}$ (or scale factor = 2.5) **[M1]**

15 **[A1]**

(b) $AC=12.5\times\frac{4}{10}$ (or $12.5\div 2.5$) **[M1]**

5 **[A1]**

(c) Area scale factor $=\left(\frac{10}{4}\right)^2=6.25$ (or $\frac{25}{4}$) **[M1]**

$10\times 6.25=62.5$ **[A1]**

[Total Marks: 6]

1001 **(a)** $\frac{6}{9}=\frac{8}{QR}$ or scale factor $=\frac{9}{6}$ (or 1.5) **[M1]**

12 **[A1]**

(b) $\frac{6}{9}=\frac{PT}{12}$ or $PT=12\div 1.5$ **[M1]**

8 **[A1]**

(c) Area scale factor $=\left(\frac{9}{6}\right)^2$ or 1.5^2 or 2.25 **[M1]**

Area of triangle $PQR=20\times 2.25=45$ **[M1]**

25 **[A1]**

[Total Marks: 7]

1002 **(a)** Area scale factor $= \frac{45}{20}$ or $\frac{9}{4}$ **[M1]**

Linear scale factor $= \sqrt{\frac{9}{4}} = \frac{3}{2}$ **[M1]**

Volume $B = 16 \times \left(\frac{3}{2}\right)^3 = 16 \times \frac{27}{8} = 54$ **[A1]**

(b) Height $A = 12 \div \frac{3}{2}$ or $12 \times \frac{2}{3}$ **[M1]**

8 **[A1]**

[Total Marks: 5]

1003 **(a)** Volume scale factor $= \frac{1350}{400} = \frac{27}{8}$ (or 3.375) **[M1]**

Linear scale factor $= \sqrt[3]{\frac{27}{8}} = \frac{3}{2}$ (or 1.5) **[M1]**

18 **[A1]**

(b) Area scale factor $= \left(\frac{3}{2}\right)^2 = \frac{9}{4}$ (or 2.25) **[M1]**

$\frac{360}{2.25} = 160$ **[A1]**

[Total Marks: 5]

1004 **(a)** Angle AEB = angle CED (vertically opposite angles) **[B1]**

Angle BAE = angle DCE (alternate angles, AB parallel to DC) **[B1]**

Angle ABE = angle CDE (alternate angles, AB parallel to DC), so all three angles are equal **[B1]**

(b) **(i)** $\frac{EC}{4} = \frac{9}{6}$ **[M1]**

6 **[A1]**

(ii) $\frac{BE}{12.5 - BE} = \frac{6}{9}$ (or $BE = \frac{2}{5} \times 12.5$) **[M1]**

5 **[A1]**

(c) $12 \times \left(\frac{9}{6}\right)^2$ **[M1]**

27 **[A1]**

[Total Marks: 9]

1005 **(a)** Angle A (or $\angle DAE$ and $\angle CAB$) is common **[B1]**

Angle ADE = Angle ACB (given) **[B1]**

Therefore the third angles are equal ($\angle AED = \angle ABC$) and the triangles are similar (AAA) **[B1]**

(b) Sets up correct ratio: $\frac{AD}{AC} = \frac{AE}{AB}$ or $\frac{x}{4+5} = \frac{4}{x+(x+1)}$ **[M1]**

$\frac{x}{9} = \frac{4}{2x+1}$ **[M1]**

$2x^2 + x - 36 = 0$ **[M1]**

$(2x+9)(x-4) = 0$ implies $x = 4$ (ignoring negative root) **[A1]**

(c) Area scale factor $= \left(\frac{4}{9}\right)^2 = \frac{16}{81}$ **[M1]**

16 : 65 **[A1]**

[Total Marks: 9]

1006 **(a)** Linear scale factor $= 14 \div 4$ or 3.5 **[M1]**

21 **[A1]**

(b) Area scale factor $= 3.5^2$ or 12.25 **[M1]**

104.125 **[A1]**

[Total Marks: 4]

1007 **(a)** Area scale factor $= \frac{800}{450}$ (or $\frac{16}{9}$) **[M1]**

Length scale factor $= \sqrt{\frac{16}{9}} = \frac{4}{3}$ and Volume scale factor $= \left(\frac{4}{3}\right)^3 = \frac{64}{27}$ **[M1]**

3200 **[A1]**

(b) Uses length scale factor: $48 \div \frac{4}{3}$ (or $48 \times \frac{3}{4}$) **[M1]**

36 **[A1]**

[Total Marks: 5]

1008 **(a)** Volume scale factor $= \frac{1200}{150} = 8$ **[M1]**

Length scale factor $= \sqrt[3]{8} = 2$ **[M1]**

24 **[A1]**

(b) Area scale factor $= 2^2 = 4$ **[M1]**

110 **[A1]**

[Total Marks: 5]

1009 **(a)** Volume scale factor $= \frac{1080}{320}$ or $\frac{27}{8}$ or 3.375 **[M1]**

Linear scale factor $= \sqrt[3]{\frac{27}{8}}$ or 1.5 **[M1]**

27 **[A1]**

(b) Area scale factor $= 1.5^2$ or 2.25 or $\frac{9}{4}$ **[M1]**

Surface area of $A = 630 \div 2.25$ or $630 \times \frac{4}{9}$ **[M1]**

280 **[A1]**

[Total Marks: 6]

1010 **(a)** Volume scale factor $= \left(\frac{30}{12}\right)^3$ or 2.5^3 or 15.625 **[M1]**

5000 **[A1]**

(b) Area scale factor $= \left(\frac{30}{12}\right)^2$ or 2.5^2 or 6.25 **[M1]**

$1125 \div 6.25 = 180$ **[A1]**

(c) $4.80 \times \left(\frac{30}{12}\right)^3$ or 4.80×15.625 **[M1]**

75 or 75.00 **[A1]**

[Total Marks: 6]

1011 **(a)** Volume scale factor $= \frac{8640}{320} = 27$ **[M1]**

Length scale factor $= \sqrt[3]{27} = 3$ **[M1]**

36 **[A1]**

(b) Area scale factor $= 3^2 = 9$ **[M1]**

250 **[A1]**

(c) Mass $= 8.5 \times 320 = 2720\,\text{g}$ **[M1]**

2.72 **[A1]**

[Total Marks: 7]

1012 **(a)** Angle A is common to both triangles **[B1]**

Angle ABC = Angle ADE (corresponding angles as BC is parallel to DE) **[B1]**

(b) $\frac{DE}{7.5} = \frac{6+4}{6}$ **[M1]**

12.5 **[A1]**

(c) Area scale factor $= \left(\frac{10}{6}\right)^2 = \frac{25}{9}$ **[M1]**

Area of $ADE = 18 \times \dfrac{25}{9} = 50$ **[M1]**

Area of $BCED = 50 - 18 = 32$ **[A1]**

[Total Marks: 7]

1013 **(a)** States that angle A is common to both triangles. **[B1]**

States that angle ADE = angle ABC (and/or angle AED = angle ACB) because they are corresponding angles (due to parallel lines). **[B1]**

(b) Calculates the full length of AB: $6 + 4 = 10$ **[M1]**

Finds the linear scale factor: $\dfrac{10}{6} = \dfrac{5}{3}$ and sets up equation $BC = 9 \times \dfrac{5}{3}$ **[M1]**

15 **[A1]**

(c) Finds the area scale factor: $\left(\dfrac{5}{3}\right)^2 = \dfrac{25}{9}$ **[M1]**

Calculates the area of triangle ABC: $27 \times \dfrac{25}{9} = 75$ **[M1]**

Calculates the area of the trapezium: $75 - 27 = 48$ **[A1]**

[Total Marks: 8]

1014 **(a)** Linear scale factor $= \dfrac{45}{15} = 3$ **[M1]**

Mass (Volume) scale factor $= 3^3 = 27$ **[M1]**

13.5 **[A1]**

(b) Mass scale factor from P to $R = \dfrac{4000}{500} = 8$ **[M1]**

Linear scale factor $= \sqrt[3]{8} = 2$ **[M1]**

30 **[A1]**

(c) Area scale factor from P to $Q = 3^2 = 9$ **[M1]**

270 **[A1]**

[Total Marks: 8]

1015 **(a)** Calculates the volume ratio: $\dfrac{1080}{320} = \dfrac{27}{8}$ or 3.375 **[M1]**

Calculates the length scale factor: $\sqrt[3]{\dfrac{27}{8}} = \dfrac{3}{2}$ or 1.5 **[M1]**

18 **[A1]**

(b) Uses the area scale factor: $450 \div 1.5^2$ or $450 \times \left(\dfrac{2}{3}\right)^2$ **[M1]**

200 **[A1]**

(c) Calculates the volume of cone R: $320 + 1080 = 1400$ **[B1]**

Finds the length scale factor from P to R: $\sqrt[3]{\frac{1400}{320}}$ or $\sqrt[3]{4.375}$ **[M1]**

Calculates the height of R: $12 \times \sqrt[3]{4.375} = 19.626\ldots$ and rounds to 19.6 **[A1]**

[Total Marks: 8]

1016 **(a)** Angle AXB = Angle CXD (vertically opposite) **[B1]**

Angle ABX = Angle CDX (alternate angles, AB parallel to DC) **[B1]**

(Accept Angle BAX = Angle DCX for the second B1 mark)

(b) Length scale factor $= \frac{15}{9} = \frac{5}{3}$ **[M1]**

$DX = 6 \times \frac{5}{3} = 10$ **[M1]**

$BD = 6 + 10 = 16$ **[A1]**

(c) Area scale factor $= \left(\frac{15}{9}\right)^2 = \frac{25}{9}$ **[M1]**

Area $= 27 \times \frac{25}{9} = 75$ **[A1]**

(d) Ratio of bases $\frac{CX}{AX} = \frac{15}{9} = \frac{5}{3}$ **[M1]**

Area $= 27 \times \frac{5}{3} = 45$ **[A1]**

[Total Marks: 9]

1017 **(a)** $\frac{4}{4+x} = \frac{5}{7+x}$ OR $\frac{4}{x} = \frac{5}{x+2}$ **[M1]**

$4(x+7) = 5(x+4)$ OR $4(x+2) = 5x$ **[M1]**

$x = 8$ **[A1]**

(b) Linear scale factor $= \frac{4+8}{4} = 3$ **[M1]**

$BC = 3.5 \times 3 = 10.5$ **[A1]**

(c) Area of $\triangle ABC = 3^2 \times A = 9A$ **[M1]**

Area of trapezium $= 9A - A = 8A$ **[A1]**

[Total Marks: 7]

1018 **(a)** Volume ratio $A : B = \frac{108}{500} = \frac{27}{125}$ **[M1]**

Area ratio $= \left(\sqrt[3]{\frac{27}{125}}\right)^2 = \left(\frac{3}{5}\right)^2 = \frac{9}{25}$ **[M1]**

Let surface areas be $9x$ and $25x$. $9x + 25x = 340 \implies 34x = 340 \implies x = 10$ **[M1]**

$25 \times 10 = 250$ **[A1]**

(b) Volume of small cone $= 1 - \frac{7}{8} = \frac{1}{8}$ of Cone B **[M1]**

Ratio of heights $= \sqrt[3]{\frac{1}{8}} = \frac{1}{2}$ **[M1]**

Height of small cone $= 20 \times \frac{1}{2} = 10$ **[A1]**

[Total Marks: 7]

1019 **(a)** $\angle AXB = \angle DXC$ (vertically opposite angles) **[B1]**

$\angle BAX = \angle CDX$ OR $\angle ABX = \angle DCX$ (angles in the same segment) **[B1]**

Third angle in triangle OR states the other pair of angles in the same segment, concluding triangles are similar (AAA) **[B1]**

(b) Linear scale factor $= \frac{7.5}{5} = 1.5$ OR $\frac{DX}{4} = \frac{7.5}{5}$ **[M1]**

$DX = 6$ **[A1]**

$DC = 9$ **[A1]**

(c) Area scale factor $= 1.5^2 = 2.25$ **[M1]**

26.55 **[A1]**

[Total Marks: 8]

1020 **(a)** $\angle AXB = \angle DXC$ because they are vertically opposite angles **[B1]**

$\angle BAX = \angle CDX$ (or $\angle ABX = \angle DCX$) because they are angles subtended by the same arc in the same segment **[B1]**

Therefore, the triangles are similar by AAA similarity **[B1]**

(b) Uses property of intersecting chords (or similar triangle ratios):
$(x+2)(x-2) = (x-3)(x+4)$ **[M1]**

Expands brackets correctly: $x^2 - 4 = x^2 + x - 12$ **[M1]**

8 **[A1]**

(c) Calculates side lengths $AX = 10$ and $DX = 12$, and sets up area scale factor: $45 \times \left(\frac{10}{12}\right)^2$ **[M1]**

31.25 **[A1]**

[Total Marks: 8]

1021 **(a)** $\frac{x+2}{(x+2)+(x-1)} = \frac{8}{8+x}$ or $\frac{x+2}{x-1} = \frac{8}{x}$ **[M1]**

$(x+2)(x+8) = 8(2x+1)$ or $x(x+2) = 8(x-1)$ **[M1]**

$x^2 + 10x + 16 = 16x + 8$ or $x^2 + 2x = 8x - 8$ leading to $x^2 - 6x + 8 = 0$ **[A1]**

(b) $(x-2)(x-4) = 0$ **[M1]**

2 and 4 **[A1]**

(c) When $x = 4$, length ratio $\dfrac{AD}{AB} = \dfrac{6}{9} = \dfrac{2}{3}$ **[M1]**

Area of $ABC = 24 \div \left(\dfrac{2}{3}\right)^2 = 54$ **[M1]**

30 **[A1]**

[Total Marks: 8]

1022 **(a)** Angle A is common to both triangles **[B1]**

Angle AED = Angle ABC (given in the question) **[B1]**

Therefore, Angle ADE = Angle ACB (sum of angles in a triangle is 180°), so the triangles are similar (AAA) **[B1]**

(b) $AB = 6 + 9 = 15$ cm **[M1]**

$\dfrac{AC}{AD} = \dfrac{AB}{AE} \implies \dfrac{AC}{6} = \dfrac{15}{5}$ **[M1]**

$AC = 18$, so $EC = 18 - 5 = 13$ **[A1]**

(c) Area scale factor for the triangles $= \left(\dfrac{5}{15}\right)^2 = \dfrac{1}{9}$ **[M1]**

1 : 8 **[A1]**

[Total Marks: 8]

1023 **(a)** States that angle BAX = angle DCX OR angle ABX = angle CDX due to alternate angles (between parallel lines) **[B1]**

States a second valid pair of angles are equal (e.g. vertically opposite angles at X or the other pair of alternate angles) concluding the triangles are similar (AAA) **[B1]**

(b) Identifies the length scale factor: $\dfrac{10.5}{7}$ or 1.5 **[M1]**

Calculates $CX = 4 \times 1.5 = 6$ **[M1]**

10 **[A1]**

(c) Calculates the area of triangle CDX: $16 \times 1.5^2 = 36$ **[M1]**

Recognises the ratio of areas of triangle BCX to triangle ABX is equal to the ratio of bases $CX : AX = 1.5 : 1$ (as they share the same height from B) **[M1]**

Calculates the area of triangle BCX (or ADX): $16 \times 1.5 = 24$ **[M1]**

Total area $= 16 + 36 + 24 + 24 = 100$ **[A1]**

[Total Marks: 9]

1024 **(a)** Angle A is common to both triangles **[B1]**

Angle ABC = Angle AED (given), so the third angles must be equal, hence AA similarity **[B1]**

(b) 2.5 or $\frac{5}{2}$ **[B1]**

(c) $\frac{AB}{AE} = 2.5 \implies \frac{2x + 2.5}{x} = 2.5$ (or equivalent valid ratio) **[M1]**

$2x + 2.5 = 2.5x$ **[M1]**

$x = 5$ **[A1]**

(d) Area scale factor $= 2.5^2 = 6.25$ **[M1]**

Formulates correct equation for areas, e.g., $6.25A - A = 42$ or $5.25A = 42$ **[M1]**

8 **[A1]**

[Total Marks: 9]

1025 (a) $\angle AXB = \angle CXD$ because they are vertically opposite angles **[B1]**

$\angle BAX = \angle DCX$ (or $\angle ABX = \angle CDX$) because they are alternate angles, $AB \parallel DC$ **[B1]**

All three pairs of corresponding angles are equal, so the triangles are similar (AAA) **[B1]**

(b) $\frac{2x - 3}{x} = \frac{6}{4}$ or equivalent **[M1]**

$4(2x - 3) = 6x$ or $2x - 3 = 1.5x$ **[M1]**

$0.5x = 3$ leading to $x = 6$ **[A1]**

(c) Area of triangle $CDX = 24 \times 1.5^2$ or 54 **[M1]**

Recognises triangle BCX shares altitude with ABX and base ratio is 6 : 4 **[M1]**

Area of triangle BCX (and triangle ADX) $= 24 \times 1.5$ or 36 **[M1]**

Total area $= 24 + 54 + 36 + 36 = 150$ **[A1]**

[Total Marks: 10]

1026 (a) Finds the volume scale factor: $\frac{2700}{800} = 3.375$ (or $\frac{27}{8}$) **[M1]**

Finds the linear scale factor by cube rooting: $\sqrt[3]{3.375} = 1.5$ **[M1]**

$12 \times 1.5 = 18$ **[A1]**

(b) Finds the area scale factor from the linear scale factor: $1.5^2 = 2.25$ **[M1]**

Sets up the multiplication for the cost: 18×2.25 **[M1]**

40.50 **[A1]**

(c) Identifies the area scale factor between C and A is 1.44 **[B1]**

Finds the linear scale factor: $\sqrt{1.44} = 1.2$ **[M1]**

Finds the volume scale factor: $1.2^3 = 1.728$ **[M1]**

Calculates the volume of C: $800 \times 1.728 = 1382.4$ **[A1]**

[Total Marks: 10]

1027 (a) Slant height $l = \sqrt{10^2 + 24^2} = 26$ **[M1]**

Surface Area $= \pi(10)^2 + \pi(10)(26)$ **[M1]**

360π **[A1]**

(b) $V_A = \frac{1}{3}\pi(10)^2(24) = 800\pi$ **[M1]**

Volume Ratio $= \frac{100\pi}{800\pi} = \frac{1}{8}$ **[M1]**

Area Ratio $= \left(\frac{1}{8}\right)^{\frac{2}{3}} = \frac{1}{4}$ **[M1]**

Surface Area of $B = 360\pi \times \frac{1}{4} = 90\pi$ **[A1]**

(c) Volume of frustum $= 800\pi - 100\pi = 700\pi$ **[M1]**

$\frac{4}{3}\pi R^3 = 700\pi$ **[M1]**

$R^3 = 700 \times \frac{3}{4} = 525$ **[M1]**

$R = \sqrt[3]{525} = 8.07$ (accept $8.066\ldots$) **[A1]**

[Total Marks: 11]

1028 **(a)** $320 \times \left(\frac{24}{15}\right)^2$ **[M1]**

819.2 **[A1]**

(b) $\frac{V_X}{2048} = \left(\frac{15}{24}\right)^3$ **[M1]**

Volume scale factor $= 1.6^3 = 4.096$ (or $\frac{512}{125}$) **[M1]**

500 **[A1]**

(c) $\frac{h_Z}{15} = \sqrt[3]{\frac{1372}{500}}$ **[M1]**

Scale factor $= \sqrt[3]{2.744} = 1.4$ **[A1]**

$15 \times 1.4 = 21$ **[A1]**

(d) Number of complete copies $= 4$ (from $\frac{2048}{500} = 4.096$) **[M1]**

Remaining volume $= 2048 - 4 \times 500 = 48$ cm^3 **[A1]**

$\frac{4}{3}\pi r^3 = 48$ giving $r^3 = \frac{36}{\pi}$ $(= 11.459\ldots)$ **[M1]**

2.25 **[A1]**

[Total Marks: 12]

1029 **(a)** 9 **[B1]**

(b) 2 **[B1]**

(c) Isosceles triangle **[B1]**

[Total Marks: 3]

1030 **(a)** 10 **[B1]**

(b) Rhombus OR Rectangle **[B1]**

(c) 72 **[B1]**

[Total Marks: 3]

1031 **(a)** **(i)** 6 **[B1]**

(ii) 6 **[B1]**

(b) **(i)** 1 **[B1]**

(ii) 1 **[B1]**

[Total Marks: 4]

1032 **(a)** 4 **[B1]**

(b) 0 **[B1]**

(c) Correct reflection of at least two 'petals' (e.g. vertices at $(1, 4)$ and $(4, 3)$) **[M1]**

Fully correct shape drawn **[A1]**

[Total Marks: 4]

1033 **(a)** **(i)** 15 **[B1]**

(ii) 15 **[B1]**

(b) Rectangle OR Rhombus **[B1]**

(c) 5 **[B1]**

[Total Marks: 4]

1034 Parallelogram: 0 lines, order 2 **[B2, B1 for each correct value]**

Regular Pentagon: 5 lines, order 5 **[B2, B1 for each correct value]**

Isosceles Trapezium: 1 line, order 1 **[B2, B1 for each correct value]**

[Total Marks: 6]

1035 **(a)** **(i)** 0 **[B1]**

(ii) 2 **[B1]**

(b) Both squares in the left column row 2 and right column row 3 shaded (or top row column 2 and bottom row column 3 shaded) **[B2]**

(c) Lines of symmetry = 12 **[B1]**

Exterior angle = 30° **[B1]**

[Total Marks: 6]

1036 **(a)** Identifies 6 vertical planes of symmetry or 1 horizontal plane of symmetry **[M1]**

7 **[A1]**

(b) 6 **[B1]**

(c) 4 **[B1]**

[Total Marks: 4]

1037 **(a)** **(i)** 6 **[B1]**

(ii) 4 **[B1]**

(iii) Infinite **[B1]**

(b) Exterior angle $= \dfrac{360}{12} = 30^\circ$ **[M1]**

Interior angle $= 180 - 30 = 150$ **[A1]**

[Total Marks: 5]

1038 **(a)** 4 **[B1]**

(b) 6 **[B2]**

Strictly 5 (vertical) or 1 (horizontal) planes identified **[B1]**

(c) A straight line passing through the apex (top vertex) **[B1]**

and the centre of the circular base. **[B1]**

[Total Marks: 5]

1039 **(a)** **(i)** 6 longitudinal planes and 1 perpendicular bisecting plane seen or implied **[M1]**

7 **[A1]**

(ii) 6 **[B1]**

(b) Planes of symmetry = 5 **[B1]**

Order of rotational symmetry = 5 **[B1]**

(c) Planes of symmetry = infinite (or ∞) **[B1]**

Slant height $l = \sqrt{6^2 + 8^2} = 10$ cm **[M1]**

Total surface area $= \pi(6)(10) + \pi(6)^2 = 96\pi$ **[A1]**

[Total Marks: 8]

1040 **(a)** Recognises that order 4 symmetry implies point symmetry (order 2) about M, so M is the midpoint of AC **[M1]**

$(5, -2)$ **[A1]**

(b) Gradient of $AC = \dfrac{-2-6}{5-1}$ or $\dfrac{-8}{4} = -2$ **[M1]**

Substitute $(1, 6)$ or $(5, -2)$ or $(3, 2)$ into $y = -2x + c$ OR $y - 6 = -2(x - 1)$ **[M1]**

$-2x + 8$ **[A1]**

(c) Deduces that $ABCD$ must be a square because it has order 4 rotational symmetry (area is $|AB|^2$) **[M1]**

$|AB|^2 = (7-1)^2 + (4-6)^2$ **[M1]**

40 **[A1]**

[Total Marks: 8]

1041 **(a)** 4 **[B2]**

If 0 scored, allow [B1] for identifying either 3 (vertical) OR 1 (horizontal/mid-plane).

(b) 3 **[B1]**

(c) 7 **[B2]**

If 0 scored, allow [B1] for 6.

[Total Marks: 5]

1042 **(a)** **(i)** $\dfrac{360}{15}$ **[M1]**

24 **[A1]**

(ii) 24 **[B1]**

(b) Kite OR Isosceles Trapezium (accept Isosceles Trapezoid) **[B1]**

(c) Infinite (accept an infinite number) **[B1]**

[Total Marks: 5]

1043 **(a)** **(i)** 4 **[B1]**

(ii) Infinite (or infinitely many) **[B1]**

(b) Uses volume formula: $\dfrac{1}{3} \times 6^2 \times h = 48$ **[M1]**

Solves for height: $h = 4$ **[A1]**

Calculates distance from centre of base to a corner $= \sqrt{3^2 + 3^2} = \sqrt{18}$ (or $3\sqrt{2}$ or 4.24...) **[M1]**

Uses Pythagoras' theorem for sloping edge $l = \sqrt{4^2 + (\sqrt{18})^2}$ **[M1]**

5.83 (or 5.830...) **[A1]**

[Total Marks: 7]

1044 **(a)** Understanding that diagonals of a rhombus intersect at 90°, hence gradient $m = -\frac{1}{3}$ **[M1]**

$-\frac{1}{3}x$ (accept equivalent forms) **[A1]**

(b) 2 **[B1]**

(c) Exterior angle $= 180° - 165° = 15°$ **[M1]**

$n = \frac{360}{15}$ **[M1]**

$n = 24$ **[A1]**

[Total Marks: 6]

1045 **(a)** **(i)** 8 **[B1]**

(ii) 9 **[B2]**

If B0, score B1 for 8 (identifying only the vertical planes)

(b) **(i)** 12 **[B1]**

(ii) $(12 - 2) \times 180$ **[M1]**

1800 **[A1]**

(c) $(-2, 3)$ **[B2]**

If B0, score M1 for point plotted correctly on grid OR one coordinate correct

[Total Marks: 8]

1046 **(a)** Calculates length $AB = \sqrt{(5-1)^2 + (4-2)^2} = \sqrt{20}$ **[M1]**

Calculates length $BC = \sqrt{(3-5)^2 + (8-4)^2} = \sqrt{20}$ **[M1]**

States $AB = BC$ therefore the triangle is isosceles **[A1]**

(b) Identifies midpoint of $AC = (\frac{1+3}{2}, \frac{2+8}{2}) = (2, 5)$ **[M1]**

Calculates gradient of line of symmetry $= \frac{4-5}{5-2} = -\frac{1}{3}$ (or uses perpendicular to AC) **[M1]**

Substitutes $(5, 4)$ or $(2, 5)$ into $y = -\frac{1}{3}x + c$ to find c **[M1]**

$-\frac{1}{3}x + \frac{17}{3}$ **[A1]**

(c) Identifies $n = 15$ and uses $\frac{(15-2) \times 180}{15}$ OR $180 - (360 \div 15)$ **[M1]**

156 **[A1]**

[Total Marks: 9]

1047 **(a)** 2 **[B1]**

(b) Gradient of $BD = -\frac{1}{2}$ (using $m_1 \times m_2 = -1$) **[M1]**

$-0.5x$ or $-\frac{1}{2}x$ **[A1]**

(c) **(i)** Uses Area $= \frac{1}{2} \times d_1 \times d_2$: $\frac{1}{2} \times AC \times 4\sqrt{5} = 40$ **[M1]**

$AC = \frac{80}{4\sqrt{5}} = 4\sqrt{5}$ **[A1]**

(ii) $OA = 2\sqrt{5}$, so $k^2 + (2k)^2 = (2\sqrt{5})^2$ or $5k^2 = 20$ **[M1]**

$(2, 4)$ **[A1]**

(iii) Sets up distance equation for $B(x, -0.5x)$ with $OB = 2\sqrt{5}$: $x^2 + (-0.5x)^2 = 20$ **[M1]**

$1.25x^2 = 20$ giving $x^2 = 16$ **[M1]**

$(4, -2)$ (rejects negative x due to visual quadrant) **[A1]**

[Total Marks: 10]

1048 **(a)** States that a right prism with a regular n-gon base has n vertical planes of symmetry and 1 horizontal plane of symmetry (halfway along its length) **[B1]**

Sets up the equation $n + 1 = 7$, yielding $n = 6$ **[B1]**

(b) **(i)** Recognises the side length of the hexagon $= 12 \div 2 = 6$ cm **[B1]**

Calculates area of the hexagonal cross-section $= 6 \times \left(\frac{1}{2} \times 6 \times 6 \times \sin(60°)\right) = 54\sqrt{3}$ (or 93.53...) **[M1]**

Uses Volume = Area × Length to find Length $= \frac{450}{54\sqrt{3}}$ **[M1]**

4.81 (or 4.811...) **[A1]**

(ii) Method for total area = 2 × Base Area + 6 × Lateral Rectangular Area **[M1]**

Substitutes values: $2 \times 54\sqrt{3} + 6 \times (6 \times 4.811...)$ **[M1]**

Simplifies to $108\sqrt{3} + 100\sqrt{3}$ (or 187.06... + 173.2...) **[M1]**

360 (or 360.2... to 360.3...) **[A1]**

[Total Marks: 10]

1049 **(a)** Name: kite **[B1]**

Order of rotational symmetry = 1 **[B1]**

(b) The line of symmetry AC is the perpendicular bisector of BD (or B is the reflection of D in AC) **[B1]**

(c) **(i)** Angle $BAM = \frac{106°}{2} = 53°$ **[B1]**

$BM = 9 \sin 53°$ $(= 7.1877\ldots)$ **[M1]**

$BD = 2 \times 7.1877\cdots = 14.4$ **[A1]**

(ii) $\sin(\text{angle } BCM) = \dfrac{7.1877\ldots}{15}$ $(= 0.47918\ldots)$ **[M1]**

Angle $BCM = 28.63°$ **[A1]**

Angle $BCD = 2 \times 28.63° = 57.3$ **[A1]**

(d) One plane along AC through the length of the prism and one plane bisecting the prism perpendicular to its length **[M1]**

2 **[A1]**

[Total Marks: 11]

1050 **(a)** 48 **[B1]**

(b) 56 **[B1]**

(c) $180 - 48 - 56 = 76$ **[B1]**

[Total Marks: 3]

1051 **(a)** 125 **[B1]**

(b) Corresponding angles (are equal) **[B1]**

(c) 144 **[B1]**

[Total Marks: 3]

1052 **(a)** Identifies alternate angles by drawing a parallel line through E OR forms a triangle by extending a line **[M1]**

85 **[A1]**

(b) $\dfrac{360}{24}$ **[M1]**

15 **[A1]**

[Total Marks: 4]

1053 **(a)** Angle $PRC = 54°$ **[B1]**

Reason: alternate angles (AB is parallel to CD) **[B1]**

(b) Angle $PQR = 180° - 112° = 68°$ **[M1]**

Angle $PRQ = 180° - 54° - 68° = 58$ **[A1]**

(c) 58 **[B1]**

[Total Marks: 5]

1054 **(a)** 52 **[B1]**

(b) Angles on a straight line sum to 180°: $180 - (52 + 61)$ **[M1]**

67 **[A1]**

(c) Identifies angle $TVU = 61°$ (alternate angle to QTV) or angle $VTW + TVW$ exterior relationship **[M1]**

Angle $TVW = 180 - 61 = 119$ **[A1]**

[Total Marks: 5]

1055 **(a)** 118 **[B1]**

Alternate angles (are equal) **[B1]**

(b) $180° - 118° = 62$ **[B1]**

(c) Uses isosceles triangle GHK: $\angle HGK = 62°$ and $\angle BGH = 180° - 118° = 62°$ (or $\angle BGK = \angle GKD = 180° - 56°$) **[M1]**

124 **[A1]**

[Total Marks: 5]

1056 **(a)** $360 \div 24$ **[M1]**

15 **[A1]**

(b) $(15 - 2) \times 180$ **[M1]**

2340 **[A1]**

(c) Exterior angle $= 180 - 165 = 15°$ **[M1]**

$360 \div 15 = 24$ **[A1]**

[Total Marks: 6]

1057 **(a)** $180 - (55 + 70)$ **[M1]**

55 **[A1]**

(b) 55 **[B1]**

(c) Angle $DAE = 55°$ (from part a) AND Angle $ADE = 55°$ **[B1]**

Two angles are equal, so the triangle is isosceles **[B1]**

[Total Marks: 5]

1058 **(a)** Exterior angle $= \dfrac{360°}{5} = 72°$ or interior sum $= (5 - 2) \times 180° = 540°$ **[M1]**

108 **[A1]**

(b) Uses angles at point B: $\angle ABC = 360° - 108° - 102°$ **[M1]**

$\angle ABC = 150°$ **[A1]**

Exterior angle of regular n-gon $= 180° - 150° = 30°$ **[M1]**

$n = \dfrac{360^\circ}{30^\circ} = 12$ **[A1]**

[Total Marks: 6]

1059 **(a)** **(i)** $\dfrac{(6-2)\times 180^\circ}{6}$ or $180^\circ - \dfrac{360^\circ}{6}$ **[M1]**

120 **[A1]**

(ii) 108 **[B1]**

(b) Angle $ABG = 360^\circ - 120^\circ - 108^\circ = 132^\circ$ **[M1]**

Triangle ABG is isosceles ($AB = BG$) **[M1]**

$\dfrac{180^\circ - 132^\circ}{2} = 24$ **[A1]**

(c) $(2x + 10) + (3x - 15) + (x + 25) + 100 = 360$ **[M1]**

$6x + 120 = 360$ giving $x = 40$ **[A1]**

Largest angle $= 3(40) - 15 = 105$ **[A1]**

[Total Marks: 9]

1060 **(a)** Equates sum of angles to formula: $3 \times 140 + 165(n-3) = 180(n-2)$ **[M1]**

Expands brackets correctly: $420 + 165n - 495 = 180n - 360$ **[M1]**

Rearranges to isolate n: $15n = 285$ (or equivalent) **[M1]**

19 **[A1]**

(b) States $a + b + c = 180$ **[B1]**

Substitutes $a + b = c$ to get $c + c = 180 \implies 2c = 180 \implies c = 90$ completing the proof **[B1]**

(c) **(i)** Calculates exterior angle of $P = \dfrac{360}{12} = 30^\circ$ so exterior angle of $Q = 15^\circ$ **[M1]**

24 **[A1]**

(ii) Calculates interior angles of P and Q: $180 - 30 = 150^\circ$ and $180 - 15 = 165^\circ$ **[M1]**

$360 - (150 + 165) = 45$ **[A1]**

[Total Marks: 10]

1061 **(a)** Sum of interior angles of a pentagon $= (5-2) \times 180^\circ = 540^\circ$ **[M1]**

$x + (x + 20) + (2x - 10) + (2x + 30) + 140 = 540$ **[M1]**

$6x + 180 = 540 \implies 6x = 360 \implies x = 60$ **[A1]**

(b) Substituting $x = 60$ into $(2x + 30)$ or checking all angle sizes $(60^\circ, 80^\circ, 110^\circ, 150^\circ, 140^\circ)$ **[M1]**

150 **[A1]**

[Total Marks: 5]

1062 **(a)** Recognises the sum of the interior angles of a pentagon is $540°$ **[B1]**

Sets up a correct equation: $(2x + 10) + 100 + 80 + 3x + (x + 50) = 540$ **[M1]**

50 **[A1]**

(b) Sets up a relationship for the angles, e.g., $4E + E = 180$ where E is the exterior angle **[M1]**

$E = 36°$ **[M1]**

10 **[A1]**

[Total Marks: 6]

1063 **(a)** $5x + 10 = 7x - 30$ **[M1]**

$x = 20$ **[A1]**

Corresponding angles (are equal) **[B1]**

(b) Angle $GHD = 7(20) - 30 = 110°$ **[B1]**

Angle $GHK = 180 - 110 = 70°$ (Angles on a straight line) **[B1]**

Angle $GKH = 180 - (70 + 35) = 75°$ (Angles in a triangle) **[B1]**

[Total Marks: 6]

1064 **(a)** **(i)** Equates the sum of the angles to the interior angle sum of a hexagon ($720°$):
$x + 2x + x + 30 + 2x + 10 + 3x - 20 + 160 = 720$ **[M1]**

Simplifies the equation to $9x + 180 = 720$ (or $9x = 540$) **[M1]**

60 **[A1]**

(ii) Sets up an equation for the exterior angle e:
$6.5e + e = 180°$ **[M1]**

$7.5e = 180° \implies e = 24°$ **[M1]**

Number of sides $n = \dfrac{360}{24} = 15$ **[A1]**

(iii) Let exterior angles be E_m and E_n. Sets up the first equation:
$E_m + E_n = 54$ **[B1]**

Relates interior difference to exterior difference:
$(180 - E_m) - (180 - E_n) = 18 \implies E_n - E_m = 18$ **[M1]**

Solves the simultaneous equations by addition (or substitution):
$2E_n = 72 \implies E_n = 36°$ **[M1]**

Calculates n: $n = \dfrac{360}{36} = 10$ **[A1]**

Calculates m: $E_m = 18° \implies m = \dfrac{360}{18} = 20$ **[A1]**

[Total Marks: 11]

1065 **(a)** Recognises that $\angle PXQ = \angle BPX + \angle DQX$ (e.g. by drawing a parallel line through X to form alternate angles) **[M1]**

$(3x + 15) + (2x + 10) = 75$ **[M1]**

$5x + 25 = 75 \implies 5x = 50 \implies x = 10$ **[A1]**

(b) Expression for exterior or interior angle of P: $\dfrac{360}{n}$ or $180 - \dfrac{360}{n}$ **[B1]**

Expression for exterior or interior angle of Q: $\dfrac{360}{3n}$ or $180 - \dfrac{120}{n}$ **[B1]**

Forms a correct equation: $\left(180 - \dfrac{120}{n}\right) - \left(180 - \dfrac{360}{n}\right) = 16$ OR $\dfrac{360}{n} - \dfrac{120}{n} = 16$ **[M1]**

$\dfrac{240}{n} = 16 \implies n = 15$ **[A1]**

[Total Marks: 7]

1066 (a) $\dfrac{(5-2) \times 180}{5}$ OR $180 - \dfrac{360}{5}$ **[M1]**

108 **[A1]**

(b) Interior angle of hexagon $= 120°$ **[B1]**

Angle $AEI = 360 - 108 - 120 = 132$ **[A1]**

(c) $AE = EI$ (sides of regular polygons sharing an edge are equal) **[M1]**

Triangle AEI is isosceles **[M1]**

Angle $EAI = \dfrac{180 - 132}{2} = 24$ **[A1]**

[Total Marks: 7]

1067 (a) $(5 - 2) \times 180 \div 5$ OR $180 - (360 \div 5)$ **[M1]**

108 **[A1]**

(b) Interior angle of square $= 90°$ **[B1]**

Interior angle of n-gon $= 360 - 90 - 108 = 162°$ **[M1]**

$n = 360 \div (180 - 162) = 20$ **[A1]**

(c) Recognises triangle PFA is isosceles (since $PA = PF$ from shared sides of regular polygons) **[M1]**

Angle $PFA = (180 - 162) \div 2 = 9$ **[A1]**

[Total Marks: 7]

1068 (a) Calculates interior angle sum of a hexagon: $(6 - 2) \times 180° = 720°$ **[B1]**

Sums algebraic expressions:
$(2x + 15) + (3x - 20) + (2x + 35) + (4x - 45) + (x + 50) + (3x + 10) = 720$ **[M1]**

Simplifies to $15x + 45 = 720$ **[M1]**

45 **[A1]**

(b) Calculates $\angle Q = 3(45) - 20 = 115°$, $\angle R = 2(45) + 35 = 125°$, and $\angle S = 4(45) - 45 = 135°$

[M1]

Finds sum of interior angles along QRS: $115° + 125° + 135° = 375°$ (or constructs line through R parallel to PQ, giving co-interior angles $65° + 60° = 125° e 135°$) **[M1]**

States No, because the sum of interior angles $\angle Q + \angle R + \angle S = 375° e 360°$ (or corresponding/alternate angles are not equal) **[A1]**

[Total Marks: 7]

1069 **(a)** Finds an expression for the interior angle ABC using the sum of angles at a point:
Interior angle $ABC = 360° - (300 - 2x)° = (60 + 2x)°$ **[M1]**

Draws a parallel line through B and uses alternate angles to form an equation:
Interior angle $ABC = (3x + 10)° + (2x + 20)°$ **[M1]**

Equates the two expressions for the interior angle ABC:
$5x + 30 = 60 + 2x$ **[M1]**

Solves the equation correctly to show the result:
$3x = 30 \implies x = 10$ **[A1]**

(b) Calculates the interior angle ABC using $x = 10$:
Angle $ABC = 60 + 2(10) = 80°$ **[M1]**

Recognises triangle ABC is isosceles and sets up the base angle calculation:
$\frac{180 - 80}{2}$ **[M1]**

50 **[A1]**

[Total Marks: 7]

1070 **(a)** Angle $PTS = 66°$ **[B1]**

Reason: co-interior angles sum to $180°$ (PW is parallel to US) **[B1]**

(b) Sum of interior angles of a pentagon $= (5 - 2) \times 180° = 540°$ **[B1]**

$114 + (3x + 16) + (4x - 14) + (2x + 34) + 66 = 540$ **[M1]**

$9x + 216 = 540$ leading to $9x = 324$ and $x = 36$ **[A1]**

(c) **(i)** 114 **[B1]**

(ii) Angle $PQR = 3(36) + 16 = 124°$ and angle $WQR = 180° - 124° = 56°$ **[M1]**

56 **[A1]**

(d) Exterior angle $= \frac{180°}{12} = 15°$ **[M1]**

$n = \frac{360}{15}$ **[M1]**

24 **[A1]**

[Total Marks: 11]

1071 **(a)** 65 **[B1]**

(b) Uses isosceles triangle OAB, e.g. $\dfrac{180-130}{2}$ **[M1]**

25 **[A1]**

[Total Marks: 3]

1072 **(a)** 116 **[B1]**

(b) $\dfrac{180-116}{2}$ **[M1]**

32 **[A1]**

[Total Marks: 3]

1073 **(a)** 34 **[B1]**

(b) Recognises angle at centre is twice angle at circumference $\implies 2 \times 34$ **[M1]**

68 **[A1]**

[Total Marks: 3]

1074 **(a)** 55 **[B1]**

(b) 55 **[B1]**

(c) Angles in the same segment (are equal) **[B1]**

[Total Marks: 3]

1075 **(a)** **(i)** 55 **[B1]**

(ii) Recognising triangle OAB is isosceles (radii) OR $(180-110) \div 2$ **[M1]**

35 **[A1]**

[Total Marks: 3]

1076 **(a)** 112 **[B1]**

(b) 124 **[B1]**

(c) Opposite angles of a cyclic quadrilateral sum to 180° **[B1]**

[Total Marks: 3]

1077 **(a)** **(i)** 48 **[B1]**

(ii) Angles in the same segment **[B1]**

(b) 96 **[B1]**

[Total Marks: 3]

1078 **(a)** 65 **[B1]**

(b) $180 - 65$ **[M1]**

115 **[A1]**

[Total Marks: 3]

1079 **(a)** 42 **[B1]**

(b) $180 - (42 + 65) = 73$ **[B1]**

(c) Angle BAD + Angle $BCD = 180$ (cyclic quadrilateral) **[M1]**

$180 - 73 = 107$ **[A1]**

[Total Marks: 4]

1080 **(a)** **(i)** 58 **[B1]**

(ii) Alternate segment theorem **[B1]**

(b) Angle $AOB = 2 \times 58 = 116°$ **[M1]**

Angle $OBA = \dfrac{180 - 116}{2} = 32°$ **[A1]**

[Total Marks: 4]

1081 **(a)** 55 **[B1]**

(b) Angle at the centre is twice the angle at the circumference. **[B1]**

(c) 30 **[B1]**

(d) $(180 - 110) \div 2$ **[M1]**

35 **[A1]**

[Total Marks: 5]

1082 **(a)** Angle at centre is twice the angle at the circumference **[M1]**

62 **[A1]**

(b) Opposite angles in a cyclic quadrilateral sum to 180° **[M1]**

$180 - 62 = 118$ **[A1]**

(c) Triangle OAC is isosceles ($OA = OC$) **[M1]**

$(180 - 124) \div 2 = 28$ **[A1]**

[Total Marks: 6]

1083 **(a)** **(i)** 42 **[B1]**

(ii) 55 **[B1]**

(b) $90 - 42$ **[M1]**

48 **[A1]**

(c) $180 - (55 + 42)$ **[M1]**

83 **[A1]**

[Total Marks: 6]

1084 **(a)** Angle $ABC = 65°$ **[B1]**

Reason: the angle at the centre is twice the angle at the circumference **[B1]**

(b) Angle $ADC = 115°$ **[B1]**

Reason: opposite angles of a cyclic quadrilateral sum to $180°$ **[B1]**

(c) Angle $OBA = 22°$ ($OA = OB$), so angle $OBC = 65° - 22° = 43°$ **[M1]**

Angle $OCB = 43$ ($OB = OC$) **[A1]**

[Total Marks: 6]

1085 **(a)** **(i)** 50 **[B1]**

(ii) $180 - 115 = 65$ **[B1]**

(iii) $180 - (50 + 65)$ **[M1]**

65 **[A1]**

[Total Marks: 4]

1086 **(a)** 55 **[B1]**

(b) Uses opposite angles of a cyclic quadrilateral (e.g., $180 - 100$) **[M1]**

80 **[A1]**

(c) Uses angles in a triangle sum to $180°$ (e.g., $180 - (55 + 80)$) **[M1]**

45 **[A1]**

[Total Marks: 5]

1087 **(a)** 42 **[B1]**

(b) 75 **[B1]**

(c) Uses sum of angles in triangle PRS equal to $180°$ ($180 - 75 - 48$) **[M1]**

57 **[A1]**

(d) 57 **[B1]**

[Total Marks: 5]

1088 **(a)** **(i)** 90 **[B1]**

(ii) Angle in a semicircle is 90° **[B1]**

(b) 54 **[B1]**

(c) $180 - 54$ **[M1]**

126 **[A1]**

[Total Marks: 5]

1089 **(a)** Angle at centre = 2× angle at circumference **[M1]**

$118 \times 2 = 236$ **[A1]**

(b) $360 - 236 = 124$ **[B1]**

(c) Triangle POR is isosceles, so angle OPR = angle $ORP = \frac{180 - 124}{2} = 28°$ **[M1]**

Angle $ORS = 90°$ (tangent-radius), so angle $PRS = 90 - 28 = 62$ **[A1]**

[Total Marks: 5]

1090 **(a)** Angle $OAT = 90°$ and Angle $OBT = 90°$ **[B1]**

$360 - 90 - 90 - 44 = 136$ **[A1]**

(b) 68 **[B1]**

(c) $180 - 68$ **[M1]**

112 **[A1]**

[Total Marks: 5]

1091 **(a)** **(i)** 62 **[B1]**

(ii) Angle in a semicircle (is 90°) **[B1]**

(b) 28 **[B1]**

(c) Angle $ABC = 62°$ OR Angle $ADB = 90°$ **[M1]**

118 **[A1]**

[Total Marks: 5]

1092 **(a)** **(i)** 58 **[B1]**

(ii) Alternate segment theorem **[B1]**

(b) Angle in a semicircle is 90° (Angle $ACB = 90°$) **[M1]**

$180 - 90 - 58$ **[M1]**

32 **[A1]**

(c) $180 - 32$ **[M1]**

148 **[A1]**

[Total Marks: 7]

1093 **(a)** Angle $TAC = 52°$ **[B1]**

Reason: alternate segment theorem **[B1]**

(b) **(i)** Angle $ATC = 180° - 54° - 52° = 74°$ **[M1]**

Angle $ABC = 74$ (angles in the same segment) **[A1]**

(ii) Angle $ACT = 54°$ (alternate segment theorem) **[M1]**

Angle $ACB = 86° - 54° = 32$ **[A1]**

(iii) Angle $OTP = 90°$ (tangent to radius $= 90°$) **[B1]**

Angle $OTA = 90° - 54° = 36°$ **[M1]**

Angle $OAT = 36$ ($OA = OT$) **[A1]**

[Total Marks: 9]

1094 **(a)** Angle $PAD = 90°$ OR Angle $DAB = 90 - 58 = 32°$ **[M1]**

58 **[A1]**

(b) Angle $ADC = 58 + 22 = 80°$ **[M1]**

$180 - 80 = 100$ **[A1]**

(c) Angle $BOC = 2 \times 22 = 44°$ **[M1]**

$\frac{180 - 44}{2}$ **[M1]**

68 **[A1]**

[Total Marks: 7]

1095 **(a)** 65 **[B1]**

(b) 115 **[B1]**

(c) Opposite angles of a cyclic quadrilateral sum to 180°. **[B1]**

[Total Marks: 3]

1096 **(a)** 48 **[B1]**

(b) Angle $BCA = 48°$ (Alternate interior angles, since $BC \parallel T_1T_2$) **[M1]**

$180 - 48 - 48 = 84°$ **[A1]**

(c) Angle $BOC = 2 \times 84 = 168°$ **[M1]**

Triangle OBC is isosceles ($OB = OC$), so Angle $OBC = (180 - 168) \div 2 = 6°$ **[A1]**

[Total Marks: 5]

1097 **(a)** Forms correct equation using angle at centre theorem: $4x - 20 = 2(x + 32)$ **[M1]**

$4x - 20 = 2x + 64 \implies 2x = 84$ **[M1]**

$x = 42$ **[A1]**

(b) Recognises angle CDE = angle ABC (exterior angle of cyclic quadrilateral) OR finds angle $ADC = 180 - (42 + 32) = 106°$ **[M1]**

74 **[A1]**

[Total Marks: 5]

1098 **(a)** Recognition that tangent meets radius at 90°, so $\angle OAT = \angle OCT = 90°$ **[M1]**

$180 - 48 = 132$ **[A1]**

(b) 66 **[B1]**

(c) Use of isosceles triangle OAC ($OA = OC$ radii): $(180 - 132) \div 2$ **[M1]**

OR $\angle TAC = (180 - 48) \div 2 = 66°$ leading to $\angle OAC = 90 - 66$ **[M1]**

24 **[A1]**

[Total Marks: 5]

1099 **(a)** Angle $PQR = 90°$ **[M1]**

$180 - 90 - 32 = 58$ **[A1]**

(b) Angle $TPR = 90°$ **[M1]**

$90 - 32 = 58$ **[A1]**

(c) Angle $PRS = 180 - 90 - 41 = 49°$ **[M1]**

49 **[A1]**

[Total Marks: 6]

1100 **(a)** Angle $ACD = 90°$ (angle in a semicircle) **[M1]**

Angle $ADC = 180 - 90 - 24 = 66°$ **[A1]**

(b) $180 - 66 = 114°$ (opposite angles of cyclic quadrilateral) **[B1]**

(c) 24° (angles in same segment as angle CAD) **[B1]**

(d) Angle $ABD = 90°$ (angle in a semicircle) **[M1]**

In triangle ABX, angle $AXB = 180 - 90 - 32 = 58°$ **[A1]**

[Total Marks: 6]

1101 **(a)** Angle $ACB = 62$ **[B1]**

Alternate segment theorem **[B1]**

(b) $180 - 62$ **[M1]**

118 **[A1]**

(c) Angle $OAB = 90 - 62 = 28°$ **[M1]**

28 (since triangle OAB is isosceles) **[A1]**

[Total Marks: 6]

1102 **(a)** Angle $AOB = 2(x + 20) = 2x + 40$ **[M1]**

Angle AOB + Angle $ATB = 180° \implies (2x + 40) + (3x - 10) = 180$ **[M1]**

$5x + 30 = 180 \implies 5x = 150 \implies x = 30$ **[A1]**

(b) Angle $AOB = 2(30) + 40 = 100°$ **[M1]**

Angle $OAB = (180 - 100) \div 2 = 40$ **[A1]**

[Total Marks: 5]

1103 **(a)** Sets up equation for opposite angles:
$(3x + 2y) + (9x + 2y) = 180 \implies 12x + 4y = 180 \implies 3x + y = 45$ **[M1]**

Sets up second equation:
$(5x + 3y) + (4x + 3y) = 180 \implies 9x + 6y = 180 \implies 3x + 2y = 60$ **[M1]**

Correct method to solve linear simultaneous equations (e.g. subtracting first from second to get $y = 15$) **[M1]**

$x = 10$, $y = 15$ **[A1]**

(b) Finds $\angle DAB = 3(10) + 2(15) = 60°$ **[M1]**

Angle at centre is twice angle at circumference: Obtuse $\angle BOD = 2 \times 60 = 120°$ **[A1]**

[Total Marks: 6]

1104 **(a)** Angle $BAD = 180° - 115° = 65°$ **[B1]**

$(2x + 15) + 65 + (x + 10) = 180$ **[M1]**

$3x + 90 = 180$ **[M1]**

30 **[A1]**

(b) Angle $ABD = (x + 10)°$ OR Angle $ADB = (2x + 15)°$ **[M1]**

40 **[A1]**

[Total Marks: 6]

1105 **(a)** **(i)** 58 **[B1]**

(ii) Alternate segment theorem **[B1]**

(b) Angle $AQB = 180 - 58 - 58 = 64°$ or Angle $BAQ = 58°$ **[M1]**

128 **[A1]**

(c) Angle $ACB = 180 - 64$ (using cyclic quadrilateral $ACBQ$) **[M1]**

116 **[A1]**

[Total Marks: 6]

1106 **(a)** States Angle CAB = Angle BCQ because of the alternate segment theorem **[B1]**

$2x - 5 = x + 17$ **[M1]**

$x = 22$ **[A1]**

(b) Identifies angle $ACB = 90°$ (angle in a semicircle) or calculates angle $CAB = 39°$ **[M1]**

51 **[A1]**

(c) Angle $ADC = 180 - 51$ (Opposite angles of cyclic quadrilateral sum to 180°) **[M1]**

129 **[A1]**

[Total Marks: 7]

1107 **(a)** Identifies angle ABC = angle CAQ (alternate segment theorem) OR angle ACB = angle PAB **[M1]**

Sets up the equation: $3x - 5 = x + 25$ **[M1]**

$x = 15$ **[A1]**

(b) Calculates angle $ABC = 40°$ and angle $ACB = 40°$ (or uses angles on straight line PQ) **[M1]**

100 **[A1]**

(c) Uses opposite angles of cyclic quadrilateral $ABDC$ sum to 180° $(180 - 100)$ **[M1]**

80 **[A1]**

[Total Marks: 7]

1108 **(a)** Angle TAB = angle ACB (alternate segment theorem) **[M1]**

$2y - 10 = 3y - 40$ **[M1]**

30 **[A1]**

(b) $x + 10 = 180 - 2(50)$ **[M1]**

70 **[A1]**

(c) Angle $ADB = 180 - 50$ (opposite angles of cyclic quadrilateral) **[M1]**

130 **[A1]**

[Total Marks: 7]

1109 **(a)** **(i)** Angle ACT = Angle $CAT = \dfrac{180 - 48}{2} = 66°$ **[M1]**

Angle ABC = Angle $ACT = 66$ (alternate segment theorem) **[A1]**

(ii) Cyclic quadrilateral $ABCD$, so angle $ADC = 180 - 66$ **[M1]**

114 **[A1]**

(iii) $180 - (114 + 31) = 35$ **[B1]**

(iv) Angle at centre $= 2\times$ angle at circumference **[M1]**

$2 \times 66 = 132$ **[A1]**

[Total Marks: 7]

1110 **(a)** $180 - 2x$ **[B1]**

(b) $\dfrac{180 - 2x}{2} = 90 - x$ **[B1]**

(c) Angle $ADB = 180 - (90 - x)$ (opposite angles of cyclic quadrilateral) **[M1]**

$90 + x$ **[A1]**

(d) $90 + x = 3(90 - x)$ **[M1]**

$90 + x = 270 - 3x \implies 4x = 180$ **[M1]**

$x = 45$ **[A1]**

[Total Marks: 7]

1111 **(a)** **(i)** $2x + 3y = x + 2y + 10$ (from Alternate Segment Theorem) **[B1]**

$3x - y = x + y + 8$ (from Angles in Same Segment) **[B1]**

(ii) Rearranges to linear equations: $x + y = 10$ and $2x - 2y = 8 \implies x - y = 4$ **[M1]**

Valid method to solve simultaneous equations (e.g. adds equations to get $2x = 14$) **[M1]**

$x = 7$ and $y = 3$ **[A1]**

(b) Uses opposite angles of cyclic quadrilateral: angle $ADC = 180° - 95° = 85°$ **[B1]**

Calculates angle $BDC = (7) + (3) + 8 = 18°$ OR angle $BAC = 18°$ **[B1]**

Finds angle $ADB = 85° - 18° = 67°$ and uses alternate segment theorem
$\implies$ angle $QAB = 67°$ **[B1]**

[Total Marks: 8]

1112 **(a)** Sets up equation for x: $2x + 25 = 4x - 5$ **[M1]**

$x = 15$ **[A1]**

Sets up equation for y: $y + 12 = 3y$ **[M1]**

$y = 6$ **[A1]**

(b) Evaluates angles on straight line: $180 - (4(15) - 5) - 3(6)$ OR evaluates angles in triangle $180 - (2(15) + 25) - (6 + 12)$ **[M1]**

107 **[A1]**

(c) Recognises $ABTC$ is a cyclic quadrilateral and uses $180 - \angle ABT$: $180 - (2(15) + 25)$ **[M1]**

125 **[A1]**

[Total Marks: 8]

1113 **(a)** **(i)** Angle $OAT = 90°$ and angle $OBT = 90°$ **[B1]**

Angle $AOB = 180 - x$ **[M1]**

$180 - x = 2(2x - 30)$ **[M1]**

$180 - x = 4x - 60 \implies 5x = 240 \implies x = 48$ **[A1]**

(ii) Angle $AOB = 180 - 48 = 132°$ **[M1]**

Angle $OAB = (180 - 132) \div 2 = 24$ **[A1]**

(b) Angle $ACB = 2(48) - 30 = 66°$ **[M1]**

Angle $ADB = 180 - 66 = 114$ (opposite angles of cyclic quad) **[A1]**

[Total Marks: 8]

1114 **(a)** $3x - 11 = 2x + 14$ **[M1]**

$x = 25$ **[A1]**

Theorem: alternate segment theorem **[B1]**

(b) Angle $ABD = 3(25) - 11 = 64°$, so angle $ABC = 64° + 48° = 112°$, giving angle $ADC = 68°$ **[B1]**

Reason: opposite angles of a cyclic quadrilateral sum to 180° (or angle in a semicircle = 90°) **[B1]**

(c) Central angle $AOC = 2 \times 68° = 136°$ **[B1]**

$AC = 2 \times 9.4 \sin 68°$ (or $\sqrt{9.4^2 + 9.4^2 - 2(9.4)(9.4)\cos 136°}$) **[M1]**

17.4 **[A1]**

(d) Central angle $AOD = 2 \times 64° = 128°$ **[B1]**

Area $= \dfrac{128}{360} \times \pi \times 9.4^2 - \dfrac{1}{2} \times 9.4^2 \sin 128°$ $(= 98.70 - 34.81)$ **[M1]**

63.9 **[A1]**

[Total Marks: 11]

1115 **(a)** Triangle PAB is isosceles $(PA = PB)$ **[M1]**

$\dfrac{180 - 4x}{2}$ **[M1]**

$90 - 2x$ **[A1]**

(b) Angle $ACB =$ Angle PAB (Alternate segment theorem) **[M1]**

$3x - 10 = 90 - 2x$ **[M1]**

$5x = 100 \implies x = 20$ **[A1]**

(c) $x = 20 \implies$ Angle $APB = 80°$ **[M1]**

Angles in quadrilateral $PAOB$ sum to 360°, with $OAP = OBP = 90°$ **[M1]**

$360 - 90 - 90 - 80 = 100$ **[A1]**

(d) Angle $ACB = 3(20) - 10 = 50°$ **[M1]**

$180 - 50$ (opposite angles of cyclic quad sum to 180°) **[M1]**

130 **[A1]**

[Total Marks: 12]

1116 (a) Recognises that tangents from an external point are equal in length, so triangle PAB is isosceles: $180 - 44 = 136$ **[M1]**

$136 \div 2 = 68$ **[A1]**

(b) Recognises angle between tangent and radius is 90°, setting up sum of angles in quadrilateral $OAPB$: $360 - 90 - 90 - 44$ **[M1]**

136 **[A1]**

[Total Marks: 4]

1117 (a) Tangents from an external point are equal in length **[B1]**

(b) 70 **[B1]**

(c) Angle $OAP = 90°$ OR Angle $OBP = 90°$ **[M1]**

140 **[A1]**

[Total Marks: 4]

1118 (a) Angle $PTO = 56 \div 2 = 28°$ **[M1]**

$180 - 90 - 28 = 62$ **[A1]**

(b) Tangents from an external point are equal in length (so $TP = TQ$) **[B1]**

(c) 62 **[B1]**

[Total Marks: 4]

1119 (a) Uses angle sum of quadrilateral $OATB$ with angle $OAT = 90°$ and angle $OBT = 90°$: $360° - 90° - 90° - 44°$ OR $180° - 44°$ **[M1]**

136 **[A1]**

(b) Recognises triangle OAB is isosceles ($OA = OB$) and uses $(180° - 136°) \div 2$ **[M1]**

22 **[A1]**

(c) Tangents from an external point are equal in length **[B1]**

[Total Marks: 5]

1120 (a) 90 **[B1]**

The perpendicular from the centre of a circle to a chord bisects the chord (or perpendicular bisector of a chord passes through the centre) **[B1]**

(b) Uses midpoint M: $AM = \dfrac{16}{2} = 8$ **[B1]**

Applies Pythagoras' theorem: $OM^2 + 8^2 = 10^2$ or $OM = \sqrt{10^2 - 8^2}$ **[M1]**

6 **[A1]**

[Total Marks: 5]

1121 **(a)** 24 **[B1]**

(b) Identifies $\angle OAP = 90°$ (tangent perpendicular to radius) **[M1]**

$OP^2 = 7^2 + 24^2 = 625 \implies OP = 25$ **[A1]**

(c) Let M be the intersection of OP and AB. Identifies $\triangle OAM \sim \triangle OPA$ OR uses Area of $\triangle OAP$ OR uses $\cos(\angle AOP)$ **[M1]**

$\cos(\angle AOP) = \dfrac{7}{25}$ and $OM = 7 \times \cos(\angle AOP)$ **[M1]**

$OM = \dfrac{49}{25}$ or 1.96 **[A1]**

[Total Marks: 6]

1122 **(a)** $AM = 12$ cm **[B1]**

Reason: the perpendicular from the centre of a circle to a chord bisects the chord **[B1]**

(b) $OM^2 = 13^2 - 12^2$ $(= 169 - 144 = 25)$ **[M1]**

5 **[A1]**

(c) $ON = 5$ cm **[B1]**

Reason: equal chords are equidistant from the centre of the circle **[B1]**

[Total Marks: 6]

1123 **(a)** $x + 3 = 3x - 5$ **[M1]**

$x = 4$ **[A1]**

(b) Perpendicular distance = 7 (can be implied) **[B1]**

$\sqrt{25^2 - 7^2}$ OR 24 **[M1]**

48 **[A1]**

[Total Marks: 5]

1124 **(a)** 12 **[B1]**

Tangents from an external point to a circle are equal in length **[B1]**

(b) Uses $\angle OAP = 90°$ and Pythagoras' theorem: $OP = \sqrt{5^2 + 12^2}$ **[M1]**

13 **[A1]**

(c) Calculates area of two congruent right-angled triangles: $2 \times \left(\frac{1}{2} \times 5 \times 12\right)$ **[M1]**

60 **[A1]**

[Total Marks: 6]

1125 (a) $OA = OB$ (radii of the circle) **[B1]**

$PA = PB$ (tangents from an external point are equal in length) **[B1]**

OP is common to both triangles (or angle OAP = angle $OBP = 90°$, tangent to radius $= 90°$), so triangles are congruent (SSS or RHS) **[B1]**

(b) (i) Angle OAP = angle $OBP = 90°$ **[M1]**

Angle $AOB = 360° - 90° - 90° - 52° = 128$ **[A1]**

(ii) $OA = OB$ so triangle OAB is isosceles **[M1]**

Angle $OAB = \frac{180° - 128°}{2} = 26$ **[A1]**

(c) 14.3 **[B1]**

[Total Marks: 8]

1126 (a) Angle $OAP = 90°$ AND Angle $OBP = 90°$ (tangent perpendicular to radius) **[B1]**

Angle $AOB = 360 - 90 - 90 - 46 = 134°$ **[B1]**

(b) Triangle OAB is isosceles ($OA = OB$) **[M1]**

Angle $OAB = (180 - 134) \div 2 = 23°$ **[A1]**

[Total Marks: 4]

1127 (a) Angle $OMB = 90°$ (line from centre to midpoint of chord is perpendicular to chord) **[B1]**

$MB = 7$ **[B1]**

$OM = \sqrt{8.5^2 - 7^2}$ **[M1]**

4.77 or 4.769... **[A1]**

(b) Half chord $= \sqrt{8.5^2 - 3^2}$ **[M1]**

$2 \times \sqrt{63.25} = 15.9$ or 15.90... **[A1]**

[Total Marks: 5]

1128 (a) Identifies right-angled triangle OAM (or OBM) and states $AM = 24 \div 2 = 12$ **[B1]**

Applies Pythagoras' theorem: $OM^2 + 12^2 = 13^2$ **[M1]**

$OM = \sqrt{169 - 144} = \sqrt{25} = 5$ **[A1]**

(b) 24 (using property that chords equidistant from the centre are equal in length) **[B1]**

(c) $\frac{1}{2} \times 24 \times 5$ **[M1]**

60 **[A1]**

[Total Marks: 6]

1129 **(a)** The line from the centre of a circle to the midpoint of a chord is perpendicular to the chord. **[B1]**

(b) $AM = 24 \div 2 = 12$ cm **[B1]**

$OM^2 + 12^2 = 13^2$ **[M1]**

$OM = \sqrt{169 - 144} = 5$ **[A1]**

(c) $\sin(\angle AOM) = \frac{12}{13}$ OR $\cos(\angle AOM) = \frac{5}{13}$ OR $\tan(\angle AOM) = \frac{12}{5}$ **[M1]**

$\angle AOM = 67.38...^\circ$ **[A1]**

Angle $AOB = 2 \times 67.38... = 134.8^\circ$ **[A1]**

[Total Marks: 7]

1130 **(a)** States or uses $OM \perp AB$ (perpendicular bisector of chord passes through centre), so $AM = 7$ **[B1]**

Uses Pythagoras' Theorem: $OM = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$ **[B1]**

(b) States $ON = OM = 24$ (equal chords are equidistant from centre) **[B1]**

Uses Cosine rule: $MN^2 = 24^2 + 24^2 - 2(24)(24)\cos 100^\circ$ OR uses $MN = 2 \times 24 \times \sin 50^\circ$ **[M1]**

36.8 (accept $36.77\ldots$) **[A1]**

(c) $\frac{1}{2} \times 24 \times 24 \times \sin 100^\circ$ **[M1]**

284 (accept $283.6\ldots$) **[A1]**

[Total Marks: 7]

1131 **(a)** Forms a right-angled triangle with hypotenuse 13 and base $\frac{24}{2} = 12$ **[M1]**

Uses Pythagoras' theorem: $\sqrt{13^2 - 12^2}$ **[M1]**

5 **[A1]**

(b) 5 **[B1]**

(c) Uses Pythagoras' theorem: $\sqrt{13^2 - 12^2}$ **[M1]**

Half of chord $EF = 5$ **[A1]**

10 **[A1]**

(d) The perpendicular from the centre to a chord bisects the chord (accept "perpendicular bisector of a chord passes through the centre") **[B1]**

[Total Marks: 8]

1132 **(a)** Equal chords are equidistant from the centre, so $OM = ON$. **[B1]**

$5x - 7 = 2x + 2$ **[M1]**

$3x = 9 \implies x = 3$ **[A1]**

(b) $OM = 5(3) - 7 = 8$ cm **[B1]**

Uses Pythagoras' theorem: $AM^2 + 8^2 = 17^2$ **[M1]**

$AM = \sqrt{289 - 64} = \sqrt{225} = 15$ **[A1]**

$AB = 2 \times 15 = 30$ **[A1]**

[Total Marks: 7]

1133 **(a)** Angle $OTA = 90°$ (tangent perpendicular to radius) **[B1]**

$AT = 12$ (perpendicular from centre to chord bisects chord) **[B1]**

In triangle OTA, $r^2 + 12^2 = R^2$ (Pythagoras theorem) **[M1]**

$R^2 - r^2 = 144$ **[A1]**

(b) Area $= \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$ **[M1]**

$k = 144$ **[A1]**

(c) Distance $= r$ (or same distance as chord AB) **[B1]**

Equal chords are equidistant from the centre **[B1]**

[Total Marks: 7]

1134 **(a)** The perpendicular bisector of a chord passes through the centre O, so $OP \perp AB$ and $OQ \perp CD$ **[B1]**

Since AB and CD are parallel, both OP and OQ are perpendicular to the same parallel lines through O, forming a single straight line POQ **[B1]**

(b) Identifies $AP = 12$, $CQ = 5$, and $OQ = 17 - x$ **[B1]**

Sets up Pythagoras' equations for both triangles: $r^2 = x^2 + 12^2$ and $r^2 = (17 - x)^2 + 5^2$ **[M1]**

Equates expressions for r^2: $x^2 + 144 = 289 - 34x + x^2 + 25$ **[M1]**

Solves linear equation: $34x = 170$ giving $x = 5$ **[A1]**

Substitutes $x = 5$ to find $r = \sqrt{5^2 + 12^2} = 13$ **[A1]**

[Total Marks: 7]

1135 **(a)** The perpendicular bisector of a chord passes through the centre of the circle **[B1]**

(b) $MX = 16 \div 2 = 8$ **[B1]**

$OX = \sqrt{10^2 - 8^2} = 6$ **[M1]**

$10 - 6 = 4$ **[A1]**

(c) Distance of PQ from $O = 11 - 6 = 5$ **[B1]**

Half length of $PQ = \sqrt{10^2 - 5^2}$ **[M1]**

$\sqrt{75}$ or 8.66... **[A1]**

$2 \times \sqrt{75} = 17.3$ (17.32...) **[A1]**

[Total Marks: 8]

1136 **(a)** Identifies $\angle OAT = 90°$ and $\angle AOT = 90° - x$ **[M1]**

Uses $AM = r\sin(90° - x) = r\cos x$ and perpendicular bisector property $AB = 2AM$ to get $2r\cos x$ **[A1]**

(b) $r\sin x$ (equal chords are equidistant from centre, distance $= OM = r\cos(90° - x)$) **[B1]**

(c) Area of $OAB = \dfrac{1}{2} \times AB \times OM = \dfrac{1}{2}(2r\cos x)(r\sin x) = r^2 \sin x \cos x$ **[M1]**

$TM = \dfrac{AM}{\tan x} = \dfrac{r\cos x}{\tan x} = \dfrac{r\cos^2 x}{\sin x}$ **[M1]**

Area of $ATB = \dfrac{1}{2} \times AB \times TM = \dfrac{1}{2}(2r\cos x)\left(\dfrac{r\cos^2 x}{\sin x}\right) = \dfrac{r^2\cos^3 x}{\sin x}$ **[M1]**

Equates areas: $r^2 \sin x \cos x = \dfrac{r^2\cos^3 x}{\sin x} \implies \sin^2 x \cos x = \cos^3 x \implies \tan^2 x = 1$ **[M1]**

45 **[A1]**

[Total Marks: 8]

1137 **(a)** Tangents from an external point are equal in length **[B1]**

(b) Recognises $BP = BQ$, $CR = CQ$, and $DR = DS$ **[M1]**

$AB + CD = AP + PB + CR + RD$ **[M1]**

Substitutes equal tangent segments:
$AS + BQ + CQ + DS = (AS + DS) + (BQ + CQ) = AD + BC$ **[A1]**

(c) Substitutes expressions into the relationship from part (b):
$(2x + 5) + (x + 8) = (2x + 1) + (3x - 2)$ **[M1]**

Correctly expands and simplifies to a linear equation, e.g., $3x + 13 = 5x - 1$ **[M1]**

7 **[A1]**

(d) Substitutes $x = 7$ into the lengths of the sides (at least two) OR uses $2(3x + 13)$ **[M1]**

68 **[A1]**

[Total Marks: 9]

1138 **(a)** Recognises angle $OPT = 90°$ and sets up Pythagoras: $TP^2 + 7^2 = 25^2$ **[M1]**

$TP = \sqrt{625 - 49} = \sqrt{576} = 24$ **[A1]**

(b) Uses trigonometry in triangle OPT: $\sin(\text{angle } PTO) = \dfrac{7}{25}$ or $\tan(\text{angle } PTO) = \dfrac{7}{24}$ **[M1]**

Angle $PTO = 16.26°$ **[A1]**

Angle $PTQ = 2 \times 16.26° = 32.5°$ (accept $32.52°$) **[A1]**

(c) Calculates total area of kite $OPTQ = 2 \times (\frac{1}{2} \times 7 \times 24) = 168$ **[M1]**

Calculates angle $POQ = 360 - 90 - 90 - 32.52 = 147.48°$ (or uses $180 - 32.52$) **[M1]**

Calculates area of sector $OPQ = \frac{147.48}{360} \times \pi \times 7^2 = 63.07\ldots$ **[M1]**

Shaded area $= 168 - 63.07 = 104.93$ (accept 104.9 to 105.0) **[A1]**

[Total Marks: 9]

1139 **(a)** $AM = 8$ cm and $CN = 6$ cm (perpendicular from centre bisects chord), giving $r^2 = x^2 + 8^2$ **[B1]**

$r^2 = (14 - x)^2 + 6^2$ **[B1]**

$x^2 + 64 = 196 - 28x + x^2 + 36$ leading to $28x = 168$ and $x = 6$ **[M1]**

$r = \sqrt{6^2 + 8^2} = 10$ **[A1]**

(b) Angle $OAT = 90°$ (tangent to radius $= 90°$) and $\tan(\text{angle } AOM) = \frac{8}{6}$ **[M1]**

$TA = 10 \times \frac{8}{6}$ (or similar triangles OMA and OAT used correctly) **[M1]**

$\frac{40}{3}$ **[A1]**

(c) $TB = \frac{40}{3}$ cm (or 13.3 cm) **[B1]**

Perimeter $= 10 + 10 + \frac{40}{3} + \frac{40}{3} = 46.7$ cm **[B1]**

(d) $\tan(\text{angle } ATO) = \frac{10}{40/3} = \frac{3}{4}$ giving angle $ATO = 36.87°$ **[M1]**

Angle $ATB = 2 \times 36.87° = 73.7$ **[A1]**

[Total Marks: 11]

1140 **(a)** 5.2 **[B1]**

(b) 4700 **[B1]**

(c) 850 **[B1]**

[Total Marks: 3]

1141 **(a)** $80 \times 45 = 3600$ **[M1]**

0.36 **[A1]**

(b) 3200 **[B1]**

[Total Marks: 3]

1142 **(a)** 35000 **[B1]**

(b) $60 \times 45 \times 30$ OR 2700×30 OR 81000 **[M1]**

their $81000 \div 1000$ **[M1]**

81 **[A1]**

[Total Marks: 4]

1143 **(a)** 4750 **[B1]**

(b) 0.32 **[B1]**

(c) 24 000 **[B1]**

(d) 6.5 **[B1]**

[Total Marks: 4]

1144 **(a)** Converts units correctly to cm: 250 mm = 25 cm AND 0.2 m = 20 cm **[B1]**

20000 **[A1]**

(b) 20 **[B1]**

(c) Uses Mass = Density × Volume: 0.6×20000 (= 12000 g) **[M1]**

12 **[A1]**

[Total Marks: 5]

1145 **(a)** **(i)** $2(45 \times 30 + 45 \times 20 + 30 \times 20) = 2(1350 + 900 + 600)$ **[M1]**

5700 **[A1]**

(ii) 0.57 **[B1]**

(b) Volume $= 45 \times 30 \times 20 = 27\,000\ cm^3$ **[B1]**

Capacity = 27 litres **[B1]**

(c) 650 g = 0.65 kg and $18 \times 1.15 = 20.7$ kg **[M1]**

21.35 **[A1]**

[Total Marks: 7]

1146 **(a)** Perimeter on plan $= 2(12 + 8) = 40$ cm and actual perimeter $= 40 \times 25\,000 = 1\,000\,000$ cm **[M1]**

10 **[A1]**

(b) **(i)** $(12 \times 8) - (4 \times 1.5) = 96 - 6$ **[M1]**

90 **[A1]**

(ii) 1 cm represents 0.25 km, so 1 cm^2 represents $0.25^2 = 0.0625\ km^2$ (or $90 \times 25\,000^2 = 5.625 \times 10^{10}\ cm^2$) **[M1]**

5.625 **[A1]**

(c) Actual pond dimensions: $4 \times 250 = 1000$ m and $1.5 \times 250 = 375$ m **[M1]**

Volume $= 1000 \times 375 \times 1.6 = 600\,000\ m^3$ **[M1]**

6×10^8 **[A1]**

[Total Marks: 9]

1147 **(a)** Converts all dimensions to cm: 250, 140, 0.8 **[M1]**

$250 \times 140 \times 0.8 = 28000$ **[A1]**

(b) Multiplies volume by 7.8: $28000 \times 7.8 = 218400$ **[M1]**

218.4 **[A1]**

[Total Marks: 4]

1148 **(a)** Converts dimensions consistently, e.g. 1.8 m = 180 cm (or 0.75 m and 0.6 m) **[B1]**

Calculates volume: $180 \times 75 \times 60 = 810\,000$ cm^3 (or 0.81 m^3) **[M1]**

810 **[A1]**

(b) Converts 810 litres = 810 000 ml (or 450 ml/s = 0.45 l/s) **[M1]**

Calculates time in seconds: $\dfrac{810\,000}{450} = 1800$ seconds **[M1]**

30 **[A1]**

[Total Marks: 6]

1149 **(a)** Converts dimensions to consistent units, e.g. 150 cm, 80 cm, 60 cm OR 1.5 m, 0.8 m, 0.6 m **[M1]**

$150 \times 80 \times 60 = 720000$ OR $1.5 \times 0.8 \times 0.6 = 0.72$ **[M1]**

720 **[A1]**

(b) $0.024 \times 1000000 = 24000$ **[M1]**

Mass $= 24000 \times 7.8 = 187200$ **[M1]**

187.2 **[A1]**

[Total Marks: 6]

1150 **(a)** Conversion of units: 15 $mm = 0.015$ m OR 15 $m = 1500$ cm etc. **[M1]**

Volume $= 15 \times 12 \times 0.015$ (or eq) **[M1]**

2700 **[A1]**

(b) Use of consistent units for tank volume equation (e.g. $r = 0.6$ m and $V = 2.7$ m^3 OR $V = 2700000$ cm^3 and $r = 60$ cm) **[M1]**

$\pi \times 0.6^2 \times h = 2.7$ OR $\pi \times 60^2 \times h = 2700000$ **[M1]**

$h = \dfrac{2.7}{0.36\pi}$ (m) OR $h = \dfrac{2700000}{3600\pi}$ (cm) **[M1]**

238.7 **[A1]**

[Total Marks: 7]

1151 **(a)** $\frac{4}{3} \times \pi \times 15^3$ OR 4500π OR 14137 **[M1]**

their $14137 \div 1000000$ OR $\frac{4}{3} \times \pi \times 0.15^3$ **[M1]**

1.41×10^{-2} **[A1]**

(b) 125 kg = 125000 g **[B1]**

$\frac{125000}{4500\pi}$ OR $\frac{125000}{14137}$ **[M1]**

8.84 (8.841 . . .) **[A1]**

(c) **(i)** 24 mm = 2.4 cm **[B1]**

$\frac{1}{3} \times \pi \times 2.4^2 \times 5.5$ **[M1]**

33.2 (33.17 . . .) OR 10.56π **[A1]**

(ii) $\frac{4500\pi}{\text{their } 33.17}$ OR 426.1 . . . **[M1]**

426 **[A1]**

[Total Marks: 11]

1152 **(a)** Identifies consistent units for radius and length (e.g. $r = 1.5$ cm and length = 80 cm) **[M1]**

Volume in one second $= \pi \times 1.5^2 \times 80 = 180\pi\ cm^3$ $(= 565.48 \ldots\ cm^3)$ **[M1]**

Converts to litres $(180\pi \div 1000)$ and confirms $0.56548 \cdots \approx 0.565$ **[A1]**

(b) Volume of pool required $= 4 \times 2.5 \times 1.2 = 12\ m^3$ OR 12000 litres **[M1]**

Time in seconds $= 12000 \div 0.56548 \ldots$ OR $12000 \div \frac{180\pi}{1000}$ **[M1]**

$= 21220.6 \ldots$ seconds $(353.6 \ldots$ minutes) **[A1]**

5 hours 54 minutes **[A1]**

[Total Marks: 7]

1153 **(a)** Converts speed to cm/min: $5.4 \times 1000 \times 100 \div 60 = 9000$ **[M1]**

Calculates volume in 1 minute: $\pi \times 2.5^2 \times 9000 = 56250\pi\ cm^3$ **[M1]**

Converts cm^3 to litres and evaluates: $56250\pi \div 1000 = 176.714 \ldots$ **[A1]**

(b) Calculates total volume in 45 minutes: $56250\pi \times 45 = 2531250\pi\ cm^3$ (or uses litres: $176.714 \cdots \times 45 = 7952.1 \ldots$ litres) **[M1]**

Converts tank base dimensions to cm and finds area: $300 \times 200 = 60000\ cm^2$ **[M1]**

Sets up equation for depth: Depth $= \frac{2531250\pi}{60000}$ **[M1]**

133 (or 132.5 to 132.6) **[A1]**

[Total Marks: 7]

1154 **(a)** Calculates area in m^2: $450 \times 320 = 144\,000$ m^2 (or 0.45×0.32) **[M1]**

0.144 **[A1]**

(b) **(i)** Total volume in ml $= 144\,000 \times 25 = 3\,600\,000$ ml **[M1]**

Converts ml to litres or cm^3: 3600 litres or $3\,600\,000$ cm^3 **[M1]**

3.6 **[A1]**

(ii) Mass of liquid in one drum $= 216 - 9 = 207$ kg $= 207\,000$ g **[B1]**

Volume of one drum $= 0.18$ $m^3 = 180\,000$ cm^3 **[B1]**

$\dfrac{207\,000}{180\,000} = 1.15$ **[A1]**

[Total Marks: 8]

1155 **(a)** Volume $= 15.7$ m^3 **[B1]**

$\pi r^2(2.5r) = 15.7$ giving $r^3 = \dfrac{15.7}{2.5\pi}$ $(= 1.99898\ldots)$ **[M1]**

$r = 1.26$ **[A1]**

(b) Internal surface area $= 2\pi(1.26)^2 + 2\pi(1.26)(3.15) = 34.913\ldots$ m^2 $(= 349\,131\ldots$ $cm^2)$ **[M1]**

Volume $= 349\,131\cdots \times 0.15$ (or $34.913\cdots \times 0.0015 \times 10^6$) **[M1]**

52 400 **[A1]**

(c) 15 700 litres $= 15\,700\,000$ ml and time $= \dfrac{15\,700\,000}{450} = 34\,888.88\ldots$ seconds **[M1]**

$\dfrac{34\,888.88\ldots}{60} = 581.48\ldots$ minutes **[M1]**

9 hours 41 minutes **[A1]**

(d) Mass $= 15\,700\,000 \times 1.02 = 16\,014\,000$ g $(= 16\,014$ kg) **[M1]**

16.014 **[A1]**

[Total Marks: 11]

1156 **(a)** Cross-sectional area $= \pi \times 2.5^2$ $(= 6.25\pi)$ **[M1]**

Speed conversion: 18 $km/h = \dfrac{18 \times 1000 \times 100}{60}$ cm/min $(= 30000$ $cm/min)$ **[M1]**

Volume per minute $= 6.25\pi \times 30000$ $(= 187500\pi$ $cm^3)$ **[M1]**

Convert to litres: $\dfrac{187500\pi}{1000}$ **[M1]**

589 or 589.0 to 589.1 (187.5π) **[A1]**

(b) Capacity of reservoir $= 50 \times 30 \times 4 = 6000$ m^3 OR 6000000 litres **[M1]**

Time in minutes $= \dfrac{6000000}{187.5\pi}$ (or 6000000÷ their (a)) $(= 10185.9...)$ **[M1]**

Time in hours $= \frac{10185.9...}{60}$ $(= 169.76...)$ **[M1]**

$169.76... \div 24 = 7.07...$ days so 7 full days **[M1]**

$0.07... \times 24 = 1.76...$ hours (or $169.76... - 7 \times 24$) **[M1]**

7 **[A1]**

2 **[A1]**

[Total Marks: 12]

1157 **(a)** $\pi \times 2.5^2 \times 160$ OR $\pi \times 0.025^2 \times 1.6$ **[M1]**

$\times 60$ **[M1]**

$\div 1000$ (if using cm) OR $\times 1000$ (if using m) **[M1]**

188 $(188.4\ldots)$ **[A1]**

(b) Volume of tank $= 4.2 \times 3.5 \times 2.4 = 35.28\ m^3$ **[M1]**

$35.28 \times 1000 = 35280$ litres **[M1]**

Time $= \frac{35280}{\text{their } 188.49}$ OR $\frac{35280}{60\pi}$ **[M1]**

187 minutes $(187.1\ldots)$ **[A1]**

3 hours 7 minutes **[A1]**

(c) Calculates lateral area correctly: $2 \times 4.2 \times 2.4 + 2 \times 3.5 \times 2.4$ OR $36.96\ m^2$ [M2]
(If incorrect, M1 for one correct face area e.g., 4.2×2.4)

$36.96 \times 10000 = 369600\ cm^2$ OR $45000 \div 10000 = 4.5\ m^2$ **[M1]**

Litres of paint required $= \frac{369600}{45000}$ OR $\frac{36.96}{4.5}$ **[M1]**

Number of tins $= \frac{8.21\ldots}{2.5}$ $(= 3.28\ldots)$ **[M1]**

4 **[A1]**

[Total Marks: 15]

1158 **(a)** $2 \times (12 + 7)$ **[M1]**

38 **[A1]**

(b) 12×5 **[M1]**

60 **[A1]**

[Total Marks: 4]

1159 **(a)** Calculates area of triangle: $\frac{1}{2} \times 12 \times 8 = 48$ **[M1]**

$4x = 48$

x = 12[A1]

(b) Uses Pythagoras to find slanted side: $\sqrt{6^2 + 8^2}$ **[M1]**

Slanted side = 10 **[A1]**

Perimeter = $12 + 10 + 10 = 32$ **[A1]**

[Total Marks: 5]

1160 **(a)** $15 \times h = 120$ **[M1]**

8 **[A1]**

(b) $2(15 + AD) = 54$ OR $15 + AD = 27$ **[M1]**

12 **[A1]**

(c) 60 **[B1]**

[Total Marks: 5]

1161 **(a)** $2(2x + 3) + 2(x - 1) = 34$ OR $2x + 3 + x - 1 = 17$ **[M1]**

$6x + 4 = 34$ OR $3x + 2 = 17$ **[M1]**

$x = 5$ **[A1]**

(b) Substitution of $x = 5$: $(2(5) + 3) \times (5 - 1)$ OR 13×4 **[M1]**

52 **[A1]**

[Total Marks: 5]

1162 **(a)** $\frac{1}{2}(10 + 16) \times 8$ **[M1]**

104 **[A1]**

(b) Base of right-angled triangle = $16 - 10 = 6$ **[M1]**

Slanted edge = $\sqrt{6^2 + 8^2} = 10$ **[M1]**

$8 + 10 + 16 + 10 = 44$ **[A1]**

[Total Marks: 5]

1163 **(a)** $\frac{1}{2}(9 + 15) \times 8$ **[M1]**

96 **[A1]**

(b) $BC^2 = (15 - 9)^2 + 8^2 = 36 + 64 = 100$ **[M1]**

$BC = 10$ cm **[A1]**

Perimeter = $9 + 10 + 15 + 8 = 42$ cm **[B1]**

(c) $DE = 15 - 9 = 6$ cm and area = $\frac{1}{2} \times 6 \times 8$ **[M1]**

24 **[A1]**

[Total Marks: 7]

1164 **(a)** $2((3x+2)+(x+4))$ or $2(4x+6)$ **[M1]**

$8x+12$ **[A1]**

(b) Area of rectangle $PQRS = (3x+2)(x+4) = 3x^2+12x+2x+8 = 3x^2+14x+8$ **[M1]**

Area of parallelogram $ABCD = x(2x-1) = 2x^2-x$ **[B1]**

$A = (3x^2+14x+8)-(2x^2-x) = x^2+15x+8$ **[A1]**

(c) **(i)** $x^2+15x+8 = 62$ giving $x^2+15x-54 = 0$ **[M1]**

$(x+18)(x-3) = 0$ **[M1]**

3 (rejecting $x = -18$ as lengths must be positive) **[A1]**

(ii) $3(2(3)-1) = 15$ **[B1]**

[Total Marks: 9]

1165 **(a)** Uses Area $= \frac{1}{2} \times$ base $\times$ height $= \frac{1}{2} \times 14 \times 9$ **[M1]**

63 **[A1]**

(b) Finds $AE = \sqrt{9^2+9^2} = \sqrt{162}$ or $12.7\ldots$ **[M1]**

Finds $BE = \sqrt{5^2+9^2} = \sqrt{106}$ or $10.2\ldots$ **[M1]**

Perimeter $= 14+\sqrt{162}+\sqrt{106}$ **[M1]**

37.0 (or $37.02\ldots$) **[A1]**

[Total Marks: 6]

1166 **(a)** $\frac{1}{2} \times (15+24) \times 8$ **[M1]**

156 **[A1]**

(b) Horizontal distance from D to the altitude from $A = \sqrt{10^2-8^2} = 6$ **[M1]**

Horizontal distance for the right-angled triangle under $B = 24-15-6 = 3$ **[M1]**

$BC = \sqrt{8^2+3^2}$ $(= \sqrt{73} \approx 8.544)$ **[M1]**

57.5 OR 57.54 to 57.55 **[A1]**

[Total Marks: 6]

1167 **(a)** $\frac{1}{2} \times (10+22) \times 8$ **[M1]**

128 **[A1]**

(b) Finds horizontal base projection of slanted side: $\frac{22-10}{2} = 6$ **[B1]**

Applies Pythagoras' theorem: $PS = \sqrt{6^2+8^2}$ **[M1]**

$PS = 10$ **[A1]**

Perimeter $= 10+22+10+10 = 52$ **[A1]**

[Total Marks: 6]

1168 **(a)** Area of rectangle $= 4(2x + 3)$ OR $8x + 12$ **[M1]**

Area of triangle $= \frac{1}{2} \times 6 \times (3x - 1)$ OR $9x - 3$ **[M1]**

$8x + 12 = 9x - 3$ **[M1]**

$x = 15$ **[A1]**

(b) Length $= 2(15) + 3 = 33$ **[M1]**

Perimeter $= 2(4 + 33) = 74$ **[A1]**

[Total Marks: 6]

1169 **(a)** $\frac{1}{2} \times 8 \times ((x + 4) + (3x - 2)) = 120$ **[M1]**

$4(4x + 2) = 120$ OR $4x + 2 = 30$ **[M1]**

7 **[A1]**

(b) Evaluates parallel sides: 11 and 19 **[B1]**

Uses Pythagoras' theorem to find the slanted side: $\sqrt{8^2 + \left(\frac{19 - 11}{2}\right)^2}$ OR $\sqrt{8^2 + 4^2}$ [M2]

(M1 for finding the horizontal base of 4)

47.9 (47.88 . . .) **[A1]**

[Total Marks: 7]

1170 **(a)** **(i)** Sets up equation for area: $\frac{1}{2}((x + 4) + (3x - 2))(x - 1) = 20$ **[M1]**

Simplifies expression to $\frac{1}{2}(4x + 2)(x - 1) = 20$ or $(2x + 1)(x - 1) = 20$ **[M1]**

Expands to $2x^2 - 2x + x - 1 = 20$ leading to $2x^2 - x - 21 = 0$ with no errors seen **[A1]**

(ii) Factorises $(2x - 7)(x + 3) = 0$ or uses quadratic formula to find $x = 3.5$ **[M1]**

Calculates straight sides: $AD = 7.5$, $BC = 8.5$, $AB = 2.5$ **[M1]**

Sets up Pythagoras for the slanted side CD: $\sqrt{(8.5 - 7.5)^2 + 2.5^2}$ **[M1]**

Calculates $CD = \sqrt{1^2 + 2.5^2} = \sqrt{7.25}$ (or 2.69 . . .) **[A1]**

Perimeter $= 7.5 + 8.5 + 2.5 + 2.69 \cdots = 21.2$ (or 21.19 . . .) **[A1]**

[Total Marks: 8]

1171 **(a)** Equates areas: $\frac{1}{2}(2x + 1)(3x - 2) = (x + 4)(2x - 1)$ **[M1]**

Expands brackets: $\frac{1}{2}(6x^2 - x - 2) = 2x^2 + 7x - 4$ **[M1]**

Multiplies by 2 and simplifies to $6x^2 - x - 2 = 4x^2 + 14x - 8$, leading to $2x^2 - 15x + 6 = 0$ **[A1]**

(b) Uses quadratic formula: $x = \frac{-(-15) \pm \sqrt{(-15)^2 - 4(2)(6)}}{2(2)}$ **[M1]**

$x = \dfrac{15 \pm \sqrt{177}}{4}$ **[M1]**

7.08 and 0.42 **[A1]**

(c) Selects valid root $x = 7.076...$ (since $x = 0.42$ gives $EF < 0$ and $BC < 0$) and substitutes into $2(3x + 3)$ **[M1]**

48.5 (accept 48.46 to 48.5) **[A1]**

[Total Marks: 8]

1172 **(a)** $\dfrac{1}{2}(x + x + 4)(x - 1) = 40$ **[M1]**

$\dfrac{1}{2}(2x + 4)(x - 1) = 40$ OR $(x + 2)(x - 1) = 40$ **[M1]**

$x^2 + 2x - x - 2 = 40 \implies x^2 + x - 42 = 0$ **[A1]**

(b) $(x + 7)(x - 6) = 0$ **[M1]**

6 **[A1]**

(c) Identifies lengths are Top = 6, Bottom = 10, Height = 5 **[M1]**

Horizontal difference = 4, so slanted side = $\sqrt{5^2 + 4^2}$ OR $\sqrt{41}$ **[M1]**

Perimeter = $5 + 6 + 10 + \sqrt{41}$ **[M1]**

27.4 (or $27.40\ldots$) **[A1]**

[Total Marks: 9]

1173 **(a)** Area of rectangle = $(2x + 1)(x + 3) = 2x^2 + 6x + x + 3$ **[M1]**

Area of triangle = $\dfrac{1}{2}(3x + 2)(2x - 1) = \dfrac{1}{2}(6x^2 - 3x + 4x - 2)$ **[M1]**

Equates areas: $2x^2 + 7x + 3 = \dfrac{1}{2}(6x^2 + x - 2)$ **[M1]**

Multiplies by 2 and simplifies correctly to $2x^2 - 13x - 8 = 0$ with no errors **[A1]**

(b) $\dfrac{-(-13) \pm \sqrt{(-13)^2 - 4(2)(-8)}}{2(2)}$ [M2]

(M1 for correct substitution into quadratic formula)

$\dfrac{13 \pm \sqrt{233}}{4}$ **[M1]**

7.07 and -0.57 **[A1]**

(c) Substitutes the positive value of x into $2(2x + 1 + x + 3)$ OR calculates sides 15.14 and 10.07 **[M1]**

50.4 (accept 50.39 to 50.43) **[A1]**

[Total Marks: 10]

1174 **(a)** Perimeter of rectangle = $6y$ OR Perimeter of square = $60 - 6y$ **[M1]**

$\frac{60-6y}{4}$ OR $15-1.5y$ **[A1]**

(b) $(15-1.5y)^2+y(2y)=113$ **[M1]**

$225-45y+2.25y^2+2y^2=113$ **[M1]**

$4.25y^2-45y+112=0$ leading to $17y^2-180y+448=0$ with no errors seen **[A1]**

(c) $\frac{-(-180)\pm\sqrt{(-180)^2-4(17)(448)}}{2(17)}$ **[M1]**

$\frac{180\pm\sqrt{1936}}{34}$ OR $\frac{180\pm44}{34}$ **[M1]**

$y=4$ and $y=6.59$ (or 6.588... or $\frac{112}{17}$) **[A1]**

(d) Identifies $y=4$ as the valid solution (since $15-1.5(6.59)<6.59$) leading to side length $=9$ **[M1]**

81 **[A1]**

[Total Marks: 10]

1175 **(a)** $BD^2=14.5^2+9.2^2-2(14.5)(9.2)\cos 118°$ **[M1]**

$BD^2=210.25+84.64-(-125.258\ldots)=420.148\ldots$ **[M1]**

20.5 **[A1]**

(b) $\frac{\sin\angle ABD}{9.2}=\frac{\sin 118°}{20.497\ldots}$ giving angle $ABD=23.347\ldots°$ **[M1]**

Angle BDC = angle $ABD=23.3°$ (alternate angles, AB is parallel to DC) **[A1]**

(c) $DC=20.497\ldots\cos 23.347\ldots°=18.819\ldots$ cm **[M1]**

$BC=20.497\ldots\sin 23.347\ldots°=8.123\ldots$ cm **[M1]**

Perimeter $=14.5+8.123\cdots+18.819\cdots+9.2=50.6$ **[A1]**

(d) $\frac{1}{2}(14.5+18.819\ldots)\times 8.123\ldots$ (or $\frac{1}{2}(14.5)(9.2)\sin 118°+\frac{1}{2}(18.819\ldots)(8.123\ldots)$) **[M1]**

135 **[A1]**

[Total Marks: 10]

1176 **(a)** $\frac{45}{360}\times 2\times\pi\times 12$ **[M1]**

3π **[A1]**

(b) $\frac{45}{360}\times\pi\times 12^2$ **[M1]**

18π **[A1]**

[Total Marks: 4]

1177 **(a)** $\frac{60}{360}\times 2\times\pi\times 12$ **[M1]**

4π **[A1]**

(b) $\frac{60}{360} \times \pi \times 12^2$ **[M1]**

24π **[A1]**

[Total Marks: 4]

1178 **(a)** $\frac{45}{360} \times \pi \times 12^2$ **[M1]**

18π **[A1]**

(b) $\frac{45}{360} \times 2 \times \pi \times 12$ **[M1]**

3π **[A1]**

$3\pi + 24$ **[A1]**

[Total Marks: 5]

1179 **(a)** $\frac{150}{360} \times \pi \times 6^2$ **[M1]**

15π **[A1]**

(b) $\frac{150}{360} \times 2 \times \pi \times 6$ **[M1]**

5π **[A1]**

$5\pi + 12$ **[A1]**

[Total Marks: 5]

1180 **(a)** $\frac{120}{360} \times 2 \times \pi \times 9$ **[M1]**

6π **[A1]**

(b) $\frac{120}{360} \times \pi \times 9^2$ **[M1]**

Simplifies $\frac{1}{3} \times 81\pi$ **[M1]**

27π **[A1]**

[Total Marks: 5]

1181 **(a)** **(i)** $\frac{80}{360} \times 2\pi(9)$ **[M1]**

4π **[A1]**

(ii) $18 + 4\pi$ **[B1]**

(b) Reflex angle $AOB = 360° - 80° = 280°$ **[B1]**

Area $= \frac{280}{360} \times \pi \times 9^2$ **[M1]**

63π **[A1]**

[Total Marks: 6]

1182 **(a)** **(i)** $\frac{120}{360} \times 2\pi(6)$ **[M1]**

4π **[A1]**

(ii) Length of arc $CD = \frac{120}{360} \times 2\pi(9) = 6\pi$ **[M1]**

Perimeter $= 4\pi + 6\pi + 3 + 3$ **[M1]**

$6 + 10\pi$ **[A1]**

(b) $\frac{120}{360} \times \pi \times 9^2 - \frac{120}{360} \times \pi \times 6^2$ **[M1]**

$27\pi - 12\pi$ **[M1]**

15π **[A1]**

(c) $\frac{150}{360}\pi r^2 = 15\pi$ giving $r^2 = 36$ **[M1]**

6 **[A1]**

[Total Marks: 10]

1183 **(a)** $\frac{\theta}{360} \times \pi r^2 = 10\pi$ **[M1]**

$\frac{\theta}{360} \times 2\pi r = 2\pi$ **[M1]**

Valid method to solve simultaneously (e.g. dividing the area equation by the arc length equation to get $\frac{r}{2} = 5$) **[M1]**

$r = 10$ **[A1]**

$\theta = 36$ **[A1]**

(b) $\frac{x + 20}{360} \times 2\pi x = \frac{16\pi}{5}$ **[M1]**

Simplifying to form a 3-term quadratic equation, e.g. $x^2 + 20x - 576 = 0$ **[M1]**

Valid method to solve the quadratic (e.g. $(x + 36)(x - 16) = 0$) **[M1]**

$x = 16$ (with -36 rejected or simply 16 as the final answer) **[A1]**

[Total Marks: 9]

1184 **(a)** $\frac{72}{360} \times \pi \times r^2 = 85$ **[M1]**

$r^2 = \frac{85 \times 5}{\pi}$ or $135.28\ldots$ **[M1]**

11.6 or $11.63\ldots$ **[A1]**

(b) Arc length $= \frac{72}{360} \times 2 \times \pi \times 11.63$ or $14.6\ldots$ **[M1]**

Perimeter $= 2 \times 11.63 + 14.61$ **[M1]**

37.9 or 37.86 to 37.87 **[A1]**

[Total Marks: 6]

1185 **(a)** $\frac{250}{360} \times 2 \times \pi \times r = 35$ **[M1]**

$r = \frac{35 \times 360}{250 \times 2 \times \pi}$ **[M1]**

8.02 or 8.021... **[A1]**

(b) $\frac{250}{360} \times \pi \times (\text{their } 8.02)^2$ **[M1]**

OR $\frac{1}{2} \times \text{their } 8.02 \times 35$ **[M1]**

140 or 140.3 to 140.4 **[A1]**

[Total Marks: 6]

1186 **(a)** Calculates major arc length: $\frac{248}{360} \times 2 \times \pi \times 8.5$ $(= 36.79...)$ **[M1]**

Adds two radii: $36.79... + 2(8.5)$ **[M1]**

53.8 (accept 53.79 to 53.8) **[A1]**

(b) Finds minor sector angle $360° - 248° = 112°$ and calculates minor sector area:
$\frac{112}{360} \times \pi \times 8.5^2 = 70.616...$ **[M1]**

Calculates area of triangle OPQ: $\frac{1}{2} \times 8.5^2 \times \sin(112°) = 33.495...$ **[M1]**

37.1 (accept 37.12) **[A1]**

[Total Marks: 6]

1187 **(a)** $\frac{120}{360} \times \pi \times 9^2$ OR $\frac{120}{360} \times \pi \times 5^2$ **[M1]**

$\frac{120}{360} \times \pi \times 9^2 - \frac{120}{360} \times \pi \times 5^2$ **[M1]**

58.6 $(58.64\ldots)$ OR $\frac{56\pi}{3}$ **[A1]**

(b) $\frac{120}{360} \times 2 \times \pi \times 9$ OR $\frac{120}{360} \times 2 \times \pi \times 5$ **[M1]**

$\frac{120}{360} \times 2 \times \pi \times 9 + \frac{120}{360} \times 2 \times \pi \times 5 + 4 + 4$ **[M1]**

37.3 $(37.32\ldots)$ OR $\frac{28\pi}{3} + 8$ **[A1]**

(c) $\pi r^2 = 58.64\ldots$ (or their area from (a)) **[M1]**

4.32 $(4.320\ldots)$ **[A1]**

[Total Marks: 8]

1188 **(a)** $\frac{110}{360} \times \pi \times 8.5^2$ **[M1]**

69.4 (or 69.35 . . .) **[A1]**

(b) Area of triangle $OAB = \frac{1}{2} \times 8.5^2 \times \sin(110°)$ (=33.94 . . .) **[M1]**

Area of segment = $69.35 \cdots - 33.94 \ldots$ **[M1]**

35.4 (or 35.40 . . .) **[A1]**

(c) Arc length $AB = \frac{110}{360} \times 2 \times \pi \times 8.5$ (=16.31 . . .) **[M1]**

Chord $AB = \sqrt{8.5^2 + 8.5^2 - 2(8.5)(8.5)\cos(110°)}$ OR $2 \times 8.5 \sin(55°)$ **[M1]**

Chord $AB = 13.92 \ldots$ **[A1]**

Perimeter = $16.31 \cdots + 13.92 \cdots = 30.2$ (or 30.24 . . .) **[A1]**

[Total Marks: 9]

1189 **(a)** Area of sector = $\frac{110}{360} \times \pi \times 8^2$ (= 61.43...) **[M1]**

Area of triangle = $\frac{1}{2} \times 8 \times 8 \times \sin(110°)$ (= 30.07...) **[M1]**

Area of segment = Area of sector − Area of triangle **[M1]**

31.4 or 31.36... **[A1]**

(b) Arc length = $\frac{110}{360} \times 2 \times \pi \times 8$ (= 15.35...) **[M1]**

Chord $AB = \sqrt{8^2 + 8^2 - 2 \times 8 \times 8 \times \cos(110°)}$ OR $2 \times 8 \times \sin(55°)$ (= 13.11...) **[M1]**

28.5 or 28.46... **[A1]**

[Total Marks: 7]

1190 **(a)** Writes area of shaded region as difference of two sectors: $\frac{\theta}{360}\pi(r+6)^2 - \frac{\theta}{360}\pi r^2 = 54\pi$ **[M1]**

Expands $(r+6)^2 - r^2$ to obtain $12r + 36$ **[M1]**

Simplifies equation: $\frac{\theta}{360}(12r + 36) = 54$ or $\theta \times 12(r+3) = 19\,440$ **[M1]**

Divides by $12(r+3)$ to show $\theta = \frac{1620}{r+3}$ **[A1]**

(b) **(i)** Writes perimeter equation: $\frac{\theta}{360}(2\pi r) + \frac{\theta}{360}(2\pi(r+6)) + 6 + 6 = 54$ **[M1]**

Substitutes $\theta = \frac{1620}{r+3}$: $\frac{1620}{360(r+3)} \times 2\pi(2r+6) + 12 = 54$, simplifying to $18\pi + 12 = 54$ — wait: adjust perimeter to $(18\pi + 12)$ or specific numerical value: using $r = 15$ gives $\theta = 90°$ and perimeter = $\frac{90}{360}(2\pi)(15+21) + 12 = 18\pi + 12 \approx 68.55$ cm; if perimeter is 68.55 cm, solves $\frac{\theta}{360}\pi(4r + 12) = 42$ **[M1]**

15 (accept 14.9 to 15.0) **[A1]**

(ii) $\dfrac{1620}{15+3} = 90$ **[B1]**

[Total Marks: 8]

1191 **(a)** **(i)** $\dfrac{x}{360} \times 2\pi r = 6\pi$ OR $\dfrac{x}{360} \times \pi r^2 = 27\pi$ **[M1]**

Valid method to eliminate x, e.g. dividing the two equations: $\dfrac{\frac{x}{360}\pi r^2}{\frac{x}{360}2\pi r} = \dfrac{27\pi}{6\pi}$ leading to $\dfrac{r}{2} = 4.5$ **[M1]**

9 **[A1]**

(ii) Substitutes their r back into an equation, e.g. $\dfrac{x}{360} \times 2\pi(9) = 6\pi$ **[M1]**

120 **[A1]**

(b) Perimeter of sector $= \dfrac{y}{360} \times 2\pi R + 2R$ **[M1]**

Equates to circumference: $\dfrac{y}{360} \times 2\pi R + 2R = 2\pi R$ **[M1]**

Rearranges to isolate terms with y: $\dfrac{y}{360} \times 2\pi R = 2\pi R - 2R$ **[M1]**

$\dfrac{360(\pi - 1)}{\pi}$ OR $360 - \dfrac{360}{\pi}$ **[A1]**

[Total Marks: 9]

1192 **(a)** Uses cosine rule: $\cos(AOB) = \dfrac{15^2 + 15^2 - 24^2}{2 \times 15 \times 15}$ or $24^2 = 15^2 + 15^2 - 2(15)(15)\cos(AOB)$ **[M1]**

$\cos(AOB) = -0.28$ **[M1]**

$106.26\ldots$ which rounds to $106.3°$ with no errors seen **[A1]**

(Alternative: $\sin(\frac{AOB}{2}) = \dfrac{12}{15}$ [M1], $\frac{AOB}{2} = 53.13\ldots°$ [M1], $AOB = 106.3°$ [A1])

(b) **(i)** Arc length $= \dfrac{106.3}{360} \times 2 \times \pi \times 15$ $(= 27.8\ldots)$ **[M1]**

Perimeter $= 24 + 27.8\ldots$ **[M1]**

51.8 (or 51.81 to 51.83) **[A1]**

(ii) Area of sector $= \dfrac{106.3}{360} \times \pi \times 15^2$ $(= 208.6\ldots$ to $208.7\ldots)$ **[M1]**

Area of triangle $= \dfrac{1}{2} \times 15 \times 15 \times \sin(106.3°)$ $(= 108.0\ldots)$ **[M1]**

Subtracts triangle area from sector area: $208.7 - 108.0$ **[M1]**

101 or 100.6 to 100.8 **[A1]**

[Total Marks: 10]

1193 **(a)** Angle $OAP = 90°$ (tangent to radius $= 90°$) **[B1]**

$\cos \angle AOP = \dfrac{8.5}{15.2}$ giving angle $AOP = 55.998\ldots°$ **[M1]**

Obtuse angle $AOB = 2 \times 55.998\ldots° = 111.997\ldots° = 112.0°$ **[A1]**

(b) Reflex angle $AOB = 360° - 112.0° = 248.0°$ **[B1]**

Major arc $AB = \dfrac{248.0}{360} \times 2\pi(8.5)$ **[M1]**

36.8 **[A1]**

(c) $PA = \sqrt{15.2^2 - 8.5^2} = 12.601\ldots$ cm **[M1]**

Area of quadrilateral $OAPB = 2 \times \dfrac{1}{2} \times 8.5 \times 12.601\cdots = 107.11\ldots\ cm^2$ **[M1]**

Area of minor sector $OAB = \dfrac{112.0}{360} \times \pi \times 8.5^2 = 70.61\ldots\ cm^2$ **[M1]**

Shaded area $= 107.11\cdots - 70.61\cdots = 36.5$ **[A1]**

(d) $\dfrac{1}{2}R(18.4) = 115$ giving $R = 12.5$ cm **[M1]**

$\dfrac{\theta}{360} \times 2\pi(12.5) = 18.4$ **[M1]**

$R = 12.5$, $\theta = 84.3°$ **[A1]**

[Total Marks: 13]

1194 **(a)** $\dfrac{4}{3} \times \pi \times 3^3$ **[M1]**

36π **[A1]**

(b) $4 \times \pi \times 3^2$ **[M1]**

36π **[A1]**

[Total Marks: 4]

1195 **(a)** $\pi \times 4^2 \times 9$ **[M1]**

144π **[A1]**

(b) Curved surface area $= 2 \times \pi \times 4 \times 9 = 72\pi$ **[M1]**

Area of two circular ends $= 2 \times \pi \times 4^2 = 32\pi$ **[M1]**

104π **[A1]**

[Total Marks: 5]

1196 **(a)** Volume of cylinder $= \pi \times 3^2 \times 10\ (= 90\pi)$ **[M1]**

Volume of hemisphere $= \dfrac{2}{3} \times \pi \times 3^3\ (= 18\pi)$ **[M1]**

108π **[A1]**

(b) Curved surface area of cylinder $= 2 \times \pi \times 3 \times 10\ (= 60\pi)$ **[M1]**

Area of circular base $= \pi \times 3^2\ (= 9\pi)$ AND Curved surface area of hemisphere =

$2 \times \pi \times 3^2$ $(= 18\pi)$ **[M1]**

87π **[A1]**

[Total Marks: 6]

1197 **(a)** **(i)** $\frac{120}{360} \times \pi \times 6^2$ **[M1]**

12π **[A1]**

(ii) $12\pi \times 10 = 120\pi$ **[B1]**

(b) Curved arc length $= \frac{120}{360} \times 2\pi(6) = 4\pi$ cm **[M1]**

Curved surface area $= 4\pi \times 10 = 40\pi\ cm^2$ **[M1]**

Area of two rectangular faces $= 2 \times (6 \times 10) = 120\ cm^2$ **[B1]**

Total surface area $= 120 + 40\pi + 2(12\pi) = 120 + 64\pi$ **[A1]**

[Total Marks: 7]

1198 **(a)** Area of base $= \pi \times 6^2 = 36\pi$ **[M1]**

Curved surface area $= \pi \times 6 \times l = 6\pi l$ **[M1]**

$36\pi + 6\pi l = 96\pi$ leading to $6l = 60$ so $l = 10$ **[A1]**

(b) $h^2 + 6^2 = 10^2$ **[M1]**

$h = \sqrt{100 - 36} = \sqrt{64} = 8$ **[A1]**

Volume $= \frac{1}{3} \times \pi \times 6^2 \times 8$ **[M1]**

96π **[A1]**

[Total Marks: 7]

1199 **(a)** Use of Pythagoras to find the height of the cone: $\sqrt{5^2 - 3^2}$ $(= 4)$ **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 3^2 \times 4$ $(= 12\pi)$ **[M1]**

Volume of hemisphere $= \frac{1}{2} \times \frac{4}{3} \times \pi \times 3^3$ $(= 18\pi)$ **[M1]**

30π **[A1]**

(b) Curved surface area of cone $= \pi \times 3 \times 5$ $(= 15\pi)$ **[M1]**

Surface area of hemisphere $= 2 \times \pi \times 3^2$ $(= 18\pi)$ **[M1]**

33π **[A1]**

[Total Marks: 7]

1200 **(a)** **(i)** $\frac{1}{2} \times \frac{4}{3}\pi \times 6^3$ **[M1]**

144π **[A1]**

(ii) Volume of cylinder $= \pi \times 6^2 \times 5 = 180\pi$ **[M1]**

Total volume $= 144\pi + 180\pi = 324\pi$ **[A1]**

(b) $\frac{1}{3}\pi \times 9^2 \times h = 324\pi$ giving $27\pi h = 324\pi$ **[M1]**

$h = \frac{324}{27} = 12$ cm **[A1]**

(c) Slant height $l = \sqrt{9^2 + 12^2} = 15$ cm **[M1]**

Total surface area $= \pi(9)(15) + \pi(9^2) = 135\pi + 81\pi$ **[M1]**

216π **[A1]**

[Total Marks: 9]

1201 **(a)** $\frac{1}{2} \times 6 \times 8 \times 15$ **[M1]**

360 **[A1]**

(b) Area of two triangular ends $= \frac{1}{2} \times 6 \times 8 \times 2$ $(= 48)$ **[M1]**

Area of three rectangular faces $= 15 \times (6 + 8 + 10)$ or $(15 \times 6) + (15 \times 8) + (15 \times 10)$ $(= 360)$ **[M1]**

408 **[A1]**

[Total Marks: 5]

1202 **(a)** Curved surface area $= 2 \times \pi \times 5 \times 12$ **[M1]**

Total surface area $= 2 \times \pi \times 5 \times 12 + 2 \times \pi \times 5^2$ **[M1]**

534 or 534.07... or 170π **[A1]**

(b) Volume of cylinder $= \pi \times 5^2 \times 12 = 300\pi$ (942...) **[M1]**

$\frac{4}{3}\pi R^3 = 300\pi$ **[M1]**

$R = \sqrt[3]{225} = 6.08$ or 6.082... **[A1]**

[Total Marks: 6]

1203 **(a)** $\frac{1}{3} \times 10^2 \times h = 400$ **[M1]**

12 **[A1]**

(b) Distance from centre of base to midpoint of side $= 5$ cm **[B1]**

Slant height of a triangular face $= \sqrt{12^2 + 5^2}$ $(= 13)$ **[M1]**

Area of one triangular face $= \frac{1}{2} \times 10 \times 13$ $(= 65)$ **[M1]**

360 **[A1]**

[Total Marks: 6]

1204 **(a)** Calculates cross-sectional sector area: $\frac{110}{360} \times \pi \times 6^2 = 11\pi$ $(= 34.557...)$ **[M1]**

Multiplies by length: $11\pi \times 15$ **[M1]**

518 (accept 518.3 to 518.4) **[A1]**

(b) Calculates curved arc length: $\frac{110}{360} \times 2 \times \pi \times 6 = \frac{11\pi}{3}$ $(= 11.519...)$ **[M1]**

Sums all five faces: $2(11\pi) + 15 \times \left(\frac{11\pi}{3} + 6 + 6\right)$ **[M1]**

422 (accept 421.9 to 422.0) **[A1]**

[Total Marks: 6]

1205 **(a)** Height of cone $= 24 - 12 = 12$ cm **[B1]**

Volume of cylinder $= \pi \times 5^2 \times 12$ $(=300\pi$ or $942.4\ldots)$ **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 5^2 \times 12$ $(=100\pi$ or $314.1\ldots)$ **[M1]**

1260 (or 1256 to 1257 or 400π) **[A1]**

(b) Slant height of cone $(l) = \sqrt{5^2 + 12^2} = 13$ **[M1]**

Curved surface area of cone $= \pi \times 5 \times 13$ $(=65\pi$ or $204.2\ldots)$ **[M1]**

Curved surface area of cylinder $= 2 \times \pi \times 5 \times 12$ $(=120\pi$ or $376.9\ldots)$ **[M1]**

Area of circular base $= \pi \times 5^2$ $(=25\pi$ or $78.5\ldots)$ **[M1]**

660 (or $659.7\ldots$ or 210π) **[A1]**

[Total Marks: 9]

1206 **(a)** $\pi \times (x)^2 \times 3x$ **[M1]**

$3\pi x^3$ **[A1]**

(b) $\frac{4}{3} \times \pi \times \left(\frac{x}{2}\right)^3$ **[M1]**

$\frac{\pi x^3}{6}$ **[A1]**

(c) 18 **[B1]**

(d) Total surface area $= 2\pi x^2 + 2\pi x(3x)$ **[M1]**

$8\pi x^2 = 140$ **[M1]**

$x = 2.36$ (or $2.360\ldots$) **[A1]**

[Total Marks: 8]

1207 **(a)** Volume of cone $= \frac{1}{3}\pi r^2(2r) = \frac{2}{3}\pi r^3$ **[B1]**

Volume of hemisphere $= \frac{1}{2} \times \frac{4}{3}\pi r^3 = \frac{2}{3}\pi r^3$ **[B1]**

Equates total volume: $\frac{2}{3}\pi r^3 + \frac{2}{3}\pi r^3 = 450\pi$ leading to $\frac{4}{3}r^3 = 450$ **[M1]**

Solves for r: $r^3 = 337.5 = 27 \times 12.5$, therefore $r = 3\sqrt[3]{12.5}$ **[A1]**

(b) Finds slant height of cone: $l = \sqrt{r^2 + (2r)^2} = r\sqrt{5}$ **[M1]**

Sets up total surface area expression: $\pi rl + 2\pi r^2 = \pi r^2(\sqrt{5} + 2)$ **[M1]**

Evaluates $r = 3\sqrt[3]{12.5} = 6.96238...$ and substitutes into area formula **[M1]**

645 (accept 645.0 to 645.4) **[A1]**

[Total Marks: 8]

1208 **(a)** Area of trapezium $= \frac{1}{2} \times (15 + 25) \times 18$ **[M1]**

360 **[A1]**

Volume $= 360 \times 120 = 43200$ cm^3 **[M1]**

43.2 **[A1]**

(b) Volume of water after 30 seconds $= 0.8 \times 30 = 24$ litres $= 24000$ cm^3 **[M1]**

Let depth be h and the width of water surface be w.
$w = 15 + \frac{10}{18}h = 15 + \frac{5}{9}h$ **[M2]**

Area of cross-section of water $= \frac{1}{2} \times (15 + 15 + \frac{5}{9}h) \times h = 15h + \frac{5}{18}h^2$ **[M1]**

Volume $= (15h + \frac{5}{18}h^2) \times 120 = 24000$
$1800h + \frac{100}{3}h^2 = 24000$
$h^2 + 54h - 720 = 0$ **[M1]**

$(h - 11.23...)(h + 65.23...) = 0$
$h = 11.2$ or $11.23...$ **[A1]**

[Total Marks: 9]

1209 **(a)** Volume of large cone $= \frac{1}{3} \times \pi \times 10^2 \times 24$ $(= 800\pi)$ **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 5^2 \times 12$ $(= 100\pi)$ **[M1]**

2200 or 2199.1 to 2199.2 **[A1]**

(b) Slant height of large cone $= \sqrt{10^2 + 24^2}$ $(= 26)$ AND Slant height of small cone $= \sqrt{5^2 + 12^2}$ $(= 13)$ **[M1]**

Curved surface area of frustum $= \pi \times 10 \times 26 - \pi \times 5 \times 13$ $(= 195\pi)$ **[M1]**

Total surface area $= 195\pi + \pi \times 10^2 + \pi \times 5^2$ **[M1]**

1010 or 1005.3 . . . **[A1]**

(c) Volume of one sphere $= \frac{4}{3} \times \pi \times 3^3$ $(= 36\pi$ or $113.1...)$ **[M1]**

$\frac{700\pi}{36\pi}$ or $\frac{2199.1}{113.1}$ **[M1]**

19 **[A1]**

[Total Marks: 10]

1210 **(a)** $\frac{1}{3} \times 12^2 \times 10$ **[M1]**

480 **[A1]**

(b) Use of Pythagoras to find slant height: $\sqrt{10^2 + 6^2}$ **[M1]**

$\sqrt{136}$ or 11.6... or 11.7 **[A1]**

Area of four triangular faces $= 4 \times \frac{1}{2} \times 12 \times$ their $\sqrt{136}$ $(= 279.8...)$ **[M1]**

424 or 423.8... to 423.9 **[A1]**

(c) Volume of one sphere $= \frac{4}{3} \times \pi \times 2.5^3$ $(= 65.4...)$ **[M1]**

$\frac{480}{\text{their } 65.4...}$ **[M1]**

7 **[A1]**

(d) $\frac{1620}{480}$ or 3.375 **[M1]**

$\sqrt[3]{3.375}$ or 1.5 **[M1]**

15 **[A1]**

[Total Marks: 12]

1211 **(a)** $\frac{1}{3} \times 16^2 \times 15$ **[M1]**

1280 **[A1]**

(b) Slant height of triangular face $= \sqrt{15^2 + 8^2} = 17$ cm **[M1]**

Total surface area $= 16^2 + 4 \times \left(\frac{1}{2} \times 16 \times 17\right) = 256 + 544$ **[M1]**

800 **[A1]**

(c) **(i)** Height of removed pyramid $= 15 - 9 = 6$ cm, giving linear scale factor $= \frac{6}{15} = 0.4$ **[M1]**

Volume of frustum $= 1280 - 0.4^3 \times 1280 = 1280 - 81.92$ **[M1]**

1198.08 (or 1200) **[A1]**

(ii) Top square side $= 0.4 \times 16 = 6.4$ cm and trapezoidal slant height $= 0.6 \times 17 = 10.2$ cm **[M1]**

Total surface area $= 16^2 + 6.4^2 + 4 \times \frac{1}{2}(16 + 6.4) \times 10.2 = 256 + 40.96 + 456.96$ **[M1]**

753.92 (or 754) **[A1]**

(d) $OA = \sqrt{8^2 + 8^2} = \sqrt{128}$ $(= 11.313\ldots)$ cm **[M1]**

$\tan \angle VAO = \frac{15}{\sqrt{128}}$ **[M1]**

53.0 **[A1]**

[Total Marks: 14]

1212 Identifies the three straight edges as $10 + 6 + 10 = 26$ **[M1]**

Calculates the arc length of the semicircle as $\frac{1}{2} \times \pi \times 6$ (or 3π) **[M1]**

$26 + 3\pi$ **[A1]**

[Total Marks: 3]

1213 **(a)** Arc length $BD = \frac{1}{4} \times 2 \times \pi \times 10 = 5\pi$ **[M1]**

Perimeter $= 10 + 10 + 5\pi = 20 + 5\pi$ **[A1]**

(b) Area of quarter circle $= \frac{1}{4} \times \pi \times 10^2 = 25\pi$ **[M1]**

Area of shaded region $= 100 - 25\pi$ **[A1]**

[Total Marks: 4]

1214 **(a)** Curved edge of semi-circle $= \frac{1}{2} \times \pi \times 4 = 2\pi$ **[M1]**

$20 + 2\pi$ **[A1]**

(b) $(8 \times 4) + \left(\frac{1}{2} \times \pi \times 2^2\right)$ **[M1]**

$32 + 2\pi$ **[A1]**

[Total Marks: 4]

1215 **(a)** Perimeter $= 12 + 12 + 8 + \frac{1}{2} \times \pi \times 8$ **[M1]**

$32 + 4\pi$ **[A1]**

(b) Area $= 12 \times 8 + \frac{1}{2} \times \pi \times 4^2$ **[M1]**

$96 + 8\pi$ **[A1]**

[Total Marks: 4]

1216 **(a)** Calculates semicircular arc length: $\frac{1}{2} \times \pi \times 14 = 7\pi$ **[M1]**

$50 + 7\pi$ **[A1]**

(b) Area of rectangle $= 18 \times 14 = 252$ **[B1]**

Area of semicircle $= \frac{1}{2} \times \pi \times 7^2 = 24.5\pi$ (or $\frac{49\pi}{2}$) **[M1]**

$252 + 24.5\pi$ (or $252 + \frac{49\pi}{2}$) **[A1]**

[Total Marks: 5]

1217 **(a)** Arc length $= \frac{1}{2} \times \pi \times 8 = 4\pi$ **[M1]**

$32 + 4\pi$ **[A1]**

(b) Area of rectangle $= 12 \times 8 = 96$ **[M1]**

Area of semicircle $= \frac{1}{2} \times \pi \times 4^2 = 8\pi$ **[M1]**

$96 + 8\pi$ **[A1]**

[Total Marks: 5]

1218 **(a)** Area of rectangle $= 10 \times 8 = 80$ **[M1]**

Area of semicircle $= \frac{1}{2} \times \pi \times 4^2 = 8\pi$ **[M1]**

$80 + 8\pi$ **[A1]**

(b) Identifies the three straight edges: $10 + 8 + 10 = 28$ **[M1]**

Perimeter of semicircle arc $= \frac{1}{2} \times \pi \times 8 = 4\pi$ **[M1]**

$28 + 4\pi$ **[A1]**

[Total Marks: 6]

1219 **(a)** Sum of three straight edges: $10 + 6 + 10 = 26$ **[M1]**

Arc length of semicircle $= \frac{1}{2} \times \pi \times 6$ $(= 3\pi)$ **[M1]**

$26 + 3\pi$ **[A1]**

(b) Area of rectangle $= 10 \times 6$ $(= 60)$ **[M1]**

Area of removed semicircle $= \frac{1}{2} \times \pi \times 3^2$ $(= 4.5\pi)$ **[M1]**

$60 - 4.5\pi$ **[A1]**

[Total Marks: 6]

1220 **(a)** Area of square $= 4 \times 4 = 16$ **[B1]**

Area of quarter circle $= \frac{1}{4} \times \pi \times 4^2 = 4\pi$ **[M1]**

$16 + 4\pi$ **[A1]**

(b) Arc length $= \frac{1}{4} \times 2 \times \pi \times 4 = 2\pi$ **[M1]**

Perimeter $= 4 + 4 + 4 + 2\pi$ **[M1]**

$12 + 2\pi$ **[A1]**

[Total Marks: 6]

1221 **(a)** Arc length $= \frac{1}{2} \times \pi \times 10 \ (= 5\pi)$ **[M1]**

Perimeter $= 12 + 12 + 10 + 5\pi$ **[M1]**

$34 + 5\pi$ **[A1]**

(b) Area of rectangle $= 12 \times 10 \ (= 120)$ **[M1]**

Area of semicircle $= \frac{1}{2} \times \pi \times 5^2 \ (= 12.5\pi)$ **[M1]**

$120 + 12.5\pi$ **[A1]**

[Total Marks: 6]

1222 **(a)** Calculates the area of the rectangle: $10 \times 6 = 60$ **[M1]**

Calculates the area of the semicircle: $\frac{1}{2} \times \pi \times 3^2 \ (= 4.5\pi)$ **[M1]**

$60 - 4.5\pi$ **[A1]**

(b) Calculates the sum of the three straight edges: $10 + 6 + 10 = 26$ **[M1]**

Calculates the arc length of the semicircle: $\frac{1}{2} \times \pi \times 6 \ (= 3\pi)$ **[M1]**

$26 + 3\pi$ **[A1]**

[Total Marks: 6]

1223 **(a)** **(i)** Semicircular arc length $= \frac{1}{2} \times 2\pi(6) = 6\pi$ cm **[M1]**

Outer perimeter $= 16 + 12 + 16 + 6\pi = 44 + 6\pi$ **[A1]**

(ii) Area of rectangle $= 16 \times 12 = 192 \ cm^2$ **[B1]**

Area of semicircle $= \frac{1}{2}\pi(6^2) = 18\pi \ cm^2$ and area of hole $= \pi(4^2) = 16\pi \ cm^2$ **[M1]**

Shaded area $= 192 + 18\pi - 16\pi = 192 + 2\pi$ **[A1]**

(b) $5(192 + 2\pi)$ **[M1]**

$960 + 10\pi$ **[A1]**

[Total Marks: 7]

1224 **(a)** $h^2 = 13^2 - 5^2 = 169 - 25 = 144$ **[M1]**

$h = \sqrt{144} = 12$ cm **[A1]**

(b) Curved surface area of cone $= \pi(5)(13) = 65\pi \ cm^2$ **[B1]**

Curved surface area of cylinder $= 2\pi(5)(12) = 120\pi \ cm^2$ **[B1]**

Inner curved surface area of hemisphere $= \frac{1}{2} \times 4\pi(5^2) = 50\pi \ cm^2$ **[B1]**

Total surface area $= 65\pi + 120\pi + 50\pi = 235\pi$ **[A1]**

(c) Volume of cone $= \frac{1}{3}\pi(5^2)(12) = 100\pi$ and volume of cylinder $= \pi(5^2)(12) = 300\pi$ **[M1]**

Volume of hemisphere $= \frac{1}{2} \times \frac{4}{3}\pi(5^3) = \frac{250}{3}\pi$ **[B1]**

Total volume $= 100\pi + 300\pi - \frac{250}{3}\pi$ **[M1]**

$\frac{950}{3}\pi$ **[A1]**

[Total Marks: 10]

1225 (a) Volume of cylinder $= \pi \times 5^2 \times 12 = 300\pi$ **[M1]**

Volume of hemisphere $= \frac{2}{3} \times \pi \times 5^3 = \frac{250\pi}{3}$ **[M1]**

1200 (or 1204 or 1204.2 . . .) **[A1]**

(b) Curved surface of cylinder $= 2 \times \pi \times 5 \times 12 = 120\pi$ **[M1]**

Curved surface of hemisphere + Base of cylinder $= 2 \times \pi \times 5^2 + \pi \times 5^2 = 75\pi$ **[M1]**

613 (or 612.6 . . .) **[A1]**

[Total Marks: 6]

1226 (a) Volume of cylinder $= \pi \times 4^2 \times 10 = 160\pi$ $(= 502.65...)$ **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 4^2 \times 7.5 = 40\pi$ $(= 125.66...)$ **[M1]**

628 (accept 628.3 or 200π) **[A1]**

(b) Finds slant height of cone: $l = \sqrt{4^2 + 7.5^2} = 8.5$ **[B1]**

Sums base area, curved cylinder area, and curved cone area:
$\pi(4)^2 + 2\pi(4)(10) + \pi(4)(8.5) = 130\pi$ **[M1]**

408 (accept 408.4 or 130π) **[A1]**

[Total Marks: 6]

1227 (a) Height of cylinder $= 25 - 5 = 20$ **[B1]**

Total Volume $= \pi \times 5^2 \times 20 + \frac{2}{3} \times \pi \times 5^3$ **[M1]**

1830 or 1832.5 . . . **[A1]**

(b) Curved surface area of cylinder $= 2 \times \pi \times 5 \times 20$ $(= 200\pi)$ **[M1]**

Area of circular base $= \pi \times 5^2$ $(= 25\pi)$ **[M1]**

Curved surface area of hemisphere $= 2 \times \pi \times 5^2$ $(= 50\pi)$ **[M1]**

864 or 863.9 . . . **[A1]**

[Total Marks: 7]

1228 **(a)** Volume of cylinder $= \pi \times 5^2 \times 12$ $(= 300\pi$ or 942.47...) **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 5^2 \times 6$ $(= 50\pi$ or 157.07...) **[M1]**

1100 or 1099.5... to 1099.6 **[A1]**

(b) Slant height of cone $= \sqrt{5^2 + 6^2}$ $(= \sqrt{61}$ or 7.81...) **[M1]**

Curved surface area of cone $= \pi \times 5 \times$ their $\sqrt{61}$ $(= 122.68...)$ **[M1]**

Curved surface area of cylinder + base $= (2 \times \pi \times 5 \times 12) + (\pi \times 5^2)$ $(= 145\pi$ or 455.53...) **[M1]**

578 or 578.2... **[A1]**

[Total Marks: 7]

1229 **(a)** Volume of cylinder $= \pi \times 6^2 \times 14$ **[M1]**

Volume of hemisphere $= \frac{2}{3} \times \pi \times 6^3$ **[M1]**

2040 or 2035.7... **[A1]**

(b) Curved surface area of cylinder + base area $= 2 \times \pi \times 6 \times 14 + \pi \times 6^2$ (or $168\pi + 36\pi$) **[M1]**

Curved surface area of hemisphere $= 2 \times \pi \times 6^2$ (or 72π) **[M1]**

Correct addition of all three parts **[M1]**

867 or 867.0... to 867.1 **[A1]**

[Total Marks: 7]

1230 **(a)** Volume of cylinder $= \pi \times 6^2 \times 15 = 540\pi$ **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 6^2 \times 8 = 96\pi$ **[M1]**

1998 or 2000 (to 3 sf) **[A1]**

(b) Slant height of cone $l = \sqrt{6^2 + 8^2} = 10$ cm **[M1]**

Curved surface area of cone $= \pi \times 6 \times 10 = 60\pi$ **[M1]**

Curved surface area of cylinder + base area $=$
$2 \times \pi \times 6 \times 15 + \pi \times 6^2 = 180\pi + 36\pi = 216\pi$ **[M1]**

867 (to 3 sf) **[A1]**

[Total Marks: 7]

1231 **(a)** Volume of cylinder $= \pi \times 3^2 \times 10 = 90\pi$ (282.7...) **[M1]**

Volume of hemisphere $= \frac{2}{3} \times \pi \times 3^3 = 18\pi$ (56.5...) **[M1]**

108π or 339 or 339.2 to 339.3 **[A1]**

(b) Area of circular base $= \pi \times 3^2 = 9\pi$ (28.2...) **[B1]**

Curved area of cylinder $= 2 \times \pi \times 3 \times 10 = 60\pi$ (188.4...) **[M1]**

Curved area of hemisphere $= 2 \times \pi \times 3^2 = 18\pi$ (56.5...) **[M1]**

87π or 273 or 273.3 to 273.4 **[A1]**

[Total Marks: 7]

1232 **(a)** $\pi \times 4^2 \times 12$ (or 192π) **[M1]**

$\frac{2}{3} \times \pi \times 4^3$ (or $\frac{128\pi}{3}$) **[M1]**

737 or 737.2... or $\frac{704\pi}{3}$ **[A1]**

(b) Calculates curved surface area of cylinder: $2 \times \pi \times 4 \times 12$ (or 96π) **[M1]**

Calculates curved surface area of hemisphere: $2 \times \pi \times 4^2$ (or 32π) **[M1]**

Adds all three external faces including the flat circular base: $96\pi + 32\pi + \pi(4^2)$ **[M1]**

452 or 452.3... to 452.4 or 144π **[A1]**

[Total Marks: 7]

1233 **(a)** Volume of large cone $= \frac{1}{3} \times \pi \times 8^2 \times 24$ (=512π) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 5^2 \times 15$ (=125π) **[M1]**

1215.8 (or 1220 or 387π) **[A1]**

(b) Slant height of large cone $= \sqrt{8^2 + 24^2}$ (=$8\sqrt{10}$ or 25.298...) **[M1]**

Slant height of small cone $= \sqrt{5^2 + 15^2}$ (=$5\sqrt{10}$ or 15.811...) **[M1]**

Curved SA of frustum $= \pi \times 8 \times 8\sqrt{10} - \pi \times 5 \times 5\sqrt{10}$ (=387.4...) **[M1]**

Total SA = Curved SA + $\pi(8^2) + \pi(5^2)$ **[M1]**

667 (or 667.0...) **[A1]**

[Total Marks: 8]

1234 **(a)** Calculates the height of the cylinder: $18 - 5 = 13$ cm **[B1]**

Calculates the volume of the cylinder: $\pi \times 5^2 \times 13$ ($= 325\pi$) **[M1]**

Calculates the volume of the hemisphere: $\frac{2}{3} \times \pi \times 5^3$ ($= \frac{250\pi}{3}$) **[M1]**

1280 (accept 1280 to 1284) **[A1]**

(b) Calculates the curved surface area of the cylinder: $2 \times \pi \times 5 \times 13$ ($= 130\pi$) **[M1]**

Calculates the curved surface area of the hemisphere: $2 \times \pi \times 5^2$ ($= 50\pi$) **[M1]**

Calculates the area of the circular base: $\pi \times 5^2$ ($= 25\pi$) **[M1]**

644 (accept 643 to 645) **[A1]**

[Total Marks: 8]

1235 **(a)** Height of cone $= 17 - 5 = 12$ cm **[M1]**

Volume of cone $= \frac{1}{3} \times \pi \times 5^2 \times 12$ (100π or 314.1...) AND Volume of hemisphere $= \frac{2}{3} \times \pi \times 5^3$ ($\frac{250}{3}\pi$ or 261.7...) **[M1]**

576 (accept 575.9 to 576.0) **[A1]**

(b) Slant height of cone $l = \sqrt{5^2 + 12^2} = 13$ **[M1]**

Curved SA of cone $= \pi \times 5 \times 13$ (65π) AND Curved SA of hemisphere $= 2 \times \pi \times 5^2$ (50π) **[M1]**

361 (accept 361.2 to 361.3) **[A1]**

(c) $\pi \times 4^2 \times h = 575.9...$ (or $\frac{550}{3}\pi$) **[M1]**

$16h = 183.3...$ (or $\frac{550}{3}$) **[M1]**

11.5 (accept 11.45 to 11.46) **[A1]**

[Total Marks: 9]

1236 **(a)** Uses similar triangles: $\frac{h}{24} = \frac{4}{10}$ **[M1]**

$h = \frac{24 \times 4}{10} = 9.6$ cm **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 10^2 \times 24 = 800\pi$ **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 4^2 \times 9.6 = 51.2\pi$ **[M1]**

2350 (or 2352.4 . . .) **[A1]**

(c) Sets up equation: $\frac{4}{3} \times \pi \times r^3 = 748.8\pi$ (or 2352.4) **[M1]**

$r^3 = 561.6$ **[M1]**

8.25 (or 8.250 . . .) **[A1]**

[Total Marks: 8]

1237 **(a)** Total height of large cone $= 20 + 10 = 30$ cm **[B1]**

Volume of large cone $= \frac{1}{3} \times \pi \times 12^2 \times 30$ ($= 1440\pi$ or 4523.89...) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 4^2 \times 10$ ($= 53.33\pi$ or 167.55...) **[M1]**

4360 or 4356... **[A1]**

(b) Slant height of large cone, $L = \sqrt{12^2 + 30^2}$ ($= \sqrt{1044}$ or 32.31...) AND slant height of small cone, $l = \sqrt{4^2 + 10^2}$ ($= \sqrt{116}$ or 10.77...) **[M1]**

Curved surface area of frustum $= \pi \times 12 \times \sqrt{1044} - \pi \times 4 \times \sqrt{116}$ ($= 1082.7...$) **[M1]**

Total surface area $= 1082.7... + \pi \times 12^2 + \pi \times 4^2$ **[M1]**

1585 or 1585.3... to 1585.4 **[A1]**

[Total Marks: 8]

1238 **(a)** Uses similar triangles: $\frac{r}{15} = \frac{16}{40}$ **[M1]**

$r = 15 \times \frac{16}{40} = 6$ leading correctly to given answer **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 15^2 \times 40 = 3000\pi$ (9424.7...) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 6^2 \times 16 = 192\pi$ (603.1...) **[M1]**

2808π or 8820 or 8821.5 to 8822 **[A1]**

(c) Equating volumes: $\frac{4}{3} \times \pi \times R^3 = 2808\pi$ **[M1]**

$R^3 = 2106$ **[M1]**

$R = \sqrt[3]{2106} = 12.8$ or 12.81 to 12.82 **[A1]**

[Total Marks: 8]

1239 **(a)** Uses similar triangles: $\frac{h-24}{h} = \frac{9}{15}$ (or $\frac{x}{x+24} = \frac{9}{15}$) **[M1]**

Solves equation to obtain $h = 60$ **[A1]**

(b) Volume of large cone $= \frac{1}{3}\pi(15)^2(60) = 4500\pi$ and volume of small cone $= \frac{1}{3}\pi(9)^2(36) = 972\pi$ **[M1]**

Volume of frustum $= 4500\pi - 972\pi = 3528\pi = 11\,083.5...$ cm^3 **[M1]**

11.1 (accept 11.08) **[A1]**

(c) Finds slant heights: $L = \sqrt{15^2 + 60^2} = 61.846...$ and $l = \sqrt{9^2 + 36^2} = 37.107...$ (or $l_{frustum} = \sqrt{6^2 + 24^2} = 24.738...$) **[M1]**

Calculates curved surface area plus circular base: $\pi(15)(61.846...) - \pi(9)(37.107...) + \pi(9)^2$ **[M1]**

2120 (accept 2120 to 2121) **[A1]**

[Total Marks: 8]

1240 **(a)** Let x be the height of the removed cone. $\frac{x}{4} = \frac{x+15}{8}$ **[M1]**

$8x = 4x + 60 \implies 4x = 60 \implies x = 15$. Total height $= 15 + 15 = 30$ cm. **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 8^2 \times 30 = 640\pi$ **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 4^2 \times 15 = 80\pi$ **[M1]**

1760 (to 3 sf) **[A1]**

(c) Slant height of large cone $L = \sqrt{8^2 + 30^2} \approx 31.05$ cm and small cone $l = \sqrt{4^2 + 15^2} \approx 15.52$ cm **[M1]**

Curved surface area of frustum $= \pi \times 8 \times \sqrt{964} - \pi \times 4 \times \sqrt{241} \approx 585.25$ **[M1]**

Area of top and bottom bases $= \pi \times 4^2 + \pi \times 8^2 = 80\pi \approx 251.33$ **[M1]**

837 (to 3 sf) **[A1]**

[Total Marks: 9]

1241 **(a)** Identifies that the centers form an equilateral triangle of side $2r$ **[M1]**

Area of triangle $= \frac{1}{2} \times 2r \times 2r \times \sin(60°) = r^2\sqrt{3}$ **[M1]**

Identifies 3 sectors inside the triangle, each with angle 60°: Total sector area $= 3 \times \frac{60}{360} \times \pi r^2 = \frac{\pi}{2}r^2$ **[M1]**

Area $= r^2\sqrt{3} - \frac{\pi}{2}r^2 = r^2\left(\sqrt{3} - \frac{\pi}{2}\right)$ with no errors seen **[A1]**

(b) $r^2\left(\sqrt{3} - \frac{\pi}{2}\right) = 15$ **[M1]**

$r^2 = \frac{15}{\sqrt{3} - \pi/2}$ (≈ 93.02) **[M1]**

9.64 (or 9.644...) **[A1]**

(c) Perimeter consists of 3 arcs, each with angle 60° **[M1]**

Total perimeter $= 3 \times \frac{60}{360} \times 2\pi(9.644\ldots) = \pi(9.644\ldots)$ **[M1]**

30.3 (or 30.29 to 30.30) **[A1]**

[Total Marks: 10]

1242 **(a)** $\frac{r}{10} = \frac{24 - 15}{24}$ or $\frac{r}{10} = \frac{9}{24}$ **[M1]**

3.75 **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 10^2 \times 24$ (or 800π) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 3.75^2 \times 9$ (or 42.1875π) **[M1]**

2380 or 2381 (accept 2380.7...) **[A1]**

(c) Slant height of large cone $L = \sqrt{10^2 + 24^2} = 26$ **[M1]**

Slant height of small cone $l = \sqrt{3.75^2 + 9^2} = 9.75$ **[M1]**

Curved surface area of frustum $= \pi \times 10 \times 26 - \pi \times 3.75 \times 9.75$ **[M1]**

Adding base and top areas: $+\pi \times 10^2 + \pi \times 3.75^2$ **[M1]**

1060 or 1060.2... to 1060.3 **[A1]**

[Total Marks: 10]

1243 **(a)** Uses similar triangles: $\frac{h}{24} = \frac{3}{8}$ **[M1]**

$h = \frac{3 \times 24}{8} = 9$ **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 8^2 \times 24$ (or 512π) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 3^2 \times 9$ (or 27π) **[M1]**

1520 or 1524 or 485π **[A1]**

(c) Calculates slant height of large cone $L = \sqrt{8^2 + 24^2}$ (or $\sqrt{640}$) and small cone $l = \sqrt{3^2 + 9^2}$ (or $\sqrt{90}$) **[M1]**

Curved surface area of frustum $= \pi(8)(\sqrt{640}) - \pi(3)(\sqrt{90})$ **[M1]**

Calculates area of both flat circular bases: $\pi(8^2) + \pi(3^2)$ (or 73π) **[M1]**

Adds curved surface area and both base areas together **[M1]**

776 (or 775.5... to 775.6) **[A1]**

[Total Marks: 10]

1244 (a) Let h be the height of the small cone. $\frac{h}{4} = \frac{h+15}{12}$ **[M1]**

$12h = 4h + 60$ **[M1]**

$h = 7.5$ **[A1]**

(b) Volume of large cone $= \frac{1}{3} \times \pi \times 12^2 \times 22.5$ (1080π or 3392.9...) **[M1]**

Volume of small cone $= \frac{1}{3} \times \pi \times 4^2 \times 7.5$ (40π or 125.6...) **[M1]**

3270 (accept 3267 to 3268) **[A1]**

(c) Slant height of large cone $= \sqrt{12^2 + 22.5^2} = 25.5$ **[M1]**

Slant height of small cone $= \sqrt{4^2 + 7.5^2} = 8.5$ **[M1]**

Curved surface area of frustum $= \pi \times 12 \times 25.5 - \pi \times 4 \times 8.5$ ($306\pi - 34\pi = 272\pi$) **[M1]**

Area of both circular bases $= \pi \times 12^2 + \pi \times 4^2$ ($144\pi + 16\pi = 160\pi$) **[M1]**

1360 (accept 1357 to 1358) **[A1]**

[Total Marks: 11]

1245 (a) Let H be the height of the large cone. Height of small removed cone $= H - 8$ **[M1]**

Uses similar triangles: $\frac{H}{12} = \frac{H-8}{6}$ (or equivalent ratio) **[M1]**

$6H = 12H - 96$

6H = 96

H = $16 with no errors seen [A1]$

(b) Volume of frustum $= \frac{1}{3} \times \pi \times 12^2 \times 16 - \frac{1}{3} \times \pi \times 6^2 \times 8$ ($= 768\pi - 96\pi = 672\pi$)
[M2, M1 for one correct cone volume]

Volume of hemisphere $= \frac{2}{3} \times \pi \times 6^3$ ($= 144\pi$) **[M1]**

Total volume $= 672\pi + 144\pi = 816\pi = 2560$ or $2563.5\ldots$ **[A1]**

(c) Slant height of large cone $= \sqrt{12^2 + 16^2} = 20$ AND Slant height of small cone $= \sqrt{6^2 + 8^2} = 10$ **[M1]**

Curved surface area of frustum $= \pi \times 12 \times 20 - \pi \times 6 \times 10$ $(= 180\pi)$ **[M1]**

Total surface area $= 180\pi$ (frustum curved) $+\pi \times 12^2$ (base) $+2 \times \pi \times 6^2$ (hemisphere) **[M1]**

1240 or 1244 to 1244.2 **[A1]**

[Total Marks: 11]

1246 **(a)** Sets up an equation using similar triangles (e.g. let x be the height of the smaller cone): $\frac{x}{4} = \frac{x+10}{6}$ (or equivalent) **[M1]**

Solves for the height of the smaller cone: $6x = 4x + 40 \implies 2x = 40 \implies x = 20$ **[M1]**

Calculates the height of the larger cone: $20 + 10 = 30$ **[A1]**

(b) Calculates the volume of the larger cone: $\frac{1}{3} \times \pi \times 6^2 \times 30$ $(= 360\pi)$ **[M1]**

Calculates the volume of the smaller cone: $\frac{1}{3} \times \pi \times 4^2 \times 20$ $(= \frac{320\pi}{3})$ **[M1]**

796 (accept 795 to 797) **[A1]**

(c) Calculates the slant height of the larger cone: $\sqrt{6^2 + 30^2} = \sqrt{936}$ $(= 30.59...)$ **[M1]**

Calculates the slant height of the smaller cone: $\sqrt{4^2 + 20^2} = \sqrt{416}$ $(= 20.39...)$ **[M1]**

Calculates the curved surface area of the frustum: $\pi \times 6 \times \sqrt{936} - \pi \times 4 \times \sqrt{416}$ $(= 320.38...)$ **[M1]**

Calculates the sum of the areas of the circular top and base: $\pi \times 6^2 + \pi \times 4^2$ $(= 52\pi$ or $163.36...)$ **[M1]**

484 (accept 483.7 to 484.5) **[A1]**

[Total Marks: 11]

1247 **(a)** $\frac{h}{h+10} = \frac{4.5}{12}$ $(= \frac{3}{8})$ **[M1]**

$8h = 3h + 30$ giving $5h = 30$ and $h = 6$ cm **[A1]**

(b) Slant height of full cone $= \sqrt{12^2 + 16^2} = 20$ cm and slant height of removed cone $= \sqrt{4.5^2 + 6^2} = 7.5$ cm **[M1]**

Curved surface area of frustum $= \pi(12)(20) - \pi(4.5)(7.5) = 206.25\pi\ cm^2$ **[M1]**

Total surface area $= 206.25\pi + \pi(12^2) + \pi(4.5^2) = 370.5\pi$ $(= 1163.96\ldots)$ **[M1]**

1160 **[A1]**

(c) Volume $= \frac{1}{3}\pi(12^2)(16) - \frac{1}{3}\pi(4.5^2)(6)$ **[M1]**

$768\pi - 40.5\pi = 727.5\pi$ $(= 2285.508\ldots)$ **[M1]**

2290 **[A1]**

(d) Volume of one sphere $= \frac{4}{3}\pi(2.4^3) = 18.432\pi$ $(= 57.9058\ldots)\ cm^3$ **[M1]**

$n = \frac{727.5\pi}{18.432\pi} = 39.469\ldots$ **[M1]**

39 **[A1]**

[Total Marks: 12]

1248 Half of the chord length $= 24 \div 2 = 12$ **[M1]**

Uses Pythagoras' theorem: $r^2 = 5^2 + 12^2$ **[M1]**

13 **[A1]**

[Total Marks: 3]

1249 Uses Pythagoras' theorem: $(8 - -4)^2 + (1 - 6)^2$ or $12^2 + (-5)^2$ **[M1]**

$144 + 25 = 169$ **[M1]**

13 **[A1]**

[Total Marks: 3]

1250 $\Delta x = 4 - (-2) = 6$ and $\Delta y = 9 - 1 = 8$ **[M1]**

$\sqrt{6^2 + 8^2}$ **[M1]**

10 **[A1]**

[Total Marks: 3]

1251 Use of Pythagoras' theorem: $(k - 1)^2 + (19 - 4)^2 = 17^2$ **[M1]**

$(k - 1)^2 + 15^2 = 289$ or $(k - 1)^2 = 64$ **[M1]**

$k - 1 = 8$ and $k - 1 = -8$ (or $k^2 - 2k - 63 = 0$

(k-9)(k+7) = 0)[$M1$]

9 and -7 **[A1]**

[Total Marks: 4]

1252 (a) Finds horizontal and vertical differences: $8 - (-4) = 12$ and $4 - (-1) = 5$ **[B1]**

Applies Pythagoras' theorem: $\sqrt{12^2 + 5^2}$ **[M1]**

13 **[A1]**

(b) Calculates $PR^2 = (8 - (-4))^2 + (-6 - (-1))^2 = 144 + 25 = 169$ and $QR = 4 - (-6) = 10$ **[M1]**

Checks converse of Pythagoras' theorem: $PQ^2 + PR^2 = 169 + 169 = 338e100$ (or $10^2 + 13^2 = 269e169$), so No, triangle PQR is not right-angled at P **[A1]**

[Total Marks: 5]

1253 (a) Horizontal distance $PR = 8 - (-4) = 12$ and vertical distance $QR = 3 - (-2) = 5$ **[B1]**

$PQ^2 = 12^2 + 5^2 = 144 + 25 = 169$ **[M1]**

13 **[A1]**

(b) $PS^2 = (2 - (-4))^2 + (4 - (-2))^2$ **[M1]**

$PS^2 = 6^2 + 6^2 = 36 + 36 = 72$ **[M1]**

$6\sqrt{2}$ **[A1]**

(c) $PT^2 = (-3 - (-4))^2 + (3 - (-2))^2 = 1^2 + 5^2 = 26$, so $PT = \sqrt{26}$ **[M1]**

$ST^2 = (-3 - 2)^2 + (3 - 4)^2 = (-5)^2 + (-1)^2 = 26$, so $ST = \sqrt{26}$ **[M1]**

$PT = ST = \sqrt{26}$, so triangle PST has two equal sides and is isosceles **[A1]**

[Total Marks: 9]

1254 **(a)** $(x - 7)^2 + x^2 = (x + 1)^2$ **[M1]**

$x^2 - 14x + 49 + x^2 = x^2 + 2x + 1$ **[M1]**

$2x^2 - 14x + 49 = x^2 + 2x + 1$

$x^2 - 16x + 48 = 0$ **[A1]**

(b) $(x - 4)(x - 12) = 0$ **[M1]**

$x = 4$ or $x = 12$ **[A1]**

Rejects $x = 4$ because $x - 7$ would be -3 (length must be positive), so $x = 12$ **[B1]**

Perimeter $= 5 + 12 + 13 = 30$ **[A1]**

[Total Marks: 7]

1255 **(a)** Angle $OMA = 90°$ and $AM^2 = 25^2 - 7^2 = 625 - 49 = 576$ **[M1]**

$AM = 24$ cm **[A1]**

$AB = 2 \times 24 = 48$ **[A1]**

(b) **(i)** $CN = \dfrac{30}{2} = 15$ cm **[B1]**

$ON^2 = 25^2 - 15^2 = 625 - 225 = 400$ **[M1]**

20 **[A1]**

(ii) Perpendicular distance between chords $= 20 - 7 = 13$ cm **[B1]**

Area $= \dfrac{1}{2}(48 + 30) \times 13$ **[M1]**

507 **[A1]**

[Total Marks: 9]

1256 **(a)** $AM = 16 \div 2 = 8$ **[M1]**

$PM = \sqrt{10^2 - 8^2} = \sqrt{36} = 6$ with no errors seen **[A1]**

(b) In triangle AMQ, $QM^2 = 17^2 - 8^2$ or $289 - 64$ **[M1]**

$QM = \sqrt{225} = 15$ **[A1]**

$PQ = 6 + 15 = 21$ **[A1]**

[Total Marks: 5]

1257 **(a)** $(2x+1)^2 = (x-1)^2 + (x+6)^2$ **[M1]**

$4x^2 + 4x + 1 = (x^2 - 2x + 1) + (x^2 + 12x + 36)$ **[M1]**

$4x^2 + 4x + 1 = 2x^2 + 10x + 37$ **[M1]**

$2x^2 - 6x - 36 = 0$

$x^2 - 3x - 18 = 0$ with no errors seen [A1]

(b) $(x-6)(x+3) = 0$ **[M1]**

$x = 6$ or $x = -3$ **[A1]**

Selects $x = 6$ and rejects $x = -3$ because lengths must be strictly positive **[B1]**

(c) Base $= 6 - 1 = 5$ and Height $= 6 + 6 = 12$ **[M1]**

Area $= \frac{1}{2} \times 5 \times 12$ **[M1]**

30 **[A1]**

[Total Marks: 10]

1258 Let M be the midpoint of AB. $AM^2 = 8.5^2 - 4^2$ **[M1]**

$AM = \sqrt{72.25 - 16} = \sqrt{56.25} = 7.5$ **[M1]**

Chord $AB = 2 \times 7.5 = 15$ **[A1]**

[Total Marks: 3]

1259 **(a)** $AM = 15 \div 2 = 7.5$ **[B1]**

$d^2 + 7.5^2 = 8.5^2$ **[M1]**

$d = \sqrt{72.25 - 56.25} = \sqrt{16} = 4$ **[A1]**

(b) Let half-chord be x. $x^2 + 7.5^2 = 8.5^2$ **[M1]**

$x = \sqrt{72.25 - 56.25} = 4$ **[A1]**

Length of chord $= 2 \times 4 = 8$ **[A1]**

[Total Marks: 6]

1260 **(a)** Uses midpoint property: $AM = \frac{18}{2} = 9$ **[B1]**

Applies Pythagoras' theorem: $OM = \sqrt{11.5^2 - 9^2}$ **[M1]**

7.16 (accept 7.158 to 7.16) **[A1]**

(b) Applies Pythagoras' theorem to find half-chord length: $\sqrt{11.5^2 - 6.4^2}$ **[M1]**

Half-chord $= \sqrt{91.29} = 9.55458...$ **[M1]**

19.1 (accept 19.11) **[A1]**

[Total Marks: 6]

1261 **(a)** Let half the chord length be x. Uses $12.4^2 = x^2 + 5.2^2$ **[M1]**

$x = \sqrt{12.4^2 - 5.2^2}$ or $x = 11.257\ldots$ **[M1]**

22.5 (or 22.51 to 22.52) **[A1]**

(b) Let perpendicular distance from O to CD be y. Uses $12.4^2 = y^2 + 9^2$ **[M1]**

$y = \sqrt{12.4^2 - 9^2}$ or $y = 8.5299\ldots$ **[A1]**

Total distance $= 5.2 + 8.5299\ldots$ **[M1]**

13.7 (or 13.72 to 13.73) **[A1]**

[Total Marks: 7]

1262 **(a)** Uses distance formula: $\sqrt{(6-(-2))^2 + (y-(-1))^2} = 10$ **[M1]**

$64 + (y+1)^2 = 100$ **[M1]**

$(y+1)^2 = 36$

$y + 1 = 6$ (since $y > 0$)

5 **[A1]**

(b) Let C be $(x, 0)$.

$AC^2 = (x-(-2))^2 + (0-(-1))^2 = (x+2)^2 + 1$ **[M1]**

$BC^2 = (x-6)^2 + (0-5)^2 = (x-6)^2 + 25$ **[M1]**

Equating distances squared: $x^2 + 4x + 5 = x^2 - 12x + 61$ **[M1]**

$16x = 56$

3.5 **[A1]**

[Total Marks: 7]

1263 **(a)** $x^2 + (x+7)^2 = (x+8)^2$ **[M1]**

$x^2 + x^2 + 14x + 49 = x^2 + 16x + 64$ **[M1]**

$x^2 - 2x - 15 = 0$ **[A1]**

(b) $(x-5)(x+3) = 0$ or correct use of the quadratic formula **[M1]**

$x = 5$ (accept value 5 only, ignoring -3 if appropriately discarded) **[A1]**

(c) Base $= 5 + 7 = 12$ and Height $= 5$ **[M1]**

Area $= \frac{1}{2} \times 12 \times 5 = 30$ **[A1]**

[Total Marks: 7]

1264 **(a)** Applies Pythagoras' theorem: $(x+2)^2 + (3x-1)^2 = (3x+1)^2$ **[M1]**

Expands brackets: $(x^2+4x+4)+(9x^2-6x+1)=9x^2+6x+1$ **[M1]**

Simplifies $10x^2-2x+5=9x^2+6x+1$ to obtain $x^2-8x+4=0$ **[A1]**

(b) Uses quadratic formula or completing the square: $x=\dfrac{8\pm\sqrt{(-8)^2-4(1)(4)}}{2}$ or $(x-4)^2=12$ **[M1]**

Simplifies surd: $\sqrt{48}=4\sqrt{3}$ or $\sqrt{12}=2\sqrt{3}$ **[M1]**

$4\pm2\sqrt{3}$ **[A1]**

(c) Evaluates $x=4-2\sqrt{3}\approx0.536$ **[M1]**

Substitutes into $BC=3x-1\approx0.608>0$ — wait: $BC=3(0.5359)-1=0.608$, check $3x-1$: if $x=0.536$, $3x-1>0$; adjust side BC to $(2x-2)$ or explain $BC=3x-1$ requires $x>\dfrac{1}{3}$ — use $BC=(3x-2)$ so $3(0.536)-2=-0.39<0$, stating a length cannot be negative **[A1]**

[Total Marks: 8]

1265 **(a)** $(x-6)^2+(x-5)^2=(x+3)^2$ **[M1]**

$x^2-12x+36+x^2-10x+25=x^2+6x+9$ **[M1]**

$2x^2-22x+61=x^2+6x+9$ simplifying to $x^2-28x+52=0$ **[A1]**

(b) $(x-2)(x-26)=0$ **[M1]**

$x=26$ (rejecting $x=2$ as $x-6$ and $x-5$ must be positive) **[A1]**

$AB=20$ cm, $BC=21$ cm, $AC=29$ cm **[A1]**

(c) **(i)** $AG^2=29^2+12^2=841+144=985$ **[M1]**

31.4 **[A1]**

(ii) Triangle ACG is right-angled at C, so $\dfrac{1}{2}\times AG\times d=\dfrac{1}{2}\times29\times12$ **[M1]**

$d=\dfrac{29\times12}{\sqrt{985}}$ $(=11.088\ldots)$ **[M1]**

11.1 **[A1]**

[Total Marks: 11]

1266 **(a)** $\tan60^\circ=\dfrac{h}{x}$ giving $x=\dfrac{h}{\sqrt{3}}$ **[M1]**

$\tan30^\circ=\dfrac{h}{x+12}$ giving $\dfrac{1}{\sqrt{3}}=\dfrac{h}{\frac{h}{\sqrt{3}}+12}$ **[M1]**

$\dfrac{h}{\sqrt{3}}+12=h\sqrt{3}$ multiplying by $\sqrt{3}$ gives $h+12\sqrt{3}=3h$ **[M1]**

$2h=12\sqrt{3}$ giving $h=6\sqrt{3}$ m **[A1]**

(b) $x=6$ m and $PR=\dfrac{6\sqrt{3}}{\sin60^\circ}=12$ m **[B1]**

$PQ=\dfrac{18}{\cos30^\circ}=\dfrac{36}{\sqrt{3}}=12\sqrt{3}$ m **[M1]**

Perimeter $= 12 + 12 + 12\sqrt{3} = 24 + 12\sqrt{3}$ **[A1]**

(c) $\frac{1}{2} \times PR \times d = \frac{1}{2} \times RT \times PT$ (or $d = 6\sin 60°$) **[M1]**

$d = \frac{6 \times 6\sqrt{3}}{12}$ **[M1]**

$3\sqrt{3}$ **[A1]**

[Total Marks: 10]

1267 Identifies cosine ratio: $\cos(\theta) = \frac{2.1}{6.5}$ **[M1]**

$\theta = \cos^{-1}\left(\frac{2.1}{6.5}\right)$ **[M1]**

71.2 **[A1]**

[Total Marks: 3]

1268 $\cos(58°) = \frac{9.4}{XZ}$ **[M1]**

$XZ = \frac{9.4}{\cos(58°)}$ **[M1]**

17.7 or 17.73... **[A1]**

[Total Marks: 3]

1269 (a) $BC^2 + 8.5^2 = 14.2^2$ **[M1]**

$BC = \sqrt{14.2^2 - 8.5^2}$ OR $BC = \sqrt{129.39}$ **[M1]**

11.4 (accept $11.37\ldots$) **[A1]**

(b) $\cos(\angle BAC) = \frac{8.5}{14.2}$ **[M1]**

53.2 (accept $53.23\ldots$) **[A1]**

[Total Marks: 5]

1270 (a) Uses tangent ratio: $\tan(34°) = \frac{AB}{14}$ or $AB = 14\tan(34°)$ **[M1]**

9.44 (accept 9.443) **[A1]**

(b) Recognises shortest distance is perpendicular to AC, so $\angle BDC = 90°$ **[B1]**

Uses sine ratio in right-angled triangle BDC: $\sin(34°) = \frac{BD}{14}$ **[M1]**

7.83 (accept 7.828 to 7.83) **[A1]**

[Total Marks: 5]

1271 **(a)** $\tan(38°) = \dfrac{TB}{45}$ **[M1]**

35.2 (or 35.15 to 35.16) **[A1]**

(b) $\cos(38°) = \dfrac{45}{PT}$ or $PT^2 = 45^2 + 35.15^2$ **[M1]**

57.1 (or 57.10 to 57.11) **[A1]**

(c) $\tan(XPB) = \dfrac{20}{45}$ **[M1]**

Angle $= \tan^{-1}\left(\dfrac{20}{45}\right)$ **[M1]**

24.0 (or $23.96\ldots$) **[A1]**

[Total Marks: 7]

1272 **(a)** $\tan 54° = \dfrac{AB}{6.5}$ giving $AB = 6.5\tan 54°$ $(= 8.946\ldots)$ **[M1]**

8.95 **[A1]**

(b) $BD = 6.5 + 8.2 = 14.7$ m **[B1]**

$\tan \angle ADB = \dfrac{8.946\ldots}{14.7}$ $(= 0.6086\ldots)$ **[M1]**

31.3 **[A1]**

(c) Shortest distance $d = 14.7\sin 31.32\ldots°$ (or $AD = \sqrt{8.946\ldots^2 + 14.7^2} = 17.208\ldots$ m) **[M1]**

$d = 7.642\ldots$ (or $\dfrac{8.946\cdots \times 14.7}{17.208\ldots}$) **[M1]**

7.64 **[A1]**

[Total Marks: 8]

1273 **(a)** Uses $\sin(52°) = \dfrac{\text{distance}}{18}$ or $18 \times \sin(52°)$ **[M1]**

14.18... or 14.2 **[A1]**

(b) Distance to closest point $= 18 \times \cos(52°) = 11.082...$ km **[M1]**

Time $= \dfrac{11.082...}{14}$ $(= 0.7915...$ hours) **[M1]**

$0.7915... \times 60 = 47.49... = 47$ **[A1]**

[Total Marks: 5]

1274 **(a)** $\tan(38°) = \dfrac{h}{50}$ **[M1]**

$h = 50\tan(38°)$ OR $39.06\ldots$ **[A1]**

Total height $= 39.06\cdots + 1.6 = 40.7$ (accept $40.66\ldots$) **[A1]**

(b) Identifies the angle of elevation of T from X as $25°$ OR $\angle BTX = 65°$ **[B1]**

$\tan(25°) = \dfrac{40.66}{BX}$ OR $\tan(65°) = \dfrac{BX}{40.66}$ **[M1]**

$BX = 87.2$ (accept $87.19\ldots$) **[A1]**

[Total Marks: 6]

1275 **(a)** Uses $\angle TPB = 42°$ (or $\angle BTP = 48°$): $\tan(42°) = \dfrac{65}{BP}$ or $BP = 65\tan(48°)$ **[M1]**

72.2 (accept 72.18 to 72.2) **[A1]**

(b) Uses $\angle TQB = 27°$ (or $\angle BTQ = 63°$): $\tan(27°) = \dfrac{65}{BQ}$ **[M1]**

Calculates $BQ = \dfrac{65}{\tan(27°)} = 127.57...$ **[A1]**

Subtracts distances: $PQ = 127.57... - 72.189...$ **[M1]**

55.4 (accept 55.38 to 55.4) **[A1]**

[Total Marks: 6]

1276 **(a)** Calculates $\angle PAB = 360° - (180° - 40°) - 130° = 90°$ (or similar valid reasoning to show right angle) **[M1]**

Uses Pythagoras' theorem: $PB^2 = 80^2 + 60^2$ **[M1]**

100 **[A1]**

(b) Uses trigonometry to find $\angle APB$: $\tan(APB) = \dfrac{60}{80}$ **[M1]**

$\angle APB = \tan^{-1}(0.75) = 36.869...°$ **[A1]**

Bearing $= 040° + 36.869...° = 076.9$ (accept 077) **[A1]**

(c) Recognises the shortest distance d is the perpendicular height of $\triangle PAB$ from A to PB. **[M1]**

Uses area equation: $\dfrac{1}{2} \times 100 \times d = \dfrac{1}{2} \times 80 \times 60$ OR uses trigonometry $80\sin(53.13°)$ or $60\sin(36.87°)$ **[M1]**

48 **[A1]**

[Total Marks: 9]

1277 **(a)** Interior angle between parallel North lines $= 180° - 52° = 128°$ **[M1]**

Angles around point $B = 360° - 128° - 142° = 90°$ with no errors seen **[A1]**

(b) $AC = \sqrt{15^2 + 20^2}$ **[M1]**

$AC = \sqrt{625}$ **[M1]**

25 **[A1]**

(c) $\tan(BAC) = \dfrac{20}{15}$ **[M1]**

Angle $BAC = 53.13°$ **[A1]**

Bearing $= 052° + 53.13°$ **[M1]**

105.1 (or $105.13\ldots$) **[A1]**

[Total Marks: 9]

1278 **(a)** Bearing of P from $L = 058° + 180° = 238°$ **[B1]**

Angle $PLQ = 238° - 148° = 90°$ **[A1]**

(b) **(i)** $PQ^2 = 24^2 + 32^2 = 576 + 1024 = 1600$ **[M1]**

40 **[A1]**

(ii) $\tan \angle LPQ = \dfrac{32}{24}$ giving angle $LPQ = 53.130\ldots°$ **[M1]**

Bearing $= 058° + 53.130\ldots°$ **[M1]**

111.1 **[A1]**

(c) $\dfrac{1}{2} \times 40 \times d = \dfrac{1}{2} \times 24 \times 32$ (or $d = 24 \sin 53.130\ldots°$) **[M1]**

19.2 **[A1]**

[Total Marks: 9]

1279 **(a)** $\angle APL = 90° - 65° = 25°$ **[B1]**

$\cos(25°) = \dfrac{PL}{120}$ OR $\sin(65°) = \dfrac{PL}{120}$ **[M1]**

109 or 108.8 (accept $108.75\ldots$) **[A1]**

(b) Identifies the shortest distance as the perpendicular distance from L to PA, creating a new right-angled triangle with hypotenuse PL **[B1]**

$\sin(25°) = \dfrac{d}{108.76}$ OR Area method: $0.5 \times 120 \times d = 0.5 \times 108.76 \times (120 \sin 25°)$ **[M1]**

46.0 (accept $45.96\ldots$) **[A1]**

[Total Marks: 6]

1280 **(a)** $PR = \sqrt{8.5^2 + 11.2^2}$ or $\sqrt{197.69}$ **[M1]**

14.06... or 14.1 **[A1]**

(b) $\sin(41°) = \dfrac{14.06...}{PS}$ or $PS = \dfrac{14.06...}{\sin(41°)}$ **[M1]**

21.43... or 21.4 **[A1]**

(c) Angle $RPS = 180° - 90° - 41° = 49°$ AND distance $= 14.06... \times \sin(49°)$
OR area method: $0.5 \times 21.43... \times h = 0.5 \times 14.06... \times (14.06.../\tan(41°))$ **[M1]**

10.60... or 10.6 **[A1]**

[Total Marks: 6]

1281 **(a)** Sets up equation for $\triangle BCD$: $\tan(42°) = \dfrac{CD}{BC}$

BC $= \dfrac{CD}{\tan(42°)}$ [M1]

Sets up equation for $\triangle ACD$: $\tan(25°) = \dfrac{CD}{BC + 50}$

BC + 50 $= \dfrac{CD}{\tan(25°)}$ [M1]

Equates or substitutes to eliminate BC: $50 = \dfrac{CD}{\tan(25°)} - \dfrac{CD}{\tan(42°)}$ **[M1]**

Factorises to solve for CD: $CD\left(\dfrac{1}{\tan(25°)} - \dfrac{1}{\tan(42°)}\right) = 50$ **[M1]**

48.4 (accept 48.36...) **[A1]**

(b) Recognises the shortest distance d is the perpendicular from B to AD, forming a right-angled triangle with hypotenuse $AB = 50$. **[M1]**

Sets up trigonometric ratio: $\sin(25°) = \dfrac{d}{50}$ **[M1]**

21.1 (accept 21.13...) **[A1]**

[Total Marks: 8]

1282 **(a)** $\angle CAB = 90° - 58° = 32°$ **[B1]**

$\angle CBA = 315° - 270° = 45°$ **[B1]**

(b) Expresses AD and BD in terms of CD: $AD = \dfrac{CD}{\tan(32°)}$ and $BD = \dfrac{CD}{\tan(45°)} = CD$ **[M1]**

Sets up sum equation: $\dfrac{CD}{\tan(32°)} + CD = 48$ **[M1]**

Factors out CD: $CD\left(\dfrac{1}{\tan(32°)} + 1\right) = 48$ giving $CD(2.6003...) = 48$ **[M1]**

18.5 (accept 18.45 to 18.46) **[A1]**

(c) Uses sine ratio in triangle ACD: $\sin(32°) = \dfrac{18.459...}{AC}$ **[M1]**

34.8 (accept 34.83 to 34.9) **[A1]**

[Total Marks: 8]

1283 **(a)** $AC = \sqrt{8^2 + 15^2}$ **[M1]**

17 **[A1]**

(b) $AD^2 + 12^2 = 17^2$ (or their AC^2) **[M1]**

$AD = \sqrt{289 - 144} = \sqrt{145}$ **[M1]**

12.0 (or 12.04...) **[A1]**

(c) Area of $\triangle ADC = \dfrac{1}{2} \times 12 \times \sqrt{145} = 72.25...$ **[M1]**

$\frac{1}{2} \times 17 \times h = 72.25\ldots$ **[M1]**

8.50 (or $8.499\ldots$) **[A1]**

(d) $\tan(BAC) = \frac{15}{8} \implies$ Angle $BAC = 61.93°$ **[M1]**

$\tan(CAD) = \frac{12}{\sqrt{145}} \implies$ Angle $CAD = 44.90°$ **[M1]**

Angle $DAB = 61.93° + 44.90°$ **[M1]**

106.8 (or 106.82 to 106.83) **[A1]**

[Total Marks: 12]

1284 (a) $\frac{\sqrt{3}}{2}$ **[B1]**

(b) $\sin 30° = \frac{1}{2}$ and $\tan 60° = \sqrt{3}$ **[M1]**

$\frac{1 - 1/2}{(\sqrt{3})^2} = \frac{1/2}{3} = \frac{1}{6}$ with no errors seen **[A1]**

[Total Marks: 3]

1285 (a) 2 **[B1]**

(b) $\frac{1}{2}$ **[B1]**

(c) $\frac{\sqrt{3}}{\sqrt{3}/2}$ **[M1]**

2 **[A1]**

[Total Marks: 4]

1286 (a) Substitutes $\sin 60° = \frac{\sqrt{3}}{2}$ and $\tan 30° = \frac{1}{\sqrt{3}}$ (or $\frac{\sqrt{3}}{3}$) **[M1]**

1 **[A1]**

(b) Substitutes $\tan 45° = 1$ leading to $\cos x = \frac{1}{\sqrt{2}}$ **[M1]**

45 **[A1]**

[Total Marks: 4]

1287 (a) $\frac{1}{2} + \frac{1}{2} = 1$ **[B1]**

(b) States at least two exact values correctly: $\tan 60° = \sqrt{3}$, $\sin 60° = \frac{\sqrt{3}}{2}$, $\cos 0° = 1$ **[B1]**

Substitutes into expression: $\sqrt{3} \times \frac{\sqrt{3}}{2} - 1 = \frac{3}{2} - 1$ **[M1]**

$\frac{1}{2}$ (or 0.5) **[A1]**

[Total Marks: 4]

1288 **(a)** **(i)** $\sin 30° = \frac{1}{2}$ and $\cos 60° = \frac{1}{2}$ **[M1]**

1 **[A1]**

(ii) $\tan 60° = \sqrt{3}$ and $\cos 30° = \frac{\sqrt{3}}{2}$, giving $\frac{\sqrt{3}}{\frac{\sqrt{3}}{2}} = 2$ **[B1]**

$(\sin 45°)^2 = \left(\frac{\sqrt{2}}{2}\right)^2 = \frac{1}{2}$ **[B1]**

$\frac{3}{2}$ (or 1.5 or $1\frac{1}{2}$) **[A1]**

(b) $\cos x = \frac{\sqrt{3}}{2}$ **[M1]**

30 **[A1]**

(c) $\tan y = 1 - 1 = 0$ **[M1]**

0 **[A1]**

[Total Marks: 9]

1289 **(a)** $\sin 45° = \frac{6}{PR}$ (or $\cos 45° = \frac{6}{PR}$) **[M1]**

$PR = 6 \div \frac{\sqrt{2}}{2} = 6\sqrt{2}$ **[A1]**

(b) $\sin 60° = \frac{6\sqrt{2}}{PS}$ **[M1]**

$PS = 6\sqrt{2} \div \frac{\sqrt{3}}{2}$ **[M1]**

$PS = \frac{12\sqrt{2}}{\sqrt{3}} = \frac{12\sqrt{6}}{3} = 4\sqrt{6}$ **[A1]**

[Total Marks: 5]

1290 **(a)** $\sin 45° = \frac{PQ}{8\sqrt{2}}$ **[M1]**

$PQ = 8\sqrt{2} \times \frac{1}{\sqrt{2}} = 8$ **[A1]**

(b) $\cos 30° = \frac{8\sqrt{2}}{PS}$ or $PS = \frac{8\sqrt{2}}{\cos 30°}$ **[M1]**

$PS = \frac{8\sqrt{2}}{\sqrt{3}/2} = \frac{16\sqrt{2}}{\sqrt{3}}$ **[M1]**

Rationalises the denominator to get $\frac{16\sqrt{6}}{3}$ **[A1]**

[Total Marks: 5]

1291 In $\triangle ABD$, $AD = 8\sqrt{2}\cos 45° = 8$ **[M1]**

$BD = 8\sqrt{2}\sin 45° = 8$ **[A1]**

In $\triangle BDC$, $BC = \frac{BD}{\sin 30°} = \frac{8}{0.5} = 16$ **[M1]**

$DC = \frac{BD}{\tan 30°} = 8\sqrt{3}$ **[M1]**

Perimeter $= AB + BC + AD + DC = 8\sqrt{2} + 16 + 8 + 8\sqrt{3}$ **[M1]**

$24 + 8\sqrt{2} + 8\sqrt{3}$ **[A1]**

[Total Marks: 6]

1292 **(a)** Uses $\sin 45° = \frac{BD}{8}$ with $\sin 45° = \frac{\sqrt{2}}{2}$ (or $\frac{1}{\sqrt{2}}$) **[M1]**

$4\sqrt{2}$ **[A1]**

(b) Uses $\sin 30° = \frac{4\sqrt{2}}{BC}$ with $\sin 30° = \frac{1}{2}$ **[M1]**

$8\sqrt{2}$ **[A1]**

(c) Finds $AD = 8\cos 45° = 4\sqrt{2}$ and $DC = \frac{4\sqrt{2}}{\tan 30°}$ (or $8\sqrt{2}\cos 30°$) **[M1]**

Evaluates $DC = 4\sqrt{2} \times \sqrt{3} = 4\sqrt{6}$ and sums $AC = AD + DC = 4\sqrt{2} + 4\sqrt{6}$ **[A1]**

[Total Marks: 6]

1293 **(a)** $BD = 8\sqrt{2}\sin 45° = 8\sqrt{2} \times \frac{\sqrt{2}}{2} = 8$ cm **[M1]**

$AD = 8\sqrt{2}\cos 45° = 8\sqrt{2} \times \frac{\sqrt{2}}{2} = 8$ cm **[A1]**

(b) $\tan 30° = \frac{8}{CD}$ giving $\frac{1}{\sqrt{3}} = \frac{8}{CD}$ **[M1]**

$8\sqrt{3}$ **[A1]**

(c) Base $AC = 8 + 8\sqrt{3}$ cm **[B1]**

Area $= \frac{1}{2}(8 + 8\sqrt{3}) \times 8$ **[M1]**

$32 + 32\sqrt{3}$ **[A1]**

(d) $\sin 30° = \frac{8}{BC}$ giving $BC = \frac{8}{\frac{1}{2}} = 16$ cm **[M1]**

Perimeter $= 8\sqrt{2} + 16 + (8 + 8\sqrt{3})$ **[M1]**

$24 + 8\sqrt{2} + 8\sqrt{3}$ **[A1]**

[Total Marks: 10]

1294 **(a)** Substitutes $\tan 30° = \dfrac{1}{\sqrt{3}}$ to get $\tan^2 30° = \dfrac{1}{3}$ **[B1]**

Evaluates numerator as $1 - \dfrac{1}{3} = \dfrac{2}{3}$ AND denominator as $1 + \dfrac{1}{3} = \dfrac{4}{3}$ **[M1]**

Simplifies $\dfrac{2/3}{4/3} = \dfrac{1}{2}$ which is equal to $\cos 60°$ **[A1]**

(b) Substitutes exact values: $x\left(\dfrac{\sqrt{3}}{2}\right) = \sqrt{3} - x\left(\dfrac{1}{2}\right)$ **[M1]**

Rearranges to collect x terms: $x\left(\dfrac{\sqrt{3}}{2} + \dfrac{1}{2}\right) = \sqrt{3}$ **[M1]**

Isolates x: $x = \dfrac{2\sqrt{3}}{\sqrt{3}+1}$ **[M1]**

Rationalises denominator: $\dfrac{2\sqrt{3}(\sqrt{3}-1)}{(\sqrt{3}+1)(\sqrt{3}-1)} = \dfrac{6-2\sqrt{3}}{2} = 3 - \sqrt{3}$ **[A1]**

[Total Marks: 7]

1295 **(a)** $\cos 30° = \dfrac{AD}{12}$ **[M1]**

$AD = 12 \times \dfrac{\sqrt{3}}{2} = 6\sqrt{3}$ with no errors seen **[A1]**

(b) Uses Area $= AB \times AD = 60\sqrt{3}$ to find $AB = 10$ **[B1]**

$DE = 12\sin 30° = 6$ **[M1]**

$CE = CD - DE = 10 - 6 = 4$ **[A1]**

(c) Identifies $BC = 6\sqrt{3}$ and $CE = 4$ **[M1]**

$\tan(\angle EBC) = \dfrac{CE}{BC} = \dfrac{4}{6\sqrt{3}}$ **[M1]**

$\dfrac{2}{3\sqrt{3}} = \dfrac{2\sqrt{3}}{9}$ **[A1]**

[Total Marks: 8]

1296 **(a)** Uses $h = 4\sqrt{3}\sin 60°$ with $\sin 60° = \dfrac{\sqrt{3}}{2}$ **[M1]**

6 **[A1]**

(b) Finds left base segment: $4\sqrt{3}\cos 60° = 4\sqrt{3} \times \dfrac{1}{2} = 2\sqrt{3}$ **[B1]**

Finds right base segment: $\dfrac{6}{\tan 45°} = 6$, giving total base $SR = 2\sqrt{3} + 6 + 6 = 12 + 2\sqrt{3}$ **[B1]**

Substitutes into trapezium area formula: $\frac{1}{2}(6+12+2\sqrt{3})\times 6$ **[M1]**

$54+6\sqrt{3}$ **[A1]**

(c) Finds slanted side QR: $\frac{6}{\sin 45^\circ}=\frac{6}{\frac{\sqrt{2}}{2}}=6\sqrt{2}$ **[M1]**

Sums all four sides: $6+6\sqrt{2}+(12+2\sqrt{3})+4\sqrt{3}=18+6\sqrt{3}+6\sqrt{2}$ **[A1]**

[Total Marks: 8]

1297 **(a)** Uses cosine rule: $7^2=x^2+(x+3)^2-2(x)(x+3)\cos 60^\circ$ **[M1]**

Substitutes $\cos 60^\circ=0.5$: $49=x^2+x^2+6x+9-x(x+3)$ **[M1]**

Expands and simplifies: $49=2x^2+6x+9-x^2-3x$ **[M1]**

$x^2+3x-40=0$ **[A1]**

(b) $(x+8)(x-5)=0$ **[M1]**

$x=5$ (rejects $x=-8$) **[A1]**

Perimeter $=5+(5+3)+7=20$ **[A1]**

(c) Uses sine area formula: Area $=\frac{1}{2}\times 5\times 8\times\sin 60^\circ$ **[M1]**

Substitutes $\sin 60^\circ=\frac{\sqrt{3}}{2}$ **[M1]**

$10\sqrt{3}$ **[A1]**

[Total Marks: 10]

1298 **(a)** Angle $OAP=90^\circ$ (tangent to radius $=90^\circ$) and angle $APO=30^\circ$ **[B1]**

$\tan 30^\circ=\frac{6}{PA}$ giving $\frac{1}{\sqrt{3}}=\frac{6}{PA}$ **[M1]**

$6\sqrt{3}$ **[A1]**

(b) Area of quadrilateral $OAPB=2\times\frac{1}{2}\times 6\times 6\sqrt{3}=36\sqrt{3}\ cm^2$ **[M1]**

Angle $AOB=360^\circ-90^\circ-90^\circ-60^\circ=120^\circ$ **[B1]**

Area of minor sector $OAB=\frac{120}{360}\times\pi\times 6^2=12\pi\ cm^2$ **[M1]**

$36\sqrt{3}-12\pi$ **[A1]**

(c) **(i)** Half-chord $=6\sin 60^\circ=6\times\frac{\sqrt{3}}{2}=3\sqrt{3}$ cm **[M1]**

$6\sqrt{3}$ **[A1]**

(ii) $OP=\frac{6}{\sin 30^\circ}=12$ cm and distance from O to $AB=6\cos 60^\circ=3$ cm **[M1]**

$12-3=9$ **[A1]**

[Total Marks: 11]

1299 $\sin x = \frac{1}{\sqrt{2}}$ **[M1]**

$x = 45°$ **[B1]**

$x = 135°$ **[B1]**

[Total Marks: 3]

1300 **(a)** $(180, -1)$ **[B1]**

(b) $60°$ **[B1]**

$300°$ **[B1]**

[Total Marks: 3]

1301 **(a)** Smooth sine wave shape passing through $(0, 0)$, $(180, 0)$ and $(360, 0)$ **[B1]**

Maximum turning point at $(90, 1)$ and minimum turning point at $(270, -1)$ **[B1]**

(b) $x = 30$ **[B1]**

$x = 150$ **[B1]**

[Total Marks: 4]

1302 **(a)** $\cos x = 0.5$ **[M1]**

$x = 60°$ **[B1]**

$x = 300°$ **[B1]**

(b) 0 (since $\sin x = \frac{5}{3} > 1$) **[B1]**

[Total Marks: 4]

1303 **(a)** Smooth cosine wave starting at $(0°, 1)$, passing through $(90°, 0)$, reaching minimum at $(180°, -1)$ **[B1]**

Passing through $(270°, 0)$ and ending at $(360°, 1)$ with correct turning point symmetry **[B1]**

(b) Rearranges to $\cos x = -\frac{1}{2}$ **[M1]**

$120°$ **[A1]**

$240°$ **[A1]**

[Total Marks: 5]

1304 **(a)** Smooth sine curve passing through $(0°, 0)$, $(180°, 0)$, and $(360°, 0)$ **[B1]**

Maximum point at $(90°, 1)$ and minimum point at $(270°, -1)$ **[B1]**

(b) Maximum $= (90°, 1)$ **[B1]**

Minimum $= (270°, -1)$ **[B1]**

(c) **(i)** $60°$ **[B1]**

$120°$ **[B1]**

(ii) $\cos x = -\frac{1}{2}$ **[M1]**

$120°$ **[A1]**

$240°$ **[A1]**

[Total Marks: 9]

1305 **(a)** Smooth U-shaped cosine curve starting at $(0, 1)$ and ending at $(4, 1)$ representing $360°$ **[B1]**

Curve passes through $(1, 0)$, $(3, 0)$ and has a minimum at $(2, -1)$ **[B1]**

(b) $(180°, -1)$ **[B1]**

(c) $\cos x = -0.5$ (or $-\frac{1}{2}$) **[M1]**

$x = 120°$ **[B1]**

$x = 240°$ **[B1]**

[Total Marks: 6]

1306 **(a)** Smooth sine curve passing through $(0, 0), (90°, 1), (180°, 0), (270°, -1), (360°, 0)$ **[B1]**

Smooth cosine curve passing through $(0, 1), (90°, 0), (180°, -1), (270°, 0), (360°, 1)$ **[B1]**

Intersection points shown approximately accurately at $x = 45°$ and $x = 225°$ **[B1]**

(b) $\tan x = 1$ **[M1]**

$x = 45°, 225°$ (Both required) **[A1]**

(c) Identifies the region between the two intersection points **[M1]**

$45° < x < 225°$ **[A1]**

[Total Marks: 7]

1307 **(a)** $\sin x = -\frac{\sqrt{3}}{2}$ **[M1]**

$240°$ **[A1]**

$300°$ **[A1]**

(b) $\cos^2 x = \frac{3}{4}$ **[M1]**

$\cos x = \pm\frac{\sqrt{3}}{2}$ **[M1]**

$30°$ and $150°$ **[A1]**

$210°$ and $330°$ **[A1]**

(Award maximum of 3 marks if $\pm$ is omitted, resulting in only two solutions)

[Total Marks: 7]

1308 **(a)** Dashed vertical asymptotes drawn at $x = 90°$ and $x = 270°$ **[B1]**

Correct increasing branches in $0° \leq x < 90°$ and $270° < x \leq 360°$ passing through $(0°, 0)$ and $(360°, 0)$ **[B1]**

Correct increasing branch in $90° < x < 270°$ passing through $(180°, 0)$ **[B1]**

(b) $90°$ **[B1]**

$270°$ **[B1]**

(c) **(i)** $\tan x = -\dfrac{1}{\sqrt{3}}$ **[M1]**

$150°$ **[A1]**

$330°$ **[A1]**

(ii) $(2\sin x + 1)(\sin x - 1) = 0$ **[M1]**

$\sin x = -\dfrac{1}{2}$ or $\sin x = 1$ **[A1]**

$x = 90°$ **[B1]**

$x = 210°$ and $x = 330°$ **[A1]**

[Total Marks: 12]

1309 **(a)** Sine wave passing through $(0, 0)$, $(180, 0)$, and $(360, 0)$ **[B1]**

Smooth curve with a maximum at $(90, 1)$ and a minimum at $(270, -1)$ **[B1]**

(b) $\arcsin(0.7)$ OR $44.42\ldots$ **[M1]**

$44.4°$ **[A1]**

$135.6°$ **[A1]**

[Total Marks: 5]

1310 **(a)** $\sin x = -0.75$ **[M1]**

Reference angle $= 48.6°$ (or $48.59...°$) **[M1]**

$x = 228.6°, 311.4°$ (Both required for the final mark) **[A1]**

(b) x-coordinate is $270°$ **[B1]**

y-coordinate is -1 **[B1]**

[Total Marks: 5]

1311 **(a)** Correct basic shape with multiple branches passing through $(0, 0)$, $(180, 0)$, and $(360, 0)$ **[B1]**

Asymptotes at $x = 90$ and $x = 270$ clearly indicated or graph showing asymptotic behaviour **[B1]**

(b) $\tan x = -\frac{5}{3}$ or $-1.66\ldots$ **[M1]**

$121.0°$ **[A1]**

$301.0°$ **[A1]**

[Total Marks: 5]

1312 **(a)** $\cos x = -0.25$ **[M1]**

$x = 104.5$ **[A1]**

$x = 360 - 104.47... = 255.5$ **[A1]**

(b) $\tan x = 2.5$ **[M1]**

$x = 68.2$ **[A1]**

$x = 68.19... + 180 = 248.2$ **[A1]**

[Total Marks: 6]

1313 **(a)** Finds acute reference angle: $\sin^{-1}(0.64) = 39.79...°$ **[M1]**

$219.8°$ **[A1]**

$320.2°$ **[A1]**

(b) Rearranges to $\tan x = -\frac{5}{3}$ and finds reference angle $\tan^{-1}\left(\frac{5}{3}\right) = 59.036...°$ **[M1]**

$121.0°$ (accept $121°$) **[A1]**

$301.0°$ (accept $301°$) **[A1]**

[Total Marks: 6]

1314 **(a)** $\cos x = -\frac{1}{3}$ **[M1]**

$x = 109.5°$ (accept $109.47\ldots$) **[A1]**

$360° - 109.47°$ **[M1]**

$x = 250.5°$ (accept $250.52\ldots$) **[A1]**

(b) Identifies that intersection with x-axis means $y = 0$, so x-coordinates are the solutions from part (a) **[M1]**

Distance $= 250.53° - 109.47°$ **[M1]**

141.1 (accept 141.0 to 141.1) **[A1]**

[Total Marks: 7]

1315 **(a)** $\cos^{-1}(-0.35) = 110.487\ldots°$ **[M1]**

x-coordinate of $A = 110.5°$ **[A1]**

x-coordinate of $B = 360° - 110.487\ldots° = 249.5°$ **[A1]**

(b) $\sin x = -0.6$ **[M1]**

$216.9°$ **[A1]**

$323.1°$ **[A1]**

(c) $\tan x = \dfrac{7}{3}$ **[M1]**

$66.8°$ **[A1]**

$246.8°$ **[A1]**

[Total Marks: 9]

1316 **(a)** $\sin 30° = \dfrac{1}{2}$ leading to $a\left(\dfrac{1}{2}\right) - b = 0$ or $\dfrac{b}{a} = \dfrac{1}{2}$ **[M1]**

$a = 2$ and $b = 1$ **[A1]**

(b) 150 **[B1]**

(c) $k = \cos(242°) = -0.469$ **[B1]**

Uses symmetry of cosine curve: $360° - 242°$ **[M1]**

$x = 118°$ **[A1]**

[Total Marks: 6]

1317 **(a)** $\tan x = -2.8$ **[M1]**

$x = 109.7°$ (accept $109.6°$ to $109.7°$) **[B1]**

$x = 289.7°$ (accept $289.6°$ to $289.7°$) **[B1]**

(b) Factorises to $\sin x(2\sin x - 1) = 0$ **[M1]**

Identifies $\sin x = 0$ or $\sin x = 0.5$ (or $\dfrac{1}{2}$) **[M1]**

$x = 0°, 180°, 360°$ (all three correct for the mark) **[A1]**

$x = 30°, 150°$ (both correct for the mark) **[A1]**

[Total Marks: 7]

1318 $5\sin x = 2\dfrac{\sin x}{\cos x}$ OR $5\sin x\cos x = 2\sin x$ **[M1]**

$\sin x(5\cos x - 2) = 0$ [M2] (Award [M1] for correctly gathering terms on one side: $5\sin x\cos x - 2\sin x = 0$)

$\sin x = 0$ **[B1]**

$0°, 180°, 360°$ [A1] (Must have all three solutions for the mark)

$\cos x = 0.4$ **[B1]**

$66.4°$ (accept $66.42\ldots$) **[A1]**

$293.6°$ (accept $293.57\ldots$) **[A1]**

[Total Marks: 7]

1319 **(a)** Divides by $\cos x$ to obtain $\tan x = \frac{2}{5}$ (or 0.4) **[M1]**

$21.8°$ **[A1]**

$201.8°$ **[A1]**

(b) Factorises quadratic in $\sin x$: $(2\sin x - 1)(\sin x + 1) = 0$ **[M1]**

Solves linear factors: $\sin x = \frac{1}{2}$ and $\sin x = -1$ **[M1]**

$30°$ **[A1]**

$150°$ **[A1]**

$270°$ **[A1]**

[Total Marks: 8]

1320 **(a)** $\frac{1}{2} \times 8 \times 5 \times \sin 60°$ **[M1]**

$10\sqrt{3}$ **[A1]**

(b) $QR^2 = 8^2 + 5^2 - 2(8)(5)\cos 60°$ **[M1]**

$QR^2 = 64 + 25 - 80\left(\frac{1}{2}\right)$ **[M1]**

7 **[A1]**

[Total Marks: 5]

1321 **(a)** $7^2 = 5^2 + x^2 - 2(5)(x)\cos 60°$ **[M1]**

$\cos 60° = \frac{1}{2}$ or 0.5 seen **[B1]**

$49 = 25 + x^2 - 5x \implies x^2 - 5x - 24 = 0$ (with no errors seen) **[A1]**

(b) $(x - 8)(x + 3) = 0$ **[M1]**

$x = 8$ **[A1]**

(c) Area $= \frac{1}{2} \times 5 \times 8 \times \sin 60°$ **[M1]**

$10\sqrt{3}$ **[A1]**

[Total Marks: 7]

1322 **(a)** $x^2 = (x - 6)^2 + (x - 5)^2 - 2(x - 6)(x - 5)\cos 120°$ **[M1]**

Using $\cos 120° = -\frac{1}{2}$: $x^2 = (x^2 - 12x + 36) + (x^2 - 10x + 25) + (x^2 - 11x + 30)$ **[M1]**

$x^2 = 3x^2 - 33x + 91$ simplifying to $2x^2 - 33x + 91 = 0$ **[A1]**

(b) $(2x - 7)(x - 13) = 0$ **[M1]**

$x = 13$ (rejecting $x = \frac{7}{2}$ as side lengths must be positive) **[A1]**

$XY = 7$ cm, $YZ = 8$ cm, $XZ = 13$ cm **[A1]**

(c) **(i)** Area $= \frac{1}{2} \times 7 \times 8 \times \sin 120° = 28 \times \frac{\sqrt{3}}{2}$ **[M1]**

$14\sqrt{3}$ **[A1]**

(ii) $\frac{1}{2} \times 13 \times d = 14\sqrt{3}$ **[M1]**

$\frac{28\sqrt{3}}{13}$ **[A1]**

(d) $\frac{\sin(\angle YXZ)}{8} = \frac{\sin 120°}{13}$ giving $\sin(\angle YXZ) = \frac{8 \times \frac{\sqrt{3}}{2}}{13}$ **[M1]**

$\frac{4\sqrt{3}}{13}$ **[A1]**

[Total Marks: 12]

1323 **(a)** $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos 72°$ **[M1]**

$BC^2 = 208 - 59.33\ldots$ OR $148.66\ldots$ **[M1]**

12.2 (accept $12.19\ldots$) **[A1]**

(b) Area $= \frac{1}{2}(8)(12)\sin 72°$ **[M1]**

45.7 (accept $45.65\ldots$) **[A1]**

[Total Marks: 5]

1324 **(a)** $0.5 \times 8 \times 12 \times \sin(42°)$ **[M1]**

32.1 (or 32.11 to 32.12) **[A1]**

(b) $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos(42°)$ **[M1]**

$BC^2 = 65.3...$ **[M1]**

8.08 (or 8.081 to 8.082) **[A1]**

[Total Marks: 5]

1325 **(a)** $\frac{1}{2} \times 8 \times 12 \times \sin 40°$ **[M1]**

30.9 (accept $30.85\ldots$) **[A1]**

(b) $8^2 + 12^2 - 2 \times 8 \times 12 \times \cos 40°$ **[M1]**

$60.9\ldots$ **[A1]**

7.80 or 7.81 **[A1]**

[Total Marks: 5]

1326 **(a)** $0.5 \times 8.5 \times 11.2 \times \sin(54°)$ **[M1]**

38.5 (38.50 to 38.51) **[A1]**

(b) $BC^2 = 8.5^2 + 11.2^2 - 2(8.5)(11.2)\cos(54°)$ [M2]
(or [M1] for correct cosine rule formula with one substitution error)
$BC^2 = 85.77...$ **[A1]**
9.26 (9.261...) **[A1]**

[Total Marks: 5]

1327 **(a)** $\frac{1}{2} \times 8 \times 12 \times \sin(50°)$ **[M1]**
36.77... so 36.8 **[A1]**

(b) $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos(50°)$ **[M1]**
$BC^2 = 208 - 123.415... = 84.58...$ **[M1]**
9.197... so 9.20 **[A1]**

[Total Marks: 5]

1328 **(a)** $\frac{1}{2} \times 7.2 \times 9.5 \times \sin 64°$ **[M1]**
30.7 or 30.73... **[A1]**

(b) $7.2^2 + 9.5^2 - 2(7.2)(9.5)\cos 64°$ **[M2]**
(or $M1$ for correct implicit cosine rule equation)
9.06 or 9.062... **[A1]**

[Total Marks: 5]

1329 **(a)** Uses Area $= \frac{1}{2} \times 8.5 \times 12.2 \times \sin 53°$ **[M1]**
41.4 (or 41.40 to 41.41) **[A1]**

(b) Uses cosine rule: $QR^2 = 8.5^2 + 12.2^2 - 2 \times 8.5 \times 12.2 \times \cos 53°$ **[M1]**
$QR^2 = 72.25 + 148.84 - 207.4 \times \cos 53° = 96.27...$ **[A1]**
9.81 **[A1]**

[Total Marks: 5]

1330 **(a)** Uses cosine rule: $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos 65°$ **[M1]**
$BC^2 = 126.85...$ **[M1]**
11.3 **[A1]**

(b) Uses area formula: $\frac{1}{2} \times 8 \times 12 \times \sin 65°$ **[M1]**
43.5 **[A1]**

[Total Marks: 5]

1331 **(a)** $\frac{1}{2} \times 8 \times 11 \times \sin(65°)$ **[M1]**

39.9 **[A1]**

(b) $BC^2 = 8^2 + 11^2 - 2(8)(11)\cos(65°)$ **[M2]**

($M1$ for correct substitution into implicit cosine rule)

10.5 **[A1]**

[Total Marks: 5]

1332 **(a)** $\frac{BC}{\sin 45°} = \frac{12}{\sin 60°}$ **[M1]**

$BC = \frac{12 \sin 45°}{\sin 60°}$ **[M1]**

9.80 **[A1]**

(b) Angle $ACB = 180° - (45° + 60°) = 75°$ **[B1]**

Area $= \frac{1}{2} \times 12 \times 9.798 \times \sin 75°$ **[M1]**

56.8 **[A1]**

[Total Marks: 6]

1333 **(a)** $\frac{BC}{\sin 40°} = \frac{12}{\sin 60°}$ **[M1]**

$BC = \frac{12 \sin 40°}{\sin 60°}$ **[M1]**

8.91 (accept $8.907\ldots$) **[A1]**

(b) Angle $BCA = 180° - (40° + 60°) = 80°$ **[B1]**

$\frac{1}{2} \times 12 \times 8.91 \times \sin 80°$ **[M1]**

52.6 (accept $52.62\ldots$ to $52.65\ldots$) **[A1]**

[Total Marks: 6]

1334 **(a)** $AC^2 = 9.4^2 + 12.6^2 - 2 \times 9.4 \times 12.6 \times \cos 118°$ **[M1]**

$AC^2 = 88.36 + 158.76 - (-111.208\ldots) = 358.328\ldots$ **[M1]**

18.9 **[A1]**

(b) $\frac{\sin \angle BAC}{12.6} = \frac{\sin 118°}{18.929\ldots}$ **[M1]**

$\sin \angle BAC = \frac{12.6 \sin 118°}{18.929\ldots} = 0.5877\ldots$ **[M1]**

36.0 **[A1]**

(c) Area $= \frac{1}{2} \times 9.4 \times 12.6 \times \sin 118°$ $(= 52.288\ldots)$ **[M1]**

52.3 **[A1]**

[Total Marks: 8]

1335 **(a)** $\cos Q = \dfrac{7^2 + 8^2 - 12^2}{2 \times 7 \times 8}$ [M2] (M1 for correct implicit cosine rule equation e.g. $12^2 = 7^2 + 8^2 - 2(7)(8)\cos Q$)

106.1 (accept 106.07...) **[A1]**

(b) $\dfrac{1}{2} \times 7 \times 8 \times \sin 106.1°$ **[M1]**

26.9 (accept 26.89...) **[A1]**

[Total Marks: 5]

1336 **(a)** $\dfrac{\sin P}{12} = \dfrac{\sin 35°}{8}$ **[M1]**

$\sin P = \dfrac{12 \sin 35°}{8} = 0.8603...$ **[M1]**

Acute angle $= 59.35...°$ **[A1]**

Obtuse angle $P = 180° - 59.35...° = 120.64...°$ so $120.6°$ **[A1]**

(b) Angle $R = 180° - 35° - 120.64...° = 24.36...°$ **[M1]**

Area $= \dfrac{1}{2} \times 8 \times 12 \times \sin(24.36...°) = 19.79...$ so 19.8 **[A1]**

[Total Marks: 6]

1337 **(a)** $\dfrac{\sin R}{15} = \dfrac{\sin 42°}{12}$ **[M1]**

$\sin R = \dfrac{15 \sin 42°}{12}$ (evaluates to 0.836...) **[M1]**

123.2 (accept 123.2 to 123.3) **[A1]**

(b) Angle $P = 180° - 42° - 123.24° = 14.76°...$ **[M1]**

$\dfrac{QR}{\sin 14.76°} = \dfrac{12}{\sin 42°}$ (or equivalent application of sine/cosine rule) **[M1]**

4.57 (accept 4.56 to 4.58) **[A1]**

[Total Marks: 6]

1338 **(a)** $\dfrac{\sin F}{15} = \dfrac{\sin 110°}{20}$ **[M1]**

$\sin F = \dfrac{15 \sin 110°}{20}$ OR 0.7047... **[M1]**

44.8 (accept 44.80 to 44.81) **[A1]**

(b) Angle $D = 180° - 110° - 44.81° = 25.19°$ **[B1]**

$\dfrac{EF}{\sin 25.19°} = \dfrac{20}{\sin 110°}$ OR $EF^2 = 15^2 + 20^2 - 2(15)(20)\cos 25.19°$ **[M1]**

9.06 (accept 9.05 to 9.06) **[A1]**

[Total Marks: 6]

1339 **(a)** Uses sine rule: $\dfrac{\sin(\text{angle } LNM)}{15} = \dfrac{\sin 45^\circ}{12}$ **[M1]**

$\sin(\text{angle } LNM) = 0.8838...$ leading to acute angle $62.11...^\circ$ **[M1]**

117.9 (or 117.88 to 117.89) **[A1]**

(b) Finds angle $LMN = 180^\circ - 45^\circ - 117.89^\circ = 17.11^\circ$ **[B1]**

Uses Area $= \dfrac{1}{2} \times 15 \times 12 \times \sin 17.11^\circ$ **[M1]**

26.5 (or 26.47 to 26.49) **[A1]**

[Total Marks: 6]

1340 **(a)** $\dfrac{\sin R}{14} = \dfrac{\sin 35^\circ}{10}$ **[M1]**

$\sin^{-1}\left(\dfrac{14\sin 35^\circ}{10}\right)$ leading to 53.4° $(53.41...)$ **[M1]**

126.6 (126.58...) **[A1]**

(b) Angle $PQR = 180^\circ - 35^\circ - 126.58^\circ = 18.4^\circ$ **[M1]**

Area $= \dfrac{1}{2} \times 14 \times 10 \times \sin(18.4^\circ)$ **[M1]**

22.1 (22.11...) **[A1]**

[Total Marks: 6]

1341 **(a)** $\dfrac{\sin Q}{14} = \dfrac{\sin 32^\circ}{9}$ **[M1]**

$\sin Q = \dfrac{14\sin 32^\circ}{9}$ or 0.824... **[M1]**

Acute angle $= 55.5^\circ$ $(55.51...)$ **[A1]**

$180 - 55.5... = 124.5^\circ$ (124.48 to 124.5) **[A1]**

(b) Angle $PRQ = 180 - 32 - 124.48 = 23.52^\circ$ **[B1]**

$0.5 \times 14 \times 9 \times \sin(23.52^\circ)$ **[M1]**

25.1 (25.12 to 25.14) **[A1]**

[Total Marks: 7]

1342 **(a)** $\cos Q = \dfrac{15^2 + 12^2 - 18^2}{2 \times 15 \times 12}$ **[M1]**

$\cos Q = \dfrac{45}{360}$ OR 0.125 **[M1]**

$Q = \cos^{-1}(0.125) = 82.819...^\circ$ **[A1]**

(b) $0.5 \times 15 \times 12 \times \sin(82.8°)$ **[M1]**

89.3 (or 89.28 to 89.30) **[A1]**

(c) Area $= 0.5 \times 15 \times h = 89.3$ OR $h = 12 \times \sin(82.8°)$ **[M1]**

11.9 (or 11.90 to 11.91) **[A1]**

[Total Marks: 7]

1343 **(a)** $\dfrac{\sin C}{12} = \dfrac{\sin 35°}{9}$ **[M1]**

$\sin C = \dfrac{12 \sin 35°}{9}$ $(= 0.764...)$ **[M1]**

49.9° (or 49.88...) **[A1]**

130.1° (or 130.11...) **[A1]**

(b) Angle $ABC = 180 - (35 + 130.1) = 14.9°$ **[M1]**

Area $= \dfrac{1}{2} \times 12 \times 9 \times \sin(14.9°)$ **[M1]**

13.9 (accept 13.87... to 13.89...) **[A1]**

[Total Marks: 7]

1344 **(a)** Uses sine rule: $\dfrac{\sin R}{15.2} = \dfrac{\sin 72°}{18.5}$ **[M1]**

$\sin R = 0.781...$ **[M1]**

51.4 **[A1]**

(b) Angle $P = 180° - 72° - 51.38° = 56.62°$ **[B1]**

$\dfrac{QR}{\sin 56.62°} = \dfrac{18.5}{\sin 72°}$ **[M1]**

16.2 **[A1]**

(c) Shortest distance $= 15.2 \times \sin 72°$ **[M1]**

14.5 **[A1]**

[Total Marks: 8]

1345 **(a)** $BD^2 = 8^2 + 10^2 - 2(8)(10)\cos 110°$ **[M1]**

$BD^2 = 164 - 160(-0.342) = 218.72$ **[M1]**

14.8 **[A1]**

(b) $\cos C = \dfrac{14^2 + 15^2 - 14.79^2}{2(14)(15)}$ **[M1]**

$\cos C = \dfrac{202.26}{420} = 0.4816$ **[M1]**

61.2 **[A1]**

(c) Area $ABD = \dfrac{1}{2}(8)(10)\sin 110° = 37.59$ **[M1]**

Area $BCD = \frac{1}{2}(14)(15)\sin 61.2° = 92.01$ **[M1]**

130 (or 129.6) **[A1]**

[Total Marks: 9]

1346 **(a)** $AC^2 = 12.4^2 + 9.8^2 - 2(12.4)(9.8)\cos 105°$ **[M1]**

$AC^2 = 153.76 + 96.04 - (-62.90)$ or 312.7 **[M1]**

17.7 (accept 17.68 . . .) **[A1]**

(b) $\frac{CD}{\sin 41°} = \frac{17.68}{\sin 68°}$ **[M1]**

$CD = \frac{17.68 \times \sin 41°}{\sin 68°}$ **[M1]**

12.5 (accept 12.51 . . .) **[A1]**

(c) Area $ABC = \frac{1}{2} \times 12.4 \times 9.8 \times \sin 105° = 58.68 \ldots$ **[B1]**

Area $ACD = \frac{1}{2} \times 17.68 \times 12.51 \times \sin(180° - 41° - 68°) = 104.6 \ldots$ (accept correct alternative using $AD = 18.0$) **[M1]**

163 (accept 163.2 to 163.3) **[A1]**

[Total Marks: 9]

1347 **(a)** $\frac{\sin \angle PRQ}{11} = \frac{\sin 42°}{8}$ **[M1]**

$\sin \angle PRQ = \frac{11 \sin 42°}{8} = 0.92005 \ldots$ **[M1]**

Acute angle $= 66.9°$ **[A1]**

Obtuse angle $= 180° - 66.934 \ldots° = 113.1°$ **[A1]**

(b) Angle $QPR_1 = 180° - 42° - 66.934 \ldots° = 71.1°$ **[B1]**

Angle $QPR_2 = 180° - 42° - 113.066 \ldots° = 24.9°$ **[B1]**

(c) Area of triangle $PQR_1 = \frac{1}{2} \times 11 \times 8 \times \sin 71.066 \ldots° = 41.619 \ldots \ cm^2$ **[M1]**

Area of triangle $PQR_2 = \frac{1}{2} \times 11 \times 8 \times \sin 24.934 \ldots° = 18.549 \ldots \ cm^2$ **[M1]**

Difference $= 41.619 \cdots - 18.549 \cdots = 23.070 \ldots$ **[M1]**

23.1 **[A1]**

[Total Marks: 10]

1348 **(a)** $\frac{\sin C}{15} = \frac{\sin 48°}{12}$ **[M1]**

$\sin C = 0.9289 \ldots$ **[A1]**

68.3 and 111.7 **[A1]**

(b) Calculates either possible angle B: $180° - 48° - 68.27° = 63.73°$ OR $180° - 48° - 111.73° = 20.27°$ **[M1]**

Area $1 = \frac{1}{2} \times 15 \times 12 \times \sin 63.73° = 80.7\ldots$ **[M1]**

Area $2 = \frac{1}{2} \times 15 \times 12 \times \sin 20.27° = 31.2\ldots$ **[M1]**

49.5 (accept 49.5 to 49.6) **[A1]**

[Total Marks: 7]

1349 (a) $\frac{\sin C}{10.5} = \frac{\sin 35°}{8.2}$ **[M1]**

$\sin C = \frac{10.5 \times \sin 35°}{8.2}$ OR 0.734... **[M1]**

$C = 180° - 47.3° = 132.7°$ (or 132.73 to 132.74) **[A1]**

(b) Angle $B = 180° - 35° - 132.7° = 12.3°$ **[M1]**

$\frac{AC}{\sin 12.3°} = \frac{8.2}{\sin 35°}$ OR $AC^2 = 10.5^2 + 8.2^2 - 2(10.5)(8.2)\cos 12.3°$ **[M1]**

3.04 (or 3.035 to 3.045) **[A1]**

(c) $0.5 \times 3.04 \times 6.4 \times \sin(ACD) = 8.5$ **[M1]**

$\sin(ACD) = \frac{8.5}{9.728}$ OR Angle $ACD = \sin^{-1}\left(\frac{8.5}{9.728}\right) = 60.9°$ (or 60.8... to 61.1°) **[A1]**

[Total Marks: 8]

1350 (a) $\frac{\sin F}{10.5} = \frac{\sin 42°}{8.2}$ **[M1]**

$\sin F = 0.8568\ldots$ yielding acute angle $58.96\ldots°$ **[A1]**

$180° - 58.96° = 121.0$ (accept $121.04\ldots$) **[A1]**

(b) Angle $EDF = 180° - 42° - 121.04° = 16.96°$ **[B1]**

$\frac{EF}{\sin 16.96°} = \frac{8.2}{\sin 42°}$ **[M1]**

3.57 (accept $3.56\ldots$ to $3.58\ldots$) **[A1]**

(c) Shortest distance $= 10.5 \times \sin 42°$ **[M1]**

7.03 (accept $7.025\ldots$ to $7.027\ldots$) **[A1]**

[Total Marks: 8]

1351 (a) $BD^2 = 6^2 + 8^2 - 2(6)(8)\cos(110°)$ **[M1]**

$BD^2 = 100 - 96(-0.342...) = 132.83...$ **[M1]**

$BD = 11.52...$ so 11.5 **[A1]**

(b) $\cos C = \frac{9^2 + 5^2 - (11.525...)^2}{2(9)(5)}$ **[M1]**

$\cos C = \dfrac{106 - 132.83...}{90} = -0.2981...$ **[M1]**

$C = 107.34...^\circ$ so 107.3° **[A1]**

(c) Area of triangle $ABD = \dfrac{1}{2} \times 6 \times 8 \times \sin(110^\circ) = 22.55...$ **[M1]**

Area of triangle $BCD = \dfrac{1}{2} \times 9 \times 5 \times \sin(107.34...^\circ) = 21.47...$ **[M1]**

Total Area $= 22.55... + 21.47... = 44.02...$ so 44.0 **[A1]**

[Total Marks: 9]

1352 **(a)** $7^2 + 9^2 - 2 \times 7 \times 9 \times \cos 110^\circ$ **[M1]**

$173.09\ldots$ **[A1]**

13.2 (accept 13.15 to 13.2) **[A1]**

(b) $13.156^2 = 10^2 + 14^2 - 2 \times 10 \times 14 \times \cos(\angle BCD)$ **[M1]**

$\cos(\angle BCD) = \dfrac{10^2 + 14^2 - 173.09}{2 \times 10 \times 14}$ **[M1]**

64.0 (accept $63.9\ldots$ to 64.0) **[A1]**

(c) $\dfrac{1}{2} \times 7 \times 9 \times \sin 110^\circ$ (evaluates to $29.6\ldots$) **[M1]**

$\dfrac{1}{2} \times 10 \times 14 \times \sin 64.0^\circ$ (evaluates to $62.9\ldots$) **[M1]**

92.5 (accept 92.4 to 92.6) **[A1]**

[Total Marks: 9]

1353 **(a)** $AC^2 = 7.4^2 + 9.1^2 - 2(7.4)(9.1)\cos(112^\circ)$ [M2]
(or [M1] for correct cosine rule formula with one error)

$AC = 13.7$ (13.71 to 13.72) **[A1]**

(b) $\dfrac{\sin D}{13.712} = \dfrac{\sin 43^\circ}{10.5}$ **[M1]**

$\sin D = \dfrac{13.712 \sin 43^\circ}{10.5}$ or 0.890... **[M1]**

63.0 (62.95 to 63.0) **[A1]**

(c) Angle $ACD = 180 - 43 - 62.95 = 74.05^\circ$ **[M1]**

Shortest distance $= 10.5 \times \sin(74.05^\circ)$ **[M1]**

10.1 (10.09 to 10.1) **[A1]**

[Total Marks: 9]

1354 **(a)** $BD^2 = 7.5^2 + 9.2^2 - 2(7.5)(9.2)\cos(105^\circ)$ **[M2]**

($M1$ for correct substitution into implicit cosine rule)

13.3 (13.28...) **[A1]**

(b) $\cos C = \dfrac{13^2 + 10.5^2 - 13.289^2}{2(13)(10.5)}$ **[M2]**

($M1$ for correct substitution into implicit cosine rule)

67.9 (67.91...) **[A1]**

(c) Area $ABD = \dfrac{1}{2} \times 7.5 \times 9.2 \times \sin(105°)$ (33.32...) **[M1]**

Area $BCD = \dfrac{1}{2} \times 13 \times 10.5 \times \sin(67.9°)$ (63.24...) **[M1]**

96.6 (96.5... to 96.6) **[A1]**

[Total Marks: 9]

1355 **(a)** $PR^2 = 9^2 + 14^2 - 2(9)(14)\cos 105°$ **[M1]**

$PR^2 = 277 - (-65.22\ldots)$ OR $342.22\ldots$ **[M1]**

18.5 (accept 18.49 to 18.50) **[A1]**

(b) Identifies correct use of cosine rule for angle PSR: $\cos S = \dfrac{12^2 + 10^2 - PR^2}{2(12)(10)}$ **[M1]**

$\cos S = \dfrac{144 + 100 - 342.22}{240}$ **[M1]**

$\cos S = -0.409\ldots$ **[M1]**

114.2 (accept 114.1 to 114.2) **[A1]**

(c) Area of triangle $PQR = \dfrac{1}{2}(9)(14)\sin 105°$ OR $60.85\ldots$ **[M1]**

Area of triangle $PRS = \dfrac{1}{2}(10)(12)\sin 114.16°$ OR $54.74\ldots$ **[M1]**

Total Area $= 60.85 + 54.74 = 115.6$ (accept 115.5 to 115.6) **[A1]**

[Total Marks: 10]

1356 **(a)** Uses cosine rule: $BD^2 = 8.5^2 + 11.2^2 - 2(8.5)(11.2)\cos 105°$ **[M1]**

$BD^2 = 246.96...$ **[A1]**

15.7 **[A1]**

(b) Uses sine rule: $\dfrac{\sin C}{15.715} = \dfrac{\sin 38°}{11.0}$ **[M1]**

Acute angle $C = 61.59...°$ **[M1]**

Obtuse angle $BCD = 180° - 61.59...° = 118.4°$ **[A1]**

(c) Area of triangle $ABD = \dfrac{1}{2} \times 8.5 \times 11.2 \times \sin 105°$ $(= 45.97...)$ **[M1]**

Angle $DBC = 180° - 38° - 118.41° = 23.59°$ **[B1]**

Area of triangle $BCD = \dfrac{1}{2} \times 11.0 \times 15.715 \times \sin 23.59°$ $(= 34.59...)$ **[M1]**

80.6 **[A1]**

[Total Marks: 10]

1357 **(a)** Uses cosine rule: $AC^2 = 8^2 + 11^2 - 2 \times 8 \times 11 \times \cos 75°$ **[M1]**

$AC^2 = 64 + 121 - 176 \times 0.2588... = 139.448...$ **[A1]**

11.8 (or 11.80 to 11.81) **[A1]**

(b) Uses sine rule in triangle ADC: $\dfrac{\sin(\text{angle } CAD)}{9} = \dfrac{\sin 68°}{11.81}$ **[M1]**

$\sin(\text{angle } CAD) = \dfrac{9 \times \sin 68°}{11.81} = 0.7066...$ **[M1]**

45.0 (or 44.95 to 44.96) **[A1]**

(c) Calculates area of triangle $ABC = \dfrac{1}{2} \times 8 \times 11 \times \sin 75° = 42.50...$ **[M1]**

Finds angle $ACD = 180° - 68° - 44.96° = 67.04°$ **[M1]**

Calculates area of triangle $ADC = \dfrac{1}{2} \times 11.81 \times 9 \times \sin 67.04° = 48.93...$ **[M1]**

Total Area $= 42.50 + 48.93 = 91.4$ (or 91.42 to 91.44) **[A1]**

[Total Marks: 10]

1358 **(a)** Angle $BAL = 085° - 050° = 35°$ **[B1]**

$\dfrac{\sin(ALB)}{40} = \dfrac{\sin 35°}{35}$ **[M1]**

$\sin(ALB) = \dfrac{40 \sin 35°}{35} = 0.6555\ldots$ **[M1]**

Acute angle $= 40.96°$, Obtuse angle $ALB = 180° - 40.96° = 139.0$ **[A1]**

(b) Angle $ABL = 180° - 35° - 139.04° = 5.96°$ **[B1]**

$\dfrac{AL}{\sin 5.96°} = \dfrac{35}{\sin 35°}$ **[M1]**

6.33 **[A1]**

(c) Bearing of A from $B = 180° + 050° = 230°$ **[M1]**

Bearing of L from $B = 230° -$ Angle $ABL = 230° - 5.96°$ **[M1]**

224.0 **[A1]**

[Total Marks: 10]

1359 **(a)** Uses Pythagoras' theorem: $AC^2 = 8^2 + 6^2$ **[M1]**

$AC = 10$ **[A1]**

(b) Uses 3D Pythagoras' theorem: $AG^2 = 10^2 + 5^2$ (or $8^2 + 6^2 + 5^2$) **[M1]**

$AG = \sqrt{125}$ **[A1]**

$5\sqrt{5}$ **[A1]**

[Total Marks: 5]

1360 **(a)** $BF^2 = 6^2 + 8^2 = 36 + 64 = 100$ **[M1]**

10 **[A1]**

(b) $AF^2 = (10\sqrt{3})^2 + 10^2 = 300 + 100 = 400$ **[M1]**

20 **[A1]**

(c) $\tan\angle AFB = \dfrac{10\sqrt{3}}{10} = \sqrt{3}$ (or $\sin\angle AFB = \dfrac{10\sqrt{3}}{20} = \dfrac{\sqrt{3}}{2}$) **[M1]**

60 **[A1]**

(d) $AC^2 = (10\sqrt{3})^2 + 6^2 = 300 + 36 = 336$ **[M1]**

$AC = \sqrt{336} = 4\sqrt{21}$ cm **[A1]**

Area $= 4\sqrt{21} \times 8 = 32\sqrt{21}$ **[A1]**

[Total Marks: 9]

1361 **(a)** $AC^2 = 8^2 + 6^2$ **[M1]**

$AC = 10$ **[A1]**

(b) $AG^2 = 10^2 + 5^2$ (or $AG^2 = 8^2 + 6^2 + 5^2$) **[M1]**

$AG = \sqrt{125} = 11.18...$ so 11.2 **[A1]**

(c) $\tan\theta = \dfrac{5}{10}$ (or $\sin\theta = \dfrac{5}{\sqrt{125}}$, etc.) **[M1]**

$\theta = 26.56...$ so $26.6°$ **[A1]**

[Total Marks: 6]

1362 **(a)** $AC^2 = 12^2 + 5^2 = 144 + 25 = 169$ **[M1]**

13 **[A1]**

(b) $AG^2 = 13^2 + 8^2 = 169 + 64 = 233$ **[M1]**

15.3 **[A1]**

(c) Angle between AG and base $ABCD$ is angle GAC **[B1]**

$\tan\angle GAC = \dfrac{8}{13}$ **[M1]**

31.6 **[A1]**

[Total Marks: 7]

1363 **(a)** Uses Pythagoras to find half diagonal AM: $AM = \dfrac{1}{2}\sqrt{10^2 + 10^2} = \sqrt{50}$ (or 7.07...) **[M1]**

$VA^2 = 12^2 + (\sqrt{50})^2 = 144 + 50 = 194$ **[M1]**

$VA = \sqrt{194} = 13.92...$ so 13.9 **[A1]**

(b) $\tan(\angle VAM) = \dfrac{12}{\sqrt{50}}$ **[M1]**

$\angle VAM = 59.49...$ so $59.5°$ **[A1]**

[Total Marks: 5]

1364 **(a)** $BD^2 = 14^2 + 8^2 = 196 + 64 = 260$ **[M1]**

16.1 (or 16.12...) **[A1]**

(b) Identifies right-angled triangle FBD with the right angle at B (implied by using tan) **[M1]**

$\tan(\angle FDB) = \dfrac{6}{\sqrt{260}}$ (or $\dfrac{6}{16.12...}$) **[M1]**

20.4 (or 20.41 to 20.42) **[A1]**

(c) Identifies the horizontal distances from A to M as 7 cm and 8 cm, and vertical distance as 6 cm **[M1]**

Uses 3D Pythagoras' theorem: $AM^2 = 7^2 + 8^2 + 6^2$ **[M1]**

12.2 (or 12.20...) **[A1]**

[Total Marks: 8]

1365 **(a)** Uses Pythagoras to find AC: $AC = \sqrt{12^2 + 5^2} = 13$ **[M1]**

$AF^2 = 13^2 + 8^2 = 169 + 64 = 233$ **[M1]**

$AF = \sqrt{233} = 15.26...$ so 15.3 **[A1]**

(b) Identifies required angle as $\angle FAC$ and uses trig, e.g. $\tan(\angle FAC) = \dfrac{8}{13}$ **[M1]**

$\angle FAC = 31.60...$ so $31.6°$ **[A1]**

(c) Recognises the required angle is $\angle FAE$ because FE is perpendicular to plane $ABED$ **[B1]**

Uses trig in triangle FAE (where $\angle FEA = 90°$): $\sin(\angle FAE) = \dfrac{5}{\sqrt{233}}$ OR finds $AE = \sqrt{12^2 + 8^2} = \sqrt{208}$ and uses $\tan(\angle FAE) = \dfrac{5}{\sqrt{208}}$ **[M1]**

$\angle FAE = 19.11...$ so $19.1°$ **[A1]**

[Total Marks: 8]

1366 **(a)** Uses Volume $= \dfrac{1}{3} \times$ base area $\times$ height $\implies 384 = \dfrac{1}{3} \times 12^2 \times VO$ **[M1]**

$48 \times VO = 384 \implies VO = 8$ **[A1]**

(b) Calculates $AO = \dfrac{1}{2}\sqrt{12^2 + 12^2} = 6\sqrt{2}$ (or 8.48...) **[M1]**

Uses Pythagoras' theorem in triangle VOA: $VA^2 = 8^2 + (6\sqrt{2})^2 = 64 + 72 = 136$ **[M1]**

11.7 (or 11.66...) **[A1]**

(c) $\tan(\angle VAO) = \dfrac{8}{6\sqrt{2}}$ (or $\dfrac{8}{8.485}$) **[M1]**

43.3 (or 43.31 to 43.32) **[A1]**

(d) Identifies the distance from O to the midpoint of AB is 6 cm **[B1]**

Let M be the midpoint of AB. $\tan(\angle VMO) = \dfrac{8}{6}$ **[M1]**

53.1 (or 53.13 to 53.14) **[A1]**

[Total Marks: 10]

1367 **(a)** $AC = \sqrt{16^2 + 12^2} = \sqrt{400} = 20$ cm, giving $OA = 10$ cm **[M1]**

$VO^2 = 26^2 - 10^2 = 676 - 100 = 576$ **[M1]**

$VO = \sqrt{576} = 24$ cm **[A1]**

(b) $\tan \angle VAO = \dfrac{24}{10}$ (or $\sin \angle VAO = \dfrac{24}{26}$) **[M1]**

67.4 **[A1]**

(c) **(i)** Angle required is angle VMO, where $OM = \dfrac{16}{2} = 8$ cm **[B1]**

$\tan \angle VMO = \dfrac{24}{8} = 3$ **[M1]**

71.6 **[A1]**

(ii) $VM = \sqrt{24^2 + 8^2} = \sqrt{640}$ $(= 25.298\ldots)$ cm **[M1]**

Shortest distance $d = 8 \sin 71.565\ldots^\circ$ (or $\dfrac{24 \times 8}{\sqrt{640}} = 7.589\ldots$) **[M1]**

7.59 **[A1]**

[Total Marks: 11]

1368 **(a)** Correct triangle B with vertices at $(-3, 3), (-1, 3), (-1, 6)$ **[B2]**

Translation by vector $\begin{pmatrix} -4 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 2 \end{pmatrix}$ **[B1]**

(b) Correct triangle C with vertices at $(-3, 1), (-5, 1), (-5, 4)$ **[B2]**

Reflection in $y = -1$ or correct orientation but wrong position **[B1]**

[Total Marks: 4]

1369 **(a)** Correct translation of Shape A **[B2]**

If incorrect, translation of $\begin{pmatrix} -4 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 2 \end{pmatrix}$ where $k \neq 0$ **[B1]**

(b) Correct reflection of Shape A to form Shape C with vertices at $(1, -3)$, $(3, -3)$, and $(1, -6)$ **[B2]**

If incorrect, reflection in $x = -1$ or $y = 1$ **[B1]**

[Total Marks: 4]

1370 **(a)** Correct triangle with vertices at $(-3, 3), (-1, 3)$ and $(-1, 6)$ **[B2]**

Translation by correct vector but incorrect shape/orientation OR correct size, shape and

orientation but wrong translation **[B1]**

(b) Reflection **[B1]**

in the y-axis OR $x = 0$ **[B1]**

[Total Marks: 4]

1371 **(a)** Vertices of image plotted correctly at $(-1, -4)$, $(1, -4)$ and $(1, -1)$ **[B2]**

If not B2, award B1 for correct size and orientation, or 2 vertices correct **[B1]**

(b) Reflection **[B1]**

in the y-axis (or $x = 0$) **[B1]**

[Total Marks: 4]

1372 **(a)** Vertices of the image at $(3, -2), (1, -2), (2, -1)$ **[B2]**

Correct orientation but wrong position, or translation by $\begin{pmatrix} 4 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -3 \end{pmatrix}$ **[B1]**

(b) Vertices of the image at $(1, -1), (1, -3), (2, -2)$ **[B2]**

Reflection in $y = -x$ or reflection in x-axis or y-axis **[B1]**

(c) Vertices of the image at $(-1, 2), (-5, 2), (-3, 4)$ **[B2]**

Correct size and orientation but wrong position, OR enlargement with SF 2 from a different centre **[B1]**

[Total Marks: 6]

1373 **(a)** Image B drawn correctly at $(-4, 3), (-2, 3), (-2, 6)$ **[B2]**

If incorrect, 1 mark for translation by $\begin{pmatrix} -5 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 2 \end{pmatrix}$ **[M1]**

(b) Image C drawn correctly at $(1, -1), (3, -1), (3, -4)$ **[B2]**

If incorrect, 1 mark for reflection in y-axis or in a line $y = k$ **[M1]**

(c) Image D drawn correctly at $(-1, -1), (-3, -1), (-3, -4)$ **[B2]**

If incorrect, 1 mark for rotation of 180° about the wrong centre, or 90° about $(0, 0)$ **[M1]**

[Total Marks: 6]

1374 **(a)** Translation **[B1]**

By vector $\begin{pmatrix} 6 \\ -4 \end{pmatrix}$ **[B1]**

(b) Correct reflection drawn with vertices at $(-4, -2)$, $(-4, 0)$, and $(-1, 0)$ **[B2]**

Award [B1] if reflected correctly in any horizontal line $y = k$ where $ke1$

(c) Correct rotation drawn with vertices at $(4, 4)$, $(2, 4)$, and $(2, 1)$ **[B3]**

Award [B2] for correct orientation and size in wrong position, or 90° anticlockwise rotation about $(0, 0)$

Award [B1] for any two vertices plotted correctly

[Total Marks: 7]

1375 **(a)** Correct shape with vertices at $(-2, -4), (-6, -4), (-6, -6), (-4, -6), (-4, -8)$ and $(-2, -8)$ **[B2]**

Enlargement, scale factor 2 centre $(0, 0)$ OR correct scale factor and orientation but wrong centre **[B1]**

(b) Correct shape with vertices at $(4, -1), (4, -3), (5, -3), (5, -2), (6, -2)$ and $(6, -1)$ **[B2]**

Rotation of 90° clockwise about the wrong centre OR rotation of 90° anti-clockwise about $(1, -1)$ **[B1]**

[Total Marks: 4]

1376 **(a)** Rotation **[B1]**

90° clockwise OR 270° anti-clockwise **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Correct triangle drawn with vertices at $(0, 2)$, $(1.5, 2)$, and $(1.5, 3)$ **[B2]**

If incorrect, correct size and orientation but wrong position OR correct centre of enlargement but scale factor 2 **[B1]**

[Total Marks: 5]

1377 **(a)** Image B drawn correctly at $(2, -4), (2, -2), (6, -2)$ **[B2]**

If incorrect, 1 mark for reflection in $y = -x$ or correct orientation but wrong position **[M1]**

(b) Image C drawn correctly at $(-4, 2), (-2, 2), (-2, -2)$ **[B2]**

If incorrect, 1 mark for rotation 90° anti-clockwise about $(2, 2)$ or 90° clockwise about the wrong centre **[M1]**

(c) Reflection **[B1]**

in the line $y = 2$ **[B1]**

[Total Marks: 6]

1378 **(a)** Vertices of image plotted correctly at $(-1, 2)$, $(-2, 2)$ and $(-2, 0.5)$ **[B2]**

If not B2, award B1 for enlargement SF $-\frac{1}{2}$ from wrong centre, or correct centre but SF $\frac{1}{2}$ **[B1]**

(b) Rotation **[B1]**

180° (direction not required) **[B1]**

centre $(1, 2)$ **[B1]**

(c) Calculates area of $T = \frac{1}{2} \times 2 \times 3 = 3$ **[M1]**

Sets up $3 \times k^2 = 48 \implies k^2 = 16$ **[M1]**

$k = 4$ **[A1]**

[Total Marks: 7]

1379 **(a)** Enlargement **[B1]**

Scale factor -2 **[B1]**

Centre $(0, 1)$ **[B1]**

(b) Enlargement scale factor $-\frac{1}{2}$ **[B1]**

Centre $(0, 1)$ **[B1]**

(c) Translating Shape T gives at least two correct coordinates of V: $(-2, 3)$, $(-1, 3)$, $(-1, 5)$ **[M1]**

Recognising reflection in $y = x$ swaps coordinates (x, y)

(y, x)$[M1]$

Final coordinates: $(3, -2)$, $(3, -1)$, $(5, -1)$ **[A1]**

[Total Marks: 8]

1380 **(a)** Enlargement **[B1]**

Scale factor -2 **[B1]**

Centre $(1, 0)$ **[B1]**

(b) **(i)** Intermediate triangle V located with vertices at $(0, 1)$, $(-2, 1)$, and $(0, 2)$ **[M1]**

Correct triangle W drawn with vertices at $(0, 3)$, $(-2, 3)$, and $(0, 2)$ **[A2]**

Award [A1] for any two vertices of W plotted correctly

(ii) Rotation **[B1]**

Through $180°$ **[B1]**

About centre $(1, 2)$ **[B1]**

(c) Reflection of (a, b) in $y = x$ gives (b, a) **[B1]**

$(b + 2, a - 4) = (5, -3)$ giving $b = 3$ and $a = 1$ **[M1]**

$(1, 3)$ **[A1]**

[Total Marks: 12]

1381 **(a)** Correct triangle drawn with vertices at $(-5, -1)$, $(-2, -1)$, $(-2, 1)$ **[B2]**

If not B2, B1 for correct size and orientation but wrong position OR two correct vertices **[B1]**

(b) Correct triangle drawn with vertices at $(1, 0)$, $(4, 0)$, $(4, -2)$ **[B2]**

If not B2, B1 for reflection in $x = 1$ OR reflection in $y = k$ where $k \neq 1$ **[B1]**

(c) Correct triangle drawn with vertices at $(-1, -2)$, $(-4, -2)$, $(-4, -4)$ **[B2]**

If not B2, B1 for rotation of 90° about $(0, 0)$ OR rotation of 180° about a different centre **[B1]**

[Total Marks: 6]

1382 **(a)** Rotation **[B1]**

90° anticlockwise OR 270° clockwise **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Correct shape R with vertices at $(4, 4), (5, 4), (5, 5), (4.5, 5), (4.5, 4.5), (4, 4.5)$ **[B2]**

Any enlargement scale factor $\frac{1}{2}$ OR correct centre $(6, 6)$ with incorrect scale factor **[B1]**

[Total Marks: 5]

1383 **(a)** Vertices of image plotted correctly at $(1, 1)$, $(2, 1)$, $(2, 1.5)$, $(1.5, 1.5)$, $(1.5, 2)$ and $(1, 2)$ **[B2]**

If not B2, award B1 for correct size and orientation, or correct enlargement from the wrong centre **[B1]**

(b) Rotation **[B1]**

90° clockwise (or $-90°$) **[B1]**

centre $(0, 0)$ (or origin) **[B1]**

[Total Marks: 5]

1384 **(a)** Enlargement **[B1]**

Scale factor $\frac{1}{2}$ (or 0.5) **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Rotation **[B1]**

90° anticlockwise OR 270° clockwise **[B1]**

Centre $(0, 0)$ **[B1]**

(c) Reflection **[B1]**

in the x-axis OR $y = 0$ **[B1]**

[Total Marks: 8]

1385 **(a)** Reflection **[B1]**

In the line $y = -x$ **[B1]**

(b) Correct enlargement drawn with vertices at $(0, 0)$, $(2, 0)$, $(2, 1)$, and $(1, 1)$ **[B3]**

Award [B2] for correct enlargement with scale factor $\frac{1}{2}$ from an incorrect centre

Award [B1] for any two vertices plotted correctly

(c) Correct rotation drawn with vertices at $(0, 2)$, $(0, 6)$, $(-2, 6)$, and $(-2, 4)$ **[B3]**

Award [B2] for correct orientation and size in wrong position, or 90° clockwise rotation about $(1, 1)$

Award [B1] for any two vertices plotted correctly

(d) $1 : 4$ **[B1]**

[Total Marks: 9]

1386 **(a)** Enlargement **[B1]**

Scale factor -0.5 (or $-\frac{1}{2}$) **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Rotation **[B1]**

90° clockwise (or 270° anti-clockwise) **[B1]**

Centre $(2, 2)$ **[B1]**

(c) Enlargement **[B1]**

Scale factor 0.5 (or $\frac{1}{2}$) **[B1]**

Centre $(0, 0)$ **[B1]**

[Total Marks: 9]

1387 **(a)** Enlargement **[B1]**

Centre $(0, 0)$ or origin **[B1]**

Scale factor $-\frac{1}{2}$ **[B1]**

(b) Image R drawn correctly at $(1, 3), (3, 3), (3, 4)$ **[B3]**

If incorrect, 2 marks for correct size and orientation but wrong position **[M2]**

1 mark for correct centre used but wrong scale factor **[M1]**

(c) Rotation **[B1]**

180° **[B1]**

Centre $(0, 1)$ **[B1]**

OR Enlargement [B1], Scale factor -1 [B1], Centre $(0, 1)$ **[B1]**

[Total Marks: 9]

1388 **(a)** Enlargement **[B1]**

Scale factor $-\frac{1}{2}$ **[B1]**

Centre $(0, 0)$ **[B1]**

(b) Reflection **[B1]**

in the x-axis OR $y = 0$ **[B1]**

[Total Marks: 5]

1389 **(a)** Correct triangle U with vertices at $(1, 2), (3, 2)$ and $(3, 0)$ **[B2]**

Enlargement scale factor $\frac{1}{2}$ from wrong centre OR ray lines drawn from $(0, 4)$ but scale factor incorrect **[B1]**

(b) Correct triangle V with vertices at $(2, 1), (2, 3)$ and $(0, 3)$ **[B2]**

Reflection of U in the line $y = -x$ OR reflection of T in the line $y = x$ **[B1]**

(c) Enlargement **[B1]**

Scale factor 2 **[B1]**

Centre $(0, 4)$ **[B1]**

[Total Marks: 7]

1390 **(a)** Correct triangle drawn with vertices at $(-1, -2)$, $(-7, -2)$, $(-7, -6)$ **[B3]**

If not B3, B2 for a correct enlargement of SF -2 but wrong centre **[B2]**

If not B2, B1 for enlargement of SF 2 from centre $(1, 0)$ OR two correct vertices **[B1]**

(b) $(-1, -2)$ **[B1]**

$(-1, -5)$ and $(-3, -5)$ **[B1]**

(c) Rotation **[B1]**

90° clockwise OR 270° anticlockwise **[B1]**

Centre $(7, 3)$ **[B1]**

[Total Marks: 8]

1391 **(a)** Subtracts $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ from at least one vertex of U **[M1]**

$(2, 4), (5, 4), (4, 8)$ **[A1]**

(b) Uses the transformation (x, y)

(y, x)$for at least one e coordinate [M1]$

$(2, 5), (2, 8), (6, 7)$ **[A1]**

(c) Identifies $(2, 5)$ maps to itself **[B1]**

Applies rotation (x, y)

(y, -x)$relative to the centre (2,5) [M1]$

$(2, 5), (5, 5), (4, 1)$ **[A1]**

(d) Enlargement **[B1]**

Scale factor -2 **[B1]**

Centre $(0, 0)$ **[B1]**

[Total Marks: 10]

1392 **(a)** $\begin{pmatrix} 8 \\ -6 \end{pmatrix} + \begin{pmatrix} -2 \\ 5 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 6 \\ -1 \end{pmatrix}$ **[A1]**

(b) $(5, 1)$ **[B1]**

[Total Marks: 3]

1393 **(a)** $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ **[B1]**

(b) $\begin{pmatrix} 6 \\ 4 \end{pmatrix}$ seen or $2\begin{pmatrix} 3 \\ 2 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 7 \\ 0 \end{pmatrix}$ **[A1]**

[Total Marks: 3]

1394 **(a)** $\begin{pmatrix} 4 \\ -3 \end{pmatrix} + \begin{pmatrix} -4 \\ 10 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 0 \\ 7 \end{pmatrix}$ **[A1]**

(b) $\begin{pmatrix} 3 \\ -1 \end{pmatrix} + \begin{pmatrix} 4 \\ -3 \end{pmatrix} = \begin{pmatrix} 7 \\ -4 \end{pmatrix}$ [M1]

$(7, -4)$ **[A1]**

[Total Marks: 4]

1395 **(a)** $3 \times \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 12 \\ -9 \end{pmatrix}$ **[A1]**

(b) $\begin{pmatrix} 4 \\ -3 \end{pmatrix} - \begin{pmatrix} -4 \\ 10 \end{pmatrix}$ or $\begin{pmatrix} 4 - 2(-2) \\ -3 - 2(5) \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 8 \\ -13 \end{pmatrix}$ **[A1]**

(c) $\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -2 \\ 5 \end{pmatrix} = \begin{pmatrix} 6 \\ -1 \end{pmatrix}$ or $(6 - (-2), -1 - 5)$ **[M1]**

$(8, -6)$ **[A1]**

[Total Marks: 6]

1396 **(a)** Horizontal component $2 - (-3) = 5$ and vertical component $4 - 1 = 3$ **[M1]**

$\begin{pmatrix} 5 \\ 3 \end{pmatrix}$ **[A1]**

(b) **(i)** $2\mathbf{q} = \begin{pmatrix} -2 \\ 8 \end{pmatrix}$ **[B1]**

$\begin{pmatrix} 1 \\ 6 \end{pmatrix}$ **[B1]**

(ii) $3\mathbf{p} = \begin{pmatrix} 9 \\ -6 \end{pmatrix}$ **[B1]**

$\begin{pmatrix} 10 \\ -10 \end{pmatrix}$ **[B1]**

(c) $\vec{OE} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} -4 \\ -6 \end{pmatrix} = \begin{pmatrix} -2 \\ -2 \end{pmatrix}$ **[M1]**

$(-2, -2)$ **[A1]**

[Total Marks: 8]

1397 **(a)** **(i)** $\vec{OA} + \frac{1}{2}\vec{AB} = \mathbf{a} + \frac{1}{2}\mathbf{c}$ **[M1]**

$\mathbf{a} + \frac{1}{2}\mathbf{c}$ **[A1]**

(ii) $\vec{MB} + \vec{BN} = \frac{1}{2}\mathbf{c} - \frac{2}{3}\mathbf{a}$ **[M1]**

$\frac{1}{2}\mathbf{c} - \frac{2}{3}\mathbf{a}$ **[A1]**

(b) $\vec{ON} = \vec{OC} + \vec{CN} = \mathbf{c} + \frac{1}{3}\mathbf{a}$ **[M1]**

$\frac{1}{3}$ **[A1]**

[Total Marks: 6]

1398 **(a)** **(i)** $\mathbf{b} - \mathbf{a}$ **[B1]**

(ii) $\vec{OP} = \vec{OA} + \frac{1}{4}\vec{AB} = \mathbf{a} + \frac{1}{4}(\mathbf{b} - \mathbf{a})$ **[M1]**

$\frac{3}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}$ **[A1]**

(iii) $\vec{MP} = \vec{MO} + \vec{OP} = -\frac{1}{2}\mathbf{a} + \left(\frac{3}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}\right)$ **[M1]**

$\frac{1}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}$ **[A1]**

(b) $\vec{OC} = \vec{OB} + \vec{BC} = \mathbf{b} + k\mathbf{a}$ **[B1]**

$\mathbf{b} + k\mathbf{a} = m\left(\frac{3}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}\right) = \frac{3m}{4}\mathbf{a} + \frac{m}{4}\mathbf{b}$ **[M1]**

Comparing coefficients of $\mathbf{b}$ gives $\frac{m}{4} = 1$, so $m = 4$ **[A1]**

Comparing coefficients of $\mathbf{a}$ gives $k = \frac{3(4)}{4} = 3$ **[A1]**

[Total Marks: 9]

1399 **(a)** $\begin{pmatrix} 8 \\ -1 \end{pmatrix}$ **[B1]**

(b) Deduces correct column vectors for $\mathbf{a}$ and $\mathbf{b}$: $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ **[B1]**

Sets up correct simultaneous equations from components: $2p - q = 8$ and $p + 2q = -1$ **[M1]**

Valid method to solve the simultaneous equations (e.g. multiplying first by 2 and adding: $4p - 2q = 16$

5p = 15)$[M1]$

$p = 3$ **[A1]**

$q = -2$ **[A1]**

[Total Marks: 5]

1400 **(a)** Sets up an equation for components, e.g. $2(3) - n = 10$ or $2m - (-2) = 7$ **[M1]**

$n = -4$ **[A1]**

$m = 2.5$ (or $\frac{5}{2}$) **[A1]**

(b) Sets up scalar multiplication equation, e.g. $\begin{pmatrix} k \\ 12 \end{pmatrix} = c\begin{pmatrix} 3 \\ m \end{pmatrix}$ **[M1]**

Finds scalar multiplier $c = \frac{12}{m}$ **[M1]**

$k = \frac{36}{m}$ **[A1]**

[Total Marks: 6]

1401 **(a)** $\begin{pmatrix} 12 \\ -9 \end{pmatrix}$ or $\begin{pmatrix} -4 \\ 10 \end{pmatrix}$ seen **[M1]**

$\begin{pmatrix} 16 \\ -19 \end{pmatrix}$ **[A1]**

(b) $\vec{AB} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ **[M1]**

$(5, 1)$ **[A1]**

[Total Marks: 4]

1402 **(a)** $\begin{pmatrix}12\\-9\end{pmatrix}$ **[B1]**

(b) $\begin{pmatrix}4\\-3\end{pmatrix} - \begin{pmatrix}-4\\10\end{pmatrix}$ **[M1]**

$\begin{pmatrix}8\\-13\end{pmatrix}$ **[A1]**

(c) $(3+4, -1-3)$ **[M1]**

$(7, -4)$ **[A1]**

[Total Marks: 5]

1403 **(a)** $\vec{AB} = \mathbf{b} - \mathbf{a}$ or $\vec{BA} = \mathbf{a} - \mathbf{b}$ **[B1]**

$\vec{OP} = \mathbf{a} + \frac{3}{4}(\mathbf{b} - \mathbf{a})$ or $\vec{OP} = \mathbf{b} + \frac{1}{4}(\mathbf{a} - \mathbf{b})$ **[M1]**

$\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$ **[A1]**

(b) $\vec{MP} = -\frac{1}{2}\mathbf{a} + \vec{OP}$ (oe) **[M1]**

$-\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$ **[A1]**

[Total Marks: 5]

1404 **(a)** $\begin{pmatrix}6\\-4\end{pmatrix} + \begin{pmatrix}x\\5\end{pmatrix} = \begin{pmatrix}-1\\y\end{pmatrix}$ **[M1]**

$6 + x = -1 \implies x = -7$ **[A1]**

$-4 + 5 = y \implies y = 1$ **[A1]**

(b) **(i)** $(4-2, 7+6)$ **[M1]**

$(2, 13)$ **[A1]**

(ii) $\begin{pmatrix}2\\-6\end{pmatrix}$ **[B1]**

[Total Marks: 6]

1405 **(a)** Forms a correct equation for the horizontal components: $3(4) - k = 10$ **[M1]**

$k = 2$ **[A1]**

Forms a correct equation for the vertical components: $3(-3) + 2y = 6$

-9 + 2y = 6[$M1$]

$y = 7.5$ **[A1]**

(b) **(i)** $\begin{pmatrix}3\\6\end{pmatrix}$ **[B1]**

(ii) Finds vector $\vec{QR} = \begin{pmatrix} 6 \\ 12 \end{pmatrix}$ OR $\vec{PR} = \begin{pmatrix} 9 \\ 18 \end{pmatrix}$ **[M1]**

States that $\vec{QR} = 2\vec{PQ}$ (or equivalent scalar multiplier), showing the vectors are parallel, AND mentions that they share a common point Q (or P), concluding they are collinear **[A1]**

[Total Marks: 6]

1406 **(a)** $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$ **[B1]**

(b) $\vec{BC} = \begin{pmatrix} -9 \\ 6 \end{pmatrix}$ **[M1]**

Identifies method to find C, e.g. $(-1, 5) + \begin{pmatrix} -9 \\ 6 \end{pmatrix}$ **[M1]**

$(-10, 11)$ **[A1]**

(c) $\mathbf{q} = \begin{pmatrix} 4 \\ -1 \end{pmatrix} - \begin{pmatrix} -3 \\ 2 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 7 \\ -3 \end{pmatrix}$ **[A1]**

[Total Marks: 6]

1407 **(a)** $\begin{pmatrix} 7 \\ 10 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 6 \\ 12 \end{pmatrix}$ **[A1]**

(b) $\vec{AC} = \begin{pmatrix} 4-1 \\ 4-(-2) \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$ **[M1]**

$\frac{1}{2}\vec{AB} = \frac{1}{2}\begin{pmatrix} 6 \\ 12 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$, hence $\vec{AC} = \frac{1}{2}\vec{AB}$ **[A1]**

(c) $(4-5, 4+2)$ **[M1]**

$(-1, 6)$ **[A1]**

(d) $\begin{pmatrix} -1 \\ 6 \end{pmatrix} - \begin{pmatrix} 7 \\ 10 \end{pmatrix}$ or equivalent path method $-\vec{AC} + \vec{CD}$ **[M1]**

$\begin{pmatrix} -8 \\ -4 \end{pmatrix}$ **[A1]**

[Total Marks: 8]

1408 **(a)** $\vec{AX} = \frac{1}{3}\mathbf{c}$ **[M1]**

$\mathbf{a} + \frac{1}{3}\mathbf{c}$ **[A1]**

(b) $B\vec{Y} = 2\mathbf{a}$ or $C\vec{Y} = 3\mathbf{a}$ **[M1]**

$3\mathbf{a} + \mathbf{c}$ **[A1]**

(c) $O\vec{Y} = 3\mathbf{a} + \mathbf{c} = 3(\mathbf{a} + \frac{1}{3}\mathbf{c}) = 3O\vec{X}$ **[M1]**

Since $O\vec{Y}$ is a scalar multiple of $O\vec{X}$ and they share the common point O, they are collinear. **[A1]**

(d) $1 : 2$ **[B1]**

[Total Marks: 7]

1409 **(a)** $\begin{pmatrix}6\\7\end{pmatrix}$ - $\begin{pmatrix}2\\-1\end{pmatrix}[M1]$

$\begin{pmatrix}4\\8\end{pmatrix}$ **[A1]**

(b) $A\vec{C} = \begin{pmatrix}k-1\\3k-1\end{pmatrix}$ **[B1]**

Gradient or ratio condition: $\frac{3k-1}{k-1} = \frac{8}{4}$ or $A\vec{C} = mA\vec{B}$ **[M1]**

$3k - 1 = 2(k-1) \implies k = -1$ **[A1]**

(c) $A\vec{C} = \begin{pmatrix}-2\\-4\end{pmatrix}$ **[M1]**

$-\frac{1}{2}A\vec{B}$ **[A1]**

[Total Marks: 7]

1410 **(a)** $P\vec{Q} = 2\begin{pmatrix}3\\-2\end{pmatrix} - \begin{pmatrix}-1\\4\end{pmatrix} = \begin{pmatrix}7\\-8\end{pmatrix}$ **[M1]**

$\begin{pmatrix}x\\y\end{pmatrix} - \begin{pmatrix}5\\-1\end{pmatrix} = \begin{pmatrix}7\\-8\end{pmatrix}$ or $(5+7, -1-8)$ **[M1]**

$(12, -9)$ **[A1]**

(b) $P\vec{R} = \begin{pmatrix}14-5\\3-(-1)\end{pmatrix} = \begin{pmatrix}9\\4\end{pmatrix}$ **[B1]**

$3h - k = 9$ and $-2h + 4k = 4$ **[M1]**

Rearranging to get $k = 3h - 9$ and substituting: $-2h + 4(3h - 9) = 4$ or equivalent valid method to solve simultaneous equations **[M1]**

$10h = 40 \implies h = 4$ **[A1]**

$k = 3(4) - 9 = 3$ **[A1]**

[Total Marks: 8]

1411 **(a)** $-\mathbf{a} + \mathbf{b}$ OR $\mathbf{b} - \mathbf{a}$ **[B1]**

(b) $\vec{OD} = \vec{OA} + \vec{AD}$ OR $\vec{OD} = \vec{OB} + \vec{BD}$ **[M1]**

$\vec{AD} = \frac{2}{3}(\mathbf{b} - \mathbf{a})$ OR $\vec{BD} = \frac{1}{3}(\mathbf{a} - \mathbf{b})$ **[M1]**

$\mathbf{a} + \frac{2}{3}(\mathbf{b} - \mathbf{a}) = \frac{1}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}$ **[A1]**

(c) $\vec{CE} = \vec{CO} + \vec{OE}$ OR $-\frac{1}{2}\mathbf{a} + 2\mathbf{b}$ **[M1]**

$-\frac{1}{2}\mathbf{a} + 2\mathbf{b}$ **[A1]**

(d) Finds $\vec{CD} = \vec{OD} - \vec{OC} = (\frac{1}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}) - \frac{1}{2}\mathbf{a} = -\frac{1}{6}\mathbf{a} + \frac{2}{3}\mathbf{b}$ **[M1]**

Shows $\vec{CE} = 3\vec{CD}$ (or equivalent scalar multiple) and states they share a common point C, concluding they form a straight line. **[A1]**

[Total Marks: 8]

1412 **(a)** **(i)** $\mathbf{a} + \mathbf{b}$ **[B1]**

(ii) $\vec{BC} = \vec{BO} + \vec{OC} = -(\mathbf{a} + \mathbf{b}) + 2\mathbf{b}$ **[M1]**

$\mathbf{b} - \mathbf{a}$ **[A1]**

(iii) $\vec{OM} = \vec{OB} + \frac{1}{2}\vec{BC} = (\mathbf{a} + \mathbf{b}) + \frac{1}{2}(\mathbf{b} - \mathbf{a})$ **[M1]**

$\frac{1}{2}\mathbf{a} + \frac{3}{2}\mathbf{b}$ **[A1]**

(b) $\vec{AC} = \vec{AO} + \vec{OC} = 2\mathbf{b} - \mathbf{a}$ **[B1]**

$\vec{ON} = \vec{OA} + \frac{3}{5}\vec{AC} = \mathbf{a} + \frac{3}{5}(2\mathbf{b} - \mathbf{a})$ **[M1]**

$\frac{2}{5}\mathbf{a} + \frac{6}{5}\mathbf{b}$ **[A1]**

(c) **(i)** $\vec{ON} = \frac{2}{5}(\mathbf{a} + 3\mathbf{b})$ and $\vec{OM} = \frac{1}{2}(\mathbf{a} + 3\mathbf{b})$, so $\vec{ON} = \frac{4}{5}\vec{OM}$ **[M1]**

$\vec{ON}$ is a scalar multiple of $\vec{OM}$ and both vectors share the common point O, so O, N, and M are collinear **[A1]**

(ii) $\vec{ON} = \frac{4}{5}\vec{OM}$ gives $ON : OM = 4 : 5$ **[M1]**

$4 : 1$ **[A1]**

[Total Marks: 12]

1413 $\sqrt{(-12)^2 + 5^2}$ or $\sqrt{144 + 25}$ **[M1]**

13 **[A1]**

[Total Marks: 2]

1414 $\sqrt{5^2 + (-12)^2}$ or $\sqrt{25 + 144}$ **[M1]**

13 **[A1]**

[Total Marks: 2]

1415 **(a)** Determines column vector $\mathbf{a} = \begin{pmatrix} -6 \\ 8 \end{pmatrix}$ **[M1]**

10 **[A1]**

(b) Determines column vector $\mathbf{b} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ **[B1]**

Calculates resultant vector $2\begin{pmatrix} -6 \\ 8 \end{pmatrix} + 3\begin{pmatrix} 3 \\ -4 \end{pmatrix} = \begin{pmatrix} -12 \\ 16 \end{pmatrix} + \begin{pmatrix} 9 \\ -12 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ **[M1]**

5 **[A1]**

[Total Marks: 5]

1416 **(a)** $\sqrt{(-8)^2 + 15^2}$ or $\sqrt{64 + 225}$ **[M1]**

17 **[A1]**

(b) $\mathbf{p} + 2\mathbf{q} = \begin{pmatrix} -8 \\ 15 \end{pmatrix} + \begin{pmatrix} 4 \\ -8 \end{pmatrix} = \begin{pmatrix} -4 \\ 7 \end{pmatrix}$ **[M1]**

$\sqrt{(-4)^2 + 7^2}$ or $\sqrt{16 + 49}$ **[M1]**

$\sqrt{65}$ **[A1]**

[Total Marks: 5]

1417 **(a)** $\begin{pmatrix} 6 \\ 8 \end{pmatrix}$ **[B1]**

(b) $|\vec{AB}| = \sqrt{6^2 + 8^2} = \sqrt{36 + 64}$ **[M1]**

10 **[A1]**

(c) **(i)** $|\mathbf{v}| = \sqrt{(-12)^2 + (-5)^2} = \sqrt{144 + 25}$ **[M1]**

13 **[A1]**

(ii) $\vec{AB} + \mathbf{v} = \begin{pmatrix} 6 \\ 8 \end{pmatrix} + \begin{pmatrix} -12 \\ -5 \end{pmatrix} = \begin{pmatrix} -6 \\ 3 \end{pmatrix}$ **[M1]**

$\sqrt{(-6)^2 + 3^2} = \sqrt{45} = 3\sqrt{5}$ **[A1]**

[Total Marks: 7]

1418 **(a)** Sets up equation: $p^2 + p^2 = 18$ **[M1]**

3 **[A1]**

(b) Finds expression for $\mathbf{w}$: $\begin{pmatrix} 4 - t \\ 1 + 2t \end{pmatrix}$ **[M1]**

Forms magnitude squared equation: $(4 - t)^2 + (1 + 2t)^2 = 29$ **[M1]**

Expands brackets correctly: $16 - 8t + t^2 + 1 + 4t + 4t^2 = 29$ **[M1]**

Rearranges to a 3-term quadratic: $5t^2 - 4t - 12 = 0$ **[M1]**

$t = 2$, or $t = -1.2$ (or $-\frac{6}{5}$) **[A1]**

[Total Marks: 7]

1419 **(a)** $\sqrt{x^2 + (x+2)^2} = 10$ or $x^2 + (x+2)^2 = 10^2$ **[M1]**

$x^2 + x^2 + 4x + 4 = 100$ **[M1]**

$2x^2 + 4x - 96 = 0$ leading to $x^2 + 2x - 48 = 0$ with no errors seen **[A1]**

(b) $(x+8)(x-6) = 0$ or correct substitution into quadratic formula **[M1]**

$x = -8, x = 6$ **[A1]**

(c) $\mathbf{u} = \begin{pmatrix} -8 \\ -6 \end{pmatrix}$ **[B1]**

$\mathbf{u} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$ **[B1]**

[Total Marks: 7]

1420 **(a)** **(i)** $\sqrt{6^2 + 2^2} = \sqrt{40} = 2\sqrt{10}$ **[B1]**

(ii) $\vec{AB} = \begin{pmatrix} -2 \\ 6 \end{pmatrix} - \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} -8 \\ 4 \end{pmatrix}$ **[M1]**

$|\vec{AB}| = \sqrt{(-8)^2 + 4^2} = \sqrt{80} = 4\sqrt{5}$ **[A1]**

(b) $|\vec{OB}| = \sqrt{(-2)^2 + 6^2} = \sqrt{40} = 2\sqrt{10}$ **[M1]**

$|\vec{OA}| = |\vec{OB}| = 2\sqrt{10}$, so triangle OAB is isosceles **[A1]**

$|\vec{OA}|^2 + |\vec{OB}|^2 = 40 + 40 = 80 = |\vec{AB}|^2$, so angle $AOB = 90°$ **[A1]**

(c) **(i)** $\vec{OP} = \vec{OA} + \frac{3}{4}\vec{AB} = \begin{pmatrix} 6 \\ 2 \end{pmatrix} + \frac{3}{4}\begin{pmatrix} -8 \\ 4 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 0 \\ 5 \end{pmatrix}$ **[A1]**

(ii) $\vec{PQ} = \begin{pmatrix} 8 \\ m \end{pmatrix} - \begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 8 \\ m-5 \end{pmatrix}$ **[M1]**

$8^2 + (m-5)^2 = 17^2$ giving $(m-5)^2 = 225$ **[M1]**

$m - 5 = -15$ (since $m < 0$), so $m = -10$ **[A1]**

[Total Marks: 11]

1421 **(a)** $\begin{pmatrix} -5 \\ 12 \end{pmatrix}$ **[B1]**

(b) $\sqrt{(-5)^2 + 12^2}$ or $\sqrt{25 + 144}$ **[M1]**

13 **[A1]**

(c) $3^2 + c^2 = 13^2$ or $9 + c^2 = 169$ **[M1]**

$c^2 = 160$ **[M1]**

12.6 and -12.6 (or $\pm 4\sqrt{10}$) **[A1]**

[Total Marks: 6]

1422 **(a)** $\vec{AC} = \begin{pmatrix} 7-3 \\ 1-(-2) \end{pmatrix}$ or $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$ **[M1]**

5 **[A1]**

(b) $\vec{AB} = \begin{pmatrix} -8 \\ y+2 \end{pmatrix}$ **[B1]**

$\sqrt{(-8)^2 + (y+2)^2} = 17$ **[M1]**

$64 + (y+2)^2 = 289$ or $(y+2)^2 = 225$ **[M1]**

$y + 2 = 15$ or $y + 2 = -15$ **[M1]**

$y = 13$, $y = -17$ **[A1]**

[Total Marks: 7]

1423 **(a)** Expresses vector $\vec{PQ}$ as $\begin{pmatrix} k-2 \\ 5-(-3) \end{pmatrix} = \begin{pmatrix} k-2 \\ 8 \end{pmatrix}$ **[M1]**

Sets up magnitude equation: $(k-2)^2 + 8^2 = 10^2$ **[M1]**

Simplifies to solve for k: $(k-2)^2 = 36$ **[M1]**

$k = 8$, or $k = -4$ **[A1]**

(b) Finds $\vec{QR}$ using $Q(8, 5)$ and $R(4, 1)$: $\vec{QR} = \begin{pmatrix} 4-8 \\ 1-5 \end{pmatrix} = \begin{pmatrix} -4 \\ -4 \end{pmatrix}$ **[M1]**

Calculates magnitude: $\sqrt{(-4)^2 + (-4)^2} = \sqrt{32}$ **[M1]**

5.66 **[A1]**

[Total Marks: 7]

1424 **(a)** $\begin{pmatrix} 9-2 \\ 4-(-3) \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 7 \\ 7 \end{pmatrix}$ **[A1]**

(b) $\sqrt{7^2 + 7^2}$ or $\sqrt{98}$ **[M1]**

9.90 **[A1]**

(c) $\sqrt{k^2 + 4^2} = 5$ or $k^2 + 4^2 = 5^2$ **[M1]**

$k^2 = 25 - 16$ or $k^2 = 9$ **[M1]**

$k = 3$, $k = -3$ **[A1]**

[Total Marks: 7]

1425 **(a)** $2\mathbf{p}-\mathbf{q}=\begin{pmatrix}8\\-6\end{pmatrix}-\begin{pmatrix}-1\\5\end{pmatrix}=\begin{pmatrix}9\\-11\end{pmatrix}$ **[M1]**

$|2\mathbf{p}-\mathbf{q}|=\sqrt{9^2+(-11)^2}=\sqrt{202}$ **[M1]**

14.2 **[A1]**

(b) $|\mathbf{p}|=\sqrt{4^2+(-3)^2}=5$ **[B1]**

$k=-\dfrac{35}{5}=-7$ **[M1]**

$\begin{pmatrix}-28\\21\end{pmatrix}$ **[A1]**

(c) $(t+3)^2+(2t-1)^2=34$ **[M1]**

$t^2+6t+9+4t^2-4t+1=34$ simplifying to $5t^2+2t-24=0$ **[M1]**

$(5t+12)(t-2)=0$ **[M1]**

$t=2$ and $t=-2.4$ (or $-\dfrac{12}{5}$) **[A1]**

[Total Marks: 10]

1426 **(a)** $(k+3)^2+(2k-1)^2=98$ **[M1]**

$k^2+6k+9+4k^2-4k+1=98$ **[M1]**

$5k^2+2k+10=98$ leading to $5k^2+2k-88=0$ with no errors seen **[A1]**

(b) $(5k+22)(k-4)=0$ or $\dfrac{-2\pm\sqrt{2^2-4(5)(-88)}}{2(5)}$ **[M2]**

If M0, allow [M1] for a correct partial factorisation e.g. $(5k+a)(k+b)$ where $ab=-88$ or correct substitution into quadratic formula with one sign error

4 and -4.4 **[A1]**

(c) $\begin{pmatrix}7\\7\end{pmatrix}$ **[B1]**

[Total Marks: 7]

1427 **(a)** $\mathbf{u}+\mathbf{v}=\begin{pmatrix}3k+1\\k-1\end{pmatrix}$ **[M1]**

$(3k+1)^2+(k-1)^2=(\sqrt{50})^2$ **[M1]**

$9k^2+6k+1+k^2-2k+1=50$ **[M1]**

$10k^2+4k+2=50 \implies 5k^2+2k-24=0$ **[A1]**

(b) $(5k+a)(k+b)=0$ where $ab=-24$ or $5b+a=2$, or correct substitution into quadratic formula **[M1]**

$(5k+12)(k-2)=0$ **[M1]**

$k=2$ **[A1]**

$k=-2.4$ or $-\dfrac{12}{5}$ **[A1]**

(c) Substituting $k = 2$ and $k = -2.4$ into $\begin{pmatrix} 2k \\ k-3 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 4 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -4.8 \\ -5.4 \end{pmatrix}$ **[A1]**

[Total Marks: 10]

1428 **(a)** 0.17 **[B1]**

(b) The train does not arrive on time (or equivalent) **[B1]**

[Total Marks: 2]

1429 **(a)** $1 - \left(\frac{2}{7} + \frac{1}{3}\right)$ or equivalent **[M1]**

$\frac{8}{21}$ **[A1]**

(b) $\frac{5}{7}$ **[B1]**

[Total Marks: 3]

1430 **(a)** $0.15 + 2x + 0.31 + x + 0.24 = 1$ or $3x + 0.7 = 1$ **[M1]**

0.1 **[A1]**

(b) $P(2) = 2(0.1) = 0.2$ **[M1]**

0.8 **[A1]**

[Total Marks: 4]

1431 **(a)** Sum of given probabilities: $0.15 + 0.3 + 0.45 = 0.9$ **[M1]**

0.1 **[A1]**

(b) 0.45 **[B1]**

(c) Uses $1 - P(G)$ formula: $1 - 0.45$ **[M1]**

0.55 **[A1]**

[Total Marks: 5]

1432 **(a)** $0.15 + 2x + 0.35 + 3x = 1$ **[M1]**

$5x = 0.50$, so $x = 0.1$ (or 0.10) **[A1]**

(b) **(i)** $P(A) = P(2) + P(4) = 0.20 + 0.30$ **[M1]**

0.5 (or 0.50 or $\frac{1}{2}$) **[A1]**

(ii) $1 - 0.5 = 0.5$ (or 0.50 or $\frac{1}{2}$) **[B1]**

(c) $P(2) + P(3) = 0.20 + 0.35 = 0.55$ **[M1]**

$200 \times 0.55 = 110$ **[A1]**

[Total Marks: 7]

1433 **(a)** $\frac{1}{5} + 0.35 + x + 2x = 1$ or $0.55 + 3x = 1$ **[M1]**

0.15 **[A1]**

(b) $1 - 2(0.15)$ or $\frac{1}{5} + 0.35 + 0.15$ **[M1]**

0.7 or $\frac{7}{10}$ **[A1]**

[Total Marks: 4]

1434 **(a)** $p^2 + p + 0.15 + 0.1 = 1$ **[M1]**

$p^2 + p - 0.75 = 0$ leading to $4p^2 + 4p - 3 = 0$ with no errors seen **[A1]**

(b) $(2p - 1)(2p + 3) = 0$ or correct substitution into the quadratic formula **[M1]**

0.5 (or $\frac{1}{2}$) **[A1]**

(c) $P(E) = 0.5 + 0.1 = 0.6$ or $P(E') = P(1) + P(3)$ **[M1]**

0.4 (or $\frac{2}{5}$) **[A1]**

[Total Marks: 6]

1435 **(a)** Places $2x$ in the intersection **[B1]**

Places $38 - 2x$ in T only region and $24 - 2x$ in C only region **[B1]**

Places x in the region outside both circles **[B1]**

(b) Forms correct equation: $(38 - 2x) + 2x + (24 - 2x) + x = 60$ or $62 - x = 60$ **[M1]**

2 **[A1]**

(c) Finds the number of people in $T \cap C'$: $38 - 2(2) = 34$ **[M1]**

$\frac{34}{60}$ or $\frac{17}{30}$ **[A1]**

[Total Marks: 7]

1436 **(a)** $x^2 + 3x + 2$ **[B1]**

(b) $\frac{x^2}{x^2 + 3x + 2} = \frac{8}{15}$ **[M1]**

$15x^2 = 8(x^2 + 3x + 2) \implies 15x^2 = 8x^2 + 24x + 16$ **[M1]**

(c) $(7x + 4)(x - 4) = 0$ or correct substitution into the quadratic formula [M2]
(Award M1 for $(7x + a)(x + b) = 0$ where $ab = -16$ or $7b + a = -24$)

4 **[A1]**

(d) Total counters $= 4^2 + 3(4) + 2 = 30$ and number of blue counters $= 3(4) = 12$ **[M1]**

$\frac{18}{30} = \frac{3}{5}$ **[A1]**

[Total Marks: 8]

1437 **(a)** Total number of counters $= x + (2x - 1) + (x^2 - 4) = x^2 + 3x - 5$ **[B1]**

$\frac{x^2 - 4}{x^2 + 3x - 5} = \frac{3}{5}$ giving $5(x^2 - 4) = 3(x^2 + 3x - 5)$ **[M1]**

$5x^2 - 20 = 3x^2 + 9x - 15$ simplifying to $2x^2 - 9x - 5 = 0$ **[A1]**

(b) $(2x + 1)(x - 5) = 0$ **[M1]**

5 (rejecting $x = -\frac{1}{2}$ since $x > 2$) **[A1]**

(c) **(i)** Total counters $= 5 + 9 + 21 = 35$, giving $P(R) = \frac{5}{35}$ **[M1]**

$\frac{1}{7}$ **[A1]**

(ii) $P(G') = 1 - P(G) = 1 - \frac{3}{5}$ **[M1]**

$\frac{2}{5}$ **[A1]**

(d) Number of counters that are not red $= 9 + 21 = 30$, giving $\frac{30}{35 + k} = \frac{3}{4}$ **[M1]**

$3(35 + k) = 120$ giving $35 + k = 40$ **[M1]**

5 **[A1]**

[Total Marks: 12]

1438 **(a)** $1 - \left(\frac{2}{5} + \frac{1}{3}\right)$ or $1 - \frac{11}{15}$ **[M1]**

$\frac{4}{15}$ **[A1]**

(b) Equates their fraction to the number of green marbles: $\frac{4}{15}$ of total $= 12$ **[M1]**

Total $= 12 \times \frac{15}{4}$ **[M1]**

45 **[A1]**

(c) $\frac{2}{3}$ **[B1]**

[Total Marks: 6]

1439 **(a)** $14 + x + 9 + 5 = 36$ or equivalent **[M1]**

8 **[A1]**

(b) $\frac{14+8}{36}$ or $\frac{22}{36}$ **[M1]**

$\frac{11}{18}$ **[A1]**

(c) $1 - P(B)$ or $\frac{14+5}{36}$ **[M1]**

$\frac{19}{36}$ **[A1]**

[Total Marks: 6]

1440 **(a)** 21 in the intersection $C \cap D$ **[B1]**

34 in C only and 47 in D only **[B1]**

18 outside both circles **[B1]**

(b) $\frac{21}{120}$ or $\frac{7}{40}$ **[B1]**

(c) $1 - \frac{68}{120}$ or $\frac{34+18}{120}$ **[M1]**

$\frac{52}{120}$ or $\frac{13}{30}$ **[A1]**

(d) $\frac{34+47}{120}$ or $\frac{120-21-18}{120}$ **[M1]**

$\frac{81}{120}$ or $\frac{27}{40}$ **[A1]**

[Total Marks: 8]

1441 **(a)** 36 **[B1]**

(b) **(i)** $n(M) = 38 + 17 = 55$, giving $\frac{55}{120}$ **[M1]**

$\frac{11}{24}$ **[A1]**

(ii) $n(M' \cap D) = 29$ **[B1]**

$\frac{29}{120}$ **[B1]**

(iii) $1 - \frac{17}{120}$ (or $\frac{38+29+36}{120}$) **[M1]**

$\frac{103}{120}$ **[A1]**

(c) Total studying Music is 55, and 17 also study Drama **[M1]**

$\frac{17}{55}$ **[A1]**

[Total Marks: 9]

1442 **(a)** $(x+4) + x + (2x-1) + (3x+5) = 50$ or $7x + 8 = 50$ **[M1]**

6 **[A1]**

(b) $(2(6)-1)+(3(6)+5)$ or $50-(6+4+6)$ or $\dfrac{11+23}{50}$ **[M1]**

$\dfrac{34}{50}$ or $\dfrac{17}{25}$ or 0.68 **[A1]**

(c) $(6+4)+(2(6)-1)$ or $10+11$ or $\dfrac{21}{50}$ **[M1]**

$\dfrac{21}{50}$ or 0.42 **[A1]**

[Total Marks: 6]

1443 **(a)** $0.15+0.25+x+0.2+3x=1$ **[M1]**

$x=0.1$ **[A1]**

(b) $P(>3)=P(4)+P(5)=0.2+3(0.1)=0.5$ **[M1]**

Expected number $=250\times0.5=125$ **[A1]**

[Total Marks: 4]

1444 **(a)** $1-(0.15+0.3+0.25+0.1)$ or $1-0.8$ **[M1]**

0.2 **[A1]**

(b) 200×0.3 **[M1]**

60 **[A1]**

[Total Marks: 4]

1445 **(a)** $\dfrac{70}{200}$ or $\dfrac{7}{20}$ or 0.35 **[B1]**

(b) $1000\times\dfrac{45}{200}$ **[M1]**

225 **[A1]**

(c) No, because the relative frequencies are not all close to 0.25 (or 50 out of 200) **[B1]**

(d) A larger number of trials leads to a relative frequency that is closer to the theoretical probability **[B1]**

[Total Marks: 5]

1446 **(a)** 0.27 **[B1]**

(b) If the dice were fair, each score would be expected to occur 40 times (or with relative frequency 0.20), whereas the observed frequencies differ considerably from 40 **[B1]**

(c) Total frequency of odd scores $=32+54+40=126$ **[M1]**

Relative frequency of an odd score $=\dfrac{126}{200}=0.63$ **[M1]**

$0.63 \times 500 = 315$ **[A1]**

(d) $\frac{40}{200} \times n = 36$ (or $0.2n = 36$) **[M1]**

180 **[A1]**

[Total Marks: 7]

1447 **(a)** $x + 2x + 5 + 3x - 2 + x + 8 = 60$ **[M1]**

$7x + 11 = 60 \implies 7x = 49$

$x = 7$ **[A1]**

(b) Frequency of $3 = 3(7) - 2 = 19$

$\frac{19}{60}$ **[B1]**

(c) Frequency of $2 = 2(7) + 5 = 19$

Relative frequency of $2 = \frac{19}{60}$

$1200 \times \frac{19}{60}$ **[M1]**

380 **[A1]**

[Total Marks: 5]

1448 **(a)** $\frac{2x+4}{x^2+10x} = \frac{1}{6}$ giving $x^2 + 10x = 6(2x + 4)$ **[M1]**

$x^2 + 10x = 12x + 24$ simplifying to $x^2 - 2x - 24 = 0$ **[A1]**

(b) $(x - 6)(x + 4) = 0$ **[M1]**

$x = 6$ (rejecting $x = -4$ since $x > 0$) **[A1]**

Total spins $= 6^2 + 10(6) = 96$ **[A1]**

(c) **(i)** Sector B occurs $5(6) + 2 = 32$ times, giving $\frac{32}{96}$ **[M1]**

$\frac{1}{3}$ **[A1]**

(ii) Sector C occurs $96 - 16 - 32 = 48$ times, giving $\frac{48}{96}$ **[M1]**

$\frac{1}{2}$ **[A1]**

(d) Expected number of spins: $A = 600 \times \frac{1}{6} = 100$, $B = 600 \times \frac{1}{3} = 200$, $C = 600 \times \frac{1}{2} = 300$ **[M1]**

Expected net profit $= (100 \times 6) + (200 \times 2) - (300 \times 3) = 600 + 400 - 900$ **[M1]**

100 **[A1]**

[Total Marks: 12]

1449 **(a)** $x = 1 - (0.15 + 0.45 + 0.10)$

$x = 0.30$ **[B1]**

(b) 300×0.45 **[M1]**

135 **[A1]**

(c) Biased, because the relative frequencies are not all close to 0.25 (or specifically Blue is 0.45, which is much higher than 0.25) **[B1]**

[Total Marks: 4]

1450 **(a)** 75 **[B1]**

(b) $\frac{30}{75}$ or $\frac{30}{30+45}$ **[M1]**

0.4 or $\frac{2}{5}$ **[A1]**

(c) $P(\text{tail}) = 1 - 0.4 = 0.6$ or $\frac{45}{75}$ **[B1]**

300×0.6 **[M1]**

180 **[A1]**

[Total Marks: 6]

1451 **(a)** $\frac{x}{300} = 0.22 \implies x = 300 \times 0.22$ **[M1]**

$y = 300 - (84 + 105 + 66)$ **[M1]**

$x = 66$, $y = 45$ **[A1]**

(b) Relative frequency of Yellow $= \frac{45}{300}$ or 0.15 **[M1]**

Total beads $= \frac{15}{0.15} = 100$ **[A1]**

(c) Probability of Blue or Green $= \frac{105+66}{300} = \frac{171}{300}$ or 0.57 **[M1]**

$800 \times 0.57 = 456$ **[A1]**

[Total Marks: 7]

1452 **(a)** $\frac{260}{800}$ **[M1]**

$\frac{13}{40}$ **[A1]**

(b) $260 + 180 + w + w + 40 = 800 \implies 480 + 2w = 800$ **[M1]**

$w = 160$ **[A1]**

(c) $\frac{160}{800}$ or $\frac{1}{5}$ or 0.2 **[B1]**

(d) $P(\text{Black}) = \dfrac{180}{800} = \dfrac{9}{40}$ **[M1]**

Expected number $= 12000 \times \dfrac{9}{40} = 2700$ **[A1]**

[Total Marks: 7]

1453 **(a)** $\dfrac{y+35}{250} = 0.42$ giving $y + 35 = 105$ **[M1]**

$y = 70$ **[A1]**

$x = 250 - (15 + 95 + 70 + 35) = 35$ **[A1]**

(b) Number of light bulbs with $t \leq 1000$ in sample $= 15 + 35 = 50$ **[M1]**

Estimated number $= \dfrac{50}{250} \times 4800$ **[M1]**

960 **[A1]**

(c) To avoid bias so that the sample is representative of all light bulbs produced by the factory **[B1]**

(d) $\dfrac{35 + 115}{250 + 750} = \dfrac{150}{1000}$ **[M1]**

0.15 (or $\dfrac{3}{20}$) **[A1]**

[Total Marks: 9]

1454 **(a)** $0.025 \times N = 40$ or $N = \dfrac{40}{0.025}$ **[M1]**

1600 **[A1]**

(b) Expected defective $= 2400 - 2364 = 36$ **[B1]**

New probability of defective $= \dfrac{36}{2400} = 0.015$ **[B1]**

Decrease in probability $= 0.025 - 0.015 = 0.01$ **[M1]**

Percentage decrease $= \dfrac{0.01}{0.025} \times 100$ **[M1]**

40 **[A1]**

[Total Marks: 7]

1455 **(a)** $0.15 + 2x + x + 0.05 + 0.2 + x = 1$ **[M1]**

$4x + 0.4 = 1 \implies 4x = 0.6 \implies x = 0.15$ **[A1]**

(b) $P(2, 3, 5) = 2(0.15) + (0.15 + 0.05) + 0.15 = 0.3 + 0.2 + 0.15 = 0.65$ **[M1]**

$600 \times 0.65 = 390$ **[A1]**

(c) Sets up both equations: $N \times p = 54$ and $N \times (1 - 3p) = 36$ **[M1]**

Expands the second equation: $N - 3Np = 36$ **[M1]**

Substitutes $Np = 54$ into the expanded equation: $N - 3(54) = 36$ **[M1]**

$N = 198$, $p = \dfrac{54}{198} = \dfrac{3}{11}$ (or 0.272...) **[A1]**

[Total Marks: 8]

1456 **(a)** Sum of probabilities: $3x + 2y = 1 \implies y = \dfrac{1 - 3x}{2}$ **[B1]**

$E_1 = 400x$ and $E_3 = 400y \implies 400x \times 400y = 5600 \implies xy = \dfrac{7}{200}$ **[M1]**

Substitute for y: $x\left(\dfrac{1 - 3x}{2}\right) = \dfrac{7}{200} \implies 100x(1 - 3x) = 7 \implies 300x^2 - 100x + 7 = 0$ **[A1]**

(b) Factorising $(30x - 7)(10x - 1) = 0$ or correct substitution into the quadratic formula **[M1]**

$x = \dfrac{7}{30}$ and $x = \dfrac{1}{10}$ (or 0.1) **[A1]**

When $x = \dfrac{7}{30}$, $y = 0.15$ (reject as $y < x$). When $x = 0.1$, $y = 0.35$ (accept as $y > x$) **[M1]**

$x = 0.1, y = 0.35$ **[A1]**

(c) $P(\text{even}) = 2x + y = 2(0.1) + 0.35 = 0.55$ **[M1]**

Expected number $= 400 \times 0.55 = 220$ **[A1]**

[Total Marks: 9]

1457 **(a)** $300p + 200(p + 0.15) = 255$ **[M1]**

$300p + 200p + 30 = 255$ **[M1]**

$500p = 225 \implies p = \dfrac{225}{500}$

$p = 0.45$ **[A1]**

(b) Probability Coin B is Heads $= 0.45 + 0.15 = 0.60$

Probability of different outcomes = Probability (H, T) + Probability (T, H)

$(0.45 \times 0.40) + (0.55 \times 0.60)$ **[M1]**

$0.18 + 0.33 = 0.51$ **[M1]**

$1000 \times 0.51 = 510$ **[A1]**

(c) **(i)** $\dfrac{72}{N} = 0.48$ **[M1]**

$N = \dfrac{72}{0.48}$

$N = 150$ **[A1]**

(ii) Probability of Tails $= 1 - 0.45 = 0.55$

150×0.55 **[M1]**

82.5 **[A1]**

[Total Marks: 10]

1458 **(a)** $\frac{3}{8}$ correctly placed on the first branch for Blue **[B1]**

$\frac{5}{8}$ and $\frac{3}{8}$ correctly placed on all second branches **[B1]**

(b) $\frac{5}{8} \times \frac{5}{8}$ **[M1]**

$\frac{25}{64}$ **[A1]**

[Total Marks: 4]

1459 **(a)** $\frac{3}{8}$ correctly placed for top branch (Red given Red) **[B1]**

$\frac{4}{8}$ correctly placed for bottom branches (Red given Blue, and Blue given Blue) **[B1]**

(b) $\frac{4}{9} \times \frac{5}{8}$ or $\frac{5}{9} \times \frac{4}{8}$ **[M1]**

$\frac{4}{9} \times \frac{5}{8} + \frac{5}{9} \times \frac{4}{8}$ **[M1]**

$\frac{5}{9}$ or equivalent (e.g. $\frac{40}{72}$) **[A1]**

[Total Marks: 5]

1460 **(a)** $\frac{6}{9}$ on the Green branch from Green **[B1]**

$\frac{2}{9}$ and $\frac{7}{9}$ on the branches from Red **[B1]**

(b) $\frac{7}{10} \times \frac{6}{9}$ or $\frac{3}{10} \times \frac{2}{9}$ **[M1]**

$\frac{7}{10} \times \frac{6}{9} + \frac{3}{10} \times \frac{2}{9}$ **[M1]**

$\frac{48}{90}$ or $\frac{8}{15}$ **[A1]**

[Total Marks: 5]

1461 **(a)** Intersection region = 6 **[B1]**

French only = 11, Spanish only = 8, outside = 5 **[B1]**

(b) **(i)** $\frac{6}{30}$ or $\frac{1}{5}$ **[B1]**

(ii) $\frac{11}{30}$ **[B1]**

(iii) $\frac{25}{30}$ or $\frac{5}{6}$ **[B1]**

[Total Marks: 5]

1462 **(a)** Number of students studying both B and C: $(18 + 15) - (30 - 4) = 7$ **[M1]**

Fully correct Venn diagram: Only $B = 11$, $B \cap C = 7$, Only $C = 8$, Neither $= 4$ **[A1]**

(b) **(i)** $\frac{8}{30}$ or $\frac{4}{15}$ **[B1]**

(ii) $\frac{26}{30}$ or $\frac{13}{15}$ **[B1]**

(iii) $\frac{7}{18}$ **[B1]**

[Total Marks: 5]

1463 **(a)** $\frac{3}{8}$ correctly placed for 1st ball Blue **[B1]**

$\frac{3}{8}$, $\frac{5}{8}$, and $\frac{3}{8}$ correctly placed on all three remaining 2nd ball branches **[B1]**

(b) $\frac{25}{64}$ **[B1]**

(c) $\left(\frac{5}{8} \times \frac{3}{8}\right) + \left(\frac{3}{8} \times \frac{5}{8}\right)$ **[M1]**

$\frac{30}{64}$ or $\frac{15}{32}$ **[A1]**

[Total Marks: 5]

1464 **(a)** First counter branches: Red $\frac{5}{8}$, Blue $\frac{3}{8}$ **[B1]**

Second counter branches: Red $\frac{5}{8}$, Blue $\frac{3}{8}$ on both sets of branches **[B1]**

(b) $\frac{5}{8} \times \frac{5}{8} = \frac{25}{64}$ **[B1]**

(c) $\left(\frac{5}{8} \times \frac{3}{8}\right) + \left(\frac{3}{8} \times \frac{5}{8}\right)$ **[M1]**

$= \frac{30}{64}$ (or $\frac{15}{32}$) **[A1]**

[Total Marks: 5]

1465 **(a)** $\frac{6}{9}$ and $\frac{3}{9}$ correctly placed on the top branches **[B1]**

$\frac{7}{9}$ and $\frac{2}{9}$ correctly placed on the bottom branches **[B1]**

(b) $\frac{7}{10} \times \frac{6}{9}$ or $\frac{3}{10} \times \frac{2}{9}$ **[M1]**

$\left(\frac{7}{10} \times \frac{6}{9}\right) + \left(\frac{3}{10} \times \frac{2}{9}\right)$ **[M1]**

$\frac{48}{90}$ or $\frac{8}{15}$ **[A1]**

[Total Marks: 5]

1466 **(a)** Number studying both $= (18 + 15) - (30 - 5) = 8$ **[M1]**

Fully correct Venn diagram: B only $= 10$, $B \cap C = 8$, C only $= 7$, Outside $= 5$ **[A1]**

(b) **(i)** $\frac{10}{30}$ or $\frac{1}{3}$ **[B1]**

(ii) $\frac{25}{30}$ or $\frac{5}{6}$ **[B1]**

(c) Identifies total number of Chemistry students as denominator: 15 **[M1]**

$\frac{8}{15}$ **[A1]**

[Total Marks: 6]

1467 **(a)** 8 in the intersection **[B1]**

16 in F only, 7 in C only, and 9 outside the circles **[B1]**

(b) Identifies 9 as the number of students playing neither **[M1]**

$\frac{9}{40}$ **[A1]**

(c) $16 + 8 + 7$ or $40 - 9$ **[M1]**

31 **[A1]**

[Total Marks: 6]

1468 **(a)** $\frac{4}{7}$ correctly placed on first Blue branch **[B1]**

$\frac{3}{7}$ and $\frac{4}{7}$ correctly placed on all second branches **[B1]**

(b) $\frac{3}{7} \times \frac{3}{7}$ **[M1]**

$\frac{9}{49}$ **[A1]**

(c) $\frac{3}{7} \times \frac{4}{7} + \frac{4}{7} \times \frac{3}{7}$ or $2 \times \left(\frac{3}{7} \times \frac{4}{7}\right)$ **[M1]**

$\frac{24}{49}$ **[A1]**

[Total Marks: 6]

1469 **(a)** First stage branches correctly labeled: $\frac{7}{12}$ for R and $\frac{5}{12}$ for B **[B1]**

Second stage branches from R correctly labeled: $\frac{6}{11}$ for R and $\frac{5}{11}$ for B **[B1]**

Second stage branches from B correctly labeled: $\frac{7}{11}$ for R and $\frac{4}{11}$ for B **[B1]**

(b) Multiplies probabilities along the R, R path: $\frac{7}{12} \times \frac{6}{11}$ **[M1]**

$\frac{42}{132}$ or $\frac{7}{22}$ **[A1]**

(c) Identifies and sums both correct combinations: $\left(\frac{7}{12} \times \frac{5}{11}\right) + \left(\frac{5}{12} \times \frac{7}{11}\right)$ or $2 \times \left(\frac{35}{132}\right)$ **[M1]**

$\frac{70}{132}$ or $\frac{35}{66}$ **[A1]**

[Total Marks: 7]

1470 **(a)** $\frac{3}{8}$ placed correctly on the lower first branch **[B1]**

$\frac{5}{8}, \frac{3}{8}$ and $\frac{5}{8}, \frac{3}{8}$ placed correctly on the second branches **[B1]**

(b) $\frac{5}{8} \times \frac{5}{8}$ **[M1]**

$\frac{25}{64}$ **[A1]**

(c) $\frac{5}{8} \times \frac{3}{8}$ or $\frac{15}{64}$ **[M1]**

$2 \times \left(\frac{5}{8} \times \frac{3}{8}\right)$ **[M1]**

$\frac{15}{32}$ **[A1]**

[Total Marks: 7]

1471 **(a)** All remaining rows completed correctly: row 1: $5, 6, 7$; row 2: $5, 6, 7, 8$; row 3: $5, 6, 7, 8, 9$; row 4: $5, 6, 7, 8, 9, 10$ **[B2]**

Award [B1] if at least 12 of the 18 missing entries are correct

(b) **(i)** $\frac{4}{24} = \frac{1}{6}$ **[B1]**

(ii) Prime outcomes $(2, 3, 5, 7)$ occur $1 + 2 + 4 + 4 = 11$ times **[M1]**

$\frac{11}{24}$ **[A1]**

(iii) Outcomes greater than 6 and even (8 and 10) occur $3 + 1 = 4$ times, giving $\frac{4}{24}$ **[M1]**

$\frac{1}{6}$ **[A1]**

(c) $\frac{1}{6} \times \frac{1}{6}$ **[M1]**

$\frac{1}{36}$ **[A1]**

[Total Marks: 9]

1472 **(a)** $\dfrac{x}{x+4} \times \dfrac{x-1}{x+3}$ **[M1]**

$= \dfrac{x(x-1)}{(x+4)(x+3)} = \dfrac{x(x-1)}{x^2+7x+12}$ **[A1]**

(b) **(i)** $\dfrac{x(x-1)}{x^2+7x+12} = \dfrac{1}{3}$ **[M1]**

$3(x^2-x) = x^2+7x+12$ or $3x^2-3x = x^2+7x+12$ **[M1]**

$2x^2-10x-12 = 0 \implies x^2-5x-6 = 0$ **[A1]**

(ii) $(x-6)(x+1) = 0$ **[M1]**

$x = 6$ or $x = -1$ **[A1]**

Rejects $x = -1$ to conclude there are 6 blue counters **[B1]**

[Total Marks: 8]

1473 **(a)** $\dfrac{5}{x+5} \times \dfrac{4}{x+4} = \dfrac{2}{15}$ **[M1]**

$20 \times 15 = 2(x+5)(x+4)$ or $300 = 2(x^2+9x+20)$ **[M1]**

$150 = x^2+9x+20$

$x^2+9x-130 = 0$ (No errors seen) [A1]

(b) $(x+13)(x-10) = 0$ **[M1]**

$x = 10$ **[A1]**

(c) Total counters = 15 **[B1]**

$\left(\dfrac{10}{15} \times \dfrac{5}{14}\right) + \left(\dfrac{5}{15} \times \dfrac{10}{14}\right)$ **[M1]**

$\dfrac{100}{210}$ or $\dfrac{10}{21}$ **[A1]**

[Total Marks: 8]

1474 **(a)** Sets up correct equation for probability of two red balls: $\dfrac{x}{x+4} \times \dfrac{x-1}{x+3} = \dfrac{1}{3}$ **[M1]**

Cross-multiplies correctly to clear fractions: $3x(x-1) = (x+4)(x+3)$ **[M1]**

Expands and rearranges correctly with no errors:
$3x^2-3x = x^2+7x+12 \implies 2x^2-10x-12 = 0 \implies x^2-5x-6 = 0$ **[A1]**

(b) Factorises or solves the quadratic equation: $(x-6)(x+1) = 0$ **[M1]**

6 (and explicitly or implicitly rejects $x = -1$) **[A1]**

(c) Uses $x = 6$ to determine probabilities for one red and one blue ball: $\dfrac{6}{10} \times \dfrac{4}{9}$ **[M1]**

Accounts for both (R, B) and (B, R) combinations: $2 \times \left(\dfrac{6}{10} \times \dfrac{4}{9}\right)$ or $\dfrac{24}{90} + \dfrac{24}{90}$ **[M1]**

$\dfrac{48}{90}$ or $\dfrac{8}{15}$ **[A1]**

[Total Marks: 8]

1475 **(a)** $\frac{3}{n} \times \frac{2}{n-1}$ **[M1]**

$\frac{6}{n(n-1)} = \frac{1}{15}$ **[M1]**

$n^2 - n = 90$

$\mathrm{n}^2 - n - 90 = 0$ with no errors seen [A1]

(b) $(n-10)(n+9) = 0$ leading to $n = 10$ **[M1]**

7 **[A1]**

(c) $\frac{3}{10} \times \frac{7}{9}$ or $\frac{7}{10} \times \frac{3}{9}$ **[M1]**

$2 \times \left(\frac{3}{10} \times \frac{7}{9}\right)$ **[M1]**

$\frac{7}{15}$ or $\frac{42}{90}$ **[A1]**

[Total Marks: 8]

1476 **(a)** $\frac{3}{n} \times \frac{2}{n-1}$ **[M1]**

$\frac{6}{n(n-1)} = \frac{1}{15}$ **[M1]**

$90 = n^2 - n$

$\mathrm{n}^2 - n - 90 = 0$ [A1]

(b) $(n-10)(n+9) = 0$ **[M1]**

10 **[A1]**

(c) $n = 10$, giving 3 Red and 7 Blue balls **[B1]**

$2 \times \left(\frac{3}{10} \times \frac{7}{9}\right)$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ **[A1]**

[Total Marks: 8]

1477 **(a)** $\frac{3}{n} \times \frac{2}{n-1} = \frac{1}{15}$ **[M1]**

$\frac{6}{n(n-1)} = \frac{1}{15}$ **[M1]**

$n^2 - n = 90 \implies n^2 - n - 90 = 0$ **[A1]**

(b) $(n-10)(n+9) = 0$ **[M1]**

$n = 10$ **[A1]**

(c) Number of Blue counters = 7 **[B1]**

$\left(\frac{3}{10} \times \frac{7}{9}\right) + \left(\frac{7}{10} \times \frac{3}{9}\right)$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ **[A1]**

[Total Marks: 8]

1478 **(a)** $\frac{x}{x+3} \times \frac{x-1}{x+2} = \frac{5}{14}$ **[M1]**

$14x(x-1) = 5(x+3)(x+2)$ **[M1]**

$14x^2 - 14x = 5(x^2 + 5x + 6)$ **[M1]**

$14x^2 - 14x = 5x^2 + 25x + 30$ leading to $3x^2 - 13x - 10 = 0$ with no errors seen **[A1]**

(b) $(3x+2)(x-5) = 0$ or correct substitution into the quadratic formula **[M1]**

5 **[A1]**

(c) $\frac{5}{8} \times \frac{3}{7} + \frac{3}{8} \times \frac{5}{7}$ or $2 \times \left(\frac{5}{8} \times \frac{3}{7}\right)$ **[M1]**

$\frac{30}{56}$ or $\frac{15}{28}$ **[A1]**

[Total Marks: 8]

1479 **(a)** $\frac{n}{n+6} \times \frac{n-1}{n+5} = \frac{1}{2}$ **[M1]**

$2n(n-1) = (n+6)(n+5)$ giving $2n^2 - 2n = n^2 + 11n + 30$ **[M1]**

$n^2 - 13n - 30 = 0$ **[A1]**

(b) $(n-15)(n+2) = 0$ **[M1]**

15 (rejecting $n = -2$ since $n > 0$) **[A1]**

(c) **(i)** $\left(\frac{15}{21} \times \frac{6}{20}\right) + \left(\frac{6}{21} \times \frac{15}{20}\right) = \frac{180}{420}$ **[M1]**

$\frac{3}{7}$ **[A1]**

(ii) $1 - \left(\frac{6}{21} \times \frac{5}{20}\right) = 1 - \frac{1}{14}$ **[M1]**

$\frac{13}{14}$ **[A1]**

(d) $3 \times \frac{15}{21} \times \frac{14}{20} \times \frac{6}{19}$ **[M2]**

$\frac{9}{19}$ **[A1]**

[Total Marks: 12]

1480 **(a)** $\frac{2}{5}$ correctly placed on all three Tails branches **[B1]**

$\frac{3}{5}$ correctly placed on the remaining Heads branch **[B1]**

(b) $\frac{3}{5} \times \frac{2}{5} + \frac{2}{5} \times \frac{3}{5}$ or $2 \times \left(\frac{3}{5} \times \frac{2}{5}\right)$ **[M1]**

$\frac{12}{25}$ or 0.48 **[A1]**

[Total Marks: 4]

1481 **(a)** 9 in the intersection of H and G **[B1]**

14, 10, and 7 in the correct regions **[B1]**

(b) $\frac{14}{40}$ or $\frac{7}{20}$ **[B1]**

(c) $\frac{9}{23} \times \frac{8}{22}$ **[M1]**

$\frac{72}{506}$ or $\frac{36}{253}$ **[A1]**

[Total Marks: 5]

1482 **(a)** 6 placed correctly in the intersection **[B1]**

12 in F only, 8 in T only, and 4 outside the circles **[B1]**

(b) $\frac{6}{30}$ or $\frac{1}{5}$ or 0.2 **[B1]**

(c) $\frac{12}{30}$ seen as the probability for the first student **[M1]**

$\frac{12}{30} \times \frac{11}{29}$ **[M1]**

$\frac{132}{870}$ or $\frac{22}{145}$ **[A1]**

[Total Marks: 6]

1483 **(a)** $2x + x + (x + 4) + 8 = 40$ **[M1]**

$4x + 12 = 40 \implies x = 7$ **[A1]**

(b) $\frac{32}{40}$ or $\frac{4}{5}$ **[B1]**

(c) Number of students playing Football only = 14 **[B1]**

$\frac{14}{40} \times \frac{13}{39}$ **[M1]**

$\frac{182}{1560}$ or $\frac{7}{60}$ **[A1]**

[Total Marks: 6]

1484 **(a)** 10 in the intersection **[B1]**

14 in F only, 9 in T only, and 7 outside the circles **[B1]**

(b) $\frac{10}{40}$ or $\frac{1}{4}$ **[B1]**

(c) $\frac{10}{24} \times \frac{14}{23}$ or $\frac{14}{24} \times \frac{10}{23}$ **[M1]**

$\left(\frac{10}{24} \times \frac{14}{23}\right) + \left(\frac{14}{24} \times \frac{10}{23}\right)$ **[M1]**

$\frac{35}{69}$ or $\frac{280}{552}$ **[A1]**

[Total Marks: 6]

1485 **(a)** Places 6 outside both circles **[B1]**

Calculates intersection: $32 + 21 + 6 - 50 = 9$ and places 9 in $H \cap G$ **[B1]**

Places 23 in H only and 12 in G only **[B1]**

(b) **(i)** $\frac{9}{50}$ **[B1]**

(ii) $\frac{38}{50}$ or $\frac{19}{25}$ **[B1]**

(c) Recognises the reduced sample space and uses probability without replacement: $\frac{9}{32} \times \frac{8}{31}$ **[M1]**

$\frac{72}{992}$ or $\frac{9}{124}$ **[A1]**

[Total Marks: 7]

1486 **(a)** Intersection $n(G \cap H) = 8$ correctly placed **[B1]**

Remaining regions: $n(G \cap H') = 16$, $n(H \cap G') = 10$, outside region $= 6$ **[B1]**

(b) **(i)** $\frac{10}{40}$ (or $\frac{1}{4}$ or 0.25) **[B1]**

(ii) Identifies regions $16 + 8 + 6 = 30$ or calculates $1 - \frac{10}{40}$ **[M1]**

$\frac{30}{40}$ (or $\frac{3}{4}$ or 0.75) **[A1]**

(c) $\frac{24}{40} \times \frac{23}{39}$ **[M1]**

$= \frac{552}{1560}$ (or $\frac{23}{65}$ or 0.354) **[A1]**

[Total Marks: 7]

1487 **(a)** 7 correctly placed in the intersection of T and G **[B1]**

17, 11, and 5 correctly placed in the remaining regions **[B1]**

(b) Identifies 17 students play tennis but not golf **[B1]**

$\frac{17}{40} \times \frac{16}{39}$ **[M1]**

$\frac{34}{195}$ or 0.174 (0.1743...) **[A1]**

(c) Denominator 18 seen **[M1]**

$\frac{7}{18}$ or 0.389 (0.3888...) **[A1]**

[Total Marks: 7]

1488 **(a)** First branch $B = \frac{3}{10}$ **[B1]**

Top branch after R: $B = \frac{3}{9}$ **[B1]**

Bottom branches after B: $R = \frac{7}{9}$ and $B = \frac{2}{9}$ **[B1]**

(b) $\frac{7}{10} \times \frac{6}{9}$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ or 0.467 **[A1]**

(c) $\frac{7}{10} \times \frac{3}{9} + \frac{3}{10} \times \frac{7}{9}$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ or 0.467 **[A1]**

[Total Marks: 7]

1489 **(a)** First pick: $\mathrm{P}(R) = \frac{5}{9}$ and $\mathrm{P}(Y) = \frac{4}{9}$ **[B1]**

Second pick: After R, $\mathrm{P}(R) = \frac{4}{8}$, $\mathrm{P}(Y) = \frac{4}{8}$; After Y, $\mathrm{P}(R) = \frac{5}{8}$, $\mathrm{P}(Y) = \frac{3}{8}$ **[B1]**

(b) $\mathrm{P}(RR) + \mathrm{P}(YY) = \left(\frac{5}{9} \times \frac{4}{8}\right) + \left(\frac{4}{9} \times \frac{3}{8}\right)$ **[M1]**

$\frac{20}{72} + \frac{12}{72} = \frac{32}{72} = \frac{4}{9}$ **[A1]**

(c) Method 1: $1 - \mathrm{P}(RR)$ or Method 2: $\mathrm{P}(RY) + \mathrm{P}(YR) + \mathrm{P}(YY)$ **[M1]**

$1 - \left(\frac{5}{9} \times \frac{4}{8}\right) = 1 - \frac{20}{72}$ **[M1]**

$\frac{52}{72} = \frac{13}{18}$ **[A1]**

[Total Marks: 7]

1490 **(a)** First branches correct: $\mathrm{P}(R_1) = \frac{5}{8}$ and $\mathrm{P}(B_1) = \frac{3}{8}$ **[B1]**

Second branches after R_1 correct: $\mathrm{P}(R_2) = \frac{4}{7}$ and $\mathrm{P}(B_2) = \frac{3}{7}$ **[B1]**

Second branches after B_1 correct: $\mathrm{P}(R_2) = \frac{5}{7}$ and $\mathrm{P}(B_2) = \frac{2}{7}$ **[B1]**

(b) $\mathrm{P}(RR) + \mathrm{P}(BB) = \left(\frac{5}{8} \times \frac{4}{7}\right) + \left(\frac{3}{8} \times \frac{2}{7}\right)$ **[M1]**

$\frac{26}{56}$ or $\frac{13}{28}$ (or 0.464 or 0.4642...) **[A1]**

(c) $1 - \mathrm{P}(BB) = 1 - \left(\frac{3}{8} \times \frac{2}{7}\right)$ or $\left(\frac{5}{8} \times \frac{4}{7}\right) + \left(\frac{5}{8} \times \frac{3}{7}\right) + \left(\frac{3}{8} \times \frac{5}{7}\right)$ **[M1]**

$\frac{50}{56}$ or $\frac{25}{28}$ (or 0.893 or 0.8928...) **[A1]**

[Total Marks: 7]

1491 **(a)** 3 placed correctly outside the circles **[B1]**

x, $24 - x$, and $18 - x$ placed correctly in the regions **[B1]**

(b) $(24 - x) + x + (18 - x) + 3 = 40$ or equivalent **[M1]**

5 **[A1]**

(c) $\frac{19}{40} \times \frac{13}{39}$ **[M1]**

$2 \times \left(\frac{19}{40} \times \frac{13}{39}\right)$ **[M1]**

$\frac{19}{60}$ or $\frac{494}{1560}$ or 0.316... **[A1]**

[Total Marks: 7]

1492 **(a)** $18 + 14 + 3 - x = 30$ or $35 - x = 30$ **[M1]**

5 **[A1]**

(b) 5 placed correctly in the intersection **[B1]**

13 in F only, 9 in S only, and 3 outside the circles **[B1]**

(c) Identifies number of students studying only French is 13 **[M1]**

$\frac{13}{30} \times \frac{12}{29}$ **[M1]**

$\frac{156}{870}$ or $\frac{26}{145}$ **[A1]**

[Total Marks: 7]

1493 **(a)** $\frac{3}{9}$ on the Red

Bluebranch[*B*1]

$\frac{7}{9}$ on the Blue

Redbranch[*B*1]

$\frac{2}{9}$ on the Blue

$Bluebranch[B1]$

(b) $\frac{7}{10} \times \frac{6}{9}$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ **[A1]**

(c) $\left(\frac{7}{10} \times \frac{3}{9}\right) + \left(\frac{3}{10} \times \frac{7}{9}\right)$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ **[A1]**

[Total Marks: 7]

1494 **(a)** 7 in the intersection **[B1]**

11, 8 and 4 in the correct regions **[B1]**

(b) $\frac{11}{30}$ **[B1]**

(c) $\frac{18}{30} \times \frac{17}{29}$ **[M1]**

$\frac{306}{870}$ or $\frac{51}{145}$ **[A1]**

(d) $\frac{7}{18} \times \frac{11}{17} + \frac{11}{18} \times \frac{7}{17}$ or $2 \times \left(\frac{7}{18} \times \frac{11}{17}\right)$ **[M2]**

Allow [M1] for one correct product $\frac{7}{18} \times \frac{11}{17}$

$\frac{154}{306}$ or $\frac{77}{153}$ **[A1]**

[Total Marks: 8]

1495 **(a)** First disc Blue $= \frac{5}{12}$ **[B1]**

Second disc Blue (after Red) $= \frac{5}{11}$ **[B1]**

Second disc Red (after Blue) $= \frac{7}{11}$ and Blue (after Blue) $= \frac{4}{11}$ **[B1]**

(b) **(i)** $\frac{7}{12} \times \frac{6}{11}$ **[M1]**

$\frac{7}{22}$ **[A1]**

(ii) $\left(\frac{7}{12} \times \frac{5}{11}\right) + \left(\frac{5}{12} \times \frac{7}{11}\right)$ **[M2]**

$\frac{35}{66}$ **[A1]**

(c) $\left(\frac{7}{12}\right)^2 + \left(\frac{5}{12}\right)^2$ **[M1]**

$\frac{49}{144} + \frac{25}{144} = \frac{74}{144}$ **[M1]**

$\frac{37}{72}$ **[A1]**

[Total Marks: 11]

1496 (a) $\frac{5}{n} \times \frac{4}{n-1} = \frac{2}{21}$ [M2]

(Award M1 for $\frac{5}{n} \times \frac{4}{n}$ or $\frac{5}{n} + \frac{4}{n-1}$)

$2(n^2 - n) = 420$ or $n(n-1) = 210$ **[M1]**

$n^2 - n - 210 = 0$ reached with no errors seen **[A1]**

(b) $(n-15)(n+14) = 0$ or correct substitution into quadratic formula **[M1]**

$n = 15$ **[A1]**

(c) $\frac{5}{15} \times \frac{10}{14}$ or $\frac{10}{15} \times \frac{5}{14}$ seen **[M1]**

Multiplies their probability by 2 (i.e. $2 \times \frac{50}{210}$) **[M1]**

$\frac{10}{21}$ or 0.476 (0.4761...) **[A1]**

[Total Marks: 8]

1497 (a) P(Blue from A) $= \frac{x}{x+3}$ and P(Red from A) $= \frac{3}{x+3}$ **[B1]**

If Blue transferred to B, Box B has 5 Blue and 2 Red (total 7), so P(Blue from B) $= \frac{5}{7}$ **[B1]**

If Red transferred to B, Box B has 4 Blue and 3 Red (total 7), so P(Blue from B) $= \frac{4}{7}$ **[B1]**

Total probability $= \left(\frac{x}{x+3} \times \frac{5}{7}\right) + \left(\frac{3}{x+3} \times \frac{4}{7}\right) = \frac{5x+12}{7(x+3)}$ **[A1]**

(b) (i) $\frac{5x+12}{7(x+3)} = \frac{3}{4}$ **[M1]**

$4(5x+12) = 21(x+3) \implies 20x + 48 = 21x + 63$ **[M1]**

$x = 15$ **[A1]**

(ii) $\frac{\text{P(Red from A and Blue from B)}}{\text{P(Blue from B)}} = \frac{\frac{3}{18} \times \frac{4}{7}}{\frac{3}{4}} = \frac{\frac{1}{14}}{\frac{3}{4}} = \frac{2}{21}$ **[B1]**

[Total Marks: 8]

1498 (a) $\frac{4}{x} \times \frac{3}{x-1}$ **[M1]**

$\dfrac{12}{x(x-1)} = \dfrac{1}{11}$ or $12 \times 11 = x(x-1)$ **[M1]**

$132 = x^2 - x \implies x^2 - x - 132 = 0$ **[A1]**

(b) $(x-12)(x+11) = 0$ or correct substitution into the quadratic formula **[M1]**

$x = 12$ **[A1]**

Number of blue discs $= 12 - 4 = 8$ **[A1]**

(c) $\dfrac{4}{12} \times \dfrac{8}{11}$ or $\dfrac{32}{132}$ **[M1]**

$\left(\dfrac{4}{12} \times \dfrac{8}{11}\right) \times 2$ **[M1]**

$\dfrac{64}{132}$ or $\dfrac{16}{33}$ **[A1]**

[Total Marks: 9]

1499 **(a)** $\dfrac{4}{n} \times \dfrac{3}{n-1} = \dfrac{1}{14}$ **[M1]**

$\dfrac{12}{n^2 - n} = \dfrac{1}{14}$ **[M1]**

$168 = n^2 - n$ leading to $n^2 - n - 168 = 0$ with no errors seen **[A1]**

(b) $(n-14)(n+12) = 0$ or correct use of quadratic formula **[M1]**

$n = 14$ (rejecting $n = -12$) **[A1]**

Number of dark chocolates $= 14 - 4 = 10$ **[A1]**

(c) $\dfrac{4}{14} \times \dfrac{10}{13}$ or $\dfrac{10}{14} \times \dfrac{4}{13}$ **[M1]**

$2 \times \left(\dfrac{4}{14} \times \dfrac{10}{13}\right)$ **[M1]**

$\dfrac{80}{182}$ or $\dfrac{40}{91}$ **[A1]**

[Total Marks: 9]

1500 **(a)** **(i)** Sets up equation $\dfrac{6}{n} \times \dfrac{5}{n-1} = \dfrac{1}{3}$ **[M1]**

Simplifies to $\dfrac{30}{n(n-1)} = \dfrac{1}{3}$ or $30 \times 3 = n(n-1)$ **[M1]**

$n^2 - n = 90$

$\mathrm{n}^2 - n - 90 = 0$ with no steps omitted [A1]

(ii) Factorises $(n-10)(n+9) = 0$ or uses quadratic formula correctly **[M1]**

Obtains $n = 10$ and $n = -9$ **[A1]**

Rejects negative root and states total number of sweets $= 10$ **[B1]**

(b) Calculates $P(\text{all 3 lemon}) = \dfrac{6}{10} \times \dfrac{5}{9} \times \dfrac{4}{8}$ **[M1]**

Identifies $P(\text{at least 1 orange}) = 1 - P(\text{all 3 lemon})$ **[M1]**

$= 1 - \frac{120}{720} = 1 - \frac{1}{6} = \frac{5}{6}$ (or 0.833) **[A1]**

[Total Marks: 9]

1501 **(a)** Sets up correct initial probability equation: $\frac{4}{n} \times \frac{3}{n-1} = \frac{2}{15}$ **[M1]**

Simplifies equation: $\frac{12}{n(n-1)} = \frac{2}{15} \implies 2n(n-1) = 180$ **[M1]**

Expands and rearranges correctly to establish $n^2 - n - 90 = 0$ with no missing algebraic steps **[A1]**

(b) Factorises $(n-10)(n+9) = 0$ or uses quadratic formula correctly **[M1]**

Rejects negative root $n = -9$ and states $n = 10$ **[A1]**

(c) Identifies total number of chocolates $n = 10$, dark $= 4$, milk $= 6$ **[B1]**

Calculates probability of one correct arrangement (e.g., DD M): $\frac{4}{10} \times \frac{3}{9} \times \frac{6}{8} = \frac{72}{720}$ or $\frac{1}{10}$ **[M1]**

Multiplies single selection probability by 3 (accounting for DD M, D MD, M DD) **[M1]**

$\frac{216}{720}$ or $\frac{3}{10}$ (or 0.3) **[A1]**

[Total Marks: 9]

1502 **(a)** $\frac{x}{x+4} \times \frac{x-1}{x+3} = \frac{5}{18}$ **[M1]**

$18x(x-1) = 5(x+4)(x+3)$ **[M1]**

$18x^2 - 18x = 5(x^2 + 7x + 12)$ **[M1]**

$13x^2 - 53x - 60 = 0$ obtained with no errors **[A1]**

(b) $(13x+12)(x-5) = 0$ or correct use of the quadratic formula [M2] (allow [M1] for one error in formula)

$x = 5$ or $x = -\frac{12}{13}$ **[A1]**

Selects $x = 5$ as the only valid solution for the number of apples **[B1]**

(c) Total apples $= 9$, green $= 5$, red $= 4$. Evaluates one branch: $\frac{5}{9} \times \frac{4}{8}$ or $\frac{4}{9} \times \frac{5}{8}$ **[M1]**

$\left(\frac{5}{9} \times \frac{4}{8}\right) + \left(\frac{4}{9} \times \frac{5}{8}\right)$ or multiplies one branch by 2 **[M1]**

$\frac{40}{72}$ or $\frac{5}{9}$ **[A1]**

[Total Marks: 10]

1503 **(a)** Boys Total $= 40$ and Girls Football $= 8$ **[B1]**

Girls Basketball $= 8$ and Total Tennis $= 27$ **[B1]**

Boys Tennis $= 13$ **[B1]**

(b) $\frac{13}{70}$ **[B1]**

[Total Marks: 4]

1504 **(a)** Tennis Total = 15, Girls Total = 22 **[B1]**

Boys Football = 12, Boys Swimming = 9 **[B1]**

Girls Football = 8 **[B1]**

(b) $\frac{12}{50}$ or $\frac{6}{25}$ **[B1]**

[Total Marks: 4]

1505 **(a)** 5 in Boys/Swimming cell **[B1]**

5 in Girls/Football, 9 in Girls/Swimming, 22 in Girls/Total **[B1]**

16 in Total/Tennis, 14 in Total/Swimming **[B1]**

(b) $\frac{9}{50}$ **[B1]**

[Total Marks: 4]

1506 **(a)** Boys Basketball = 8 **[B1]**

Girls Football = 6 AND Girls Total = 22 **[B1]**

Total Basketball = 18 AND Total Tennis = 11 **[B1]**

(b) Denominator of 11 seen or fraction $\frac{5}{n}$ where $n > 5$ **[M1]**

$\frac{5}{11}$ **[A1]**

[Total Marks: 5]

1507 **(a)** 6 or more correct missing entries **[B3]**

4 or 5 correct missing entries **[B2]**

2 or 3 correct missing entries **[B1]**

(Correct Boys row: 24, 10, 11, 45)

(Correct Girls row: 14, 12, 9, 35)

(Correct Total row: 38, 22, 20, 80)

(b) $\frac{10}{80}$ or $\frac{1}{8}$ **[B1]**

(c) Denominator of 35 seen **[M1]**

$\frac{9}{35}$ **[A1]**

[Total Marks: 6]

1508 **(a)** **(i)** Qualitative **[B1]**

(ii) Continuous **[B1]**

(iii) Discrete **[B1]**

(b) Row $60 \leq r < 70$: correct tally of 8 and frequency 8 **[B1]**

Row $70 \leq r < 80$: correct tally of 8 and frequency 8 **[B1]**

Row $80 \leq r < 90$: correct tally of 4 and frequency 4 **[B1]**

(c) $\dfrac{4+8}{24} \times 100$ **[M1]**

50 **[A1]**

[Total Marks: 8]

1509 **(a)** $x + (2x + 10) + (x + 5) + (3x - 11)$ **[M1]**

$7x + 4$ **[A1]**

(b) $7x + 4 = 130$ **[M1]**

$x = 18$ **[A1]**

(c) Correct Male row: 18, 46, 64 **[B1]**

Correct Female row: 23, 43, 66 **[B1]**

Correct Total row: 41, 89, 130 **[B1]**

(d) Denominator of 41 seen **[M1]**

$\dfrac{23}{41}$ **[A1]**

[Total Marks: 9]

1510 **(a)** **(i)** $(2x + y) + (x + 6) = 28$ **[M1]**

$22 - 3x$ **[A1]**

(ii) $(x^2 + 2x) + (2x + y) + (4x - y) + (2x + 3y) + (x + 6) + (3x + 3y) = 150$ simplifying to $x^2 + 14x + 6y + 6 = 150$ **[M1]**

$x^2 + 14x + 6(22 - 3x) - 144 = 0$ **[M1]**

$x^2 + 14x + 132 - 18x - 144 = 0$ simplifying to $x^2 - 4x - 12 = 0$ **[A1]**

(b) $(x - 6)(x + 2) = 0$ **[M1]**

$x = 6$ (rejecting $x = -2$ since $x > 0$) **[A1]**

$y = 22 - 3(6) = 4$ **[A1]**

(c) **(i)** Row Yes: $48, 16, 20$, Total $= 84$ **[B1]**

Row No: $24, 12, 30$, Total $= 66$; Column Totals: $72, 28, 50$ **[B1]**

(ii) $\dfrac{84}{150} \times 100$ **[M1]**

56 **[A1]**

[Total Marks: 12]

1511 **(a)** Total Women = 55, Total Tea = 70, Overall Total = 120 placed correctly **[B1]**

Men Coffee = 25 and Total Coffee = 50 or Men Total = 65 placed correctly **[B1]**

Men Tea = 40, Women Tea = 30, Women Coffee = 25 **[B1]**

(b) $\frac{30}{120} \times 100$ **[M1]**

25 **[A1]**

[Total Marks: 5]

1512 **(a)** Total column correct: 75 (Men), 45 (Women), 120 (Total) **[B1]**

Bus column correct: 15 (Men), 15 (Women), 30 (Total) **[B1]**

Train column correct: 24 (Men), 14 (Women), 38 (Total) **[B1]**

Car column correct: 36 (Men), 16 (Women), 52 (Total) **[B1]**

(b) Denominator of 52 seen or fraction $\frac{36}{n}$ **[M1]**

$\frac{36}{52}$ or $\frac{9}{13}$ **[A1]**

[Total Marks: 6]

1513 **(a)** Total Juniors = 240, Total Seniors = 160 **[B1]**

Total Bus = 180, Total Car = 110, Total Walk = 110 **[B1]**

Juniors: Bus = 100, Car = 80, Walk = 60 AND Seniors: Bus = 80, Car = 30, Walk = 50 **[B1]**

(b) $\frac{60}{400}$ or $\frac{3}{20}$ **[B1]**

(c) $\frac{80}{160} \times \frac{79}{159}$ **[M1]**

$\frac{79}{318}$ **[A1]**

[Total Marks: 6]

1514 **(a)** $x + 2x - 5 + x + 6 + x - 2 + x + 5 + 2x + 1 = 85$ or $8x + 5 = 85$ **[M1]**

$8x = 80$ **[M1]**

10 **[A1]**

(b) Man who prefers Water $= 10 + 6 = 16$ **[M1]**

$\frac{16}{85}$ **[A1]**

(c) Total who prefer Coffee $= 15 + 15 = 30$ **[M1]**

$\frac{15}{30}$ or $\frac{1}{2}$ **[A1]**

[Total Marks: 7]

1515 **(a)** Row totals: Year 9 = 96, Year 10 = 114, Year 11 = 90 **[B1]**

Column total Bus = 100 **[B1]**

Year 9 row completed correctly: Walk = 35, Bus = 36, Car = 25 **[B1]**

Year 11 row completed correctly: Walk = 30, Bus = 42 **[B1]**

Year 10 row completed correctly: Walk = 50, Bus = 22, Car = 42 **[B1]**

(b) $\frac{42}{114} \times 100$ $(= 36.842\ldots)$ **[M1]**

36.8 **[A1]**

(c) $\frac{42}{100}$ **[M1]**

$\frac{21}{50}$ **[A1]**

[Total Marks: 9]

1516 **(a)** $x + 2x + (x + 5) + (3x - 2) = 80$ **[M1]**

$7x + 3 = 80$ **[M1]**

$x = 11$ **[A1]**

(b) Boys Bus = 11, Boys Walk = 22, Boys Total = 33 **[B1]**

Girls Bus = 16, Girls Walk = 31, Girls Total = 47 and Column totals (Bus = 27, Walk = 53) correct **[B1]**

(c) Denominator of 53 seen **[M1]**

$\frac{31}{53}$ **[A1]**

[Total Marks: 7]

1517 **(a)** Boys total = 120 and Girls total = 180 **[B1]**

Boys/Art = 24 **[B1]**

Boys/Drama = 48 and Boys/Music = 48 **[B1]**

Girls/Drama = 60 **[B1]**

Girls/Art = 45 and Girls/Music = 75 AND all totals correct $(69, 123, 108, 300)$ **[B1]**

(b) Girls who do not choose Drama = 45 + 75 = 120 **[M1]**

$\frac{120}{300}$ or $\frac{2}{5}$ **[A1]**

[Total Marks: 7]

1518 **(a)** Correct Managers row: 48, 12, 60 **[B1]**

Correct Office Workers row: 90, 60, 150 **[B1]**

Correct Technicians row: 78, 12, 90 **[B1]**

Correct Total row: 216, 84, 300 **[B1]**

(b) Denominator of 84 seen **[M1]**

$\frac{60}{84}$ or $\frac{5}{7}$ **[A1]**

(c) $\frac{12}{60} \times \frac{48}{59}$ **[M1]**

$\frac{12}{60} \times \frac{48}{59} + \frac{48}{60} \times \frac{12}{59}$ or $2 \times \left(\frac{12}{60} \times \frac{48}{59}\right)$ **[M1]**

$\frac{96}{295}$ or 0.325 (0.3254...) **[A1]**

[Total Marks: 9]

1519 **(a)** $\frac{3}{4}x$ or $0.75x$ **[B1]**

(b) Forms correct equation: $\frac{3}{4}x + \left(\frac{1}{4}x + 30\right) = 110$ **[M1]**

Simplifies to $x + 30 = 110$ leading to $x = 80$ with no errors **[A1]**

(c) Male row correct: 20 (Golf), 60 (Tennis), 80 (Total) **[B1]**

Female row correct: 70 (Golf), 50 (Tennis), 120 (Total) **[B1]**

Total row correct: 90 (Golf), 110 (Tennis), 200 (Total) **[B1]**

(d) Identifies correct fraction for first female: $\frac{70}{90}$ **[M1]**

Multiplies by correct fraction for second female: $\frac{70}{90} \times \frac{69}{89}$ **[M1]**

$\frac{4830}{8010}$ or $\frac{161}{267}$ **[A1]**

[Total Marks: 9]

1520 **(a)** Total number of left-handed students $= 3x$ **[B1]**

$\frac{3x}{120} \times \frac{3x-1}{119} = \frac{3}{34}$ **[M1]**

$\frac{9x^2 - 3x}{14280} = \frac{3}{34}$ or $3x(3x-1) = 1260$ **[M1]**

$9x^2 - 3x = 1260 \implies 3x^2 - x - 420 = 0$ (with no errors seen) **[A1]**

(b) $(3x - 36)(x + 11.66...) = 0$ or correct substitution into quadratic formula **[M1]**

$x = 12$ **[A1]**

Total students equation: $x + y + 2x + y - 6 = 120$ **[M1]**

$3(12) + 2y - 6 = 120 \implies 2y = 90 \implies y = 45$ **[A1]**

(c) Total right-handed $= 2y - 6 = 2(45) - 6 = 84$ OR Right-handed girls $= 45 - 6 = 39$ **[M1]**

$\frac{39}{84}$ or $\frac{13}{28}$ **[A1]**

[Total Marks: 10]

1521 **(a)** Class B, because the mean (or median) is higher. **[B1]**

(b) Class B, because the range (or interquartile range) is smaller. **[B1]**

(c) Identifies that the table only gives summary statistics / does not show individual marks **[B1]**

Mentions overlapping ranges / a specific example (e.g. the lowest mark in Class B could be lower than the highest mark in Class A) **[B1]**

[Total Marks: 4]

1522 **(a)** Median $= 2$ **[B1]**

Range $= 4$ **[B1]**

(b) Team A generally scored more goals because their median is higher $(2 > 1.5)$ **[B1]**

Team A was more consistent because their range is smaller $(4 < 5)$ **[B1]**

[Total Marks: 4]

1523 **(a)** B **[B1]**

(b) 3 **[B1]**

(c) Total students $= 5+10+3+2+7+8+3+2 = 40$ AND Total Grade A $= 5+7 = 12$ **[M1]**

30 **[A1]**

[Total Marks: 4]

1524 **(a)** 27 **[B1]**

(b) Upper quartile $= 35$ OR Lower quartile $= 20$ **[M1]**

15 **[A1]**

(c) $\frac{7}{15}$ **[B1]**

[Total Marks: 4]

1525 **(a)** 27 **[B1]**

(b) 31 **[B1]**

(c) 36 and 20 identified as quartiles **[M1]**

16 **[A1]**

(d) $\frac{6}{15}$ or $\frac{2}{5}$ **[B1]**

[Total Marks: 5]

1526 **(a)** Median $= 32$ **[B1]**

Interquartile range $= 40 - 25 = 15$ **[B1]**

(b) Class A has a higher median score ($32 > 29$), so on average Class A performed better than Class B **[B1]**

Class A has a smaller interquartile range ($15 < 18$), so Class A's scores are more consistent (less spread out) than Class B's **[B1]**

(c) In Class B, 3 students scored above 40 ($41, 46, 49$), and in Class A, 3 students also scored above 40 ($42, 45, 47$, as 40 is not above 40) **[M1]**

No, the claim is incorrect because both classes had the same number (3 students) scoring above 40 **[A1]**

(d) The results are from only one test (or a small sample of 15 students per class) and may not represent overall mathematical ability across all topics **[B1]**

[Total Marks: 7]

1527 **(a)** Identifies the middle value is 7 **[M1]**

Identifies that 3 must appear at least twice as the lowest values: $3, 3$ **[M1]**

Uses the range to find the maximum value: $3 + 11 = 14$ **[M1]**

Uses the mean to find the missing value: $40 - (3 + 3 + 7 + 14) = 13$, yielding $3, 3, 7, 13, 14$ **[A1]**

(b) New sum $= 6 \times 9 = 54$ AND Old sum $= 5 \times 8 = 40$ (or $54 - 40$) **[M1]**

14 **[A1]**

(c) No, because standard deviation measures the spread of all data points around the mean, not just the difference between the extreme values **[B1]**

[Total Marks: 7]

1528 **(a)** $y = 80 - (12 + 28 + 14 + 6) = 20$ **[B1]**

Calls lasting more than 8 minutes in Branch South $= 35 + 20 + 10 = 65$, giving $\frac{65}{80 + x} = \frac{13}{20}$ **[M1]**

$13(80 + x) = 1300$ giving $80 + x = 100$ **[M1]**

$x = 20$ **[A1]**

(b) Branch North: $\sum fm = 12(2) + 28(6) + 20(10) + 14(14) + 6(18) = 696$ **[M1]**

Branch North estimated mean $= \frac{696}{80} = 8.7$ minutes **[A1]**

Branch South: $\sum fm = 15(2) + 20(6) + 35(10) + 20(14) + 10(18) = 960$ **[M1]**

Branch South estimated mean $= \frac{960}{100} = 9.6$ minutes **[A1]**

(c) Statement 1 is valid because the shortest possible call duration is just above 0 minutes and the longest possible duration is 20 minutes, giving a maximum possible range of $20 - 0 = 20$ minutes **[B2]**

Statement 2 is not valid because the 35 calls in the interval $8 < t \leq 12$ are grouped, so all 35 calls could lie outside $9 \leq t \leq 11$ (for example, between 8 and 9 minutes) **[B2]**

[Total Marks: 12]

1529 **(a)** 40 **[B1]**

(b) 25 (from $50 - 25$) **[B1]**

(c) Group Y was slower on average / had a higher median time ($40 > 35$) **[B1]**

Group Y's times were more consistent / had a smaller IQR ($15 < 25$) or smaller range ($40 < 60$) **[B1]**

(d) Identifies that exactly 25% of Group Y took less than 30 minutes (since lower quartile is 30) **[M1]**

States Group X had a larger proportion, because 30 is between their LQ (25) and median (35), meaning more than 25% took less than 30 minutes **[A1]**

[Total Marks: 6]

1530 **(a)** 45 **[B1]**

(b) 10 **[B1]**

(c) Group A students were faster on average because their median time (45 min) is less than Group B (50 min). **[B1]**

Group B times are more consistent / less spread out because their interquartile range (10 min) is smaller than Group A (20 min). **[B1]**

(d) Recognises that 55 is the upper quartile, representing 25% or $\frac{1}{4}$ of the students **[M1]**

20 **[A1]**

[Total Marks: 6]

1531 **(a)** 65 **[B1]**

(b) $75 - 40$ **[M1]**

35 **[A1]**

(c) Class X has a higher average score (or median is higher: $65 > 55$) **[B1]**

Class X scores are less spread out / more consistent (or IQR is lower: $30 < 35$, or Range is lower: $60 < 65$) **[B1]**

(d) No, because the ranges overlap (e.g., the maximum score in Class Y is 90, which is higher than the lower quartile of Class X) **[B1]**

[Total Marks: 6]

1532 **(a)** $120 < m \leq 140$ **[B1]**

(b) Use of correct midpoints: $110, 130, 150, 170, 190$ **[M1]**

$110 \times 10 + 130 \times 20 + 150 \times 15 + 170 \times 10 + 190 \times 5$ **[M1]**

$= 1100 + 2600 + 2250 + 1700 + 950 = 8600$ **[M1]**

$\frac{8600}{60} = 143.33\ldots$ so 143 (to 3 sf) **[A1]**

(c) The exact masses of the individual mangoes are unknown because the data is grouped **[B1]**

The 15 mangoes could have any combination of masses between 140 g and 160 g, not necessarily exactly at the midpoint **[B1]**

[Total Marks: 7]

1533 **(a)** Sum of goals $= (0 \times 8) + (1 \times 12) + (2 \times x) + (3 \times 6) + (4 \times y) + (5 \times 2)$ OR $12 + 2x + 18 + 4y + 10$ **[M1]**

Equates sum of goals to 40×1.7 OR $\frac{40 + 2x + 4y}{40} = 1.7$ **[M1]**

$40 + 2x + 4y = 68$

2x + 4y = 28$(Correct completion with no errors)[A1]$

(b) $8 + 12 + x + 6 + y + 2 = 40$ **[M1]**

$x + y = 12$ **[A1]**

Solves simultaneous equations: e.g. $2x + 4(12 - x) = 28$

2x + 48 - 4x = 28

2x = 20$[M1]$

$x = 10, y = 2$ **[A1]**

[Total Marks: 7]

1534 **(a)** **(i)** 34 **[B1]**

(ii) $48 - 26$ **[M1]**

22 **[A1]**

(b) **(i)** 0.25×240 **[M1]**

60 **[A1]**

(ii) For Company X, 22 and 38 are the lower and upper quartiles, so 50% of 160 = 80 deliveries took between 22 and 38 minutes **[B1]**

For Company Y, 22 and 38 are not quartiles, so the exact proportion of deliveries in this interval cannot be determined from the box plot **[B1]**

Since Company Y had more total deliveries (240 > 160), Company Y could have more than 80 deliveries in that interval **[B1]**

(c) Company X **[B1]**

Company X has a lower median delivery time (30 minutes $<$ 34 minutes), so it is faster on average **[B1]**

Company X has a smaller interquartile range (16 minutes $<$ 22 minutes), so its delivery times are more consistent **[B1]**

[Total Marks: 11]

1535 **(a)** Upper Quartile = 44 and Lower Quartile = 26 **[M1]**

$44 - 26 = 18$ **[A1]**

(b) The Adults generally completed the puzzle faster because their median time is lower ($32 < 36$) **[B1]**

The Students' times were more consistent (less spread out) because their interquartile range (12) is smaller than the Adults' interquartile range (18) **[B1]**

(c) The reasoning is invalid because range only measures spread, not the actual values or maximum times **[B1]**

However, the conclusion happens to be true because the maximum time for Students (48) is less than the maximum time for Adults (55) **[B1]**

[Total Marks: 6]

1536 **(a)** Forms a correct equation: $x + 2x + 120 + x = 360$ **[M1]**

Solves to find $4x = 240$

x = 60$[M1]$

Calculates number of students: $\frac{60}{360} \times 120 = 20$ **[A1]**

(b) Calculates the percentage for Class P: $\frac{120^\circ}{360^\circ} \times 100 = 33.3\%$ (or $\frac{1}{3}$) **[M1]**

Identifies total ratio parts for Class Q $= 1 + 3 + 2 + 2 = 8$ **[M1]**

Calculates percentage for Class Q: $\frac{3}{8} \times 100 = 37.5\%$ and concludes $37.5\% > 33.3\%$ **[A1]**

(c) The two classes have different total numbers of students (so proportions/percentages must be used for a fair comparison). **[B1]**

[Total Marks: 7]

1537 **(a)** 112 **[B1]**

(b) 120 and 104 seen as quartiles **[M1]**

16 **[A1]**

(c) Cumulative frequency of 110 seen, leading to $120 - 110 = 10$ apples **[M1]**

8.33 (8.333...) **[A1]**

(d) Farm X apples are heavier on average as their median (112 g) is greater than Farm Y (105

g). **[B1]**

Farm X apples are more consistent in mass / less spread out as their interquartile range (16 g) is smaller than Farm Y (22 g). **[B1]**

[Total Marks: 7]

1538 **(a)** **(i)** 136 to 138 **[B1]**

(ii) Upper quartile (90th) = 146 to 148 AND Lower quartile (30th) = 126 to 128 **[M1]**

18 to 22 **[A1]**

(b) Reads cumulative frequency at 150g as 98 (allow 97 to 99) **[M1]**

$120 - 98 = 22$ (allow 21 to 23) **[A1]**

(c) Number of apples $\leq$ 150g is 98 (or their read value from (b)) **[B1]**

$\frac{98}{120} \times \frac{97}{119}$ **[M1]**

$\frac{9506}{14280}$ or $\frac{4753}{7140}$ or 0.666 $(0.6656\ldots)$ **[A1]**

[Total Marks: 8]

1539 **(a)** A and D **[B1]**

(b) B and C **[B1]**

[Total Marks: 2]

1540 **(a)** Continuous **[B1]**

(b) Discrete **[B1]**

(c) Continuous **[B1]**

[Total Marks: 3]

1541 **(a)** **(i)** Discrete **[B1]**

(ii) Continuous **[B1]**

(iii) Continuous **[B1]**

(iv) Discrete **[B1]**

[Total Marks: 4]

1542 **(a)** Discrete **[B1]**

(b) Continuous **[B1]**

(c) Continuous **[B1]**

(d) Continuous **[B1]**

[Total Marks: 4]

1543 **(a)** Discrete **[B1]**

(b) Continuous **[B1]**

(c) Continuous **[B1]**

(d) Discrete **[B1]**

[Total Marks: 4]

1544 **(a)** **(i)** Discrete **[B1]**

(ii) Continuous **[B1]**

(iii) Continuous **[B1]**

(iv) Discrete **[B1]**

(b) Width is measured (rather than counted) and can take any numerical value within a given interval **[B1]**

(c) A measurement of 6 metres would belong to both intervals (overlapping boundary) **[B1]**

$4 \leq w < 6$ (or $4 < w \leq 6$) **[B1]**

[Total Marks: 7]

1545 **(a)** Lower bound = 37.5 **[B1]**

Upper bound = 42.5 **[B1]**

(b) Smallest possible number = 1248 **[B1]**

Largest possible number = 1252 **[B1]**

(Note: Continuous boundaries are 1247.5 and 1252.5, but as a discrete integer variable, it must take the integer values strictly within this range.)

[Total Marks: 4]

1546 **(a)** Data Set A (time) is continuous, so it can take any decimal value and inequalities are needed to define exact boundary limits without gaps. **[B1]**

Data Set B (number of push-ups) is discrete, so it only takes integer values and a hyphen clearly defines the inclusive start and end integers. **[B1]**

(b) **(i)** 19.5 and 29.5 **[B1]**

(ii) 10 **[B1]**

(c) Age (usually given as an integer in years) OR Money/Currency (rounded to nearest cent/pence) OR Shoe size **[B1]**

[Total Marks: 5]

1547 **(a)** k is discrete **[B1]**

m is continuous **[B1]**

(b) For one rod, $1.95 \leq r < 2.05$, and for the total mass, $19.95 \leq m < 20.05$ **[B1]**

$1.95k < 20.05$ giving $k < \dfrac{401}{39} = 10\dfrac{11}{39}$ **[M1]**

$2.05k > 19.95$ giving $k > \dfrac{399}{41} = 9\dfrac{30}{41}$ **[M1]**

Since k is discrete, k must be an integer, so $k = 10$ **[A1]**

(c) **(i)** Mean of $d = \dfrac{0(24) + 1(15) + 2(7) + 3(3) + 4(1)}{50} = \dfrac{42}{50}$ **[M1]**

0.84 **[A1]**

d is discrete with exact individual values known, whereas L is continuous data grouped into intervals so individual rod lengths are unknown and mid-interval values must be used **[B1]**

(ii) $\dfrac{18 + 17}{50} \times 100 = 70\%$ **[B1]**

L is continuous data grouped in the interval $50.0 \leq L < 50.2$, so the exact distribution of the 17 lengths within that interval is unknown **[B1]**

[Total Marks: 11]

1548 **(a)** Continuous **[B1]**

(b) Height is continuous / can take decimal values. **[B1]**

A plant with a height between the intervals (e.g. 19.5 cm) would not fit into any class. **[B1]**

(c) Discrete **[B1]**

[Total Marks: 4]

1549 **(a)** Discrete data can only take specific or exact values (often counted) **[B1]**

Continuous data can take any value within a given range (often measured) **[B1]**

(b) **(i)** Continuous **[B1]**

(ii) Rounding limits the precision of the recording, but the actual true length could still be any value between the rounded boundaries **[B1]**

[Total Marks: 4]

1550 **(a)** Continuous **[B1]**

(b) Uses midpoints: 90, 110, 130, 150 **[M1]**

$\Sigma fx = 90 \times 12 + 110 \times 26 + 130 \times 15 + 150 \times 7 = 6940$ **[M1]**

$\dfrac{6940}{60} = 115.66... = 116$ or 115.7 **[A1]**

(c) The exact masses of the apples in each class are not known (or similar reference to data

being grouped) **[B1]**

[Total Marks: 5]

1551 **(a)** Length **[B1]**

(b) **(i)** 14.45 **[B1]**

(ii) 14.55 **[B1]**

(c) Identifies that the discrete values include $5, 6, 7, 8, 9$ OR uses continuous boundaries $9.5 - 4.5$ **[M1]**

5 **[A1]**

[Total Marks: 5]

1552 **(a)** Continuous **[B1]**

(b) Identifies midpoints: $150, 250, 350, 450$ **[M1]**

1553 **(a)** N is discrete (integer values only), so there are no values between 12 and 13 and inclusive inequalities ($10 \leq N \leq 12$, $13 \leq N \leq 15$) cover all counts without gaps **[B1]**

T is continuous, so strict upper inequalities ($48 \leq T < 50$, $50 \leq T < 52$) are needed to include all decimal values up to 50 without overlapping **[B1]**

(b) For N: $\sum fm = 6(11) + 14(14) + 12(17) + 8(20) = 626$ **[M1]**

Estimated mean of $N = \dfrac{626}{40} = 15.65$ **[A1]**

For T: $\sum fm = 5(49) + 15(51) + 13(53) + 7(55) = 2084$ **[M1]**

Estimated mean of $T = \dfrac{2084}{40} = 52.1$ seconds **[A1]**

(c) **(i)** The mean is a measure of average and does not have to be a possible value of a discrete variable **[B1]**

No, the student is incorrect; 15.65 is a valid mean even though N takes only integer values **[B1]**

(ii) Diagram for N: Bar chart (or vertical line graph) **[B1]**

Diagram for T: Histogram (or cumulative frequency curve / frequency polygon) **[B1]**

[Total Marks: 10]

1554 **(a)** **(i)** Discrete **[B1]**

(ii) Continuous **[B1]**

(b) The variable can still only take specific, fixed intervals/values (e.g., jumping by 0.5) **[B1]**

It cannot take any value within a range (e.g., a shoe size of 8.23 does not exist) **[B1]**

[Total Marks: 4]

1555 **(a)** Continuous data can take any value within a given range (measured). **[B1]**

Discrete data can only take specific, exact values (usually counted). **[B1]**

(b) **(i)** Continuous **[B1]**

(ii) Shoe size can only take specific, set values (there are gaps between sizes like 7 and 7.5). **[B1]**

It cannot take every possible value (e.g. you cannot have a shoe size of 7.23), so it must be discrete. **[B1]**

[Total Marks: 5]

1556 **(a)** **(i)** Discrete **[B1]**

(ii) Continuous **[B1]**

(b) Upper bound of mass of one book = 855 g **[M1]**

$24 \times 855 = 20520$ g $\implies$ 20.52 **[A1]**

(c) Lower bound of empty box = 1.15 kg **[B1]**

Lower bound of mass of one book = 845 g = 0.845 kg seen **[B1]**

Total lower bound $= 1.15 + (24 \times 0.845) = 21.43$ **[A1]**

[Total Marks: 7]

1557 **(a)** **(a)** 4 **[B1]**

(b) Orders the data: $4, 4, 5, 7, 8, 9, 12$ **[M1]**

7 **[A1]**

(c) $12 - 4 = 8$ **[B1]**

[Total Marks: 3]

1558 **(a)** 1 **[B1]**

(b) Ordered list: $0, 0, 1, 1, 1, 1, 2, 2, 3, 4$ **[M1]**

1 **[A1]**

(c) $\dfrac{2+0+1+4+1+0+3+1+2+1}{10} = \dfrac{15}{10}$ **[M1]**

1.5 **[A1]**

[Total Marks: 5]

1559 **(a)** 16 **[B1]**

(b) 21 **[B1]**

(c) Ordered list seen: $5, 8, 10, 13, 14, 16, 18, 20, 22, 24, 26$ **[M1]**

12 **[A1]**

(d) $(14 + 20 + 8 + 24 + 16 + 5 + 26 + 13 + 22 + 10 + 18) \div 11$ or $176 \div 11$ **[M1]**

16 **[A1]**

[Total Marks: 6]

1560 **(a)** 7 **[B1]**

(b) $12 - 7 = 5$ **[B1]**

(c) 8 **[B1]**

(d) $(7 + 7 + 7 + 8 + 10 + 12 + 12) \div 7$ or $\frac{63}{7}$ **[M1]**

9 **[A1]**

(e) $\frac{63 + y}{8} = 9.5$ or 9.5×8 **[M1]**

$76 - 63 = 13$ **[A1]**

[Total Marks: 7]

1561 **(a)** 15 **[B1]**

(b) **(i)** Ordered list: $8, 10, 11, 13, 14, 15, 15, 18, 21, 23, 50$ **[M1]**

15 **[A1]**

(ii) Lower quartile $Q_1 = 11$ and upper quartile $Q_3 = 21$ **[M1]**

$21 - 11 = 10$ **[A1]**

(c) $\frac{8 + 10 + 11 + 13 + 14 + 15 + 15 + 18 + 21 + 23 + 50}{11} = \frac{198}{11}$ **[M1]**

18 **[A1]**

(d) The value 50 is an extreme outlier that increases the mean, whereas the median is not affected by extreme values **[B1]**

[Total Marks: 8]

1562 **(a)** $4 + x + 18 + y + 6 = 50$ simplifying to $x + y = 22$ **[B1]**

(b) **(i)** $\sum fm = 4(5) + 15x + 18(25) + 35y + 6(45) = 15x + 35y + 740$ **[M1]**

$\frac{15x + 35y + 740}{50} = 27$ giving $15x + 35y + 740 = 1350$ **[M1]**

$15x + 35y = 610$ dividing by 5 to give $3x + 7y = 122$ **[A1]**

(ii) Eliminating one variable, for example $(3x + 7y) - 3(x + y) = 122 - 66$ giving $4y = 56$ **[M1]**

$y = 14$ **[A1]**

$x = 8$ **[A1]**

(c) $20 < h \leq 30$ **[B1]**

(d) **(i)** $30 < h \leq 40$ **[B1]**

(ii) New $\sum fm = 1350 + 10(35) = 1700$ **[M1]**

$\frac{1700}{60} = \frac{85}{3}$ **[M1]**

$28\frac{1}{3}$ **[A1]**

[Total Marks: 12]

1563 **(a)** $30 < h \leq 40$ **[B1]**

(b) Uses midpoints $15, 25, 35, 45, 55$ **[M1]**

$\Sigma fx = (15 \times 12) + (25 \times 25) + (35 \times 28) + (45 \times 11) + (55 \times 4)$ or
$180 + 625 + 980 + 495 + 220 = 2500$ **[M1]**

$\frac{2500}{80}$ **[M1]**

31.25 **[A1]**

[Total Marks: 5]

1564 **(a)** **(a)** $20 < h \leq 30$ **[B1]**

(b) Uses midpoints: $5, 15, 25, 35, 45$ **[M1]**

$\Sigma fx = (5 \times 4) + (15 \times 12) + (25 \times 20) + (35 \times 10) + (45 \times 4)$ **[M1]**

$\Sigma fx = 20 + 180 + 500 + 350 + 180 = 1230$ **[M1]**

$\frac{1230}{50} = 24.6$ **[A1]**

[Total Marks: 5]

1565 **(a)** $100 < m \leq 110$ **[B1]**

(b) Midpoints used: $85, 95, 105, 115, 125$ **[M1]**

$\Sigma fx = 85 \times 8 + 95 \times 22 + 105 \times 31 + 115 \times 14 + 125 \times 5$ **[M1]**

$\Sigma fx = 680 + 2090 + 3255 + 1610 + 625 = 8260$ **[M1]**

Mean $= \frac{8260}{80} = 103.25$ **[A1]**

(c) The exact masses of individual apples are not known **[B1]**

The calculation uses the midpoints of the class intervals, assuming the data is evenly distributed within each class **[B1]**

[Total Marks: 7]

1566 **(a)** $30 < t \leq 40$ **[B1]**

(b) Midpoints: $15, 25, 35, 45, 55$ **[M1]**

$\Sigma fx = 15(14) + 25(28) + 35(32) + 45(18) + 55(8)$ **[M1]**

$= 210 + 700 + 1120 + 810 + 440 = 3280$ **[M1]**

$\frac{3280}{100} = 32.8$ **[A1]**

(c) $18 + 8 = 26$ **[M1]**

26 **[A1]**

(d) $40 < t \leq 50$ **[B1]**

[Total Marks: 8]

1567 **(a)** $140 \leq m < 160$ **[B1]**

(b) Mid-interval values: $110, 130, 150, 170, 190$ **[B1]**

$\sum fm = 10(110) + 22(130) + 28(150) + 14(170) + 6(190) = 11680$ **[M1]**

$\frac{11680}{80}$ **[M1]**

146 **[A1]**

(c) Total mass of second crate $= 40 \times 155 = 6200$ **[M1]**

$\frac{11680 + 6200}{120} = \frac{17880}{120}$ **[M1]**

149 **[A1]**

(d) $140 \leq m < 160$ **[B1]**

[Total Marks: 9]

1568 **(a)** $p + q = 25$ **[B1]**

$\frac{10p + 420 + 800 + 70q + 450}{60} = 48$ or $10p + 70q = 1210$ [B2] (Allow B1 for unsimplified $\Sigma fx = 10p + 420 + 800 + 70q + 450$ seen)

Correctly sets up and solves simultaneous equations to eliminate one variable **[M1]**

$p = 9$, $q = 16$ **[A1]**

(b) Identifies median is between 30th and 31st value and shows cumulative frequencies 23 and 39 (or $9 + 14 = 23$ and $23 + 16 = 39$) **[M1]**

$40 < m \leq 60$ **[A1]**

[Total Marks: 7]

1569 **(a)** Since the mode is 8, there must be at least two 8s. **[M1]**

Since the median is 11, the middle number when ordered is 11. The first three numbers are $8, 8, 11$. **[M1]**

The mean is 12, so the sum of the five numbers is $5 \times 12 = 60$. **[M1]**

The range is 15, so the largest number is $8 + 15 = 23$. **[M1]**

Let the fourth number be x. Then $8 + 8 + 11 + x + 23 = 60 \implies 50 + x = 60 \implies x = 10$. **[A1]**

(b) New sum $= 6 \times 13 = 78$ **[M1]**

$k = 78 - 60 = 18$ **[A1]**

[Total Marks: 7]

1570 **(a)** **(a)** Sets up equation for the mean: $\dfrac{(x-3)+x+(x+2)+2x+(2x+5)+3x}{6} = 9$ **[M1]**

Simplifies numerator: $10x + 4 = 54$ **[M1]**

$10x = 50 \implies x = 5$ **[A1]**

(b) Substitutes $x = 5$ to find the ordered list: $2, 5, 7, 10, 15, 15$ **[M1]**

Lower quartile $= 5$ and Upper quartile $= 15$, so IQR $= 15 - 5 = 10$ **[A1]**

(c) Recognises that for 7 numbers, the median is the 4th value in the ordered list. **[M1]**

The current values are $2, 5, 7, 10, 15, 15$. For the 4th value to be 8, y must be placed exactly at the 4th position, giving $2, 5, 7, 8, 10, 15, 15$. Thus $y = 8$. **[A1]**

[Total Marks: 7]

1571 **(a)** $15 + x + y + 8 + 2 = 50$ leading to $x + y = 25$ **[B1]**

(b) Midpoints: $1, 3, 5, 7, 9$ **[M1]**

$\dfrac{1(15) + 3(x) + 5(y) + 7(8) + 9(2)}{50} = 3.8$ **[M1]**

$15 + 3x + 5y + 56 + 18 = 190$ **[M1]**

$3x + 5y + 89 = 190$ leading to $3x + 5y = 101$ **[A1]**

(c) $3x + 3y = 75$ or substitution method $3(25 - y) + 5y = 101$ **[M1]**

$2y = 26 \implies y = 13$ **[A1]**

$x = 12$ **[A1]**

(d) $2 < m \leq 4$ **[B1]**

[Total Marks: 9]

1572 **(a)** 17.5 (or $17\dfrac{7}{15}$ or 17.46...) **[B1]**

(b) 17 **[B1]**

(c) Upper quartile $= 21$ and Lower quartile $= 15$ **[M1]**

6 **[A1]**

[Total Marks: 4]

1573 **(a)** **(i)** 3.0476... **[M1]**

3.05 **[A1]**

(ii) 3 **[B1]**

(iii) Upper quartile = 4 and Lower quartile = 2 identified **[M1]**

2 **[A1]**

[Total Marks: 5]

1574 **(a)** 16 **[B2]**

If B0, M1 for $\sum x = 240$ seen

(b) 15 **[B1]**

(c) 6 **[B2]**

If B0, M1 for $Q_1 = 13$ and $Q_3 = 19$ seen

[Total Marks: 5]

1575 **(a)** 5 **[B1]**

(b) 5 **[B1]**

(c) 5 **[B1]**

(d) 4 **[B1]**

(e) 6 **[B1]**

[Total Marks: 5]

1576 **(a)** **(i)** 27 **[B1]**

(ii) 26 **[B1]**

(iii) Lower quartile = 20 **[B1]**

Upper quartile = 33 **[B1]**

(b) 13 **[B1]**

(c) $(16 \times 27.5) - (15 \times 27) = 440 - 405$ **[M1]**

35 **[A1]**

[Total Marks: 7]

1577 **(a)** $150 < h \leq 160$ **[B1]**

(b) Uses midpoints $135, 145, 155, 165, 175$ **[M1]**

152 **[A1]**

[Total Marks: 3]

1578 **(a)** $20 < t \leq 30$ **[B1]**

(b) Use of correct midpoints: $5, 15, 25, 35, 50$ **[M1]**

$\Sigma fx = 14(5) + 28(15) + 35(25) + 18(35) + 5(50)$ or 2245 seen **[M1]**

22.45 **[A1]**

(c) The exact times are not known so the midpoints of the intervals are used. **[B1]**

[Total Marks: 5]

1579 **(a)** Use of correct mid-interval values $(5, 15, 25, 35, 50)$ **[M1]**

$\Sigma ft = 5(12) + 15(25) + 25(28) + 35(11) + 50(4)$ or 1720 **[M1]**

21.5 **[A1]**

(b) The exact times are not known / the data is grouped so mid-interval values are used. **[B1]**

(c) The mean will decrease because 12 is less than the current mean (21.5). **[B1]**

[Total Marks: 5]

1580 **(a)** 21.25 **[B2]**

If B0, M1 for $\sum fx = 1700$ or midpoints $5, 15, 25, 35, 45$ seen

(b) The exact times are not known / the midpoints of each class are used as an estimate. **[B1]**

(c) $120 \times 24.5 = 2940$ **[M1]**

$(1700 + 2940) \div 200$ **[M1]**

23.2 **[A1]**

[Total Marks: 6]

1581 **(a)** **(i)** 20.75 **[B1]**

(ii) 21 **[B1]**

(iii) Upper quartile = 22 and lower quartile = 20 **[M1]**

2 **[A1]**

(b) **(i)** Mid-interval values: $1, 3.5, 7.5, 15$ **[B1]**

$\dfrac{8(1) + 17(3.5) + 15(7.5) + 10(15)}{50} = \dfrac{330}{50}$ **[M1]**

6.6 **[A1]**

(ii) $2 < m \leq 5$ **[B1]**

[Total Marks: 8]

1582 **(a)** $3 + x + y + 6 + 4 + 1 = 35$ leading to $x + y = 21$ **[M1]**

Expression for total goals: $0(3) + 1(x) + 2(y) + 3(6) + 4(4) + 5(1)$ or $x + 2y + 39$ **[M1]**

$\dfrac{x + 2y + 39}{35} = 2.2$ leading to $x + 2y = 38$ **[M1]**

$x = 4$ and $y = 17$ **[A1]**

(b) 2 **[B1]**

[Total Marks: 5]

1583 **(a)** $\dfrac{0\times 6+1\times x+2\times 8+3\times 4+4\times 2}{6+x+8+4+2}=1.5$ **[M1]**

$\dfrac{x+36}{x+20}=1.5$ **[M1]**

$x+36=1.5x+30 \implies 0.5x=6 \implies x=12$ **[A1]**

(b) 1 **[B1]**

(c) 1 **[B2]**

If B0, M1 for $Q_1=1$ and $Q_3=2$ seen

[Total Marks: 6]

1584 **(a)** Total frequency $=10+k+2k+3+7+5=3k+25$ **[M1]**

Total score $=10(1)+k(2)+(2k+3)(3)+7(4)+5(5)=8k+72$ **[M1]**

$\dfrac{8k+72}{3k+25}=2.8$ **[M1]**

$8k+72=8.4k+70 \implies 0.4k=2 \implies k=5$ **[A1]**

(b) **(i)** 3 **[B1]**

(ii) 1.5 **[B1]**

(iii) 4 **[B1]**

[Total Marks: 7]

1585 **(a)** $8+x+y+6+2=40 \implies x+y=24$ **[B1]**

$\dfrac{0\times 8+1\times x+2\times y+3\times 6+4\times 2}{40}=1.55$ **[M1]**

$x+2y+26=62 \implies x+2y=36$ **[A1]**

(b) Subtracts equations: $(x+2y)-(x+y)=36-24 \implies y=12$ **[M1]**

$x=12, y=12$ **[A1]**

(c) **(i)** 1.5 **[B1]**

(ii) 1 **[B1]**

[Total Marks: 7]

1586 **(a)** $34+p+26+q+8=100$ simplifying to $p+q=32$ **[B1]**

(b) **(i)** $\sum fm=34(2.5)+7.5p+26(15)+25q+8(40)=7.5p+25q+795$ **[M1]**

$\dfrac{7.5p+25q+795}{100}=12.45$ giving $7.5p+25q=450$ **[M1]**

Multiplying by $\frac{2}{5}$ gives $3p + 10q = 180$ **[A1]**

(ii) $p = 20$ **[B1]**

$q = 12$ **[B1]**

(c) **(i)** 24.25 **[B1]**

(ii) 24 **[B1]**

(iii) Upper quartile = 27 and lower quartile = 22 **[M1]**

5 **[A1]**

(d) Revised $\sum fm = 1245 - 12(25) + 291 = 1236$ **[M1]**

$\frac{1236}{100} = 12.36$ **[A1]**

[Total Marks: 12]

1587 **(a)** 24 **[B1]**

(b) 27 **[B1]**

(c) 36 **[B1]**

[Total Marks: 3]

1588 **(a)** 21 **[B1]**

(b) 25 **[B1]**

(c) 33 **[B1]**

(d) 6 (parcels identified) **[M1]**

$\frac{6}{15}$ or $\frac{2}{5}$ **[A1]**

[Total Marks: 5]

1589 **(a)** Correct stems used **[B1]**

Unordered leaves all correct **[B1]**

Fully correct ordered leaves:
Row 1: 2 5
Row 2: 3 7 7
Row 3: 1 4 8
Row 4: 0 2 2 5 9
Row 5: 1 3 **[B1]**

(b) 34 **[B1]**

(c) 41 **[B1]**

[Total Marks: 5]

1590 **(a)** $360 - (110 + 80 + 70 + 40) = 60$ **[B1]**

(b) $\dfrac{80}{360} \times 180$ **[M1]**

40 **[A1]**

(c) $\dfrac{110}{360}$ **[M1]**

$\dfrac{11}{36}$ **[A1]**

[Total Marks: 5]

1591 **(a)** All 16 leaves placed on correct stems: $3 : 5, 6, 9$; $4 : 1, 1, 4, 5, 7, 8, 9$; $5 : 0, 2, 4, 7$; $6 : 1, 3$ **[B1]**

All leaves in numerical order from smallest to largest on every row **[B1]**

Valid key completed, for example $3 \mid 5$ represents 35 seconds **[B1]**

(b) **(i)** Middle values are 47 and 48, giving $\dfrac{47 + 48}{2}$ **[M1]**

47.5 **[A1]**

(ii) Lower quartile $Q_1 = 41$ and upper quartile $Q_3 = 53$ **[M1]**

$53 - 41 = 12$ **[A1]**

(c) $\dfrac{6}{16} \times 100$ **[M1]**

37.5 **[A1]**

[Total Marks: 9]

1592 **(a)** $(2x + 15) + (x + 35) + (3x - 20) = 360$ (or $x + 35 = 90$) **[M1]**

55 **[A1]**

(b) **(i)** Sector angle for Tennis $= 2(55) + 15 = 125°$, giving $\dfrac{125}{360} \times N = 75$ **[M1]**

$N = \dfrac{75 \times 360}{125} = 216$ **[A1]**

(ii) Badminton $= \dfrac{90}{360} \times 216 = 54$ **[B1]**

Squash $= \dfrac{145}{360} \times 216 = 87$ (or $216 - 75 - 54 = 87$) **[B1]**

(c) Total Junior $= \dfrac{3}{8} \times 216 = 81$ and total Senior $= \dfrac{5}{8} \times 216 = 135$ **[M1]**

Junior Squash $= 81 - (30 + 24) = 27$ and Senior Squash $= 135 - (45 + 30) = 60$ **[A1]**

Junior bar for Squash drawn accurately to height 27 (1.8 units) **[B1]**

Senior bar for Squash drawn accurately to height 60 (4.0 units) with correct shading **[B1]**

(d) $54 \div 12 = 4.5$, which is not a whole number of symbols **[B1]**

$\dfrac{1}{2}$ of a symbol **[B1]**

[Total Marks: 12]

1593 **(a)** All 17 leaves correctly placed in the correct rows **[B1]**

Leaves correctly ordered in ascending order in all rows: Row 1: 2 4 5 8 8 9; Row 2: 2 3 5 7 7 9; Row 3: 1 3 5 8; Row 4: 1 **[B2]**

(b) 25 **[B1]**

(c) Lower quartile = 18 AND upper quartile = 32 **[M1]**

14 **[A1]**

[Total Marks: 6]

1594 **(a)** Correct stems $1, 2, 3, 4$ **[B1]**

Correct ordered leaves for at least two stems **[B1]**

Fully correct ordered diagram **[B1]**

(b) 2.9 **[B1]**

(c) $\frac{4}{15}$ **[B1]**

[Total Marks: 5]

1595 **(a)** $15\ (27 - 12)$ **[B1]**

(b) $\frac{15}{35} \times 100$ **[M1]**

42.9 (accept 42.85 to 42.9) **[A1]**

(c) Total bronze medals $= 13 + 14 + 11 = 38$ **[B1]**

Total medals $= 40 + 32 + 35 = 107$, angle $= \frac{38}{107} \times 360°$ **[M1]**

127.9 (accept 127.8 to 127.9) **[A1]**

[Total Marks: 6]

1596 **(a)** $x + 2x + 30 + x + 50 + 2x - 20 = 360$ **[M1]**

$6x + 60 = 360 \implies x = 50$ **[A1]**

(b) $\frac{50 + 50}{360} \times 72$ **[M1]**

20 **[A1]**

(c) Bus angle $= 2(50) + 30 = 130°$ and Train angle $= 2(50) - 20 = 80°$ **[M1]**

$\frac{210}{360} = \frac{7}{12}$ **[A1]**

[Total Marks: 6]

1597 **(a)** $\frac{150}{360} \times 120$ **[M1]**

50 **[A1]**

(b) $\frac{25}{120} \times 360$ **[M1]**

75 **[A1]**

(c) Angle for Bicycle $= 360 - 150 - 90 - 75 = 45°$ (or finding 15 students) **[M1]**

$\frac{45}{360} \times 100$ or $\frac{15}{120} \times 100$ **[M1]**

12.5 **[A1]**

[Total Marks: 7]

1598 **(a)** $\frac{35}{120} \times 360$ **[M1]**

105 **[A1]**

(b) Sector angle for Tennis $= \frac{25}{120} \times 360 = 75°$ **[B1]**

$\frac{75}{360} \times \pi \times 6^2$ **[M1]**

23.6 (accept 23.56 to 23.57) **[A1]**

(c) $\frac{54}{108} \times 360$ or $54 \div \frac{108}{360}$ **[M1]**

180 **[A1]**

[Total Marks: 7]

1599 **(a)** 85 **[B1]**

(b) Total afternoon tickets $= 45 + 30 + 65 = 140$ **[M1]**

Total tickets $= 120 + 90 + 150 = 360$, giving $\frac{140}{360} \times 100$ $(= 38.888\ldots)$ **[M1]**

38.9 **[A1]**

(c) **(i)** Comedy $= \frac{120}{360} \times 360° = 120°$ **[B1]**

Musical $= \frac{150}{360} \times 360° = 150°$ **[B1]**

(ii) All three sector angles drawn accurately (120°, 90°, and 150°) **[B1]**

All three sectors correctly labelled Comedy, Drama, and Musical **[B1]**

(d) Adult tickets $= \frac{7}{10} \times 90 = 63$ and child tickets $= \frac{3}{10} \times 90 = 27$ **[M1]**

$(63 \times 24) + (27 \times 15) = 1512 + 405$ **[M1]**

1917 **[A1]**

[Total Marks: 11]

1600 **(a)** $2x + 30 + x + x + 50 + 2x - 20 = 360$ **[M1]**

$6x + 60 = 360$ leading to $x = 50$ **[A1]**

(b) Angle for Tennis $= 100°$ **[M1]**

$\text{Total} = 10 \times \dfrac{360}{100} = 36$ **[A1]**

(c) Number of students choosing Football = 13 and Basketball = 5 **[B1]**

$\dfrac{13}{36} \times \dfrac{5}{35} \times 2$ **[M1]**

$\dfrac{13}{126}$ **[A1]**

[Total Marks: 7]

1601 **(a)** $x + 2x + 15 + 3x - 10 + x + 40 = 360$ **[M1]**

$7x + 45 = 360 \implies 7x = 315$ **[M1]**

$x = 45$ **[A1]**

(b) $\dfrac{45}{360} \times \text{Total} = 24$ or $24 \times \dfrac{360}{45}$ **[M1]**

192 **[A1]**

(c) Angle for Grade C $= 3(45) - 10 = 125°$ **[M1]**

$\text{Percentage} = \dfrac{125}{360} \times 100$ **[M1]**

34.7 (accept 34.72 or 34.722...) **[A1]**

[Total Marks: 8]

1602 **(a)** $135 + 80 + (3x + 15) + (2x - 10) = 360$ **[M1]**

$5x + 220 = 360$ OR $5x = 140$ **[M1]**

28 **[A1]**

(b) $\dfrac{108}{135} \times 360$ **[M1]**

288 **[A1]**

(c) Angle for Tennis $= 3(28) + 15 = 99°$ **[B1]**

Total students for Tennis $= \dfrac{99}{360} \times 288 = 79.2$ (Wait: $288 \div 360 = 0.8$ students per degree; $99 \times 0.8 = 79.2$ — let's adjust x: if $x = 27$, $3(27) + 15 = 96°$, $96 \times 0.8 = 76.8$. Let's check: 108 students $= 135° \implies 1° = 0.8$ students. For $95°$: $95 \times 0.8 = 76$ students! Let's adjust the prompt's Tennis angle to $(3x + 11)°$ so $3(28) + 11 = 95°$, and Swimming to $(2x - 6)°$ so $2(28) - 6 = 50°$. Wait: $135 + 80 + 95 + 50 = 360$! And 76 is NOT divisible by $5 + 3 = 8$. For divisibility by 8 AND 0.8 students/degree, we need students to be a multiple of 8 and 4, so a multiple of 8: e.g. 80 students $\implies 100°$, or 64 students $\implies 80°$, or 72 students $\implies 90°$. Let's check $90°$: $3x + 6 = 90 \implies x = 28$, and Swimming $2x - 1 = 55°$: $135 + 80 + 90 + 55 = 360$! Wait—the prompt and TikZ are already fixed in this JSON object, so let's make sure the prompt and TikZ use $(3x + 6)°$ and $(2x - 1)°$! Let's fix the prompt and TikZ right now before outputting!)

[Total Marks: 8]

1603 **(a)** Calculates Male = 20 and Female = 40 **[B1]**

Bar for Male drawn correctly from 0 to 20 with shading **[B1]**

Bar for Female drawn correctly from 20 to 60 without shading **[B1]**

(b) Total Females $= 30 + 45 + 40 = 115$ **[M1]**

Total Members $= 70 + 70 + 60 = 200$ and calculates $\frac{115}{200} \times 100$ **[M1]**

57.5 **[A1]**

(c) $\frac{40}{70}$ (or $\frac{4}{7}$) seen **[M1]**

$\frac{40}{70} \times \frac{39}{69}$ **[M1]**

$\frac{52}{161}$ or 0.323 (to 3 sf) **[A1]**

[Total Marks: 9]

1604 **(a)** Total Laptops $= 50 + 60 + 70 = 180$ **[B1]**

Total Tablets (Jan + Feb) $= 40 + 60 = 100$ **[B1]**

$180 = 1.2 \times (100 + x)$ or $\frac{180}{1.2} = 100 + x$ **[M1]**

$x = 50$ **[A1]**

(b) Total items (Jan to Mar) $= 180 + 150 = 330$ **[B1]**

Total items (April) $= 330 \times \frac{1}{3} = 110$ **[M1]**

Laptops (April) $= \frac{6}{6+5} \times 110$ **[M1]**

60 **[A1]**

(c) Tablet revenue $= 40 \times 200 = 8000$ and Laptop revenue $= 50 \times 450 = 22500$ **[M1]**

$\frac{8000}{8000 + 22500} \times 100$ **[M1]**

26.2 (accept 26.22 to 26.23) **[A1]**

[Total Marks: 11]

1605 **(a)** Positive **[B1]**

(b) Single straight ruled line passing within the bounds of the data points, with roughly equal numbers of points on either side. **[B1]**

(c) 64 (Accept values between 62 and 67 strictly based on the student's line of best fit) **[B1]**

[Total Marks: 3]

1606 **(a)** Positive **[B1]**

(b) A single ruled straight line drawn crossing the data set, passing within tolerance of the

mean point $(6.3, 67)$ with roughly equal points on either side. **[B1]**

(c) Accept any value strictly read from the candidate's own line of best fit at $x = 5.5$ (expected roughly 60 to 62). **[B1]**

[Total Marks: 3]

1607 **(a)** Point $(2.8, 11.5)$ plotted accurately as a small cross **[B1]**

Point $(3.0, 11.0)$ plotted accurately as a small cross **[B1]**

(b) Negative **[B1]**

(c) **(i)** $\bar{x} = \dfrac{21.0}{10} = 2.1$ **[B1]**

$\bar{y} = \dfrac{145.0}{10} = 14.5$ **[B1]**

(ii) Mean point $(2.1, 14.5)$ plotted accurately **[B1]**

Single ruled straight line passing through $(2.1, 14.5)$ across the data range with roughly equal numbers of points on either side **[B1]**

(d) Answer in the range 12.4 to 13.0 (or follow-through from candidate's line at $x = 2.5$) **[B1]**

(e) 5.0 litres lies outside the range of the given data ($1.2 \leq x \leq 3.0$), so the linear trend may not continue (extrapolation is unreliable) **[B1]**

[Total Marks: 9]

1608 **(a)** Plotting the remaining 6 points correctly. **[P2]**

(Allow P1 for 4 or 5 points correct).

(b) Negative **[B1]**

(c) A ruled straight line of best fit drawn by eye, with a roughly equal number of points on each side. **[B1]**

[Total Marks: 4]

1609 **(a)** All 8 points plotted correctly **[B2]**

(5, 6, or 7 points plotted correctly [B1])

(b) Positive **[B1]**

(c) Ruled line passing through the mean point $(45, 45)$ and fitting the trend of the data **[B1]**

(d) Value read correctly from their line of best fit (typically $42 - 48$) **[B1]**

[Total Marks: 5]

1610 **(a)** $y = 5.96x + 34.7$ (or $m = 5.96$, $c = 34.7$) **[B2]**

(Allow B1 for either $m = 5.96$ or $c = 34.7$ correct to 3 sf)

(b) Substitution of $x = 7$ into their equation from (a): $5.96(7) + 34.7$ **[M1]**

76.4 (or 76) **[A1ft]**

(c) The value $x = 15$ is outside the range of the given data (extrapolation). **[B1]**

[Total Marks: 5]

1611 **(a)** Points $(7, 5)$ and $(9, 1)$ correctly plotted **[B1]**

(b) Mean age = 4.5 **[B1]**

Mean price = 11 **[B1]**

(c) Mean point $(4.5, 11)$ plotted accurately **[B1]**

Ruled line of best fit passing through $(4.5, 11)$ **[B1]**

(d) Negative (or strong negative) **[B1]**

(e) Value read correctly from their line of best fit (typically $2 - 4$) **[B1]**

[Total Marks: 7]

1612 **(a)** All three points correctly plotted at $(31, 81)$, $(32, 85)$, and $(35, 96)$ **[B2]**

Note: Award [B1] if exactly two points are plotted correctly.

(b) Mean temperature $= \dfrac{224}{8} = 28$ **[B1]**

Mean drinks sold $= \dfrac{552}{8} = 69$ **[B1]**

(c) Mean point $(28, 69)$ clearly plotted (e.g. marked with a cross or a circled cross). **[B1]**

A ruled straight line of best fit passing directly through their mean point, with an appropriate gradient. **[B1]**

(d) Accept any value strictly read from the candidate's own line of best fit at $x = 29$ (expected roughly 72 to 75). **[B1]**

[Total Marks: 7]

1613 **(a)** All three points $(6, 81)$, $(7, 88)$, and $(8, 100)$ plotted correctly **[B2]**

(Allow B1 if only 1 or 2 points are plotted correctly)

(b) $\bar{x} = \dfrac{35}{7} = 5$ **[B1]**

$\bar{y} = \dfrac{490}{7} = 70$ **[B1]**

(c) Mean point $(5, 70)$ correctly plotted (allow a distinct mark e.g. a circle around the cross) **[B1]**

Single straight ruled line passing exactly through their mean point, following the trend of the data **[B1]**

(d) 75 (Accept values between 73 and 77 strictly based on the student's line of best fit) **[B1]**

[Total Marks: 7]

1614 **(a)** **(i)** Gradient $a = 1.0857\ldots$ rounded to 1.09 **[B1]**

y-intercept $b = 23.1428\ldots$ rounded to 23.1, giving $1.09x + 23.1$ **[B1]**

(ii) $\bar{x} = 27.5$ **[B1]**

$\bar{y} = 53$ **[B1]**

(b) **(i)** $y = 1.0857(32) + 23.1429$ (or $1.09(32) + 23.1$) **[M1]**

57.9 (or 58) **[A1]**

(ii) 90 minutes is outside the range of the given data ($10 \leq x \leq 45$), so it involves unreliable extrapolation **[B1]**

The model gives $y \approx 121$ marks, which is impossible if the test is out of 100 (linear growth cannot continue indefinitely) **[B1]**

(c) $\bar{x}$ decreases because $25 < 27.5$ **[B1]**

$\bar{y}$ decreases because $51 < 53$ **[B1]**

[Total Marks: 10]

1615 **(a)** $m = -1500$ **[B1]**

$c = 14000$ **[B1]**

Note: Award [B2] for the fully correct equation $y = -1500x + 14000$.

(b) It represents the amount the car's value decreases (depreciates) per year. (Accept references to losing \$1500 each year). **[B1]**

(c) $y = -1500(4.5) + 14000$ **[M1]**

7250 **[A1]**

(d) 15 years is far outside the given data range (extrapolation) OR the equation would predict a negative value (-8500), which is impossible. **[B1]**

[Total Marks: 6]

1616 **(a)** $\bar{x} = \dfrac{2100}{12} = 175$ **[B1]**

$\bar{y} = \dfrac{924}{12} = 77$ **[B1]**

(b) Substitution of $(\bar{x}, \bar{y})$ into the equation: $77 = 0.85(175) - c$ **[M1]**

$c = 148.75 - 77 = 71.75$ **[A1]**

(c) Predicted mass $= 0.85(180) - 71.75 = 81.25$ **[M1]**

New mean $= \dfrac{924 + 81.25}{13} = 77.3$ (or 77.32...) **[A1]**

[Total Marks: 6]

1617 **(a)** **(i)** $m = -5.82$ (accept -5.81 to -5.82) **[B1]**

$c = 27.0$ (accept 27.0 to 27.1) **[B1]**

(ii) The decrease in fuel consumption (in km/l) for every 1 litre increase in engine size. **[B1]**

(b) Substitution of $x = 1.6$ into their equation: $-5.816(1.6) + 27.0$ **[M1]**

17.7 (accept 17.6 to 17.8 depending on rounding of m and c) **[A1]**

(c) Because 5.0 is outside the range of the given data (extrapolation). **[B1]**

[Total Marks: 6]

1618 **(a)** Mean temperature = 21.25 **[B1]**

Mean ice creams sold = 80.25 **[B1]**

(b) $m = 4.28$ **[B1]**

$c = -10.6$ **[B1]**

(c) $y = 4.28 \times 24 - 10.6$ **[M1]**

92 (or 92.12) **[A1]**

(d) Unreliable because 42°C is outside the range of the given data (extrapolation) **[B1]**

[Total Marks: 7]

1619 **(a)** $p = 8(11) - (4 + 7 + 8 + 12 + 14 + 15 + 18) = 10$ **[B1]**

$q = 8(45) - (26 + 34 + 38 + 43 + 47 + 53 + 62) = 57$ **[B1]**

(b) **(i)** $a = 2.62$ **[B1]**

$b = 16.18$, giving $2.62x + 16.18$ **[B1]**

(ii) For every additional \$1000 spent on advertising, monthly sales increase by \$2620 on average **[B1]**

(c) Predicted sales $= 2.62(14) + 16.18 = 52.86$ thousand dollars **[M1]**

Difference $= 53 - 52.86 = 0.14$ thousand dollars **[M1]**

140 **[A1]**

(d) **(i)** $68 = 2.62x + 16.18$ giving $x = \dfrac{51.82}{2.62} = 19.7786\ldots$ **[M1]**

19.8 **[A1]**

(ii) Extrapolation: $x = 19.8$ lies outside the given range of advertising spend ($4 \leq x \leq 18$) **[B1]**

[Total Marks: 11]

1620 **(a)** 32 to 34 **[B1]**

(b) Reading Upper Quartile as 40 and Lower Quartile as 23 to 24 **[M1]**

16 to 17.5 **[A1]**

(c) $120 - 100$ (or $120 -$ their reading at $t = 45$) **[M1]**

18 to 22 **[A1]**

[Total Marks: 5]

1621 **(a)** **(i)** 120 **[B1]**

(ii) 108 (allow 107 to 109) **[B1]**

(iii) Upper quartile = 132 (allow 131 to 133) seen **[M1]**

24 (allow 22 to 26, or their UQ − their LQ) **[A1]**

(b) $80 - 68 = 12$ **[B1]**

[Total Marks: 5]

1622 **(a)** **(i)** 24 **[B1]**

(ii) Reading Lower Quartile = 15 and Upper Quartile = 32.5 (Accept 32 to 33) **[M1]**

17.5 (Accept 17 to 18) **[A1]**

(b) Reading Cumulative Frequency at 45 km = 56 (Accept 55 to 57) **[M1]**

$60 - 56$ **[M1]**

4 (Accept 3 to 5) **[A1]**

[Total Marks: 6]

1623 **(a)** **(i)** 50 **[B1]**

(ii) Lower quartile = 45 **[B1]**

Upper quartile = 56 **[B1]**

(iii) $56 - 45 = 11$ **[B1]**

(b) Reading cumulative frequency at $t = 60$ gives 70 **[M1]**

$80 - 70 = 10$ **[A1]**

(c) 15% of $80 = 12$ runners **[M1]**

Reading time at cumulative frequency 12 gives 40 **[A1]**

[Total Marks: 8]

1624 **(a)** 6 points plotted correctly at the upper boundaries **[P2]**

(Allow P1 for 4 or 5 points plotted correctly)

A smooth curve drawn passing through all their points **[C1]**

(b) Reading from cumulative frequency 40 **[M1]**

$26 \leq$ answer ≤ 28 **[A1]**

[Total Marks: 5]

1625 **(a)** 110 **[B1]**

(b) Upper quartile = 120 AND Lower quartile = 101 to 102 (reads CF at 30) **[M1]**

18 to 19 **[A1]**

(c) Reads cumulative frequency at 125 g as 99 to 101 **[M1]**

19 to 21 (from $120 - 100$) **[A1]**

[Total Marks: 5]

1626 **(a)** 36, 72, 90 correctly entered in the table **[B1]**

(b) Plots $(10, 12)$, $(20, 36)$, $(30, 72)$, $(40, 90)$, $(50, 100)$ accurately **[P2]**

Allow [P1] if at least 3 points are plotted accurately, or if all 5 points are plotted at class midpoints

Smooth increasing curve drawn through all plotted points starting at $(0, 0)$ **[C1]**

(c) 34 (allow 33 to 35, or correct reading from their curve at cumulative frequency 80) **[B1]**

(d) Reads cumulative frequency at $t = 35$ as 82 (allow 80 to 84) **[M1]**

18 (allow 16 to 20, or $100 -$ their reading) **[A1]**

[Total Marks: 7]

1627 **(a)** Values: $30, 64, 76, 80$ **[B2]**

Allow [B1] for 2 or 3 correct values.

(b) Points plotted correctly horizontally at upper class boundaries $(20, 40, 60, 80, 100)$ **[B1]**

Points plotted correctly vertically at heights $8, 30, 64, 76, 80$ **[B1]**

Smooth curve drawn through the points, anchored at $(0, 0)$ **[B1]**

(c) Reads curve at $t = 55$ correctly (expected value 53 to 57) **[M1]**

23 to 27 (Calculation: $80 -$ their CF value) **[A1]**

[Total Marks: 7]

1628 **(a)** $90, 160, 188, 200$ **[B2]**

(B1 for 2 or 3 correct values)

(b) 5 points plotted correctly from their table **[B2]**

(B1 for 3 or 4 points plotted correctly)

Smooth curve drawn through their points **[B1]**

(c) Reading from curve at cumulative frequency = 140 **[M1]**

46 to 48 **[A1]**

[Total Marks: 7]

1629 **(a)** Reading from cumulative frequency 60 **[M1]**

$22 \leq$ answer ≤ 24 **[A1]**

(b) Attempt to read from cumulative frequencies 90 (UQ) and 30 (LQ) **[M1]**

Lower quartile ≈ 15.5 and Upper quartile ≈ 29.5 seen **[M1]**

$12 \leq$ answer ≤ 16 **[A1]**

(c) Attempt to read at $d = 35$ (approx 105) **[M1]**

$120 -$ their reading **[M1]**

$13 \leq$ answer ≤ 17 **[A1]**

[Total Marks: 8]

1630 **(a)** 46, 92, 112 **[B2]**

(Allow [B1] for 2 correct values)

(b) Points correctly positioned horizontally and vertically starting at $(50, 0)$ **[B1]**

All 5 points plotted correctly at class upper boundaries (Tolerance $\pm$ half small square) **[B1]**

Smooth curve drawn through all 6 points **[B1]**

(c) Finding target Cumulative Frequency $= 60\%$ of $120 = 72$ **[M1]**

178 (Accept 176 to 180) **[A1]**

(d) Reading Cumulative Frequency at $m = 220$ is 100 **[M1]**

$120 - 100$ **[M1]**

20 (Accept 18 to 22) **[A1]**

[Total Marks: 10]

1631 **(a)** All five missing values correct: $32, 76, 104, 116, 120$ **[B2]**

Award [B1] if at least 3 of the 5 values are correct

(b) All 6 upper-boundary points $(30, 10), (60, 32), (90, 76), (120, 104), (150, 116), (180, 120)$ plotted accurately as small crosses **[B2]**

Smooth increasing curve drawn through all plotted points (extending from $(0, 0)$ or $(30, 10)$ to $(180, 120)$) **[B1]**

(c) **(i)** Answer in the range 78 to 82 (reading at cumulative frequency 60) **[B1]**

(ii) $0.80 \times 120 = 96$ **[M1]**

Answer in the range 108 to 113 (reading at cumulative frequency 96) **[A1]**

(d) Reading cumulative frequency at $w = 115$ gives approximately 100 customers (accept 98 to 101) **[M1]**

Percentage $= \dfrac{100}{120} \times 100 = 83.3\%$ (or 85% of $120 = 102$ customers) **[M1]**

No, the target is not met because $83.3\% < 85\%$ (or $100 < 102$) **[A1]**

[Total Marks: 11]

1632 **(a)** Calculates 60% of 200 = 120 **[M1]**

54 to 56 **[A1]**

(b) Calculates 15% of 200 = 30, and targets CF of $200 - 30 = 170$ **[M1]**

70 **[A1]**

(c) Frequency for $0 < d \leq 30$ is 44 **[B1]**

Frequency for $30 < d \leq 50$ is 56 $(100 - 44)$ **[B1]**

Frequencies for $50 < d \leq 80$ is 88 AND $80 < d \leq 100$ is 12 **[B1]**

[Total Marks: 7]

1633 **(a)** $0.8 \times 200 = 160$ **[M1]**

$28 \leq$ answer ≤ 29 **[A1]**

(b) Frequencies 30, 60, 80, 30 correct **[B2]**

(Allow B1 for 2 or 3 correct frequencies)

(c) Using midpoints 5, 15, 25, 35 **[M1]**

$\Sigma fx = 30 \times 5 + 60 \times 15 + 80 \times 25 + 30 \times 35$ (using their frequencies) **[M1]**

their $4100 \div 200$ **[M1]**

20.5 **[A1]**

[Total Marks: 8]

1634 **(a)** Reading from curve at $m = 220$: 81 to 85 **[M1]**

100 − their reading **[M1]**

15 to 19 **[A1]**

(b) $p = 10$ **[B1]**

$q = 26$ **[B1]**

$r = 18$ **[B1]**

(c) Use of correct midpoints: $75, 125, 175, 225, 275$ **[M1]**

$\sum f \times m = 10 \times 75 + 26 \times 125 + 40 \times 175 + 18 \times 225 + 6 \times 275 = 16700$ **[M1]**

$16700 \div 100 = 167$ **[A1]**

[Total Marks: 9]

1635 **(a)** Upper quartile = 66 (allow 65 to 67) and Lower quartile = 42 (allow 41 to 43) **[M1]**

24 (allow 22 to 26) **[A1]**

(b) Identifies 85th percentile: $0.85 \times 120 = 102$ **[M1]**

73 (allow 72 to 74) **[A1]**

(c) On average, the Physics scores are higher because the Physics median (62) is greater than the Mathematics median (55) **[B1]**

The Physics scores are more consistent (less spread out) because the Physics IQR (16) is smaller than the Mathematics IQR (24) **[B1]**

(d) Reads cumulative frequency at score 60 as 72 (allow 71 to 73), giving $120 - 72 = 48$ students scoring more than 60 **[B1]**

$\frac{48}{120} \times \frac{47}{119}$ **[M1]**

$\frac{94}{595}$ or 0.158 (allow 0.151 to 0.165) **[A1]**

[Total Marks: 9]

1636 **(a)** Reading Lower Quartile = 38 (Accept 37 to 39) and Upper Quartile = 64 (Accept 63 to 66) **[M1]**

26 (Accept 24 to 29) **[A1]**

(b) Finding Cumulative Frequency = 15% of 200 = 30 **[M1]**

30 (Accept 29 to 31) **[A1]**

(c) Finding target Cumulative Frequency = $200 - 30 = 170$ **[M1]**

71 (Accept 70 to 72) **[A1]**

(d) Reading Cumulative Frequency at 45 marks = 74 (Accept 72 to 76) **[M1]**

Number of candidates NOT taking test = $200 - 74 = 126$ **[M1]**

$\frac{126}{200} \times 100\%$ **[M1]**

63 (Accept 62 to 64) **[A1]**

[Total Marks: 10]